

Solutions to Lars Ljungqvist and Thomas J. Sargent,
Recursive Macroeconomic Theory

Fourth edition (MIT Press, 2018)

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All 254 exercises of the 23 chapters that have exercises (Chapters 1, 4, 10, 12, 20, 25, 30 and 31 have none).

Solutions only: the exercise statements are not reproduced. Read them in the book.

Authorship

These solutions were produced by **Zijun Meng** and **Claude Opus 5**.

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The exercise statements are not reproduced here: read them in Ljungqvist and Sargent (2018). Notation, parameter values, part labels such as (a) or (ii), and equation numbers such as (1) that belong to an exercise are those of the exercise in the book. Equations, sections, figures and programs cited as (16.5.9) or Figure 11.9.4 refer to Ljungqvist and Sargent (2018). The figures of these solutions are numbered $SN.k$ (chapter N). Numerical parts are computed by the Python scripts in the `code/` folder that accompanies this document, which replace the book's Matlab programs. Where an exercise leaves parameters open, the solution states its choices. Apparent errors in the book are flagged in remarks where they arise.

Chapter 2

Time Series

Solutions to all 30 exercises of Chapter 2 of Lars Ljungqvist and Thomas J. Sargent, *Recursive Macroeconomic Theory* (4th edition, MIT Press, 2018). Equation numbers such as (2.4.10) refer to the book. Wherever an exercise asks for a Matlab program from the book's web site, we use the Python program `code/ch2.py`, which implements the same algorithms (`doublej`, the Howard improvement algorithm, the Kalman filter). Every number reported below is produced by that program.

Notation: a Markov chain (P, π_0) lives on the unit vectors $e_1, \dots, e_n$; $P_{ij} = \mathbb{P}(x_{t+1} = e_j \mid x_t = e_i)$ and $\pi'_t = \pi'_0 P^t$. A linear state-space system is $x_{t+1} = Ax_t + Cw_{t+1}$, $y_t = Gx_t$, with $Ew_{t+1}w'_{t+1} = I$.

Exercise 2.1 *Likelihood of short histories*

Since $\bar{y}$ has distinct entries, a history of y reveals the history of states, and $\mathbb{P}(x_0, \dots, x_4) = \pi_{0,x_0} \prod_{t=1}^4 P_{x_{t-1}x_t}$.

- (a) States 1, 2, 1, 2, 1: $.5 \times P_{12}P_{21}P_{12}P_{21} = .5 \times (.1 \times .3)^2 = 0.00045$.
- (b) States 1, 1, 1, 1, 1: $.5 \times .9^4 = 0.32805$.
- (c) States 2, 2, 2, 2, 2: $.5 \times .7^4 = 0.12005$.

The chain is persistent (large diagonal elements), so constant histories are far more likely than alternating ones.

Exercise 2.2 *Recovering P from conditional moments*

By (2.2.8), $E(h(x_{t+1}) \mid x_t) = Ph$ for any function h of the state. With $h_1 = \bar{y} = (1, 5)'$ and $h_2 = (1, 25)'$ (the squares), $P[h_1 \ h_2] = J$ where $J = \begin{bmatrix} 1.8 & 5.8 \\ 3.4 & 15.4 \end{bmatrix}$. The matrix $h = [h_1 \ h_2] = \begin{bmatrix} 1 & 1 \\ 5 & 25 \end{bmatrix}$ is nonsingular ($\det h = 20$), so, as in section 2.2.6,

$$P = Jh^{-1} = \begin{bmatrix} 1.8 & 5.8 \\ 3.4 & 15.4 \end{bmatrix} \frac{1}{20} \begin{bmatrix} 25 & -1 \\ -5 & 1 \end{bmatrix} = \begin{bmatrix} .8 & .2 \\ .4 & .6 \end{bmatrix}.$$

Its rows are nonnegative and sum to one, so it is a transition matrix. It is *unique*: $P = Jh^{-1}$ is the only solution of $Ph = J$. In fact the first moments alone already pin P down, because each row has two unknowns and satisfies two equations, $P_{i1} + P_{i2} = 1$ and $P_{i1} + 5P_{i2} = E(y_{t+1} \mid x_t = e_i)$. The second moments are then an overidentifying check, and they pass it ($.8 + 25 \times .2 = 5.8$, $.4 + 25 \times .6 = 15.4$).

Exercise 2.3 *Valuing Markov consumption processes; Bayesian model choice*

(a) By the law of iterated expectations and the Markov property, $v_i = u_i + \beta E[\sum_{t \geq 1} \beta^{t-1} u(c_t) \mid x_0 = e_i] = u_i + \beta \sum_j P_{ij} v_j$, i.e. $v = u + \beta P v$. Because P is stochastic, $\|\beta P\|_\infty = \beta < 1$, so $I - \beta P$ is invertible, with $(I - \beta P)^{-1} = \sum_t \beta^t P^t$, and $v = (I - \beta P)^{-1} u$. (Equivalently, $E[u(c_t) \mid x_0 = e_i] =$

$(P^t u)_i$ by (2.2.10), and summing gives the same formula.) Averaging over the initial state gives $V = E v(x_0) = \sum_i \pi_{0,i} v_i$.

(b) With $u_1 = u(1)$ and $u_2 = u(5)$:

	$\gamma = 2.5$		$\gamma = 4$	
	v'	V	v'	V
process 1 ($P = I$)	$(-13.333, -1.193)$	-7.2630	$(-6.667, -0.053)$	-3.3600
process 2 (i.i.d.)	$(-7.566, -6.959)$	-7.2630	$(-3.525, -3.195)$	-3.3600

The consumer is *indifferent* in both cases, and this is no accident. $V = \sum_t \beta^t E u(c_t)$ depends only on the *marginal* distributions of c_t . With $\pi_0 = (.5, .5)'$, both chains have $\pi_t = (.5, .5)'$ for every t (both matrices leave $(.5, .5)$ invariant). So $V = \frac{1}{1-\beta} \cdot \frac{u(1)+u(5)}{2}$ for both, whatever γ and β . Time-separable expected utility is blind to the serial correlation of consumption: process 1 is a single gamble resolved at date 0, and process 2 is a fresh gamble every period. Their conditional values differ (state 1 is much worse under the permanent process), but ex ante they are the same.

(c) Under process 1 the chain never leaves its initial state: $\mathbb{P}(\text{data} \mid M_1) = \pi_{0,1} \cdot 1^9 = .5$. Under process 2 each observation is an independent coin flip: $\mathbb{P}(\text{data} \mid M_2) = .5^{10} = 1/1024 \approx 0.000977$.

(d) $\mathbb{P}(M_1 \mid \text{data}) = \frac{.5 \cdot .5}{.5 \cdot .5 + .5 \cdot .5^{10}} = \frac{512}{513} = 0.99805$ and $\mathbb{P}(M_2 \mid \text{data}) = 1/513 = 0.00195$.

(e) The sample switches states, which is impossible under $P = I$, so $\mathbb{P}(\text{data} \mid M_1) = 0$ while $\mathbb{P}(\text{data} \mid M_2) = .5^{10}$. The posterior puts probability 1 on process 2 and 0 on process 1.

Exercise 2.4 A fourth-order autoregression

(a) Let $x_t = (y_t, y_{t-1}, y_{t-2}, y_{t-3}, 1)'$. Then $x_{t+1} = Ax_t + Cw_{t+1}$ and $y_t = Gx_t$ with

$$A = \begin{bmatrix} \rho_1 & \rho_2 & \rho_3 & \rho_4 & \alpha \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} c \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

(b) Apart from the unit root that carries the constant, the eigenvalues of A are those of the 4×4 companion block. When they lie inside the unit circle, the stationary mean is the eigenvector of A for the unit eigenvalue, normalized to have last entry 1. Because $\alpha = \mu(1 - \sum \rho_j)$, it is $(\mu, \mu, \mu, \mu, 1)'$, so $Ey = \mu$. The stationary covariance Σ_∞ solves $\Sigma = A\Sigma A' + CC'$ ((2.4.10); computed with a doubling algorithm, as `doublej.m` does). Covariance stationarity requires the initial condition $(y_0, y_{-1}, y_{-2}, y_{-3})$ to have mean $(\mu, \dots, \mu)$ and covariance equal to the upper-left 4×4 block of Σ_∞ , i.e. $\text{cov}(y_{-i}, y_{-j}) = \gamma(|i - j|)$.

case	largest $ \lambda $ (AR block)	stationary?	Ey_t	$\text{var}(y_t)$
(i)	0.845	yes	10	7.4286
(ii)	0.845	yes	10	29.7143
(iii)	0.900	yes	5	5.2632
(iv)	0.896	yes	5	1.4764
(v)	1.078	no	—	—

Checks: for an AR(2), $\text{var}(y) = \frac{(1-\rho_2)c^2}{(1+\rho_2)[(1-\rho_2)^2-\rho_1^2]}$, which gives $1.3/(0.7 \times 0.25) = 7.4286$ in case (i), and four times that in case (ii). For an AR(1), $\text{var}(y) = 1/(1 - .81) = 5.2632$. In case (v), $\rho_1 + \rho_2 = 1.1 > 1$ and the companion block has the root 1.078. The shock loads on the explosive direction, so the variance grows without bound for *every* initial condition, and no covariance-stationary solution exists. (Case (v) violates the exercise's own maintained assumption that A is stable.)

(c) $E_t y_{t+5} = GA^5 x_t$, so $(h_0, h_1, h_2, h_3, \gamma_0) = GA^5$:

case	h_0	h_1	h_2	h_3	γ_0
(i), (ii)	0.73872	-0.26028	0	0	5.2156
(iii)	0.59049	0	0	0	2.0475
(iv)	0.20032	0.02000	0.00400	0.25080	2.6244
(v)	1.15808	0.32268	0	0	-2.4038

(The forecast does not depend on c . As a check, at $y_{t-j} \equiv \mu$ the forecast equals μ : $(0.73872 - 0.26028) \times 10 + 5.2156 = 10$.)

(d) By (2.4.18), $E_t \sum_k .95^k y_{t+k} = G(I - .95A)^{-1} x_t$, provided the eigenvalues of $.95A$ are inside the unit circle. The last component of $G(I - .95A)^{-1}$ is a constant. The exercise's formula omits it, but it is needed unless $\mu = 0$.

case	$\tilde{h}_0$	$\tilde{h}_1$	$\tilde{h}_2$	$\tilde{h}_3$	constant
(i), (ii)	7.64818	-2.17973	0	0	145.3155
(iii)	6.89655	0	0	0	65.5172
(iv)	2.48295	1.06441	1.12043	1.17940	70.7641

At $y \equiv \mu$ each row gives $\mu/(1 - .95)$; e.g. $(7.64818 - 2.17973) \times 10 + 145.3155 = 200$. In case (v), $.95 \times 1.078 = 1.024 > 1$, so the expected discounted sum diverges and $\tilde{h}$ does not exist.

(e) Under the stationary distribution, $E(x_t - \mu)(x_{t-k} - \mu)' = A^k \Sigma_\infty$ ((2.4.11)), so the autocovariance of y is $GA^k \Sigma_\infty G'$:

case	$k = 1$	$k = 5$	$k = 10$
(i)	6.8571	3.7029	1.5976
(ii)	27.4286	14.8114	6.3903
(iii)	4.7368	3.1078	1.8351
(iv)	0.4159	0.3652	0.1316
(v)	not defined (no stationary distribution)		

In case (iv) the coefficient $\rho_4 = .5$ creates a strong four-period “seasonal” echo, so the autocovariance at lag 5 is almost as large as at lag 1.

Exercise 2.5 *Expected discounted quadratic utility*

(a) Stack $x_t = (c_t, c_{t-1}, z_t, z_{t-1}, 1)'$, so that $x_{t+1} = Ax_t + Cw_{t+1}$ with

$$A = \begin{bmatrix} \rho_1 & \rho_2 & \delta_1 & \delta_2 & \alpha_c \\ 1 & 0 & 0 & 0 & 0 \\ \gamma_1 & \gamma_2 & \phi_1 & \phi_2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} \psi_1 & 0 \\ 0 & 0 \\ 0 & \psi_2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Write $c_t - 60 = Hx_t$ with $H = (1, 0, 0, 0, -60)$, so $u(c_t) = -.5 x_t' H' H x_t$. By the geometric-sum formulas (2.4.19)–(2.4.20) with $Y = H' H$,

$$V_0 = -.5(x_0' \nu x_0 + \sigma), \quad \nu = H' H + .95 A' \nu A, \quad \sigma = \frac{.95}{1 - .95} \text{tr}(\nu C C').$$

The Lyapunov equation has a unique solution because the eigenvalues of $\sqrt{.95}A$ lie inside the unit circle (the unit eigenvalue of the constant is discounted by $\sqrt{.95}$). ν depends only on A and the utility function. σ is a risk term that vanishes when $C = 0$.

(b) The formula depends on x_0 , which the exercise leaves unspecified. We report V_0 at x_0 equal to the stationary mean, and also its expectation when x_0 is drawn from the stationary distribution ($E x_0' \nu x_0 = \mu' \nu \mu + \text{tr}(\nu \Sigma_\infty)$). In both cases the stationary means are $Ec = 2$ and $Ez = 0$: z is an exogenous AR(2) with mean zero, and $\bar{c} = 1 + .5\bar{c}$.

case	σ	V_0 at $x_0 = \mu$	$E V_0, x_0 \sim \text{stationary}$
(i) $\psi_1 = \psi_2 = 1$	36.144	−33658.07	−33659.70
(ii) $\psi_1 = 2, \psi_2 = 1$	134.340	−33707.17	−33712.76

The value is dominated by the distance of mean consumption from bliss: $-.5 \times (60 - 2)^2 / (1 - .95) = -33640$. A larger ψ_1 raises the volatility of consumption, and with it the risk term σ , so case (ii) is worse. Check: in case (i) the stationary variance of c is 1.97007, and $-.5(58^2 + 1.97007)/.05 = -33659.70$, as in the last column.

Exercise 2.6 *A population regression*

(a) The stationary mean of $x_t = (c_t, c_{t-1}, z_t, z_{t-1}, 1)'$ is $(2, 2, 0, 0, 1)'$, and the stationary covariance of $(c_t, c_{t-1}, z_t, z_{t-1})$ solves $\Sigma = A\Sigma A' + CC'$:

$$\Sigma_{\infty} = \begin{bmatrix} 1.97007 & 1.24935 & 0.24052 & 0.48691 \\ 1.24935 & 1.97007 & 0.07098 & 0.24052 \\ 0.24052 & 0.07098 & 1.57895 & 0.92105 \\ 0.48691 & 0.24052 & 0.92105 & 1.57895 \end{bmatrix}.$$

Starting the system from x_0 with this mean and covariance makes the process covariance stationary. (Check: z is an AR(2) with $\phi = (.7, -.2)$, so $\text{var}(z) = \frac{1.2}{0.8(1.44-.49)} = 1.57895$.)

(b) With $E(x_{t+j} - \mu)(x_t - \mu)' = A^j \Sigma_{\infty}$, the needed moments are $\text{cov}(c_{t+2}, z_t) = (A^2 \Sigma_{\infty})_{cz}$, $\text{cov}(c_{t+2}, z_{t-4}) = (A^6 \Sigma_{\infty})_{cz}$, $\text{var}(z_t)$ and $\text{cov}(z_t, z_{t-4}) = (A^4 \Sigma_{\infty})_{zz}$. The normal equations give

$$\alpha_1 = 0.31714, \quad \alpha_2 = -0.02472, \quad \alpha_0 = Ec - (\alpha_1 + \alpha_2)Ez = 2.$$

z_t raises c two periods ahead through z_{t+1} (via $\phi_1 = .7$) and the consumption lag. z_{t-4} enters with a small negative coefficient, because given z_t it partly corrects for the oscillatory (complex-root) dynamics of z .

Exercise 2.7 *Reading impulse responses and spectra*

For $a(L)y_t = b(L)w_t$ with unit-variance w , the spectrum is $S_y(\omega) = |b(e^{-i\omega})|^2 / |a(e^{-i\omega})|^2$, and the impulse response is the sequence of coefficients of $b(L)/a(L)$. Figure S2.1 shows all three objects for the seven processes (Python version of `bigshow3`).

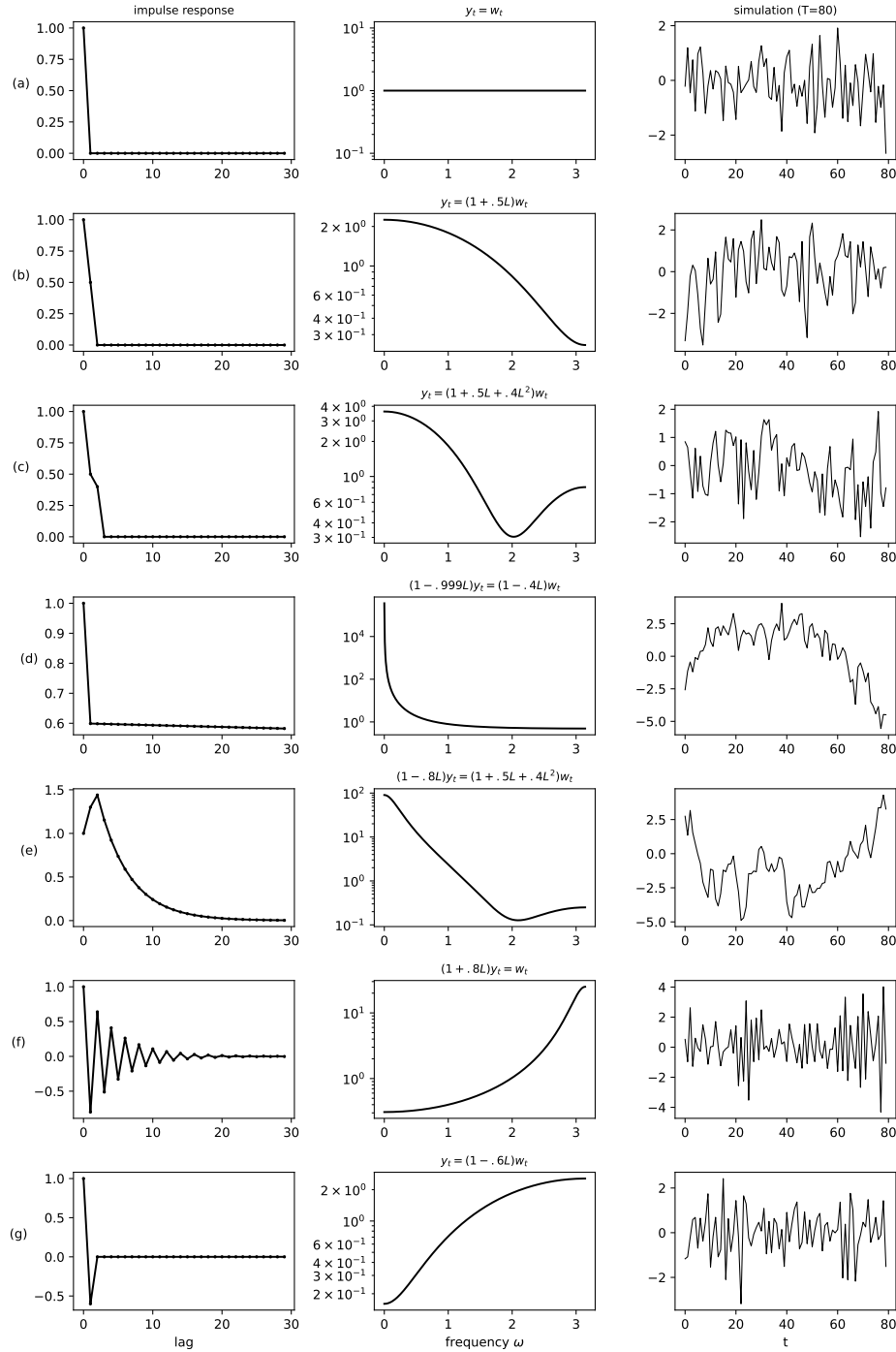


Figure S2.1. Impulse responses, spectra (log scale, $0 \leq \omega \leq \pi$) and simulated paths.

Patterns:

- White noise (a) has a flat spectrum: all frequencies contribute equally, and the path has no visible structure.
- Positive moving-average coefficients, as in (b) and (c), smooth the process. They move variance toward low frequencies: $S(0) = b(1)^2$ (2.25 and 3.61) exceeds $S(\pi) = b(-1)^2$ (0.25 and 0.81). In

- (c) the zero of $|b(e^{-i\omega})|$ near $\omega \approx 2$ carves a trough. A negative coefficient, as in (g), does the opposite: $S(0) = 0.16$, $S(\pi) = 2.56$, and the path looks choppy than white noise.
- A near unit root, as in (d), produces an enormous spectral peak at frequency zero ($S(0) \approx 3.6 \times 10^5$). The impulse response is 1 on impact and then an almost permanent 0.6. The simulated path wanders like a random walk. This is Granger's "typical spectral shape".
 - Combining a persistent AR with a smoothing MA, as in (e), gives a hump-shaped impulse response (it rises before it decays) and a spectrum that declines steeply in ω .
 - A negative AR root, as in (f), produces an oscillating impulse response $((-0.8)^j)$ and a spectrum that rises toward $\omega = \pi$, the two-period cycle. The path alternates in sign.

In short: the spectrum records at which frequencies a process has its variance, and the impulse response tells the same story in the time domain. Smooth, persistent responses mean low-frequency power; oscillating responses mean high-frequency power.

Exercise 2.8 *Cagan's money demand under perfect foresight*

Equation (4) refers to a numbered display of the exercise in the book.

Let $\lambda = \alpha/(1 + \alpha) \in (0, 1)$. The demand function can be written $p_t = (1 - \lambda)m_t + \lambda p_{t+1}$.

(a) *A particular solution.* Try $p_t = w_0 m_t + w_1 m_{t-1}$. Substituting, and using $m_{t+1} = (1 + \rho)m_t - \rho m_{t-1}$, gives

$$(1 - w_0)m_t - w_1 m_{t-1} = -\alpha(w_0 \rho + w_1)(m_t - m_{t-1}).$$

Matching coefficients: $w_0 + w_1 = 1$ and $w_1(1 + \alpha) = -\alpha \rho w_0$, i.e. $w_1 = -\lambda \rho w_0$. Hence

$$w_0 = \frac{1}{1 - \lambda \rho}, \quad w_1 = -\frac{\lambda \rho}{1 - \lambda \rho}, \quad w_j = 0 \ (j \geq 2), \quad p_t^* = \frac{m_t - \lambda \rho m_{t-1}}{1 - \lambda \rho}.$$

The same formula follows from solving forward: $p_t^* = (1 - \lambda) \sum_{j \geq 0} \lambda^j m_{t+j}$. Write $m_{t+j} = k_1 + k_2 \rho^{t+j}$ ($\rho \neq 1$), with the constants expressed through m_t, m_{t-1} , and sum the two geometric series.

All solutions. The difference $f_t = p_t - p_t^*$ of any two solutions satisfies the homogeneous equation $-f_t = -\alpha(f_{t+1} - f_t)$, i.e. $f_{t+1} = \lambda^{-1} f_t = \frac{1+\alpha}{\alpha} f_t$. So every equilibrium is

$$p_t = \frac{m_t - \lambda \rho m_{t-1}}{1 - \lambda \rho} + f_0 \left(\frac{1 + \alpha}{\alpha} \right)^t, \quad f_0 \in \mathbb{R} \text{ arbitrary.}$$

(b) There is a continuum of equilibria, one for each f_0 (equivalently, each initial price level p_0). Every equilibrium with $f_0 \neq 0$ contains a "bubble" growing at the gross rate $(1 + \alpha)/\alpha > 1$.

(c) Yes: $f_0 = 0$ gives $p_t = p_t^*$. It is the unique equilibrium in which the price level equals the discounted value of current and future money supplies, and hence the only one without an explosive bubble relative to m .

(d) The undetermined-coefficients formula for (w_0, w_1) only needs $\lambda \rho \neq 1$. Condition (4), $|\lambda \rho| < 1$, is what makes the forward sum $(1 - \lambda) \sum_j \lambda^j m_{t+j}$ converge when money grows like ρ^t . So it is needed to interpret p_t^* as a present value, and to single out $f_t = 0$ as the "no-bubble" solution by a transversality-type argument.

(e) With $\alpha = 1$, $\lambda = \frac{1}{2}$, and $\rho = 1$ (condition (4) holds: $\lambda \rho = \frac{1}{2}$):

$$p_t = 2m_t - m_{t-1} + f_0 2^t = m_t + (m_t - m_{t-1}) + f_0 2^t.$$

Money grows linearly, $m_t = m_{-1} + (t+1)(m_{-1} - m_{-2})$. The fundamental price level is money plus α times the (constant) inflation rate, since real balances equal $-\alpha$ times inflation. Figure S2.2 plots the fundamental solution (solid) and four bubble equilibria ($f_0 = \pm 0.01, \pm 0.02$, dashed) for $m_{-2} = 0$, $m_{-1} = 1$, so that $m_t = t + 2$ and $p_t^* = t + 3$.

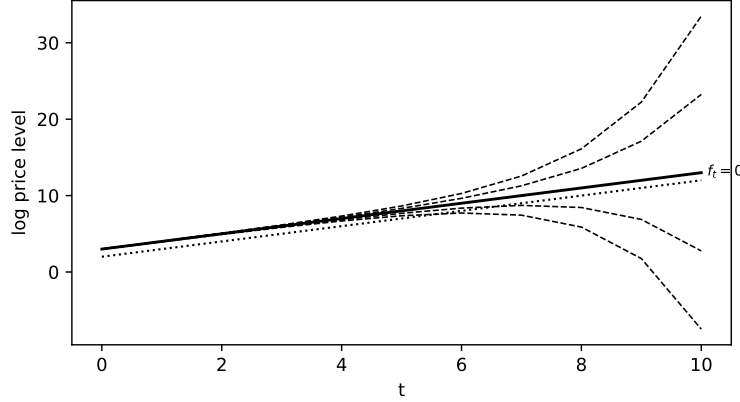


Figure S2.2. Equilibria of the Cagan model with $\alpha = \rho = 1$; dotted: m_t .

Exercise 2.9 *Pricing a one-period payoff with a linear stochastic discount factor*

(a) $m_{t+1}p_{t+1} = \frac{(Mx_{t+1})(Px_{t+1})}{Mx_t} = \frac{x'_{t+1}M'Px_{t+1}}{Mx_t}$. Using $E_t x_{t+1} = Ax_t$ and $E_t(x_{t+1} - Ax_t)(x_{t+1} - Ax_t)' = C_t C_t'$,

$$q_t = f_t(x_t) = \frac{(MAx_t)(PAx_t) + PC_t C_t' M'}{Mx_t} = \underbrace{\frac{MAx_t}{Mx_t}}_{E_t m_{t+1}} \underbrace{PAx_t}_{E_t p_{t+1}} + \underbrace{\frac{PC_t C_t' M'}{Mx_t}}_{\text{cov}_t(m_{t+1}, p_{t+1})}.$$

(This requires $Mx_t \neq 0$.) The price is the risk-neutral value $E_t m \cdot E_t p$ plus a risk adjustment equal to the conditional covariance of the discount factor with the payoff. A enters through the conditional means and C_t through the covariance, and the time variation of C_t is why the pricing function carries a t subscript.

(b) Let $e_{t+1} = m_{t+1}p_{t+1} - q_t$. The model says $E(e_{t+1} | J_t) = 0$. The econometrician's information $z^t = (z_t, z_{t-1}, \dots)$ is contained in J_t , so by the law of iterated expectations $E(e_{t+1} | z^t) = 0$. Hence $E[h(z^t)e_{t+1}] = 0$ for every function h of the econometrician's information. Choose a vector of instruments h_t (a constant, z_t , lags, squares), form the sample moments $g_T = T^{-1} \sum_t h_t e_{t+1}$, and test $Eh_t e_{t+1} = 0$ with Hansen's J statistic $Tg_T' \hat{S}^{-1} g_T \rightarrow \chi^2_{\dim h}$, where $\hat{S} = T^{-1} \sum_t h_t h_t' e_{t+1}^2$. Because e_{t+1} is a martingale difference, no autocorrelation correction is needed. Equivalently, regress e_{t+1} on h_t and test that all coefficients are zero with robust standard errors. The econometrician does not need to see all of x_t : omitted state variables only reduce the power of the test, not its validity.

Exercise 2.10 *Mixtures of invariant distributions*

Let $\pi_\alpha = \alpha\pi_1 + (1-\alpha)\pi_2$. By linearity, $P'\pi_\alpha = \alpha P'\pi_1 + (1-\alpha)P'\pi_2 = \alpha\pi_1 + (1-\alpha)\pi_2 = \pi_\alpha$. Its components are nonnegative, since they are convex combinations of nonnegative numbers, and they sum to $\alpha \cdot 1 + (1-\alpha) \cdot 1 = 1$. So π_α is a probability vector satisfying $\pi_\alpha = P'\pi_\alpha$, i.e. an invariant distribution. The set of invariant distributions is convex.

Exercise 2.11 Distributions over time for a chain with two absorbing states

From $\pi'_{t+1} = \pi'_t P$, component by component: $\pi_{1,t+1} = \pi_{1t} + .2\pi_{2t}$, $\pi_{2,t+1} = .5\pi_{2t}$, $\pi_{3,t+1} = \pi_{3t} + .3\pi_{2t}$. The middle equation gives $\pi_{2t} = .5^t \pi_{2,0}$. Substituting into the first and summing,

$$\pi_{1t} = \pi_{1,0} + .2 \sum_{s=0}^{t-1} \pi_{2s} = \pi_{1,0} + .2 \pi_{2,0} \sum_{s=0}^{t-1} .5^s = \pi_{1,0} + .2 \frac{1 - .5^t}{1 - .5} \pi_{2,0},$$

and identically for π_{3t} with .3 in place of .2. (Formally, each formula is proved by induction on t , using the one-step recursion.) As $t \rightarrow \infty$ the transient state 2 empties, $\pi_{1\infty} = \pi_{1,0} + .4\pi_{2,0}$ and $\pi_{3\infty} = \pi_{3,0} + .6\pi_{2,0}$.

Exercise 2.12 Columns of powers of P'

The distribution evolves as $\pi_{t+1} = P'\pi_t$, so $\pi_t = (P')^t \pi_0$ by induction. With $\pi_0 = e_j$, $\pi_t = (P')^t e_j$, which is the j th column of $(P')^t$. Equivalently, the (i, j) element of $(P')^t$ is $(P^t)_{ji} = \mathbb{P}(x_t = e_i \mid x_0 = e_j)$.

Exercise 2.13 A permanent income problem solved by Howard's algorithm

(a) The budget constraint gives $b_{t+1} = \beta^{-1}(u_t + 30 + b_t - y_t)$, and $y_{t+1} = .5 + 1.3y_t - .4y_{t-1} + .05w_{t+1}$. Hence $x_{t+1} = Ax_t + Bu_t + Cw_{t+1}$ with

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 30/\beta & 1/\beta & -1/\beta & 0 \\ .5 & 0 & 1.3 & -.4 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1/\beta \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 \\ 0 \\ .05 \\ 0 \end{bmatrix}.$$

A is block triangular. Its eigenvalues are 1 (the constant), $1/\beta = 1.0526$ (debt compounds at the gross interest rate) and the roots of $\lambda^2 - 1.3\lambda + .4 = 0$, namely 0.8 and 0.5. The zeros of $1 - 1.3z + .4z^2$ are 1.25 and 2, the reciprocals of 0.8 and 0.5. The one-period loss is $x'_t R x_t + u'_t Q u_t$ (scaled by .5) with $R = \text{diag}(0, 10^{-6}, 0, 0)$ and $Q = 1$.

(b) Howard's algorithm needs an initial rule F_0 for which $\sqrt{\beta}(A - BF_0)$ is stable. We use $u_t = -(1 - \beta)b_t$, which makes debt a unit-root process ($b_{t+1} = b_t + (30 - y_t)/\beta$). Then $\sqrt{\beta}(A - BF_0)$ has spectral radius 0.975. Iterating on (2.4.30), alternately solving the Lyapunov equation for P_j and updating $F_{j+1} = \beta(Q + \beta B' P_j B)^{-1} B' P_j A$, converges in **3 improvement steps** with the criterion $\max |F_{j+1} - F_j| < 10^{-10} \max |F_{j+1}|$. The same answer (to 6×10^{-14}) takes 820 iterations of the Riccati recursion (2.4.31), which illustrates the speed of Howard's algorithm. The optimal rule is

$$c_t = 30 - F^* x_t = 3.7695 - 0.050019 b_t + 0.396944 y_t - 0.150836 y_{t-1}.$$

At $b_t = 0$ and $y_t = y_{t-1} = 5$ this gives $c_t = 5$, and the coefficient on debt is essentially $-(1 - \beta) = -0.05$, the permanent income rule.

(c) The eigenvalues of $A - BF^*$ are 0.5, 0.8, 0.99998, 1. The explosive debt root has been replaced by a root just below one. The tiny penalty $10^{-6}b_t^2$ makes debt, and hence consumption, very slowly mean-reverting. Figure S2.3 compares the impulse responses with the theoretical ones from

(2.12.18), where $z_t = (1, y_t, y_{t-1})'$, $U_y = (0, 1, 0)$ and $(1 - \beta)U_y(I - \beta A_{22})^{-1}C_2 = (1 - \beta) \cdot 0.05 / (1 - 1.3\beta + .4\beta^2) = 0.019841$.

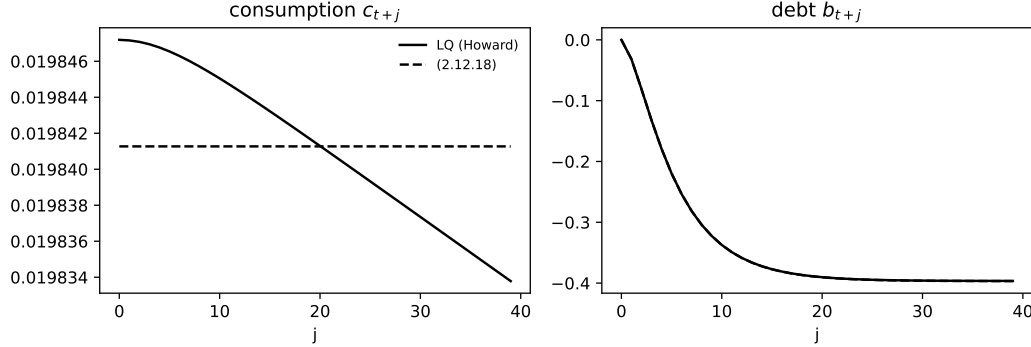


Figure S2.3. Responses of c_{t+j} and b_{t+j} to $w_{t+1} = 1$: optimal LQ rule (solid) and formula (2.12.18) (dashed).

Consumption jumps by 0.019847 on impact (theory: 0.019841) and stays there: 0.019834 after 40 periods. Debt falls by the amounts theory predicts (-0.03174 after one period, -0.21971 after five, -0.39643 after 39, versus -0.03175 , -0.21974 , -0.39673). Up to the 10^{-6} penalty, the LQ solution *is* the permanent income model: consumption is a random walk that moves by $(1 - \beta)$ times the revision in the present value of income.

Exercise 2.14 *A chain with a closed class, a transient state and an absorbing state*

(a) States $\{1, 2\}$ form a closed class, state 3 is transient (it leaks to 4) and state 4 is absorbing. On $\{1, 2\}$, $\pi_1 = .5\pi_1 + .1\pi_2$ gives $\pi_2 = 5\pi_1$, i.e. $(1/6, 5/6)$. State 3 gets no mass in any stationary distribution, since $\pi_3 = .9\pi_3$. The stationary distributions are

$$\pi^{(\gamma)} = \gamma \left(\frac{1}{6}, \frac{5}{6}, 0, 0 \right)' + (1 - \gamma) (0, 0, 0, 1)', \quad \gamma \in [0, 1].$$

(b) Invariant functions solve $P\bar{y} = \bar{y}$: $\bar{y}_1 = \bar{y}_2$ and $\bar{y}_3 = \bar{y}_4$, so $\bar{y} = (a, a, b, b)'$. The chain is ergodic under $\pi^{(\gamma)}$ if every invariant function is constant on the support of $\pi^{(\gamma)}$. This holds for $\gamma = 1$ (support $\{1, 2\}$) and $\gamma = 0$ (support $\{4\}$), but not for $\gamma \in (0, 1)$, where (a, a, b, b) with $a \neq b$ is invariant and non-constant. So yes: the extreme points $(\frac{1}{6}, \frac{5}{6}, 0, 0)$ and $(0, 0, 0, 1)$ make the chain ergodic.

(c) A path that starts in $\{1, 2\}$ stays there, and by the ergodic theorem its sample mean converges to $1 \cdot \frac{1}{6} + 2 \cdot \frac{5}{6} = \frac{11}{6}$. A path that starts in 3 or 4 is absorbed in state 4 in finite time (with probability one), so its sample mean converges to 4. The possible limits are $11/6$ and 4. The limit is $11/6$ with probability $\pi_{0,1} + \pi_{0,2}$ and 4 with probability $\pi_{0,3} + \pi_{0,4}$.

Exercise 2.15 *The Slutsky effect*

$|1 + e^{-i\omega}|^2 = 2 + 2\cos\omega = 4\cos^2(\omega/2)$ and $|1 - e^{-i\omega}|^2 = 2 - 2\cos\omega = 4\sin^2(\omega/2)$, so

$$S_y(\omega) = 4^{m+n} \cos^{2n}(\omega/2) \sin^{2m}(\omega/2).$$

Setting the derivative of $n \log \cos^2(\omega/2) + m \log \sin^2(\omega/2)$ to zero gives $\tan^2(\omega/2) = m/n$: the spectrum has a single peak (on each side of zero) at

$$\omega^* = 2 \arctan \sqrt{m/n}.$$

(m, n)	(10, 10)	(10, 40)	(40, 10)	(120, 30)
peak ω^*	1.571	0.927	2.214	2.214
period $2\pi/\omega^*$	4.00	6.78	2.84	2.84
width at half power	0.523	0.330	0.330	0.190

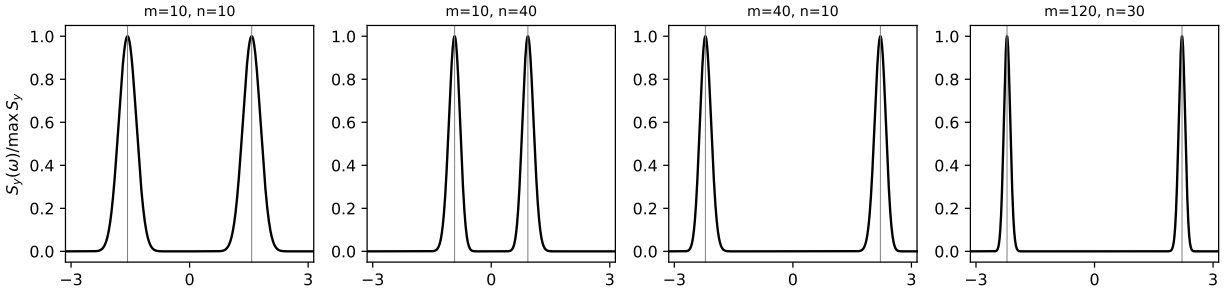


Figure S2.4. Normalized spectra $S_y(\omega)/\max S_y$ on $[-\pi, \pi]$; vertical lines at $\pm\omega^*$.

Comment. Summing $(1 + L)$ is a low-pass filter and differencing $(1 - L)$ a high-pass filter. Their product is a band-pass filter. Applied many times, it concentrates almost all the variance of pure white noise in a narrow band around ω^* , set by the ratio m/n . The band narrows as $m + n$ grows (compare (40, 10) with (120, 30), which share ω^*). The filtered series therefore displays strikingly regular “cycles” of period $2\pi/\omega^*$, although the input has no cyclical structure at all. This is the Slutsky–Yule effect: cycles in data can be artifacts of the filters (sums and differences) applied to random shocks.

Exercise 2.16 A law of large numbers for indicator functions

(a) Under the stationary distribution, $\mathbb{P}(x_t = e_i) = \pi_i$ for every t , so $E I_{it} = 1 \cdot \pi_i + 0 \cdot (1 - \pi_i) = \pi_i$.

(b) $I_{it} = \bar{y}' x_t$ with $\bar{y} = e_i$ is a function of the state. Since (P, π) is stationary and ergodic, the law of large numbers THEOREM 2.2.5 gives $\frac{1}{T} \sum_t I_{it} \rightarrow E I_{i0} = \pi_i$ with probability one. For a self-contained argument:

- If $\pi_i = 0$, then $\mathbb{P}(x_t = e_i) = 0$ for all t , so $I_{it} = 0$ almost surely and the average is $0 = \pi_i$.
- If $\pi_i > 0$: with a unique stationary distribution, a finite chain has exactly one closed communicating class, which is the support of π . Started from π , the chain lives on that class, where it is irreducible and positive recurrent. The times between successive visits to e_i are then i.i.d. with some finite mean m_i . The strong law of large numbers applied to these cycle lengths gives $\frac{1}{T} \#\{t < T : x_t = e_i\} \rightarrow 1/m_i$ almost surely. By bounded convergence, $1/m_i = \lim_T \frac{1}{T} \sum_t E I_{it} = \pi_i$ (Kac’s formula).

Exercise 2.17 The lake model

(a) $P = \begin{bmatrix} 1-\lambda & \lambda \\ \delta & 1-\delta \end{bmatrix}$. Stationarity requires $\pi_u \lambda = \pi_e \delta$ (flows into and out of unemployment balance), so the unique stationary distribution is

$$\pi_u = \frac{\delta}{\lambda + \delta}, \quad \pi_e = \frac{\lambda}{\lambda + \delta}.$$

All entries of P are positive, so by THEOREM 2.2.1 the chain is asymptotically stationary. The only invariant functions are constants, since $P\bar{y} = \bar{y}$ forces $\bar{y}_1 = \bar{y}_2$, so the chain is ergodic under π .

(b) $\pi_u = .25/.30 = 0.8333$ and $\pi_e = 0.1667$. By the ergodic theorem, the fraction of an infinitely long life spent unemployed is $\pi_u = 5/6$.

(c) $g_{ij} = \pi_j P_{ji}$: $g_{11} = \pi_u(1-\lambda) = 0.79167$, $g_{21} = \pi_u \lambda = 0.04167$, $g_{12} = \pi_e \delta = 0.04167$, $g_{22} = \pi_e(1-\delta) = 0.125$. In a stationary lake, the flows out of and into unemployment are equal ($g_{21} = g_{12}$).

(d) The pair (x_t, x_{t-1}) is itself a stationary ergodic Markov chain, so by the ergodic theorem $\frac{1}{T} \sum_t I_{ij,t} \rightarrow g_{ij}$ almost surely, with the values in (c).

(e) Since $\lambda = g_{21}/(g_{11} + g_{21})$ and $\delta = g_{12}/(g_{12} + g_{22})$, replace population moments by sample frequencies. Let n_{ij} be the number of transitions from j to i ; for instance, n_{21} counts transitions from unemployment into employment. Then

$$\hat{\lambda} = \frac{n_{21}}{n_{11} + n_{21}}, \quad \hat{\delta} = \frac{n_{12}}{n_{12} + n_{22}}, \quad \hat{w} = F^{-1}(1 - \hat{\lambda}),$$

the last because $\lambda = 1 - F(\bar{w})$. It is unique if F is strictly increasing near $\bar{w}$.

(f) Conditional on x_0 , the likelihood of a sample is $(1-\lambda)^{n_{11}} \lambda^{n_{21}} \delta^{n_{12}} (1-\delta)^{n_{22}}$. Maximizing, $\hat{\lambda}^{ML} = n_{21}/(n_{11} + n_{21})$ and $\hat{\delta}^{ML} = n_{12}/(n_{12} + n_{22})$.

(g) They are identical. (If the likelihood also includes $\mathbb{P}(x_0) = \pi_{x_0}$, the MLE differs from the method of moments estimator by a term of order $1/T$.)

(h) The per-period log likelihood contributes information $\pi_u/[\lambda(1-\lambda)]$ about λ and $\pi_e/[\delta(1-\delta)]$ about δ , with no cross term. Hence

$$\sqrt{T} \begin{pmatrix} \hat{\lambda} - \lambda \\ \hat{\delta} - \delta \end{pmatrix} \rightarrow \mathcal{N} \left(0, \begin{bmatrix} \lambda(1-\lambda)/\pi_u & 0 \\ 0 & \delta(1-\delta)/\pi_e \end{bmatrix} \right) = \mathcal{N} \left(0, \begin{bmatrix} 0.0570 & 0 \\ 0 & 1.1250 \end{bmatrix} \right)$$

at $\lambda = .05$, $\delta = .25$. δ is estimated much less precisely: a worker spends only one-sixth of the time employed, which is when firing can be observed.

Exercise 2.18 Gambler's ruin

(a) States 1 and 4 are absorbing, and 2 and 3 are transient. The stationary distributions are $\gamma e_1 + (1-\gamma)e_4$, $\gamma \in [0, 1]$.

(b) $P\bar{y} = \bar{y}$ requires $\bar{y}_2 = \frac{1}{2}(\bar{y}_1 + \bar{y}_3)$ and $\bar{y}_3 = \frac{1}{2}(\bar{y}_2 + \bar{y}_4)$, i.e. $\bar{y}$ is affine in the index. The invariant functions form the two-dimensional space $\bar{y}_i = a + bi$. The chain is ergodic only for $\gamma \in \{0, 1\}$: for $\gamma \in (0, 1)$ the invariant function $\bar{y}_i = i$ is not constant on the support $\{1, 4\}$.

(c) $P\bar{y} = (1, .5 \cdot 1 + .5 \cdot 3, .5 \cdot 2 + .5 \cdot 4, 4)' = (1, 2, 3, 4)'$. So $E[y_{t+1} | x_t] = y_t$ in every state: y_t is a martingale.

(d) Condition (2.2.12) holds, and $\bar{y}$ is a right eigenvector of P with unit eigenvalue ((2.2.13)). Under *any stationary* distribution, the chain sits in state 1 or 4 forever, so y_t is constant along every path of positive probability: it is invariant, as THEOREM 2.2.4 asserts. But from the transient states 2 or 3, y_t moves until absorption. Such initial conditions are not stationary distributions, and stationarity of (P, π) is the hypothesis of THEOREM 2.2.4 that fails. A martingale need not be constant if the chain is not started from its stationary distribution.

(e) y_t is a bounded martingale, so it converges almost surely by the martingale convergence theorem. Directly: from state 2 or 3 the chain is absorbed next period with probability 1/2, so absorption happens in finite time almost surely, and $y_t \rightarrow y_\infty \in \{1, 4\}$. Optional stopping gives $E[y_\infty | x_0 = e_i] = i$, so $\mathbb{P}(y_\infty = 4 | x_0 = e_i) = (i - 1)/3$ and $\mathbb{P}(y_\infty = 1 | x_0 = e_i) = (4 - i)/3$. Starting from state 2, for instance, $y_t \rightarrow 4$ with probability 1/3 and $\rightarrow 1$ with probability 2/3.

Exercise 2.19 *Learning a person's IQ*

(a) This is the Kalman filter (2.10.8) with $A = 1$, $C = 0$, $G = 1$, $R = 100$, $\hat{x}_0 = 100$, $\Sigma_0 = 100$. With $\Sigma_t = E(IQ_t - \theta)^2$,

$$IQ_{t+1} = IQ_t + K_t(y_t - IQ_t), \quad K_t = \frac{\Sigma_t}{\Sigma_t + 100}, \quad \Sigma_{t+1} = \frac{100 \Sigma_t}{\Sigma_t + 100}.$$

In precision form, $1/\Sigma_{t+1} = 1/\Sigma_t + 1/100$, so $\Sigma_t = 100/(t + 1)$ and $K_t = 1/(t + 2)$. Solving the recursion,

$$IQ_t = \frac{100 + \sum_{s=0}^{t-1} y_s}{t + 1} :$$

the prior mean counts as one extra test score, because the prior variance equals the variance of a single test.

(b) See Figure S2.5 (code/ch2.py).

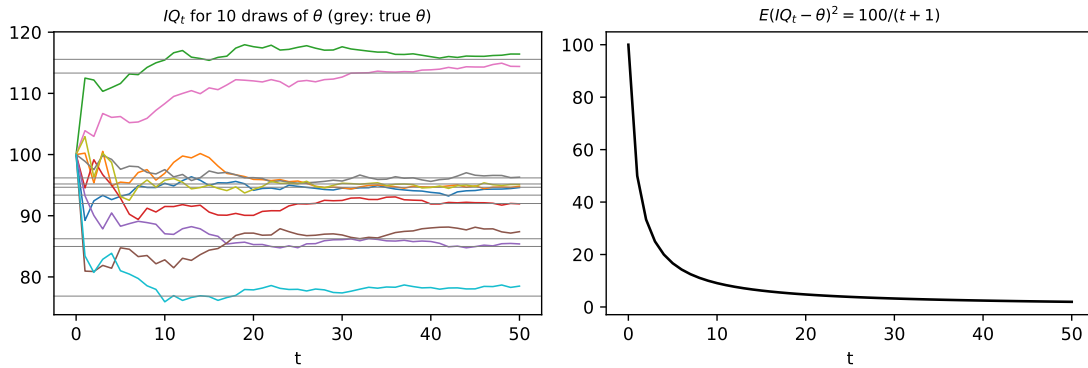


Figure S2.5. Simulated IQ_t paths for 10 persons (left) and the mean squared error $100/(t + 1)$ (right).

(c) $E(IQ_t - \theta)^2 = \Sigma_t = 100/(t + 1) \rightarrow 0$. The estimate is consistent in mean square, although it converges only at rate $t^{-1/2}$.

Exercise 2.20 Two noisy signals on a random walk

(a) With $A = 1$, $C = 1$, $G = \iota = (1, 1)'$ and $R = I$, we have $(\Sigma \iota \iota' + I)^{-1} = I - \frac{\Sigma}{1+2\Sigma} \iota \iota'$. The Riccati equation (2.7.15) becomes

$$\Sigma_{t+1} = \Sigma_t + 1 - \Sigma_t^2 \iota' (\Sigma_t \iota \iota' + I)^{-1} \iota = \frac{1 + 3\Sigma_t}{1 + 2\Sigma_t}.$$

Its fixed point solves $2\Sigma^2 - 2\Sigma - 1 = 0$, i.e. $\Sigma = (1 + \sqrt{3})/2 = 1.36602540378444$, exactly the given Σ_0 . So the filter is time invariant from the start, with gain

$$K = \Sigma \iota' (\Sigma \iota \iota' + I)^{-1} = \frac{\Sigma}{1 + 2\Sigma} (1, 1) = \frac{\sqrt{3}-1}{2} (1, 1) = (0.36603, 0.36603).$$

Hence $\hat{x}_{t+1} = (1 - 2k)\hat{x}_t + k(y_{1t} + y_{2t})$ with $k = (\sqrt{3} - 1)/2$ and $1 - 2k = 2 - \sqrt{3}$, and

$$E[x_t | y^{t-1}] = \hat{x}_t = (2 - \sqrt{3})^t \hat{x}_0 + \frac{\sqrt{3}-1}{2} \sum_{j=0}^{t-1} (2 - \sqrt{3})^j (y_{1,t-1-j} + y_{2,t-1-j}).$$

The two signals enter symmetrically through their sum, and $E[x_{t+j} | y^{t-1}] = \hat{x}_t$ for all $j \geq 0$ because x is a random walk.

(b) $A - KG = 1 - 2k = 2 - \sqrt{3} = 0.26795$, inside the unit circle (confirmed numerically by `ch2.py`).

(c) From (2.9.3), $y_t = G \sum_{j \geq 0} (A - KG)^j K y_{t-j-1} + a_t$, i.e.

$$y_t = \sum_{j=1}^{\infty} \frac{\sqrt{3}-1}{2} (2 - \sqrt{3})^{j-1} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} y_{t-j} + a_t, \quad E a_t a_t' = \Sigma \iota \iota' + I = \begin{bmatrix} 2.36603 & 1.36603 \\ 1.36603 & 2.36603 \end{bmatrix}.$$

Exercise 2.21 Impulse response of a VAR

Iterating $\hat{x}_{t+1} = A\hat{x}_t + K a_t$ and $y_t = G\hat{x}_t + a_t$ gives the moving-average representation

$$y_t = a_t + \sum_{j \geq 1} G A^{j-1} K a_{t-j}, \quad \text{i.e.} \quad y_t = [I + G(I - AL)^{-1} KL] a_t.$$

The response of y_{t+j} to a_t is I for $j = 0$ and $G A^{j-1} K$ for $j \geq 1$. For the state, the response of $\hat{x}_{t+j}$ is $A^{j-1} K$. If one wants responses to orthogonal shocks, write $a_t = \Omega^{1/2} \varepsilon_t$ with $\Omega = E a_t a_t'$ (e.g. a Cholesky factor) and post-multiply the responses by $\Omega^{1/2}$.

Exercise 2.22 Kalman filter with correlated shocks

(a) $E(C w_{t+1})(D w_{t+1})' = CD'$. Partition $w = (w_1', w_2')'$ and set $C = [C_1 \ 0]$, $D = [0 \ D_2]$. Then $CD' = 0$: the state shocks and measurement errors are driven by different components of w , as in the setup of section 2.7. *Correlated example*: an MA(1), $y_{t+1} = w_{t+1} + \theta w_t$. Take $x_t = w_t$, $A = 0$, $C = 1$, $G = \theta$, $D = 1$. Then $C w_{t+1} = D w_{t+1} = w_{t+1}$ and $CD' = 1 \neq 0$.

(b) Let $\tilde{x}_t = x_t - \hat{x}_t$. Because w_{t+1} is independent of (x_t, y^t) , $E[y_{t+1} | y^t] = G\hat{x}_t$, so

$$a_{t+1} = y_{t+1} - G\hat{x}_t = G\tilde{x}_t + D w_{t+1}, \quad E a_{t+1} a_{t+1}' = G \Sigma_t G' + D D'.$$

The information in y^{t+1} equals the information in (y^t, a_{t+1}) , with a_{t+1} orthogonal to y^t . So, by the population regression formula (with Gaussian shocks, projections are conditional expectations),

$$\hat{x}_{t+1} = E[x_{t+1} | y^t] + \text{cov}(x_{t+1}, a_{t+1}) [\text{var}(a_{t+1})]^{-1} a_{t+1}.$$

Here $E[x_{t+1} | y^t] = A\hat{x}_t$ and $\text{cov}(x_{t+1}, a_{t+1}) = E(A\tilde{x}_t + Cw_{t+1})(G\tilde{x}_t + Dw_{t+1})' = A\Sigma_t G' + CD'$, since $E\tilde{x}_t w_{t+1}' = 0$. Hence $\hat{x}_{t+1} = A\hat{x}_t + K_t a_{t+1}$ with $K_t = (CD' + A\Sigma_t G')(DD' + G\Sigma_t G')^{-1}$, and $y_{t+1} = G\hat{x}_t + a_{t+1}$ by definition of a_{t+1} . The new estimation error is

$$\tilde{x}_{t+1} = A\tilde{x}_t + Cw_{t+1} - K_t(G\tilde{x}_t + Dw_{t+1}) = (A - K_t G)\tilde{x}_t + (C - K_t D)w_{t+1},$$

and because $\tilde{x}_t$ and w_{t+1} are uncorrelated, $\Sigma_{t+1} = (A - K_t G)\Sigma_t(A - K_t G)' + (C - K_t D)(C - K_t D)'$. (We assume $DD' + G\Sigma_t G'$ is nonsingular.) With $CD' = 0$ these formulas reduce to the standard filter.

Exercise 2.23 *A monopolist who learns the demand curve*

(a) Since Q_t is chosen before p_t is seen, $E_{t-1}p_t Q_t = (E_{t-1}a - bQ_t)Q_t$, which is maximized by $Q_t = E_{t-1}a/(2b)$. In particular $Q_0 = \mu_a/(2b)$. The firm knows b and Q_s , so observing p_s is equivalent to observing

$$s_s \equiv p_s + bQ_s = a + \sigma_p \epsilon_s,$$

a noisy signal of a whose information content does not depend on Q_s . This is the signal-extraction problem (2.10.8). With $m_t = E[a | s^{t-1}]$ and $\Sigma_t = \text{var}(a | s^{t-1})$, starting from $m_0 = \mu_a$ and $\Sigma_0 = \sigma_a^2$,

$$m_{t+1} = m_t + K_t(s_t - m_t), \quad K_t = \frac{\Sigma_t}{\Sigma_t + \sigma_p^2}, \quad \frac{1}{\Sigma_{t+1}} = \frac{1}{\Sigma_t} + \frac{1}{\sigma_p^2},$$

so $m_t = \frac{\mu_a/\sigma_a^2 + \sum_{s < t} (p_s + bQ_s)/\sigma_p^2}{1/\sigma_a^2 + t/\sigma_p^2}$, and $Q_t = m_t/(2b)$. Because $m_t = 2bQ_t$, the rule can be written in terms of last period's price and quantity alone:

$$Q_{t+1} = Q_t + \frac{K_t}{2b} (p_t - bQ_t).$$

(b) Given (p_t, Q_t) , the next quantity Q_{t+1} is determined, and $p_{t+1} = a - bQ_{t+1} + \sigma_p \epsilon_{t+1}$. Conditional on a , this is a Markov process. Under the firm's own (Bayesian) predictive distribution it is too, since $p_{t+1} | \text{history} \sim \mathcal{N}(m_{t+1} - bQ_{t+1}, \Sigma_{t+1} + \sigma_p^2)$ with $m_{t+1} = 2bQ_{t+1}$. But the transition law depends on calendar time through K_t and Σ_t . So (p_t, Q_t) is Markov with a time-varying transition law, and (p_t, Q_t, Σ_t) is a time-homogeneous Markov process.

(c) State (m_t, Σ_t) , or equivalently (Q_t, Σ_t) :

$$Q_t = \frac{m_t}{2b}, \quad m_{t+1} = m_t + \frac{\Sigma_t}{\Sigma_t + \sigma_p^2} (p_t + bQ_t - m_t), \quad \Sigma_{t+1} = \frac{\Sigma_t \sigma_p^2}{\Sigma_t + \sigma_p^2}.$$

m_t is the firm's best estimate of the demand intercept, and Σ_t measures how uncertain it still is. Σ_t governs how strongly the firm reacts to a surprise in price.

(d) $E_{t-1}p_t = m_t - bQ_t = m_t/2$. Since $\Sigma_t = (1/\sigma_a^2 + t/\sigma_p^2)^{-1} \rightarrow 0$ and $\frac{1}{t} \sum_{s < t} s_s \rightarrow a$ by the strong law of large numbers, $m_t \rightarrow a$ almost surely. Hence $E_{t-1}p_t \rightarrow a/2$, and output converges to the full-information monopoly output $a/(2b)$. The limit is not a constant: it is the random variable

$a/2$, which differs across realizations of the demand intercept. Sample averages of prices converge to $a/2$ path by path, not to the unconditional mean $\mu_a/2$. The process is not ergodic, because what is learned depends on the draw of a .

(e) Write the objective as $E_{-1} \sum_t \beta^t E_{t-1}[p_t Q_t] = E_{-1} \sum_t \beta^t (m_t - bQ_t)Q_t$. The sequence $\{m_t\}$ depends only on the signals $s_t = a + \sigma_p \epsilon_t$, which are the same whatever quantities are chosen: output has no effect on what the firm learns. So the problem separates date by date, and each term is maximized by $Q_t = m_t/(2b)$. The infinitely lived monopolist uses the same myopic rule as in (a)–(c), for any β . There is no motive to experiment, because the information flow is independent of actions. Experimentation would matter if, for example, b were also unknown, since then p_t observed at different Q_t would carry different information about (a, b) .

Exercise 2.24 *A noisy AR(1)*

(a) Yes. z_{t+1} given the whole past depends only on z_t : $z_{t+1} | z^t \sim \mathcal{N}(.9z_t, 1)$.

(b) No. y is a noisy measure of a hidden Markov state, so its conditional distribution depends on the entire history. From the innovations representation in (f), $E[y_t | y^{t-1}] = \hat{z}_t = K \sum_{j \geq 0} (.9 - K)^j y_{t-1-j}$ in the stationary limit, with $.9 - K = .3623 \neq 0$: every lag matters. (y is an ARMA(1,1) process.)

(c) z_t is covariance stationary if Ez_t is the same for all t and $\text{cov}(z_t, z_{t-j})$ depends only on j , for all t and j .

(d) $\hat{z}_0 = 0$ and $\Sigma_0 = 1/(1 - .81) = 5.2632$ (the fixed points of $\mu_{t+1} = .9\mu_t$ and $\Sigma_{t+1} = .81\Sigma_t + 1$).

(e) The same definition applied to y . If z is covariance stationary, so is y , with $Ey_t = 0$, $\text{var}(y_t) = \Sigma_0 + 1 = 6.2632$ and $\text{cov}(y_t, y_{t-j}) = .9^j \Sigma_0$ for $j \geq 1$.

(f) Run the Kalman filter (2.7.14) with $A = .9$, $C = 1$, $G = 1$, $R = 1$, from $(\hat{z}_0, \Sigma_0)$:

$$a_t = y_t - \hat{z}_t, \quad K_t = \frac{.9\Sigma_t}{\Sigma_t + 1}, \quad \hat{z}_{t+1} = .9\hat{z}_t + K_t a_t, \quad \Sigma_{t+1} = .81\Sigma_t + 1 - \frac{.81\Sigma_t^2}{\Sigma_t + 1}.$$

Then $y_t | y^{t-1} \sim \mathcal{N}(\hat{z}_t, \Omega_t)$ with $\Omega_t = \Sigma_t + 1$. Starting from the stationary $\Sigma_0 = 5.263$, Ω_t falls quickly (6.263, 2.681, 2.508, 2.487, ...) to the fixed point of the Riccati equation. That fixed point solves $\Sigma^2 - .81\Sigma - 1 = 0$: $\bar{\Sigma} = 1.4839$, $\bar{\Omega} = 2.4839$, $\bar{K} = 0.5377$, and $\hat{z}_{t+1} = 0.3623 \hat{z}_t + 0.5377 y_t$. Note that y_t can be covariance stationary while the conditional variances Ω_t vary over time: the unconditional distribution is stationary, but the conditioning information grows.

Exercise 2.25 *Two classic permanent income examples*

(a) *Example 1* (the consumer sees z_t): $A_{22} = \text{diag}(1, 0)$, $C_2 = \text{diag}(\sigma_1, \sigma_2)$, $U_y = (1, 1)$. Then $(I - \beta A_{22})^{-1} = \text{diag}(\frac{1}{1-\beta}, 1)$ and (2.12.18A) gives

$$c_{t+1} - c_t = (1 - \beta)U_y(I - \beta A_{22})^{-1}C_2 w_{t+1} = \sigma_1 w_{1,t+1} + (1 - \beta)\sigma_2 w_{2,t+1},$$

which is (2.12.21). *Example 2* (the consumer sees only y): the state is (y_t, a_t) with $A_{22} = \begin{bmatrix} 1 & -(1-K) \\ 0 & 0 \end{bmatrix}$, $C_2 = (1, 1)'$, $U_y = (1, 0)$. Then $U_y(I - \beta A_{22})^{-1} = (\frac{1}{1-\beta}, -\frac{\beta(1-K)}{1-\beta})$, so

$$c_{t+1} - c_t = (1 - \beta) \frac{1 - \beta(1 - K)}{1 - \beta} a_{t+1} = [1 - \beta(1 - K)] a_{t+1},$$

which is (2.12.23). The first row of the state equation is $y_{t+1} = y_t - (1 - K)a_t + a_{t+1}$, which is (2.12.24).

(b), (c) Example 1 needs no filtering: $\Delta c_{t+1} = w_{1,t+1} + .05w_{2,t+1}$ in (b) and $2w_{1,t+1} + .05w_{2,t+1}$ in (c). For Example 2, the steady state of (2.10.3) with $Q = \sigma_1^2$, $R = \sigma_2^2$ solves $\Sigma^2 - Q\Sigma - QR = 0$, so $\Sigma = (Q + \sqrt{Q^2 + 4QR})/2$ and $K = \Sigma/(\Sigma + R)$:

	Σ	K	Δc_{t+1}	$\text{var}(a) = \Sigma + R$	$b_{t+1} - b_t$
(b) $\sigma_1 = \sigma_2 = 1$	1.6180	0.6180	$0.6371 a_{t+1}$	2.6180	$-0.3820 a_t$
(c) $\sigma_1 = 2, \sigma_2 = 1$	4.8284	0.8284	$0.8370 a_{t+1}$	5.8284	$-0.1716 a_t$

(For $\sigma_1 = \sigma_2$, $K = (\sqrt{5} - 1)/2$ is the golden-ratio conjugate.) A larger permanent shock variance raises K : more of each income surprise is read as permanent, so consumption responds more and less is saved.

(d) (2.12.20): $b_{t+1} - b_t = U_y(I - \beta A_{22})^{-1}(A_{22} - I)z_t$. In Example 1, $A_{22} - I = \text{diag}(0, -1)$, so $b_{t+1} - b_t = -z_{2t} = -\sigma_2 w_{2t}$, which is (2.12.22). In Example 2,

$$U_y(I - \beta A_{22})^{-1}(A_{22} - I) = \left(\frac{1}{1-\beta}, -\frac{\beta(1-K)}{1-\beta} \right) \begin{bmatrix} 0 & -(1-K) \\ 0 & -1 \end{bmatrix} = (0, -(1-K)),$$

so $b_{t+1} - b_t = (K - 1)a_t$, which is (2.12.25).

Exercise 2.26 Math and verbal IQ

(a) This is a time-varying Kalman filter with $A = I$, $C = 0$. With $\Sigma_0 = 100I$,

$$IQ_{t+1} = IQ_t + K_t(y_t - G_t IQ_t), \quad K_t = \Sigma_t G_t' (G_t \Sigma_t G_t' + 50)^{-1},$$

$$\Sigma_{t+1} = \Sigma_t - \Sigma_t G_t' (G_t \Sigma_t G_t' + 50)^{-1} G_t \Sigma_t.$$

(b) See Figure S2.6.

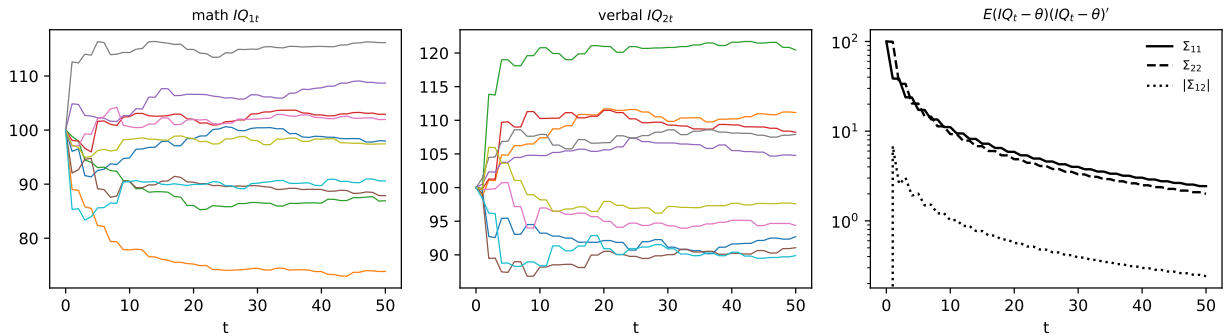


Figure S2.6. Simulated math and verbal IQ paths for 10 persons and the elements of Σ_t (log scale).

(c) In information (precision) form, $\Sigma_{t+1}^{-1} = \Sigma_t^{-1} + G'_t G_t / 50$, so

$$\Sigma_t^{-1} = \frac{I}{100} + \frac{1}{50} \sum_{s=0}^{t-1} G'_s G_s.$$

Every two periods the sum grows by $M = G'_e G_e + G'_o G_o = \begin{bmatrix} .8101 & .0999 \\ .0999 & .9901 \end{bmatrix}$, which is positive definite because $G_e = (.9, .1)$ and $G_o = (.01, .99)$ are linearly independent. Hence $\lambda_{\min}(\Sigma_t^{-1}) \geq \lfloor t/2 \rfloor \lambda_{\min}(M)/50 \rightarrow \infty$, and $\Sigma_t \rightarrow 0$. Each test mostly measures one ability, but because the two tests load on the abilities in linearly independent ways, alternating them identifies both. Numerically, $\Sigma_{50} = \begin{bmatrix} 2.438 & -0.241 \\ -0.241 & 2.004 \end{bmatrix}$, and it falls roughly like $1/t$.

Exercise 2.27 A two-person consumption loans economy

(a) By (2.12.9), $c_t = (1 - \beta)[E_t \sum_j \beta^j y_{t+j} - b_t]$. Since y^i is i.i.d. with mean .75, $E_t \sum_j \beta^j y_{t+j}^i = y_t^i + \frac{.75\beta}{1-\beta}$, so

$$c_t^i = (1 - \beta)(y_t^i - b_t^i) + .75\beta.$$

The budget constraint gives $b_{t+1}^i = \beta^{-1}(c_t^i + b_t^i - y_t^i) = b_t^i + .75 - y_t^i$. For consumer 1, $y_t^1 = 1$ when $s_t = 0$ and .5 when $s_t = 1$, so $b_{t+1}^1 = b_t^1 - .25$ or $b_t^1 + .25$. For consumer 2 the signs are reversed. A consumer saves (reduces debt by) the excess of current income over its mean.

(b) $b_{t+1}^i - b_t^i = .75 - y_t^i$ and $c_{t+1}^i - c_t^i = (1 - \beta)(y_{t+1}^i - .75)$ are i.i.d. with mean zero, so b_t^i and c_t^i are random walks (unit roots). But $c_t^i + (1 - \beta)b_t^i = (1 - \beta)y_t^i + .75\beta$ is i.i.d. and hence stationary. So (c_t^i, b_t^i) is cointegrated with cointegrating vector $(1, 1 - \beta)$, as in (2.12.19).

(c) $b_{t+1}^i = b_t^i + .75 - y_t^i$ is a function of (b_t^i, s_t) , both known at t . It does not depend on s_{t+1} , so it is independent of news arriving at $t + 1$: the payoff of a one-period IOU is risk-free.

(d) $(b^1 + b^2)_{t+1} - (b^1 + b^2)_t = 1.5 - (y_t^1 + y_t^2) = 0$, so $b_t^1 + b_t^2 = b_0^1 + b_0^2 = 0$ for all t . Then $c_t^1 + c_t^2 = (1 - \beta)[(y_t^1 + y_t^2) - (b_t^1 + b_t^2)] + 1.5\beta = 1.5$ (this holds from $t = 0$ on). Aggregate consumption equals the constant aggregate endowment, and one consumer's debt is the other's asset. The decision rules therefore describe a closed pure consumption-loans economy: in a state where y^1 is high, consumer 1 lends .25 to consumer 2 by buying a one-period IOU at price β , and the interest rate $R = \beta^{-1}$ clears the loan market.

(e) $u'(c) = \gamma - c$ and $c_{t+1}^i = c_t^i + (1 - \beta)(y_{t+1}^i - .75)$, so $m_{t+1}^i = \beta[1 - \frac{(1-\beta)(y_{t+1}^i - .75)}{\gamma - c_t^i}]$. Consumer 1 has high income ($y^1 = 1$) when $s_{t+1} = 0$, so

$$m_{t+1}^1 = \begin{cases} \beta - \frac{.25\beta(1-\beta)}{\gamma - c_t^1}, & s_{t+1} = 0, \\ \beta + \frac{.25\beta(1-\beta)}{\gamma - c_t^1}, & s_{t+1} = 1, \end{cases} \quad m_{t+1}^2 = \begin{cases} \beta + \frac{.25\beta(1-\beta)}{\gamma - c_t^2}, & s_{t+1} = 0, \\ \beta - \frac{.25\beta(1-\beta)}{\gamma - c_t^2}, & s_{t+1} = 1. \end{cases}$$

Remark. The book's display has the two branches of each discount factor interchanged. In the state where a consumer's income is high, consumption rises and marginal utility falls, so the discount factor is below β , not above. (Also, "Prob($s_1 = 0$)" in the statement should read Prob($s_t = 0$).)

The discount factors are *not* equal: in each state one is above and the other below β . With only a risk-free bond, markets are incomplete, and marginal rates of substitution are not equalized state by state.

(f) In both cases the two branches are $\beta \pm x$ with probability $\frac{1}{2}$ each, so $E_t m_{t+1}^1 = E_t m_{t+1}^2 = \beta = R^{-1}$. Both consumers agree on the price of the only traded asset, the risk-free bond, though not on the prices of state-contingent claims.

Exercise 2.28 *A non-invertible moving average*

(a) Formally inverting $1 - 2L$ gives $\epsilon_t = \sum_{j \geq 0} 2^j y_{t-j}$, whose weights explode. To prove that no square-summable representation exists, let $H(y^t)$ be the closed linear span of $\{y_s, s \leq t\}$ and $a_t = y_t - E[y_t \mid H(y^{t-1})]$. Since ϵ_t is independent of $\epsilon_{t-1}, \epsilon_{t-2}, \dots$, it is orthogonal to $H(y^{t-1})$. Hence its projection on $H(y^t) = H(y^{t-1}) \oplus \{a_t\}$ is

$$E[\epsilon_t \mid H(y^t)] = \frac{E\epsilon_t a_t}{Ea_t^2} a_t = \frac{E\epsilon_t y_t}{4} a_t = \frac{a_t}{4},$$

using $Ea_t^2 = 4$ from (c). Its variance is $1/4 < 1 = \text{var } \epsilon_t$, so $\epsilon_t \notin H(y^t)$. Even the whole past and present of y leave $3/4$ of the variance of ϵ_t unexplained.

(b) $x_t = (\epsilon_t, \epsilon_{t-1})'$, $x_{t+1} = Ax_t + Cw_{t+1}$ and $y_t = Gx_t$ with $A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $G = \begin{bmatrix} 1 & -2 \end{bmatrix}$, $w_{t+1} = \epsilon_{t+1}$, and no measurement error.

(c) Iterating the Kalman filter (the Python analogue of `kfilter.m`) converges to $K = (0, 0.25)'$ and $\Omega = G\Sigma G' = 4$. The innovations representation $\hat{x}_{t+1} = A\hat{x}_t + Ka_t$, $y_t = G\hat{x}_t + a_t$ has moving-average coefficients 1, $GK = -0.5$, $GAK = 0, \dots$, so $y_t = a_t - .5a_{t-1}$. As a check, both representations have the same autocovariances: $\gamma_0 = 1 + 4 = 5 = 1.25 \cdot 4$ and $\gamma_1 = -2 = -.5 \cdot 4$. The innovation variance is $\text{var}(a_t) = 4$, four times $\text{var}(\epsilon_t)$. Flipping the root from 2 to $1/2$ makes the representation invertible, at the cost of a larger shock variance.

(d) $(1 - .5L)^{-1}y_t = a_t$, i.e. $\sum_{j \geq 0} .5^j y_{t-j} = a_t$, so

$$y_t = - \sum_{j \geq 1} (.5)^j y_{t-j} + a_t, \quad A_j = -(.5)^j,$$

with $Ea_t y_{t-j} = 0$ for $j \geq 1$.

(e) None of them. Since y is Gaussian, the distribution of y_{t+1} given its past is normal with variance 4 and mean $-\sum_{j \geq 0} .5^{j+1} y_{t-j}$, which depends on every lag (all $A_j \neq 0$). So no finite window $(y_t, \dots, y_{t-k})$ is a sufficient statistic, and none of the three vectors is Markov.

(f) From $y_{t+1} = \epsilon_{t+1} - 2\epsilon_t$, $\epsilon_t = \frac{1}{2}(\epsilon_{t+1} - y_{t+1})$. Iterating forward, $\epsilon_t = -\sum_{j=1}^k 2^{-j} y_{t+j} + 2^{-k} \epsilon_{t+k}$, and the remainder vanishes in mean square as $k \rightarrow \infty$. Hence $\epsilon_t = -\sum_{j \geq 1} 2^{-j} y_{t+j}$, with square-summable weights. The shock ϵ_t lies in the span of the *future* of y , not of its past.

Exercise 2.29 *Pure prediction and the Lyapunov equation*

Equation (3) refers to a numbered display of the exercise in the book.

(a) Iterating the equation forward k times, $\mu_t = \sum_{j=0}^{k-1} (A')^j R x_{t+j} + (A')^k \mu_{t+k}$. With A stable and bounded μ , the remainder vanishes, so

$$\mu_t = \sum_{j=0}^{\infty} (A')^j R x_{t+j} = \left(\sum_{j \geq 0} (A')^j R A^j \right) x_t.$$

Remark. The exercise writes the solution as $R \sum_j (A')^j x_{t+j}$, with R in front of the sum. That expression satisfies the equation only if R commutes with A' : substituting it gives $R x_t + A' R \sum_j (A')^j x_{t+1+j}$, which equals $R \sum_j (A')^j x_{t+j}$ iff $A' R = R A'$. The correct solution puts R between $(A')^j$ and x_{t+j} , as above. (The second part is also mislabelled “a.”.)

(b) If $\mu_t = P x_t$ for all x_t , then $P x_t = R x_t + A' P A x_t$, so $P = R + A' P A$. The unique solution is $P = \sum_j (A')^j R A^j$, which converges because A is stable. This is the matrix found in (a).

(c) (2.4.10), $\Sigma_\infty = A \Sigma_\infty A' + C C'$, is the same Lyapunov equation with A replaced by A' and R by $C C'$: $\Sigma_\infty = \sum_j A^j C C' (A')^j$ versus $P = \sum_j (A')^j R A^j$. Covariances propagate forward in time through A , while values (shadow prices) cumulate backward from the future through A' . The two equations are duals of each other.

(d) (2.4.25), $P_0 = R + F_0' Q F_0 + \beta(A - B F_0)' P_0 (A - B F_0)$, is (3) with $A \rightarrow \sqrt{\beta}(A - B F_0)$ and $R \rightarrow R + F_0' Q F_0$. Evaluating a fixed decision rule, the policy-evaluation step of Howard’s algorithm, is a pure prediction problem: summing discounted quadratic forms along the closed-loop path.

(e) Use the duality $A \leftrightarrow A'$, $B \leftrightarrow G'$, $R \leftrightarrow C C'$, $Q \leftrightarrow R_v$ (the measurement-error covariance) and $F \leftrightarrow K'$ between the regulator and the filter. Start from a gain K_0 with $A - K_0 G$ stable, and iterate:

1. (evaluate) given K_j , solve the Lyapunov equation $\Sigma_j = (A - K_j G) \Sigma_j (A - K_j G)' + C C' + K_j R_v K_j'$ for the estimation-error covariance obtained by using the fixed gain K_j forever;
2. (improve) set $K_{j+1} = A \Sigma_j G' (G \Sigma_j G' + R_v)^{-1}$.

This is exactly (2.4.30) applied to the dual regulator problem (with $\beta = 1$). Like Howard’s algorithm, it is a Newton method and converges in a handful of steps. For the model of exercise 2.24, starting from $K_0 = 0$, it reaches $K = 0.53767$ and $\Sigma = 1.48390$ in 6 iterations (`ch2.py`), the same as iterating the Riccati equation to convergence.

Exercise 2.30 *Quasi-hyperbolic discounting: Phelps and Pollak meet Howard*

(a) $V(x) = \sum_{j \geq 0} \beta^j r(x_j, -F x_j)$ with $x_0 = x$ and $x_{j+1} = (A - B F) x_j$. $V(x)$ is the value, discounted exponentially at rate β , of the return stream generated by using the rule $u = -F x$ forever, starting from x .

(b) Guess $V(x) = -x' P x$. Matching quadratic forms gives the Lyapunov equation

$$P = R + F' Q F + \beta(A - B F)' P (A - B F),$$

i.e. (2.4.25). It has a unique positive definite solution if $\sqrt{\beta}(A - B F)$ is stable. Compute it by iterating $P_{k+1} = R + F' Q F + \beta(A - B F)' P_k (A - B F)$ from $P_0 = 0$, or faster with the doubling algorithm (`doublej`).

(c) $U(x)$ is the value to the *current* decision maker, who gives weight 1 to today and $\delta\beta^j$ to date $t+j$, when he chooses today's action optimally, taking as given that all future decision makers use $u = -Fx$. The first-order condition gives $G = \delta\beta(Q + \delta\beta B'PB)^{-1}B'PA$.

(d) A (stationary linear) Markov perfect equilibrium is a rule F such that, if all future decision makers use $u = -Fx$, the current decision maker's best response is also $u = -Fx$: $G(F) = F$. Equivalently, a pair (F, P) with

$$P = R + F'QF + \beta(A - BF)'P(A - BF), \quad F = \delta\beta(Q + \delta\beta B'PB)^{-1}B'PA.$$

(e) Iterate on best responses: given F_j , solve the Lyapunov equation for P_j , set $F_{j+1} = \delta\beta(Q + \delta\beta B'P_jB)^{-1}B'P_jA$, and repeat to convergence. (Convergence is not guaranteed in general, but the iteration converges quickly in examples; see below.)

(f) With $\delta = 1$ this is exactly Howard's policy improvement algorithm (2.4.30). With $\delta < 1$ the evaluation step discounts at β but the improvement step discounts at $\delta\beta$. The fixed point is then an equilibrium of a game among the successive selves, not the maximum of any single objective, and the monotone-improvement property of Howard's algorithm is lost.

(g) Yes. The equilibrium strategy is stationary and depends only on the current state, and the conditions defining it are the same at every date. So the continuation from any date and state is again a Markov perfect equilibrium (it is subgame perfect).

(h) From date 1 on, the dictator's weights $\delta\beta, \delta\beta^2, \dots$ are proportional to $1, \beta, \beta^2, \dots$. So the continuation is a standard exponential-discounting LQ problem, with value $W(x) = -x'P^*x$ solving the Riccati equation

$$P^* = R + \beta A'P^*A - \beta^2 A'P^*B(Q + \beta B'P^*B)^{-1}B'P^*A,$$

$$u_t = -F^*x_t \quad (t \geq 1), \quad F^* = \beta(Q + \beta B'P^*B)^{-1}B'P^*A.$$

At date 0, $\max_u \{r(x_0, u) + \delta\beta W(Ax_0 + Bu)\}$ gives

$$u_0 = -G^*x_0, \quad G^* = \delta\beta(Q + \delta\beta B'P^*B)^{-1}B'P^*A.$$

The dictator's plan is $u_0 = -G^*x_0$, then $u_t = -F^*x_t$.

(i) No, unless $\delta = 1$. The time-1 dictator weights date 1 by 1 and later dates by $\delta\beta^j$, so she would choose $u_1 = -G^*x_1$ rather than the $u_1 = -F^*x_1$ planned at date 0. The preferences are time inconsistent.

(j) With $\delta = 1$: then $G^* = F^*$, the dictator's plan is time consistent, and the Markov perfect equilibrium coincides with the optimal exponential-discounting rule F^* . For $\delta < 1$ they differ generically. An example (`ch2.py`) with $A = \begin{bmatrix} 1 & .1 \\ 0 & .9 \end{bmatrix}$, $B = (0, 1)'$, $R = I$, $Q = 1$, $\beta = .95$ gives:

	MPE F	dictator, date 0: G^*	dictator, dates ≥ 1 : F^*
$\delta = 1$	(0.4562, 0.5965)	(0.4562, 0.5965)	(0.4562, 0.5965)
$\delta = .7$	(0.3938, 0.5146)	(0.3912, 0.5114)	(0.4562, 0.5965)

(The MPE iteration converged in 7 and 14 steps.) With present bias ($\delta < 1$), each self acts less aggressively than the exponential discounter would, and the equilibrium rule is not the continuation of anyone's optimal plan.

Chapter 3

Dynamic Programming

Solution to the single exercise of Chapter 3 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018).

Exercise 3.1 *Howard's policy iteration algorithm*

Write $y_t = Ak_t^\alpha \theta_t$ for output and $z_t = \ln y_t$.

Step 1: evaluating an arbitrary policy. Under $k_{t+1} = hy_t$ with $h \in (0, 1)$, consumption is $c_t = (1 - h)y_t > 0$ and

$$z_{t+1} = \ln A + \alpha \ln k_{t+1} + \ln \theta_{t+1} = \ln A + \alpha \ln h + \alpha z_t + \ln \theta_{t+1}.$$

Since $E \ln \theta = 0$, $E_0 z_t = \alpha^t z_0 + (\ln A + \alpha \ln h) \frac{1 - \alpha^t}{1 - \alpha}$. Therefore

$$\begin{aligned} J_h(k_0, \theta_0) &= \sum_{t \geq 0} \beta^t [\ln(1 - h) + E_0 z_t] = \frac{\ln(1 - h)}{1 - \beta} + \frac{z_0}{1 - \alpha\beta} + \frac{\ln A + \alpha \ln h}{1 - \alpha} \left(\frac{1}{1 - \beta} - \frac{1}{1 - \alpha\beta} \right) \\ &= \underbrace{\frac{\ln(1 - h)}{1 - \beta} + \frac{\beta(\ln A + \alpha \ln h)}{(1 - \beta)(1 - \alpha\beta)} + \frac{\ln A}{1 - \alpha\beta}}_{\equiv \kappa(h)} + \frac{\alpha}{1 - \alpha\beta} \ln k_0 + \frac{1}{1 - \alpha\beta} \ln \theta_0. \end{aligned}$$

The key fact is that the policy parameter h affects only the constant $\kappa(h)$. The coefficient on $\ln k$, $\alpha/(1 - \alpha\beta)$, is the same for every h , because under any constant-savings-rate policy the elasticity of future output with respect to current capital is α^t .

Step 2: the improvement step. With $J_0 = J_{h_0}$, $EJ_0(k', \theta') = \kappa(h_0) + \frac{\alpha}{1 - \alpha\beta} \ln k'$ (using $E \ln \theta' = 0$), so the improvement problem is

$$\max_{k'} \left\{ \ln(y - k') + \frac{\alpha\beta}{1 - \alpha\beta} \ln k' \right\} + \text{constant}.$$

The objective is strictly concave, and the first-order condition $\frac{1}{y - k'} = \frac{\alpha\beta}{1 - \alpha\beta} \cdot \frac{1}{k'}$ gives $k' = \alpha\beta y$. Hence

$$h_1 = \alpha\beta \quad \text{for every initial guess } h_0 \in (0, 1).$$

Step 3: convergence and optimality. Repeating the improvement step from $J_1 = J_{\alpha\beta}$ again gives $h_2 = \alpha\beta$, because the coefficient on $\ln k'$ is the same. So the iterations have converged after one step. The limit is optimal: $J_{\alpha\beta}$ satisfies the Bellman equation $J(k, \theta) = \max_{k'} \{ \ln(Ak^\alpha \theta - k') + \beta EJ(k', \theta') \}$, since the maximum is attained at $k' = \alpha\beta Ak^\alpha \theta$, the policy that generates $J_{\alpha\beta}$. It coincides with the solution obtained by guess and verify in section 3.1.2, $k_{t+1} = \alpha\beta Ak_t^\alpha \theta_t$, extended to the multiplicative shock. (The unbounded log return does not cause trouble here: along any constant-savings policy the value is finite, and the guess-and-verify argument of section 3.1.2 delivers the same function.)

Why one step? Policy iteration is Newton's method on the Bellman equation. In this model the Bellman equation is "linear" in the right coordinates: every constant-savings-rate policy produces a value function in the same two-parameter family (affine in $\ln k$ and $\ln \theta$), with the payoff-relevant slope already correct. So a single Newton step lands on the fixed point.

Chapter 5

Linear Quadratic Dynamic Programming

Solutions to all 14 exercises of Chapter 5 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). The Matlab programs the exercises refer to (`olrp.m`, `policyi.m`, `doublej.m`, `dlqrmon.m`) are replaced by the Python functions `olrp`, `lq_howard` and `doublej` in `code/rmtlib.py`; all numbers come from `code/ch5.py`.

Conventions: a linear regulator *minimizes* $\sum_t \beta^t (x'_t R x_t + u'_t Q u_t + 2u'_t H x_t)$ subject to $x_{t+1} = A x_t + B u_t (+ C w_{t+1})$. The value of the corresponding maximization problem is $v(x) = -x' P x (-d)$, and the optimal rule is $u_t = -F x_t$.

Exercise 5.1 *A regulator with cross products*

(a) Guess $v(x) = -x' P x$ with P symmetric. The Bellman equation is

$$-x' P x = \max_u -\{x' R x + u' Q u + 2u' H x + \beta(Ax + Bu)' P (Ax + Bu)\}.$$

Collect terms: the braces equal $x'(R + \beta A' P A)x + u' M u + 2u' N x$ with $M = Q + \beta B' P B$ and $N = \beta B' P A + H$. M is positive definite (as Q is and $P \succeq 0$), so the objective is strictly concave in u . The first-order condition $M u + N x = 0$ gives

$$u = -M^{-1} N x = -(Q + \beta B' P B)^{-1} (\beta B' P A + H) x.$$

At the optimum, $u' M u + 2u' N x = x' N' M^{-1} N x - 2x' N' M^{-1} N x = -x' N' M^{-1} N x$. Matching quadratic forms then gives

$$P = R + \beta A' P A - (\beta A' P B + H')(Q + \beta B' P B)^{-1} (\beta B' P A + H).$$

Equivalently, the change of variables $u^* = u + Q^{-1} H x$ removes the cross product: it turns the problem into a standard regulator with $\tilde{R} = R - H' Q^{-1} H$, $\tilde{A} = A - B Q^{-1} H$, and then $F = \tilde{F} + Q^{-1} H$.

(b) `olrp(beta, A, B, Q, R, H)` in `code/rmtlib.py` iterates $P_{j+1} = R + \beta A' P_j A - (\beta A' P_j B + H')(Q + \beta B' P_j B)^{-1} (\beta B' P_j A + H)$ from $P_0 = 0$ until successive iterates differ by less than 10^{-12} (relative), and returns $F = (Q + \beta B' P B)^{-1} (\beta B' P A + H)$. On a random 3×3 test problem it converges in 16 iterations, with a Riccati residual of 3×10^{-13} . It reproduces the transformed problem's solution to machine precision ($|P - \tilde{P}| < 10^{-15}$, $|F - (\tilde{F} + Q^{-1} H)| < 10^{-16}$).

Exercise 5.2 *Policy improvement for the linear regulator*

The algorithm alternates between evaluating a policy and improving it.

Evaluation. The value of using $u = -F_j x$ forever is $v_j(x) = -\sum_t \beta^t x'_t (R + F'_j Q F_j) x_t$ with $x_{t+1} = (A - B F_j) x_t$. It satisfies $v_j(x) = -x' (R + F'_j Q F_j) x + \beta v_j((A - B F_j) x)$. With $v_j(x) = -x' P_j x$ this is exactly (5.2.10):

$$P_j = R + F'_j Q F_j + \beta (A - B F_j)' P_j (A - B F_j),$$

a Lyapunov equation with solution $P_j = \sum_k \beta^k (A - BF_j)'^k (R + F_j' Q F_j) (A - BF_j)^k$. It exists when the eigenvalues of $A - BF_j$ are below $\beta^{-1/2}$ in modulus.

Improvement. F_{j+1} solves the one-period problem $\max_u \{-x' R x - u' Q u + \beta v_j (A x + B u)\}$, whose first-order condition $(Q + \beta B' P_j B) u = -\beta B' P_j A x$ is (5.2.11): $F_{j+1} = \beta (Q + \beta B' P_j B)^{-1} B' P_j A$.

Why it improves. Let $T_F v(x) = -x' (R + F' Q F) x + \beta v((A - BF)x)$ and $Tv = \max_F T_F v$. Both operators are monotone. By construction $T_{F_{j+1}} v_j = T v_j \geq T_{F_j} v_j = v_j$. Applying $T_{F_{j+1}}$ repeatedly and using monotonicity gives $v_{j+1} = \lim_k T_{F_{j+1}}^k v_j \geq v_j$, i.e. $P_{j+1} \preceq P_j$. The value never deteriorates, and every rule the algorithm visits keeps $\sqrt{\beta}(A - BF_j)$ stable.

Fixed point. If $F_{j+1} = F_j = F$, then $(Q + \beta B' P B) F = \beta B' P A$. Expanding (5.2.10),

$$P = R + \beta A' P A - \beta (A' P B F + F' B' P A) + F' (Q + \beta B' P B) F = R + \beta A' P A - \beta F' B' P A,$$

because $F' (Q + \beta B' P B) F = \beta F' B' P A$ and $A' P B F = (F' B' P A)' = F' B' P A$ (a symmetric matrix). Substituting F gives $P = R + \beta A' P A - \beta^2 A' P B (Q + \beta B' P B)^{-1} B' P A$, the algebraic Riccati equation of the discounted problem. So the limit of the policy improvement algorithm is the optimal rule.

Exercise 5.3 A household's saving problem

(a) State $x_t = (a_t, y_t, y_{t-1}, 1)'$ and control $u_t = i_t$. Then $c_t - b = h' x_t - u_t$ with $h = (r, 1, 0, -b)'$, so $(c_t - b)^2 + \gamma i_t^2 = x_t' h h' x_t - 2 u_t h' x_t + (1 + \gamma) u_t^2$. Hence

$$R = h h', \quad Q = 1 + \gamma, \quad H = -h', \quad A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \rho_1 & \rho_2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

(b) `olrp` converges in 496 iterations; Howard's algorithm gives the same F to 3×10^{-10} . The optimal rules are

$$i_t = 0.316713 y_t + 0.055891 y_{t-1}, \quad c_t = r a_t + y_t - i_t = 0.052632 a_t + 0.683287 y_t - 0.055891 y_{t-1}.$$

Three features stand out.

- The bliss level $b = 30$ does not appear. With $\beta(1 + r) = 1$, a constant consumption gap $c - b$ satisfies the Euler equation: the marginal cost $b - c$ of saving a unit today equals the discounted value $r \sum_{s \geq 1} \beta^s (b - c)$ of the extra consumption it finances forever. So the level of b cannot affect saving.
- The household consumes exactly the interest on its assets, $r a_t$, and never draws down wealth: investment does not depend on a_t .
- Assets therefore follow a unit-root process: the closed-loop matrix $A - BF$ has eigenvalues 0.355, 0.845 (those of the income process), 1 (assets) and 1 (the constant). Because income is transitory here (it decays to zero), the household saves a large share, 31.7%, of current income.

Exercise 5.4 Durable goods and habits

(a) State $x_t = (a_t, y_t, y_{t-1}, h_t, 1)'$, control $u_t = i_t$. With $c_t = ra_t + y_t - u_t$, $s_t - b = \tilde{h}'x_t - \pi u_t$ where $\tilde{h} = (\pi r, \pi, 0, \lambda, -b)'$. So

$$R = \tilde{h}\tilde{h}', \quad Q = \pi^2 + \gamma, \quad H = -\pi\tilde{h}', \quad A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \rho_1 & \rho_2 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ \theta r & \theta & 0 & \delta & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -\theta \\ 0 \end{bmatrix}.$$

(b), (c) The Riccati iteration and Howard's algorithm agree to 4×10^{-10} . Writing both rules in terms of $x_t = (a_t, y_t, y_{t-1}, h_t)$ (the constant has a zero coefficient again, for the reason given in exercise 5.3):

		a_t	y_t	y_{t-1}	h_t
(b) durables	i_t	-0.2991	-1.5205	0.6121	0.2842
	c_t	0.3517	2.5205	-0.6121	-0.2842
(c) habits	i_t	-0.1792	-0.6881	0.4254	0.1702
	c_t	0.2318	1.6881	-0.4254	-0.1702
exercise 5.3	c_t	0.0526	0.6833	-0.0559	—

The closed-loop eigenvalues are $\{0.355, 0.367, 0.845, 1, 1\}$ in (b) and $\{0.355, 0.601, 0.845, 1, 1\}$ in (c).

(d) *Durables* ($\lambda = 1$, small π , $\delta = .95$, $\theta = 1$): utility comes almost entirely from the stock h_t of durables, which depreciates at 5% and is replenished by purchases c_t . The household smooths services s_t , not purchases. A rise in income raises the desired stock, and since purchases are the only way to adjust the stock, expenditures jump far more than one for one: the coefficient on y_t is 2.52, versus 0.68 for nondurable consumption in exercise 5.3. The household dissaves ($i_t < 0$) to finance the purchase. A larger existing stock h_t reduces purchases. This is the familiar excess volatility of durable expenditures, an accelerator.

Habits ($\lambda = -1$, $\pi = 1$): services are $s_t = c_t - h_t$, where the habit stock h_t accumulates past consumption. Consumption today raises utility today but also raises the habit, which lowers future services. The household internalizes this cost: consumption again responds to income more than in the model without habits, and it falls with the habit stock. Note that with $\delta = .95$, $\theta = 1$ the habit stock is twenty times consumption in a steady state, so $s = c - h = -19c$. Bliss $s = b = 30$ would require negative consumption. That is allowed in this LQ setting, but it shows these particular parameters are not a realistic habit specification (a small θ , such as .05, would be).

Exercise 5.5 A geometric sum of forecasts

(a) With $x_t = (y_t, y_{t-1}, w_t)'$, $x_{t+1} = Ax_t + Cw_{t+1}$ where $A = \begin{bmatrix} \rho_1 & \rho_2 & \gamma \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $C = (1, 0, 1)'$. With

$y_t = Gx_t$, $G = (1, 0, 0)$, (2.4.18) gives $E_t \sum_j \beta^j y_{t+j} = G(I - \beta A)^{-1} x_t$ (valid since the eigenvalues of βA are inside the unit circle). Solving $v(I - \beta A) = G$ for the row vector v gives a closed form:

$$E \left[\sum_{j \geq 0} \beta^j y_{t+j} \mid y^t, w^t \right] = \frac{y_t + \beta \rho_2 y_{t-1} + \beta \gamma w_t}{1 - \beta \rho_1 - \beta^2 \rho_2}.$$

(b) $1 - \beta \rho_1 - \beta^2 \rho_2 = 1 - 1.14 + .361 = 0.221$, so

$$E_t \sum_j \beta^j y_{t+j} = 4.52489 y_t - 1.71946 y_{t-1} + 2.14932 w_t.$$

ch5.py confirms this by summing the forecasts $GA^j x$ directly.

Exercise 5.6 Finding the state is an art

(a) The summand $s_t d_t = \beta^t (b_0 d_t - b_1 d_t^2)$ is a quadratic form in the state $x_t = (d_t, d_{t-1}, 1)'$:

$$x_{t+1} = \underbrace{\begin{bmatrix} \rho_1 & \rho_2 & \rho_0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_A x_t + \underbrace{\begin{bmatrix} \sigma_d \\ 0 \\ 0 \end{bmatrix}}_C \epsilon_{t+1}, \quad b_0 d_t - b_1 d_t^2 = x_t' Y x_t, \quad Y = \begin{bmatrix} -b_1 & 0 & b_0/2 \\ 0 & 0 & 0 \\ b_0/2 & 0 & 0 \end{bmatrix}.$$

By (2.4.19)–(2.4.20), $v_0 = x_0' \nu x_0 + \sigma$, where

$$\nu = Y + \beta A' \nu A, \quad \sigma = \frac{\beta}{1 - \beta} \text{tr}(\nu C C').$$

Algorithm: (1) solve the Lyapunov equation for ν , either by iterating $\nu_{k+1} = Y + \beta A' \nu_k A$ from $\nu_0 = 0$ or with the doubling algorithm ($\nu_{k+1} = \nu_k + \beta^{2^k} A_k' \nu_k A_k$, $A_{k+1} = A_k^2$); (2) compute σ ; (3) evaluate at $x_0 = (d_0, d_{-1}, 1)'$. The same ν, σ give the value $v_t = x_t' \nu x_t + \sigma$ at every date, because the problem is time invariant. (Recursively, $v_t \equiv E_t \sum_{j \geq 0} \beta^j (b_0 - b_1 d_{t+j}) d_{t+j}$ satisfies $v_t = (b_0 - b_1 d_t) d_t + \beta E_t v_{t+1}$.)

(b) The series defining ν is $\sum_k \beta^k A'^k Y A^k$, which converges if every eigenvalue of $\sqrt{\beta} A$ lies inside the unit circle. The eigenvalues of A are 1 (the constant, harmless since $\sqrt{\beta} < 1$) and the roots λ_1, λ_2 of $\lambda^2 - \rho_1 \lambda - \rho_2 = 0$. So a sufficient condition is

$$|\lambda_i| < \beta^{-1/2}, \quad i = 1, 2.$$

It is needed because $E_0 d_t^2$ grows like $|\lambda_{\max}|^{2t}$, so $\beta^t E_0 d_t^2 \rightarrow 0$ requires $\beta |\lambda_{\max}|^2 < 1$ (it then also covers the linear term, $\beta |\lambda_{\max}| < 1$). A simple sufficient condition on the coefficients is $|\rho_1| + |\rho_2| < \beta^{-1/2}$, because $|\lambda| \geq 1$ with $|\lambda|^2 = |\rho_1 \lambda + \rho_2|$ would require $|\lambda| \leq |\rho_1| + |\rho_2|$. In particular, d may even have a unit root: $\beta < 1$ still makes the sums converge.

Exercise 5.7 *Dynamic Laffer curves*

(a) *Approach.* Eliminate M : the demand equation gives $M_t = \gamma_1 p_t - \gamma_2 p_{t+1}$, and the budget constraint gives $M_t = M_{t-1} + g p_t$. Hence

$$\gamma_2 p_{t+1} = (\gamma_1 + \gamma_2 - g) p_t - \gamma_1 p_{t-1} \quad (t \geq 1), \quad p_1 = \frac{(\gamma_1 - g) p_0 - M_{-1}}{\gamma_2} \quad (t = 0).$$

This is a second-order linear difference equation, so any p_0 determines the whole path, with $M_t = \gamma_1 p_t - \gamma_2 p_{t+1}$. The program (`ch5.py`, function `path`) iterates this recursion and checks positivity. Its characteristic polynomial, $\gamma_2 \lambda^2 - (\gamma_1 + \gamma_2 - g)\lambda + \gamma_1 = 0$, is exactly the condition for a constant gross inflation rate π to be a stationary equilibrium, $g = (\gamma_1 - \gamma_2 \pi)(1 - 1/\pi)$. With $g = .05$ its roots are

$$\lambda_L = 1.001002, \quad \lambda_H = 1.997998.$$

The general solution is $p_t = c_L \lambda_L^t + c_H \lambda_H^t$, where (c_L, c_H) are determined by p_0 and $p_1(p_0)$.

(b) Positivity of M_t requires $p_{t+1}/p_t < \gamma_1/\gamma_2 = 2$. Positivity of p_t requires $c_H \geq 0$: otherwise the dominant root λ_H eventually drives p_t negative. Now $c_H = (p_1 - \lambda_L p_0)/(\lambda_H - \lambda_L)$, so $c_H \geq 0$ iff $(\gamma_1 - g - \gamma_2 \lambda_L) p_0 \geq M_{-1}$, i.e.

$$p_0 \geq p_0^{\min} = \frac{M_{-1}}{\gamma_1 - g - \gamma_2 \lambda_L} = 2.004012.$$

For every $p_0 \geq p_0^{\min}$ the path is an equilibrium. Gross inflation p_{t+1}/p_t is a weighted average of λ_L and λ_H that moves monotonically toward $\lambda_H < 2$, and $\pi_1 = (\gamma_1 - g - M_{-1}/p_0)/\gamma_2 < 1.999$, so $M_t > 0$ throughout. So there is a continuum of equilibria indexed by $p_0 \in [2.004012, \infty)$. Numerically, $p_0 = p_0^{\min}(1 - 10^{-6})$ already produces a negative price within 80 periods.

(c) For $p_0 > p_0^{\min}$, $c_H > 0$, so $p_{t+1}/p_t \rightarrow \lambda_H$. Since $M_t = \gamma_1 p_t - \gamma_2 p_{t+1} \approx c_H(\gamma_1 - \gamma_2 \lambda_H) \lambda_H^t$, the money growth rate also converges to λ_H . Both converge to $\lambda_H = 1.997998$, i.e. almost 100% inflation per period (Figure S5.1, left).

(d) The minimal-price equilibrium has $c_H = 0$: $p_t = p_0^{\min} \lambda_L^t$ from the start, and inflation and money growth are constant at $\lambda_L = 1.001002$ (0.1% per period). It is the unique equilibrium whose inflation does not converge to λ_H . (The exercise's lettering is slightly off: part (b) computes $p_0^{\min}$, and the two inflation rates are those of parts (c) and (d).)

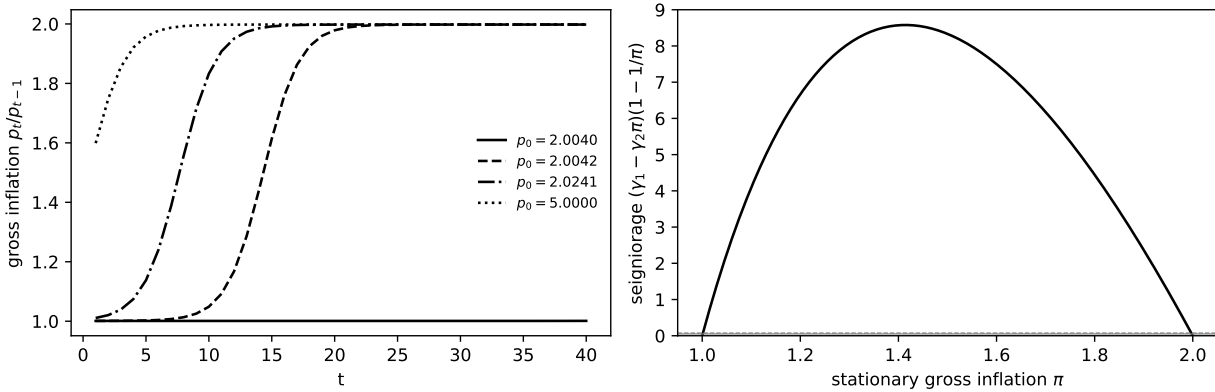


Figure S5.1. Left: gross inflation along four equilibria ($g = .05$). Right: stationary seigniorage $(\gamma_1 - \gamma_2\pi)(1 - 1/\pi)$; the deficits $g = .05$ and $.075$ are the nearly invisible horizontal lines near zero.

(e) With $g = .075$ the roots are $\lambda_L = 1.001505$ and $\lambda_H = 1.996995$, and $p_0^{\min} = 2.006027$. The larger deficit *raises* the low stationary inflation rate ($1.001002 \rightarrow 1.001505$), consistent with the view that larger deficits cause more inflation. But it *lowers* the high rate ($1.997998 \rightarrow 1.996995$), toward which all other equilibria converge. Along those equilibria, larger deficits mean lower eventual inflation.

(f) Stationary seigniorage $S(\pi) = (\gamma_1 - \gamma_2\pi)(1 - 1/\pi)$ is a Laffer curve: zero at $\pi = 1$ (no inflation tax) and at $\pi = \gamma_1/\gamma_2 = 2$ (no money demand), with a maximum at $\pi^* = \sqrt{\gamma_1/\gamma_2} = \sqrt{2}$, where $S(\pi^*) = (\sqrt{\gamma_1} - \sqrt{\gamma_2})^2 = 8.579$. Any deficit $g < 8.579$ can be financed at two stationary inflation rates, one on each side of the peak. Raising g moves the low-inflation root up (the “good” side, where a higher tax rate raises revenue) and the high-inflation root down (the “wrong” side of the Laffer curve). The continuum of equilibria converging to λ_H lives on the wrong side: there, a larger deficit is financed with a smaller inflation tax rate and a larger tax base. These are the “perverse” comparative statics familiar from the literature on deficits financed by seigniorage (e.g. Sargent and Wallace, 1981; Marcet and Sargent, 1989).

Exercise 5.8 *Optimal tax smoothing and what a VAR can reveal*

Here b_t is the government’s holding of one-period loans (assets) maturing at t . The budget constraint says $T_t + b_t = g_t + qb_{t+1}$.

(a) State $x_t = (b_t, g_{Pt}, g_{Tt}, 1)'$ (the constant carries μ_T and the linear term w_1T). Control $u_t = T_t$. The transition law is

$$x_{t+1} = \underbrace{\begin{bmatrix} q^{-1} & -q^{-1} & -q^{-1} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \rho & (1-\rho)\mu_T \\ 0 & 0 & 0 & 1 \end{bmatrix}}_A x_t + \underbrace{\begin{bmatrix} q^{-1} \\ 0 \\ 0 \\ 0 \end{bmatrix}}_B T_t + \underbrace{\begin{bmatrix} 0 & 0 \\ c_1 & 0 \\ 0 & c_2 \\ 0 & 0 \end{bmatrix}}_C \epsilon_{t+1},$$

and the Bellman equation is

$$V(x) = \min_T \{w_1T + .5w_2T^2 + \beta EV(Ax + BT + C\epsilon')\}.$$

This is a stochastic linear regulator with $R = 0$, $Q = .5w_2$ and cross-product term $H = (0, 0, 0, .5w_1)$. The value function is quadratic, $V(x) = x'Px + d$ with $d = \frac{\beta}{1-\beta} \text{tr}(PCC')$, and the decision rule is linear, $T_t = -Fx_t$ (certainty equivalence: F does not depend on c_1, c_2).

Computational strategy, and a pitfall. Because $R = 0$, iterating the Riccati equation from $P = 0$ converges immediately to $P \approx 0$ and the rule $T_t = -w_1/w_2$: tax at the loss-minimizing level forever and borrow without limit. That violates the no-Ponzi condition. The terminal condition selects the *stabilizing* solution, for which $\sqrt{\beta}(A - BF)$ is stable. It can be computed by Howard’s algorithm started from a rule that keeps assets from exploding, such as $T_t = g_t - (1 - q)b_t$ (which keeps b_t constant), or by the invariant-subspace method of section 5.5.

Closed form when $\beta = q$. The Euler equation $W'(T_t) = \frac{\beta}{q} E_t W'(T_{t+1})$ becomes $E_t T_{t+1} = T_t$: taxes are a martingale (Barro's tax smoothing). Combining this with the present-value budget constraint $E_t \sum_j q^j T_{t+j} = E_t \sum_j q^j g_{t+j} - b_t$ gives

$$T_t = (1 - q) \left[E_t \sum_{j \geq 0} q^j g_{t+j} - b_t \right] = g_{Pt} + \frac{1 - q}{1 - q\rho} g_{Tt} + \left(1 - \frac{1 - q}{1 - q\rho} \right) \mu_T - (1 - q)b_t.$$

With $\beta = q = .95$, $\rho = .8$, $\mu_T = 1$, Howard's algorithm converges in two steps to exactly $T_t = -0.05 b_t + g_{Pt} + 0.2083 g_{Tt} + 0.7917$, for any (w_1, w_2) . If $\beta \neq q$, taxes drift ($E_t T_{t+1} = \frac{q}{\beta} T_t + \frac{w_1}{w_2} (\frac{q}{\beta} - 1)$), and w_1/w_2 enters the constant of the rule but not its slopes. (Numerically, with $\beta = .96$ the slopes are the same for $(w_1, w_2) = (1, 2)$ and $(3, 5)$, and the constants are 1.047 versus 1.067.)

(b) With $T_t = -F x_t$, the closed-loop system is $x_{t+1} = (A - BF)x_t + C\epsilon_{t+1}$, and $(g_{Tt}, g_{Pt}, T_t, b_{t+1}) = S x_t$ for a selection-and-rule matrix S (rows $e'_3, e'_2, -F$, and the first row of $A - BF$). When $\beta = q$,

$$T_{t+1} - T_t = c_1 \epsilon_{1,t+1} + \frac{1 - q}{1 - q\rho} c_2 \epsilon_{2,t+1}, \quad b_{t+1} = b_t + q^{-1} \left[\left(\frac{1 - q}{1 - q\rho} - 1 \right) (g_{Tt} - \mu_T) \right].$$

Taxes are a random walk. A permanent spending shock is taxed one for one immediately, while only the annuity value of a transitory shock is taxed and the rest is financed by running down assets. Assets are a unit-root process driven by the transitory component. This is exactly the permanent income model of section 2.12 with taxes in place of consumption.

(c) Let $y_t = (g_t, T_t)'$. The model implies a state-space system $x_{t+1} = A^o x_t + C\epsilon_{t+1}$, $y_t = G^o x_t$, with $A^o = A - BF$ and $G^o = \begin{bmatrix} 0 & 1 & 1 & 0 \\ & -F & & \end{bmatrix}$. The population VAR is the infinite-order projection

$$y_t = \sum_{j \geq 1} \mathcal{A}_j y_{t-j} + a_t, \quad E a_t y'_{t-j} = 0 \quad (j \geq 1),$$

obtained from the steady-state Kalman filter for this system as in (2.9.3): $\mathcal{A}_j = G^o (A^o - K G^o)^{j-1} K$ with $E a_t a'_t = G^o \Sigma G^{o'}$.

(d) The VAR is a function of (A^o, C, G^o) only. The exogenous parameters ρ, c_1, c_2, μ_T enter A and C directly. q and β enter through the budget constraint and the rule F . Because of certainty equivalence and the linear-quadratic structure, w_1 and w_2 enter only through the constant of F , and only via w_1/w_2 when $\beta \neq q$. When $\beta = q$ they do not enter at all. So the VAR's slopes and innovation covariance depend on $(\rho, q, \beta, c_1, c_2)$ and its intercepts on $(\mu_T, w_1/w_2)$. The loss parameters w_1, w_2 are not separately identified, and when $\beta = q$ not at all. Also, because g and T both inherit unit roots (and assets have a unit root), the VAR is in levels of nonstationary variables; this is harmless for the population projection but matters for estimation.

(e) Adding $\tilde{b}_t$ gives a third observable with measurement error: $y_t = (g_t, T_t, \tilde{b}_t)' = G^o x_t + v_t$ with G^o augmented by the row e'_1 and $E v_t v'_t = \text{diag}(0, 0, c_3^2)$. The VAR is again the innovations representation delivered by the Kalman filter (with $R = \text{diag}(0, 0, c_3^2)$), so it is a known function of $(\rho, \beta, \mu_T, q, w_1, w_2, c_1, c_2, c_3)$. It helps with inference: the budget identity links $\tilde{b}_{t+1}$, T_t , g_t and $\tilde{b}_t$ up to measurement error, which pins down q and c_3 . The best way to use it is to estimate the parameters by maximum likelihood, evaluating the Gaussian likelihood recursively with the Kalman filter (2.8.3) and solving the government's problem for each trial parameter vector. Equivalently,

choose the parameters so that the model's implied VAR matches the estimated one (minimum distance). w_1 and w_2 remain unidentified beyond the ratio w_1/w_2 , and only when $\beta \neq q$.

Exercise 5.9 *A permanent income planning problem and its econometrics*

Remark. The exercise states the permanent income condition as $\beta\gamma(1-\delta) = 1$. Since $k_{t+1} = (1-\delta+\gamma)k_t + d_t - c_t$, the gross return on capital is $1-\delta+\gamma$, so the Hall–Friedman condition $\beta R = 1$ reads $\beta(\gamma+1-\delta) = 1$.

Part I (a) State $x_t = (k_t, b_t, d_t, 1)'$, control $u_t = i_t$ (then $c_t = \gamma k_t + d_t - i_t$). The transition law is

$$x_{t+1} = \begin{bmatrix} 1-\delta & 0 & 0 & 0 \\ 0 & \rho_b & 0 & \mu_b(1-\rho_b) \\ 0 & 0 & \rho_d & \mu_d(1-\rho_d) \\ 0 & 0 & 0 & 1 \end{bmatrix} x_t + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} i_t + \begin{bmatrix} 0 & 0 \\ \sigma_b & 0 \\ 0 & \sigma_d \\ 0 & 0 \end{bmatrix} \epsilon_{t+1}.$$

The return is $-.5[(\gamma k_t + d_t - b_t - i_t)^2 + e i_t^2]$, i.e. $R = .5hh'$ with $h = (\gamma, -1, 1, 0)'$, $Q = .5(1+e)$, $H = -.5h'$. The Bellman equation is $V(x) = \max_i \{-.5[(\gamma k + d - b - i)^2 + e i^2] + \beta EV(x')\}$. $V(x) = -x'Px - d$ is quadratic, and $i_t = -F x_t$.

Part I (b) None on the decision rules. By the certainty equivalence principle (section 5.3), F solves the same Riccati equation as the nonstochastic problem, which does not involve C . Increases in σ_b or σ_d only lower the value function, through the constant $d = \frac{\beta}{1-\beta} \text{tr}(PCC')$. The *outcomes* (the variability of consumption and capital) of course become more volatile.

Part I (c) Iterate the Riccati equation with cross products (exercise 5.1) from $P = 0$, or use Howard's algorithm or the Schur (invariant-subspace) method of section 5.5. Because $\beta R = 1$ and adjustment costs are quadratic, capital has a unit root under the optimal rule (as in exercises 5.3 and 5.13). If the Riccati iteration from zero picks a non-stabilizing solution, start Howard's algorithm from a stabilizing rule.

Part II (a) For a trial θ , solve the planning problem to get $F(\theta)$. Then the model is the state-space system

$$x_{t+1} = (A - BF)x_t + C\epsilon_{t+1}, \quad Y_t \equiv \begin{bmatrix} c_t \\ i_t \end{bmatrix} = \begin{bmatrix} (\gamma, 0, 1, 0) + F \\ -F \end{bmatrix} x_t \equiv Gx_t,$$

with the unobserved $(k_0, b_0, d_0) \sim \mathcal{N}(\mu_0, \Sigma_0)$. (The row for c_t uses $c_t = \gamma k_t + d_t - i_t$ and $i_t = -F x_t$.) Two shocks drive two observables and there is no measurement error, so the system is not stochastically singular. The Kalman filter (2.7.14) with $R = 0$, started from $(\hat{x}_0, \Sigma_0) = (\mu_0, \Sigma_0)$, delivers innovations $a_t = Y_t - G\hat{x}_t$ and covariances $\Omega_t = G\Sigma_t G'$. The log likelihood is then built recursively as in (2.8.2)–(2.8.3):

$$\log L(\theta) = \sum_{t=0}^T \left[-\log 2\pi - \frac{1}{2} \log \det \Omega_t - \frac{1}{2} a_t' \Omega_t^{-1} a_t \right].$$

Maximize $\log L$ numerically over θ (excluding the known β , and imposing $\beta(\gamma+1-\delta) = 1$, which pins down γ given δ). Every evaluation re-solves the Riccati equation, which is how the cross-equation restrictions of rational expectations enter.

Part II (b) The posterior is $p(\theta | Y^T) \propto L(\theta)p(\theta)$, and the likelihood is available pointwise from the Kalman filter. A random-walk Metropolis–Hastings sampler draws from it: given $\theta^{(j)}$, propose

$\theta^* = \theta^{(j)} + \eta$ with $\eta \sim \mathcal{N}(0, \Delta)$, compute $L(\theta^*)$ (solve the regulator, run the filter), accept with probability $\min\{1, \frac{L(\theta^*)p(\theta^*)}{L(\theta^{(j)})p(\theta^{(j)})}\}$, and repeat. After a burn-in, the draws approximate the posterior; Δ can be tuned from the inverse Hessian at the posterior mode.

Exercise 5.10 *Evaluating and improving a consumption rule*

(a) State $x_t = (k_t, y_t, y_{t-1}, 1)'$. Under the rule, $k_t + y_t - c_t = (2 - R)k_t + (1 - \alpha)y_t$, so

$$x_{t+1} = A^o x_t + C \epsilon_{t+1}, \quad A^o = \begin{bmatrix} R(2-R) & R(1-\alpha) & 0 & 0 \\ 0 & \rho_1 & \rho_2 & \mu_y(1-\rho_1-\rho_2) \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 \\ \sigma_y \\ 0 \\ 0 \end{bmatrix},$$

and $c_t - b = h'x_t$ with $h = (R-1, \alpha, 0, -b)'$. The value $v(x) = -.5E \sum_t \beta^t x_t' h h' x_t$ satisfies the Bellman equation $v(x) = -.5x' h h' x + \beta E v(A^o x + C \epsilon')$. Guessing $v(x) = -.5(x' \nu x + \sigma)$ gives

$$\nu = h h' + \beta A^{o'} \nu A^o, \quad \sigma = \frac{\beta}{1-\beta} \text{tr}(\nu C C'),$$

a Lyapunov equation that has a unique solution when the eigenvalues of $\sqrt{\beta} A^o$ are inside the unit circle. These eigenvalues are $R(2-R) = 1 - (R-1)^2 \leq 1$, the income roots, and 1, so the condition holds for R near one and stationary income. Solve it by doubling.

(b) Treat $u_t = c_t$ as the control: $x_{t+1} = A x_t + B u_t + C \epsilon_{t+1}$ with $B = (-R, 0, 0, 0)'$ and the first row of A equal to $(R, R, 0, 0)$. The loss $.5(u_t - b)^2$ has $Q = .5$, $H = (0, 0, 0, -.5b)$, $R_x = .5b^2 e_4 e_4'$. The given rule is $u_t = -F_0 x_t$ with $F_0 = -(R-1, \alpha, 0, 0)$. Howard's algorithm (exercise 5.2, with cross products) then alternates

$$P_j = R_x + F_j' Q F_j - F_j' H - H' F_j + \beta (A - B F_j)' P_j (A - B F_j), \quad F_{j+1} = (Q + \beta B' P_j B)^{-1} (\beta B' P_j A + H).$$

The first step, F_1 , is the best rule when one may deviate from the given rule for one period only. It already improves the value ($P_1 \preceq P_0$), and iterating to convergence yields the optimal rule. With $\beta R = 1$ that is the permanent income rule. Here k_t already includes interest, since $k_t = \sum_j \beta^j (c_{t+j} - y_{t+j})$, so the rule is $c_t = (1 - \beta)[k_t + E_t \sum_j \beta^j y_{t+j}]$. It puts weight $1 - \beta = (R-1)/R$ on k_t , close to the given rule's $R-1$, and replaces αy_t by the annuity value of the income forecast. (`lq_howard` in `code/rmtlib.py` implements this.)

Exercise 5.11 *Firm-level adjustment costs*

(a) $v(y, Y) = \max_{y'} \{(A_0 - A_1 Y)y - .5d(y' - y)^2 + \beta v(y', H_0 + H_1 Y)\}$. As a regulator, take $x = (y, Y, 1)'$ and $u = y' - y$:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & H_1 & H_0 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & A_1/2 & -A_0/2 \\ A_1/2 & 0 & 0 \\ -A_0/2 & 0 & 0 \end{bmatrix}, \quad Q = \frac{d}{2},$$

so the return is $-(x' R x + u' Q u)$ and $v(x) = -x' P x$.

(b) **olrp** converges in 562 iterations to

$$P = \begin{bmatrix} 0 & 2.0833 & 6916.67 \\ 2.0833 & -6.3953 & -50010.98 \\ 6916.67 & -50010.98 & -1.3556 \times 10^9 \end{bmatrix}, \quad y_{t+1} = y_t - 1.58333 Y_t - 6966.67.$$

So $v(y, Y) = -4.1667 yY - 13833.3 y + 6.3953 Y^2 + 100022 Y + 1.3556 \times 10^9$. Interpretation: the firm's return is linear in its own output, so $\partial v / \partial y = -4.1667 Y - 13833.3$ is simply the present value of prices, $\sum_j \beta^j p_{t+j} = \frac{A_0}{1-\beta} - \frac{A_1 Y_t}{1-\beta H_1} - \frac{A_1 \beta H_0}{(1-\beta)(1-\beta H_1)}$. The Euler equation $d(y_{t+1} - y_t) = \sum_{j \geq 1} \beta^j p_{t+j}$ then gives the decision rule: output changes do not depend on the level y_t (the closed-loop matrix has eigenvalues 1, 0.8, 1).

Remark. With these parameter values the perceived market output converges to $H_0/(1 - H_1) = 1000$, so the price converges to $A_0 - 1000A_1 = -900$. The firm therefore plans to drive its output far below zero. The numbers are correct for the stated problem, but more sensible beliefs would have $H_0/(1 - H_1) < A_0/A_1$ (compare the values used in chapter 7).

Exercise 5.12 *Firm-level adjustment costs, II*

(a) Since p_t and Y_t are observed, so is u_t . The Bellman equation is $v(y, Y, u) = \max_{y'} \{ (A_0 - A_1 Y + u)y - .5d(y' - y)^2 + \beta E v(y', H_0 + H_1 Y + H_2 u, \rho u + \sigma_u \epsilon') \}$. In regulator form, $x = (y, Y, u, 1)'$ and $u^c = y' - y$, with

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & H_1 & H_2 & H_0 \\ 0 & 0 & \rho & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 \\ 0 \\ \sigma_u \\ 0 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & A_1/2 & -1/2 & -A_0/2 \\ A_1/2 & 0 & 0 & 0 \\ -1/2 & 0 & 0 & 0 \\ -A_0/2 & 0 & 0 & 0 \end{bmatrix},$$

B and Q as before. The value function is $v(x) = -x' P x - d$ with $d = \frac{\beta}{1-\beta} \text{tr}(P C C')$.

(b) The decision rule is

$$y_{t+1} = y_t - 1.58333 Y_t - 24.3506 u_t - 6966.67,$$

with the same P as in exercise 5.11 in the $(y, Y, 1)$ block, plus $P_{yu} = 23.851$, $P_{Yu} = -152.77$, $P_{uu} = -4944.6$, $P_{u1} = -2.0222 \times 10^6$, and $d = -234.87$. By certainty equivalence, σ_u does not affect the rule; it enters the value only through d . (Here $d < 0$: the value is *increasing* in demand uncertainty, because future prices enter the firm's value linearly and the value is a convex function of the shock.) The response to u_t is negative. A demand shock raises the current price, which has already been earned. Under the firm's beliefs it also raises future market output by $H_2 = 2$ per unit, and that depresses future prices by more than the shock's persistence raises them: the response of p_{t+j} to u_t is $-19(.9)^j + 20(.8)^j < 0$ for $j \geq 1$.

Exercise 5.13 *Permanent income model*

(a) State $x_t = (a_t, y_t, y_{t-1}, 1)'$, control $u_t = c_t - b$:

$$v(x) = \max_u \{ -.5u^2 - .5\epsilon a^2 + \beta E v(Ax + Bu + C\epsilon') \},$$

$$A = \begin{bmatrix} R & 1 & 0 & -b \\ 0 & \rho_1 & \rho_2 & 1 - \rho_1 - \rho_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 \\ \sigma_y \\ 0 \\ 0 \end{bmatrix},$$

a regulator with $Q = .5$, $R_x = .5\epsilon e_1 e_1'$. Then $v(x) = -x'Px - d$ and $u_t = -Fx_t$.

(b) Both the Riccati iteration from $P = 0$ (682 iterations) and Howard's algorithm give

$$c_t = 0.052650 a_t + 0.226314 y_t - 0.085998 y_{t-1} + 0.8597,$$

$$P = \begin{bmatrix} 0.02771 & 0.11911 & -0.04526 & -525.86 \\ 0.11911 & 0.51202 & -0.19456 & -2260.50 \\ -0.04526 & -0.19456 & 0.07393 & 858.99 \\ -525.86 & -2260.50 & 858.99 & 9.9828 \times 10^6 \end{bmatrix}, \quad d = 0.024321.$$

The huge constant reflects the permanent shortfall of consumption from bliss: $.5 \cdot 999^2 / (1 - \beta) \approx 10^7$.

(c) $|\text{eig}(A - BF)| = \{0.632456, 0.632456, 0.999982, 1\}$: a complex pair from the income process (modulus $\sqrt{.4}$), a root just below one for assets, and the constant.

(d) With $\epsilon = 0$ nothing penalizes debt, and iterating the Riccati equation from $P = 0$ stops immediately at $P = 0$ and $F = 0$: consume at bliss, $c_t = b$, forever. Assets then follow $a_{t+1} = Ra_t + y_t - b$, and $A - BF$ has the explosive root $R = 1.0526$. This is a Ponzi scheme that the household's problem implicitly rules out ($E_0 \sum_t \beta^t a_t^2 < \infty$, as in (2.12.4)). The solution that respects that condition is the stabilizing one. Howard's algorithm started from $c_t - b = (R - 1)a_t + y_t - b$ (which keeps assets constant) finds it in two steps:

$$c_t = 0.052632 a_t + 0.226244 y_t - 0.085973 y_{t-1} + 0.8597, \quad d = 0.024314,$$

with closed-loop eigenvalues $\{0.632, 0.632, 1, 1\}$. This is the exact permanent income rule: the coefficient on assets is $R - 1 = (1 - \beta)R = 0.052632$ exactly, and assets have a unit root. The $\epsilon = 10^{-6}$ solution in (b) differs only in the fifth decimal. The tiny penalty replaces the unit root by 0.999982, which makes the Riccati iteration from zero converge to the right (stabilizing) solution. That is the purpose of the ϵa_t^2 term.

Exercise 5.14 *Permanent income model again*

(a) Now the state must carry four lags of income: $x_t = (a_t, y_t, y_{t-1}, y_{t-2}, y_{t-3}, 1)'$, with $y_{t+1} = (1 - \rho_1 - \rho_2) + \rho_1 y_t + \rho_2 y_{t-3} + \sigma_y \epsilon_{t+1}$ in the second row of A and shifts in rows 3–5. The Bellman equation has the same form as in exercise 5.13.

(b) With $\epsilon = 10^{-6}$ (682 Riccati iterations; Howard agrees):

$$c_t = 0.052650 a_t + 0.214505 y_t + 0.055170 y_{t-1} + 0.058075 y_{t-2} + 0.061133 y_{t-3} + 0.6111,$$

and the value function has $d = 0.021850$, $P_{aa} = 0.02771$, $P_{yy} = 0.45999$ and $P_{11} = 9.9878 \times 10^6$ (full matrix printed by `ch5.py`). Because y_{t-3} forecasts future income, all four lags now enter the consumption rule with positive weights.

(c) $|\text{eig}(A - BF)| = \{0.633, 0.715, 0.715, 0.927, 0.999982, 1\}$. The first four are the roots of $\lambda^4 - .55\lambda^3 - .3 = 0$ (the income process).

(d) With $\epsilon = 0$, iteration from $P = 0$ again yields the Ponzi rule $c_t = b$ (explosive root 1.0526). Howard's algorithm from a stabilizing start gives the permanent income rule $c_t = 0.052632 a_t + 0.214456 y_t + 0.055161 y_{t-1} + 0.058064 y_{t-2} + 0.061120 y_{t-3} + 0.6112$, with $d = 0.021846$ and a unit root in assets. As in exercise 5.13, the two solutions agree to four or five decimals; the penalty merely selects the stabilizing solution.

Chapter 6

Search and Unemployment

Solutions to all 27 exercises of Chapter 6 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Numerical results come from `code/ch6.py`.

Notation: F is the wage-offer c.d.f. with $F(0) = 0$, $F(B) = 1$; Q denotes the value of an unemployed worker *before* drawing an offer; $\bar{w}$ is a reservation wage; $S(x) \equiv \int_x^B (w' - x) dF(w') = \int_x^B [1 - F(w')] dw'$, a decreasing, convex function with $S'(x) = -[1 - F(x)]$. In McCall's model (6.3.1), $\bar{w}$ solves $\bar{w} - c = \frac{\beta}{1-\beta} S(\bar{w})$ ((6.3.3)). The left side increases in $\bar{w}$ and the right side decreases, so the solution is unique.

A basic lemma used repeatedly: if a problem's value of accepting an offer w is $w/(1 - \delta)$ for some effective discount factor δ , and the value of rejecting is a constant x that does not depend on the current offer, then the optimal rule is “accept iff $w \geq \bar{w} \equiv (1 - \delta)x$.”

Exercise 6.1 *Being unemployed with a chance of an offer*

$$v(w) = \max \left\{ \frac{w}{1-\beta}, c + \beta \left[\phi v(0) + (1-\phi) \int_0^B v(w') dF(w') \right] \right\}.$$

The value of rejecting is a constant, Q^r say. Since $c > 0$ it exceeds $0/(1 - \beta)$, so $v(0) = Q^r$, and the policy is a reservation wage $\bar{w} = (1 - \beta)Q^r$. Substituting $v(w') = \max\{w'/(1 - \beta), Q^r\}$ gives $Q^r = c + \beta Q^r + \beta(1 - \phi) \int (w'/(1 - \beta) - Q^r)^+ dF$, i.e.

$$\bar{w} - c = \frac{\beta(1 - \phi)}{1 - \beta} \int_{\bar{w}}^B (w' - \bar{w}) dF(w').$$

The chance of not getting an offer multiplies the option value of search by $1 - \phi$, so the reservation wage falls as ϕ rises.

Exercise 6.2 *Two offers per period*

(a) The best of two independent draws has c.d.f. F^2 . So

$$v(w) = \max \left\{ \frac{w}{1-\beta}, c + \beta \int_0^B v(w') dF^2(w') \right\},$$

and the reservation wage $\bar{w}_2$ solves $\bar{w}_2 - c = \frac{\beta}{1-\beta} S_2(\bar{w}_2)$ with $S_2(x) = \int_x^B [1 - F^2(w)] dw$.

(b) $1 - F^2 = (1 - F)(1 + F) \geq 1 - F$, so $S_2(x) - S_1(x) = \int_x^B F(w)[1 - F(w)] dw \geq 0$ for every x : the best of two offers first-order stochastically dominates a single offer. Let $\bar{w}_1$ solve $\bar{w}_1 - c = \frac{\beta}{1-\beta} S_1(\bar{w}_1)$. Then at $x = \bar{w}_1$ the right side of the two-offer equation is at least the left side, $\frac{\beta}{1-\beta} S_2(\bar{w}_1) \geq \bar{w}_1 - c$. Since $x - c - \frac{\beta}{1-\beta} S_2(x)$ is strictly increasing, its zero satisfies $\bar{w}_2 \geq \bar{w}_1$. The inequality is strict whenever F puts positive mass on $(\bar{w}_1, B)$ with $0 < F < 1$ there, i.e. whenever search is nontrivial. More offers per period raise the option value of waiting and so make the worker choosier.

Exercise 6.3 A random number of offers per period

Conditional on n offers, the best has c.d.f. F^n . Averaging over n , the best offer has c.d.f. $H(w) = \sum_{n=1}^N \pi_n F(w)^n$, and

$$v(w) = \max \left\{ \frac{w}{1-\beta}, c + \beta \int_0^B v(w') dH(w') \right\} = \max \left\{ \frac{w}{1-\beta}, c + \beta \sum_{n=1}^N \pi_n \int v(w') dF^n(w') \right\}.$$

The reservation wage solves $\bar{w} - c = \frac{\beta}{1-\beta} \int_{\bar{w}}^B [1 - H(w)] dw$.

Exercise 6.4 Cyclical fluctuations in the number of offers

(a) $v(w, n) = \max \left\{ \frac{w}{1-\beta}, c + \beta \sum_{m=1}^N \pi_{mn} \int v(w', m) dF^m(w') \right\}.$

(b) The value of rejecting, $Q(n) \equiv c + \beta \sum_m \pi_{mn} \int v(w', m) dF^m(w')$, depends on n (which forecasts next period's number of offers) but not on w , while the value of accepting, $w/(1-\beta)$, is increasing in w . Hence the worker accepts iff $w \geq \bar{w}(n) \equiv (1-\beta)Q(n)$. If $[\pi_{mn}]$ is monotone, so that a high n today makes high m tomorrow more likely, then $Q(n)$ and $\bar{w}(n)$ are higher in "good times." Workers are choosier when offers are plentiful and expected to remain so.

Exercise 6.5 Choosing the number of offers

$$Q = \max_{n \geq 1} \left\{ -k(n) + \int_0^B \max \left\{ \frac{w}{1-\beta}, c + \beta Q \right\} dF^n(w) \right\}.$$

After the offers arrive, the worker accepts the best one iff $w \geq \bar{w} = (1-\beta)(c + \beta Q)$. The reservation wage does not depend on n , since the cost $k(n)$ is sunk by then. The problem is stationary, so the optimal n^* is the same every period. It trades off the marginal cost $k(n+1) - k(n)$ against the gain in $\int \max\{w/(1-\beta), c + \beta Q\} dF^n$ from one more draw.

Exercise 6.6 The Mortensen externality

Let $\Psi(n, Q) \equiv \int \max\{\theta/(1-\beta), \beta Q\} dF^n(\theta)$: with n draws, the pair takes the best match if its value exceeds the value βQ of searching again.

(a) $Q_i = \max_{c_i \geq 0} \{-c_i + \Psi(f(c_i + c_j), Q_i)\}$, $i = 1, 2$. Both parties value a match identically, so they have the same reservation match $\bar{\theta} = (1-\beta)\beta Q$ and there is no unrequited love. An interior choice satisfies

$$1 = f'(c_1 + c_2) \Psi_n(n, Q_i).$$

(b) Both parties receive the same θ , so the planner's weighted benefit of a match is $\lambda\theta + (1-\lambda)\theta = \theta$:

$$Q(\lambda) = \max_{c_1, c_2 \geq 0} \left\{ -\lambda c_1 - (1-\lambda)c_2 + \Psi(f(c_1 + c_2), Q(\lambda)) \right\}.$$

(c) Only total search $c = c_1 + c_2$ affects the number of offers. The planner therefore assigns search to the party with the smaller weight and sets $\min\{\lambda, 1-\lambda\} = f'(c)\Psi_n$. In the Nash equilibrium each party equates its *own* marginal cost, 1, to $f'(c)\Psi_n$. So in equilibrium the marginal benefit of search equals 1, while at the planner's optimum it equals $\min\{\lambda, 1-\lambda\} \leq \frac{1}{2} < 1$. With f concave and Ψ_n decreasing in n (additional draws improve the best match by less and less), the planner

chooses more total search. The source of the inefficiency is the externality: one party's search creates matches that benefit the other party too, but each party counts only its own gain. (Only the sum $c_1 + c_2$ is pinned down in the Nash equilibrium.)

Exercise 6.7 Variable labor supply

Once employed at w , the worker faces the same static problem every period, $U(w) \equiv \max_{l \in [0,1]} u(w(1-l), l)$, with no link between periods (no assets and no state). So the optimal leisure $l^*(w)$, and hence hours $1 - l^*(w)$, are the same in every period. The value of accepting is $U(w)/(1 - \beta)$. By the envelope theorem $U'(w) = u_1(w(1 - l^*), l^*)(1 - l^*) \geq 0$, so U is nondecreasing in w (strictly if the worker works). The Bellman equation is

$$v(w) = \max \left\{ \frac{U(w)}{1 - \beta}, u(c, 1) + \beta \int v(w') dF(w') \right\},$$

taking an unemployed worker to enjoy full leisure. The value of rejecting is a constant, so the worker accepts iff $U(w) \geq (1 - \beta)[u(c, 1) + \beta \int v dF]$, i.e. iff $w \geq \bar{w}$: the reservation wage property.

Exercise 6.8 Wage growth and the reservation wage

Accepting w yields $\sum_t \beta^t \phi^t w = w/(1 - \beta\phi)$, so

$$v(w) = \max \left\{ \frac{w}{1 - \beta\phi}, c + \beta \int v(w') dF(w') \right\}, \quad \bar{w} = (1 - \beta\phi) Q^r.$$

Proceeding as in (6.3.3) with $Q^r = c + \beta \int \max\{w'/(1 - \beta\phi), Q^r\} dF$ gives $(1 - \beta)Q^r = c + \frac{\beta}{1 - \beta\phi} S(\bar{w})$, i.e.

$$(1 - \beta) \bar{w} = (1 - \beta\phi) c + \beta S(\bar{w}).$$

The left side increases in $\bar{w}$ and the right side decreases, so $\bar{w}$ is unique. Raising ϕ lowers the right side by $\beta c d\phi$ at every $\bar{w}$, so the solution falls: $\phi_1 > \phi_2 \Rightarrow \bar{w}(\phi_1) < \bar{w}(\phi_2)$ when $c > 0$. Faster wage growth makes every job worth more relative to the fixed unemployment compensation, so an early start is more valuable. (With $c = 0$ all values scale with $1/(1 - \beta\phi)$ and the reservation wage is independent of ϕ .)

Exercise 6.9 Search with a finite horizon

Period 2. Accept w iff $w \geq c$: $\bar{w}_2 = c$, and the value of entering period 2 unemployed is $E \max\{w, c\}$.

Period 1. Accepting yields $(1 + \beta)w$, increasing in w . Rejecting yields the constant $c + \beta E \max\{w, c\}$. So the worker accepts iff

$$w \geq \bar{w}_1 = \frac{c + \beta E \max\{w, c\}}{1 + \beta} \geq \frac{c + \beta c}{1 + \beta} = c = \bar{w}_2,$$

with strict inequality if $\mathbb{P}(w > c) > 0$. The reservation wage declines because the option value of continued search, $E \max\{w, c\} - c$, is available only while there is time left, and a job accepted earlier lasts longer.

Exercise 6.10 *Finite horizon and a mean-preserving spread*

Let $A_t = \sum_{s=t}^T \beta^{s-t}$ be the annuity factor, so accepting at t yields $A_t w$. Let U_t be the value of entering t unemployed, before drawing, with $U_{T+1} = 0$. Then

$$U_t = E \max\{A_t w, y^u + \beta U_{t+1}\}, \quad \bar{w}_t = \frac{y^u + \beta U_{t+1}}{A_t},$$

where y^u is the payoff of an unemployed period (0 in (a), (b); c in (c)).

(a) With $y^u = 0$: $\bar{w}_T = 0$ for both distributions. At $T-1$, $\bar{w}_{T-1} = \beta U_T / A_{T-1}$ with $U_T = E \max\{w, 0\} = Ew$, which is the same under F_1 and F_2 (equal means). So $\bar{w}_{T-1}$ coincides too.

(b) The function $x \mapsto \max\{A_t x, k\}$ is convex, so a mean-preserving spread raises its expectation: $E_1 \max\{A_t w, k\} \geq E_2 \max\{A_t w, k\}$ for every k . (Integrating by parts twice and using the equal means (i),

$$E_1 g - E_2 g = \int_0^B g''(y) \left[\int_0^y (F_1 - F_2) dw \right] dy \geq 0$$

for smooth convex g by (iii) of section 6.2.2; kinked convex functions are limits of smooth ones.) Now suppose $U_{t+1}^1 \geq U_{t+1}^2$. Then

$$U_t^1 = E_1 \max\{A_t w, \beta U_{t+1}^1\} \geq E_1 \max\{A_t w, \beta U_{t+1}^2\} \geq E_2 \max\{A_t w, \beta U_{t+1}^2\} = U_t^2.$$

Starting from $U_T^1 = U_T^2$ gives $U_{T-1}^1 \geq U_{T-1}^2$. The inequality is strict when $\beta U_T / A_{T-1} = \bar{w}_{T-1}$ lies where $\int_0^y (F_1 - F_2) > 0$. By induction $U_{t+1}^1 \geq U_{t+1}^2$ for all $t \leq T-2$, so $\bar{w}_t^1 \geq \bar{w}_t^2$ for $t \leq T-2$. The riskier distribution raises the option value of search.

(c) With $y^u = c > 0$: $\bar{w}_T = c$ for both, but now $U_T = E \max\{w, c\}$, the expectation of a convex function with a kink at c . So $U_T^1 > U_T^2$ whenever $\int_0^c (F_1 - F_2) dw > 0$, which holds for c inside the region where the spread moves mass. Hence $\bar{w}_{T-1}^1 = (c + \beta U_T^1) / A_{T-1} > \bar{w}_{T-1}^2$. Unemployment compensation creates the kink that makes the spread valuable one period earlier.

Exercise 6.11 *Pissarides: taxation and search intensity*

(a) Let $D \equiv 1 - \beta(1 - \alpha)$ and let U be the value of an unemployed worker. Employment at w is worth $V(w) = w - T(w) + \beta[(1 - \alpha)V(w) + \alpha U]$, so $V(w) = [w - T(w) + \beta \alpha U] / D$, increasing in w . The effort cost e is paid whether or not an offer is accepted, so

$$U = \max_e \left\{ -e + \pi(e) \int \max\{V(w), 1 + z + \beta U\} dF(w) + [1 - \pi(e)](1 + z + \beta U) \right\}.$$

Given an offer, the worker compares $V(w)$ with $1 + z + \beta U$; neither depends on the e already sunk. So she accepts iff $w \geq \bar{w}$, where $V(\bar{w}) = 1 + z + \beta U$, independently of e . Writing the gain from an offer as $\int (V(w) - V(\bar{w}))^+ dF = \frac{1-t}{D} S(\bar{w})$ in the linear-tax case, or in general $\int_{\bar{w}}^B \frac{[w - T(w)] - [\bar{w} - T(\bar{w})]}{D} dF$, the first-order condition for effort is

$$1 = \pi'(e) \int_{\bar{w}}^B (V(w) - V(\bar{w})) dF(w).$$

(b) With $T(w) = t(w - a)$, $V(w) - V(\bar{w}) = (1 - t)(w - \bar{w}) / D$. Eliminating U between the reservation condition and the Bellman equation, one finds, after simplification using $D(1 + z) + \beta(1 - \alpha)(1 + z) =$

$1 + z$,

$$(1 - t)\bar{w} = 1 + z + \beta(1 - \alpha)\Phi(\bar{w}) - ta, \quad \Phi(\bar{w}) \equiv \max_e \left\{ -e + \pi(e) \frac{1-t}{D} S(\bar{w}) \right\}.$$

The left side increases in $\bar{w}$; the right side decreases in $\bar{w}$, since $\Phi' = \pi(e) \frac{1-t}{D} S'(\bar{w}) < 0$ by the envelope theorem. A higher a shifts the right side down, so $\bar{w}$ falls. The effort condition $1 = \pi'(e) \frac{1-t}{D} S(\bar{w})$ then gives a larger $S(\bar{w})$, so $\pi'(e)$ must fall, i.e. e rises ($\pi'' < 0$). With more effort and a lower reservation wage, the probability of leaving unemployment, $\pi(e)[1 - F(\bar{w})]$, rises. Intuitively, a larger exemption a is a lump-sum subsidy to employment.

(c) Differentiating the reservation condition with respect to t (with $\Phi_t = -\pi(e)S(\bar{w})/D$):

$$[(1 - t) - \beta(1 - \alpha)\Phi'(\bar{w})] \frac{d\bar{w}}{dt} = \bar{w} - a - \beta(1 - \alpha) \frac{\pi(e)S(\bar{w})}{D}.$$

The bracket is positive. So a higher tax rate raises the reservation wage, the opposite of a higher a , iff

$$\bar{w} - a > \beta(1 - \alpha)\pi(e)S(\bar{w})/D,$$

i.e. if the marginal worker pays enough tax, $t(\bar{w} - a)$, relative to the value of the search option. In that case effort falls too, since both $(1 - t)$ and $S(\bar{w})$ fall in $1 = \pi'(e) \frac{1-t}{D} S(\bar{w})$. If $\bar{w} < a$ (the marginal worker is subsidized), a higher t lowers $\bar{w}$.

Exercise 6.12 Search and financial income

Assume the worker sees the current ϵ before deciding. Let $\mu_\epsilon = E\epsilon$. Accepting w yields $w + \phi\epsilon + \beta \frac{w + \phi\mu_\epsilon}{1 - \beta} = \frac{w}{1 - \beta} + \phi\epsilon + \frac{\beta\phi\mu_\epsilon}{1 - \beta}$, so

$$v(w, \epsilon) = \max \left\{ \frac{w}{1 - \beta} + \phi\epsilon + \frac{\beta\phi\mu_\epsilon}{1 - \beta}, c + \epsilon + \beta \iint v(w', \epsilon') dF(w') dG(\epsilon') \right\}.$$

Let $Q = \iint v dF dG$. The worker accepts iff

$$w \geq \bar{w}(\epsilon) = (1 - \beta)[c + (1 - \phi)\epsilon + \beta Q] - \beta\phi\mu_\epsilon,$$

which is increasing in ϵ with slope $(1 - \beta)(1 - \phi) > 0$. Taking a job costs part of the financial income, $(1 - \phi)\epsilon$, this period, so high current financial income raises the opportunity cost of accepting. (The assumption $0 < \mathbb{P}\{w \geq c + (1 - \phi)\epsilon\} < 1$ guarantees that the acceptance decision is nontrivial.)

Exercise 6.13 Search and asset accumulation

Measure assets after the return has been realized. Let $W(a, w)$ be the value of a worker who held a job at w last period (and may quit), and $V(a, w)$ the value of a previously unemployed worker holding offer w . Let $U(a)$ be the value of being unemployed this period:

$$\begin{aligned} U(a) &= \max_{0 \leq c \leq a+z} \left\{ u(c) + v(1) + \beta \iint V(R'(a + z - c), w') dH(R') dF(w') \right\}, \\ E(a, w) &= \max_{0 \leq c \leq a+w} \left\{ u(c) + v(0) + \beta \int W(R'(a + w - c), w) dH(R') \right\}, \\ W(a, w) &= \max\{E(a, w), U(a)\}, \quad V(a, w) = \max\{E(a, w), U(a)\}. \end{aligned}$$

Here E is the value of working this period at w , and U is the value of spending the period unemployed. Formally the two decisions coincide, but they differ in what comes next: after an unemployed period the worker draws a fresh offer. The constraint $a \geq 0$ (no borrowing) appears through $c \leq a + w$ or $c \leq a + z$. With concave u and wealth effects, the reservation wage now depends on assets, $\bar{w}(a)$: typically rising in a , because a wealthier worker can better afford to wait.

Exercise 6.14 *Temporary unemployment compensation*

(a) Let v_1 be the value in the first period of unemployment (eligible) and v_+ thereafter, each with an offer in hand, and let $Q_+ = \int v_+ dF$:

$$v_1(w) = \max\left\{\frac{w}{1-\beta}, \gamma + \beta Q_+\right\}, \quad v_+(w) = \max\left\{\frac{w}{1-\beta}, \beta Q_+\right\}.$$

Each is a comparison of an increasing function with a constant, so the policy is a reservation wage: $\bar{w}_1 = (1-\beta)(\gamma + \beta Q_+)$ in the first period and $\bar{w}_+ = (1-\beta)\beta Q_+$ afterwards. $\bar{w}_+$ solves McCall's equation with $c = 0$: $\bar{w}_+ = \frac{\beta}{1-\beta} S(\bar{w}_+)$.

(b) $\bar{w}_1 = \bar{w}_+ + (1-\beta)\gamma > \bar{w}_+$. The reservation wage is high in the first period, while compensation is still available, and drops permanently afterwards.

(c) The hazard is $1 - F(\bar{w}_1)$ in the first period and $1 - F(\bar{w}_+) \geq 1 - F(\bar{w}_1)$ afterwards. So it is nondecreasing in duration, rising once benefits are exhausted.

(d) Let $Q_1 = \int v_1^u dF$ and $Q_+ = \int v_+^u dF$. A worker who stays at w this period is again “previously employed” next period; one who quits draws an offer at once, as an eligible worker:

$$v_e(w) = \max\{w + \beta v_e(w), Q_1\},$$

$$v_1^u(w') = \max\{w' + \beta v_e(w'), \gamma + \beta Q_+\}, \quad v_+^u(w') = \max\{w' + \beta v_e(w'), \beta Q_+\}.$$

(e) Staying is a stationary decision, so $v_e(w) = \max\{w/(1-\beta), Q_1\}$, and the value of accepting an offer is $A(w) \equiv w + \beta v_e(w) = \max\{w/(1-\beta), w + \beta Q_1\}$.

- *Quitting*: an employed worker quits iff $w < w_e = (1-\beta)Q_1$. Since eligibility is valuable, Q_1 , and hence w_e , increase with γ .
- *Ineligible unemployed*: $v_1^u \geq v_+^u$ pointwise, so $Q_1 \geq Q_+$ and $A(w) \geq w + \beta Q_1 \geq \beta Q_+$ for every $w \geq 0$. An ineligible worker accepts *every* offer, $w_+^u = 0$, because one period of work restores eligibility and she can quit next period. (With $\gamma > 0$ the inequality is strict.)
- *Eligible unemployed*: she accepts iff $A(w) \geq \gamma + \beta Q_+$. This reservation wage depends on the parameters. If $w_1^u \geq w_e$, jobs she takes are kept, and $w_1^u = (1-\beta)(\gamma + \beta Q_+)$. If $w_1^u < w_e$, she accepts jobs she will quit next period, and $w_1^u = \gamma - \beta(Q_1 - Q_+)$. The second case arises when γ is small relative to the gain from resampling. The system can then generate a “revolving door”: short spells of work that requalify workers for benefits.

Exercise 6.15 *Seasons, I*

(a) With v_o (v_e) the value in an odd (even) period with best offer w in hand,

$$v_o(w) = \max\left\{\frac{w}{1-\beta}, c + \beta \int v_e dF^2\right\}, \quad v_e(w) = \max\left\{\frac{w}{1-\beta}, c + \beta \int v_o dF\right\}.$$

(b) Two reservation wages, $\bar{w}_o = (1-\beta)x_o$ and $\bar{w}_e = (1-\beta)x_e$, where $x_o = c + \beta Q_e$ and $x_e = c + \beta Q_o$ are the values of rejecting, with $Q_e = \int \max\{w/(1-\beta), x_e\} dF^2$ and $Q_o = \int \max\{w/(1-\beta), x_o\} dF$. Claim: $\bar{w}_o \geq \bar{w}_e$. Suppose instead $x_o < x_e$. Then $Q_e = E_{F^2} \max\{\cdot, x_e\} \geq E_{F^2} \max\{\cdot, x_o\} \geq E_F \max\{\cdot, x_o\} = Q_o$, using first-order dominance of F^2 over F . So $x_o = c + \beta Q_e \geq c + \beta Q_o = x_e$, a contradiction. Hence the worker is choosier in odd periods, when tomorrow brings two offers, than in even periods, when tomorrow brings only one.

Exercise 6.16 *Seasons, II*

(a) Let $W_o(w)$ be the value of a worker employed at w entering an odd period, before the firing draw, and $W_e(w)$ the value of working at w in an even period. Let $V_o(w)$, $V_e(w)$ be the values of an unemployed worker with offer w in hand in an odd or even period. A new hire in an odd period faces no firing risk that period.

$$W_o(w) = (1-\pi)[w + \beta W_e(w)] + \pi \int V_o(w') dF(w'), \quad W_e(w) = w + \beta W_o(w),$$

$$V_o(w) = \max\left\{w + \beta W_e(w), c + \beta \int V_e dF\right\}, \quad V_e(w) = \max\left\{W_e(w), c + \beta \int V_o dF\right\}.$$

(b) W_e and W_o are increasing in w , while the values of rejecting are constants. So the policy is a pair of reservation wages, $\bar{w}_o$ for odd and $\bar{w}_e$ for even periods, where the worker is indifferent. They differ because a job accepted in an odd period survives at least two periods, one accepted in an even period only one, and because the continuation values of rejecting differ. In an example with F uniform on $[0, 1]$, $\beta = .95$ and $c = .5$ (ch6.py), the difference is negligible for small π but $\bar{w}_o > \bar{w}_e$ for large π (.645 versus .635 at $\pi = .5$): the worker is a little choosier when the job she accepts is more durable.

Exercise 6.17 *Gittins indexes for beginners*

(a) Let $V(i, I)$ be the value at the end of a period, before choosing next period's job:

$$V(i, I) = \max\left\{\sum_j A_{ij}[w_a(j) + \beta V(j, I)], \sum_J B_{IJ}[w_b(J) + \beta V(i, J)]\right\}.$$

The state of the idle job is frozen, because returning to a job draws the wage from the distribution conditioned on the last wage there.

(b) Let $v_a(i)$ ($v_b(I)$) be the value of a worker last in state i on A (state I on B) who has never tried the other job. Trying the other job for the first time draws its wage from π_b (π_a):

$$v_a(i) = \max\left\{\sum_j A_{ij}[w_a(j) + \beta v_a(j)], \sum_J \pi_{bJ}[w_b(J) + \beta V(i, J)]\right\},$$

$$v_b(I) = \max\left\{\sum_J B_{IJ}[w_b(J) + \beta v_b(J)], \sum_j \pi_{aj}[w_a(j) + \beta V(j, I)]\right\},$$

and a new worker chooses $\max\{\sum_j \pi_{aj}[w_a(j) + \beta v_a(j)], \sum_J \pi_{bJ}[w_b(J) + \beta v_b(J)]\}$. This is a two-armed bandit. By Gittins' theorem, the optimal rule is to work at the job with the larger *Gittins index*. Each job's index depends only on its own chain and current state, which reduces the two-dimensional problem to two one-dimensional ones.

Exercise 6.18 Jovanovic (1979b): job-specific capital and on-the-job search

Remark. As printed, $x_{t+1} = g(x_t \phi_t) - \delta x_t$ can become negative, and “ δ is a depreciation rate” suggests that $g(x\phi) - \delta x$ is the *change* in capital. We use $x_{t+1} = x_t + g(x_t \phi_t) - \delta x_t$.

(a) With $x' = x + g(x\phi) - \delta x$,

$$V(x) = \max_{\phi, s \geq 0, \phi + s \leq 1} \left\{ x(1 - \phi - s) + \beta \left[\pi(s) \int \max\{V(\mu'), V(x')\} dF(\mu') + (1 - \pi(s)) V(x') \right] \right\}.$$

(b) V is increasing in x : more job-specific capital raises current and future earnings for any policy. So an offer μ' is accepted iff $V(\mu') \geq V(x')$, i.e. iff $\mu' \geq x'$. The reservation match equals the capital the worker would carry into next period on the current job.

(c) We use $\beta = .95$, $A = .8$, $\alpha = .5$, $\delta = .1$, and μ' uniform on $\{1, 2, \dots, 20\}$. We solve by value-function iteration on a grid of 200 points for $x \in [.5, 40]$, with (ϕ, s) on a 58×58 grid subject to $\phi + s \leq .95$. It converges in 408 iterations (Figure S6.1).

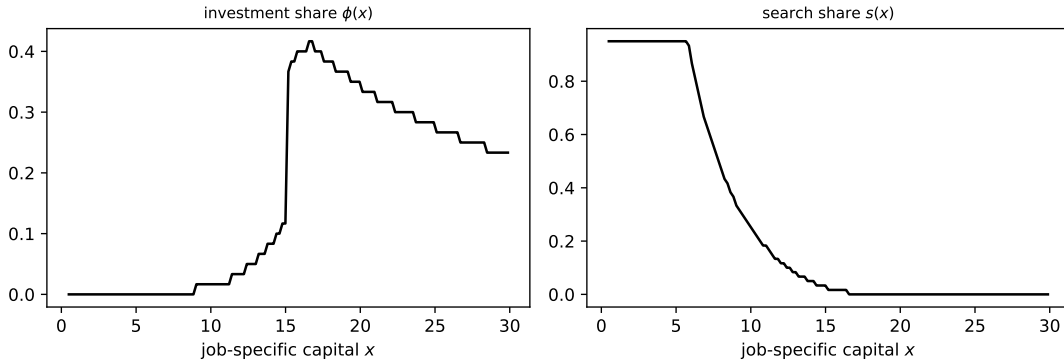


Figure S6.1. Optimal investment and search shares as functions of job-specific capital.

The policies split jobs into bad and good matches. In a poor match ($x \lesssim 6$) the worker searches as hard as possible and does not invest: the capital is not worth building, and the job is a waiting room. Search intensity falls steeply as x rises. Above $x \approx 15$ the worker stops searching and invests heavily (ϕ jumps to about .4), and capital grows toward a stable level near $x \approx 21$ where investment just offsets depreciation. Below $x \approx 15$, capital drifts down while the worker searches. Turnover is thus concentrated among young (low- x) matches, which is Jovanovic’s explanation for the declining separation rate with tenure.

Exercise 6.19 Value function iteration versus policy improvement

(a) $v(w) = \max\{w/(1 - \beta), c + \beta \int v dF\}$. The reservation wage satisfies $w^* - c = \frac{\beta}{1-\beta} S(w^*)$ (see the notation above).

(b) Start from any v_0 with $\int v_0 dF$ finite. Each iterate $v_{n+1}(w) = \max\{w/(1 - \beta), c + \beta \int v_n dF\}$ compares an increasing function with a constant, so its policy is “accept iff $w \geq w_{n+1} = (1 - \beta)(c + \beta \int v_n dF)$.” From $n \geq 1$ on, $v_n(w) = \max\{w, w_n\}/(1 - \beta)$, so

$$w_{n+1} = T(w_n) \equiv (1 - \beta)c + \beta E \max\{w, w_n\}, \quad T'(w) = \beta F(w) \leq \beta.$$

T is a contraction with modulus β whose fixed point is w^* , and

$$|w_{n+1} - w^*| \leq \beta F(\max\{w_n, w^*\}) |w_n - w^*|.$$

Convergence is linear, asymptotically at rate $\beta F(w^*) \leq \beta$.

(c) Given the policy “accept iff $w \geq w_n$,” its value before drawing solves

$$Q_n = c + \beta \left[F(w_n) Q_n + \int_{w_n}^B \frac{w'}{1 - \beta} dF(w') \right].$$

The improvement step compares $w/(1 - \beta)$ with the constant Q_n , which is again a reservation-wage policy, with

$$w_{n+1} = H(w_n) \equiv \frac{(1 - \beta)c + \beta \int_{w_n}^B w' dF(w')}{1 - \beta F(w_n)}.$$

$H(w^*) = w^*$ is equivalent to the reservation equation. Differentiating (with density f),

$$H'(w) = \frac{\beta f(w) [H(w) - w]}{1 - \beta F(w)} \Rightarrow H'(w^*) = 0, \quad H''(w^*) = -\frac{\beta f(w^*)}{1 - \beta F(w^*)}.$$

A Taylor expansion gives $w_{n+1} - w^* \cong K(w_n - w^*)^2$ with

$$K = \frac{1}{2} H''(w^*) = -\frac{\beta f(w^*)}{2[1 - \beta F(w^*)]}.$$

(Howard’s algorithm is Newton’s method, hence quadratic.) *Numerical check* (`ch6.py`; F uniform on $[0, 1]$, $\beta = .95$, $c = .5$, so $w^* = 0.817256$ and $K = -2.1243$), starting both methods from $w_0 = 0$:

n	VFI $w_n - w^*$	Howard $w_n - w^*$	$(w_n - w^*)/(w_{n-1} - w^*)^2$ (Howard)
1	-3.2×10^{-1}	-3.2×10^{-1}	—
2	-2.0×10^{-1}	-9.1×10^{-2}	-0.905
3	-1.4×10^{-1}	-1.3×10^{-2}	-1.532
4	-9.6×10^{-2}	-3.3×10^{-4}	-2.016
5	-7.0×10^{-2}	-2.2×10^{-7}	-2.121
8	-3.0×10^{-2}	$< 10^{-13}$	

The value-iteration error shrinks by a factor approaching $\beta w^* = 0.776$ per step, and the Howard ratio converges to K .

Exercise 6.20 *Riskier offer distributions*

(a) $v(w) = \max\{w/(1 - \beta), c + \beta \int v dF_\alpha\}$. The problem is stationary (the same distribution and payoffs every period, no recall), so the value of rejecting is the same constant every period. The policy is a single reservation wage $\bar{w}_\alpha$ with $\bar{w}_\alpha - c = \frac{\beta}{1 - \beta} \int_{\bar{w}_\alpha}^B (w - \bar{w}_\alpha) dF_\alpha$.

(b) Every F_α has mean $B/2$: $\int_{\alpha B}^{B - \alpha B} \frac{w}{B} dw + \alpha B = \frac{B}{2}$. Moreover $F_\alpha - F_0$ equals $\alpha - w/B \geq 0$ on $[0, \alpha B]$, 0 on the middle interval, and $1 - \alpha - w/B \leq 0$ on $[B - \alpha B, B]$. So $\int_0^y (F_\alpha - F_0) \geq 0$ for all y , with equality at $y = B$: F_α is a mean-preserving spread of the uniform F_0 (conditions (i) and (iii) of section 6.2.2). By the argument of section 6.3.2, using $w - c = \beta(Ew - c) + \beta \int_0^w F$, the spread raises the reservation wage. Here the inequality is strict, because $\int_0^y (F_{.3} - F_0) > 0$ for $0 < y < B$ and $\bar{w} \in (c, B)$. The $\alpha = .3$ worker has the higher reservation wage.

(c) Ambiguous in general. The acceptance probability is $1 - F_\alpha(\bar{w}_\alpha)$, and a higher reservation wage by itself lowers it. But under $F_{.3}$ the offers bunch at the top. With $B = 1$ and $\beta = .95$ (`ch6.py`):

c	$\bar{w}_0$	$\mathbb{P}(\text{accept}), \alpha = 0$	$\bar{w}_{.3}$	$\mathbb{P}(\text{accept}), \alpha = .3$
.05	0.732	0.268	0.858	0.300
.50	0.817	0.183	0.925	0.300
.80	0.898	0.102	0.970	0.300

The $\alpha = .3$ worker waits for the top offer $w = B$ (her reservation wage exceeds $.7B$), which arrives with probability $.3$. That is more often than the uniform worker accepts, so the riskier-distribution worker finds a job *faster*. With a much lower β (a less patient worker) $\bar{w}_0$ falls toward c , and the ranking can reverse.

Exercise 6.21 *Searching for the lowest price*

(a) Let V be the minimal expected total outlay (price plus remaining search costs) before a draw. Then $V = c + E \min\{p, V\}$: buy at p if $p \leq V$, otherwise draw again. Rearranging,

$$c = E(V - p)^+ = \int_{\underline{B}}^V (V - p) dF(p) = \int_{\underline{B}}^V F(p) dp.$$

The right side increases continuously from 0 at $V = \underline{B}$ to $\bar{B} - Ep$ at $V = \bar{B}$, so there is a unique solution $p^* = V$ if $c < \bar{B} - Ep$ (otherwise buy at the first draw). The strategy is to buy at the first price $p \leq p^*$, the reservation price.

(b) The expected price plus search costs is $V = p^*$. Equivalently, the number of draws is geometric with mean $1/F(p^*)$, the expected price paid is $E[p \mid p \leq p^*]$, and $E[p \mid p \leq p^*] + c/F(p^*) = p^*$, because $c = F(p^*)(p^* - E[p \mid p \leq p^*])$ by the equation in (a).

Exercise 6.22 *Quits*

(a) Let $J(w)$ be the value of working at w this period, and $Q = \int v dF$ the value of an unemployed worker before drawing:

$$v(w) = \max\{J(w), c + \beta Q\}, \quad J(w) = w + \beta[(1 - \alpha)J(w) + \alpha Q].$$

After a reclassification the worker faces exactly the problem of an unemployed worker holding an offer. So $J(w) = [w + \beta\alpha Q]/[1 - \beta(1 - \alpha)]$, which is increasing in w .

(b) Accept iff $J(w) \geq c + \beta Q$, i.e. iff $w \geq \bar{w}$.

(c) A reclassified worker holding w' makes the same comparison, so he quits iff $w' < \bar{w}$; a worker who is not reclassified never quits. Eliminating Q , using $(1 - \beta)Q = c + S(\bar{w})/[1 - \beta(1 - \alpha)]$, gives

$$\bar{w} - c = \frac{\beta(1 - \alpha)}{1 - \beta(1 - \alpha)} \int_{\bar{w}}^B (w - \bar{w}) dF(w),$$

McCall's equation with the effective discount factor $\beta(1 - \alpha)$: jobs are less durable, which lowers the reservation wage.

(d) Solve the last equation for $\bar{w}$. The per-period quit probability of an employed worker is $\alpha F(\bar{w})$ (reclassified, then drawing below $\bar{w}$).

Exercise 6.23 A career ladder

- (a) $V(w) = (1 - \alpha)[w + \beta V(w)] + \alpha[\gamma w + \beta V(\gamma w)]$. Guessing $V(w) = \kappa w$ gives $\kappa = (1 + \beta\kappa)m$ with $m \equiv 1 - \alpha + \alpha\gamma$ (the expected gross wage growth), so $\kappa = m/(1 - \beta m)$, provided $\beta m < 1$.
- (b) Accepting w yields $w + \beta V(w) = w/(1 - \beta m)$. So $v(w) = \max\{w/(1 - \beta m), c + \beta \int v dF\}$.
- (c) Accept iff $w \geq \bar{w} = (1 - \beta m)(c + \beta Q)$. Equivalently, $(1 - \beta)\bar{w} = (1 - \beta m)c + \beta S(\bar{w})$, the model of exercise 6.8 with growth factor m . Better promotion prospects lower the reservation wage when $c > 0$.
- (d) An employed worker never wants to leave. His wage only rises, and a job he accepted was already better than continued search. If quitting were allowed, he would never exercise the option.

Exercise 6.24 Human capital

- (a) Let $A(w, h)$ be the value of working this period at w with human capital h , $U(h)$ the value of an unemployed worker about to draw an offer, and $V^e(w, h)$ the value of a worker employed last period at w after seeing his new h :

$$A(w, h) = wh + \beta \sum_{h'} H_e(h, h') V^e(w, h'), \quad V^e(w, h) = \max\{A(w, h), U(h)\},$$

$$U(h) = \int \max\left\{A(w, h), c + \beta \sum_{h'} H_u(h, h') U(h')\right\} dF(w).$$

- (b) $A(w, h)$ is increasing in w , so there are h -dependent reservation wages. An unemployed worker accepts iff $w \geq w^u(h)$, where $A(w^u(h), h) = c + \beta E_u U(h')$. An employed worker quits iff $w < w^e(h)$, where $A(w^e(h), h) = U(h)$. Since $U(h) \geq c + \beta E_u U(h')$ (the option to draw now includes the option to reject), $w^e(h) \geq w^u(h)$. Employed workers may quit. While employed, h tends to grow, so the value $U(h)$ of drawing a fresh w (which multiplies the higher h) rises, and a worker with a low w may find it optimal to quit and redraw. A worker may even accept a low $w \in [w^u(h), w^e(h))$ as a temporary job, to keep his human capital from decaying, and quit and redraw later.
- (c) Discretize w (on the support of F) and use the grid $\bar{h}_1, \dots, \bar{h}_n$. Iterate on the pair (V^e, U) : given (V_k^e, U_k) , compute A_k from the first equation, U_{k+1} from the third and $V_{k+1}^e = \max\{A_k, U_{k+1}\}$. The mapping is a contraction with modulus β in the sup norm. Stop when successive iterates are close, then read off $w^u(h)$ and $w^e(h)$ as the smallest w on the grid satisfying the acceptance and staying conditions.

Exercise 6.25 Markov wages

- (a) $v(i) = \max\{w_i/(1 - \beta), c + \beta \sum_j P_{ij} v(j)\}$, $i = 1, \dots, 5$.
- (b) No, not in general. The value of rejecting, $c + \beta(Pv)_i$, now depends on the current offer, because today's offer forecasts tomorrow's. The acceptance set $\{i : w_i/(1 - \beta) \geq c + \beta(Pv)_i\}$ is an upper interval if $w_i/(1 - \beta) - \beta(Pv)_i$ is increasing in i . That holds, for instance, when offers are i.i.d. (identical rows of P), but nothing guarantees it in general. With the given P it fails: from $w = 3$ the next offer is 1, 2, 4 or 5 with equal probability, so state 3 is a springboard.
- (c), (d) Value iteration gives:

w_i	1	2	3	4	5
$\beta = .95: v(i)$	35.83	40.00	61.83	80.31	100.00
accept?	no	yes	no	no	yes
$\beta = .99: v(i)$	224.19	230.46	352.80	466.76	500.00
accept?	no	no	no	no	yes

(e) With $\beta = .95$ the optimal policy is *not* a reservation wage: the worker accepts 2 and 5 but rejects 3 and 4. From $w = 2$ the chain mostly stays at low wages (it moves up to 3 with probability only .02), so the worker grabs the job. From 3 the next offer is a lottery over $\{1, 2, 4, 5\}$ whose option value exceeds $3/(1 - \beta) = 60$. From 4 the chain stays near the top (4 or 5), so waiting for 5 is worth more than 80. With $\beta = .99$ the worker is patient enough to hold out for the best wage from every state: the acceptance set shrinks to $\{5\}$, which here is a reservation-wage policy. More patience raises the value of waiting for the absorbing-like top state relative to locking in a lower wage.

Exercise 6.26 Markov implications of Neal's model

(a) *Row 3:* a worker who stays put keeps (θ, ϵ) forever, so $P_{33} = 1$. *Row 2:* the new-job region lies in $\theta \geq \bar{\theta}$, where the worker never abandons the career (section 6.5). Keeping θ , he draws $\epsilon' \sim G$ and next period stays put iff $\epsilon' \geq \bar{\epsilon}$, where the reservation job $\bar{\epsilon}$ from (6.5.5) is *independent of θ* . So $P_{21} = 0$, $P_{23} = 1 - G(\bar{\epsilon})$ and $P_{22} = G(\bar{\epsilon})$, the same for every θ in the region. That is what makes the partition Markov. *Row 1:* a new pair is drawn from $F \times G$, so P_{1j} is the $F \times G$ probability of region j of Figure 6.5.2.

Solving Neal's model on the book's grid (50 equally spaced values on $[0, 5]$ for each of θ and ϵ , $\beta = .95$; `ch6.py`) reproduces Figure 6.5.2 (our Figure S6.2), with $\bar{\theta} = 3.98$ and $\bar{\epsilon} = 4.18$, and gives

$$P = \begin{bmatrix} 0.7620 & 0.1804 & 0.0576 \\ 0 & 0.82 & 0.18 \\ 0 & 0 & 1 \end{bmatrix}.$$

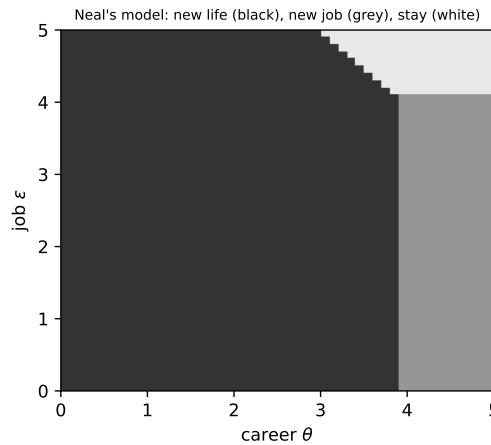


Figure S6.2. Decision regions of Neal's model (compare Figure 6.5.2 of the book).

(b) States 1 and 2 are transient ($P_{11}, P_{22} < 1$) and state 3 is absorbing. The unique invariant distribution is $(0, 0, 1)'$.

(c) Yes. The invariant distribution is unique and concentrated on state 3, so every invariant function is constant on its support. Along any path the worker settles down in finite time. The expected number of periods before settling is finite: $1/(1 - P_{11}) = 4.2$ periods in state 1, plus the time spent in state 2.

(d) Each observed transition reveals the state: a switch of both job and occupation between t and $t + 1$ means $s_t = 1$; a job switch within the occupation means $s_t = 2$; no switch means $s_t = 3$. With the counts n_{ij} of consecutive pairs $(s_t, s_{t+1}) = (i, j)$, the maximum likelihood estimator of a Markov chain is $\hat{P}_{ij} = n_{ij} / \sum_k n_{ik}$. The model implies zero restrictions ($P_{21} = P_{31} = P_{32} = 0$), which can be imposed, or tested with a likelihood ratio test: Neal's data show some late career switches, which reject them. Given parametric forms for F and G , one can go further and estimate the primitives by matching P_{12}, P_{13}, P_{23} (or by maximum likelihood).

Exercise 6.27 *Neal's model with unemployment*

(a)

$$v(\theta, \epsilon) = \max \left\{ \theta + \epsilon + \beta v(\theta, \epsilon), c + \beta \int v(\theta, \epsilon') dG(\epsilon'), c + \beta \iint v(\theta', \epsilon') dF(\theta') dG(\epsilon') \right\}.$$

(b) Let $C(\theta) = c + \beta \int v(\theta, \epsilon') dG$ and $Q = c + \beta \iint v dF dG$. As in Neal's model, v is increasing, $C(\theta)$ is increasing in θ , and the worker keeps his career iff $C(\theta) \geq Q$, i.e. iff $\theta \geq \bar{\theta}$. He stays put iff $(\theta + \epsilon)/(1 - \beta) \geq \max\{C(\theta), Q\}$. The important difference is *within* a kept career. There the worker solves $W(\theta, \epsilon) = \max\{(\theta + \epsilon)/(1 - \beta), c + \beta \int W(\theta, \epsilon') dG\}$. While searching for a job he forgoes his career income θ , so this is McCall's problem for the offer ϵ with effective compensation $c - \theta$:

$$\bar{\epsilon}(\theta) - (c - \theta) = \frac{\beta}{1 - \beta} \int_{\bar{\epsilon}(\theta)} (\epsilon' - \bar{\epsilon}(\theta)) dG(\epsilon'), \quad \frac{d\bar{\epsilon}}{d\theta} = -\frac{1}{1 + \beta[1 - G(\bar{\epsilon})]/(1 - \beta)} \in (-1, 0).$$

So the reservation job quality *falls* with the quality of the career. (In Neal's original model $\bar{\epsilon}$ is independent of θ , because a job switch costs nothing there.) Higher c lowers the cost of switching: it raises $\bar{\epsilon}(\theta)$ and $\bar{\theta}$ and enlarges the switching regions. With the grids of exercise 6.26 (`ch6.py`), $c = 0$ gives $\bar{\theta} = 3.57$, and $\bar{\epsilon}$ falls from 3.16 to 3.06 as θ rises from 3.57 to 5. With $c = 2$, $\bar{\theta} = 3.78$ and $\bar{\epsilon}$ falls from 3.47 to 3.37, in line with the formula (up to grid discreteness). Because $\bar{\epsilon}$ now depends on θ , the three-state chain of exercise 6.26 is no longer exactly Markov: the probability of settling after a job switch depends on the career.

Chapter 7

Recursive Competitive Equilibrium: I

Solutions to all 7 exercises of Chapter 7 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). The Matlab programs `olrp.m` and `nash.m` are replaced by the Python function `olrp` (`code/rmtlib.py`) and a best-response iteration in `code/ch7.py`, which prints every number below.

Throughout exercises 7.1–7.5: $A_0 = 100$, $A_1 = .05$, $\beta = .95$, $d = 10$. We solve linear regulators in the minimization form $\min \sum \beta^t (x_t' R x_t + u_t' Q u_t)$, $x_{t+1} = A x_t + B u_t$, so that a maximized value is $-x' P x$.

Exercise 7.1 A competitive firm

(a) $v(y, Y) = \max_{y'} \{A_0 y - A_1 y Y - .5d(y' - y)^2 + \beta v(y', Y')\}$ subject to $Y' = H_0 + H_1 Y$.

(b) State $x = (y, Y, 1)'$ (own output, market output and a constant), control $u = y' - y$:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & H_1 & H_0 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad R = \begin{bmatrix} 0 & A_1/2 & -A_0/2 \\ A_1/2 & 0 & 0 \\ -A_0/2 & 0 & 0 \end{bmatrix}, \quad Q = \frac{d}{2},$$

so that $-(x' R x + u' Q u) = A_0 y - A_1 y Y - .5d u^2$.

(c) `olrp` converges in 489 iterations to

$$y_{t+1} = 96.9487 + 1.0000 y_t - 0.046282 Y_t.$$

The coefficient on the firm's own output is exactly one. The firm's return is linear in its own y , so, as in exercise 5.11, the *change* in output depends only on the forecast of future prices, i.e. on Y_t , and not on y_t .

(d) With $Y_t = n y_t$, $Y_{t+1} = n(h_0 + h_1 Y_t/n + h_2 Y_t) = n h_0 + (h_1 + n h_2) Y_t$. For $n = 1$: $Y_{t+1} = 96.9487 + 0.953718 Y_t$, which differs from the perceived $(.95, .95)$, so these beliefs are not rational.

(e) The first-order condition for y_{t+1} , combined with the envelope condition $v_y(y, Y) = A_0 - A_1 Y + d(y' - y)$, gives

$$-d(y_{t+1} - y_t) + \beta [A_0 - A_1 Y_{t+1} + d(y_{t+2} - y_{t+1})] = 0,$$

to be solved with y_0 given and the transversality condition $\lim_t \beta^t y_t v_y(y_t, Y_t) = 0$.

Exercise 7.2 Rational expectations

With $y = Y$, the perceived law is reproduced iff $h_0 = H_0$ and $h_1 + h_2 = H_1$ (the map M of section 7.2).

(a)

perceived (H_0, H_1)	implied actual $(h_0, h_1 + h_2)$	equilibrium?
(i) (94.0888, .9211)	(118.4668, .964986)	no
(ii) (93.22, .9433)	(104.7364, .956861)	no
(iii) (95.08187459, .95245906)	(95.08187459, .95245906)	yes

Only (iii) is a fixed point (to 10^{-12}). Its steady state is $H_0/(1 - H_1) = 2000 = A_0/A_1$, where price is zero, as the Euler equation requires in a steady state.

(b) Start from a guess $H^{(0)}$. Given $H^{(k)}$, solve the firm's regulator (exercise 7.1) to get (h_0, h_1, h_2) , compute $M(H^{(k)}) = (h_0, h_1 + h_2)$, and update $H^{(k+1)} = (1 - \gamma)H^{(k)} + \gamma M(H^{(k)})$ with a relaxation parameter $\gamma \in (0, 1]$, until $|H^{(k+1)} - H^{(k)}|$ is negligible. M is not a contraction, so damping ($\gamma < 1$) may be needed. With $\gamma = .5$, starting from (95.5, .95), the iteration converges to (iii) in 22 steps.

Exercise 7.3 *Maximizing welfare*

(a) $V(Y) = \max_{Y'} \{A_0 Y - \frac{A_1}{2} Y^2 - .5d(Y' - Y)^2 + \beta V(Y')\}$.

(b) State $x = (Y, 1)'$, control $u = Y' - Y$, $A = I_2$, $B = (1, 0)'$, $R = \begin{bmatrix} A_1/2 & -A_0/2 \\ -A_0/2 & 0 \end{bmatrix}$, $Q = d/2$.

The solution is

$$Y_{t+1} = 95.0818745922 + 0.952459062704 Y_t,$$

with $V(Y) = -0.262705Y^2 + 1050.82Y + 949181$.

(c) (s_0, s_1) equals the rational expectations equilibrium (iii) of exercise 7.2 to every digit. The planner's Euler equation, $-d(Y_{t+1} - Y_t) + \beta[A_0 - A_1 Y_{t+1} + d(Y_{t+2} - Y_{t+1})] = 0$, is the firm's Euler equation with $y = Y$ imposed. So the competitive equilibrium maximizes the sum of consumer and producer surplus, and solving the planning problem is a reliable way to compute the equilibrium (section 7.2.1).

Exercise 7.4 *Monopoly*

(a) $V(Y) = \max_{Y'} \{A_0 Y - A_1 Y^2 - .5d(Y' - Y)^2 + \beta V(Y')\}$.

(b) The regulator of exercise 7.3 with $R = \begin{bmatrix} A_1 & -A_0/2 \\ -A_0/2 & 0 \end{bmatrix}$ gives

$$Y_{t+1} = 73.4729 + 0.926527 Y_t, \quad \text{steady state } \frac{73.4729}{1 - 0.926527} = 1000.$$

(c) The monopolist's problem is the planner's with the slope of demand doubled: it maximizes revenue, whose derivative is marginal revenue $A_0 - 2A_1 Y$, instead of total surplus, whose derivative is price. Output converges to $A_0/(2A_1) = 1000$, half the efficient steady state 2000. Output also adjusts faster, with root .9265 versus .9525, because the monopolist's objective is more concave in Y .

Exercise 7.5 *Duopoly*

(a) With state $x = (y_1, y_2, 1)'$, a (linear) Markov perfect equilibrium is a pair of rules $y_{i,t+1} = f_i(x_t)$ and value functions v_i such that, for each i , v_i solves firm i 's Bellman equation given that the

rival uses f_j , and f_i attains the maximum. Each firm's rule is a best response to the other's, and strategies depend only on the current state.

(b) $v_1(y_1, y_2) = \max_{y'_1} \{(A_0 - A_1(y_1 + y_2))y_1 - .5d(y'_1 - y_1)^2 + \beta v_1(y'_1, f_2(y_1, y_2))\}$, and symmetrically for firm 2.

(c) In regulator form $x_{t+1} = x_t + B_1 u_{1t} + B_2 u_{2t}$ with $u_i = y'_i - y_i$, $B_1 = e_1$, $B_2 = e_2$. Firm 1 has $R_1 = \begin{bmatrix} A_1 & A_1/2 & -A_0/2 \\ A_1/2 & 0 & 0 \\ -A_0/2 & 0 & 0 \end{bmatrix}$ and $Q_1 = d/2$ (firm 2 symmetric). Given $u_2 = -F_2 x$, firm 1 solves a regulator with transition matrix $A - B_2 F_2$ (equations (7.7.4)–(7.7.7) in their infinite-horizon form). Iterating best responses to convergence (12 rounds, $|F_1 - \text{BR}(F_2)| < 10^{-12}$) gives the symmetric equilibrium

$$y_{i,t+1} = 65.2742 + 0.932925 y_{it} - 0.024230 y_{jt}, \quad i \neq j.$$

Each firm reduces its output when its rival's output is high. Steady-state output is 714.90 per firm, 1429.80 for the industry, with closed-loop roots 0.957 and 0.909.

	monopoly	static Cournot	duopoly (MPE)	competition
steady-state output	1000	1333.3	1429.8	2000

The dynamic duopoly produces *more* than the static Cournot outcome. Adjustment costs make current output a commitment to future output, and a rival cuts its own output when it sees a higher y_j . So each firm expands output strategically to discourage the other (a Markov perfect “capacity” effect).

Exercise 7.6 *Self-control*

(a) A Markov perfect equilibrium is a sequence of consumption rules $\{c_t(k)\}_{t=0}^T$, each depending only on the current capital stock, such that for every t and every k , $c_t(k)$ maximizes personality t 's utility $u(c) + \delta \sum_{j \geq 1} \beta^j u(c_{t+j})$ subject to $k_{t+1} = f(k) - c$, when the later personalities use the rules $c_{t+1}(\cdot), \dots, c_T(\cdot)$. Since the definition applies at every k , it requires optimality off the equilibrium path too (subgame perfection).

(b) Index personalities by the number j of periods remaining after them, so personality $T - j$ is the j th from the end. Let $W_j(k)$ be the *exponentially discounted* value $u(c_{T-j}) + \beta u(c_{T-j+1}) + \dots + \beta^j u(c_T)$ generated by the equilibrium rules from capital k at date $T - j$. The last personality consumes everything: $c_0(k) = f(k)$ and $W_0(k) = u(f(k))$. Personality $T - j - 1$ weighs date $T - j - 1 + s$ by $\delta \beta^s$ for $s \geq 1$. Given the later rules, its objective is therefore $u(c) + \delta \beta W_j(f(k) - c)$, so its value and rule are

$$V_{j+1}(k) = \max_c \{u(c) + \delta \beta W_j(f(k) - c)\}, \quad c_{j+1}(k) = \arg \max.$$

To continue the induction, earlier personalities need the exponentially discounted continuation, which follows the rule c_{j+1} : $W_{j+1}(k) = u(c_{j+1}(k)) + \beta W_j(f(k) - c_{j+1}(k))$. The factor δ appears once, between “now” and “later”. All dates after the next are weighted relative to each other by

β alone, which is why W , not V , is the right object to carry backward. With $\delta = 1$, $V_j = W_j$ and this is ordinary dynamic programming.

(c) The time-0 dictator chooses $\{c_t\}_{t=0}^T$ to maximize $u(c_0) + \delta \sum_{t=1}^T \beta^t u(c_t)$ subject to $k_{t+1} = f(k_t) - c_t$. From date 1 on the weights $\delta \beta^t$ are proportional to β^{t-1} , so the continuation is a standard problem. Let $J_0(k) = u(f(k))$ and

$$J_{s+1}(k) = \max_c \{u(c) + \beta J_s(f(k) - c)\}, \quad s = 0, \dots, T-1,$$

and at date 0 solve $\max_{c_0} \{u(c_0) + \delta \beta J_T(f(k_0) - c_0)\}$. The plan uses the time-consistent exponential rules from date 1 on. With commitment, later personalities cannot re-optimize, and they would want to if $\delta < 1$. This is the value of a “self-control” technology.

Exercise 7.7 *Equilibrium search*

(a) Let $U' = G(U)$ be the law of motion that workers perceive. Let $v(w, U)$ be the value of an unemployed worker holding offer w , $Q(U) = \int v(w', U) dF(w')$ the value of drawing an offer, and $E(w, U)$ the value of a worker employed at w at the start of a period, before the firing draw:

$$\begin{aligned} v(w, U) &= \max \{w + \beta E(w, G(U)), c(U) + \beta Q(G(U))\}, \\ E(w, U) &= (1 - \lambda) [w + \beta E(w, G(U))] + \lambda Q(U). \end{aligned}$$

(b) $E(w, U)$ is increasing in w , while the value of rejecting does not depend on w . So the rule is a reservation wage that depends on the aggregate state: accept iff $w \geq \bar{w}(U)$, where $\bar{w}(U) + \beta E(\bar{w}(U), G(U)) = c(U) + \beta Q(G(U))$. Since $c(U)$ rises with U , unemployment is more attractive when unemployment is high. Future labor-market conditions enter through $G(U)$.

(c) With a continuum of workers and i.i.d. draws, the law of large numbers gives $\phi(U) = 1 - F(\bar{w}(U))$, the probability that a draw exceeds the reservation wage.

(d) A recursive competitive equilibrium is a set of functions $v(w, U)$, $E(w, U)$, $Q(U)$, a reservation wage $\bar{w}(U)$ and a law of motion $U' = G(U)$ such that: (i) given G , v , E , Q solve the Bellman equations in (a) and $\bar{w}(U)$ is the optimal rule; (ii) the perceived law of motion equals the actual one implied by the workers' decisions:

$$G(U) = \lambda(1 - U) + F(\bar{w}(U)) U.$$

As in section 7.2, individual workers take the aggregate law as given, and in equilibrium their collective behavior reproduces it. A stationary equilibrium unemployment rate is a fixed point $U^* = G(U^*)$.

Chapter 8

Equilibrium with Complete Markets

Solutions to all 28 exercises of Chapter 8 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Numerical results come from `code/ch8.py`, which also cross-checks them (brute-force sums over histories, bisection on budget constraints, residuals of Euler equations and budget constraints).

Notation. $q_t^0(s^t)$ is the time-0 (Arrow–Debreu) price of a unit of consumption at history s^t , normalized by $q_0^0(s_0) = 1$; $Q(s' | s)$ is the price, in units of today's good, of a one-period Arrow security paying one unit tomorrow if $s_{t+1} = s'$. With a Markov state and consumption a function of the state, (8.9.4) gives

$$Q(s' | s) = \beta \frac{u'(c(s'))}{u'(c(s))} P(s, s').$$

For two states we collect these prices in a matrix Q ; then (8.10.3) says that the h -step prices are the entries of Q^h . For a claim on a Markov dividend $y(s)$ that includes today's dividend, the value V solves $V = y + QV$, so $V = (I - Q)^{-1}y$. This is also the formula for the natural debt limits (8.9.13). In growth economies, $y_{t+1} = \lambda_{t+1}y_t$ with CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$, the one-period prices are $Q_{ij} = \beta P_{ij} \lambda_j^{-\gamma}$, and ratios of values to the current endowment depend only on the state.

Exercise 8.1 *Family economics, I*

Remark. The book says that “person 2 has no labor but is endowed with s units.” Since n_2 is person 2's labor, and exercise 8.2 starts “instead of being endowed with s units of consumption, household 1 is endowed with 1 unit,” the sentence should read “person 1 has no labor but is endowed with s units.” This matters only for the “natural” distribution of property in (d)–(f).

(a) With multiplier $\mu \geq 0$ on the resource constraint, and respecting $n_2 \geq 0$,

$$\mathcal{L} = \lambda_1 \log c_1 + \lambda_2 (\log c_2 - n_2) + \mu (n_2 + s - c_1 - c_2).$$

(b) The objective is concave and the constraint linear, so the Kuhn–Tucker conditions are sufficient:

$$\frac{\lambda_1}{c_1} = \mu, \quad \frac{\lambda_2}{c_2} = \mu, \quad \mu - \lambda_2 \leq 0 \quad (= 0 \text{ if } n_2 > 0).$$

Interior case, $s < 1 + \lambda_1/\lambda_2$: $\mu = \lambda_2$, so

$$c_1 = \frac{\lambda_1}{\lambda_2}, \quad c_2 = 1, \quad n_2 = 1 + \frac{\lambda_1}{\lambda_2} - s > 0.$$

Corner, $s \geq 1 + \lambda_1/\lambda_2$: $n_2 = 0$, the endowment is split in proportion to the weights, $c_i = \lambda_i s / (\lambda_1 + \lambda_2)$, and $\mu = (\lambda_1 + \lambda_2)/s \leq \lambda_2$, as the third condition requires.

(c) μ is the increase in family welfare from one more unit of the good: the shadow price of the consumption good (of the endowment s). When person 2 works, a unit of the good can also be

obtained with one unit of her labor, which costs λ_2 in welfare. So $\mu = \lambda_2$ is also the shadow wage of her labor. At the corner the good is abundant and $\mu < \lambda_2$: labor is not worth using.

(d) Let the good be the numeraire, $p = 1$. A competitive firm with the linear technology $y = n$ earns zero profits only if the wage is $w = 1$. Give person 1 a claim to e_1 units of the good and person 2 the rest of the endowment, $s - e_1$, plus her time. Each person then maximizes his or her own utility, $\log c_1$ and $\log c_2 - n_2$, subject to $c_1 \leq e_1$ and $c_2 \leq s - e_1 + wn_2$. By the second welfare theorem, the Pareto allocation of (b) is a competitive allocation when $e_1 = c_1$: $e_1 = \lambda_1/\lambda_2$ in the interior case and $e_1 = \lambda_1 s/(\lambda_1 + \lambda_2)$ at the corner. (If $\lambda_1/\lambda_2 > s$, person 2's nonlabor wealth $s - e_1$ is negative: she owes person 1 a lump-sum transfer.) With the natural property distribution of the remark, person 1 owns all of s , $e_1 = s$.

(e) Person 1 consumes his wealth, $c_1 = e_1$. Person 2's first-order conditions $1/c_2 = \nu$ and $\nu w = 1$ (if $n_2 > 0$) give $c_2 = w = 1$ and $n_2 = 1 - (s - e_1)$. So the equilibrium is $p = w = 1$, $c_1 = e_1$, $c_2 = 1$, $n_2 = 1 + e_1 - s$ if $s - e_1 < 1$; otherwise $n_2 = 0$ and $c_2 = s - e_1$. With $e_1 = s$: $c_1 = s$, $c_2 = 1$, $n_2 = 1$. This is the Pareto optimum with weights $\lambda_1/\lambda_2 = s$.

(f) For fixed Pareto weights, $n_2 = \max\{0, 1 + \lambda_1/\lambda_2 - s\}$: labor falls one for one with s , while consumption does not change. Utility is linear in labor, so the whole income effect of a larger endowment falls on labor. In the competitive economy in which person 1 owns s , instead, $n_2 = 1$ for every s . The extra endowment then goes to person 1 and does not affect person 2's budget. The planner, by contrast, pools the endowment and relieves person 2 of work.

Exercise 8.2 *Family economics, II*

Remark. The book says that preferences are those of “exercise 8.25”; it means exercise 8.1.

(a) $\mathcal{L} = \lambda_1 \log c_1 + \lambda_2 (\log c_2 - n_2) + \mu (sn_2 + 1 - c_1 - c_2)$, with $n_2 \geq 0$.

(b) The conditions are $\lambda_1/c_1 = \mu = \lambda_2/c_2$ and $s\mu - \lambda_2 \leq 0$, with equality if $n_2 > 0$. *Interior case*, $s(1 + \lambda_1/\lambda_2) > 1$:

$$\mu = \frac{\lambda_2}{s}, \quad c_1 = \frac{\lambda_1}{\lambda_2} s, \quad c_2 = s, \quad n_2 = 1 + \frac{\lambda_1}{\lambda_2} - \frac{1}{s}.$$

Corner, $s(1 + \lambda_1/\lambda_2) \leq 1$: $n_2 = 0$, $c_i = \lambda_i/(\lambda_1 + \lambda_2)$ and $\mu = \lambda_1 + \lambda_2 \leq \lambda_2/s$.

(c) μ is the shadow price of the good. With positive work, $s\mu = \lambda_2$: a unit of labor yields s goods and costs λ_2 in welfare, so λ_2 is the shadow wage and $\mu = \lambda_2/s$ is the labor cost of one more unit of the good.

(d) Price of the good $p = 1$, wage $w = s$ (zero profits for a firm with technology $y = sn$). Person 1 owns e_1 units and person 2 owns $1 - e_1$ plus her time; the natural distribution is $e_1 = 1$.

(e) Person 2 solves $\max \log c_2 - n_2$ subject to $c_2 = 1 - e_1 + sn_2$. If she works, $s/c_2 = 1$, so $c_2 = s$ and $n_2 = 1 - (1 - e_1)/s$. Person 1 consumes $c_1 = e_1$. With the natural distribution $e_1 = 1$: $c_1 = 1$, $c_2 = s$, $n_2 = 1$. This is the Pareto optimum for $\lambda_1/\lambda_2 = 1/s$. Any interior Pareto optimum is supported by $e_1 = \lambda_1 s/\lambda_2$.

(f) For fixed weights, $n_2 = \max\{0, 1 + \lambda_1/\lambda_2 - 1/s\}$ increases with productivity s ($dn_2/ds = 1/s^2$): a higher return to work raises labor, and both members consume more. In the competitive economy in which person 1 owns the unit endowment, $n_2 = 1$ whatever s is. With log utility and no

nonlabor income, the income and substitution effects of the wage s cancel. If person 2 owned part $e_2 = 1 - e_1 > 0$ of the endowment, $n_2 = 1 - e_2/s$ would increase with s . Labor responds in the opposite direction to exercise 8.1, where s was a pure wealth effect.

Exercise 8.3 *Existence of a representative consumer*

Existence and uniqueness. The objective $\theta u(c_1) + (1 - \theta)w(c - c_1)$ is continuous and strictly concave in c_1 on $[0, c]$, so it has a unique maximizer. We assume Inada conditions (or that the solution is interior); then $c_1(c, \theta) \in (0, c)$ solves

$$\theta u'(c_1) = (1 - \theta)w'(c - c_1).$$

Concavity. Take $c, c' > 0$ with optimal allocations (c_1, c_2) and (c'_1, c'_2) , and $\tau \in [0, 1]$. The allocation $\tau(c_1, c_2) + (1 - \tau)(c'_1, c'_2)$ is feasible for $\tau c + (1 - \tau)c'$, so

$$v_\theta(\tau c + (1 - \tau)c') \geq \theta u(\tau c_1 + (1 - \tau)c'_1) + (1 - \theta)w(\tau c_2 + (1 - \tau)c'_2) \geq \tau v_\theta(c) + (1 - \tau)v_\theta(c'),$$

by concavity of u and w (strictly if $c \neq c'$). v_θ is also increasing, since more resources can be given to either consumer.

Derivative. By the implicit function theorem applied to the first-order condition, $c_1(c, \theta)$ is differentiable, with

$$\frac{\partial c_1}{\partial c} = \frac{(1 - \theta)w''(c_2)}{\theta u''(c_1) + (1 - \theta)w''(c_2)} \in (0, 1).$$

Differentiating $v_\theta(c) = \theta u(c_1(c, \theta)) + (1 - \theta)w(c - c_1(c, \theta))$,

$$v'_\theta(c) = \theta u'(c_1) \frac{\partial c_1}{\partial c} + (1 - \theta)w'(c_2) \left(1 - \frac{\partial c_1}{\partial c}\right) = \theta u'(c_1) = (1 - \theta)w'(c_2),$$

using the first-order condition (an envelope theorem). Moreover $v''_\theta(c) = \theta u''(c_1) \partial c_1 / \partial c = \frac{\theta(1 - \theta)u''w''}{\theta u'' + (1 - \theta)w''} < 0$.

Interpretation. In a complete-markets equilibrium the allocation solves the Pareto problem with $\theta = \mu_1^{-1}/(\mu_1^{-1} + \mu_2^{-1})$. Then (8.5.4) gives $q_t^0(s^t) \propto \beta^t \pi_t(s^t) v'_\theta(Y_t(s^t))$, where Y_t is aggregate consumption: prices are those of a single consumer with utility v_θ who consumes the aggregate endowment. Since $\theta u'(c_1) = (1 - \theta)w'(c_2) = v'_\theta(c)$, each consumer's marginal utility is proportional to the representative consumer's. The representative consumer's preferences depend on the distribution of wealth through θ , unless preferences are identical and homothetic.

Exercise 8.4 *Term structure of interest rates*

(a) Let $s^t = (\lambda_0, \dots, \lambda_t)$; then $y_t(s^t) = y_0 \lambda_1 \cdots \lambda_t$. A competitive equilibrium (with time-0 trading) is a price system $\{q_t^0(s^t)\}$ and an allocation $\{c_t(s^t)\}$ such that (i) given the prices, $\{c_t\}$ maximizes $\sum_t \sum_{s^t} \beta^t u(c_t(s^t)) \pi_t(s^t)$ subject to $\sum_t \sum_{s^t} q_t^0(s^t) [c_t(s^t) - y_t(s^t)] \leq 0$, and (ii) $c_t(s^t) = y_t(s^t)$ for all t, s^t .

(b) Market clearing gives the allocation, $c_t = y_t$. The first-order condition (8.5.4), normalized by $q_0^0 = 1$, then gives the prices

$$q_t^0(s^t) = \beta^t \pi_t(s^t | s_0) \left(\frac{y_t(s^t)}{y_0} \right)^{-\gamma} = \prod_{k=1}^t \beta P(\lambda_{k-1}, \lambda_k) \lambda_k^{-\gamma}.$$

So the one-period kernel is $Q_{ij} = \beta P_{ij} \bar{\lambda}_j^{-\gamma}$, which depends on the state only, and the time-0 price of a date- j claim contingent on $\lambda_j = \bar{\lambda}_k$ is $(Q^j)_{i_0 k}$. With these prices the budget holds with equality at $c = y$, and the first-order conditions are sufficient because u is concave.

(c) The stationary distribution of P is $\bar{\pi} = (1 - p_{22}, 1 - p_{11}) / (2 - p_{11} - p_{22}) = (3/7, 4/7)$, so the unconditional mean gross growth rate is

$$E\lambda = \frac{3}{7}(.98) + \frac{4}{7}(1.03) = 1.008571,$$

i.e. 0.857% per period. (If “before observing λ_0 ” refers to the given $\pi_\lambda = (.5, .5)$, which is not stationary, the mean growth rate at t is $\pi_\lambda P^t \bar{\lambda}$: 1.005, 1.00625, 1.00706, ..., converging to 1.008571.)

(d)–(f) With $\lambda_0 = \bar{\lambda}_1$ the prices are the entries of row 1 of Q^j and their sums. Here $Q = \begin{bmatrix} 0.799667 & 0.180978 \\ 0.149938 & 0.769158 \end{bmatrix}$ and `ch8.py` gives (checked by brute-force summation over histories):

	$j = 0$	$j = 1$	$j = 2$
(d) riskless	1	0.980645	0.950526
(e) only if $\lambda_j = .98$	1	0.799667	0.666602
(f) only if $\lambda_j = 1.03$	0	0.180978	0.283923

At $j = 0$ the state is known: $\lambda_0 = .98$, so the claim in (e) is riskless and the one in (f) pays nothing.

(g) Each price in (d) is the sum of the prices in (e) and (f). A riskless claim is a portfolio of the two contingent claims, and any other price would create an arbitrage. The riskless yields are $-\log(0.980645) = 1.95\%$ for one period and $-\frac{1}{2}\log(0.950526) = 2.54\%$ per period for two: the term structure slopes up in the low-growth state, because growth is expected to rise. Per unit of probability, claims on the low-growth state are dearer ($0.7997/.8 = 1.000$ versus $0.1810/.2 = 0.905$ at $j = 1$): they pay when marginal utility is high.

Exercise 8.5 Periodic endowments

(a) A price system $\{q_t\}$ and an allocation $\{c_t^1, c_t^2\}$ such that each $\{c_t^i\}$ maximizes $\sum_t \beta^t u(c_t^i)$ subject to $\sum_t q_t c_t^i \leq \sum_t q_t y_t^i$, and $c_t^1 + c_t^2 = y_t^1 + y_t^2$ for all t .

(b) The aggregate endowment is 1 every period. The first-order conditions $\beta^t u'(c_t^i) = \mu_i q_t$ give $u'(c_t^1)/u'(c_t^2) = \mu_1/\mu_2$. The map $c \mapsto u'(c)/u'(1-c)$ is strictly decreasing, so $c_t^1 = \bar{c}^1$ is constant, and so is $c_t^2 = 1 - \bar{c}^1$ (as in section 8.6.3). Then $q_t = \beta^t u'(\bar{c}^1)/\mu_1$, and the normalization $q_0 = 1$ gives $q_t = \beta^t$. Consumer 1's budget, $\bar{c}^1 \sum_t \beta^t = \sum_{j \geq 0} \beta^{3j}$, gives

$$\bar{c}^1 = \frac{1 - \beta}{1 - \beta^3} = \frac{1}{1 + \beta + \beta^2}, \quad \bar{c}^2 = \frac{\beta + \beta^2}{1 + \beta + \beta^2}.$$

Consumer 2 has two thirds of the endowment but receives it later, so his share is below $2/3$ ($\bar{c}^2 < 2/3$ because $\beta < 1$).

(c) By (8.7.1) the asset is worth the value of its payouts at the equilibrium prices:

$$p = \sum_{t \geq 0} .05 q_t = \frac{.05}{1 - \beta}$$

units of the time-0 good (or $.05\beta/(1-\beta)$ if the first payment is at $t = 1$). The asset is redundant, so issuing it changes neither the prices nor the allocation.

Exercise 8.6 A Mehra–Prescott economy

(a) As in exercise 8.4(a): prices $\{q_t^0(s^t)\}$ and an allocation $\{c_t(s^t)\}$ such that the allocation maximizes expected utility subject to the single time-0 budget constraint, and $c_t(s^t) = d_t(s^t)$. In equilibrium $q_t^0(s^t) = \beta^t \pi_t(s^t)(d_t/d_0)^{-\gamma}$.

(b), (c) $q_5^0 = \beta^5 \mathbb{P}(\lambda_1, \dots, \lambda_5 \mid \lambda_0) d_5^{-\gamma}$:

$$(b) \quad \mathbb{P} = P_{11}P_{11}P_{12}P_{21}P_{12} = .00256, \quad d_5 = .97^3 1.03^2 = 0.968255,$$

$$q_5^0 = .95^5 (.00256)(0.968255)^{-2} = 0.0021129;$$

$$(c) \quad \mathbb{P} = P_{12}P_{22}^3P_{21} = .01458, \quad d_5 = 1.03^4 .97 = 1.091744,$$

$$q_5^0 = .95^5 (.01458)(1.091744)^{-2} = 0.0094653.$$

(d) Let $A_{ij} = \beta P_{ij} \bar{\lambda}_j^{1-\gamma}$. Then $\beta^t E_0[(d_t/d_0)^{1-\gamma}] = e'_{\lambda_0} A^t \mathbf{1}$ and, by (8.7.1),

$$p_0 = \sum_{t \geq 0} \sum_{s^t} q_t^0(s^t) d_t(s^t) = d_0 \sum_{t \geq 0} \beta^t E_0 \left[\left(\frac{d_t}{d_0} \right)^{1-\gamma} \right] = d_0 e'_{\lambda_0} (I - A)^{-1} \mathbf{1},$$

which converges because the spectral radius of A is $0.943 < 1$. Numerically, $(I - A)^{-1} \mathbf{1} = (18.933, 16.799)'$, so the claim (including the date-0 dividend) is worth 18.933; ex dividend it is worth 17.933.

(e) With $Q_{ij} = \beta P_{ij} \bar{\lambda}_j^{-\gamma}$, summing $\beta^5 \pi_5(s^5) d_5^{-\gamma}$ over all histories with $\lambda_5 = .97$ gives

$$q = e'_1 Q^5 e_1 = (Q^5)_{11} = 0.44024,$$

confirmed by brute-force summation over the 2^5 histories.

(f) Arrow prices do not depend on d_t : $Q(\lambda' \mid \lambda) = \beta P(\lambda, \lambda') (c_{t+1}/c_t)^{-\gamma} = \beta P(\lambda, \lambda') \lambda'^{-\gamma}$. A recursive competitive equilibrium is an initial wealth $a_0 (= 0)$, natural debt limits $\bar{A}(d, \lambda)$, a pricing kernel $Q(\lambda' \mid \lambda)$, a value function $v(a, d, \lambda)$ and decision rules $c = h(a, d, \lambda)$, $\hat{a}(\lambda') = g(a, d, \lambda, \lambda')$ such that: (i) given Q and $\bar{A}$, v solves

$$v(a, d, \lambda) = \max_{c, \hat{a}(\cdot)} \left\{ u(c) + \beta \sum_{\lambda'} P(\lambda, \lambda') v(\hat{a}(\lambda'), \lambda' d, \lambda') \right\}$$

subject to $c + \sum_{\lambda'} Q(\lambda' \mid \lambda) \hat{a}(\lambda') \leq d + a$ and $-\hat{a}(\lambda') \leq \bar{A}(\lambda' d, \lambda')$, with h, g the maximizers; (ii) markets clear: $h(a_t, d_t, \lambda_t) = d_t$ and $g(\cdot) = 0$ (Arrow securities are in zero net supply). By homogeneity, v and the rules can be written in terms of a/d and λ .

(g) The natural debt limit is the value of the endowment from t on, $\bar{A}(d, \lambda_i) = d \bar{a}_i$, with $\bar{a} = (I - A)^{-1} \mathbf{1}$ from (d):

$$\bar{A} = 18.933 d_t \quad \text{if } \lambda_t = .97, \quad \bar{A} = 16.799 d_t \quad \text{if } \lambda_t = 1.03.$$

At t the consumer may promise at most $\lambda' d_t \bar{a}(\lambda')$ for delivery in state λ' at $t + 1$. The limit per unit of endowment is lower in the high-growth state because, with $\gamma = 2$, high growth lowers the price of future goods (the interest rate is high there).

(h) The allocation is $c_t = d_t$ with zero Arrow-security holdings, and

$$Q = \begin{bmatrix} \beta P_{11} \bar{\lambda}_1^{-2} & \beta P_{12} \bar{\lambda}_2^{-2} \\ \beta P_{21} \bar{\lambda}_1^{-2} & \beta P_{22} \bar{\lambda}_2^{-2} \end{bmatrix} = \begin{bmatrix} 0.807737 & 0.179093 \\ 0.100967 & 0.805920 \end{bmatrix}.$$

(i) The bond is a portfolio of one of each Arrow security, so its price is the row sum of Q : 0.98683 if $\lambda_t = .97$ and 0.90689 if $\lambda_t = 1.03$ (gross interest rates 1.0133 and 1.1027). High growth is persistent, so in the high-growth state consumption is expected to grow fast, and the interest rate is high.

Remark. A bond “issued in period t that pays at $t+1$ ” is a one-period bond, which is what we priced. If a claim paying at $t+2$ was intended, its price is the row sum of Q^2 : 0.95952 ($\lambda_t = .97$) and 0.83052 ($\lambda_t = 1.03$).

Exercise 8.7 Two log consumers, Markov endowment

(a) $\pi_t(s^t) = \pi(s_t \mid s_{t-1})\pi(s_{t-1} \mid s_{t-2}) \cdots \pi(s_1 \mid s_0)\pi_0(s_0)$, as in (8.9.1); after s_0 is observed, $\pi_0(s_0) = 1$.

(b) The Lagrangian is $\sum_t \sum_{s^t} \{\beta^t \pi_t[\theta \ln c_t^1 + (1-\theta) \ln c_t^2] + \eta_t(s^t)[Y(s_t) - c_t^1 - c_t^2]\}$. The first-order conditions $\theta \beta^t \pi_t / c_t^1 = \eta_t = (1-\theta) \beta^t \pi_t / c_t^2$ and feasibility give

$$c_t^1(s^t) = \theta Y(s_t), \quad c_t^2(s^t) = (1-\theta) Y(s_t), \quad \eta_t(s^t) = \beta^t \pi_t(s^t) / Y(s_t).$$

Each consumer bears only aggregate risk, in proportion to his weight.

(c) A price system $\{q_t^0(s^t)\}$ and an allocation $\{c_t^1(s^t), c_t^2(s^t)\}$ such that each consumer maximizes his expected utility subject to $\sum_t \sum_{s^t} q_t^0(s^t)[c_t^i(s^t) - y_t^i(s^t)] \leq 0$, and $c_t^1(s^t) + c_t^2(s^t) = Y(s_t)$ for all t, s^t .

(d) From (8.5.4), $\beta^t \pi_t / c_t^i = \mu_i q_t^0$, and $c_t^i = \alpha_i Y(s_t)$ with constant shares. Normalizing $q_0^0(s_0) = 1$,

$$q_t^0(s^t) = \beta^t \pi_t(s^t) \frac{Y(s_0)}{Y(s_t)}.$$

Consumer 1's share is his share of wealth, $\alpha_1 = \sum q_t^0 y_t^1 / \sum q_t^0 Y_t$. With log utility $\sum_t q_t^0 Y_t = Y(s_0)/(1-\beta)$, so $\alpha_1 = (1-\beta)V^1(s_0)/Y(s_0)$, where $V^1 = (I-Q)^{-1}y^1$ is the value of consumer 1's endowment (with Q from (h)).

(e) First welfare theorem: the competitive allocation is Pareto optimal. It is the Pareto optimum with $\theta = \alpha_1$, i.e. $\theta = \mu_1^{-1}/(\mu_1^{-1} + \mu_2^{-1})$. Second welfare theorem: every Pareto optimum, i.e. every $\theta \in (0, 1)$, is a competitive allocation after a suitable reallocation of initial wealth. In particular, if consumer 1 receives a share θ of the aggregate endowment claim, the Pareto allocation with weight θ is the equilibrium.

(f) Identical homothetic preferences make the consumption shares constant. So every consumer's marginal utility is proportional to $1/Y(s_t)$, and the formula in (d) gives the prices from the aggregate endowment alone. Only then are the budget constraints needed, to find α_1 .

(g) A recursive competitive equilibrium consists of initial wealths a_0^i (zero here, summing to zero); a pricing kernel $Q(s' \mid s)$; natural borrowing limits $\bar{A}_i(s)$; value functions $v_i(a, s)$; and decision rules $c = h_i(a, s)$, $\hat{a}(s') = g_i(a, s, s')$. They satisfy: (i) v_i solves (8.9.8), maximizing $\ln c + \beta \sum_{s'} v_i(\hat{a}(s'), s')\pi(s' \mid s)$ subject to $c + \sum_{s'} Q(s' \mid s)\hat{a}(s') \leq y^i(s) + a$ and $-\hat{a}(s') \leq \bar{A}_i(s')$, and h_i, g_i attain the maximum; (ii) $\sum_i h_i = Y(s)$ and $\sum_i g_i(\cdot, s') = 0$ for every s' . The natural

borrowing limit $\bar{A}_i(s)$ is the value of consumer i 's endowment from the current state on, the most he could repay by consuming nothing forever. It solves (8.9.13),

$$\bar{A}_i(s) = y^i(s) + \sum_{s'} Q(s' | s) \bar{A}_i(s'), \quad \text{i.e.} \quad \bar{A}_i = (I - Q)^{-1} y^i.$$

With the numbers of (h), $\bar{A}_1 = (3.619, 9.143)'$ and $\bar{A}_2 = (16.381, 30.857)'$ for $s = 0, 1$. Their sum is $Y/(1 - \beta) = (20, 40)'$, as it must be with log utility.

(h) By (8.9.4), $Q(s' | s) = \beta \pi(s' | s) c^i(s)/c^i(s') = \beta \pi(s' | s) Y(s)/Y(s')$: one price for each pair (s, s') , four numbers. With $Y = (1, 2)$:

$$Q = .95 \begin{bmatrix} .8 \cdot \frac{1}{1} & .2 \cdot \frac{1}{2} \\ .3 \cdot \frac{2}{1} & .7 \cdot \frac{2}{2} \end{bmatrix} = \begin{bmatrix} 0.760 & 0.095 \\ 0.570 & 0.665 \end{bmatrix}.$$

The equilibrium consumption shares are $\alpha_1 = 0.18095$ if $s_0 = 0$ and $\alpha_1 = 0.22857$ if $s_0 = 1$ (confirmed by bisection on consumer 1's budget). Consumer 1 enters each period with Arrow-security wealth $\Upsilon^1 = (I - Q)^{-1}(\alpha_1 Y - y^1)$: $(0, -1.905)'$ for $s = 0, 1$ if $s_0 = 0$. He owes goods in the state in which his endowment is high: that is how he insures.

(i) A claim to one unit at $t+1$ in every state has the same payoff as a portfolio of one Arrow security for each s' . If its price differed from the portfolio's, buying the cheaper and selling the dearer would be a riskless profit, as in (8.10.5). Equivalently, the Euler equation gives $p = E_t[\beta u'(c_{t+1})/u'(c_t)]$. So

$$p(s) = \sum_{s'} Q(s' | s) : \quad p(0) = 0.855, \quad p(1) = 1.235.$$

When $s_t = 1$ the bond sells above par, a gross interest rate of 0.810. The endowment is high today and expected to fall, and the interest rate must fall until consumers are content to consume their endowments.

(j) By (8.10.3) the two-period prices are the entries of Q^2 , so the two-period riskless claim costs the row sums of Q^2 : 0.767125 when $s_t = 0$ and 1.308625 when $s_t = 1$.

(k) The five-period Arrow prices are the entries of Q^5 (row s_t , column s_{t+5}):

$$Q^5 = \begin{bmatrix} 0.473941 & 0.149920 \\ 0.899520 & 0.324021 \end{bmatrix}.$$

Exercise 8.8 *Optimal taxation with complete markets*

(a) With $\pi_t(g^t) = P(g_0, g_1)P(g_1, g_2) \cdots P(g_{t-1}, g_t)$,

$$E_0 \sum_{t \geq 0} \beta^t W(R_t) = \sum_{t \geq 0} \sum_{g^t} \beta^t P(g_0, g_1) \cdots P(g_{t-1}, g_t) W(R_t(g^t)).$$

(b) The government minimizes this subject to $\sum_t \sum_{g^t} \beta^t \pi_t(g^t) [R_t(g^t) - g_t] \geq 0$. With multiplier μ the first-order conditions are $\beta^t \pi_t(g^t) W'(R_t(g^t)) = \mu \beta^t \pi_t(g^t)$, i.e. $W'(R_t(g^t)) = \mu$ for every t and g^t . W' is strictly increasing, so revenue is constant, $R_t(g^t) = \bar{R}$: perfect tax smoothing over time and states, because the market prices $\beta^t \pi_t(g^t)$ are proportional to the weights of the government's objective. The budget gives $\bar{R} = (1 - \beta) E_0 \sum_t \beta^t g_t = (1 - \beta) v(g_0)$ with $v = (I - \beta P)^{-1} \bar{g}$. Here

$\det(I - \beta P) = 1 - 1.3\beta + .3\beta^2 = (1 - \beta)(1 - .3\beta)$ and

$$v(\bar{g}_1) = \frac{.2 + .1\beta}{(1 - \beta)(1 - .3\beta)}, \quad v(\bar{g}_2) = \frac{1 - .7\beta}{(1 - \beta)(1 - .3\beta)}, \quad \text{so} \quad R_t(g^t) = \bar{R} = \frac{.2 + .1\beta}{1 - .3\beta}$$

for all t, g^t . The exercise leaves β unspecified. For example, $\bar{R} = 0.41259$ if $\beta = .95$ and 0.39726 if $\beta = .9$. As $\beta \rightarrow 1$, $\bar{R} \rightarrow 3/7$, the stationary mean of g ; as $\beta \rightarrow 0$, $\bar{R} \rightarrow g_0 = .2$. (Quadratic W is used only for the budget arithmetic: any strictly convex W gives the same constant $\bar{R}$.)

(c) $Q(g' | g) = p_{t+1}^0(g^{t+1})/p_t^0(g^t) = \beta P(g, g')$: the price is $\beta P(g, g')$ units of the time- t good.

(d) Let $B_t(g_t)$ be what the government owes at the start of t (payments on the Arrow securities it issued at $t-1$ for the realized state). Its budget constraint is $g_t + B_t(g_t) = \bar{R} + \sum_{g'} \beta P(g_t, g') B_{t+1}(g')$, so B_t is the present value of future surpluses, $B(g) = \bar{R}/(1 - \beta) - v(g)$:

$$B_t(\bar{g}_1) = 0, \quad B_t(\bar{g}_2) = \frac{.2 + .1\beta - (1 - .7\beta)}{(1 - \beta)(1 - .3\beta)} = -\frac{.8}{1 - .3\beta}$$

(-1.1189 if $\beta = .95$; `ch8.py` checks both budget constraints to 10^{-15}). The negative value means that the government *holds* Arrow securities that pay off in war.

(e) Taxes are constant forever. In peace the government runs a surplus $\bar{R} - .2 = .16\beta/(1 - .3\beta)$. It spends the surplus buying claims that pay $.8/(1 - .3\beta)$ if war breaks out next period, at cost $\beta(.2)(.8)/(1 - .3\beta)$: it buys war insurance, and it issues nothing contingent on peace. In war the deficit is $1 - \bar{R} = .8(1 - .5\beta)/(1 - .3\beta)$. It is covered by the insurance payoff, $.8/(1 - .3\beta)$, net of the cost $\beta(.5)(.8)/(1 - .3\beta)$ of insuring against war continuing. Complete markets let the government insure against the fiscal shock instead of adjusting taxes. (With only riskless debt, taxes would have to adjust to spending shocks, as in the martingale tax model of exercise 5.8.)

Exercise 8.9 A competitive equilibrium

(a) A price system $\{q_t\}_{t \geq 0}$ and an allocation $\{c_t^1, c_t^2\}$ such that (i) for each type, $\{c_t^i\}$ maximizes utility subject to $\sum_t q_t c_t^i \leq \sum_t q_t y_t^i$ and $c_t^i \geq 0$; (ii) $c_t^1 + c_t^2 = y_t^1 + y_t^2$ for all t . (With equal numbers, we may work with one consumer of each type.)

(b) The Kuhn–Tucker conditions are, with multipliers $\mu_1, \mu_2 > 0$ on the budgets,

$$\text{type 1: } \beta^t \leq \mu_1 q_t \quad (= \text{ if } c_t^1 > 0), \quad \text{type 2: } \beta^t / c_t^2 = \mu_2 q_t.$$

Try $q_t = \beta^t$ (type 1's condition holding with equality at all dates, $\mu_1 = 1$). Then type 2's condition makes $c_t^2 = \bar{c}^2$ constant, and her budget $\bar{c}^2/(1 - \beta) = \sum_{t \text{ odd}} \beta^t \alpha = \alpha\beta/(1 - \beta^2)$ gives

$$\bar{c}^2 = \frac{\alpha\beta}{1 + \beta} = \mu.$$

So $c_t^2 = \mu$ for all t , and $c_t^1 = 0$ in even periods and $c_t^1 = \alpha$ in odd periods. Every condition holds: $c_t^1 \geq 0$, type 1's condition holds with equality (consistent with $c_t^1 \geq 0$ everywhere), and markets clear. The value of α is exactly the one at which type 1's nonnegativity constraint is just reached in even periods; see exercise 8.10 for larger α .

(c) At $q_t = \beta^t$: type 1's wealth is $\sum_t \beta^t \mu = \mu/(1 - \beta)$, and type 2's is $\alpha\beta/(1 - \beta^2) = \mu(1 + \beta)/(1 - \beta^2) = \mu/(1 - \beta)$. They are equal. Both budgets hold with equality: type 1 consumes $\sum_{t \text{ odd}} \beta^t \alpha = \mu/(1 - \beta)$.

(d) With no uncertainty, the Arrow securities are one-period bonds. An equilibrium is a sequence of bond prices $\{Q_t\}$ (Q_t units of the time- t good buy one unit at $t+1$), natural debt limits $A_t^i = \sum_{\tau \geq t} (Q_t \cdots Q_{\tau-1}) y_\tau^i$, initial holdings $a_0^i = 0$, and consumption and bond plans $\{c_t^i, a_{t+1}^i\}$ such that each type maximizes utility subject to

$$c_t^i + Q_t a_{t+1}^i \leq y_t^i + a_t^i, \quad -a_{t+1}^i \leq A_{t+1}^i, \quad c_t^i \geq 0,$$

and markets clear: $c_t^1 + c_t^2 = y_t^1 + y_t^2$ and $a_{t+1}^1 + a_{t+1}^2 = 0$.

(e) $Q_t = q_{t+1}/q_t = \beta$, a gross interest rate $1/\beta$; the allocation is that of (b). The bond positions are the continuation wealths (8.8.1), $a_t^i = \sum_{\tau \geq t} \beta^{\tau-t} (c_\tau^i - y_\tau^i)$. For type 2, $c^2 - y^2$ alternates $\mu, -\mu/\beta, \mu, \dots$ starting in an even period, so

$$a_t^2 = 0 \text{ (} t \text{ even)}, \quad a_t^2 = -\mu/\beta \text{ (} t \text{ odd)}, \quad a_t^1 = -a_t^2.$$

In even periods type 2 borrows μ from type 1 (she sells μ/β bonds at price β) and consumes it, while type 1 lends his whole endowment and consumes nothing. In odd periods type 2 repays μ/β out of $\alpha = \mu + \mu/\beta$ and consumes μ , and type 1 consumes $\mu + \mu/\beta = \alpha$. The natural debt limit of type 2 at odd dates is $\alpha/(1 - \beta^2) = \mu/[\beta(1 - \beta)] > \mu/\beta$, so it does not bind.

Exercise 8.10 *Corners*

Remark. Part (d) says that type 1's endowment "continues to be 1 each period"; it is μ (or the author has set $\mu = 1$). We keep μ .

(a)–(c) As in exercise 8.9: $q_t = \beta^t$, $c_t^2 = \mu$, $c_t^1 \in \{0, \alpha\}$ in even/odd periods, and both wealths equal $\mu/(1 - \beta)$. The one-period gross interest rate is $q_t/q_{t+1} = 1/\beta$ from every date, odd or even. (ch8.py confirms the budgets and the Kuhn–Tucker conditions on a truncated horizon.)

(d) *No interior equilibrium.* If $c_t^1 > 0$ at every date, then $q_t = \beta^t$ (normalized) and, as in 8.9(b), $c_t^2 = \alpha\beta/(1 + \beta) > \mu$, so $c_t^1 = \mu - c_t^2 < 0$ in even periods: a contradiction. Type 1 is therefore at the corner $c_t^1 = 0$ in even periods, where type 2 consumes the whole aggregate endowment μ . Suppose $c_t^1 > 0$ in odd periods. Then:

- even t : $c_t^2 = \mu$ and $q_t = \beta^t/(\mu_2\mu)$; normalizing $q_0 = 1$ gives $\mu_2 = 1/\mu$ and $q_t = \beta^t$;
- odd t : $q_t = \beta^t/\mu_1$, so $c_t^2 = \mu_1/\mu_2 \equiv k$ is constant and $q_t = \beta^t\mu/k$.

Type 2's budget, $\sum_{\text{even}} \beta^t \mu + \sum_{\text{odd}} \beta^t (\mu/k) k = \sum_{\text{odd}} \beta^t (\mu/k) \alpha$, reads $\mu(1 + \beta)/(1 - \beta^2) = \alpha\beta\mu/[k(1 - \beta^2)]$, so $k = \alpha\beta/(1 + \beta)$. Type 1's corner condition at even dates, $\beta^t \leq \mu_1 q_t = \beta^t \mu_1$, holds iff $\mu_1 = k/\mu \geq 1$, i.e. iff $\alpha \geq \mu(1 + \beta^{-1})$, which is our assumption. The equilibrium is

$$c_t^2 = \begin{cases} \mu & t \text{ even} \\ \frac{\alpha\beta}{1 + \beta} & t \text{ odd} \end{cases}, \quad c_t^1 = \begin{cases} 0 & t \text{ even} \\ \mu + \frac{\alpha}{1 + \beta} & t \text{ odd} \end{cases}, \quad q_t = \begin{cases} \beta^t & t \text{ even} \\ \beta^t \frac{\mu(1 + \beta)}{\alpha\beta} & t \text{ odd} \end{cases}.$$

Type 1's budget then holds by Walras' law. The time-0 wealths are $\mu/(1 - \beta)$ for type 2 (independent of α) and $[\mu + \mu^2(1 + \beta)/\alpha]/(1 - \beta^2)$ for type 1. At $\alpha = \mu(1 + \beta^{-1})$ this reduces to (b). With $\mu = 1$, $\beta = .95$, $\alpha = 3$ (ch8.py): $c^1 = (0, 2.538)$, $c^2 = (1, 1.462)$ in (even, odd) periods, and the wealths are 16.923 and 20.

(e) The gross one-period rates alternate:

$$\frac{q_t}{q_{t+1}} = \frac{\alpha}{\mu(1+\beta)} > \frac{1}{\beta} \quad (t \text{ even}), \quad \frac{q_t}{q_{t+1}} = \frac{\mu(1+\beta)}{\alpha\beta^2} < \frac{1}{\beta} \quad (t \text{ odd})$$

(1.538 and 0.720 in the example), while the two-period rate is still $1/\beta^2$. Compared with (b), rates are higher from even to odd dates and lower from odd to even. The risk-neutral type 1 would like to lend more at even dates, to consume at odd dates, but he has already lent his entire even-date endowment ($c^1 = 0$). So his arbitrage no longer pins the interest rate at $1/\beta$. The rates are set by type 2's Euler equation, $q_{t+1}/q_t = \beta c_t^2/c_{t+1}^2$: her consumption rises from even to odd dates ($\mu < \alpha\beta/(1+\beta)$), and only an interest rate above $1/\beta$ makes her accept that tilt.

Exercise 8.11 *Equivalent martingale measure*

Representation. By definition, $\beta^t \frac{u'(c_t^i)}{\mu_i} \pi_t = b_t \tilde{\pi}_t$. Substituting into a_0 ,

$$a_0 = \sum_t \sum_{s^t} b_t \tilde{\pi}_t(s^t) d_t(s^t) = \sum_t b_t \sum_{s^t} d_t(s^t) \tilde{\pi}_t(s^t) = \tilde{E}_0 \sum_t b_t d_t,$$

where $\tilde{E}_0$ takes each date- t term with respect to $\tilde{\pi}_t$.

$\tilde{\pi}_t$ is a probability measure. (i) Since $u' > 0$, $\mu_i > 0$ and $\pi_t \geq 0$, every term of b_t is nonnegative, and $b_t > 0$ because some history has positive probability. So $\tilde{\pi}_t(s^t) \geq 0$. (ii) $\sum_{s^t} \tilde{\pi}_t(s^t) = b_t^{-1} b_t = 1$. Hence $\tilde{\pi}_t$ is a probability distribution over date- t histories. It is *equivalent* to π_t : $\tilde{\pi}_t(s^t) = 0$ iff $\pi_t(s^t) = 0$, because $u' > 0$. (With d_t bounded and $\sum_t b_t < \infty$, the sums converge absolutely, which justifies the rearrangements.)

Interpretation of b_t . By (8.5.4), $b_t = \sum_{s^t} q_t^0(s^t)$: it is the time-0 price of a riskless claim to one unit of consumption at t , the t -period zero-coupon bond or “riskless strip” of section 8.7.3. So a_0 discounts expected payouts at riskless rates, with the expectation taken under the twisted measure: $\tilde{\pi}_t$ puts more weight on histories with high marginal utility.

Remark. Each $\tilde{\pi}_t$ is a probability distribution on date- t histories, but the family $\{\tilde{\pi}_t\}$ is in general *not* consistent across dates (it does not define one measure on infinite histories). Summing over s_{t+1} , $\sum_{s_{t+1}} \tilde{\pi}_{t+1}(s^{t+1}) = \tilde{\pi}_t(s^t) p_{t+1}^t(s^t) b_t/b_{t+1}$, where $p_{t+1}^t(s^t)$ is the price of the one-period riskless bond at s^t . This equals $\tilde{\pi}_t(s^t)$ only if one-period interest rates are not random. In the economy of exercise 8.4 (ch8.py), $\tilde{\pi}_1 = (0.8155, 0.1845)$, but the marginal of $\tilde{\pi}_2$ on s_1 is $(0.8250, 0.1750)$. $\tilde{\pi}_t$ is what is now called the t -forward measure (the measure that makes prices in units of the date- t zero-coupon bond martingales); with deterministic interest rates it is the risk-neutral measure.

Exercise 8.12 *Harrison–Kreps prices*

For histories with $\pi_t(s^t) > 0$, define $p_t^0(s^t) = q_t^0(s^t)/[\beta^t \pi_t(s^t)]$. By (8.5.4), $q_t^0(s^t) = 0$ whenever $\pi_t(s^t) = 0$, so those histories contribute nothing to (8.7.1), and p_t^0 may be defined arbitrarily there. Then

$$a_0 = \sum_t \sum_{s^t} q_t^0(s^t) d_t(s^t) = \sum_t \sum_{s^t} \beta^t p_t^0(s^t) d_t(s^t) \pi_t(s^t) = E_0 \sum_t \beta^t p_t^0 d_t.$$

By (8.5.4), $p_t^0(s^t) = u'(c_t^i(s^t))/\mu_i = u'(c_t^i(s^t))/u'(c_0^i(s_0))$ under the normalization $q_0^0 = 1$, and this is the same for every consumer i . So $\beta^t p_t^0$ is the stochastic discount factor $m_{0,t} = \beta^t u'(c_t)/u'(c_0)$: Harrison and Kreps' state-price density, the price of a history-contingent claim per unit of probability.

It links the two representations: $\tilde{\pi}_t = \beta^t p_t^0 \pi_t / b_t$, i.e. $\beta^t p_t^0 / b_t$ is the likelihood ratio (Radon–Nikodym derivative) of the equivalent martingale measure of exercise 8.11 with respect to π_t .

Exercise 8.13 *Early resolution of uncertainty*

(a) A recursive competitive equilibrium in the sense of section 8.9.3: initial wealths $a_0^i = 0$; a pricing kernel $Q(s' | s)$; natural debt limits $\bar{A}_i(s) = y^i(s) + \sum_{s'} Q(s' | s) \bar{A}_i(s')$; value functions $v_i(a, s)$ and decision rules $h_i(a, s)$, $g_i(a, s, s')$ that solve (8.9.8) subject to $c + \sum_{s'} Q(s' | s) \hat{a}(s') \leq y^i(s) + a$ and $-\hat{a}(s') \leq \bar{A}_i(s')$; and market clearing, $\sum_i c^i = \sum_i y^i(s)$ and $\sum_i \hat{a}^i(s') = 0$.

(b) The aggregate endowment is $s/2 + 1 - s/2 = 1$ in every state, so by section 8.6.3 each consumer's consumption is constant, $\bar{c}^i = (1 - \beta)E_0 \sum_t \beta^t y_t^i$. The chain starts at $s_0 = 1$ and moves at $t = 1$ to the absorbing state 0 or 2 with probability 1/2 each, so $E_0 y_t^1 = 1/2$ for every t . Hence

$$\bar{c}^1 = \bar{c}^2 = \frac{1}{2}, \quad Q(s' | s) = \beta P(s, s').$$

The natural debt limits, $\bar{A}_i = (I - \beta P)^{-1} y^i$, are

$$\bar{A}_1(0) = 0, \quad \bar{A}_1(1) = \frac{1/2}{1 - \beta}, \quad \bar{A}_1(2) = \frac{1}{1 - \beta}; \quad \bar{A}_2(0) = \frac{1}{1 - \beta}, \quad \bar{A}_2(1) = \frac{1/2}{1 - \beta}, \quad \bar{A}_2(2) = 0.$$

The portfolios are the continuation wealths (8.8.1), $a^1(s) = E[\sum_{\tau \geq 0} \beta^\tau (\bar{c}^1 - y_{t+\tau}^1) | s_t = s]$, i.e. $a^1(0) = \frac{1/2}{1 - \beta}$, $a^1(2) = -\frac{1/2}{1 - \beta}$, $a^1(1) = 0$, and $a^2 = -a^1$. At $t = 0$ consumer 1 buys $\frac{1/2}{1 - \beta}$ claims on $s_1 = 0$ and sells as many claims on $s_1 = 2$. Their costs, $\beta(.5)\frac{1/2}{1 - \beta}$ each, cancel, so $c_0^1 = y_0^1 = 1/2$. Each position respects the debt limits: $\frac{1/2}{1 - \beta} \leq \bar{A}_1(2) = \frac{1}{1 - \beta}$.

(c) Only two histories are possible.

- *History* (1, 0, 0, 0, ...): $c_t^1 = c_t^2 = \frac{1}{2}$ for all t . Wealths: $a_0^1 = a_0^2 = 0$; $a_t^1 = \frac{1/2}{1 - \beta}$ and $a_t^2 = -\frac{1/2}{1 - \beta}$ for $t \geq 1$. Every period consumer 1 (endowment 0) consumes $\frac{1}{2}$ out of his assets and buys again $\frac{1/2}{1 - \beta}$ claims on state 0 for $\beta \frac{1/2}{1 - \beta}$: $\frac{1}{2} + \beta \frac{1/2}{1 - \beta} = \frac{1/2}{1 - \beta}$. Consumer 2 (endowment 1) consumes $\frac{1}{2}$ and uses the other $\frac{1}{2}$ to service her debt.
- *History* (1, 2, 2, 2, ...): the mirror image, $c_t^1 = c_t^2 = \frac{1}{2}$, with $a_t^1 = -\frac{1/2}{1 - \beta}$ and $a_t^2 = \frac{1/2}{1 - \beta}$ for $t \geq 1$.

All the uncertainty is resolved at $t = 1$; the trade at $t = 0$, before the resolution, is the only insurance there is, and afterwards the consumers simply roll their positions over. Each consumer's total wealth, financial plus the value of his endowment, is $\frac{1/2}{1 - \beta}$ in every state, which is what keeps consumption constant. If markets opened only after s_1 was revealed, the insurance would be impossible: the unlucky consumer would have a zero endowment and a zero debt limit forever.

Exercise 8.14 *Weill's economy*

Remark. The book writes “booms $y_t(1) = \bar{y}^0$ ”; it should be $\bar{y}^1$.

(a) A price system $\{q_t^0(s^t)\}$ and allocations $\{c_t^j(s^t)\}$ such that each type maximizes $U(c^j)$ subject to $\sum_t \sum_{s^t} q_t^j(c_t^j - y_t^j) \leq 0$, and $c_t^0 + c_t^1 = y(s_t)$. With identical CRRA preferences, (8.6.1)–(8.6.2) give constant shares, $c_t^j(s^t) = \alpha_j y(s_t)$, and

$$q_t^0(s^t) = \beta^t \pi(s^t | s_0) \left(\frac{y(s_t)}{\bar{y}^1} \right)^{-\gamma}.$$

The shares are the wealth shares. With $Q(s' | s) = \beta\pi(s' | s)[y(s')/y(s)]^{-\gamma}$, the wealths are $W_1 = [(I - Q)^{-1}]_{11}\bar{y}^1$ and $W_0 = [(I - Q)^{-1}]_{10}\bar{y}^0$ (rows and columns indexed by $s = 0, 1$; the row is $s_0 = 1$). Inverting the 2×2 matrix,

$$\alpha_1 = \frac{1 - \beta\pi(0 | 0)}{1 - \beta\pi(0 | 0) + \beta\pi(0 | 1)(\bar{y}^0/\bar{y}^1)^{1-\gamma}}, \quad \alpha_0 = 1 - \alpha_1.$$

(For $\bar{y}^0 = 1$, $\bar{y}^1 = 2$, $\beta = .95$, $\pi(0 | 0) = .6$, $\pi(1 | 1) = .7$: $\alpha_1 = 0.681, 0.430, 0.086$ for $\gamma = .5, 2, 5$; `ch8.py`.)

(b) $\bar{U}(c) = E_0 \sum_t \beta^t c_t^{1-\gamma}/(1-\gamma)$, the common utility function (up to a positive scale). A single agent who consumes y_t has first-order conditions that give exactly the prices in (a), for *any* distribution of the endowments across the types. The distribution affects only the shares α_j . This is exact (Gorman) aggregation: with identical homothetic preferences and complete markets, marginal utilities of all agents are proportional and are powers of aggregate consumption. The representative agent is a valid device for pricing and for aggregate quantities, though not for individual welfare.

(c) (i) *Claims on endowment streams*. At $t = 0$ type 1 sells a fraction α_0 of his tree and buys a fraction α_1 of type 0's tree. The trade balances, since $\alpha_1(W_0 + W_1) = W_1$. Holding α_j of both trees yields $\alpha_j y(s_t)$ forever, the complete-markets allocation, and no further trade is needed.

(ii) *Arrow securities*. The same allocation and prices, with Q as above. Type j holds $a^j(s) = [(I - Q)^{-1}(\alpha_j y - y^j)](s)$, so he enters each state with a wealth that depends only on the current state. (iii) *Two riskless bonds*. A two-period bond bought at t is worth $p_1(s_{t+1}) = \sum_{s''} Q(s'' | s_{t+1})$

at $t + 1$, while the one-period bond pays 1. If $p_1(0) \neq p_1(1)$, the payoff matrix $\begin{bmatrix} 1 & p_1(0) \\ 1 & p_1(1) \end{bmatrix}$ is nonsingular: the two bonds span both states next period, markets are dynamically complete, and the allocation is again the complete-markets one. (In the example, $p_1 = (0.665, 1.805)$ for $\gamma = 2$.) *Without aggregate uncertainty* (y constant), $Q = \beta\pi$ and $p_1(0) = p_1(1) = \beta$. Both bonds are then riskless next period, markets are incomplete, and consumers cannot insure their idiosyncratic risk: they can only self-insure by borrowing and lending, and consumption is no longer constant. (i) and (ii) still deliver full insurance, with constant consumption $\alpha_j \bar{y}$.

(d) Now the state alternates deterministically, $1, 0, 1, 0, \dots$, so $q_t^0 = \beta^t$ at even dates and $\beta^t(\bar{y}^0/\bar{y}^1)^{-\gamma}$ at odd dates. The wealths are $W_1 = \bar{y}^1/(1 - \beta^2)$ and $W_0 = \beta(\bar{y}^0)^{1-\gamma}(\bar{y}^1)^\gamma/(1 - \beta^2)$, so

$$\alpha_1 = \frac{1}{1 + \beta(\bar{y}^0/\bar{y}^1)^{1-\gamma}}, \quad c_t^1 = \alpha_1 y_t, \quad c_t^0 = (1 - \alpha_1) y_t.$$

Type 1 consumes the larger share iff $\beta(\bar{y}^0/\bar{y}^1)^{1-\gamma} < 1$. Since $\bar{y}^0 < \bar{y}^1$, this always holds for $\gamma \leq 1$. For $\gamma > 1$ it holds iff

$$\gamma < \gamma^* = 1 + \frac{\log(1/\beta)}{\log(\bar{y}^1/\bar{y}^0)}.$$

For $\bar{y}^1/\bar{y}^0 = 2$, $\beta = .95$: $\gamma^* = 1.074$, and $\alpha_1 = 0.598, 0.513, 0.345, 0.062$ for $\gamma = .5, 1, 2, 5$. Intuition: type 1's endowment is larger and arrives first, but it arrives when goods are plentiful. The price of recession goods relative to boom goods is $(\bar{y}^0/\bar{y}^1)^{-\gamma}$, which grows with risk aversion (or with the desire to smooth consumption over time). For large γ , the owner of the scarce recession goods is the richer type.

(e) At $t = 1$ (a recession) type 0 has consumed her share $\alpha_0 \bar{y}^1$ of the boom at $t = 0$ and owes deliveries in present value. She owes $\alpha_1 \bar{y}^0$ in every recession and receives $\alpha_0 \bar{y}^1$ in every boom, and the time-0 budget implies that the value of these net deliveries is $\alpha_0 \bar{y}^1 > 0$. If she defaults she consumes $\bar{y}^0$ in recessions and 0 in booms. With $u(c) = c^{1-\gamma}/(1-\gamma)$:

- If $\gamma \geq 1$, $u(0) = -\infty$, so she never defaults.
- If $\gamma < 1$, compare the two values from $t = 1$ on:

$$V^{\text{honor}} = \frac{\alpha_0^{1-\gamma}[(\bar{y}^0)^{1-\gamma} + \beta(\bar{y}^1)^{1-\gamma}]}{(1-\gamma)(1-\beta^2)}, \quad V^{\text{default}} = \frac{(\bar{y}^0)^{1-\gamma}}{(1-\gamma)(1-\beta^2)}.$$

She defaults iff $\alpha_0^{1-\gamma}[1 + \beta(\bar{y}^1/\bar{y}^0)^{1-\gamma}] < 1$. As $\gamma \rightarrow 0$ (risk neutrality) the condition becomes $\beta^2 < 1$, which is true: a risk-neutral debtor defaults, because she gains the debt and loses nothing she values from smoothing. For $\bar{y}^1/\bar{y}^0 = 2$ and $\beta = .95$ she defaults iff $\gamma < 0.061$ (`ch8.py`).

So with the usual (moderate or high) risk aversion the option is never exercised: autarky means going without consumption in every boom, and the value of insurance and smoothing exceeds the debt. This is the logic of the enforcement-constrained models of chapter 21.

Exercise 8.15 *Diverse beliefs, I*

(a) An equilibrium is an allocation $\{c_t^i(s^t)\}$, Arrow-security holdings $\{a_{t+1}^i(s_{t+1}, s^t)\}$ and prices $\{Q_t(s_{t+1} | s^t)\}$ such that:

- given the prices, each consumer i maximizes $\sum_t \sum_{s^t} \beta^t \ln c_t^i(s^t) \pi_t^i(s^t)$, using his own beliefs π^i , subject to the budget constraints $c_t^i(s^t) + \sum_{s_{t+1}} Q_t(s_{t+1} | s^t) a_{t+1}^i(s_{t+1}, s^t) \leq y_t^i(s^t) + a_t^i(s^t)$, $a_0^i = 0$, and the natural debt limits $a_{t+1}^i(s^{t+1}) \geq -A_{t+1}^i(s^{t+1})$;
- markets clear: $\sum_i c_t^i = \sum_i y_t^i$ and $\sum_i a_{t+1}^i = 0$.

(b) Solve the equivalent time-0 (Arrow-Debreu) problem, then translate. With log utility the first-order conditions are $\beta^t \pi_t^i(s^t)/c_t^i(s^t) = \mu_i q_t^0(s^t)$. The aggregate endowment is always 1, so

$$q_t^0(s^t) = \beta^t \left[\frac{\pi_t^1(s^t)}{\mu_1} + \frac{\pi_t^2(s^t)}{\mu_2} \right], \quad c_t^1(s^t) = \frac{\pi_t^1(s^t)/\mu_1}{\pi_t^1(s^t)/\mu_1 + \pi_t^2(s^t)/\mu_2}.$$

Let $r = \mu_2/\mu_1$ and normalize $q_0^0 = 1$. Then $c_0^1 = r/(1+r)$, and for $t \geq 1$ $c_t^1 = r/(2+r)$ on history 1 and $c_t^1 = 2r/(1+2r)$ on history 2. The prices are $\beta^{-t} q_t^0 = (2+r)/[3(1+r)]$ and $(1+2r)/[3(1+r)]$ on the two histories. Consumer 1's budget is

$$c_0^1 - .5 + \frac{\beta}{1-\beta} \left[\frac{2+r}{3(1+r)} \left(\frac{r}{2+r} - 1 \right) + \frac{1+2r}{3(1+r)} \frac{2r}{1+2r} \right] = 0.$$

The bracket equals $\frac{2(r-1)}{3(1+r)}$, so the budget reduces to $\frac{(r-1)(3+\beta)}{6(1+r)(1-\beta)} = 0$ and $r = 1$ for every β (`ch8.py` confirms this by bisection for $\beta = .5, .9, .95$). The equilibrium is

$$c_0^1 = c_0^2 = \frac{1}{2}; \quad \text{history 1: } (c^1, c^2) = \left(\frac{1}{3}, \frac{2}{3}\right); \quad \text{history 2: } (c^1, c^2) = \left(\frac{2}{3}, \frac{1}{3}\right),$$

with $q_t^0 = \beta^t/2$ on each history for $t \geq 1$. Each consumer consumes more on the history he considers more likely, and so ends up richer on the history in which he is *poorly* endowed. Consumer 1 thinks history 2 (where he owns nothing) is twice as likely as history 1, so he buys insurance against it.

Sequential trading. Arrow prices are $Q_0(1 \mid .5) = Q_0(0 \mid .5) = \beta/2$ at $t = 0$ and $Q_t(s \mid s) = \beta$ afterwards, since uncertainty is resolved at $t = 1$. At $t = 1$ consumer 1 needs wealth equal to the value of his excess consumption. On history 1 that is $a_1^1(1) = (\frac{1}{3} - 1)/(1 - \beta) = -\frac{2}{3(1-\beta)}$, a debt; on history 2 it is $a_1^1(0) = \frac{2}{3(1-\beta)}$. These holdings cost nothing at $t = 0$ ($\frac{\beta}{2}[a_1^1(1) + a_1^1(0)] = 0$), consistent with $c_0^1 = y_0^1$. The debt on history 1 is within the natural debt limit $1/(1 - \beta)$. Consumer 2 holds the opposite positions.

(c) Yes, relative to the consumers' own preferences. Each consumer has expected-utility preferences with respect to his own beliefs, markets are complete, and the first welfare theorem applies. The allocation satisfies the Pareto conditions $\pi_t^1 u'(c_t^1)/[\pi_t^2 u'(c_t^2)] = \lambda_2/\lambda_1$ of (8.B.2), with weights $\lambda_i = 1/\mu_i$. Nature's probabilities play no role in this statement. An outside observer who uses nature's probabilities would call the allocation inefficient, since it does not equalize consumption across the two histories, but the consumers themselves would not.

Exercise 8.16 *Diverse beliefs, II*

(a) Maximize $\sum_i \lambda_i \sum_t \sum_{s^t} \beta^t \ln c_t^i \pi_t^i$ subject to $\sum_i c_t^i(s^t) = Y_t(s^t)$. With multiplier $\theta_t(s^t)$, the first-order conditions $\lambda_i \beta^t \pi_t^i(s^t)/c_t^i(s^t) = \theta_t(s^t)$ and feasibility give

$$c_t^i(s^t) = \frac{\lambda_i \pi_t^i(s^t)}{\sum_j \lambda_j \pi_t^j(s^t)} Y_t(s^t).$$

Each consumer's share is his Pareto weight times his probability of the history, divided by the weighted sum over consumers.

(b) Complete risk sharing means constant shares, $c_t^i = \bar{\theta}_i Y_t$. That requires the ratios $\pi_t^i(s^t)/\pi_t^j(s^t)$ to be the same for all histories and dates. Since each $\pi_t^i(\cdot)$ sums to one over s^t , a constant ratio must equal 1. So the Pareto plan implies complete risk sharing if and only if all consumers have the same beliefs (on histories of positive probability).

(c) The time- t allocation depends on the history only through the likelihood ratios $\pi_t^i(s^t)/\pi_t^1(s^t)$. It is not history dependent if and only if these ratios are the same for all histories with the same $Y_t(s^t)$, i.e. they are functions of (t, Y_t) alone. The leading case is common beliefs. If consumers disagree about, say, an i.i.d. state, the likelihood ratio depends on how often each state has occurred, and so do the consumption shares.

(d) Take $q_t^0(s^t) = \theta_t(s^t)$, i.e. $q_t^0(s^t) = \beta^t \sum_j \lambda_j \pi_t^j(s^t)/Y_t(s^t)$, and set $\mu_i = 1/\lambda_i$. Then each consumer's first-order condition $\beta^t \pi_t^i/c_t^i = \mu_i q_t^0$ holds at the Pareto allocation. Consumer i 's spending is $\sum_t \sum_{s^t} q_t^0 c_t^i = \sum_t \beta^t \lambda_i \sum_{s^t} \pi_t^i(s^t) = \lambda_i/(1 - \beta)$. So any endowment distribution worth $\lambda_i/(1 - \beta)$ to each consumer supports the allocation. An example is $y_t^i(s^t) = \frac{\lambda_i}{\sum_j \lambda_j} Y_t(s^t)$, whose value is $\frac{\lambda_i}{\sum_j \lambda_j} \sum_t \beta^t \sum_j \lambda_j = \lambda_i/(1 - \beta)$. Initial wealth shares must equal $\lambda_i/\sum_j \lambda_j$.

(e) From any consumer's first-order condition, $q_t^0(s^t) = \beta^t \pi_t^i(s^t)/[\mu_i c_t^i(s^t)]$. Equivalently, writing $W_i = \lambda_i/(1 - \beta)$ for consumer i 's wealth,

$$q_t^0(s^t) = \beta^t (1 - \beta) \frac{\sum_i W_i \pi_t^i(s^t)}{Y_t(s^t)}.$$

Prices use a wealth-weighted average of the consumers' beliefs, discounted by β^t and divided by the aggregate endowment.

(f) With $I = 2$, $q_t^0(s^t) = \beta^t \lambda_2 [(\lambda_1/\lambda_2)\pi_t^1(s^t) + \pi_t^2(s^t)]/Y_t(s^t)$. As $\lambda_2/\lambda_1 \rightarrow \infty$ this becomes proportional to $\beta^t \pi_t^2(s^t)/Y_t(s^t)$, the price system of a representative consumer with consumer 2's beliefs. The initial wealths that decentralize the allocation satisfy $W_2/W_1 = \lambda_2/\lambda_1 \rightarrow \infty$. So the planner gives consumer 2 almost all the wealth, and the wealth-weighted average of beliefs in (e) converges to π^2 .

Exercise 8.17 *Diverse beliefs, III*

(a) $\pi_t(s^t) = \pi_0(s_0)P(s_0, s_1) \cdots P(s_{t-1}, s_t)$. Here $s_0 = 1$ for sure, s_1 is 0 or 2 with probability .5 each, and both 0 and 2 are absorbing. So only two histories have positive probability, $(1, 0, 0, \dots)$ and $(1, 2, 2, \dots)$, each with probability .5 for every $t \geq 1$.

(b) As in Exercise 8.15(a), with common beliefs π_t .

(c) The aggregate endowment is always 1 and beliefs are common, so consumption shares are constant and Arrow prices are $Q(s' | s) = \beta P(s, s')$. Consumer 1's endowment is worth $.5 + \sum_{t \geq 1} \beta^t (.5 \cdot 0 + .5 \cdot 1) = .5 + .5\beta/(1 - \beta)$, half of total wealth $1/(1 - \beta)$. Hence

$$c_t^1 = c_t^2 = \frac{1}{2} \quad \text{for all } t \text{ and all histories.}$$

(d) Both consumers eat .5 along every history, so the allocation is full insurance. At $t = 0$ consumer 1 buys $a_1^1(0) = .5/(1 - \beta)$ Arrow securities on $s_1 = 0$ and sells $a_1^1(2) = -.5/(1 - \beta)$ on $s_1 = 2$, both at price $\beta/2$, so the portfolio costs nothing. On history $(1, 0, 0, \dots)$ he has no endowment and lives off wealth $.5/(1 - \beta)$, which stays constant: each period he consumes .5 and rolls over $a = .5/(1 - \beta)$ at price β . On $(1, 2, 2, \dots)$ he owes $.5/(1 - \beta)$ and pays .5 per period out of his endowment of 1. Consumer 2's positions are the mirror image.

(e) For consumer 2, $\pi_t^2(1, 0, \dots, 0) = .4$ and $\pi_t^2(1, 2, \dots, 2) = .6$ for all $t \geq 1$ (and $\pi_0^2(1) = 1$).

(f) With weights λ and $1 - \lambda$, Exercise 8.16(a) gives $c_0^1 = \lambda$ and, for $t \geq 1$,

$$c_t^1 = \frac{.5\lambda}{.5\lambda + .4(1 - \lambda)} \quad \text{on } (1, 0, 0, \dots), \quad c_t^1 = \frac{.5\lambda}{.5\lambda + .6(1 - \lambda)} \quad \text{on } (1, 2, 2, \dots),$$

with $c_t^2 = 1 - c_t^1$. Consumer 2 thinks $s = 2$ (where he owns nothing) is more likely than it is, so he gets relatively more consumption there.

(g)–(h) An equilibrium with time-0 trading consists of prices $q_t^0(s^t)$ and an allocation such that each consumer maximizes his expected utility (under his own beliefs) subject to the single budget constraint $\sum_t \sum_{s^t} q_t^0(c_t^i - y_t^i) \leq 0$, and markets clear. It is the Pareto allocation of (f) with $q_t^0(s^t) = \beta^t [\lambda \pi_t^1(s^t) + (1 - \lambda) \pi_t^2(s^t)]$ ($q_0^0 = 1$). The weight λ is pinned down by consumer 1's budget. Using $c_t^1 = .5\lambda\beta^t/q_t^0$ on each branch,

$$\lambda - .5 + \frac{\beta}{1 - \beta} \left[.5\lambda + (.5\lambda - (.5\lambda + .6(1 - \lambda))) \right] = 0 \implies \lambda = \frac{5 + \beta}{10 + \beta}.$$

Hence $c_0^1 = \frac{5 + \beta}{10 + \beta}$, $c_t^1 = \frac{5 + \beta}{9 + \beta}$ on $(1, 0, \dots)$ and $c_t^1 = \frac{5 + \beta}{11 + \beta}$ on $(1, 2, \dots)$. For $\beta = .9$ these are .5413, .5960 and .4958 (ch8.py solves the budget by bisection and gets the same λ). Prices are $q_t^0 = \beta^t \frac{9 + \beta}{2(10 + \beta)}$ on $(1, 0, \dots)$ and $\beta^t \frac{11 + \beta}{2(10 + \beta)}$ on $(1, 2, \dots)$. Consumer 1 is better off than in Part I.

Consumer 2's pessimism about $s = 2$ makes him pay a high price for insurance against it, and consumer 1 sells that insurance.

(i) The Arrow prices at $t = 0$ are $Q(0 | 1) = \beta \frac{9+\beta}{2(10+\beta)}$ and $Q(2 | 1) = \beta \frac{11+\beta}{2(10+\beta)}$ (.4087 and .4913 for $\beta = .9$). Afterwards $Q(s | s) = \beta$, since both consumers agree that 0 and 2 are absorbing. Consumption is as in (h). At $t = 1$ consumer 1's wealth (his Arrow holdings) is the value of his planned excess consumption. On $(1, 0, \dots)$ it is $a_1^1(0) = \frac{5+\beta}{(9+\beta)(1-\beta)}$, and on $(1, 2, \dots)$ it is $a_1^1(2) = (\frac{5+\beta}{11+\beta} - 1)/(1 - \beta) = -\frac{6}{(11+\beta)(1-\beta)}$. Both stay constant afterwards, and consumer 2 holds the opposite positions.

Exercise 8.18 *Diverse beliefs, IV*

(a) With weights $\lambda, 1 - \lambda$, the planner maximizes $\lambda[\ln c_0^1 + \sum_{t \geq 1} \beta^t \ln c_t^1(h_1)] + (1 - \lambda)[\ln c_0^2 + \sum_{t \geq 1} \beta^t \ln c_t^2(h_2)]$. Consumer 1's consumption on history 2, and consumer 2's on history 1, has zero probability for its owner and does not enter the objective. So the planner gives all of the endowment on history 1 to consumer 1 and all of it on history 2 to consumer 2, and splits $t = 0$ as $c_0^1 = \lambda$, $c_0^2 = 1 - \lambda$. (This is formula 8.16(a) with $\pi_t^2(h_1) = \pi_t^1(h_2) = 0$.)

(b) As in Exercise 8.15(a).

(c) Yes: the equilibrium is autarky. Consumer 1 owns 1 unit per period on history 1, the only history he cares about, and nothing on history 2; symmetrically for consumer 2. So the endowment allocation is the Pareto allocation of (a) with $\lambda = \frac{1}{2}$. Supporting Arrow prices are $Q_0(1 | .5) = \beta c_0^1/c_1^1(h_1) = \frac{\beta}{2}$ (consumer 1's marginal rate of substitution), $Q_0(0 | .5) = \frac{\beta}{2}$ (consumer 2's), and $Q_t(s | s) = \beta$ afterwards. At these prices consumer 1 would like to sell claims on history 2, which he believes worthless. But his natural debt limit there is the value of his endowment on history 2, which is zero, so he cannot. Consumer 2 is in the same position on history 1, so nobody trades. Each consumer consumes nothing on the history he considers impossible; with $0 \cdot \ln 0 \equiv 0$, this costs him nothing in expected utility.

Exercise 8.19 *Diverse beliefs, V*

(a) The Pareto problem does not involve endowments, so the answer is that of Exercise 8.18(a): consumer 1 gets everything on history 1, consumer 2 everything on history 2, and $c_0^1 = \lambda$.

(b) As in Exercise 8.15(a).

(c) Yes, and it involves extreme trades. Now consumer 1 owns 1 unit per period on history 2, which he considers impossible, and nothing on history 1. Consumer 2 owns 1 per period on history 1 and nothing on history 2. At any positive prices each consumer sells his entire endowment on the history he considers impossible and buys consumption on the other. Let $q_t^0(h_1) = \beta^t a_1$ and $q_t^0(h_2) = \beta^t a_2$, with $q_0^0 = 1$. Consumer 1's first-order conditions give $c_0^1 = 1/\mu_1$ and $c_t^1(h_1) = 1/(\mu_1 a_1)$. Market clearing on history 1 requires $c_t^1(h_1) = 1$, so $a_1 = 1/\mu_1$. Symmetrically $a_2 = 1/\mu_2$, and clearing at $t = 0$ gives $a_1 + a_2 = 1$. Consumer 1's budget, $c_0^1 - .5 + \frac{\beta}{1-\beta}(a_1 - a_2) = 0$, then gives $a_1 = a_2 = \frac{1}{2}$. So

$$c_0^1 = c_0^2 = \frac{1}{2}, \quad c_t^1(h_1) = 1, \quad c_t^2(h_1) = 0, \quad c_t^1(h_2) = 0, \quad c_t^2(h_2) = 1 \quad (t \geq 1),$$

with prices $q_t^0(h) = \beta^t/2$. This is the same allocation as in Exercise 8.18, reached through trade instead of autarky.

With sequential trading, the Arrow prices are $Q_0(1 | .5) = Q_0(0 | .5) = \beta/2$, and $Q_t(s | s) = \beta$ afterwards. At $t = 0$ consumer 1 buys $a_1^1(1) = 1/(1 - \beta)$ claims on $s_1 = 1$, enough to consume 1 forever on history 1. He pays for them by selling $a_1^1(0) = -1/(1 - \beta)$ claims on $s_1 = 0$. That debt is exactly his natural debt limit on history 2, where his endowment of 1 per period services it, so the trade is allowed. On history 2 he then consumes nothing, an outcome he regarded as impossible. Consumer 2 does the mirror image. So in Exercises 8.18 and 8.19 the equilibrium allocation is the same, and endowments matter only for the prices and trades that support it. Whichever history occurs, the consumer who was right consumes everything from then on.

Exercise 8.20 *Risk-free bonds*

(a) $\pi_t(s^t) = \pi_0(s_0) \prod_{\tau=1}^t P(s_{\tau-1}, s_\tau)$.

(b) Prices $Q_t(s_{t+1} | s^t)$, holdings $a_{t+1}(s_{t+1}, s^t)$ and consumption $c_t(s^t)$ such that the consumer maximizes expected utility subject to $c_t + \sum_{s'} Q_t(s' | s^t) a_{t+1}(s')$ $\leq y_t + a_t$ (with natural debt limits), and markets clear: $c_t = y_t$ and $a_{t+1} = 0$, since a representative consumer holds no net assets.

(c) At $c_t = y_t$ the Euler equation for each Arrow security gives

$$Q_t(s_{t+1} | s^t) = \beta P(s_t, s_{t+1}) \frac{u'(y_{t+1})}{u'(y_t)} = \beta \frac{P(s_t, s_{t+1})}{\lambda(s_{t+1})}.$$

Prices depend on the history only through s_t .

(d) The same objects with the complete set of Arrow securities replaced by a one-period risk-free claim, bought at price p_t^b , paying one unit at $t + 1$ in every state. The consumer's budget is $c_t + p_t^b b_{t+1} \leq y_t + b_t$ with a debt limit, and the market clears with $c_t = y_t$, $b_{t+1} = 0$. With a representative consumer, markets are trivially complete enough: the allocation is autarky either way.

(e) The bond is a portfolio of one of each Arrow security, so

$$p_t^b = \sum_{s'} Q_t(s' | s_t) = \beta \sum_{s'} \frac{P(s_t, s')}{\lambda(s')}, \quad R_t = \left[\beta \sum_{s'} \frac{P(s_t, s')}{\lambda(s')} \right]^{-1}.$$

R_t depends on the history only through the current state, via expected growth next period.

(f) If $s_t = 1$, growth is 1 forever (state 1 is absorbing), so $p^b = .95$ and $R = 1.052632$. If $s_t = 2$, then $p^b = .95(.5/1 + .5/1.02) = 0.940686$ and $R = 1.063054$. The interest rate is higher when growth is expected, because consumers would like to borrow against higher future consumption.

(g) The equilibrium conditions (b)–(e) involve only the consumer's subjective probabilities. All formulas remain valid with P replaced by the subjective transition matrix, and so does (a) if π_t means subjective probabilities. The true probabilities matter only for the statistical behavior of the outcomes, e.g. how often R_t takes each value, and for interpreting average excess returns as risk premia. The equality of the two is used only when we say that prices reflect “the” probabilities of future states.

Exercise 8.21 Entropy

Let $m(s) = \tilde{f}(s)/f(s)$ where $f(s) > 0$. Then $\text{ent}(f, \tilde{f}) = \int m(s) \log m(s) f(s) ds$, with the convention $0 \log 0 = 0$. (If $\tilde{f}$ puts positive mass where $f = 0$, the entropy is $+\infty$ and there is nothing to prove.)

Pointwise inequality. For $x > 0$ let $g(x) = x \log x - x + 1$. Then $g'(x) = \log x$ and $g''(x) = 1/x > 0$, so g is strictly convex with its minimum at $x = 1$, where $g(1) = 0$. Hence $x \log x \geq x - 1$ for all $x \geq 0$, with equality only at $x = 1$.

Integration. Since $\int m f ds = \int \tilde{f} ds = 1$ and $\int f ds = 1$,

$$\text{ent}(f, \tilde{f}) = \int m \log m f ds \geq \int (m - 1) f ds = 1 - 1 = 0.$$

Equality requires $m = 1$ wherever $f > 0$, i.e. $\tilde{f} = f$ almost everywhere. In particular $\text{ent}(\tilde{f}, \tilde{f}) = \int \log(1) \tilde{f} ds = 0$. (Equivalently, by Jensen's inequality, $-\text{ent} = \int \log(f/\tilde{f}) \tilde{f} ds \leq \log \int f ds = 0$; this is the proof in footnote 17 of the chapter.)

Exercise 8.22 Heterogeneous beliefs again

(a) By Exercise 8.16(a) with weights λ and $1 - \lambda$,

$$c_t^2 = \frac{(1 - \lambda)\pi_t^2}{\lambda\pi_t^1 + (1 - \lambda)\pi_t^2} y_t = \frac{(1 - \lambda)L_t}{\lambda + (1 - \lambda)L_t} y_t,$$

dividing numerator and denominator by π_t^1 . The likelihood ratio L_t acts as a stochastic Pareto weight.

(b) Time-0 prices are $q_t^0(s^t) \propto \beta^t [\lambda\pi_t^1(s^t) + (1 - \lambda)\pi_t^2(s^t)]/y_t(s^t)$, so

$$\tilde{Q}(s_{t+1} | s^t) = \frac{q_{t+1}^0(s^{t+1})}{q_t^0(s^t)} = \beta \frac{y_t(s^t)}{y_{t+1}(s^{t+1})} \left[w_t^1 \pi^1(s_{t+1}) + w_t^2 \pi^2(s_{t+1}) \right],$$

where $w_t^i = c_t^i/y_t$ are the consumption shares and $\pi^i(s)$ is consumer i 's one-period probability (α_i or $1 - \alpha_i$). Arrow prices are those of a log-utility representative consumer whose beliefs are the consumption-share-weighted average of individual beliefs.

(c) Assume $0 < \alpha_1, \alpha_2 < 1$ (common support) and that s_t is truly i.i.d. with $\tilde{\alpha} = \alpha_2 \neq \alpha_1$. Then $\log L_t = \sum_{\tau \leq t} \log[\pi^2(s_\tau)/\pi^1(s_\tau)]$ is a sum of i.i.d. terms whose true mean is

$$\text{ent}(\pi, \pi^1) = \tilde{\alpha} \log \frac{\tilde{\alpha}}{\alpha_1} + (1 - \tilde{\alpha}) \log \frac{1 - \tilde{\alpha}}{1 - \alpha_1} > 0$$

(Exercise 8.21). By the strong law of large numbers, $\frac{1}{t} \log L_t \rightarrow \text{ent}(\pi, \pi^1)$ almost surely. So $L_t \rightarrow \infty$ and consumer 1's share $w_t^1 = \lambda/(\lambda + (1 - \lambda)L_t) \rightarrow 0$ exponentially fast, at the rate $\text{ent}(\pi, \pi^1)$. Hence, almost surely under the true probability law (pathwise), $\tilde{Q}_t(s_{t+1} | s^t) - \beta \frac{y_t}{y_{t+1}} \pi^2(s_{t+1}) \rightarrow 0$.

With $\alpha_1 = .4$, $\tilde{\alpha} = \alpha_2 = .6$ and $\lambda = .5$, the entropy is 0.0811. Along a simulated path consumer 1's share is .4 at $t = 0$, 2×10^{-4} at $t = 50$ and 10^{-6} at $t = 100$. The price-implied probability of $s' = 0$, $\tilde{Q}_t(0 | s^t)y(0)/[\beta y(s_t)]$, goes from .52 to .6000 (Figure S8.1).

(d) In the limit consumer 2 consumes the whole endowment and prices are those of a representative consumer with log utility and the *correct* beliefs. The consumer with wrong beliefs keeps betting

on states that occur less often than he thinks, and so loses his wealth to the consumer who knows the truth. This is the “market selection” (Friedman, Blume–Easley) argument of Appendix B: in the tail of the economy, rational expectations and the representative-agent asset pricing formulas hold.

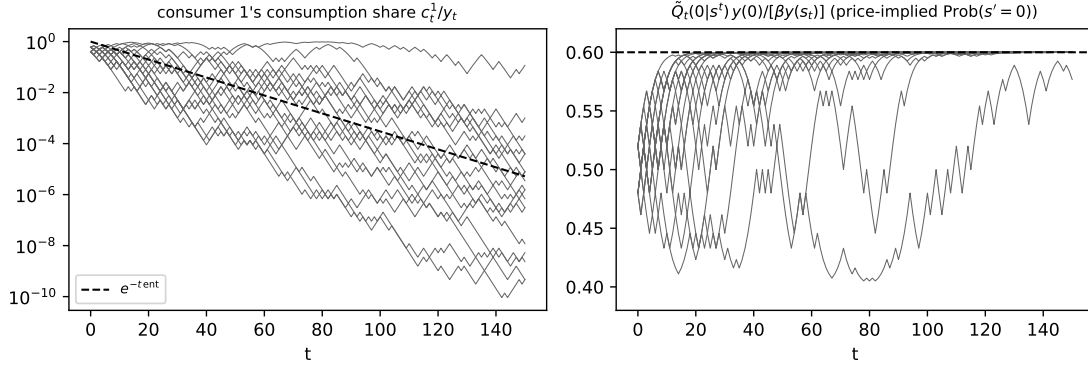


Figure S8.1. Twenty simulated paths ($\alpha_1 = .4$, $\alpha_2 = \tilde{\alpha} = .6$, $\lambda = .5$). Left: consumer 1's consumption share against $e^{-t \text{ent}}$. Right: the probability of $s' = 0$ implied by Arrow prices, which converges to the truth.

Exercise 8.23 *Recursive version of the Pareto problem*

(a)–(c) The recursive Pareto problem is the Bellman equation of the exercise, with consumer 1's promised value v as the state and (c_s, w_s) as the controls. Each consumer's expected utility is computed with his own probabilities: consumer 2's value $P(v)$ with Π^2 , and the promise-keeping constraint for consumer 1 with Π^1 .

(d) With multiplier $\theta \geq 0$ on promise keeping, the Lagrangian is $\sum_s \Pi_s^2 [u_2(1 - c_s) + \beta P(w_s)] + \theta(\sum_s \Pi_s^1 [u_1(c_s) + \beta w_s] - v)$. The first-order conditions and the envelope condition are

$$\Pi_s^2 u'_2(1 - c_s) = \theta \Pi_s^1 u'_1(c_s), \quad \Pi_s^2 P'(w_s) + \theta \Pi_s^1 = 0, \quad P'(v) = -\theta.$$

Hence

$$\frac{u'_2(1 - c_s)}{u'_1(c_s)} = -P'(v) \frac{\Pi_s^1}{\Pi_s^2}, \quad P'(w_s) = P'(v) \frac{\Pi_s^1}{\Pi_s^2}.$$

(e) The left side of the first condition is increasing in c_s , since u'_1 falls and $u'_2(1 - c_s)$ rises as c_s rises. So c_s is increasing in Π_s^1/Π_s^2 . In the second condition, $P'(v) < 0$, so a ratio above 1 makes $P'(w_s)$ more negative. Because P is concave, that means $w_s > v$; a ratio below 1 gives $w_s < v$. Thus in states that consumer 1 considers relatively more likely, he gets both more consumption now and a higher continuation value, and in the other states less of both. With common beliefs the ratio is 1, $w_s = v$ and c_s is constant: the result of Section 8.12.

(f) The relative Pareto weight $-P'(v_t)$ evolves as $-P'(v_{t+1}) = -P'(v_t) \Pi^1(s_{t+1})/\Pi^2(s_{t+1})$, so

$$-P'(v_t) = -P'(v_0) \prod_{\tau \leq t} \frac{\Pi^1(s_\tau)}{\Pi^2(s_\tau)} = -P'(v_0) \frac{\pi_t^1(s^t)}{\pi_t^2(s^t)}.$$

This is exactly the continuation Pareto weight $\lambda_i^t(s^t) \propto \pi_t^i(s^t) \lambda_i$ of Appendix B, now carried as a state variable (the promised value). If consumer 2's beliefs are closer to the truth, the likelihood ratio π_t^1/π_t^2 goes to zero almost surely, at the exponential rate of Exercise 8.22. Consumer 1's

weight then vanishes, his promised value drifts down to the bottom of the admissible set, and his consumption goes to zero. The recursive formulation shows the market-selection result of Appendix B through the drift of continuation values.

Exercise 8.24 Recovering probabilities from prices

(a) (8.10.3) says $Q_h(j | i) = \sum_k Q(k | i)Q_{h-1}(j | k)$, which is the (i, j) element of the matrix product QQ_{h-1} . Hence $Q_h = Q^h$.

(b) Take $i = 1$. The first row of $Q_2 = Q^2$ is

$$(Q^2)_{11} = Q_{11}^2 + Q_{12}Q_{21}, \quad (Q^2)_{12} = Q_{11}Q_{12} + Q_{12}Q_{22}.$$

If $Q_{12} > 0$, these give the second row: $Q_{21} = [(Q^2)_{11} - Q_{11}^2]/Q_{12}$ and $Q_{22} = (Q^2)_{12}/Q_{12} - Q_{11}$. (Row 2 of the data gives row 1 symmetrically.)

(c) Consumption equals the endowment, so the Euler equation gives

$$Q_{ij} = \beta P_{ij} \left(\frac{c_j}{c_i} \right)^{-\gamma}, \quad \text{i.e.} \quad Q = \beta D^{-1} P D, \quad D = \text{diag}(c_1^{-\gamma}, c_2^{-\gamma}).$$

(d) Let $x = (c_2/c_1)^{-\gamma}$. Then $Q_{11} = \beta P_{11}$, $Q_{12} = \beta P_{12}x$, $Q_{21} = \beta P_{21}/x$ and $Q_{22} = \beta P_{22}$. Because each row of P sums to one, $\beta = Q_{11} + Q_{12}/x = Q_{21}x + Q_{22}$, which gives the quadratic

$$Q_{21}x^2 + (Q_{22} - Q_{11})x - Q_{12} = 0.$$

The product of its roots is $-Q_{12}/Q_{21} < 0$, so it has exactly one positive root. That root gives x , then $\beta = Q_{11} + Q_{12}/x$, $P_{11} = Q_{11}/\beta$ and $P_{22} = Q_{22}/\beta$. Only the combination $x = (c_2/c_1)^{-\gamma}$ is identified, not γ itself (unless the consumption ratio is observed).

This is Ross's recovery theorem. Since $Q = \beta D^{-1} P D$ and $P\mathbf{1} = \mathbf{1}$, we have $Q(D^{-1}\mathbf{1}) = \beta D^{-1}\mathbf{1}$. So β is the Perron root of the positive matrix Q , and its positive eigenvector $z = (c_1^\gamma, c_2^\gamma)$ gives D , hence $P = \beta^{-1} D Q D^{-1}$. With $\beta = .96$, $\gamma = 2$, $P = \begin{bmatrix} .8 & .2 \\ .15 & .85 \end{bmatrix}$ and $c = (1, 1.2)$, `ch8.py` recovers row 2 of Q from row 1 of Q and Q^2 . From Q it recovers $\beta = .96$, $P_{11} = .8$, $P_{22} = .85$ and $x = 1.2^{-2} = .69444$ exactly. The Perron root is .96 and the eigenvector ratio is $1.44 = 1.2^2$.

Exercise 8.25 Recovering probabilities from prices, II

(a)–(b) Exactly as in Exercise 8.24(a)–(b).

(c) $c_{t+1}/c_t = \lambda(s_{t+1})$, so $Q_{ij} = \beta P_{ij} \lambda_j^{-\gamma}$: $Q = P \text{diag}(z)$ with $z_j = \beta \lambda_j^{-\gamma}$. The discount factor depends only on tomorrow's state.

(d) Row sums of P equal one, so $Q \text{diag}(z)^{-1} \mathbf{1} = \mathbf{1}$, i.e. $(1/z_1, 1/z_2)' = Q^{-1} \mathbf{1}$. Thus Q reveals z , and then $P = Q \text{diag}(z)^{-1}$: here the transition probabilities *are* identified. But β is not. The two numbers $z_j = \beta \lambda_j^{-\gamma}$ involve three unknowns, β , λ_1 , λ_2 , and for any β the growth rates $\lambda_j = (\beta/z_j)^{1/\gamma}$ reproduce Q exactly. So even with γ known, β (and with it the pair (β, P)) cannot be inferred from Q unless growth rates are observed. `ch8.py` illustrates with $\beta = .96$, $\gamma = 2$, P as in 8.24 and $\lambda = (.98, 1.03)$. It recovers P from Q alone, and $(\beta, \lambda) = (.90; .9489, .9973)$, $(.96; .98, 1.03)$ and $(.99; .9952, 1.0460)$ all generate the same Q .

Why Ross's method fails here. Applying the Perron–Frobenius recovery of Exercise 8.24 to this Q gives $\hat{\beta} = .9498$ and $\hat{P} = \begin{bmatrix} .842 & .158 \\ .190 & .810 \end{bmatrix}$, both wrong. In a growth economy the stochastic discount factor

$\beta(c_{t+1}/c_t)^{-\gamma}$ is not of the ratio form $\beta d(s_{t+1})/d(s_t)$. It has a permanent (martingale) component, and Perron–Frobenius recovers a “long-term risk-neutral” measure instead of the true probabilities (Borovička, Hansen and Scheinkman, 2016).

Exercise 8.26 Recovering probabilities from prices, III

(a) $m_{t,t+h} = \beta^h (c_{t+h}/c_t)^{-\gamma}$: the price of any claim to d_{t+h} is $E_t[m_{t,t+h} d_{t+h}]$. Markets are complete (one-period Arrow securities in every state, traded every period), so state prices are unique, and so is the discount factor, $m = q/\pi$ state by state.

(b) A bond pays 1 in every state, so $p_{t+h}^t = \sum_j (Q^h)_{s_t,j} = \beta^h E_t[(c_{t+h}/c_t)^{-\gamma}]$.

(c) No. Since $Q = \beta D^{-1} P D$ is similar to βP , its eigenvalues are β and $\beta\rho$ with $\rho = P_{11} + P_{22} - 1$. By the Cayley–Hamilton theorem $Q^2 = e_1 Q - e_2 I$, where $e_1 = \text{tr } Q = \beta(P_{11} + P_{22})$ and $e_2 = \det Q = \beta^2 \rho$. Hence the bond prices obey

$$p_{h+2} = e_1 p_{h+1} - e_2 p_h, \quad p_0 = 1.$$

From p_1, p_2, p_3 one obtains e_1 and e_2 , i.e. β (the larger root of $z^2 - e_1 z + e_2 = 0$) and ρ . Then p_4 is implied by the recursion and carries no new information. The remaining datum is $p_1 = \beta[P_{11} + (1 - P_{11})r]$ (for $s_t = 1$), with $r = (c(2)/c(1))^{-\gamma}$. That is one equation in the two unknowns P_{11} and r , with $P_{22} = 1 + \rho - P_{11}$. So β and $P_{11} + P_{22}$ are identified, but P_{11} , P_{22} and the marginal-utility ratio are not separately. Example (`ch8.py`): $(\beta, P_{11}, P_{22}, r) = (.95, .8, .7, .7)$ and $(.95, .7, .8, .8)$ give identical prices $p_1, \dots, p_4 = (.893, .821275, .767351, .722874)$. The recursion recovers the roots $(.95, .475)$ and predicts p_4 exactly. Bonds span only the conditional expectation of the discount factor, not its distribution across states, and that is what the Arrow prices of Exercise 8.24 add.

Exercise 8.27 Corners

(a) The problem separates by date and history: maximize $\alpha c^1 + (1 - \alpha) \ln c^2$ subject to $c^1 + c^2 \leq Y(s) \equiv y^1(s) + y^2(s)$ and $c^i \geq 0$. In an interior optimum $\alpha = (1 - \alpha)/c^2$, so the log consumer eats the constant amount $(1 - \alpha)/\alpha$ and the risk-neutral consumer absorbs all fluctuations. Hence

$$c^2(s) = \min\left\{\frac{1 - \alpha}{\alpha}, Y(s)\right\}, \quad c^1(s) = \max\left\{0, Y(s) - \frac{1 - \alpha}{\alpha}\right\}.$$

The constraint $c^2 \geq 0$ never binds, because $u'_2(0) = \infty$. The constraint $c^1 \geq 0$ binds in states where the aggregate endowment falls short of $(1 - \alpha)/\alpha$. There type 2 consumes the whole aggregate endowment, $c^2 = Y(s)$, and consumption fluctuates with Y because the risk-neutral consumer cannot absorb the shortfall.

(b) Arrow prices $Q(s' | s)$, consumption $c^i(s^t) \geq 0$ and holdings $a_{t+1}^i(s')$ such that each type maximizes V_i subject to $c_t^i + \sum_{s'} Q(s' | s_t) a_{t+1}^i(s') \leq y^i(s_t) + a_t^i$, $a_0^i = 0$ and natural debt limits, and markets clear: $\sum_i c_t^i = Y(s_t)$ and $\sum_i a_{t+1}^i(s') = 0$.

(c) *Assumption 1 (interior)*. Type 1’s first-order condition with constant marginal utility gives risk-neutral prices, $Q(s' | s) = \beta P(s, s')$. Type 2’s Euler equation $Q(s' | s) = \beta P(s, s') c^2(s)/c^2(s')$ then forces constant consumption $\bar{c}^2$. His budget fixes it at the annuity value of his endowment at

these prices,

$$\bar{c}^2 = (1 - \beta)[(I - \beta P)^{-1}y^2](s_0), \quad c^1(s) = Y(s) - \bar{c}^2.$$

This is valid if and only if $\bar{c}^2 \leq \min_s Y(s)$. Holdings are $a^i(s) = [(I - Q)^{-1}(c^i - y^i)](s)$. In the example of `ch8.py`, $\lambda = .3$, $\delta = .4$, $\beta = .95$, $y^1 = (1, 1)$, $y^2 = (0, a)$ and $s_0 = \bar{s}_1$. With $a = .5$: $\bar{c}^2 = .2588$ and $c^1 = (.7412, 1.2412)$. With $a = 1.5$: $\bar{c}^2 = .7763$. The assumption holds for $a \leq a^* = 1.9323$.

Assumption 2 (a corner). Suppose type 1's constraint binds in the low-endowment state ℓ and not in the high state h . Then $c^1(\ell) = 0$ and $c^2(\ell) = Y(\ell)$. Type 1's first-order conditions, $\beta^t \pi_t \leq \mu_1 q_t^0$ with equality where $c^1 > 0$, combined with type 2's $\beta^t \pi_t / c_t^2 = \mu_2 q_t^0$, give $c_t^2 \leq \mu_1 / \mu_2 \equiv k$, with equality where $c^1 > 0$. So $c^2(h) = k$, and prices are set by type 2:

$$Q(s' | s) = \beta P(s, s') \frac{c^2(s)}{c^2(s')} \quad (\geq \beta P(s, s') \text{ when } s' = \ell).$$

Type 2's budget pins down k . With log utility it reads $\sum_t \beta^t E_0[y_t^2 / c_t^2] = 1 / (1 - \beta)$. For $a = 4$: $c^2 = (1, 2.0700)$, $c^1 = (0, 2.9300)$, and $Q = \begin{bmatrix} .2850 & .3213 \\ 1.1799 & .3800 \end{bmatrix}$. Consumption in the low state is scarce, so its Arrow price from the high state exceeds 1. Type 1's shadow value of wealth in the low state is $k / c^2(\ell) = 2.07 > 1$; he would like to consume a negative amount there. (If “always binds” is read as binding at every date, type 1 would consume nothing ever, which is an equilibrium only if he owns nothing, $y^1 \equiv 0$. Then $c^2 = Y$ and prices are $Q = \beta P(s, s') Y(s) / Y(s')$.) This is the decentralized counterpart of the corner in (a), with $(1 - \alpha) / \alpha = k$ equal to the ratio of the consumers' marginal utilities of wealth.

Exercise 8.28 Recovering probabilities from prices, IV

(a) As in Exercise 8.24: prices $Q_{ij}^{(1)}$ and a consumption plan such that the consumer maximizes expected utility subject to the sequential budget constraints, and markets clear, $c_t = y(s_t)$ with zero asset holdings. Adding two-period securities (part (c)) adds prices $Q_{ij}^{(2)}$ and the corresponding holdings.

(b) With $x = (\bar{y}_2 / \bar{y}_1)^{-\gamma}$,

$$Q^{(1)} = \begin{bmatrix} \beta\lambda & \beta(1 - \lambda)x \\ \beta(1 - \delta)/x & \beta\delta \end{bmatrix}.$$

(c) By no arbitrage, $Q^{(2)} = (Q^{(1)})^2$:

$$Q^{(2)} = \beta^2 \begin{bmatrix} \lambda^2 + (1 - \lambda)(1 - \delta) & x(1 - \lambda)(\lambda + \delta) \\ x^{-1}(1 - \delta)(\lambda + \delta) & (1 - \delta)(1 - \lambda) + \delta^2 \end{bmatrix}.$$

(d) The four numbers are today's prices of consumption in each state one and two periods ahead. Each is a probability times a discount factor times a marginal-utility ratio: $Q^{(h)}(1, j) = \beta^h \mathbb{P}(s_{t+h} = j | s_t = 1) (\bar{y}_j / \bar{y}_1)^{-\gamma}$. Yes, λ , δ and β can be inferred. By Exercise 8.24(b), the first row of $Q^{(1)}$ and $Q^{(2)}$ gives the second row of $Q^{(1)}$, and by Exercise 8.24(d) the full $Q^{(1)}$ gives β , λ , δ and x . What cannot be inferred is γ separately from the endowment ratio: only $x = (\bar{y}_2 / \bar{y}_1)^{-\gamma}$ is identified. Check (`ch8.py`): with $(\lambda, \delta, \beta, \gamma, \bar{y}_1, \bar{y}_2) = (.7, .6, .95, 3, 1, 1.1)$, the observed row is $(.665, .214125)$ and $(.550525, .264444)$, and the procedure returns $\beta = .95$, $\lambda = .7$, $\delta = .6$ and $x = .751315$ exactly.

The argument uses the stationarity of the level economy. In a growth economy the same data would not reveal β (Exercise 8.25).

Chapter 9

Overlapping Generations

Solutions to all 13 exercises of Chapter 9 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). The exercises are mostly analytical. `code/ch9.py` checks every closed form numerically (Euler equations, difference equations for real balances, the tree economy of 9.3, the Laffer curves of 9.10 and 9.13, credit controls, the Grandmont–Hall scheme and forced saving) and draws the figures.

Throughout, R_t is the gross real return between t and $t + 1$ (on currency, $R_t = p_t/p_{t+1}$), and m_t denotes real balances per young person. With $\ln c_t^t + \ln c_{t+1}^t$ and endowment (w_1, w_2) , a young person saving s at return R solves $\max \ln(w_1 - s) + \ln(w_2 + Rs)$. The first-order condition $w_2 + Rs = R(w_1 - s)$ gives the saving function

$$s(R) = \frac{1}{2}(w_1 - w_2/R),$$

which is positive if and only if $R > w_2/w_1$, the autarky (no-trade) interest rate.

Exercise 9.1 *Equilibria with valued fiat currency*

(a) The first-order condition $u'(w_1 - s) = Ru'(w_2 + Rs)$ reads $(w_1 - s)^{-\gamma} = R(w_2 + Rs)^{-\gamma}$, i.e. $w_2 + Rs = R^{1/\gamma}(w_1 - s)$, so

$$s(R) = \frac{R^{1/\gamma}w_1 - w_2}{R^{1/\gamma} + R}, \quad s(R) > 0 \iff R > R_{aut} \equiv (w_2/w_1)^\gamma.$$

For $\gamma = 1$ this is $s(R) = \frac{1}{2}(w_1 - w_2/R)$. For $\gamma \leq 1$, s is increasing in R ; for $\gamma > 1$ the income effect can make it bend backward at high returns.

(b) A price sequence $\{p_t\}_{t \geq 1}$ with $0 < p_t < \infty$ and an allocation such that: each young agent born at $t \geq 1$ maximizes utility given the return $R_t = p_t/p_{t+1}$ on currency; the initial old consume $c_1^0 = w_2 + M/(Np_1)$; and the currency market clears, $\frac{M}{Np_t} = s(p_t/p_{t+1})$ for all $t \geq 1$. Equivalently, with $m_t = M/(Np_t)$,

$$m_t = s(R_t), \quad m_{t+1} = R_t m_t.$$

(c) An equilibrium with valued currency in which $p_t = \bar{p}$ is constant (so $R_t = 1$ and $m_t = \bar{m}$).

(d) With $R = 1$: $\bar{m} = s(1) = \frac{1}{2}(w_1 - w_2) > 0$, so

$$\bar{p} = \frac{2M}{N(w_1 - w_2)}, \quad c_t^t = c_{t+1}^t = \frac{1}{2}(w_1 + w_2).$$

This holds for every γ , since $R = 1$ makes the agent equalize consumption.

(e) A continuum. The equilibrium conditions form a first-order difference equation in real balances: m_t determines R_t through $s(R_t) = m_t$, and then $m_{t+1} = R_t m_t$. Any $m_1 \in (0, \bar{m}]$ (equivalently $p_1 \geq \bar{p}$) starts an equilibrium. With s increasing, $m_1 < \bar{m}$ gives $R_1 < 1$, so $m_2 < m_1$, and so on: m_t decreases to 0 and the value of currency vanishes. $m_1 > \bar{m}$ gives $R_t > 1$ and m_t grows until saving would have to exceed w_1 , which is impossible. So the equilibria are indexed by $p_1 \in [\bar{p}, \infty)$: the stationary one and a continuum of nonstationary ones. For $\gamma = 1$ the chapter's (9.4.8) gives

them in closed form, $p_t = \bar{p} + c(w_1/w_2)^t$ with $c \geq 0$. (For large γ the saving function can bend backward, as it does in `ch9.py` for $\gamma = 5$. Then $s(R_t) = m_t$ can have two solutions, and there are even more equilibria.)

(f) In every nonstationary equilibrium $m_t \rightarrow 0$. Since $m_t = s(R_t)$ and s is continuous with $s(R) \rightarrow 0$ only as $R \downarrow R_{aut}$,

$$\lim_{t \rightarrow \infty} R_t = R_{aut} = (w_2/w_1)^\gamma < 1.$$

The rate of return on currency converges to the autarky interest rate, and inflation to $(w_1/w_2)^\gamma$. The economy approaches the nonmonetary equilibrium, where the young's marginal rate of substitution at the endowment, $u'(w_1)/u'(w_2) = (w_2/w_1)^\gamma$, is the (shadow) interest rate. `ch9.py` iterates the difference equation from $m_1 = \bar{m}/2$ with $(w_1, w_2) = (2, 1)$ and finds $R_{60} = .7071, .5000, .2500, .0313$ for $\gamma = .5, 1, 2, 5$, equal to $(1/2)^\gamma$.

Exercise 9.2 *Heterogeneous endowments and valued currency*

(a) Type 1, endowed with $(w_1, 0)$: $\max \ln(w_1 - s) + \ln(Rs)$ gives $s_1 = w_1/2$, independent of R . Type 2, endowed with $(0, w_2)$: $\max \ln(-s) + \ln(w_2 + Rs)$ gives $s_2(R) = -w_2/(2R)$, so type 2 borrows $w_2/(2R)$ and repays $w_2/2$.

(b) Without valued currency the only trades are loans within a generation. The loan market clears when $\frac{N}{2}s_1 + \frac{N}{2}s_2(R_t) = 0$, so

$$R_t = w_2/w_1 \quad \text{for all } t,$$

and this equilibrium is unique. Each young person of either type consumes $w_1/2$ when young and $w_2/2$ when old. The initial old consume their endowments $(w_2$ or $0)$.

(c) Prices $0 < p_t < \infty$ and an allocation with each young person optimizing at $R_t = p_t/p_{t+1}$, where the young's net saving equals the real value of the currency stock: $\frac{N}{2} \cdot \frac{w_1}{2} - \frac{N}{2} \cdot \frac{w_2}{2R_t} = \frac{NM}{p_t}$ for $t \geq 1$.

(d) Dividing by $N/4$, $4M = w_1 p_t - w_2 p_{t+1}$, i.e. $p_{t+1} = \frac{w_1}{w_2} p_t - \frac{4M}{w_2}$. The general solution is

$$p_t = \bar{p} + c \left(\frac{w_1}{w_2} \right)^t, \quad \bar{p} = \frac{4M}{w_1 - w_2},$$

and $c \geq 0$ is required for $p_t > 0$ at all t , since $w_1/w_2 > 1$. So the equilibria with valued currency are indexed by $c \geq 0$ (equivalently $p_1 \geq \bar{p}$). For $c > 0$ the price level explodes, real balances go to zero, and $R_t = p_t/p_{t+1}$ falls monotonically to w_2/w_1 , the nonmonetary interest rate (Figure S9.1).

(e) A constant p_t requires $c = 0$, so $\bar{p} = 4M/(w_1 - w_2)$ is the unique stationary equilibrium, with $R = 1$.

(f) The equilibria are not Pareto ranked. A lender's utility $\ln(w_1/2) + \ln(R_t w_1/2)$ rises with R_t , while a borrower's utility $\ln \frac{w_2}{2R_t} + \ln \frac{w_2}{2}$ falls with R_t . Along each path, R_t is highest in the stationary equilibrium ($R_t = 1$) and decreasing in c . So lenders and the initial old, whose consumption M/p_1 is larger when p_1 is smaller, prefer the stationary equilibrium, while borrowers prefer equilibria with lower interest rates. (With $(w_1, w_2) = (2, 1)$: generation 1's lenders get utility 0 at $c = 0$ and -1.182 at $c = .5$; its borrowers get -1.386 and -1.204 .)

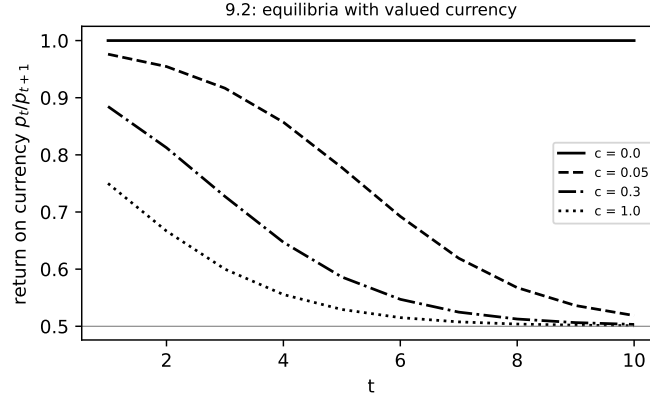


Figure S9.1. Exercise 9.2 with $(w_1, w_2) = (2, 1)$: the return on currency in the equilibria $p_t = \bar{p} + c(w_1/w_2)^t$. Only $c = 0$ is stationary; the others converge to the nonmonetary rate w_2/w_1 .

Exercise 9.3 A tree with a tiny dividend

Let q_t be the ex-dividend price of the tree per young person (normalize $N = 1$). Whoever buys it at t receives $q_{t+1} + d$ at $t + 1$.

(a) There is none. Suppose currency is valued at every date. Then currency and the tree must earn the same return, $R_t = p_t/p_{t+1} = (q_{t+1} + d)/q_t$. Real balances obey $m_{t+1} = R_t m_t$, so $R_1 R_2 \cdots R_{t-1} = m_t/m_1 \leq w_1/m_1$, because saving cannot exceed w_1 . Solving the tree's pricing equation forward,

$$q_1 \geq \sum_{t=1}^T \frac{d}{R_1 \cdots R_t} \geq T d \frac{m_1}{w_1} \rightarrow \infty,$$

which contradicts $q_1 \leq s(R_1) < w_1$. When currency has a nonnegative real rate of return in the long run, the tree's dividend stream is worth infinitely much. So any positive dividend eliminates all equilibria with valued fiat currency.

(b) The tree is the only asset: $s(R_t) = q_t$ and $R_t q_t = q_{t+1} + d$. A stationary equilibrium has $q = s(R)$ and $q(R - 1) = d$, i.e.

$$s(R)(R - 1) = d, \quad R > 1.$$

The left side is 0 at $R = 1$ and increases in R (as long as s does), so there is a unique solution $R^* > 1$, with $q^* = s(R^*)$. It is also the only equilibrium. Write the dynamics as $q_{t+1} = R(q_t)q_t - d$, where $R(q) = s^{-1}(q)$ is increasing. Then $q_{t+1} - q_t = (R(q_t) - 1)q_t - d$ is positive for $q_t > q^*$ and negative for $q_t < q^*$. A path that starts above q^* rises until the young cannot afford the tree, and one that starts below falls until the tree's price would be negative.

(c) With $w_1 = 10$, $w_2 = 5$, $d = 10^{-6}$ and log utility (the exercise leaves γ open; we take $\gamma = 1$): $s(R) = 5 - 2.5/R$, and $s(R)(R - 1) = d$ gives $R^* = 1 + 4.0 \times 10^{-7}$ and $q^* = 2.500001$. `ch9.py` starts the tree's price 10^{-3} above or below q^* and finds that it leaves the feasible range after 11 and 32 periods. The tree, with an almost worthless dividend, is priced like fiat money in the stationary monetary equilibrium of Exercise 9.1 ($\bar{m} = 2.5$), and the allocation is essentially that equilibrium's.

But whereas Exercise 9.1 has a continuum of monetary equilibria plus the nonmonetary one, the tiny dividend selects a single equilibrium, the stationary one.

Exercise 9.4 *Population growth*

(a) All young people are identical, and the old cannot repay loans. So there is no trade: each person consumes the endowment (w_1, w_2) , and the (shadow) interest rate is the marginal rate of substitution at the endowment, $R_{aut} = w_2/w_1$.

(b) In a stationary equilibrium real balances per young person are constant, $M/(N(t)p_t) = \bar{m}$, so total real balances grow with the population. Then $p_t/p_{t+1} = N(t+1)/N(t) = n$: currency earns the population growth rate. Hence $\bar{m} = s(n) = \frac{1}{2}(w_1 - w_2/n) > 0$ (since $n \geq 1 > w_2/w_1$) and

$$p_t = \frac{M}{N(t)s(n)} = \frac{2M}{N(t)(w_1 - w_2/n)}, \quad c_t^t = \frac{1}{2}\left(w_1 + \frac{w_2}{n}\right), \quad c_{t+1}^t = \frac{1}{2}(nw_1 + w_2).$$

The price level falls at rate n . Currency earns the “biological” rate of return $R = n$, which delivers the golden-rule allocation (the best stationary allocation, given the resource constraint $c_t^t + c_t^{t-1}/n = w_1 + w_2/n$).

Exercise 9.5 *Population growth and all monetary equilibria*

(a) An equilibrium without valued currency is an allocation and interest rates at which the young optimize and loans clear (and $1/p_t = 0$). All young are identical, so it is autarky, with interest rate $R_{aut} = w_2/w_1$.

(b) Prices $0 < p_t < \infty$ with $N(t)s(R_t) = N(0)/p_t$ (the total currency stock is $H = N(0)$) and $R_t = p_t/p_{t+1}$.

(c) Let $m_t = H/(N(t)p_t)$. Then $m_{t+1} = R_t m_t/n$, so $R_t = n m_{t+1}/m_t$, and $m_t = \frac{1}{2}\left(w_1 - \frac{w_2 m_t}{n m_{t+1}}\right)$. In terms of $x_t = 1/m_t$ this equation is linear:

$$x_{t+1} = \frac{nw_1}{w_2} x_t - \frac{2n}{w_2}, \quad \text{so} \quad x_t = x^* + c\left(\frac{nw_1}{w_2}\right)^t, \quad x^* = \frac{1}{m^*}, \quad m^* = \frac{1}{2}\left(w_1 - \frac{w_2}{n}\right).$$

Valued currency requires $m_t > 0$ for all t .

- If $n \leq w_2/w_1$, then $m^* \leq 0$. No equilibrium with valued currency exists: autarky already offers a return $w_2/w_1 \geq n$, at least what currency could pay.
- If $n > w_2/w_1$, then $nw_1/w_2 > 1$ and the equilibria are indexed by $c \geq 0$. $c = 0$ is the stationary one ($m_t = m^*$, $R_t = n$), and $c > 0$ gives $x_t \rightarrow \infty$, i.e. $m_t \rightarrow 0$. ($c < 0$ would make m_t negative eventually.)

(d) In the stationary equilibrium $R_t = n$. In the others $R_t = n x_t/x_{t+1} \rightarrow n \cdot \frac{w_2}{nw_1} = w_2/w_1$, the autarky rate of (a): the value of currency vanishes and the economy converges to autarky. (ch9.py: with $(w_1, w_2) = (2, 1)$ and $n = 1.2$, R_t falls from .993 to .5000.)

(e) Yes. A young person’s utility increases with the return on saving ($s > 0$ in any monetary equilibrium), and $R_t = n \frac{x^* + c\lambda^t}{x^* + c\lambda^{t+1}}$, with $\lambda = nw_1/w_2 > 1$, is decreasing in c at every t . The initial old consume $w_2 + 1/p_1$, which is also decreasing in c . So every generation prefers smaller c : the equilibria are Pareto ranked, the stationary one ($c = 0$) is best, and autarky, the limit $c \rightarrow \infty$, is worst. The difference from Exercise 9.2 is that here all young people are lenders.

Exercise 9.6 *Exchange rate determinacy*

Preliminaries. A poor young person saves $\alpha/2$ whatever the return (log utility, no endowment when old), and can save only in currency. World real money demand is therefore constant at $2N \cdot \alpha/2 = N\alpha$. The rich save $(\beta - T_i/N)/2$ in storage or bonds at R ; currency, which returns less than R , is dominated for them.

(a) Price levels $0 < p_i(t) < \infty$, government policies satisfying the budget constraints, and an allocation such that: the poor choose currency portfolios and the rich choose storage and bonds optimally; the goods markets clear; and the currency markets clear, $\frac{H_1(t)}{p_1(t)} + \frac{H_2(t)}{p_2(t)} = N\alpha$ together with a split of this demand across the currencies that poor people are willing to hold.

(b) If both currencies are valued, poor people hold both, which they do only if the returns are equal: $\frac{p_1(t)}{p_1(t+1)} = \frac{p_2(t)}{p_2(t+1)}$. If one return were higher, nobody would hold the other currency. Rearranging, $e_t = p_1(t)/p_2(t) = p_1(t+1)/p_2(t+1) = e_{t+1}$: the exchange rate is constant.

(c) Let $m_i(t) = H_i(t)/p_i(t)$ and $\sigma_i = G_i - T_i$. With $B_i = 0$, each government finances σ_i by seigniorage: $m_i(t) - R_m(t-1)m_i(t-1) = \sigma_i$ for $t \geq 2$, and $m_i(1) = \sigma_i + H_i(0)/p_i(1)$, where R_m is the common return on currency. Adding across countries and using $m_1 + m_2 = N\alpha$ gives $R_m = 1 - (\sigma_1 + \sigma_2)/(N\alpha)$ for all t . At $t = 1$ the two unknown price levels must satisfy only one equation,

$$\frac{H_1(0)}{p_1(1)} + \frac{H_2(0)}{p_2(1)} = N\alpha - \sigma_1 - \sigma_2.$$

Every exchange rate $e = p_1(1)/p_2(1)$ in a nonempty interval (with both terms positive) is consistent with equilibrium, and then stays constant by (b). Governments set their policies in real terms, and nothing pins down how the fixed world demand for real balances is divided between the two perfectly substitutable currencies. This is Kareken and Wallace's indeterminacy.

(d) For $t \geq 2$ the budget reads $G_i + RB_i(1) = B_i(1) + T_i$ (no money creation), a restriction on the constants. At $t = 1$ it reads $G_i = B_i(1) + [H_i(1) - H_i(0)]/p_i(1) + T_i$, which gives

$$p_i(1) = \frac{H_i(1) - H_i(0)}{G_i - T_i - B_i(1)}$$

(for $H_i(1) \neq H_i(0)$). Each government's nominal injection must buy a given real amount of goods, which pins down its own price level. So $e = p_1(1)/p_2(1)$ is determinate. With constant money stocks and equal returns, m_1 and m_2 are constant, so the price levels are constant as well. An equilibrium exists only if these price levels also satisfy $H_1(1)/p_1(1) + H_2(1)/p_2(1) = N\alpha$.

(e) The exercise does not say how country 2 finances the residual $G_2 + (R-1)B_2(1) - T_2(t)$. If country 2 financed it by printing money at a given real rate, both governments would be setting real seigniorage, and we would be back in case (c), with an indeterminate exchange rate. The claim holds under the reading that parallels (d): country 2 fixes its money stock after an initial issue at $t = 1$, $H_2(t) = H_2(1)$, and lets taxes absorb the residual. (Country 1 finances G_1 at $t = 1$ with $B_1(1) = 0$ and a given tax.) Then:

- country 2's $t = 1$ budget, $G_2 = B_2(1) + [H_2(1) - H_2(0)]/p_2(1) + T_2(1)$, pins down $p_2(1)$ as in (d);

- the world money market gives $m_1(1) = N\alpha - H_2(1)/p_2(1)$;
- country 1's $t = 1$ budget, $H_1(0)/p_1(1) = m_1(1) - (G_1 - T_1(1))$, then pins down $p_1(1)$.

So $e = p_1(1)/p_2(1)$ is determinate, and it is constant thereafter by (b). From $t = 2$ on, $m_1(t) = s_1 + R_m(t-1)m_1(t-1)$ and $R_m(t-1) = m_2(t)/m_2(t-1)$ determine the whole path. One can show $N\alpha - m_1(t) = (N\alpha - m_1(t-1))(1 - s_1/(N\alpha))$: country 1's currency gradually takes over the world money demand, and currency pays $R_m = 1 - s_1/(N\alpha)$. The general principle is that the exchange rate is determinate when at least one government fixes a nominal quantity that its budget constraint ties to its price level. With purely real policies, as in (c), it is not.

Exercise 9.7 Credit controls

Lenders save $y/2$ at any return. Unconstrained borrowers borrow $y/(2R)$ and repay $y/2$.

(a) Prices $1/p_t = 0$ and a loan return R_t such that each young person optimizes and the loan market clears. Only lenders and borrowers of the same generation trade: lenders lend to borrowers of their own generation.

(b) Clearing requires $\frac{N}{2} \cdot \frac{y}{2} = \frac{N}{2} \cdot \frac{y}{2R}$, so $R = 1$. Everybody consumes $(y/2, y/2)$, and the initial old consume nothing.

(c) Prices $0 < p_t < \infty$ at which lenders are willing to hold both loans and currency, so $R_t = p_t/p_{t+1}$. The young's net saving must equal the real value of the currency stock:

$$\frac{N}{2} \cdot \frac{y}{2} - \frac{N}{2} \cdot \frac{y}{2R_t} = \frac{H(0)}{p_t}.$$

The young lend to each other and buy the old's currency.

(d) With $m_t = H(0)/(Np_t) > 0$, the condition reads $m_t = \frac{y}{4}(1 - 1/R_t)$, which requires $R_t > 1$. Then $m_{t+1} = R_t m_t = m_t/(1 - 4m_t/y) > m_t$, so $m_{t+1} - m_t = \frac{4m_t^2}{y - 4m_t} \geq \frac{4m_1^2}{y} > 0$. Real balances rise by at least a fixed amount each period, so they eventually exceed $y/4$, the most the young could want to hold. (ch9.py: from $m_1 = .01$ with $y = 1$ this takes 24 periods.) A constant price level would need $R = 1$ and hence $m = 0$. So no equilibrium with valued currency exists. At $R = 1$ intragenerational lending already absorbs all saving, and currency could only be valued at a return above 1, which feeds on itself.

(e) Borrowers may promise at most $y/4$, and they want to promise $y/2$, so the limit binds: each borrows $y/(4R)$. Clearing, $\frac{y}{2} = \frac{y}{4R}$, gives $R = \frac{1}{2}$. Lenders consume $(y/2, y/4)$ and borrowers $(y/2, 3y/4)$.

(f) With a constant price level currency returns 1, so loans must return 1 too. Borrowers still want to borrow $y/2$ at $R = 1$ and are held to $y/4$. Each lender therefore lends $y/4$ and holds the other $y/4$ of saving in currency. Real balances are $\frac{N}{2} \cdot \frac{y}{4} = \frac{H(0)}{p}$, so

$$p = \frac{8}{y} \frac{H(0)}{N}, \quad q = \frac{8}{y}.$$

Lenders consume $(y/2, y/2)$, borrowers $(y/4, 3y/4)$, and each initial old person consumes $y/8$. The return on loans is $R = 1$. The credit limit leaves an excess supply of loanable funds, which currency absorbs.

(g) Lenders get $(y/2, y/2)$ in both economies, so they are indifferent. Borrowers are worse off in (f): $\ln \frac{y}{4} + \ln \frac{3y}{4} < 2 \ln \frac{y}{2}$ (-1.674 against -1.386 for $y = 1$). The initial old are better off in (f) ($y/8$ against 0). The credit control transfers resources from borrowers to the initial old by creating a demand for money.

Exercise 9.8 *Inside money and real bills*

(a) Type 1 saves $y/2$, and type 2 borrows $Y/(2R)$. Clearing, $N_1y/2 = N_2Y/(2R)$, gives $1 + r = N_2Y/(N_1y)$.

(b) A constant price level makes currency return 1. If currency is held, loans must also return 1, and then the young's net saving is $\frac{N_1y - N_2Y}{2} = H_0/p$. But $1 + r > 1$ in (a) means $N_2Y > N_1y$, so net saving would be negative, a contradiction. With $N_2Y > N_1y$, the young as a group want to borrow at $R = 1$: there is no demand for an outside asset.

(c) If $N_2Y < N_1y$, set $R = 1$ and $H_0/p = \frac{N_1y - N_2Y}{2} > 0$, i.e. $q = \frac{2}{N_1y - N_2Y}$. Everyone optimizes at $R = 1$, loans and currency both pay 1, and markets clear. The equilibrium is not Pareto superior to the nonmonetary one: type-2 people are borrowers, whose utility $\ln \frac{Y}{2R} + \ln \frac{Y}{2}$ is decreasing in R , and R rises from $N_2Y/(N_1y) < 1$ to 1. Lenders and the initial old gain, but borrowers lose.

(d) (i) Guess $p(t) = \bar{p}$ and $R = 1$. The government's loans pay 1, so the L_g it lent last period comes back as L_g and is lent again. Its budget balances, and $M_I = L_g\bar{p}$ stays constant. Type 1 holds currency, inside money and private loans, all paying 1. Borrowers' demand $N_2Y/2$ is met by the government's L_g plus private lending $N_1y/2 - (H_0 + M_I)/\bar{p}$. With $M_I/\bar{p} = L_g$, clearing reduces to $H_0/\bar{p} = (N_1y - N_2Y)/2$, the condition of (c). So $\bar{p} = qH_0$ is again an equilibrium price level, with the same allocation.

(ii) The inside money is backed one for one by private loans that earn the same return as money. The government's purchase of L_g loans displaces exactly L_g of private lending, and savers hold L_g more inside money instead. This is a pure swap of assets that savers regard as perfect substitutes, so aggregate demand for goods is unchanged. This is the real bills doctrine: money issued against sound private loans is not inflationary.

(iii) In many models, open-market operations exchange money for assets that are *not* perfect substitutes for it. Examples are government bonds backed by future taxes when money pays a lower return, or settings where money is needed for transactions (cash-in-advance, reserve requirements). Other operations change the government's net claims, so future taxes or seigniorage must change. Here there is no transaction role for money, money and loans pay the same return, and the government's loans fully back its liabilities.

Exercise 9.9 *Social security and the price level*

(a) An equilibrium is $0 < p_t < \infty$ with the young optimizing and $Ns_t = H/p_t$. With endowment $(y, 0)$ the young save $y/2$ whatever the return, so

$$\frac{H}{p_t} = \frac{Ny}{2} \implies p_t = \frac{2H}{Ny} \quad \text{for all } t.$$

The market-clearing condition pins down every price level directly, so the equilibrium exists and is unique. It is stationary, with $R = 1$ and consumption $(y/2, y/2)$.

(b) Now the effective endowment is $(y - \tau, \tau)$, and $s(R) = \frac{1}{2}(y - \tau - \tau/R)$. Valued currency needs $s > 0$ at the returns it pays. A stationary equilibrium ($R = 1$) needs $s(1) = \frac{1}{2}(y - 2\tau) > 0$, i.e.

$$\tau < \frac{y}{2}.$$

If $\tau \geq y/2$, social security already gives the old at least as much as the young, the autarky interest rate $\tau/(y - \tau)$ is at least 1, and currency cannot be valued: no equilibrium with valued currency exists.

(c) No. With $\tau > 0$ saving depends on the return, and $m_t = s(m_{t+1}/m_t)$ becomes $m_{t+1} = \tau m_t/(y - \tau - 2m_t)$. Its fixed point $\bar{m} = (y - 2\tau)/2$ is unstable, and 0 is stable (the slope there is $\tau/(y - \tau) < 1$). So every $m_1 \in (0, \bar{m}]$ starts an equilibrium. As in Exercise 9.1, there is a continuum of equilibria in which the value of currency goes to zero and the return converges to $\tau/(y - \tau)$.

(d) The stationary equilibrium is unique:

$$\bar{p} = \frac{H}{N\bar{m}} = \frac{2H}{N(y - 2\tau)},$$

which is increasing in τ . A larger social security system crowds out the demand for money, because the young need to save less for old age. The price level rises (the value of currency falls), and currency is worthless when $\tau \geq y/2$.

Exercise 9.10 *Seigniorage*

(a) As in 9.8(a), $R = N_2\beta/(N_1\alpha) < 1$.

(b) No. At the equilibrium every young person's marginal rate of substitution between youth and old age is $R < 1$. Let each young person, of either type, give ε to the old in every period. Each person loses ε when young and gains ε when old. The utility change is $\varepsilon[u'(c^{old}) - u'(c^{young})] = \varepsilon u'(c^{young})(1/R - 1) > 0$ to first order, and the initial old gain ε . So there is a Pareto improvement. (This is the Balasko–Shell criterion of Section 9.7.1: with interest rates below the growth rate, the allocation is not optimal.)

(c) Let $\mathcal{M}_t = H(t)/p(t)$ be real balances. Then the return on currency is $R_t = p_t/p_{t+1} = \mathcal{M}_{t+1}/(z\mathcal{M}_t)$, and the young's net saving must equal $\mathcal{M}_t$: $\frac{N_1\alpha}{2} - \frac{N_2\beta}{2R_t} = \mathcal{M}_t$. Hence

$$\mathcal{M}_{t+1} = f(\mathcal{M}_t) \equiv \frac{N_2\beta z \mathcal{M}_t}{N_1\alpha - 2\mathcal{M}_t}.$$

The map f is increasing and convex, with $f'(0) = N_2\beta z/(N_1\alpha)$. Under the assumption $1/z > N_2\beta/(N_1\alpha)$, $f'(0) < 1$, and f has a positive fixed point $\mathcal{M}^* = \frac{1}{2}(N_1\alpha - N_2\beta z)$ with $f'(\mathcal{M}^*) > 1$. Every $\mathcal{M}_1 \in (0, \mathcal{M}^*]$, i.e. every initial price level $p(1) \geq H(1)/\mathcal{M}^*$, starts an equilibrium, and for $\mathcal{M}_1 < \mathcal{M}^*$ real balances decline to zero. That is a continuum. (Starting above $\mathcal{M}^*$, real balances explode past $N_1\alpha/2$.)

(d) The stationary equilibrium has

$$p(t) = \frac{2}{N_1\alpha - N_2\beta z} H(t),$$

so the price level is proportional to the money stock and grows at rate z . Currency and loans both return $1/z$. Seigniorage is $G = \mathcal{M}^*(1 - 1/z) = \frac{1}{2}(N_1\alpha - N_2\beta z)(1 - 1/z)$.

(e) If $1/z < N_2\beta/(N_1\alpha)$, then $f(\mathcal{M})/\mathcal{M} = \frac{N_2\beta z}{N_1\alpha - 2\mathcal{M}} > 1$ for every $\mathcal{M} > 0$, so real balances grow from any positive start until they exceed $N_1\alpha/2$: no equilibrium with valued currency exists. Interpretation: at the money growth rate z , a stationary monetary equilibrium would pay $1/z$ on currency. That is less than the interest rate $N_2\beta/(N_1\alpha)$ that private loans pay even when nobody holds currency. Currency is dominated in rate of return, so no one will hold it and the inflation tax cannot be collected.

(f) Write $a = N_1\alpha$, $b = N_2\beta$. Then $G(z) = \frac{1}{2}(a - bz)(1 - 1/z) = \frac{1}{2}(a + b - bz - a/z)$ and $G'(z) = \frac{1}{2}(a/z^2 - b)$, so

$$z^* = \sqrt{a/b} = \sqrt{N_1\alpha/(N_2\beta)}, \quad G(z^*) = \frac{1}{2}(\sqrt{a} - \sqrt{b})^2.$$

The largest money growth compatible with valued currency is $z_{\max} = a/b = (z^*)^2 > z^*$. Revenue peaks well before the demand for currency disappears, so the Laffer curve is single-peaked on $(1, z_{\max})$ (Figure S9.2, right; with $a = 2$, $b = 1$: $z^* = 1.4142$, $z_{\max} = 2$, $G_{\max} = .0858$).

Exercise 9.11 *Unpleasant monetarist arithmetic*

In a stationary equilibrium $R_1(t) = R$ and real balances $g(R)$ are constant, so seigniorage is $p(t)[M(t) - M(t-1)] = g(R) - \frac{p(t)}{p(t-1)}p(t-1)M(t-1) = g(R)(1 - R) = h(R)$.

(a) With $B = 0$ the budget is $h(R) = D$. Since h rises from 0 at $\underline{R}$ to $h(R_m) > D$ and falls back to 0 at $R = 1$, the equation has exactly two solutions, $R_a < R_m < R_b$. These are a high-inflation and a low-inflation stationary equilibrium, on the two sides of the Laffer curve.

(b) With $B(t) = B$ the budget is $D = h(R) + B - R_2B$, i.e.

$$h(R) = D + (R_2 - 1)B.$$

Solutions exist if and only if $D + (R_2 - 1)B \leq h(R_m)$, i.e. for $0 < B \leq \bar{B} \equiv \frac{h(R_m) - D}{R_2 - 1}$, which is positive. For $B < \bar{B}$ there are again two.

(c) Assume the economy is on the “good” side of the Laffer curve, $R > R_m$, where $h'(R) < 0$: more inflation (lower R) raises revenue. The stationary root $R_b(B)$ solves $h(R_b) = D + (R_2 - 1)B$. Implicit differentiation gives $R'_b(B) = (R_2 - 1)/h'(R_b) < 0$. So a lower B means a higher return on currency, i.e. lower stationary inflation $1/R_b$. Interest on bonds is a permanent claim on seigniorage. A government that tightens money today by selling bonds, raising B and lowering $M(1)$, must collect more seigniorage forever, and so has higher inflation in the long run. (On the other side of the Laffer curve the comparative static is reversed, which is why the assumption is needed.)

Exercise 9.12 *Grandmont–Hall*

(a) Sequences $\{p(t), 1 + r(t), H(t)\}_{t \geq 1}$ with $0 < p(t) < \infty$ that satisfy: the arbitrage condition $1 + r(t) = [1 + r(t)]p(t)/\bar{p}$ (currency is valued); loan-market clearing, $f[1 + r(t)] = H(t)/p(t)$; and the government budget constraint with $H(1) = H(0)$.

(b) Arbitrage with $1 + r(t) > 0$ forces $p(t) = \bar{p}$ at every date: the interest paid on currency makes its real return equal to the loan rate exactly when the price level is at its target. With $p(t) = p(t+1) = \bar{p}$, the budget becomes $H(t+1) = [1 + r(t)]H(t)$, and loan-market clearing becomes $f[1 + r(t)] = H(t)/\bar{p}$. At $t = 1$, $f[1 + r(1)] = H(0)/\bar{p} = f(1)$, so $1 + r(1) = 1$ because f is

strictly increasing. Then $H(2) = H(1)$, so $f[1 + r(2)] = f(1)$ and $1 + r(2) = 1$, and by induction

$$p(t) = \bar{p}, \quad 1 + r(t) = 1, \quad H(t) = H(0) \quad \text{for all } t \geq 1.$$

Every monetary equilibrium must be this one, so it is unique. Paying interest on currency at the loan rate times the ratio of the price level to its target turns the price level into the stationary monetary equilibrium's value. It eliminates the continuum of nonstationary equilibria (with $p_t \rightarrow \infty$) of the laissez-faire economy, and in equilibrium no interest is actually paid. (`ch9.py` iterates the conditions with $f(R) = \frac{1}{2}(2 - 1/R)$ and finds $1 + r(t) = 1$.)

Exercise 9.13 *Bryant–Keynes–Wallace*

(a) Constant real balances $m = H(t)/(Np(t))$ and a constant return $R = p(t)/p(t+1)$ such that the young optimize ($m = s(R)$) and the government budget holds: $g = [H(t) - H(t-1)]/(Np(t)) = m - mR$. So

$$g = \phi(R) \equiv s(R)(1 - R) = \frac{1}{2} \left(w_1 - \frac{w_2}{R} \right) (1 - R), \quad R \in (w_2/w_1, 1).$$

(b) $\phi(R) = \frac{1}{2}(w_1 + w_2 - w_1R - w_2/R)$ is continuous and strictly concave on $R > 0$ ($\phi'' = -w_2/R^3 < 0$). It vanishes at $R = w_2/w_1$ and $R = 1$, and it peaks at $R^* = \sqrt{w_2/w_1}$ with $\phi(R^*) = \frac{1}{2}(\sqrt{w_1} - \sqrt{w_2})^2 > 0$. For every $g \in (0, \phi(R^*))$ there is a solution, i.e. a stationary equilibrium with valued currency.

(c) By strict concavity, for $g < \phi(R^*)$ the equation $\phi(R) = g$ has exactly two roots, one on each side of R^* . So if R_1 is a stationary equilibrium return (and g is not the maximum), the other root $R_2 \neq R_1$ is another. (With $(w_1, w_2) = (2, 1)$ and $g = .05$: $R = .5649$ and $.8851$; Figure S9.2, left.)

(d) No. After the government takes g per young person, a stationary allocation must satisfy $c_1 + c_2 = w_1 + w_2 - g$. The best one for each generation equalizes marginal utilities, $c_1 = c_2$, i.e. it has a marginal rate of substitution of 1. In the monetary equilibria the young face $R < 1$, so their marginal rate of substitution is $R \neq 1$. The inflation tax distorts saving. The high- R equilibrium Pareto dominates the low- R one (utilities .7736 against .6969 in the example), and neither is optimal.

(e) Take any stationary monetary equilibrium (m, R) with $m(1 - R) = g$. Set $F = m$ and $1 + r_2 = R$. The young, who would voluntarily save $s(R) = m$ at that return, are willing to hold F , and the government's budget holds: $B - B(1 + r_2) = m(1 - R) = g$. At $t = 1$, redeeming $H(0)$ at the price $p(1)$ with $H(0)/(Np(1)) = F - g$ gives the initial old the same consumption as in the monetary equilibrium. So a forced-saving equilibrium exists whenever a monetary one does.

(f) With a binding requirement, the stationary budget gives $1 + r_2 = 1 - g/F$, and a young person consumes $(w_1 - F, w_2 + F - g)$. Maximizing $\ln(w_1 - F) + \ln(w_2 + F - g)$ over F gives

$$F = \frac{w_1 - w_2 + g}{2}, \quad 1 + r_2 = 1 - \frac{g}{F} = \frac{w_1 - w_2 - g}{w_1 - w_2 + g}, \quad c_1 = c_2 = \frac{w_1 + w_2 - g}{2}.$$

The requirement indeed binds: at $1 + r_2 < 1$ the young would voluntarily save only $s(1 + r_2) < F$ (for $(w_1, w_2, g) = (2, 1, .05)$: $F = .525$, $1 + r_2 = .9048$, voluntary saving .4474). Utility is .7773, above

both monetary equilibria. Forced saving at a low return is a lump-sum tax in disguise, because the prohibition on intermediation keeps people from undoing it.

(g) Yes. Every generation $t \geq 1$ gets $c_1 = c_2 = (w_1 + w_2 - g)/2$, the best stationary allocation of what is left after G . The initial old receive $w_2 + H(0)/(Np(1)) = w_2 + F - g = (w_1 + w_2 - g)/2$ as well. The scheme is a way of raising G with (effectively) lump-sum taxes, which the inflation tax of (a)–(c) cannot do.

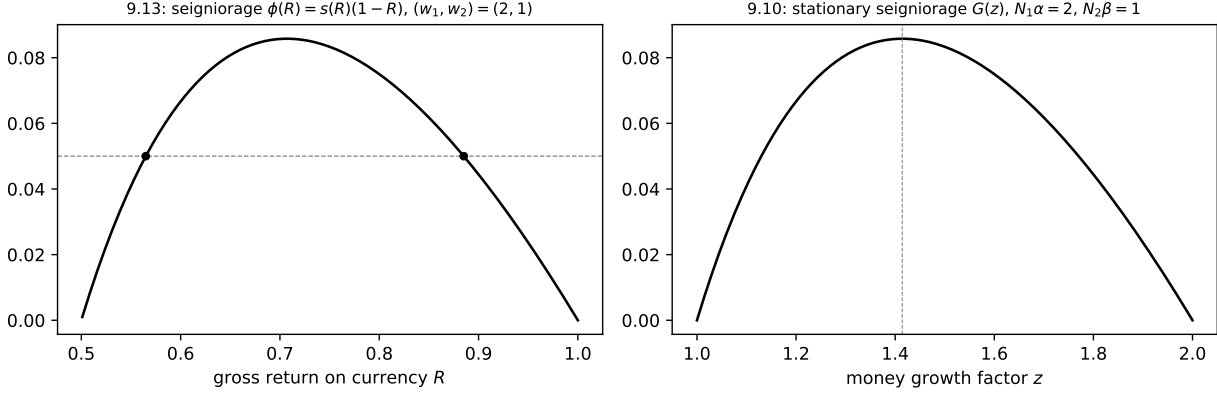


Figure S9.2. Left: stationary seigniorage $\phi(R)$ in Exercise 9.13 with $(w_1, w_2) = (2, 1)$; $g = .05$ is raised at two returns. Right: stationary seigniorage as a function of money growth in Exercise 9.10 ($N_1\alpha = 2$, $N_2\beta = 1$); revenue peaks at $z^* = \sqrt{2}$, well before $z_{\max} = 2$.

Chapter 11

Fiscal Policies in a Growth Model

Solutions to all 20 exercises of Chapter 11 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Numerical results and figures come from `code/ch11.py`, which replaces the book's Matlab and Dynare programs: it computes perfect-foresight equilibria by Newton's method on the stacked Euler equations and checks them against the book's shooting algorithm and its linear approximation.

Equilibrium conditions. Throughout, $u(c) = c^{1-\gamma}/(1-\gamma)$ ($\log c$ if $\gamma = 1$), $\beta = 1/(1+\rho)$, and, unless said otherwise, $f(k) = zk^\alpha$ and labor is supplied inelastically. In a competitive equilibrium with time-0 trading the firm sets $\eta_t = f'(k_t)$ and $w_t = f(k_t) - k_t f'(k_t)$ (11.5.7). Bounded household budget sets require the no-arbitrage conditions $q_t = q_{t+1}[(1 - \tau_{k,t+1})(\eta_{t+1} - \delta) + 1]$ (11.5.2). The household's first-order condition is $\beta^t u'(c_t) = \mu q_t(1 + \tau_{ct})$ (11.5.5A), and $\lim_T q_T k_{T+1} = 0$. Eliminating prices, an equilibrium allocation satisfies feasibility $c_t + g_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$, the initial condition, the terminal condition (11.5.3), and

$$u'(c_t) = \beta u'(c_{t+1}) \bar{R}_{t+1}, \quad \bar{R}_{t+1} = \frac{1 + \tau_{ct}}{1 + \tau_{c,t+1}} [(1 - \tau_{k,t+1})(f'(k_{t+1}) - \delta) + 1]. \quad (\text{E})$$

With constant policy the steady state solves $f'(\bar{k}) = \delta + \rho/(1 - \tau_k)$ (11.6.7), and $\bar{R} = 1 + \rho$. Lump-sum taxes appear nowhere in these conditions.

Linear approximation. Let $x_t = k_t - \bar{k}$. Around a steady state with constant taxes, feasibility gives $\hat{c}_t = Px_t - x_{t+1}$, where $P = f'(\bar{k}) + 1 - \delta$ is the pre-tax gross return, and the log of (E) gives $\gamma(\hat{c}_{t+1} - \hat{c}_t)/\bar{c} = \beta(1 - \tau_k)f''(\bar{k})x_{t+1}$. Hence

$$x_{t+2} - (1 + P + A)x_{t+1} + Px_t = 0, \quad A \equiv -\beta\bar{c}(1 - \tau_k)f''(\bar{k})/\gamma > 0.$$

The characteristic polynomial is $P > 0$ at $\lambda = 0$ and $-A < 0$ at $\lambda = 1$, so it has one root $\lambda_2 \in (0, 1)$ and one root $\lambda_1 = P/\lambda_2 > 1$ (with $\tau_k = 0$, $P = 1/\beta$ and $\lambda_1\lambda_2 = 1/\beta$). λ_2 is the feedback coefficient of (11.10.8) and $1/\lambda_1$ the rate at which foreseen future policy is discounted. Since $A \propto 1/\gamma$, λ_2 rises from 0 as $\gamma \rightarrow 0$ to 1 as $\gamma \rightarrow \infty$.

Computation. For a policy that is constant from some date on, `ch11.py` stacks (E) for $t = 0, \dots, S-2$ ($S = 300$) and solves for $k_1, \dots, k_{S-1}$ by Newton's method (the Jacobian is tridiagonal), with k_0 given and the terminal condition $k_S - \bar{k} = \lambda_2(k_{S-1} - \bar{k})$ of the terminal steady state. This is the book's two-point boundary value problem (section 11.6.3), solved in a way that stays accurate over long horizons. Checks: an unchanged policy starting from its steady state returns the constant path exactly; $S = 600$ changes no digit; the book's shooting algorithm reproduces c_0 to 5×10^{-10} , and the linear approximation (11.10.8) is within 0.015 of the nonlinear path (both in exercise 11.17). Where an exercise leaves parameters open we use those of section 11.9: $\alpha = .33$, $\delta = .2$, $\beta = .95$ ($1 + \rho = 1.0526$), $\gamma = 2$, $z = 1$, $g = .2$. Then $\bar{k} = 1.4900$, $f(\bar{k}) = 1.1406$, $\bar{c} = 0.6426$, and $\lambda_2 = 0.8524$.

Exercise 11.1 Tax reform: I

Equation (1) refers to a numbered display of the exercise in the book.

(a) A competitive equilibrium is a budget-feasible policy $\{g_t, \tau_{ct}, \tau_{ht}\}$ (one that satisfies (1)), a feasible allocation $\{c_t, k_{t+1}\}$, and a price system $\{q_t, \eta_t, w_t\}$ such that, given the prices and the policy, the allocation solves the household's problem (k_0 given) and $\{k_t, n_t = 1\}$ solves the firm's problem. As shown in the introduction, it is characterized by feasibility, the terminal condition and (E) with $\tau_{kt} = 0$: $u'(c_t)/(1 + \tau_{ct}) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta]/(1 + \tau_{c,t+1})$.

(b) With $\tau_{ct} = 0$, (E) is the planner's Euler equation. In a steady state $c_{t+1} = c_t$, so $\bar{k}_0$ solves $f'(\bar{k}_0) = \rho + \delta$ (the augmented golden rule), and $\bar{c} = f(\bar{k}_0) - \delta\bar{k}_0 - g$. The capital-labor ratio does not depend on g .

(c) Let an allocation, prices and a policy $\{g_t, 0, \tau_{ht}\}$ be an equilibrium, and let $\{\tilde{\tau}_{ht}\}$ be another sequence with $\sum_t q_t \tilde{\tau}_{ht} = \sum_t q_t \tau_{ht}$. Lump-sum taxes enter the household's budget only through $\sum_t q_t \tau_{ht}$, so its budget set and its optimal choice are unchanged. They do not enter the firm's problem or feasibility, and (1) still holds. So the same allocation and prices are an equilibrium under $\{\tilde{\tau}_{ht}\}$. Equivalently, the equilibrium conditions (E), feasibility and the terminal condition do not involve τ_h ; only the present value $\sum_t q_t \tau_{ht} = \sum_t q_t g_t$ matters, and (1) pins it down. (This is Ricardian equivalence: government borrowing and lending absorb any retiming.)

(d) A constant τ_c cancels from (E), so the equilibrium conditions are those of part (b): the allocation is the planning allocation with g deducted. From $k_0 = \bar{k}_0$ it is $k_t = \bar{k}_0$ and $c_t = \bar{c}$ for all t . The new steady state is the old one, and there is no transition. With constant consumption $q_t = \beta^t q_0$, and (1) becomes $\tau_c \bar{c} \sum_t \beta^t = g \sum_t \beta^t$:

$$\tau_c = g/\bar{c}.$$

With inelastic labor, a constant consumption tax taxes the household's initial wealth, since its budget reads

$$(1 + \tau_c) \sum_t q_t c_t = (\eta_0 + 1 - \delta) q_0 k_0 + \sum_t q_t w_t;$$

so it is equivalent to a lump-sum tax. Pre-tax prices are unchanged. With the chapter's parameters, $\bar{k}_0 = 1.4900$, $\bar{c} = 0.6426$ and $\tau_c = 0.3112$.

(e) The tax is constant from $t = 10$ on, so the steady state again has $f'(k) = \rho + \delta$: the same $\bar{k}_0$ and $\bar{c}$. But (E) at $t = 9$ reads

$$u'(c_9) = \beta u'(c_{10}) \frac{f'(k_{10}) + 1 - \delta}{1 + \tau_c}.$$

A foreseen increase of the consumption tax is a tax on saving between $t = 9$ and $t = 10$, at rate $\tau_c/(1 + \tau_c)$ on the gross return; at all other dates (E) is undistorted. The transition is that of Figure 11.9.4. Consumption jumps up at $t = 0$, and households run down capital to consume before consumption becomes expensive. The falling capital stock raises $\bar{R}_{t+1} = f'(k_{t+1}) + 1 - \delta$ above $1 + \rho$, so consumption keeps rising until $t = 9$. At $t = 10$, $\bar{R}_{10}$ collapses and consumption falls sharply. Afterwards the economy climbs back to $\bar{k}_0$ along the undistorted saddle path, with $\bar{R}_t > 1 + \rho$ and rising consumption.

The tax rate now solves a fixed point, because the path depends on it:

$$\tau_c \sum_{t \geq 10} q_t(\tau_c) c_t(\tau_c) = g \sum_{t \geq 0} q_t(\tau_c).$$

We solve it by an outer root-finder on τ_c around the Newton solution for the path. The result is $\tau_c = 0.6239$, twice the rate of policy (d): nothing is collected for ten years, so the government borrows and must service the debt forever, and the tax base after $t = 10$ is smaller. Capital falls from 1.490 to 1.127 at $t = 10$. Consumption jumps from 0.6426 to 0.6584 at $t = 0$, rises to 0.7024 at $t = 9$, drops to 0.5647 at $t = 10$ ($\bar{R}_{10} = 0.680$), and recovers. Government debt, from $B_t = R_{t-1,t}B_{t-1} + g_t - T_t$ (11.4.2) with $T_t = \tau_{ct}c_t$, reaches $B_9 = 2.80$ and converges to 3.818, where the primary surplus $\tau_c \bar{c} - g$ equals the interest ρB (a check on the computation).

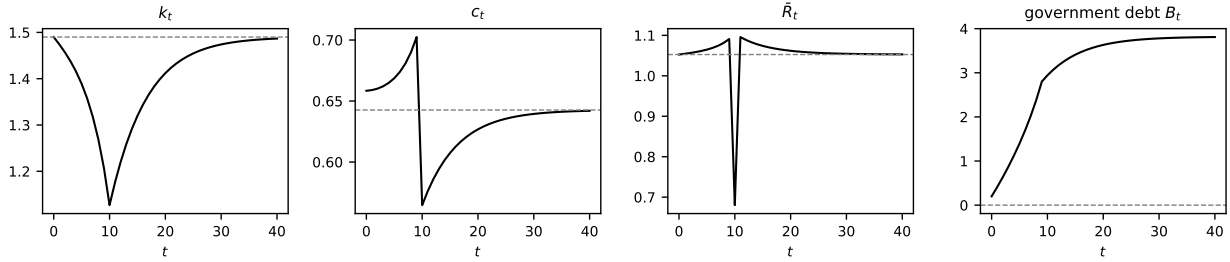


Figure S11.1. Policy (e): the consumption tax rises from 0 to 0.624 at $t = 10$. Dashed lines: the path under policy (d), which is the initial steady state.

(f) Policy (d). Its allocation satisfies the Euler equation, feasibility and the transversality condition of the planning problem $\max \sum_t \beta^t u(c_t)$ subject to $c_t + g + k_{t+1} \leq f(k_t) + (1 - \delta)k_t$, $k_0 = \bar{k}_0$. That problem is strictly concave, so this is its unique solution. The allocation under (e) is feasible but violates the planner's Euler equation at $t = 9$, so it yields strictly lower utility. Postponing the tax does not ease the pain; it creates an intertemporal distortion: too much consumption before $t = 10$, too little afterwards, and a capital stock that is run down and rebuilt. The welfare cost equals a permanent 0.435% cut in consumption.

Exercise 11.2 Tax reform: II

(a) As in exercise 11.1(a), with a budget-feasible policy $\{g_t, \tau_{ct}, \tau_{kt}\}$. The equilibrium is characterized by feasibility, the terminal condition and (E) with both taxes.

(b) In a steady state (E) gives $1 = \beta[(1 - \tau_k)(f'(\bar{k}_0) - \delta) + 1]$, so

$$f'(\bar{k}_0) = \delta + \frac{\rho}{1 - \tau_k}, \quad \bar{c}_0 = f(\bar{k}_0) - \delta \bar{k}_0 - g.$$

The pre-tax net return on capital is $f'(\bar{k}_0) - \delta = \rho/(1 - \tau_k)$; the after-tax return is ρ ($\bar{R} = 1 + \rho$). In a steady state $q_t \propto \beta^t$, and the government budget reduces to period-by-period balance, which determines τ_k :

$$g = \tau_k (f'(\bar{k}_0) - \delta) \bar{k}_0 = \frac{\tau_k \rho}{1 - \tau_k} \bar{k}_0(\tau_k).$$

The right side is a Laffer curve. It is zero at $\tau_k = 0$ and, for $f = zk^\alpha$, tends to zero as $\tau_k \rightarrow 1$, because then $\bar{k}_0 \propto (1 - \tau_k)^{1/(1-\alpha)}$. A g below the peak has two solutions, and the equilibrium tax rate is the smaller one; the larger is on the wrong side of the Laffer curve.

Remark. With the chapter's calibration the peak revenue is 0.146 (at $\tau_k = 0.907$), so a capital tax cannot finance $g = .2$: its base, capital income net of depreciation, is only $\rho\bar{k}_0/(1 - \tau_k)$. We use $g = .05$. Then $\tau_k = 0.4454$ (the other root is 0.9964), $\bar{k}_0 = 1.1827$, $\bar{c}_0 = 0.7704$, and the pre-tax net return is 0.0949.

(c) A constant consumption tax cancels from (E), so the new equilibrium is the planning allocation from $\bar{k}_0$, with g deducted. Capital rises monotonically to $\bar{k}$ with $f'(\bar{k}) = \rho + \delta$, which exceeds $\bar{k}_0$ because f' is decreasing. Steady-state consumption $\bar{c} = f(\bar{k}) - \delta\bar{k} - g$ rises too: $f(k) - \delta k$ increases below the golden rule ($f'(k) > \delta$), and $\bar{k}_0 < \bar{k}$ both lie below it. The pre-tax return on capital falls from $\rho/(1 - \tau_k)$ to ρ . The after-tax return is $1 + \rho$ in both steady states; during the transition $\bar{R}_{t+1} = f'(k_{t+1}) + 1 - \delta$ exceeds $1 + \rho$ and falls. In our example $\bar{k} = 1.4900$, $\bar{c} = 0.7926$, and the consumption tax is $\tau_c = \sum_t q_t g / \sum_t q_t c_t = 0.0646$. Consumption drops at $t = 0$ to 0.7220, below $\bar{c}_0$, because investment rises, and it stays below $\bar{c}_0$ for seven periods; $\bar{R}_1 = 1.0871$.

(d) No. The comparison ignores the transition: the larger capital stock of the new steady state must be accumulated, which requires lower consumption first. The right measure evaluates $\sum_t \beta^t u(c_t)$ along the equilibrium path from the common initial condition $\bar{k}_0$, against $u(\bar{c}_0)/(1 - \beta)$ for keeping the capital tax. In our example, the steady-state comparison implies a gain equal to a permanent 2.89% increase in consumption, while the true gain is 0.60%. The sign is right here, because the new allocation solves the planning problem from $\bar{k}_0$, for which remaining at $\bar{k}_0$ is feasible. In general, steady-state comparisons can even get the sign wrong: steady-state consumption is largest at the golden rule (exercise 11.5), yet moving there from $\bar{k}$ lowers welfare.

Exercise 11.3 *Anticipated productivity shift*

(a) Choose $\{c_t, x_t, k_{t+1}\}$ to maximize $\sum_t \beta^t u(c_t)$ subject to the two constraints, with k_0 given. The Lagrangian is

$$L = \sum_t \beta^t \{u(c_t) + \mu_t [f(k_t) - c_t - x_t] + \lambda_t [(1 - \delta)k_t + \psi_t x_t - k_{t+1}]\}.$$

The first-order conditions are $u'(c_t) = \mu_t$, $\mu_t = \psi_t \lambda_t$ and $\lambda_t = \beta[\mu_{t+1} f'(k_{t+1}) + (1 - \delta)\lambda_{t+1}]$, plus $\lim_t \beta^t \lambda_t k_{t+1} = 0$. So $\lambda_t = u'(c_t)/\psi_t$: installed capital is worth $1/\psi_t$ units of consumption, and

$$\frac{u'(c_t)}{\psi_t} = \beta u'(c_{t+1}) \left[f'(k_{t+1}) + \frac{1 - \delta}{\psi_{t+1}} \right]. \quad (1)$$

With constant ψ the steady state solves $f'(\bar{k}) = (\rho + \delta)/\psi$, with $\bar{x} = \delta\bar{k}/\psi$ and $\bar{c} = f(\bar{k}) - \delta\bar{k}/\psi$. A permanent increase in ψ lowers $f'(\bar{k})$, so $\bar{k}$ rises: $d \log \bar{k} / d \log \psi = 1/\varepsilon$, where $\varepsilon = -\bar{k} f''(\bar{k}) / f'(\bar{k})$. Consumption rises: $d\bar{c}/d\psi = [f'(\bar{k}) - \delta/\psi] d\bar{k}/d\psi + \delta\bar{k}/\psi^2 = (\rho/\psi) d\bar{k}/d\psi + \delta\bar{k}/\psi^2 > 0$. Investment in efficiency units, $\psi\bar{x} = \delta\bar{k}$, rises, and $d \log \bar{x} / d \log \psi = 1/\varepsilon - 1$, so investment in goods rises iff $\varepsilon < 1$. For Cobb–Douglas $\varepsilon = 1 - \alpha$, so $\bar{x} \propto \psi^{\alpha/(1-\alpha)}$ rises. With the chapter's parameters and no government: $\psi = 1$ gives $(\bar{k}, \bar{c}, \bar{x}) = (1.490, 0.843, 0.298)$, and $\psi = 2$ gives $(4.192, 1.186, 0.419)$.

(b) Because ψ_t depends on calendar time, the state must record the date relative to the switch. Let $s_t = \min\{t, 4\}$, with $s_{t+1} = \min\{s_t + 1, 4\}$, $\psi(s) = 1$ for $s \leq 3$ and $\psi(4) = 2$ (we take $\psi_0 = 1$; the exercise defines ψ_t only for $t \geq 1$). The state is (k_t, s_t) , the control k_{t+1} , and

$$V(k, s) = \max_{k'} \left\{ u \left(f(k) - \frac{k' - (1 - \delta)k}{\psi(s)} \right) + \beta V(k', \min\{s + 1, 4\}) \right\}.$$

For $s = 4$ this is the stationary Bellman equation of the economy with $\psi = 2$ forever; $V(\cdot, 3), \dots, V(\cdot, 0)$ follow by four backward steps. The state evolves as $k_{t+1} = h_{s_t}(k_t)$ and $s_{t+1} = \min\{s_t + 1, 4\}$.

(c) A price system is $\{q_t, \eta_t, w_t\}$: q_t is the time-0 price of a time- t good (consumption or investment), η_t the rental rate of capital and w_t the wage, both in time- t goods. The household maximizes $\sum_t \beta^t u(c_t)$ subject to $\sum_t q_t(c_t + x_t) \leq \sum_t q_t(\eta_t k_t + w_t)$ and $k_{t+1} = (1 - \delta)k_t + \psi_t x_t$, k_0 given. Substituting $x_t = [k_{t+1} - (1 - \delta)k_t]/\psi_t$, its first-order conditions are

$$\beta^t u'(c_t) = \mu q_t, \quad \frac{q_t}{\psi_t} = q_{t+1} \left[\eta_{t+1} + \frac{1 - \delta}{\psi_{t+1}} \right],$$

together with $\lim_t q_t k_{t+1}/\psi_t = 0$. The second condition is a no-arbitrage condition: a unit of capital costs q_t/ψ_t at t and returns the rental plus undepreciated capital worth $1/\psi_{t+1}$ goods at $t + 1$. The firm maximizes $\sum_t q_t [f(k_t)n_t - \eta_t k_t n_t - w_t n_t]$, so $\eta_t = f'(k_t)$ and $w_t = f(k_t) - k_t f'(k_t)$.

(d) Substituting the firm's conditions into the household's gives exactly (1). Feasibility holds because $c_t + x_t = \eta_t k_t + w_t = f(k_t)$, and the transversality conditions coincide. So the equilibrium allocation is the planning allocation, and the planning allocation with $q_t = \beta^t u'(c_t)/u'(c_0)$, $\eta_t = f'(k_t)$ and $w_t = f(k_t) - k_t f'(k_t)$ is an equilibrium. The price q_t is the (normalized) multiplier $\beta^t \mu_t/\mu_0$ on the time- t resource constraint; η_t and w_t are marginal products; and the price of installed capital, $1/\psi_t$ goods, is the ratio λ_t/μ_t of the multipliers.

(e) Yes, the jump has effects that precede it. From $t = 4$ on, a unit of goods buys two units of capital. This makes the economy richer, so consumption rises at $t = 0$ already. It also imposes a capital loss: capital carried from $t = 3$ into $t = 4$ is worth only $1/\psi_4 = 1/2$ good at $t = 4$, so the gross return between $t = 3$ and 4 collapses to $f'(k_4) + (1 - \delta)/2$. Anticipating this, the household postpones investment until it becomes cheap: investment falls below its old steady-state level from $t = 0$ on, and capital declines until $t = 4$. By (1), consumption must then fall between $t = 3$ and 4, when the investment boom starts; afterwards the return is high and consumption rises toward the new steady state. The computed path is:

t	0	1	2	3	4	5	6	∞
k_t	1.490	1.432	1.367	1.292	1.201	1.592	1.955	4.192
c_t	0.901	0.904	0.910	0.921	0.747	0.825	0.888	1.186
x_t	0.240	0.222	0.198	0.167	0.315	0.341	0.360	0.419
$R_{t,t+1}$	1.059	1.068	1.078	0.692	1.283	1.221	1.180	1.053

Here $R_{t,t+1} = \psi_t[f'(k_{t+1}) + (1 - \delta)/\psi_{t+1}]$ is the gross real interest rate. Consumption jumps from 0.843 to 0.901 at $t = 0$ and investment falls from 0.298 to 0.240. Investment stays positive throughout (its minimum is 0.167 at $t = 3$), so requiring $x_t \geq 0$ would change nothing (Figure S11.2).

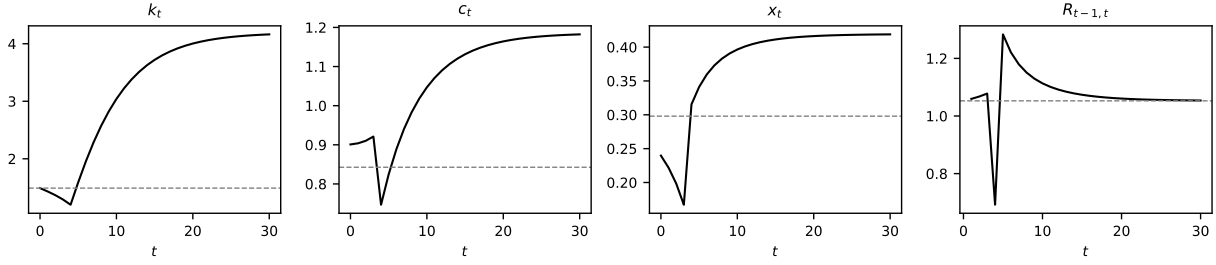


Figure S11.2. Response to the investment-technology shift at $t = 4$, foreseen at $t = 0$. Dashed lines: the initial steady state.

Exercise 11.4 A capital levy

(a) The levy is a bill of φk_T time- T goods. (If it is a fraction of the value $(\eta_T + 1 - \delta)k_T$ instead, rescale φ .) The household maximizes $\sum_t \beta^t u(c_t)$ subject to

$$\sum_t q_t [c_t + k_{t+1} - (1 - \delta)k_t] \leq \sum_t q_t (\eta_t k_t + w_t) - q_T \varphi k_T,$$

k_0 given; the government's budget is $\sum_t q_t g_t = q_T \varphi k_T$. A competitive equilibrium is a budget-feasible policy $(\{g_t\}, \varphi, T)$, a feasible allocation and prices such that household and firm optimize. It is characterized by $\eta_t = f'(k_t)$, $w_t = f(k_t) - k_t f'(k_t)$, $\beta^t u'(c_t) = \mu q_t$, the no-arbitrage conditions $q_{t-1} = q_t [\eta_t + 1 - \delta - \varphi \mathbf{1}\{t = T\}]$ for $t \geq 1$, feasibility and the terminal condition. Hence

$$u'(c_t) = \beta u'(c_{t+1}) [f'(k_{t+1}) + 1 - \delta - \varphi \mathbf{1}\{t + 1 = T\}].$$

A levy at $T = 0$ falls on the predetermined k_0 , so it is a lump-sum tax. A levy at $T \geq 1$ taxes the gross return on capital carried from $T - 1$ into T .

(b) With $g \equiv 0$ and no taxes, $f'(k_0) = \rho + \delta$, so $k_0 = [\alpha A / (\rho + \delta)]^{1/(1-\alpha)}$.

(c) The levy is lump sum, so the allocation is the planning allocation with g deducted, and the steady state is again $f'(k) = \rho + \delta$: the new steady-state capital stock is k_0 . Since the economy starts there, $k_t = k_0$, $c_t = f(k_0) - \delta k_0 - g$ (lower by exactly g), and $R_{t,t+1} = f'(k_0) + 1 - \delta = 1 + \rho$ for all t . Only consumption differs from the $g \equiv 0$ equilibrium. To compute φ : (i) compute the equilibrium allocation, which does not depend on φ ; (ii) compute $q_t = \beta^t u'(c_t) / u'(c_0)$; (iii) set $\varphi = \sum_t q_t g_t / (q_0 k_0)$. Here $q_t = \beta^t$, so

$$\varphi = \frac{g}{(1 - \beta)k_0} = \frac{(1 + \rho)g}{\rho k_0}.$$

It exceeds one when $g > (1 - \beta)k_0$, and then the household pays part of the levy out of the present value of its wages. Any $g < f(k_0) - \delta k_0$ can be financed. With $A = 1$ and the chapter's other parameters, $k_0 = 1.490$; for $g = .02$, $\varphi = 0.268$ and consumption falls from 0.843 to 0.823.

(d) After $t = 10$ there are no taxes, so the steady state is again k_0 . But the levy now distorts the decision to carry capital into $t = 10$:

$$u'(c_9) = \beta u'(c_{10}) [f'(k_{10}) + 1 - \delta - \varphi].$$

Algorithm. For a trial φ , compute the equilibrium path from k_0 to the steady state, with this distorted Euler equation at $t = 9$, by shooting or Newton's method. Compute q_t and the budget gap $\Gamma(\varphi) = q_{10}\varphi k_{10} - \sum_t q_t g$, and solve $\Gamma(\varphi) = 0$ by bisection or the secant method, taking the smallest root.

A pitfall: the Laffer curve. The household can shrink the tax base k_{10} , so the revenue of a foreseen levy is bounded. For each φ we compute (by continuation in φ) the g that the levy finances. It rises to a maximum of $g = 0.0354$ at $\varphi = 1.58$, where $k_{10} = 0.216$ and the gross return from $t = 9$ to 10 net of the levy is 0.14, and then falls slowly. So a surprise levy at $T = 0$ can finance any feasible g (up to 0.843), while a levy foreseen for $T = 10$ can finance at most about 3% of output. We take $g = .02$.

Paths (Figure S11.3). The levy rate is $\varphi = 0.494$, versus 0.268 at $T = 0$. Anticipating the levy, the household consumes more and invests less before $t = 10$: capital falls from 1.490 to 1.012 at $t = 10$. Consumption starts at 0.840 (above the 0.823 of part (c), slightly below the 0.843 of the $g \equiv 0$ equilibrium) and rises to 0.911 at $t = 9$, because the interest rate $R_{t,t+1} = f'(k_{t+1}) + 1 - \delta$ exceeds $1 + \rho$ as capital falls. Between $t = 9$ and 10 the return net of the levy is 0.633, which forces consumption down to 0.707 at $t = 10$. Then capital is rebuilt ($R_{10,11} = 1.112$) and consumption rises to 0.823, the level of part (c). The long run is the same as in (c); the transition is distorted, at a welfare cost equal to a permanent 0.53% of consumption.

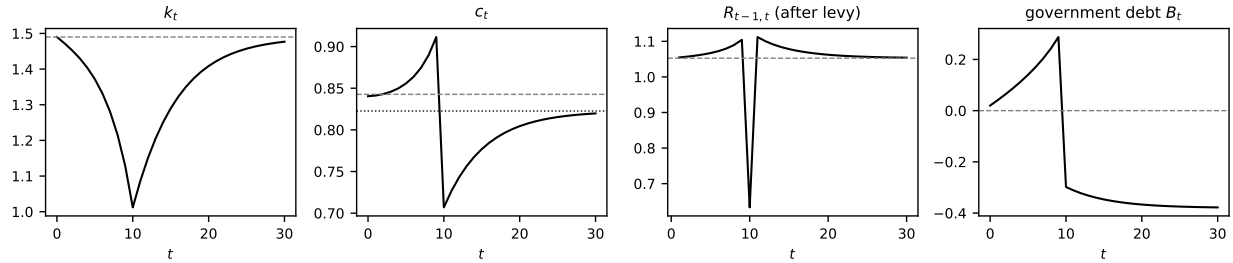


Figure S11.3. Levy at $T = 10$ foreseen at $t = 0$ ($g = .02$). Dashed lines: the $g \equiv 0$ steady state; dotted line: consumption with the levy at $T = 0$.

(e) Without uncertainty, one-period Arrow securities are one-period bonds. At each t the household chooses c_t , k_{t+1} and holdings b_{t+1} of bonds that pay one good at $t + 1$ and cost $1/R_{t,t+1}$, subject to

$$c_t + k_{t+1} + b_{t+1}/R_{t,t+1} \leq (\eta_t + 1 - \delta)k_t + w_t + b_t - \varphi k_t \mathbf{1}\{t = T\}, \quad b_0 = 0,$$

and a natural borrowing limit. The government satisfies $g_t + b_t^g = b_{t+1}^g/R_{t,t+1} + \varphi k_t \mathbf{1}\{t = T\}$ with $b_0^g = 0$, where b_t^g is its debt maturing at t . An equilibrium is a policy, an allocation, bond holdings and prices $\{R_{t,t+1}, \eta_t, w_t\}$ such that household and firm optimize and the goods and bond markets clear ($b_t = b_t^g$). To compute it, note that the allocation and $R_{t,t+1} = q_t/q_{t+1}$ are those of the time-0 equilibrium, because one-period bonds complete the markets here; the debt follows from the government's constraint. In the book's units, the value of debt issued at t satisfies $B_t = R_{t-1,t}B_{t-1} + g_t - T_t$ (11.4.2), where T_t is the levy.

Under (d) the government borrows from the household to pay for g until $t = 9$: $B_0 = 0.020$ and $B_9 = 0.288$. By no arbitrage, bonds earn the same return as capital from $t = 9$ to 10, $R_{9,10} = 0.633$,

so bondholders share in the levy. At $t = 10$ the levy (0.500) pays off the debt, and the government becomes a creditor: $B_{10} = -0.298$. Its claims converge to $-g/\rho = -0.380$, whose interest pays for g forever. The household mirrors this: it holds government bonds before $t = 10$, and afterwards it is permanently in debt to the government, paying interest of about g per period.

Exercise 11.5 *A permanent increase in government purchases*

- (a) $f'(k) = \rho + \delta$, so $k = [\alpha z / (\rho + \delta)]^{1/(1-\alpha)}$.
- (b) The golden rule solves $f'(\tilde{k}) = \delta$, so $\tilde{k} = (\alpha z / \delta)^{1/(1-\alpha)} > k$, because $\rho > 0$ and f' is decreasing. Since $\tilde{k}$ maximizes $f(k) - \delta k$ and $k \neq \tilde{k}$, $\tilde{c} > c$. Raising capital from k to $\tilde{k}$ would increase steady-state consumption, but the extra capital must be accumulated by consuming less first, and the household discounts the later gains. At k , the marginal gain in steady-state consumption, $f'(k) - \delta = \rho$, exactly compensates for impatience (the modified golden rule). With the chapter's parameters, $k = 1.490$ and $c = 0.8426$, while $\tilde{k} = 2.112$ and $\tilde{c} = 0.8574$.
- (c) $g = c/2$ is feasible. We claim $k_t = k$, $c_t = c - g = c/2$ and $\bar{R}_{t+1} = 1 + \rho$ for all t . This path satisfies feasibility ($c/2 + g + k = f(k) + (1 - \delta)k$), the Euler equation (c_t constant and $f'(k) + 1 - \delta = 1/\beta$), the initial condition and the transversality condition $\lim_t \beta^t k / c_t = 0$. With lump-sum taxes the equilibrium is the planning allocation with g deducted (section 11.8); that problem is strictly concave, so these conditions are sufficient and the path is the unique equilibrium. (`ch11.py` confirms that the Newton solver returns this constant path.) The reason: g does not enter the steady-state condition $f'(k) = \rho + \delta$, and a permanent, unforeseen increase in g is a permanent loss of resources for private use. A consumption-smoothing household absorbs it entirely by consuming less, at once and forever, as in the permanent income theory. There is nothing to gain from changing the capital stock, so the interest rate does not move.

Exercise 11.6 *Trade and growth*

- (a) $f'(\bar{k}) = \rho + \delta$, so $\bar{k} = [\alpha z / (\rho + \delta)]^{1/(1-\alpha)}$.
- (b) The value function V is strictly concave, and the optimal policy $k' = h(k)$ is strictly increasing, because the objective $u(f(k) + (1 - \delta)k - k') + \beta V(k')$ has strictly increasing differences in (k, k') . Consumption $c(k) = f(k) + (1 - \delta)k - h(k)$ is also increasing: the first-order condition $u'(c(k)) = \beta V'(h(k))$ has a right side that falls with k . The positive fixed points of h satisfy the steady-state Euler equation, so $\bar{k}$ is the only one. If $h(k) \leq k$ for some $k < \bar{k}$, then (as h is increasing and continuous) k_t would decrease monotonically to a fixed point below $\bar{k}$, which would have to be 0. That is impossible: as $k_t \rightarrow 0$, $f'(k_{t+1}) \rightarrow \infty$, and the Euler equation would make consumption grow without bound while resources shrink to zero. So $h(k) > k$ on $(0, \bar{k})$. Therefore, from $k_0 < \bar{k}$, k_t rises monotonically to $\bar{k}$, c_t rises monotonically to $\bar{c} = f(\bar{k}) - \delta \bar{k}$, and $\bar{R}_{t+1} = 1 - \delta + f'(k_{t+1})$ falls monotonically. The Euler equation $(c_{t+1}/c_t)^\gamma = \beta \bar{R}_{t+1} > 1$ confirms that consumption grows while the return exceeds $1 + \rho$. With $k_0 = .5\bar{k}$ and $\gamma = 2$ (our choice), c rises from 0.648 to 0.843 and $\bar{R}_1 = 1.166$.
- (c) $\bar{R} = 1 + \rho = 1/\beta$.
- (d) Above, since f' is decreasing.

Part II. Attach multipliers $\beta^t \lambda_t$ to the budget constraints for $t \geq 0$, and substitute $k_0 = \bar{k}_0 + B_{-1}$. The first-order conditions are

$$u'(c_t) = \lambda_t, \quad \lambda_t = R\beta\lambda_{t+1} = \lambda_{t+1} (B_t), \quad \lambda_t = \beta\lambda_{t+1}[f'(k_{t+1}) + 1 - \delta] (k_{t+1}),$$

and $f'(k_0) + 1 - \delta = R$ for B_{-1} . Hence consumption is constant, and $f'(k_t) = \rho + \delta$ for all $t \geq 0$: the planner borrows $B_{-1} = \bar{k} - \bar{k}_0$ and jumps to $k_t = \bar{k}$ at once. The present-value budget (with $\lim_t R^{-t} B_t = 0$) is $c/(1 - \beta) = [f(\bar{k}) - \delta\bar{k}]/(1 - \beta) - R(\bar{k} - \bar{k}_0)$, so

$$c = f(\bar{k}) - \delta\bar{k} - \rho(\bar{k} - \bar{k}_0) = \bar{w} + \rho\bar{k}_0, \quad \bar{w} \equiv f(\bar{k}) - \bar{k}f'(\bar{k}).$$

The debt is rolled over forever, $B_t = \bar{k} - \bar{k}_0$, with interest ρB_t . The open economy reaches the steady-state capital stock at once, and its consumption is constant: higher than the closed economy's consumption early on, but permanently below the closed economy's eventual $\bar{c} = \bar{w} + \rho\bar{k}$, the difference being interest on the foreign debt.

Welfare is higher in the open economy. Any closed-economy plan is feasible in the open economy (set $B_t = 0$), and the closed-economy path is not optimal there, because its consumption is not constant. With $\bar{k}_0 = .5\bar{k}$: $B_{-1} = 0.745$ and $c = 0.8034$. The closed economy's path is worth as much as a constant consumption of 0.7924, so opening up is worth 1.39% of consumption every period (Figure S11.4). Comparing steady-state consumption ($0.803 < 0.843$) would wrongly suggest that opening hurts (compare section 11.13.5.1).

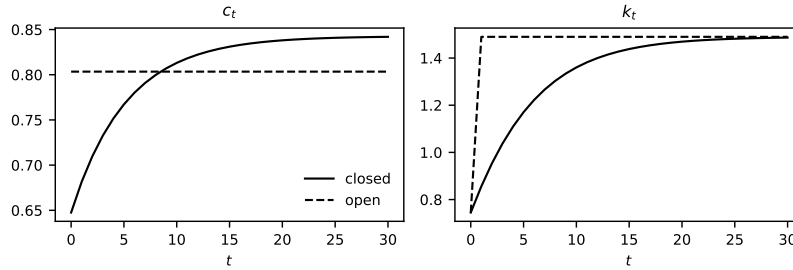


Figure S11.4. Closed and open economy from $\bar{k}_0 = .5\bar{k}$.

Exercise 11.7 A capital tax, unforeseen and foreseen

(a) $f'(\bar{k}) = \rho + \delta$, $\bar{k} = [\alpha z / (\rho + \delta)]^{1/(1-\alpha)}$, $\bar{c} = f(\bar{k}) - \delta\bar{k} - \bar{g}$ and $\bar{R} = 1 + \rho$. With the chapter's parameters: $\bar{k} = 1.490$, $\bar{c} = 0.6426$ and $\bar{R} = 1.0526$.

(b) The new steady state has $f'(\bar{k}') = \delta + \rho/(1 - \tau_k) > \rho + \delta$, so $\bar{k}' < \bar{k}$; $\bar{c}' = f(\bar{k}') - \delta\bar{k}' - \bar{g} < \bar{c}$ (both lie below the golden rule); and the after-tax return is again $\bar{R} = 1 + \rho$. The change is unforeseen, so $k_0 = \bar{k}$ is inherited, and the economy moves along the new saddle path. Since capital must fall ($k_1 < k_0 = \bar{k}$), consumption jumps up at $t = 0$: $c_0 = f(\bar{k}) + (1 - \delta)\bar{k} - \bar{g} - k_1 > \bar{c}$. After that, k falls monotonically to $\bar{k}'$. The return $\bar{R}_{t+1} = 1 + (1 - \tau_k)(f'(k_{t+1}) - \delta)$ drops below $1 + \rho$ at $t = 1$ and then rises back to $1 + \rho$ as capital falls. Since $\bar{R} < 1 + \rho$, consumption declines monotonically to $\bar{c}' < \bar{c}$. To a linear approximation $k_{t+1} - \bar{k}' = \lambda_2(k_t - \bar{k}')$, and `ch11.py` confirms that all three paths are monotone. With $\tau_k = .2$ (our choice) and $\gamma = 2$: $\bar{k}' = 1.381$, $\bar{c}' = 0.6362$, $c_0 = 0.6575$ and $\bar{R}_1 = 1.0435$ (Figure S11.5, top row).

(c) Steady states do not depend on γ ; the speed of adjustment does. Near the new steady state, $c_t - \bar{c}' \approx P(k_t - \bar{k}') - (k_{t+1} - \bar{k}') = (P - \lambda_2)(k_t - \bar{k}')$, with $P = f'(\bar{k}') + 1 - \delta$. So the initial jump is $c_0 - \bar{c}' \approx (P - \lambda_2)(\bar{k} - \bar{k}')$, which is smaller the larger is λ_2 , i.e. the larger is γ . A household with low γ is willing to let consumption vary: it eats the excess capital quickly, with a big consumption spike, fast convergence and a quick return of $\bar{R}$ to $1 + \rho$. A household with high γ spreads the adjustment over many periods. With $\gamma = .2$, $\lambda_2 = 0.599$, the jump is $+0.044$, and by $t = 10$ capital has essentially converged (1.382). With $\gamma = 2$, $\lambda_2 = 0.865$, the jump is $+0.015$, and $k_{10} = 1.406$. The formula $(P - \lambda_2)(\bar{k} - \bar{k}')$ gives 0.051 and 0.022, against exact values 0.051 and 0.021.

(d) i) As in (b): $\bar{k}' = 1.381$, $\bar{c}' = 0.636$, $\bar{R} = 1 + \rho$.

ii) Capital carried into periods $t \geq 10$ will be taxed, so households begin to reduce their capital at once. Before $t = 10$ capital income is still untaxed, so the falling capital stock raises $\bar{R}_{t+1} = 1 - \delta + f'(k_{t+1})$ above $1 + \rho$, and by (E) consumption must rise. Thus consumption jumps up slightly at $t = 0$ and keeps rising until $t = 9$ ($\gamma = 2$: from 0.6426 to 0.6449, then to 0.6502; $\bar{R}_9 = 1.0571$). At $t = 10$ the tax cuts the after-tax return to $\bar{R}_{10} = 1.0466 < 1 + \rho$, and consumption starts to decline toward $\bar{c}'$. After $t = 10$ only transient dynamics remain: capital keeps falling to $\bar{k}'$ and $\bar{R}$ rises back to $1 + \rho$ (Figure S11.5, bottom row, as in the book's Figure 11.9.5).

iii) Two decay rates govern the path: the feedback coefficient λ_2 after $t = 10$, and the discount factor $1/\lambda_1 = \lambda_2/P$ applied to future policy before $t = 10$ (11.10.8). With $\gamma = 2$, $\lambda_2 = 0.865$ and $1/\lambda_1 = 0.812$: anticipation effects reach far back, consumption responds at $t = 0$, capital starts falling at once, and convergence after $t = 10$ is slow (the computed ratios $(k_{t+1} - \bar{k}')/(k_t - \bar{k}')$ approach 0.865). With $\gamma = .2$, $\lambda_2 = 0.599$ and $1/\lambda_1 = 0.562$: almost nothing happens until a few periods before $t = 10$ ($c_0 = 0.6428$), then there is a sharp consumption boom that peaks at 0.670 at $t = 9$, and convergence after $t = 10$ is fast. Because $\lambda_1 \lambda_2 = P$, a sluggish feedback response goes together with a long anticipation horizon: this is the logic of (11.10.16).

Remark. Formula (11.10.16) is misprinted. As printed it gives $\lambda_2 = 3.53$ at $\gamma = 2$, and $\lambda_2 \rightarrow \infty$ as $\gamma \rightarrow \infty$, which is impossible for a stable root. Solving the linearization above with $\tau_k = 0$, i.e. $\lambda^2 - (1 + R + a/\gamma)\lambda + R = 0$ with $R = 1/\beta$ and $a = -\beta \bar{c} f''(\bar{k})$, gives

$$\lambda_2 = \frac{\gamma}{a_1 \gamma + a_2 + (a_3 \gamma + a_4 \gamma^2 + a_5)^{1/2}},$$

with exactly the book's constants: $a_1 = \frac{1+R}{2R} = .975$, $a_2 = \frac{a}{2R} = .0329$, $a_3 = \frac{2(1+R)a}{4R^2} = .0642$, $a_4 = \frac{(R-1)^2}{4R^2} = .000625$ and $a_5 = \frac{a^2}{4R^2} = .0011$. With these rounded constants it matches the exact root to four decimals (e.g. 0.8524 at $\gamma = 2$, 0.5779 at $\gamma = .2$) and reproduces Figure 11.10.1.

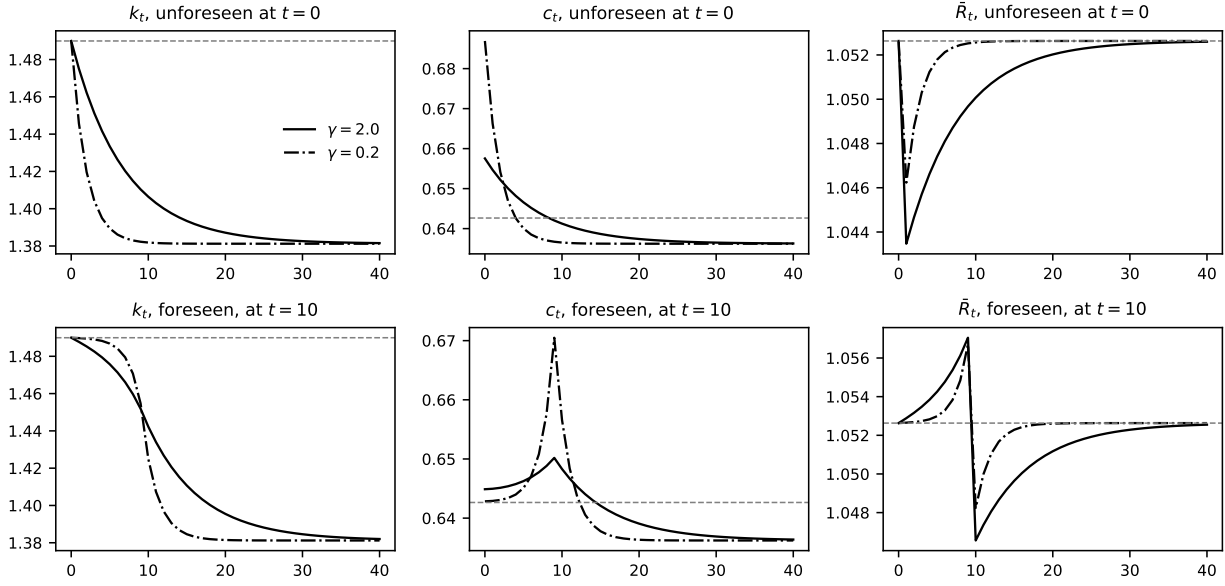


Figure S11.5. Capital tax $\tau_k = .2$, unforeseen at $t = 0$ (top) or foreseen for $t = 10$ (bottom); solid $\gamma = 2$, dash-dotted $\gamma = .2$. Dashed lines: the initial steady state.

Exercise 11.8 Trade and growth, version II

Remark. The technology is printed as “ $c_t + k_t + 1 = f(k_t) + (1 - \delta)k_t$ ” (read k_{t+1}), and the “where” after it is followed by nothing; we take u and f from exercise 11.6. Also, if $\bar{k}_0$ is sold at $t = -1$ for bonds that pay the gross rate R , then $A_0 = R\bar{k}_0$; that is what “consume the interest payments” means. The printed $A_0 = \bar{k}_0$ omits one period’s interest. We use $A_0 = R\bar{k}_0$; with $A_0 = \bar{k}_0$, replace $\rho\bar{k}_0$ below by $(1 - \beta)\bar{k}_0 = \rho\bar{k}_0/R$.

(a)–(d) These mirror exercise 11.6. $\bar{k} = [\alpha z / (\rho + \delta)]^{1/(1-\alpha)}$. From $\bar{k}_0 > \bar{k}$, the increasing policy h satisfies $h(k) < k$ on $(\bar{k}, \infty)$, so k_t falls monotonically to $\bar{k}$, c_t falls monotonically to $\bar{c}$, and $\bar{R}_{t+1} = 1 - \delta + f'(k_{t+1})$ rises monotonically to $1 + \rho$. So the steady-state return is $\bar{R} = 1 + \rho$, and $\bar{R}_{t+1} < 1 + \rho$ while $k_{t+1} > \bar{k}$. With $\bar{k}_0 = 2\bar{k}$ and $\gamma = 2$: $c_0 = 1.117$, $k_1 = 2.701$, $\bar{R}_1 = 0.970$, and c falls to 0.843.

(e) With multipliers $\beta^t \ell_t$ on the budget constraints: $u'(c_t) = \ell_t$, and the condition for A_{t+1} , $-\beta^t \ell_t \beta + \beta^{t+1} \ell_{t+1} = 0$, gives $\ell_t = \ell_{t+1}$. So consumption is constant. The budget in present value (no Ponzi games) is $\sum_t \beta^t c_t = A_0$, so $c_t = (1 - \beta)A_0 = \rho\bar{k}_0$ and $A_t = A_0$ for all t : the planner consumes the interest on its wealth.

(f) Under “sell all”, capital is zero, consumption is constant at $\rho\bar{k}_0$, and the return is $1 + \rho$. In the closed economy, capital and consumption fall toward $\bar{k}$ and $\bar{c}$, and the return rises toward $1 + \rho$ from below. The ranking depends on $\bar{k}_0$. The closed economy can keep capital at $\bar{k}_0$ and consume $f(\bar{k}_0) - \delta\bar{k}_0$ forever, and its optimal plan does strictly better than that. So it is strictly preferred whenever $f(\bar{k}_0) - \delta\bar{k}_0 \geq \rho\bar{k}_0$, i.e. $f(\bar{k}_0)/\bar{k}_0 \geq \rho + \delta = f'(\bar{k})$; for Cobb–Douglas, iff $\bar{k}_0 \leq \alpha^{-1/(1-\alpha)}\bar{k} = 5.23\bar{k}$. For very large $\bar{k}_0$ the ranking reverses. In a closed economy excess capital can only be eaten slowly, and meanwhile it earns $1 - \delta + f'(k)$, far less than the world rate $1 + \rho$. In the numerical example, “sell all” becomes better once $\bar{k}_0 > 14.6\bar{k}$. Both plans are dominated by the plan of part (g):

$\bar{k}_0/\bar{k}$	1.5	2	5	10	20	40
closed economy	-0.66%	-2.08%	-13.7%	-29.7%	-48.7%	-65.9%
sell all	-86.7%	-83.0%	-66.1%	-49.4%	-32.8%	-19.6%

The entries are welfare as a permanent percentage change of the consumption of plan (g) ($\gamma = 2$).

(g) i) This is exercise 11.6, Part II, with lending instead of borrowing ($B_{-1} < 0$). The planner sells $\bar{k}_0 - \bar{k}$ at $t = -1$, keeps $k_t = \bar{k}$ for all $t \geq 0$ (where $f'(\bar{k}) + 1 - \delta = R$), holds foreign assets $\bar{k}_0 - \bar{k}$ forever, and consumes

$$c = f(\bar{k}) - \delta\bar{k} + \rho(\bar{k}_0 - \bar{k}) = \bar{w} + \rho\bar{k}_0, \quad \bar{w} = f(\bar{k}) - \bar{k}f'(\bar{k}).$$

ii) Plan (g) beats “sell all” by exactly the wage $\bar{w} = 0.764$ per period: shutting down the technology wastes the country’s labor endowment. At the margin, starting from zero capital, a unit of capital installed at $t - 1$ returns $f'(k) + 1 - \delta$, which is infinite at $k = 0$, while it costs R in forgone bond income. So it pays to install capital until $f'(k) = \rho + \delta$. (The hint compares $z\epsilon^\alpha$ with $\rho\epsilon$ and ignores depreciation; including it, the comparison is $z\epsilon^\alpha$ with $(\rho + \delta)\epsilon$, and the conclusion is the same.)

Exercise 11.9 *Linear utility with a finite horizon*

Let $W = \sum_{t=0}^T R^{-t}y_t > 0$ and $d_t = R^{-t}c_t \geq 0$, the present value of consumption at t . The problem becomes

$$\max_{d \geq 0} \sum_{t=0}^T (\beta R)^t d_t \quad \text{subject to} \quad \sum_{t=0}^T d_t = W,$$

a linear program over a simplex. Its maximum puts all weight on the largest coefficient $(\beta R)^t$.

(a) $\beta R < 1$: the largest coefficient is at $t = 0$. So $c_0 = W$ and $c_t = 0$ for $t \geq 1$: the consumer borrows against all future income and consumes everything at once, because impatience beats the interest rate.

(b) $\beta R > 1$: the largest is at $t = T$. So $c_T = R^T W = \sum_t R^{T-t} y_t$ and $c_t = 0$ for $t < T$: the consumer saves everything and consumes at the end.

(c) $\beta R = 1$: all coefficients equal one. Every nonnegative path with $\sum_t R^{-t} c_t = W$ is optimal, for instance $c_t = y_t$ or the constant path $c_t = W(1 - \beta)/(1 - \beta^{T+1})$. With linear utility there is no motive to smooth consumption, and only the comparison of β with R^{-1} shapes the path.

Exercise 11.10 *Term structure of interest rates*

Remark. In the book’s definition of $\bar{R}_{t+1}$, “ $f'(k_{t+1} - \delta)$ ” should read $f'(k_{t+1}) - \delta$, as in (11.6.8E).

(a) Consumption is constant at $\bar{c}$, so $q_t^0 = \beta^t u'(c_t)/u'(c_0) = \beta^t$ (normalizing $q_0^0 = 1$).

(b), (c) $r_{t-1,t} = \log(q_{t-1}^0/q_t^0) = -\log \beta = \log(1 + \rho) = 0.0513$ for every t , and hence $r_{0,t} = 0.0513$ too: the yield curve is flat (the dashed lines in Figure S11.6).

(d) The equilibrium is that of exercise 11.7(b): consumption jumps up and then declines to $\bar{c}'$. From $q_t^0 = \beta^t u'(c_t)/u'(c_0)$,

$$r_{t-1,t} = -\log \beta + \gamma \log(c_t/c_{t-1}) = \log \bar{R}_t, \quad \bar{R}_t = 1 + (1 - \tau_k)(f'(k_t) - \delta),$$

where the second equality is (E). i) Consumption declines at a decreasing rate, so the short rates are below $\log(1 + \rho)$ and rising. In economic terms, the tax cuts the after-tax return at once ($\bar{R}_1 = 1.0435$), and as capital is run down the pre-tax marginal product rises, so the after-tax return gradually returns to $1 + \rho$. With $\tau_k = .2$ and $\gamma = 2$, $r_{0,1} = 0.0426$ rising to $r_{9,10} = 0.0489$. ii) Yields average the short rates, so they too lie below $\log(1 + \rho)$ and rise: the yield curve slopes upward, from $r_{0,1} = 0.0426$ to $r_{0,10} = 0.0463$ (Figure S11.6). (`ch11.py` checks that the rates computed from q_t^0 , from $\log \bar{R}_t$ and from the consumption path agree to 10^{-14} .)

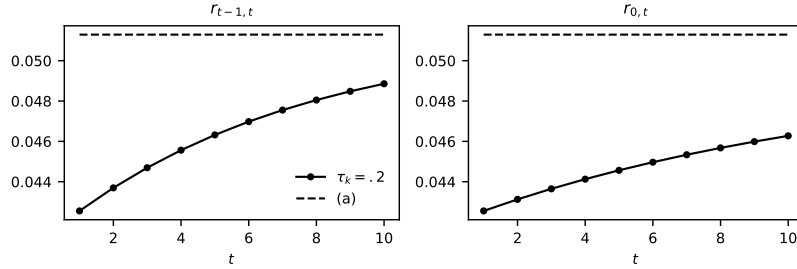


Figure S11.6. Short rates and yields after an unforeseen permanent $\tau_k = .2$ at $t = 0$; dashed: the flat curve of parts (b), (c).

Exercise 11.11 *Reading a rising consumption path*

With a constant consumption tax, (E) gives $c_{t+1}/c_t = (\beta \bar{R}_{t+1})^{1/\gamma}$. Rising consumption therefore means $\bar{R}_{t+1} > 1 + \rho$, and slowing growth means that $\bar{R}_t$ falls toward $1 + \rho$. The path is smooth after $t = 0$, without the kinks or inflection points that foreseen policy changes create (Figures 11.9.1–11.9.5). So it suggests pure transient dynamics toward a steady state, with no news after $t = 0$.

Story 1. Policy is constant ($g_t = \bar{g}$, $\tau_{kt} = \tau_k$), and the economy starts with less capital than its steady state, for example after a war destroyed part of it. Then k_t rises monotonically to $\bar{k}$, with $f'(\bar{k}) = \delta + \rho/(1 - \tau_k)$; $\bar{R}_t$ falls monotonically to $1 + \rho$; and consumption rises at a decreasing rate. With $k_0 = .5\bar{k}$ and the chapter's parameters, consumption grows by 5.4%, 4.4%, 3.6%, ..., while $\bar{R}_t$ falls from 1.172 (Figure S11.7, solid lines).

Story 2. The economy was in a steady state with a capital tax, and at $t = 0$ the tax is unexpectedly and permanently cut, with lump-sum taxes making up the revenue. The steady-state capital stock rises, and the dynamics are the same: k rises, and $\bar{R}_t = 1 + (1 - \tau_k^{\text{new}})(f'(k_t) - \delta)$ exceeds $1 + \rho$ and falls. Consumption drops at $t = 0$ relative to its old level, which the figure does not show, and then rises concavely. For a cut from .2 to 0, consumption goes from 0.636 to 0.620 at $t = 0$ and then rises to 0.643 (dashed lines).

In both stories g_t is constant. A permanent unforeseen change in g at $t = 0$ would only shift the level of consumption, and a foreseen future change would produce an inflection.

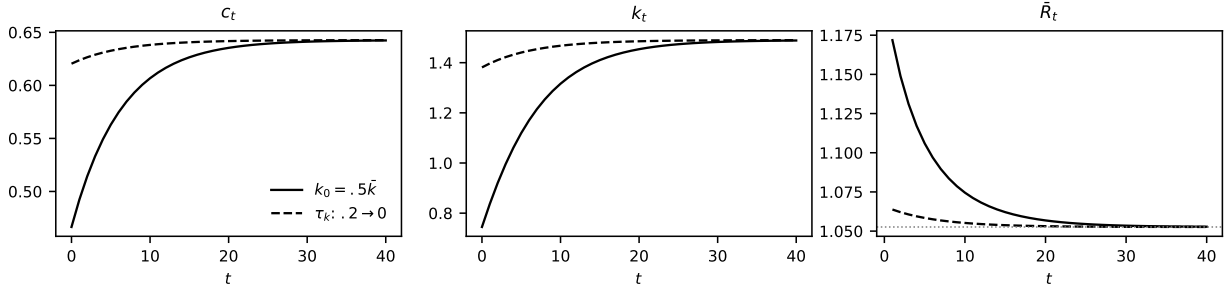


Figure S11.7. Two stories behind Figure 11.1.

Exercise 11.12 *Reading a falling consumption path*

(a) Three principles apply. (i) Under perfect foresight, consumption reacts to news when it arrives, and foreseen policy changes affect consumption before they occur. The flat path before $t = 12$ therefore means that until $t = 12$ households expected policy to stay constant and the economy was in a steady state. Whatever happens at $t = 12$ was unforeseen. (ii) A jump in consumption at $t = 12$ signals news at $t = 12$. (iii) After $t = 12$ consumption declines smoothly to a lower level, so $\bar{R}_t < 1 + \rho$ during the decline and returns to $1 + \rho$. In the long run, private consumption is lower, which requires a permanently higher g and/or a permanently higher τ_k .

Story A (our main story). At $t = 12$ the government unexpectedly and permanently raises g and finances it partly with a permanently higher capital tax, with lump-sum taxes balancing the budget. From then on the new policy is foreseen correctly. On impact, consumption falls by Δg , less the upward jump caused by the incentive to run down capital (exercise 11.7(b)). It then declines along the saddle path to $\bar{c}' = f(\bar{k}') - \delta\bar{k}' - \bar{g} - \Delta g$. With g rising from .2 to .23 and τ_k from 0 to .3, consumption drops from 0.6426 to 0.6355 at $t = 12$ and then declines, convexly, to 0.6013 (Figure S11.8, solid lines). The relative size of the drop and of the later decline depends on the relative size of the two policy changes.

(b) Under story A, $g_t = \bar{g}$ until $t = 11$ and is permanently higher from $t = 12$. Capital is constant at $\bar{k}$ until $t = 12$ and then falls monotonically to $\bar{k}' = 1.311$, where $f'(\bar{k}') = \delta + \rho/(1 - \tau_k)$. $\bar{R}_t = 1 + \rho$ until $t = 11$. The realized return on capital carried into $t = 12$ falls to $1 + (1 - \tau_k)(f'(\bar{k}) - \delta) = 1.0368$: a surprise capital loss, so the Euler equation between $t = 11$ and 12 fails ex post. Then $\bar{R}_{13} = 1.0387$, and $\bar{R}_t$ rises back to $1 + \rho$ as capital falls.

Story B is also consistent. At $t = 12$ news arrives that g will rise permanently at a later date, say $t = 22$, with lump-sum finance. Consumption drops at $t = 12$ (from 0.643 to 0.609) and declines thereafter to $\bar{c} - \Delta g$. Capital rises until $t = 22$ (to 2.10) and then falls back to $\bar{k}$, so $\bar{R}_t < 1 + \rho$ throughout, with its minimum at $t = 22$; g is constant until $t = 22$. In story B the decline accelerates until g rises and decelerates afterwards, an S-shape. In story A the decline is convex throughout, which is closer to Figure 11.2 (Figure S11.8, dashed lines).

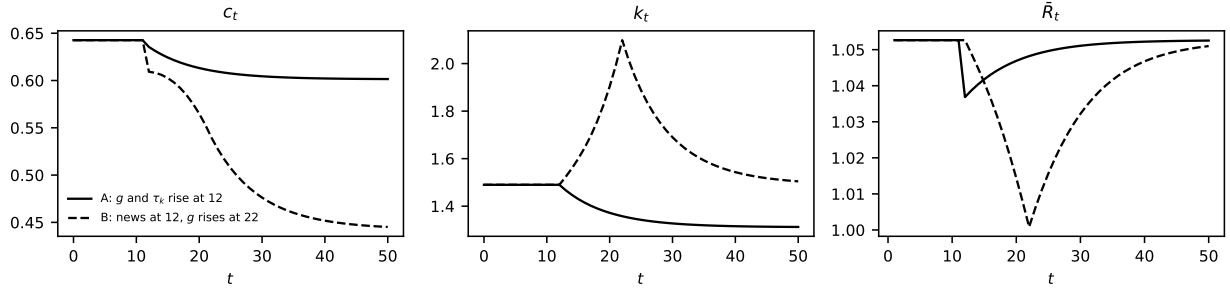


Figure S11.8. Two stories behind Figure 11.2: A, an unforeseen rise of g (.2 $\rightarrow$.23) and τ_k (0 $\rightarrow$.3) at $t = 12$; B, news at $t = 12$ that g will rise from .2 to .4 at $t = 22$.

Exercise 11.13 *Linear utility and the saving rate*

(a) Let $W_t = f(k_t) + (1 - \delta)k_t$, so $c_t = W_t - k_{t+1}$ with $0 \leq k_{t+1} \leq W_t$. Every feasible path is bounded, and summing by parts gives

$$\sum_{t=0}^{\infty} \beta^t c_t = W_0 + \sum_{t=1}^{\infty} \beta^t \pi(k_t), \quad \pi(k) \equiv f(k) - (\rho + \delta)k,$$

because $\beta^t [f(k_t) + (1 - \delta)k_t] - \beta^{t-1} k_t = \beta^t \pi(k_t)$ and $\beta^T k_{T+1} \rightarrow 0$. So the planner wants each k_t as close as possible to the maximizer of the strictly concave π , which is $\bar{k}$, where $f'(\bar{k}) = \rho + \delta$. The only link between periods is $k_{t+1} \leq W_t$. Let K_t be the largest attainable capital, $K_0 = k_0$ and $K_{t+1} = f(K_t) + (1 - \delta)K_t$. We claim the optimal policy is

$$k_{t+1} = \min\{\bar{k}, f(k_t) + (1 - \delta)k_t\}, \quad c_t = f(k_t) + (1 - \delta)k_t - k_{t+1}.$$

It generates $k_t^* = \min\{\bar{k}, K_t\}$ for $t \geq 1$, because once $K_t \geq \bar{k}$ we have $f(\bar{k}) + (1 - \delta)\bar{k} > \bar{k}$. Any feasible path has $k_t \leq K_t$. If $K_t \leq \bar{k}$, then $\pi(k_t) \leq \pi(K_t) = \pi(k_t^*)$, since π increases on $[0, \bar{k}]$; otherwise $\pi(k_t) \leq \pi(\bar{k}) = \pi(k_t^*)$. So the policy is optimal term by term, and strict concavity makes it unique.

(b) If $k_0 > \bar{k}$, the planner eats the excess at once: $c_0 = f(k_0) + (1 - \delta)k_0 - \bar{k} > \bar{c}$, and then $k_t = \bar{k}$ and $c_t = \bar{c} = f(\bar{k}) - \delta\bar{k}$ for $t \geq 1$.

(c) If $k_0 < \bar{k}$ but $f(k_0) + (1 - \delta)k_0 \geq \bar{k}$, then $k_1 = \bar{k}$ with $c_0 = f(k_0) + (1 - \delta)k_0 - \bar{k} \in [0, \bar{c})$, and the steady state follows. Otherwise the planner consumes nothing and invests everything ($c_t = 0$, $k_{t+1} = K_{t+1}$) until $\bar{k}$ becomes reachable; in that period it consumes the surplus above $\bar{k}$, and from then on it stays at $\bar{k}$. With linear utility there is no smoothing motive, so adjustment is “bang-bang” and as fast as possible: this is the case $\gamma = 0$, $\lambda_2 = 0$, of section 11.10.6.

(d) From the policy, $s(k) = [k_{t+1} - (1 - \delta)k]/f(k)$, so

$$s(k) = \min\left\{1, \frac{\bar{k} - (1 - \delta)k}{f(k)}\right\}.$$

It equals 1 for small k , when $f(k) + (1 - \delta)k \leq \bar{k}$. Where it is interior it is strictly decreasing: its derivative has the sign of $-(1 - \delta)f(k) - [\bar{k} - (1 - \delta)k]f'(k)$, which is negative when $\bar{k} \geq (1 - \delta)k$. When $\bar{k} < (1 - \delta)k$ the sign is that of $(1 - \delta)[kf'(k) - f(k)] - \bar{k}f'(k) < 0$, because $kf'(k) < f(k)$. At the steady state, $s(\bar{k}) = \delta\bar{k}/f(\bar{k})$, which equals $\alpha\delta/(\rho + \delta)$ for Cobb–Douglas. The saving rate is

negative (gross disinvestment) for $k > \bar{k}/(1 - \delta)$, and it tends to $-\infty$ as $k \rightarrow \infty$ since $f(k)/k \rightarrow 0$. With the chapter's parameters, $s = 1$ for $k \leq 0.734$ ($= .49\bar{k}$), $s(\bar{k}) = 0.261$, and $s < 0$ for $k > 1.862$. Figure S11.9 compares this kinked rule with the optimal saving rates for CRRA utility: they are flatter, and they approach the linear-utility rule as $\gamma \rightarrow 0$.

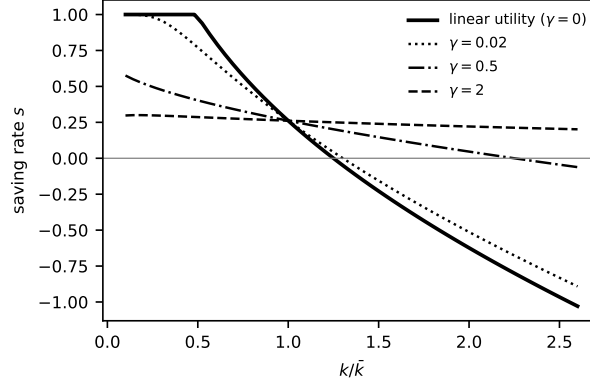


Figure S11.9. Saving rate $s = [k_{t+1} - (1 - \delta)k_t]/f(k_t)$ as a function of k_t : linear utility and CRRA utility ($g = 0$).

Exercise 11.14 *Transitions with and without government purchases*

(a) The household maximizes $\sum_t \beta^t u(c_t)$ subject to

$$\sum_t q_t [(1 + \tau_{ct})c_t + k_{t+1} - (1 - \delta)k_t] \leq \sum_t q_t [\eta_t k_t - \tau_{kt}(\eta_t - \delta)k_t + w_t - \tau_{ht}],$$

k_0 given. The firm maximizes $\sum_t q_t [f(k_t) - \eta_t k_t - w_t]$ per worker. The government satisfies $\sum_t q_t g_t = \sum_t q_t [\tau_{ct}c_t + \tau_{kt}(\eta_t - \delta)k_t + \tau_{ht}]$. A competitive equilibrium is a budget-feasible policy, a feasible allocation ($c_t + g_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$) and a price system such that, given prices and policy, the allocation solves the household's and the firm's problems. It is characterized by (E), feasibility and the terminal condition.

(b) $f'(\bar{k}) = \rho + \delta$, so $\bar{k} = [\alpha z / (\rho + \delta)]^{1/(1-\alpha)}$. The steady-state saving rate is $\bar{x}/f(\bar{k}) = \delta \bar{k}/f(\bar{k})$, which for Cobb–Douglas ($f(\bar{k})/\bar{k} = (\rho + \delta)/\alpha$) equals $\alpha \delta / (\rho + \delta) = 0.261$ with the chapter's parameters.

(c) As in exercise 11.6(b): k rises monotonically from 0.745 to 1.490, c rises from 0.648 to 0.843, and $\bar{R}_{t+1}$ falls from 1.166 to 1.0526 (Figure S11.10, solid lines; $\gamma = 2$).

(d) We take $\varphi = .1$, so $\bar{g} = 0.0907$. *Identical*: the steady-state capital stock and the steady-state return, since $\bar{g}$ does not enter $f'(\bar{k}) = \rho + \delta$, and the fact that the timing of the lump-sum taxes is irrelevant. *Different*: steady-state consumption is lower by $\bar{g}$, and the whole transition differs. Consumption is lower (at $t = 0$, 0.565 instead of 0.648, a fall smaller than $\bar{g}$, so investment also falls: $k_1 = 0.848$ instead of 0.856). Capital converges more slowly ($\lambda_2 = 0.840$ instead of 0.830; $k_{10} = 1.343$ instead of 1.360), so $\bar{R}$ stays high longer (dashed lines). The reason: in terms of total absorption $C_t = c_t + \bar{g}$, the planner maximizes $\sum_t \beta^t u(C_t - \bar{g})$, a Stone–Geary utility whose curvature $-C u''/u' = \gamma C/(C - \bar{g})$ exceeds γ . Government purchases act like a subsistence requirement, which

makes the household less willing to substitute consumption over time, so, as with a higher γ , the transition is slower. Only with linear utility would the capital paths coincide.

(e) *Algorithm.* For a trial $\bar{\tau}_k$: (i) the terminal steady state solves $f'(\bar{k}) = \delta + \rho/(1 - \bar{\tau}_k)$; (ii) solve for the equilibrium path from k_0 (by shooting or Newton's method), using (E) with $\bar{R}_{t+1} = 1 + (1 - \bar{\tau}_k)(f'(k_{t+1}) - \delta)$ and feasibility with $g_t = \bar{g}$; (iii) compute $q_t = \beta^t u'(c_t)/u'(c_0)$ and the present-value surplus

$$\Gamma(\bar{\tau}_k) = \sum_t q_t \bar{\tau}_k (f'(k_t) - \delta) k_t - \bar{g} \sum_t q_t;$$

(iv) solve $\Gamma(\bar{\tau}_k) = 0$ by bisection or the secant method, taking the smallest root if there are several (Laffer curve). An equilibrium exists only if $\bar{g}$ is small enough. For Cobb–Douglas, $(f'(k) - \delta)k = \alpha f(k) - \delta k$, so revenue per period can never exceed $\max_k [\alpha f(k) - \delta k] = 0.164$. With $\varphi = .2$ ($\bar{g} = 0.181$) no equilibrium would exist. With $\varphi = .1$, Γ increases on $[.05, .95]$ and its root is $\bar{\tau}_k = 0.641$. The steady state is then $k = 0.929$, $c = 0.699$, and capital rises only slowly from 0.745 (dash-dotted lines). The government runs primary surpluses early, when the tax base $\alpha f(k) - \delta k$ is large, and a small primary deficit in the long run (revenue $0.087 < \bar{g}$), financed by interest on the assets it has accumulated.

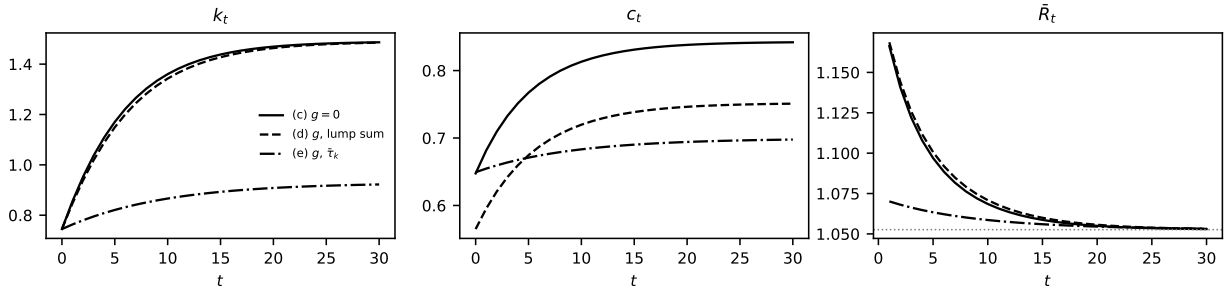
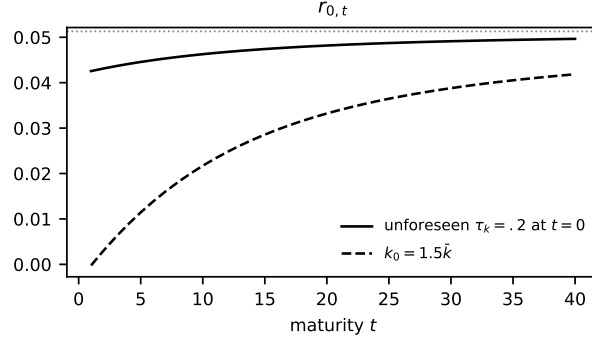


Figure S11.10. Transitions from $k_0 = .5\bar{k}$: (c) $g = 0$; (d) $\bar{g} = .1f(k_0)$ with lump-sum taxes; (e) the same $\bar{g}$ financed by $\bar{\tau}_k = 0.641$.

Exercise 11.15 *Reading the yield curve*

Assume a constant consumption tax (or none). Then $r_{t,t+1} = \log(q_t/q_{t+1}) = -\log \beta + \gamma \log(c_{t+1}/c_t) = \log \bar{R}_{t+1}$. Since $r_{0,t}$ is the average of the first t short rates, an increasing $r_{0,t}$ means the short rates increase, and a flattening curve means they converge. If policy is eventually constant, rates converge to the steady-state rate $\log(1 + \rho)$ (11.6.10). So the short rates are below $\log(1 + \rho)$ and rising, and consumption falls at a decreasing rate, $c_{t+1}/c_t = (\beta \bar{R}_{t+1})^{1/\gamma} < 1$, approaching its steady state from above. With a constant capital tax, $\bar{R}_{t+1} < 1 + \rho$ means $f'(k_{t+1}) < \delta + \rho/(1 - \tau_k)$: capital is above its steady state and falling. The smooth concave curve points to pure transient dynamics. A foreseen future policy change would bend the curve, as in the U-shaped curve of Figure 11.9.3.

Stories that fit: (i) the unforeseen permanent increase of the capital tax at $t = 0$ of exercise 11.10(d), with lump-sum rebates, for which $r_{0,t}$ rises from 0.0426 at $t = 1$ to 0.0463 at $t = 10$ and 0.0497 at $t = 40$; (ii) more generally, constant policy and an economy that starts with more capital than its steady state. With $k_0 = 1.5\bar{k}$, $r_{0,t}$ rises from 0.000 to 0.042 at $t = 40$ (Figure S11.11).

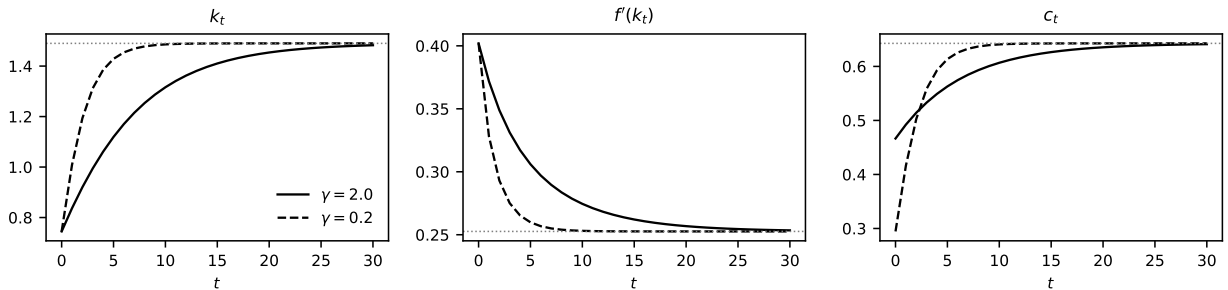
Figure S11.11. Upward-sloping, concave yield curves at $t = 0$; dotted: $\log(1 + \rho)$.**Exercise 11.16 Low versus high γ**

The steady state, $f'(\bar{k}) = \delta + \rho/(1 - \tau_k)$, does not involve γ , so both economies converge to the same $\bar{k}$ (the dashed line at $\rho + \delta$ in Figure 11.5 shows $\tau_k = 0$). The speed of convergence is λ_2 , which increases with γ . A household with low γ has a high intertemporal elasticity $1/\gamma$: when $k_0 < \bar{k}$ it accepts very low consumption now in order to exploit the high return on capital, so capital converges quickly.

(a) Panel (b), where capital jumps to the steady state almost at once, is the low- γ economy; panel (a), with gradual convergence, is the high- γ economy.

(b) $f'(k_t)$ mirrors k_t , because f' is decreasing. Panel (b), with its fast decline to $\rho + \delta$, is the low- γ economy; panel (a) is the high- γ one.

(c) By (E), $c_{t+1}/c_t = (\beta \bar{R}_{t+1})^{1/\gamma}$: for a given return above $1 + \rho$, consumption grows faster the lower is γ . In equilibrium the low- γ household starts from much lower consumption and catches up quickly; the high- γ household's path is smoother and slower. With $k_0 = .5\bar{k}$ and the chapter's parameters (Figure S11.12): for $\gamma = .2$, $\lambda_2 = 0.578$, $c_0 = 0.295$, $c_5 = 0.614$, and capital is within 1% of $\bar{k}$ after 8 periods; for $\gamma = 2$, $\lambda_2 = 0.852$, $c_0 = 0.467$, $c_5 = 0.563$, and it takes 26 periods. Both paths converge to $\bar{c} = 0.643$. (With $\tau_k > 0$ the capital-tax revenue differs along the two paths, so the lump-sum taxes that balance the budget differ, as the book's footnote notes; the allocation does not depend on their timing.)

Figure S11.12. Low ($\gamma = .2$) versus high ($\gamma = 2$) curvature, from $k_0 = .5\bar{k}$ ($g = .2$, no distorting taxes).

Exercise 11.17 Anticipated versus unanticipated government purchases

Panel (a), where consumption is constant at 0.643 until $t = 9$ and constant at 0.443 from $t = 10$, is the unanticipated jump. Panel (b), where consumption declines smoothly from $t = 0$, is the anticipated one.

Unanticipated. Until $t = 10$ households expect $g = .2$ forever, so the economy stays in its steady state ($k_t = \bar{k}$, $c_t = \bar{c}$, $\bar{R}_t = 1 + \rho$). At $t = 10$ they learn that $g = .4$ forever. Since g does not enter $f'(\bar{k}) = \rho + \delta$ and $k_{10} = \bar{k}$, the new equilibrium is $k_t = \bar{k}$ and $c_t = \bar{c} - .2 = 0.443$ for all $t \geq 10$ (exercise 11.5(c)). Consumption falls by exactly the increase in g , and capital and the interest rate never move.

Anticipated. At $t = 0$ the present value of lump-sum taxes rises, so consumption falls at once, to 0.609, and keeps falling. Households build up capital before $t = 10$, to $k_{10} = 2.098$, to smooth consumption, and run it down afterwards. Because $k_t > \bar{k}$ for all $t \geq 1$, $\bar{R}_t = 1 - \delta + f'(k_t) < 1 + \rho$ throughout; this is what makes a declining consumption path optimal. $\bar{R}_t$ reaches its minimum, 1.0008, at $t = 10$, when capital peaks, and then rises back to $1 + \rho$. The time-0 yield curve is U-shaped, as in Figure 11.9.3.

Figure S11.13 reproduces Figure 11.6. Cross-checks: the book's shooting algorithm gives $c_0 = 0.609242$, the same as Newton's method to 5×10^{-10} . The linear approximation (11.10.8) around the terminal steady state, with $\lambda_2 = 0.879$ and $\lambda_1 = 1.197$ ($\beta\lambda_1\lambda_2 = 1$), gives $k_{10} = 2.090$ and stays within 0.015 of the nonlinear path.

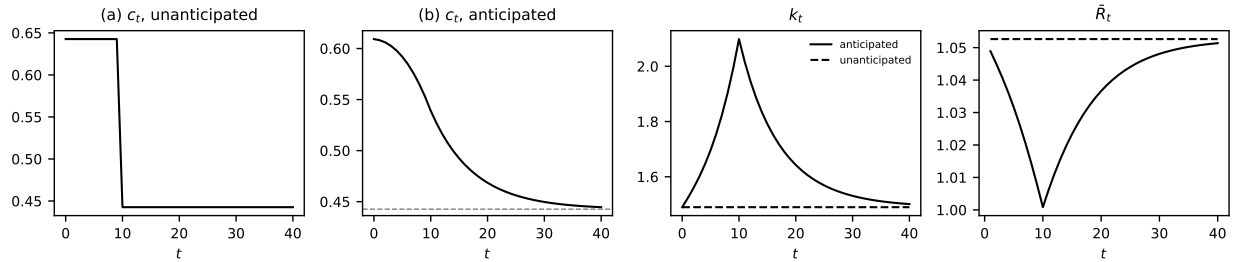


Figure S11.13. An unanticipated and an anticipated jump of g from .2 to .4 at $t = 10$ ($\gamma = 2$).

Exercise 11.18 Habit persistence and durability

Remark. The exercise states $\lim_{c \downarrow 0} u'(c) = 0$; for CRRA utility, and for the usual Inada condition, the limit is $+\infty$.

(a) Let $x_t = c_t - \alpha c_{t-1}$, and attach $\beta^t \mu_t$ to the time- t resource constraint. The first-order conditions are

$$\mu_t = u'(x_t) - \alpha \beta u'(x_{t+1}), \quad \mu_t = \beta \mu_{t+1} [f'(k_{t+1}) + 1 - \delta], \quad \lim_{t \rightarrow \infty} \beta^t \mu_t k_{t+1} = 0,$$

together with feasibility, $x_t > 0$ and the initial conditions. The first condition says that consuming more today raises today's utility, but it also raises tomorrow's habit ($\alpha > 0$) or tomorrow's services ($\alpha < 0$). Eliminating μ gives a third-order difference equation in c (fourth-order in k):

$$u'(x_t) - \alpha \beta u'(x_{t+1}) = \beta [f'(k_{t+1}) + 1 - \delta] [u'(x_{t+1}) - \alpha \beta u'(x_{t+2})].$$

(b) An optimal steady state is a constant triple (k, c, μ) that satisfies these conditions: if $(k_0, c_{-1}) = (k, c)$, the optimal path stays there.

(c) A constant $\mu > 0$ requires $1 = \beta[f'(k) + 1 - \delta]$, so

$$f'(k) = \rho + \delta, \quad c = f(k) - \delta k, \quad \mu = (1 - \alpha\beta) u'((1 - \alpha)c) > 0.$$

The steady state does not depend on α : habits and durability shape the dynamics, not the long run. With the chapter's technology, $k = 1.490$, $f'(k) = 0.2526$ and $c = 0.843$.

(d) The state is (k_t, c_{t-1}) : two predetermined variables. Linearized, the fourth-order equation has four roots. Because the problem is a planning problem they come in pairs $(\lambda, 1/(\beta\lambda))$: two stable and two unstable roots. So the two initial conditions and the requirement to converge pin down a unique path.

Algorithm (implemented in `ch11.py`). 1. Compute the steady state of (c). 2. Choose a large S , take $k_1, \dots, k_{S-1}$ as unknowns, and set $k_t = k$ for $t \geq S$ (so $c_t = c$ for $t \geq S$). 3. Given the unknowns, compute c_t from feasibility, $x_t = c_t - \alpha c_{t-1}$ using the given c_{-1} , and μ_t from the first condition. 4. Solve the $S - 1$ Euler equations $\mu_t = \beta\mu_{t+1}[f'(k_{t+1}) + 1 - \delta]$ by Newton's method; the Jacobian is banded, since each equation involves $k_{t-1}, \dots, k_{t+3}$. 5. Check that the solution does not change when S grows, and that $x_t > 0$ and $\mu_t > 0$. A shooting alternative guesses (c_0, c_1) , iterates forward (for k_{t+1} from feasibility, μ_{t+1} from the Euler equation, and then x_{t+2} from the first condition), and adjusts the two guesses by a two-dimensional Newton method until $(k_S, c_S) = (k, c)$; it becomes inaccurate for long horizons, as does all shooting. The linear approximation solves the two stable roots backward and the two unstable roots forward.

Results ($\gamma = 2$, $k_0 = .5k$, $c_{-1} = c$; Figure S11.14). For $\alpha = .5$ the stable roots are 0.516 and 0.822, for $\alpha = -.5$ they are -0.487 and 0.831 , and each root times its unstable partner equals $1/\beta = 1.0526$. The computed paths converge at the dominant stable root, and for $\alpha = 0$ the algorithm reproduces the standard model's path to 10^{-15} . With a habit, low consumption relative to last period's c_{-1} is painful, so the planner cuts consumption gradually ($c_0 = 0.734$, against 0.648 without habit) and accumulates capital more slowly. With durability, the stock inherited from c_{-1} still provides services, so purchases fall more at $t = 0$ ($c_0 = 0.565$) and capital is built faster. The negative root makes purchases of the durable oscillate.

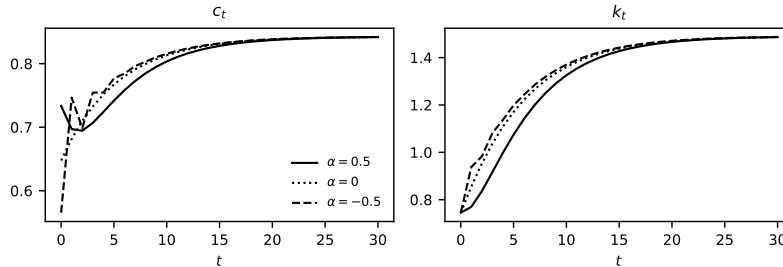


Figure S11.14. Optimal paths with habit ($\alpha = .5$), without ($\alpha = 0$) and with durability ($\alpha = -.5$), from $k_0 = .5k$ and $c_{-1} = c$.

Exercise 11.19 *The invisible hand, I*

(a) The household takes as given the price system: $\{q_t\}$ (equivalently the interest rates $R_{t,t+1} = q_t/q_{t+1}$, i.e. the whole yield curve), the rental rates $\{\eta_t\}$ and the wages $\{w_t\}$. It also takes as given the tax plan: lump-sum taxes $\{\tau_{ht}\}$, of which only the present value matters, and zero distorting taxes. Its own initial capital k_0 is an endowment, not a market signal. The household does not care about g_t as such; the increase in g reaches it only through taxes and prices. After the announcement: the present value of lump-sum taxes rises from 4.00 to 9.79 (time-0 goods); interest rates fall below $\rho = 0.0526$ and follow a U-shaped path ($R_{t,t+1} - 1 = 0.049$ at $t = 0$, 0.026 at $t = 5$, 0.0008 at $t = 9$, 0.006 at $t = 10$, 0.038 at $t = 20$); and wages and rental rates move with aggregate capital.

(b) Big K is the aggregate path. Shooting (or Newton's method) is applied to the aggregate conditions $u'(C_t) = \beta u'(C_{t+1})[f'(K_{t+1}) + 1 - \delta]$ and $K_{t+1} = f(K_t) + (1 - \delta)K_t - g_t - C_t$, with $K_0 = \bar{k}$ and $K_S \approx \bar{k}$. This gives $\{C_t, K_t\}$, and then $\eta_t = f'(K_t)$, $w_t = f(K_t) - K_t f'(K_t)$, $q_t = \beta^t u'(C_t)/u'(C_0)$, and lump-sum taxes with $\sum_t q_t \tau_{ht} = \sum_t q_t g_t$ (for instance $\tau_{ht} = g_t$).

(c) The household maximizes $\sum_t \beta^t u(c_t)$ subject to $\sum_t q_t [c_t + k_{t+1} - (1 - \delta)k_t] \leq \sum_t q_t (\eta_t k_t + w_t - \tau_{ht})$. Because prices satisfy no-arbitrage, $q_t = q_{t+1}(\eta_{t+1} + 1 - \delta)$, capital drops out and the budget becomes

$$\sum_t q_t c_t = W_0 \equiv (\eta_0 + 1 - \delta)k_0 + \sum_t q_t (w_t - \tau_{ht}), \quad q_0 = 1 :$$

the household is indifferent about how much capital it holds, as long as the budget holds. Its first-order condition $\beta^t u'(c_t) = \mu q_t$ gives

$$c_t = c_0 (q_t / \beta^t)^{-1/\gamma}, \quad c_0 = \frac{W_0}{\sum_t q_t (q_t / \beta^t)^{-1/\gamma}}.$$

Two forces are at work. A wealth effect: taxes are higher, and at the new prices the old plan ($c_t = 0.643$ forever) would cost 18.47, more than $W_0 = 14.50$. A substitution effect: interest rates below $1/\beta - 1$ tilt consumption downward over time, $c_{t+1}/c_t = (\beta R_{t,t+1})^{1/\gamma} < 1$. So the household cuts consumption at once, to 0.609, and plans a declining path. With $\tau_{ht} = g_t$, it invests the difference between income and consumption in capital.

(d) Guess c_0 . Generate $c_{t+1} = c_t (\beta R_{t,t+1})^{1/\gamma}$ from the given interest rates and $k_{t+1} = (\eta_t + 1 - \delta)k_t + w_t - \tau_{ht} - c_t$ from the budget. If k_S ends above its target (the terminal condition $\lim q_T k_{T+1} = 0$, here K_S), c_0 was too low: raise it; otherwise lower it. Since prices are given, this is a linear problem, and bisection converges to machine precision. Numerically the household's $c_0 = 0.60924195$ equals C_0 , the budget formula of (c) gives the same value, and $\max_t |c_t - C_t| = 3 \times 10^{-14}$, $\max_t |k_t - K_t| = 2 \times 10^{-13}$ for $t \leq 100$.

(e) Check that the household's optimal choice, given the prices computed from Big K , is Big K itself: $c_t = C_t$, $k_t = K_t$. Indeed C_t satisfies $u'(C_t) = \beta u'(C_{t+1})R_{t,t+1}$ with $R_{t,t+1} = f'(K_{t+1}) + 1 - \delta$, which is the household's Euler equation. With $k_t = K_t$ and $\tau_{ht} = g_t$ the household's budget gives $k_{t+1} = f'(K_t)K_t + (1 - \delta)K_t + f(K_t) - K_t f'(K_t) - g_t - C_t = K_{t+1}$. So individual optimization reproduces the aggregate path that generated the prices, and that fixed point is the competitive equilibrium.

Exercise 11.20 *The invisible hand, II*

We use the parameters of section 11.9 ($\alpha = .33$, $\delta = .2$, $\beta = .95$, $g = .2$), $B = 3$, and a rise of τ_n from 0 to .2; these reproduce Figure 11.12.3 (Figure S11.15). The steady states have $k/n = \tilde{k} = 1.490$ and $w = \bar{w} = 0.764$ both before and after, with $(c, n, k) = (0.2547, 0.540, 0.804)$ before and $(0.2038, 0.479, 0.714)$ after.

(a) The household takes as given the prices $\{q_t, \eta_t, w_t\}$, the labor tax rates $\{\tau_{nt}\}$ (0 before $t = 10$, .2 after), zero consumption and capital taxes, and the lump-sum taxes, which here are transfers rebating the labor-tax revenue (only their present value matters). The signals that matter are the after-tax wage $(1 - \tau_{nt})w_t$ and the interest rates $R_{t,t+1} = q_t/q_{t+1} = 1 + f'(K_{t+1}/N_{t+1}) - \delta$.

(b) The household maximizes $\sum_t \beta^t [\log c_t + B(1 - n_t)]$ subject to $\sum_t q_t c_t = (\eta_0 + 1 - \delta)k_0 + \sum_t q_t [(1 - \tau_{nt})w_t n_t - \tau_{ht}]$. Its first-order conditions $\beta^t/c_t = \mu q_t$ and $\beta^t B = \mu q_t (1 - \tau_{nt})w_t$ give

$$c_t = (1 - \tau_{nt})w_t/B, \quad (\text{I})$$

$$c_{t+1}/c_t = \beta q_t/q_{t+1} = \beta R_{t,t+1}. \quad (\text{II})$$

By (I), consumption is tied to the after-tax wage: given prices, it does not depend on wealth. Substituting $\mu q_t (1 - \tau_{nt})w_t = \beta^t B$ into the budget shows that the value of any labor plan is $\sum_t q_t (1 - \tau_{nt})w_t n_t = (B/\mu) \sum_t \beta^t n_t$. So utility depends on the labor plan only through $\sum_t \beta^t n_t$, which the budget pins down. At equilibrium prices the household is therefore indifferent among all labor plans with the same $\sum_t \beta^t n_t$: its intertemporal labor supply is perfectly elastic (Hansen–Rogerson). The timing of work is set by the Big K conditions, and small price changes can support large reallocations of work. That is why:

- *Before* $t = 10$ the capital-labor ratio, and hence w_t and $R_{t,t+1}$, stay almost exactly at their steady-state values (k/n between 1.4848 and 1.4898). By (I) and (II), so does consumption, which stays within 0.12% of 0.2547.
- *Work is shifted toward* $t < 10$. From $t = 10$ on, the after-tax wage is lower, so leisure is cheaper then. The household works more before $t = 10$: n rises from 0.540 to 0.614 at $t = 9$. Consumption cannot rise, since it is tied to the after-tax wage, so the extra earnings are saved, and capital rises from 0.804 to 0.976 at $t = 10$. At a constant capital-labor ratio, feasibility reads $k_{t+1} = [f(\tilde{k})/\tilde{k} + 1 - \delta]k_t - g - c_t$, an explosive equation with root 1.566. So a consumption dip of only 0.12% at $t = 0$ is amplified into a large build-up of capital and work.
- *At* $t = 10$ work falls to 0.409. The capital-labor ratio jumps to 2.39, and the pre-tax wage jumps to 0.893 (+17%), which partly offsets the tax: the after-tax wage falls only 6.5%, and by (I) so does consumption (to 0.238). The high capital-labor ratio lowers the return to $\bar{R}_{10} = 0.984 < 1$, which is what (II) requires for consumption to fall.
- *After* $t = 10$ the capital accumulated before $t = 10$ is run down to finance consumption while people work less. The capital-labor ratio returns to $\tilde{k}$, the wage to 0.764, consumption to $0.8 \times 0.2547 = 0.2038$, and (n, k) to $(0.479, 0.714)$. Along the way $\bar{R} < 1 + \rho$, consistent with declining consumption.

Big K , little k : the prices are computed from the aggregate path, and at those prices the aggregate path satisfies (I), (II) and the budget. Since the household is indifferent among labor plans with the same $\sum_t \beta^t n_t$, choosing $(c_t, n_t, k_t) = (C_t, N_t, K_t)$ is optimal for it: the fixed point.

Remark. The book says that before $t = 10$ the capital-labor ratio and consumption stay *equal* to their initial steady-state values. They cannot be exactly constant. By (I), constant consumption would require a constant capital-labor ratio $\tilde{k}$. The explosive equation above would then keep k and n at their old steady-state values until $t = 10$: no response at all before $t = 10$. From $t = 10$ on the path would be the response to an unforeseen tax increase at $t = 10$, for which $c_{10}/c_9 = 0.848$ while $\beta \bar{R}_{10} = 0.973$, violating (II) between $t = 9$ and 10, which must hold when the change is foreseen. The deviations are tiny, however: 0.12% for consumption and 0.35% for k/n .

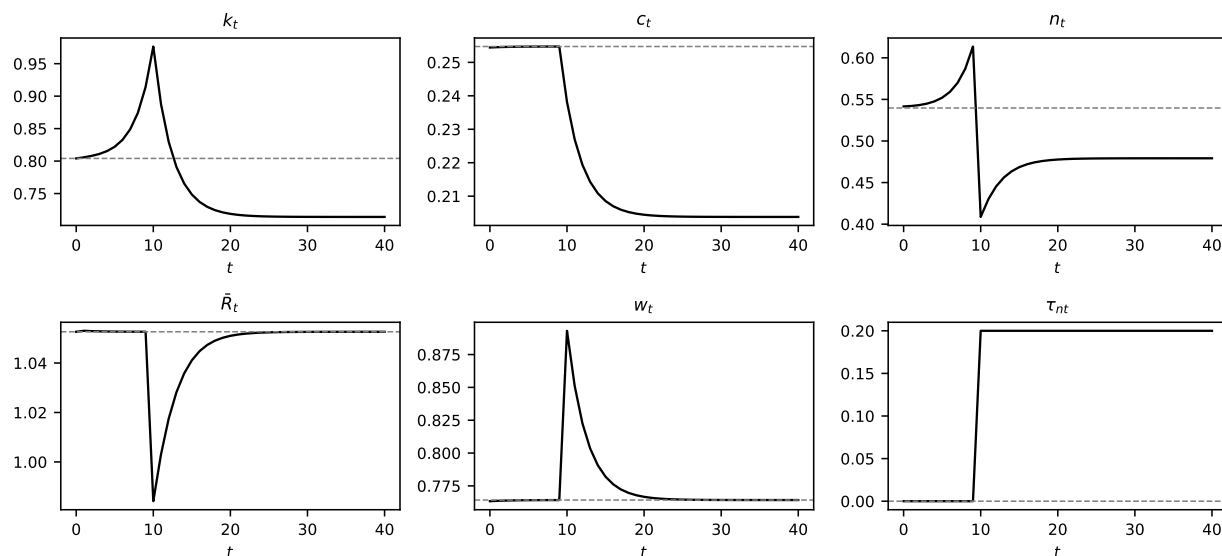


Figure S11.15. Elastic labor supply: a rise of τ_n from 0 to .2 at $t = 10$, foreseen at $t = 0$ (compare the book's Figure 11.12.3). Dashed lines: the initial steady state.

Chapter 13

Asset Pricing Theory

Solutions to all 5 exercises of Chapter 13 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Numbers come from `code/ch13.py`, which also recomputes the book's Table 13.A.1 (all six rows are confirmed).

Exercise 13.1 A Markov economy

(a) In equilibrium the consumer consumes the endowment, so the price at t (in time- t goods) of one unit of consumption at $t + 1$ in state $\bar{\lambda}_j$, given $\lambda_t = \bar{\lambda}_i$, is the marginal rate of substitution times the probability:

$$Q_{ij} = \beta P_{ij} \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} = \beta P_{ij} \bar{\lambda}_j^{-\gamma}.$$

Prices depend on the state only through i , because utility is homothetic and consumption growth is Markov. The gross risk-free rate in state i is $1 / \sum_j Q_{ij}$.

(b) A claim to one unit at $t + 2$ in state j costs $\sum_k \beta^2 P_{ik} P_{kj} (\bar{\lambda}_k \bar{\lambda}_j)^{-\gamma} = (Q^2)_{ij}$. A claim contingent on the whole path $(\bar{\lambda}_k, \bar{\lambda}_j)$ costs $Q_{ik} Q_{kj}$. Two-step prices are products of one-step prices, which is why one-step Arrow securities suffice to complete the market.

(c) The ex-dividend price satisfies $p_t = E_t [\beta (C_{t+1}/C_t)^{-\gamma} (C_{t+1} + p_{t+1})]$. Guess $p_t = v(\lambda_t) C_t$: the price-dividend ratio depends only on the growth state. Substituting and dividing by C_t ,

$$v_i = \sum_j \beta P_{ij} \bar{\lambda}_j^{1-\gamma} (1 + v_j), \quad \text{i.e.} \quad v = \tilde{Q}(\mathbf{1} + v), \quad \tilde{Q}_{ij} \equiv \beta P_{ij} \bar{\lambda}_j^{1-\gamma} = Q_{ij} \bar{\lambda}_j.$$

Hence

$$v = (I - \tilde{Q})^{-1} \tilde{Q} \mathbf{1} = \sum_{k \geq 1} \tilde{Q}^k \mathbf{1}, \quad p(C, \bar{\lambda}_i) = v_i C,$$

where $(I - \tilde{Q})^{-1}$ is the resolvent operator. It is well defined when the spectral radius of $\tilde{Q}$ is below one, a growth-adjusted discounting condition. (Illustration, `ch13.py`: $\beta = .95$, $\gamma = 2$,

$\bar{\lambda} = (.97, 1.05)$, $P = \begin{bmatrix} .6 & .4 \\ .2 & .8 \end{bmatrix}$ give $v = (13.84, 13.14)$, and iterating $v \leftarrow \tilde{Q}(\mathbf{1} + v)$ reproduces it.)

(d) The option's value depends on the level of the tree's price, $v_i C$, relative to the fixed strike, so its state is (C, λ) . At the start of a period the owner either exercises, receiving $v(\lambda)C - \tilde{p}$, or waits:

$$w(C, \bar{\lambda}_i) = \max \left\{ v_i C - \tilde{p}, \sum_j Q_{ij} w(\bar{\lambda}_j C, \bar{\lambda}_j) \right\}.$$

Algorithm: put C on a lattice closed under multiplication by the $\bar{\lambda}_j$ (e.g. a geometric grid when the $\bar{\lambda}$'s are powers of a common factor), truncated far from the region where exercise is marginal. Stack w into a vector $\mathbf{w}$ indexed by (C, λ) and let $\mathcal{Q}$ be the matrix with element Q_{ij} in position $((C, i), (\bar{\lambda}_j C, j))$. Then iterate $\mathbf{w}_{k+1} = \max\{\mathbf{g}, \mathcal{Q}\mathbf{w}_k\}$ (elementwise), with $\mathbf{g}$ the exercise payoffs, from $\mathbf{w}_0 = \max\{\mathbf{g}, 0\}$. At the fixed point, the option is exercised in the states where $\mathbf{w} = \mathbf{g}$.

(e) Let $(Tw)(C, i) = \max\{v_i C - \check{p}, \sum_j Q_{ij} w(\bar{\lambda}_j C, j)\}$. Because payoffs grow with C , use the weighted sup norm $\|w\| = \sup |w(C, i)|/(1+C)$. Then: (i) *monotonicity*: $w \leq w'$ implies $Tw \leq Tw'$, because $Q_{ij} \geq 0$ and max is monotone; (ii) *discounting*: for $a \geq 0$,

$$T(w + a(1+C))(C, i) \leq Tw(C, i) + a \sum_j Q_{ij}(1 + \bar{\lambda}_j C) \leq Tw(C, i) + a\kappa(1+C),$$

with $\kappa = \max_i \max\{\sum_j Q_{ij}, \sum_j \tilde{Q}_{ij}\}$. If $\kappa < 1$ (positive risk-free rates and a finite tree price, with row sums of $\tilde{Q}$ below one), Blackwell's sufficient conditions, in their weighted form, make T a contraction with modulus κ . So the iterations converge to the unique fixed point from any starting value. On a truncated C -grid the plain sup norm works with modulus $\max_i \sum_j Q_{ij}$.

Exercise 13.2 A Harrison–Kreps asset market

The rows of the table below are: $p_h = \beta(I - \beta P_h)^{-1} P_h d$ (homogeneous beliefs, (13.A.1)); p solving (13.A.2) (the optimist prices the asset); the reservation prices $\hat{p}_h(s) = \beta P_h(s, \cdot)(d + p)$ that type h would pay given resale at p (the book's rows 4–5); and $\check{p}$ solving (13.A.4) (the pessimist prices it). As a check, the program reproduces all six rows of the book's Table 13.A.1 exactly.

s_t	(a)		(b)	
	1	2	1	2
p_a	1.263	1.105	1.500	1.500
p_b	0.923	1.154	0.800	0.867
p (optimists)	1.500	1.500	1.500	1.500
$\hat{p}_a$	1.500	1.313	1.500	1.500
$\hat{p}_b$	1.313	1.500	1.313	1.375
$\check{p}$ (pessimists)	0.750	0.750	0.800	0.867

(a) Optimism alternates, as in the book. In state 1, type a assigns probability $\frac{1}{2}$ to the dividend state versus b 's $\frac{1}{4}$; in state 2 the roles reverse. So a holds the asset in state 1 and b in state 2, and the asset changes hands whenever the state changes. The optimists' price $p = (1.5, 1.5)$ exceeds *both* fundamental valuations in *both* states ($1.5 > 1.263, 0.923$ in state 1 and $1.5 > 1.105, 1.154$ in state 2): a speculative bubble in the sense of Harrison and Kreps and Scheinkman, driven by the option to resell to the other type. The pessimists' price 0.75 is below both fundamental values.

(b) Now type a is the optimist in *both* states: it assigns $\frac{1}{2}$ to the high-dividend state, while b assigns $\frac{1}{4}$ or $\frac{1}{3}$. Type a always holds the asset, there is never anyone to resell to at a higher valuation, and the resale option is worthless. Hence $p = p_a = (1.5, 1.5)$: no bubble and no trading volume. Likewise $\check{p} = p_b$, because b is always the pessimist. Heterogeneous beliefs alone do not create a speculative premium; what matters is that the identity of the optimist changes over time.

(c) For $p \in \mathbb{R}^2$ let $(Tp)(s) = \beta \max_{h \in \{a, b\}} \sum_j P_h(s, j)(d_j + p_j)$. *Monotonicity*: if $p \leq q$ componentwise, each $\sum_j P_h(s, j)(d_j + p_j) \leq \sum_j P_h(s, j)(d_j + q_j)$ because $P_h \geq 0$, and the maximum preserves the inequality, so $Tp \leq Tq$. *Discounting*: for a constant $a \geq 0$, $\sum_j P_h(s, j)(d_j + p_j + a) =$

$\sum_j P_h(s, j)(d_j + p_j) + a$ because rows sum to one. So $T(p + a\mathbf{1}) = Tp + \beta a\mathbf{1}$. By Blackwell's theorem T is a contraction on $(\mathbb{R}^2, \|\cdot\|_\infty)$ with modulus $\beta = .75$, so (13.A.3) converges to the unique solution of (13.A.2) from any starting point. (The same argument, with min, covers (13.A.4).)

Exercise 13.3 *The Lucas model again: a consol and options on it*

(a) The investor takes as given the price process $p_t = p(\lambda_t)$ and the equilibrium law of motion of the state. She believes she cannot influence prices, and she knows the stochastic process of her own endowment. She chooses holdings s_{t+1} of the consol subject to $c_t + p_t s_{t+1} = y_t + (\zeta + p_t)s_t$. Giving up p_t units of consumption at t to buy one more unit of the claim delivers $\zeta + p_{t+1}$ at $t + 1$, so at an optimum

$$u'(c_t)p_t = \beta E_t[u'(c_{t+1})(\zeta + p_{t+1})].$$

In equilibrium her consumption is the aggregate consumption process $c_{t+1} = \lambda_{t+1}c_t$, and she willingly holds the outstanding supply at these prices.

(b) Divide by $u'(c_t)$: $p(\lambda_i) = \sum_j \beta P_{ij} \lambda_j^{-\gamma} (\zeta + p(\lambda_j))$, i.e. with $Q_{ij} = \beta P_{ij} \lambda_j^{-\gamma}$,

$$p = Q(\zeta\mathbf{1} + p) \implies p = (I - Q)^{-1}Q\zeta\mathbf{1}.$$

Solve it directly or by iterating $p \leftarrow Q(\zeta\mathbf{1} + p)$. A unique nonnegative solution exists if the spectral radius of Q is below one. A simple sufficient condition is $\beta \max_i \sum_j P_{ij} \lambda_j^{-\gamma} < 1$, i.e. positive risk-free rates in every state. Then the iteration is a contraction.

(c) The owner of the call chooses between exercising, worth $p(\lambda) - p_S$, and waiting:

$$w(\lambda_i) = \max\left\{p(\lambda_i) - p_S, \sum_j Q_{ij} w(\lambda_j)\right\}.$$

Iterate $w_{k+1} = \max\{p - p_S\mathbf{1}, Qw_k\}$ from $w_0 = \max\{p - p_S\mathbf{1}, 0\}$. The map is a contraction with modulus $\max_i \sum_j Q_{ij} < 1$ (the argument of exercise 13.2(c)), so it converges. The exercise region is the set of states where $w = p - p_S$.

(d) Similarly, $J(\lambda_i) = \max\{p_S - p(\lambda_i), \sum_j Q_{ij} J(\lambda_j)\}$, solved in the same way. *Illustration* (`ch13.py`, the chain of exercise 13.1 with $\zeta = 1$): $p = (11.40, 10.28)$; with $p_S = 10.84$ the call is worth $(0.560, 0.364)$ and is exercised in state 1, where the consol is expensive; the put is worth $(0.490, 0.560)$ and is exercised in state 2.

Exercise 13.4 *A baby Lucas model*

(a) With the time-0 price of date- t goods $q_t^0 = \prod_{s < t} Q(s, s+1) = \delta^t (X_t/X_0)^{-\theta}$, the budget is $\sum_t q_t^0 c_t = 1$. The first-order conditions $\beta^t c_t^{-\gamma} = \mu q_t^0$ give, dividing consecutive dates,

$$\frac{c_{t+1}}{c_t} = \left(\frac{\beta}{Q(t, t+1)}\right)^{1/\gamma} = \left(\frac{\beta}{\delta}\right)^{1/\gamma} \left(\frac{X_{t+1}}{X_t}\right)^{\theta/\gamma}.$$

(b) $c_{t+1}/c_t = g \equiv (\beta/\delta)^{1/\gamma} \phi^{\theta/\gamma}$, constant.

(c) $c_t = c_0 g^t$ and $q_t^0 = (\delta \phi^{-\theta})^t$. The budget becomes $c_0 \sum_t \rho^t = 1$ with

$$\rho \equiv \delta \phi^{-\theta} g = \delta^{1-1/\gamma} \beta^{1/\gamma} \phi^{\theta(1/\gamma-1)}.$$

Well-definedness requires $\rho < 1$ (the present value of the consumption plan is finite; for $\gamma = 1$ this is simply $\beta < 1$). Then $c_0 = 1 - \rho$ and $c_t = (1 - \rho)g^t$. *Bonds*: the value at t of the remaining plan is $W_t = \sum_{s \geq t} \frac{q_s^0}{q_t^0} c_s = \frac{c_t}{1 - \rho}$. The consumer enters t holding bonds that pay $b_t = W_t$ units of time- t goods ($b_0 = 1$), consumes c_t , and buys $b_{t+1} = W_{t+1} = c_{t+1}/(1 - \rho)$ one-period bonds at price $Q(t, t+1)$. Check: $c_t + Q(t, t+1)b_{t+1} = c_t + \delta\phi^{-\theta}g c_t/(1 - \rho) = c_t/(1 - \rho) = b_t$. He always holds a positive amount of bonds, and it grows at rate g .

(d) With $\delta = \beta$, $\theta = \gamma$: $c_{t+1}/c_t = X_{t+1}/X_t$, so $c_t = c_0 X_t/X_0$ with $c_0 = 1 - \beta\phi^{1-\gamma}$. The prices $Q(t, t+1) = \beta(X_{t+1}/X_t)^{-\gamma}$ are exactly this consumer's marginal rates of substitution along a consumption path proportional to X . In that sense he is “representative”: an economy of such consumers whose aggregate endowment is proportional to X_t would support these prices, and he chooses consumption growth equal to the growth of X . He is *not* representative in levels. He owns only one unit of time-0 goods, so his consumption is proportional to, not equal to, X_t . And he holds a positive stock of bonds every period, which cannot clear a market in which he is the only agent (net bond supply is zero in a representative-agent economy).

Exercise 13.5 *Another baby Lucas model*

(a) The date-0 price of consumption after history s^t is the product of one-step prices, which telescopes:

$$q_t^0(s^t) = \prod_{k < t} Q_{s_k s_{k+1}} = \beta^t \pi_t(s^t) \left(\frac{x_t}{x_0} \right)^{-\gamma}.$$

The first-order conditions $\beta^t \pi_t(s^t) c_t(s^t)^{-\gamma} = \mu q_t^0(s^t)$ give $c_t(s^t)^{-\gamma} = \mu (x_t/x_0)^{-\gamma}$. So $c_t(s^t) = c_0 x_t/x_0$: consumption is proportional to X_t , and $c_{t+1}/c_t = x_{t+1}/x_t$. The budget $\sum_t \sum_{s^t} q_t^0 c_t = 1$ becomes $c_0 \Psi(s_0) = 1$ with

$$\Psi(s_i) = \sum_{t \geq 0} \beta^t E \left[\left(\frac{x_t}{x_0} \right)^{1-\gamma} \mid s_0 = s_i \right] = [(I - \beta \tilde{P})^{-1} \mathbf{1}]_i, \quad \tilde{P}_{ij} = P_{ij} (x_j/x_i)^{1-\gamma},$$

which is finite when the spectral radius of $\beta \tilde{P}$ is below one. So $c_0 = 1/\Psi(s_0)$. *Holdings*: the value of the remaining plan at a date with state i and consumption c is $W = c \Psi_i$. To finance next period's wealth, the consumer holds Arrow securities paying

$$a_j = c \frac{x_j}{x_i} \Psi_j \quad \text{units in state } j \text{ next period, } j = 1, \dots, n,$$

and the budget identity $c + \sum_j Q_{ij} a_j = c \Psi_i$ holds because $\Psi_i = 1 + \beta \sum_j P_{ij} (x_j/x_i)^{1-\gamma} \Psi_j$ (verified numerically in `ch13.py`). He holds more claims to states in which X (and hence his planned consumption) is high.

(b) The Arrow prices are his own marginal rates of substitution evaluated at a consumption process proportional to X . If X were the aggregate endowment and consumers were identical, each owning a share of it, every consumer would choose consumption proportional to X at these prices, and markets would clear with these prices. In that sense he is a representative consumer, and Q_{ij} are the equilibrium prices of the Lucas model with aggregate consumption X . As in exercise 13.4, he is not representative in levels: his wealth is one unit of date-0 goods, not the whole endowment,

and he holds a nonzero portfolio of Arrow securities, which a single representative agent could not do in equilibrium.

Chapter 14

Asset Pricing Empirics

Solutions to all 24 exercises of Chapter 14 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Numerical results come from `code/ch14.py`, which also checks every closed form by simulation or quadrature. The Matlab program asked for in exercise 14.13 is replaced by Python. The consumption series of exercise 14.1 (`epdata.m`) is not printed in the book; we use the moments of the same series reported in Table 14.3.1.

Notation and facts used throughout. $\rho \equiv -\log \beta$. If $\log X \sim \mathcal{N}(a, s^2)$, then $\mathbb{E}X = e^{a+s^2/2}$ and $\text{std } X = \mathbb{E}X (e^{s^2} - 1)^{1/2}$. In particular, for $\epsilon \sim \mathcal{N}(0, I)$ and a constant vector b , $\mathbb{E} \exp(b'\epsilon) = \exp(b'b/2)$. So if U is conditionally Gaussian, the risk-sensitivity operator satisfies

$$\mathsf{T}_t(U) \equiv -\theta \log \mathbb{E}_t \exp(-U/\theta) = \mathbb{E}_t U - \frac{1}{2\theta} \text{var}_t(U).$$

With CRRA utility and log consumption $c_t = \log C_t$, the stochastic discount factor is $m_{t+1} = \beta \exp[-\gamma(c_{t+1} - c_t)]$, and $r_t = -\log \mathbb{E}_t m_{t+1}$ is the one-period (continuously compounded) risk-free rate.

Exercise 14.1 *Hansen–Jagannathan bounds*

(a) Let $x = (R^s, R^b)'$, so the price vector is $q = (1, 1)' \equiv \mathbf{1}$. For a hypothesized mean $v = E(y)$, (14.5.8) gives the regression coefficient $b(v) = \Sigma^{-1}(\mathbf{1} - v\mu)$, and (14.5.9) gives

$$\sigma(y) \geq \sigma_{HJ}(v) \equiv [b(v)'\Sigma b(v)]^{1/2} = [(1 - v\mu)'\Sigma^{-1}(1 - v\mu)]^{1/2}.$$

Thus $\sigma_{HJ}(v)^2 = Av^2 - 2Bv + C$ with $A = \mu'\Sigma^{-1}\mu = 357.256$, $B = \mathbf{1}'\Sigma^{-1}\mu = 348.936$ and $C = \mathbf{1}'\Sigma^{-1}\mathbf{1} = 340.894$. The bound is smallest, .2898, at $v = B/A = .9767$. Some values:

$E(y)$	.90	.92	.94	.96	.98	1.00	1.02
σ_{HJ}	1.479	1.110	0.752	0.429	0.296	0.527	0.868

Figure S14.1 plots the bound (solid line) on $[\text{.87}, 1.02]$, slightly wider than asked so that the points of part (b) fit; the requested interval is shaded.

Checks. The discount factor $y^* = v + (x - \mu)'b(v)$ satisfies $E(y^*x) = v\mu + \Sigma b(v) = \mathbf{1}$, so it prices both returns and attains the bound. Also $\sigma_{HJ}(v)/v$ is the maximal Sharpe ratio against the riskless return $1/v$; maximizing the Sharpe ratio numerically over portfolios reproduces $\sigma_{HJ}(v)$ to 10^{-15} . The ratio $\sigma_{HJ}(v)/v$ is smallest at $v = C/B = .9770$, where it equals $(A - B^2/C)^{1/2} = .2967$. So every admissible discount factor has a market price of risk of at least .30 per year.

Remark. The exercise gives 1.02 for the mean bill return. Table 14.3.1, which reports the same Mehra–Prescott data with the same covariance matrix, gives 1.010 (Mehra and Prescott report a mean real bill return of 0.8 percent), so 1.02 is probably a misprint. The dashed line in Figure S14.1 uses 1.010: the bound moves right and up (minimum .351 at $E(y) = .986$), which only strengthens the conclusions of part (b).

(b) The file `epdata.m` is not reproduced in the book and was not available to us. Table 14.3.1 reports the two moments of the same consumption series that matter here: $E(C_t/C_{t-1}) = 1.018$ and $\text{var}(C_t/C_{t-1}) = .00127$. We assume that $g_t = \log(C_t/C_{t-1})$ is normal with these moments, so $g \sim \mathcal{N}(\mu_g, s_g^2)$ with $s_g^2 = \log(1 + .00127/1.018^2)$ and $\mu_g = \log 1.018 - s_g^2/2$, i.e. $\mu_g = .017228$, $s_g = .034996$. Then $m_t = \beta e^{-\gamma g_t}$ is lognormal and, as in (14.6.6)–(14.6.7),

$$E(m) = \beta \exp(-\gamma \mu_g + \tfrac{1}{2} \gamma^2 s_g^2), \quad \sigma(m) = E(m) [\exp(\gamma^2 s_g^2) - 1]^{1/2}.$$

As a second method we use a symmetric two-point distribution $C_t/C_{t-1} \in \{1.018 \pm .0356\}$ with equal probabilities (the marginal distribution of Mehra and Prescott's Markov chain), for which the moments of m are exact:

	lognormal		two-point		σ_{HJ} at $E(m)$	
γ	$E(m)$	$\sigma(m)$	$E(m)$	$\sigma(m)$	bills 1.02	bills 1.010
0	0.9900	0	0.9900	0	0.384	0.361
5	0.9223	0.1626	0.9223	0.1599	1.068	1.235
10	0.8860	0.3198	0.8850	0.2978	1.740	1.898

The two distributions give nearly the same numbers (a second-order Taylor expansion of $\beta G^{-\gamma}$ gives .9222/.1585 and .8841/.2899).

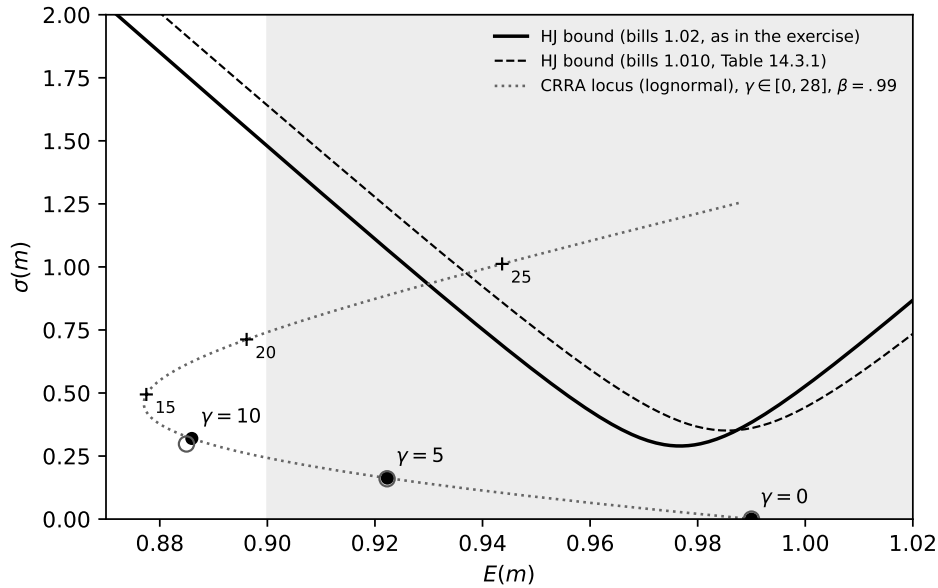


Figure S14.1. Hansen–Jagannathan bounds and CRRA discount factors ($\beta = .99$). Filled circles: lognormal consumption growth; open circles: two-point distribution; the dotted line traces the lognormal locus as γ rises from 0 to 28 (crosses at $\gamma = 15, 20, 25$).

Inference. None of the three points is admissible; all lie far below the bound. With $\gamma = 0$ the discount factor is constant. Raising γ raises $\sigma(m)$, roughly in proportion to $\gamma s_g E(m)$. But it also lowers $E(m)$, i.e. it raises the implied risk-free rate $1/E(m)$ (to 12.9% at $\gamma = 10$), because γ also governs intertemporal substitution and consumption grows on average. The point therefore moves

left, into the region where the bound is steep, and the gap widens: at $\gamma = 10$, $\sigma(m) = .32$, while the bound at $E(m) = .886$ is 1.74. This is the equity premium puzzle combined with Weil's risk-free rate puzzle (section 14.6). Two calculations make the point sharper.

- The ratio $\sigma(m)/E(m) = [\exp(\gamma^2 s_g^2) - 1]^{1/2}$ does not depend on β . Admissibility needs it to be at least .2967, i.e. $\gamma \geq 8.30$, and then β must put $E(m)$ at .977, which requires $\beta = 1.081 > 1$. This is Kocherlakota's observation (footnote 15 of the chapter).
- With $\beta = .99$, $E(m)$ falls until $\gamma = \mu_g/s_g^2 = 14.1$ (where $E(m) = .877$) and then rises, because the precautionary term $\frac{1}{2}\gamma^2 s_g^2$ takes over. The lognormal locus enters the admissible region only at $\gamma = 23.8$ (29.4 with the two-point distribution). Section 14.2 explains why such values are implausible; moreover, at such γ the answer depends on the tails of the distribution of consumption growth.

With a mean bill return of 1.010 these thresholds become $\gamma \geq 9.87$ with $\beta = 1.101$, and $\gamma = 24.5$ (30.3).

Exercise 14.2 *The term structure and regime switching*

Let $k \equiv \gamma\alpha_1 > 0$ and $\bar{r} \equiv \rho + \gamma(\alpha_0 + \mu) - \frac{1}{2}\gamma^2\tau^2$. The discount factor is $m_{t+1} = \beta \exp[-\gamma(\alpha_0 - \alpha_1 s_t + \varepsilon_{t+1})]$. Write $p_s \equiv \mathbb{P}(s_{t+1} = 1 \mid s_t = s)$, so $p_0 = 1 - \pi(0)$ and $p_1 = \pi(1)$.

(a) $E_t m_{t+1} = \beta e^{-\gamma(\alpha_0 - \alpha_1 s_t)} E e^{-\gamma\varepsilon} = \exp[-\rho - \gamma(\alpha_0 - \alpha_1 s_t) - \gamma\mu + \frac{1}{2}\gamma^2\tau^2]$. So the gross short rate is $R_1(s_t) = e^{r_1(s_t)}$ with

$$r_1(s) = \bar{r} - ks = \rho + \gamma(\alpha_0 - \alpha_1 s + \mu) - \frac{1}{2}\gamma^2\tau^2.$$

It equals ρ plus γ times expected log consumption growth, minus a precautionary term.

(b) The price of one unit at $t+2$ is $q_2(s_t) = E_t[m_{t+1}m_{t+2}]$. Given s_t , m_{t+1} depends only on ε_{t+1} , which is independent of s_{t+1} . So $q_2(s) = E_t[m_{t+1}] E_t[E_{t+1}m_{t+2}] = e^{-r_1(s)} \sum_{s'} \mathbb{P}(s' \mid s) e^{-r_1(s')}$, and the two-period rate per period, $r_2 = -\frac{1}{2} \log q_2$, is

$$r_2(s) = \frac{1}{2}r_1(s) - \frac{1}{2} \log E_s[e^{-r_1(s')}] = \bar{r} - \frac{1}{2}ks - \frac{1}{2} \log(1 - p_s + p_s e^k), \quad R_2(s) = e^{r_2(s)}.$$

(c) *All maturities.* The same argument gives $q_n(s) = e^{-n\bar{r}} h_n(s)$ with $h_n(s) = E_s \exp(k \sum_{j=0}^{n-1} s_{t+j})$, so

$$r_n(s) = \bar{r} - \frac{1}{n} \log h_n(s), \quad h_1 = D\mathbf{1}, \quad h_{n+1} = DP h_n, \quad D = \text{diag}(1, e^k),$$

where P is the transition matrix. *Claim:* $1 \leq h_n(0) \leq h_n(1) \leq e^{nk}$ for all n . The outer inequalities hold because $0 \leq \sum_j s_{t+j} \leq n$. For the middle one, the ratio $\varrho_n = h_n(1)/h_n(0)$ satisfies $\varrho_1 = e^k$ and $\varrho_{n+1} = f(\varrho_n)$ with

$$f(\varrho) = e^k \frac{1 - \pi(1) + \pi(1)\varrho}{\pi(0) + (1 - \pi(0))\varrho}, \quad f(1) = e^k.$$

f is a Möbius map: increasing if $\pi(0) + \pi(1) > 1$, decreasing if $\pi(0) + \pi(1) < 1$, and constant ($= e^k$) if $\pi(0) + \pi(1) = 1$. If it is increasing, $\varrho \geq 1$ implies $f(\varrho) \geq f(1) = e^k \geq 1$. If it is decreasing, f maps $[1, e^k]$ into $[f(e^k), e^k] \subset [1, e^k]$, because $f(e^k) \geq 1$ (its numerator is at least e^k and its denominator at most e^k). By induction $\varrho_n \geq 1$ for all n . (A grid check in `ch14.py` over 625 chains, four values of k and 119 maturities finds no violation.) Consequences:

- *Levels.* $r_n(0) - r_n(1) = \frac{1}{n} \log \varrho_n \in (0, k)$ for $n \geq 2$ (strictly, when $0 < \pi(0), \pi(1) < 1$): yields of every maturity are higher in good times, but they move less than the short rate. As $n \rightarrow \infty$, h_n becomes proportional to the Perron vector of DP , so $\frac{1}{n} \log \varrho_n \rightarrow 0$: the longest yields are acyclical, $r_\infty = \bar{r} - \log \lambda_{\max}(DP)$ in both states.
- *Slope.* $r_n(0) - r_1(0) = -\frac{1}{n} \log h_n(0) \leq 0$ and $r_n(1) - r_1(1) = k - \frac{1}{n} \log h_n(1) \geq 0$. The yield curve is inverted at peaks (rates are expected to fall) and upward sloping in troughs: the slope is countercyclical.
- *Term premia.* By Jensen's inequality, $r_n(s) \leq \bar{r} - \frac{k}{n} E_s \sum_j s_{t+j} = \frac{1}{n} \sum_{j < n} E_s r_1(s_{t+j})$: long yields lie below the average expected short rate (a convexity effect). There is no *risk* premium, however. The one-period holding return on any bond depends only on s_{t+1} , which is independent of m_{t+1} given s_t , so every bond's expected gross holding return equals $R_1(s_t)$. The expectations hypothesis holds exactly for gross holding returns.

(d) Procyclical: $r_1(0) - r_1(1) = \gamma \alpha_1 > 0$. In good times consumption is expected to grow fast; the rate must be high to dissuade the representative agent from borrowing against that growth (the inverse of the intertemporal elasticity of substitution is γ).

(e) There is a definite answer: by (c), long rates are procyclical too, whatever the parameters, but less so than short rates. How much less depends on the permanence of the cycle, which we measure by the autocorrelation $\lambda \equiv \pi(0) + \pi(1) - 1$ of s_t (equivalently, the expected durations $1/[1 - \pi(s)]$ of the regimes). For the two-period rate,

$$r_2(0) - r_2(1) = \frac{1}{2} [k + \log(1 - p_1 + p_1 e^k) - \log(1 - p_0 + p_0 e^k)],$$

and $\log(1 - p + p e^k)$ increases in p . So the long rate moves by more than half as much as the short rate iff $p_1 > p_0$, i.e. iff $\lambda > 0$. In the limit of permanent regimes ($\pi(0), \pi(1) \rightarrow 1$) it moves one for one with the short rate; with i.i.d. regimes ($\lambda = 0$) exactly half as much; when the regime alternates every period ($\pi(0), \pi(1) \rightarrow 0$) not at all. Long rates average expected future short rates, and only a persistent cycle moves those expectations. An illustration with $\beta = .99$, $\gamma = 2$, $\alpha_0 = .03$, $\alpha_1 = .02$, $\mu = .005$, $\tau = .02$ ($k = .04$; $r_1 = 7.925\%$ and 3.925% in the two states):

$(\pi(0), \pi(1)), \lambda$	$r_n(0) - r_n(1)$ (percentage points)					r_∞ (%)
	$n = 1$	2	5	10	40	
(.9, .8), +.7	4	3.403	2.231	1.310	0.338	6.487
(.6, .4), 0	4	2.000	0.800	0.400	0.100	6.306
(.3, .2), -.5	4	0.998	0.550	0.266	0.067	6.052

A Monte Carlo evaluation of the two- and five-period bond prices agrees with the formulas to within sampling error (e.g. $r_5(1) = 5.0773\%$ versus 5.0771%).

Exercise 14.3 Growth slowdowns and stock market crashes

Remark. The statement writes the growth factor as γ in “ $d_{t+1} = \gamma d_t$ ” and in “ $d_{T-1} = d_T = \gamma^{(T-1)}$ ”, but calls it ν elsewhere, and γ is also the risk-aversion parameter. The growth factor is ν throughout. We also assume that the tree is growing at date 0 ($d_{-1} = 1/\nu$).

(a) A competitive equilibrium is an ex-dividend share price p_t , a function of the history of dividends, and an allocation $\{c_t, s_{t+1}\}$ such that: (i) given prices, the representative household chooses $\{c_t, s_{t+1}\}$ to maximize $E_0 \sum \beta^t u(c_t)$ subject to $c_t + p_t s_{t+1} \leq (p_t + d_t)s_t$, $s_0 = 1$, and a no-Ponzi condition; (ii) markets clear: $c_t = d_t$ and $s_{t+1} = 1$. The Euler equation and the transversality condition at the equilibrium allocation give

$$p_t = E_t \left[\beta \left(\frac{d_{t+1}}{d_t} \right)^{-\gamma} (p_{t+1} + d_{t+1}) \right], \quad \text{so} \quad p_t = E_t \sum_{j \geq 1} \beta^j \left(\frac{d_{t+j}}{d_t} \right)^{-\gamma} d_{t+j}.$$

(b) There are two regimes: growing (G) and stopped (S). In S the dividend is constant forever, so $p_t = w_S d_t$ with $w_S = \beta/(1 - \beta)$. In G the distribution of future growth rates is the same at every date, so $p_t = w_G d_t$ with a constant price-dividend ratio w_G . Let $x \equiv \nu^{1-\gamma}$. The Euler equation gives $w_G = \beta \pi x (1 + w_G) + \beta (1 - \pi) (1 + w_S)$, so

$$w_G = \frac{\beta \pi x + \beta (1 - \pi) / (1 - \beta)}{1 - \beta \pi x}, \quad w_G - w_S = \frac{\beta \pi (x - 1)}{(1 - \beta)(1 - \beta \pi x)}.$$

w_G is finite because $\beta \pi x < \beta \nu^{1-\gamma} < 1$. (Summing $\sum_j \beta^j E(d_{t+j}/d_t)^{1-\gamma}$ directly gives the same numbers.) Hence

$$p_t = w_G \nu^t \quad (t < T), \quad p_t = w_S \nu^{T-1} \quad (t \geq T).$$

The price is *not* a function of the current dividend alone: at $T - 1$ and at T the dividend is the same, ν^{T-1} , but the prices differ unless $\gamma = 1$. It depends on (d_t, d_{t-1}) , i.e. on the current dividend and on whether it grew last period; that pair is a Markov state. With $\gamma = 1$, $w_G = w_S$ and $p_t = \beta d_t / (1 - \beta)$.

(c) $p_{T-1} - p_T = (w_G - w_S) \nu^{T-1}$, which is positive iff $x > 1$, i.e. iff $\gamma < 1$. The percentage change at T is $p_T/p_{T-1} - 1 = w_S/w_G - 1$, with $w_G/w_S - 1 = \pi(x - 1)/(1 - \beta \pi x)$. News that growth has stopped has two effects. A *cash-flow effect*: future dividends will be lower, which lowers the price. A *discount-rate effect*: expected consumption growth falls, so the interest rate falls from $R_G^f = [\beta(\pi \nu^{-\gamma} + 1 - \pi)]^{-1}$ to $R_S^f = 1/\beta$ and future dividends are discounted less. With CRRA preferences the intertemporal elasticity of substitution is $1/\gamma$. If $\gamma < 1$ the cash-flow effect dominates and the market crashes; if $\gamma > 1$ the discount-rate effect dominates and the price *jumps up*; with log utility the two cancel. An example with $\beta = .96$, $\nu = 1.02$, $\pi = .95$ ($w_S = 24$, $R_S^f = 1.0417$):

γ	w_G	$p_T/p_{T-1} - 1$	gross return at T	R_G^f
0.5	26.87	-10.7%	0.930	1.0515
1	24.00	0	1.042	1.0614
2	19.78	+21.3%	1.264	1.0816
5	12.97	+85.0%	1.927	1.1441

So in this model a growth slowdown produces a crash only if the intertemporal elasticity of substitution exceeds one. The crash is then larger the more surprising the slowdown ($w_G/w_S - 1$ increases with π). Such crashes are rational responses to news, with no bubble.

(d) After the switch the market value stops growing: it rose at the rate ν before and is flat afterwards, at $w_S \nu^{T-1}$. At the switch itself it jumps by $w_S/w_G - 1$: down if $\gamma < 1$, up if $\gamma > 1$. So a pattern of a crash followed by a long stagnation, as in Japan after 1990, is consistent with the model only if the elasticity of intertemporal substitution exceeds one. With the conventional $\gamma > 1$ the model predicts a boom when growth slows. To reconcile high risk aversion with this evidence one needs preferences that separate risk aversion from intertemporal substitution (Epstein–Zin, section 14.7), as in the long-run risk literature.

Exercise 14.4 *The term structure and consumption*

Let $c_t = \log C_t$, $\kappa = \log C^*$ and $e_{t+1} = \log \varepsilon_{t+1} \sim \mathcal{N}(-s^2/2, s^2)$ with $s^2 = \log(1 + \sigma^2)$ (this makes $E\varepsilon = 1$, $\text{var } \varepsilon = \sigma^2$). Then $c_{t+1} = \kappa + \varphi c_t + e_{t+1}$, a stationary AR(1) with mean $\bar{c} = (\kappa - s^2/2)/(1 - \varphi)$.

(a) Let p_t be the ex-dividend price of a tree and Q_{jt} the price at t of a claim to one unit of consumption at $t + j$ ($Q_{0t} \equiv 1$). A household holding s_t trees and $b_{j,t-1}$ bonds bought at $t - 1$ with j periods to maturity faces

$$c_t + p_t s_{t+1} + \sum_{j \geq 1} Q_{jt} b_{jt} \leq (p_t + C_t) s_t + \sum_{j \geq 1} Q_{j-1,t} b_{j,t-1}$$

(each bond bought as a j -period bond is a $(j - 1)$ -period bond next period), plus a no-Ponzi condition. A competitive equilibrium is a set of price functions p_t, Q_{jt} of the state C_t and an allocation such that the allocation solves every household's problem at those prices and markets clear: $c_t = C_t$, $s_{t+1} = 1$, and $b_{jt} = 0$ (bonds are in zero net supply). The Euler equations then give $Q_{jt} = E_t[\beta(C_{t+1}/C_t)^{-\gamma} Q_{j-1,t+1}] = E_t[\beta^j (C_{t+j}/C_t)^{-\gamma}]$.

(b) From the AR(1), $E_t c_{t+j} - c_t = (1 - \varphi^j)(\bar{c} - c_t)$ and $\text{var}_t(c_{t+j}) = s^2(1 - \varphi^{2j})/(1 - \varphi^2)$. With the gross yield per period defined by $Q_{jt} = \tilde{R}_{jt}^{-j}$,

$$\log \tilde{R}_{jt} = -\frac{\log Q_{jt}}{j} = \rho + \underbrace{\frac{\gamma(1 - \varphi^j)}{j}}_{\equiv b_j} (\bar{c} - c_t) - \frac{\gamma^2 s^2}{2j} \frac{1 - \varphi^{2j}}{1 - \varphi^2}.$$

For $j = 1$: $\log \tilde{R}_{1t} = \rho + \gamma E_t(c_{t+1} - c_t) - \frac{1}{2} \gamma^2 s^2$. As $j \rightarrow \infty$, $\log \tilde{R}_{jt} \rightarrow \rho$ in every state. When C_t is above its mean, consumption is expected to fall, short rates are low and the yield curve rises toward ρ ; below the mean, short rates are high and the curve falls. (`ch14.py` verifies that these prices satisfy $Q_{j+1,t} = E_t[\beta(C_{t+1}/C_t)^{-\gamma} Q_{j,t+1}]$ by quadrature, to 3×10^{-16} .)

(c) Disagree. $\text{var}(\log \tilde{R}_{jt}) = b_j^2 \text{var}(c_t) = b_j^2 s^2/(1 - \varphi^2)$, and $b_j = \gamma(1 - \varphi) \cdot \frac{1}{j} \sum_{i=0}^{j-1} \varphi^i$ is $\gamma(1 - \varphi)$ times an average of the first j terms of the decreasing sequence φ^i , so it is strictly decreasing in j (e.g. with $\gamma = 2$, $\varphi = .9$: $b_1 = .2$, $b_2 = .19$, $b_5 = .164$, $b_{10} = .130$, $b_{40} = .049$). The short rate is the *most* volatile yield. Long yields average expected future short rates, and mean reversion makes distant expectations move little. (Holding a long bond for one period is riskier than holding a short one, but that is a statement about prices and holding returns, not about the variability of yields.)

(d) Agree. $\partial \log \tilde{R}_{jt} / \partial c_t = -b_j < 0$, largest in absolute value at $j = 1$ and vanishing as $j \rightarrow \infty$.

(e) Disagree. The negative relation between rates and consumption is an equilibrium implication of the Euler equation, with causation running from consumption to rates: a high C_t means consumption is expected to revert, i.e. to grow slowly, and a low rate is needed to stop people from wanting to save. In this endowment economy consumption is exogenous; nothing the Fed does to interest rates changes it. Reading a policy lever into an equilibrium correlation is the error that the Lucas critique warns against.

(f) Agree, in the limit. As $\varphi \rightarrow 1$, $b_j \rightarrow 0$ for every fixed j , and $\log \tilde{R}_{jt} \rightarrow \rho + \gamma(\kappa - s^2/2) - \frac{1}{2}\gamma^2 s^2$ for all j : the term structure becomes flat and constant. Interest rates depend on expected consumption *growth*, not on its level. When shocks are permanent (a random walk in $\log C$), a shock moves the whole expected path of consumption in parallel and leaves expected growth unchanged.

Exercise 14.5 *Ambiguity averse multiplier preferences*

Equations (1)–(3) refer to the numbered displays of the exercise in the book.

(a) Under the guess, $U_{t+1} = k_0 + k_1(c_t + \mu) + k_1\sigma_c\varepsilon_{t+1}$ is conditionally normal, so

$$-\beta\theta \log E_t \exp(-U_{t+1}/\theta) = \beta[k_0 + k_1(c_t + \mu)] - \frac{\beta k_1^2 \sigma_c^2}{2\theta}.$$

Matching $k_0 + k_1 c_t$ with c_t plus this expression: the coefficients on c_t give $k_1 = 1 + \beta k_1$, and the constants give $(1 - \beta)k_0 = \beta k_1 \mu - \beta k_1^2 \sigma_c^2 / (2\theta)$. Hence

$$k_1 = \frac{1}{1 - \beta}, \quad k_0 = \frac{\beta}{(1 - \beta)^2} \left[\mu - \frac{\sigma_c^2}{2\theta(1 - \beta)} \right],$$

which is (14.7.9).

(b) $E_t U_{t+1} = k_0 + k_1(c_t + \mu)$ and $\text{var}_t(U_{t+1}) = k_1^2 \sigma_c^2$, so the right side of (3) is $c_t + \beta[k_0 + k_1(c_t + \mu)] - \beta k_1^2 \sigma_c^2 / (2\theta)$, identical to the right side of (1) computed in (a). More generally, (1) and (3) coincide whenever U_{t+1} is conditionally Gaussian, by the formula for T_t in the preamble.

The plans. Treat the numbers at the nodes as c_t for $t = 0, 1, 2$, with nothing after $t = 2$. In plan A, $c_0 = c_1 = 1$, and at $t = 1$ the consumer learns whether $c_2 = 2$ or $c_2 = 0$ (probability .5 each). Plan B has the same payoffs, but nothing is learned at $t = 1$, and $c_2 \in \{2, 0\}$ is revealed at $t = 2$. In plan C, c_1 and c_2 are independent draws from $\{2, 1\}$; in plan D, $c_1 \in \{2, 1\}$ and $c_2 = c_1$. All four plans have the same marginal distribution of c_t at each date within each pair, so with $\theta = \infty$ (expected utility) the consumer is indifferent within each pair. Applying (3) backward:

$$\begin{aligned} U_0^A &= 1 + \beta + \beta^2 - \frac{\beta^3}{2\theta}, & U_0^B &= 1 + \beta + \beta^2 - \frac{\beta^2}{2\theta}, \\ U_0^C &= 1 + \frac{3}{2}\beta + \frac{3}{2}\beta^2 - \frac{\beta(1 + \beta)}{8\theta}, & U_0^D &= 1 + \frac{3}{2}\beta + \frac{3}{2}\beta^2 - \frac{\beta(1 + \beta)^2}{8\theta}. \end{aligned}$$

(In A, $U_1 \in \{1 + 2\beta, 1\}$ has variance β^2 , penalized at $t = 0$. In B, $U_1 = 1 + \beta - \beta/(2\theta)$ in both nodes, because $\text{var}_1(c_2) = 1$ is penalized at $t = 1$. In C, $\text{var}_0(U_1) = \text{var}_1(U_2) = \frac{1}{4}$; in D, $U_1 \in \{2 + 2\beta, 1 + \beta\}$ has variance $(1 + \beta)^2/4$.) So

$$U_0^A - U_0^B = \frac{\beta^2(1 - \beta)}{2\theta} > 0, \quad U_0^C - U_0^D = \frac{\beta^2(1 + \beta)}{8\theta} > 0.$$

Interpretation. Representation (3) charges a penalty, $\beta/(2\theta)$ per unit, on the conditional variance of *continuation utility*. With early resolution the risk about c_2 enters U_1 multiplied by β ; its variance, β^2 , is penalized at $t = 0$, for a total of $\beta \cdot \beta^2/(2\theta)$. With late resolution the same risk is penalized at $t = 1$, where it enters undiscounted, for a total of $\beta \cdot \beta/(2\theta)$. Because the variance scales with the *square* of the discount factor, resolving risk early is cheaper. With persistent risk, one shock moves both c_1 and c_2 , so it moves U_1 by $(1 + \beta)$ times the shock, and the variance of continuation utility, $(1 + \beta)^2/4$, includes the cross term $2\beta/4$. With i.i.d. risk the two shocks are penalized separately, $(1 + \beta)/4$ in total after discounting. Risk that accumulates over time is disliked more. The exact operator (1) applied to these binary lotteries (for which (3) is only a second-order approximation) gives the same ranking. For $\beta = .95$, $\theta = 1$: $U_0^A = 2.476 > U_0^B = 2.461$ and $U_0^C = 3.556 > U_0^D = 3.385$ (with (3): 2.424, 2.401, 3.547, 3.327).

(c) With $U_t = k_0 + k_1 \log C_t$, $\partial U_t / \partial C_t = k_1 / C_t$: this is the effect of raising C_t together with the whole future consumption path, which under the random walk (2) is proportional to C_t . For the dependence on C_{t+1} , let ε take values ε_j with probabilities P_j , as in appendix B, and write $U_{t+1,j}$ for continuation utility in state j . Then

$$\frac{\partial U_t}{\partial C_{t+1,j}} = -\beta \theta \frac{P_j \exp(-U_{t+1,j}/\theta) (-1/\theta)}{\sum_k P_k \exp(-U_{t+1,k}/\theta)} \frac{\partial U_{t+1,j}}{\partial C_{t+1,j}} = \beta P_j \frac{\exp(-U_{t+1,j}/\theta)}{E_t \exp(-U_{t+1}/\theta)} \frac{k_1}{C_{t+1,j}},$$

using $\partial U_{t+1} / \partial C_{t+1} = k_1 / C_{t+1}$. Dividing by the probability P_j , i.e. expressing the derivative per unit of probability, gives the formula in the exercise. With a continuous ε the same statement reads: a perturbation $\delta C_{t+1}(\varepsilon)$ changes U_t by $E_t[\beta g k_1 \delta C_{t+1} / C_{t+1}]$, where $g = \exp(-U_{t+1}/\theta) / E_t \exp(-U_{t+1}/\theta)$. The ratio of the two derivatives is the stochastic discount factor (14.7.12), $m_{t+1} = \beta(C_t / C_{t+1})g$; the factor k_1 cancels. (Perturbing only C_t , or only C_{t+1} , gives $1/C_t$ and $\beta P_j g_j / C_{t+1,j}$ instead, and the same ratio.)

(d) By (14.7.5), $(1 - \beta)(1 - \gamma) = -1/\theta$, so $g = \exp(-U_{t+1}/\theta) / E_t \exp(-U_{t+1}/\theta)$. Since $U_{t+1} = \text{const}_t + k_1 \sigma_c \varepsilon_{t+1}$,

$$g(\varepsilon_{t+1}) = \frac{\exp(-k_1 \sigma_c \varepsilon_{t+1} / \theta)}{\exp(k_1^2 \sigma_c^2 / (2\theta^2))} = \exp\left(w \varepsilon_{t+1} - \frac{w^2}{2}\right), \quad w = -\frac{k_1 \sigma_c}{\theta} = -\frac{\sigma_c}{\theta(1 - \beta)}.$$

Equivalently $w = \sigma_c(1 - \gamma)$. (ch14.py confirms (a), (b) and (d) by Gauss–Hermite quadrature.)

Exercise 14.6 *Exponential affine stochastic discount factor*

Equation (2) refers to a numbered display of the exercise in the book.

(a) Given z_t , $\log R_{j,t+1} \sim \mathcal{N}(\nu_t - \frac{1}{2} \alpha'_t \alpha_t, \alpha'_t \alpha_t)$, so $E_t R_{j,t+1} = \exp(\nu_t - \frac{1}{2} \alpha'_t \alpha_t + \frac{1}{2} \alpha'_t \alpha_t) = e^{\nu_t}$.

(b) $\log(m_{t+1} R_{j,t+1}) = \nu_t - r_t - \frac{1}{2} \Lambda'_t \Lambda_t - \frac{1}{2} \alpha'_t \alpha_t + (\alpha_t - \Lambda_t)' \varepsilon_{t+1}$ is conditionally normal, so

$$E_t(m_{t+1} R_{j,t+1}) = \exp\left(\nu_t - r_t - \frac{1}{2} \Lambda'_t \Lambda_t - \frac{1}{2} \alpha'_t \alpha_t + \frac{1}{2} (\alpha_t - \Lambda_t)' (\alpha_t - \Lambda_t)\right) = \exp(\nu_t - r_t - \alpha'_t \Lambda_t).$$

Setting this to one gives (2).

(c) $\nu_t(j) - r_t = \log E_t R_{j,t+1} - \log R_t^f$ is the (log) expected excess return, and (2) says it equals $\sum_i \alpha_{it}(j) \Lambda_{it}$: the exposure of the return to each risk ε_i times the price of that risk. Equivalently $\nu_t - r_t = -\text{cov}_t(\log m_{t+1}, \log R_{j,t+1})$. Only exposure to shocks that move the discount factor earns a premium; Λ_t is the vector of market prices of risk (the maximal Sharpe ratio is approximately

$|\Lambda_t|$, by (14.11.6)). Risk premia vary over time if either the prices ($\Lambda_z \neq 0$) or the exposures ($\alpha_z \neq 0$) depend on the state; they are then quadratic in z_t .

(d) Let log consumption growth be affine in the state with homoskedastic shocks, $c_{t+1} - c_t = \mu_c + \iota' z_t + \sigma'_c \varepsilon_{t+1}$, with z_t following the VAR (for example z_t may contain $c_t - c_{t-1}$ itself). Then $\log m_{t+1} = \log \beta - \gamma \mu_c - \gamma \iota' z_t - \gamma \sigma'_c \varepsilon_{t+1}$, which has the required form with

$$\Lambda_0 = \gamma \sigma_c, \quad \Lambda_z = 0, \quad \delta_1 = \gamma \iota, \quad \delta_0 = \rho + \gamma \mu_c - \frac{1}{2} \gamma^2 \sigma'_c \sigma_c.$$

The i.i.d. case $\iota = 0$ is (14.6.8)–(14.6.9). CRRA delivers only *constant* risk prices. Making σ_c depend on z_t would make Λ_t affine, but then $r_t = \rho + \gamma(\mu_c + \iota' z_t) - \frac{1}{2} \gamma^2 |\sigma_c(z_t)|^2$ would be quadratic rather than affine in z_t .

Exercise 14.7 Value function for CRRA

Equation (1) refers to a numbered display of the exercise in the book.

(a) $C_t^{1-\gamma} = C_0^{1-\gamma} \exp[(1-\gamma)(\mu t + \sigma_c \sum_{s \leq t} \varepsilon_s)]$, so $E_0 C_t^{1-\gamma} = C_0^{1-\gamma} \exp[t((1-\gamma)\mu + \frac{1}{2}(1-\gamma)^2 \sigma_c^2)]$. Let

$$\kappa \equiv \beta \exp[(1-\gamma)\mu + \frac{1}{2}(1-\gamma)^2 \sigma_c^2].$$

The terms of (1) are $\kappa^t C_0^{1-\gamma} / (1-\gamma)$, so

$$U_0 = \frac{C_0^{1-\gamma}}{(1-\gamma)(1-\kappa)} \quad \text{if} \quad \log \beta + (1-\gamma)\mu + \frac{1}{2}(1-\gamma)^2 \sigma_c^2 < 0.$$

If $\kappa \geq 1$ the sum diverges, to $+\infty$ if $\gamma < 1$ and to $-\infty$ if $\gamma > 1$. For $\gamma = 0$ the condition is $\beta e^{\mu + \sigma_c^2/2} < 1$. For $\gamma = 1$ (utility $\log C$), $U_0 = \sum_t \beta^t (\log C_0 + \mu t) = \log C_0 / (1-\beta) + \beta \mu / (1-\beta)^2$, finite for every μ, σ_c .

(b) Guess $U_t = AC_t^{1-\gamma}$: $AC_t^{1-\gamma} = C_t^{1-\gamma} / (1-\gamma) + \beta AC_t^{1-\gamma} E_t e^{(1-\gamma)(\mu + \sigma_c \varepsilon_{t+1})}$ gives $A = 1 / [(1-\gamma)(1-\kappa)]$. The recursion has other solutions: $U_t + B\beta^{-t}$ satisfies it for any B . Iterating the recursion T times gives $U_0 = E_0 \sum_{t < T} \beta^t u(C_t) + \beta^T E_0 U_T$, so the solution that satisfies (1) is the one with $\lim_{T \rightarrow \infty} \beta^T E_0 U_T = 0$. For $U_t = AC_t^{1-\gamma}$ this term is $AC_0^{1-\gamma} \kappa^T \rightarrow 0$ exactly when $\kappa < 1$. The solutions with $B \neq 0$ violate the terminal condition.

Exercise 14.8 Lucas tree economy

We read the timing as usual: whoever buys the tree at t pays p_t and receives $y_{t+1} + p_{t+1}$ at $t+1$, so $p_t = E_t[m_{t+1}(y_{t+1} + p_{t+1})]$.

(a) $m_{t+1} = \beta u'(C_{t+1}) / u'(C_t) = \beta y_t / y_{t+1}$, which takes the four values $\beta y_s / y_{s'}$. A one-period risk-free bond pays 1 in both states, so its price is $E_t m_{t+1}$ and its gross return is

$$R^f(s) = \left[\beta \sum_{s'} P_{ss'} \frac{y_s}{y_{s'}} \right]^{-1}, \quad \text{i.e.} \quad R^f(L) = \left[\beta \left(\pi_L + (1-\pi_L) \frac{y_L}{y_H} \right) \right]^{-1},$$

$$R^f(H) = \left[\beta \left((1-\pi_H) \frac{y_H}{y_L} + \pi_H \right) \right]^{-1}.$$

(b) $p(s) = \beta y_s / (1-\beta)$. Check: $E_s[\beta \frac{y_s}{y'} (y' + \frac{\beta}{1-\beta} y')] = \beta y_s (1 + \frac{\beta}{1-\beta}) = \frac{\beta y_s}{1-\beta}$. With log utility the income and substitution effects of dividend news cancel, and the price-dividend ratio is constant.

(c) $R_{t+1} = (y_{t+1} + p_{t+1})/p_t = y_{t+1}/(\beta y_t)$: the return is $1/\beta$ times dividend growth. It takes the values $1/\beta$ (no change), $y_H/(\beta y_L)$ (switch from L to H) and $y_L/(\beta y_H)$ (switch from H to L), and it is serially correlated through s_t . Its conditional mean is $E_s R = \beta^{-1} E_s(y'/y_s)$, which exceeds $R^f(s) = [\beta E_s(y_s/y')]^{-1}$ by Jensen's inequality ($EX \cdot E(1/X) \geq 1$). The equity premium is positive because the return is high exactly when consumption is high.

(d) A unit of consumption at $\bar{t} + j$ is worth $E_{\bar{t}}[\beta^j y_{\bar{t}}/y_{\bar{t}+j}] = \beta^j y_s (P^j u^{-1})_s$ at $\bar{t}$, where $u^{-1} \equiv (1/y_L, 1/y_H)'$ and $s = s_{\bar{t}}$. Summing over $j \geq 1$,

$$B(s; \eta) = \eta y_s [(I - \beta P)^{-1} \beta P u^{-1}]_s, \quad (I - \beta P)^{-1} = \frac{1}{\Delta} \begin{bmatrix} 1 - \beta \pi_H & \beta(1 - \pi_L) \\ \beta(1 - \pi_H) & 1 - \beta \pi_L \end{bmatrix},$$

with $\Delta = (1 - \beta)[1 - \beta(\pi_L + \pi_H - 1)]$. B is proportional to η . Although the coupons are certain, the value of the bonds varies with the state, because interest rates do.

(e) By the same calculation, the claim to $y_t - \eta$ is worth

$$E(s; \eta) = \sum_{j \geq 1} \beta^j y_s [1 - \eta(P^j u^{-1})_s] = \frac{\beta y_s}{1 - \beta} - B(s; \eta).$$

(f) $B(s; \eta) + E(s; \eta) = p(s)$ for every η and s : the Modigliani–Miller theorem. Pricing is linear, and the bonds and the equity together are exactly the tree's dividend stream, so how the stream is split does not change its value, and the zero-net-worth entrepreneur just breaks even. Example ($\beta = .95$, $y_L = 1$, $y_H = 1.25$, $\pi_L = .8$, $\pi_H = .9$, $\eta = .5$; closed forms checked against direct solution of $p = M(y + p)$ with $M_{ss'} = \beta P_{ss'} y_s / y_{s'}$):

state	R^f	$E_s R$	p	B	E	$B + E$
L	1.0965	1.1053	19.00	8.366	10.634	19.00
H	1.0270	1.0316	23.75	10.209	13.541	23.75

Interest rates are high in the low state, where consumption is expected to rise, so the bonds are worth less there.

Exercise 14.9 Long-run risk, I

(Here ρ is the persistence of z_t , so we write $-\log \beta$ explicitly.)

(a) $m_{t+1} = \beta \exp[-\gamma(c_{t+1} - c_t)] = \beta \exp[-\gamma(\mu + z_t + \sigma_c \varepsilon_{t+1})]$.

(b) $E_t m_{t+1} = \beta \exp[-\gamma(\mu + z_t) + \frac{1}{2} \gamma^2 \sigma_c^2]$: the price of a one-period risk-free bond. The short rate is

$$r_t = -\log \beta + \gamma(\mu + z_t) - \frac{1}{2} \gamma^2 \sigma_c^2,$$

high when expected consumption growth $\mu + z_t$ is high, and it inherits the persistence of z_t . The price of risk of one-period returns is $\gamma \sigma_c$: the discount factor loads only on the shock to current consumption.

(c) $c_{t+2} - c_t = 2\mu + (1 + \rho)z_t + (\sigma_c + \sigma_z)\varepsilon_{t+1} + \sigma_c \varepsilon_{t+2}$, because ε_{t+1} moves both c_{t+1} and z_{t+1} . So

$$E_t(m_{t+1} m_{t+2}) = \beta^2 \exp\left(-\gamma[2\mu + (1 + \rho)z_t] + \frac{1}{2} \gamma^2 [\sigma_c^2 + (\sigma_c + \sigma_z)^2]\right),$$

the price of a two-period zero-coupon bond. Its yield per period is

$$y_t(2) = -\log \beta + \gamma\mu + \frac{1}{2}\gamma(1+\rho)z_t - \frac{1}{4}\gamma^2[\sigma_c^2 + (\sigma_c + \sigma_z)^2].$$

Compared with r_t , it responds less to z_t ($\frac{1}{2}(1+\rho) < 1$), and its precautionary term is larger when $\sigma_z > 0$, because a shock that lowers consumption today also lowers expected growth, so two-period consumption growth is riskier than two one-period steps.

(d) In general, $\sum_{i=1}^n (c_{t+i} - c_{t+i-1}) = n\mu + \frac{1-\rho^n}{1-\rho}z_t + \sum_{k=1}^n [\sigma_c + \sigma_z \frac{1-\rho^{n-k}}{1-\rho}] \varepsilon_{t+k}$, so the n -period yield is

$$y_t(n) = -\log \beta + \gamma\mu + \gamma \frac{1-\rho^n}{n(1-\rho)} z_t - \frac{\gamma^2}{2n} \sum_{k=0}^{n-1} \left[\sigma_c + \sigma_z \frac{1-\rho^k}{1-\rho} \right]^2.$$

(i) All yields move with z_t , with loadings $\gamma(1-\rho^n)/[n(1-\rho)]$ that decline with maturity: short rates are the most volatile. (ii) The slope $y_t(n) - y_t(1)$ falls when z_t rises: good news about growth flattens or inverts the curve. (iii) With $\sigma_z > 0$ the average curve slopes *down* toward $-\log \beta + \gamma\mu - \frac{1}{2}\gamma^2(\sigma_c + \sigma_z/(1-\rho))^2$: long bonds pay off when growth prospects deteriorate, so they hedge long-run consumption risk and carry negative term premia. With $\beta = .995$, $\gamma = 2$, $\mu = .005$, $\rho = .95$, $\sigma_c = .005$, $\sigma_z = .0005$ (quarterly), the yields at $z_t = 0$ are 1.496, 1.495, 1.486, 1.466, 1.459% for $n = 1, 4, 20, 100, 400$ (limit 1.456%), and the loadings on z_t are 2, 1.85, 1.28, 0.40, 0.10. (A Monte Carlo check reproduces $\log E_t(m_{t+1}m_{t+2})$ and $\log E_t(m_{t+1}m_{t+2}m_{t+3})$ to six digits.)

(e) By the same algebra, $E_0 C_t^{1-\gamma} = C_0^{1-\gamma} \exp(a_t + b_t z_0)$, with $b_t = (1-\gamma) \frac{1-\rho^t}{1-\rho}$ and $a_t = (1-\gamma)\mu t + \frac{1}{2}(1-\gamma)^2 \sum_{k=0}^{t-1} [\sigma_c + \sigma_z \frac{1-\rho^k}{1-\rho}]^2$. So

$$U_0 = \frac{C_0^{1-\gamma}}{1-\gamma} \sum_{t \geq 0} \beta^t \exp(a_t + b_t z_0) \equiv C_0^{1-\gamma} f(z_0).$$

The summands in a_t converge geometrically to $\sigma_\infty^2 \equiv (\sigma_c + \sigma_z/(1-\rho))^2$, and b_t is bounded. So the terms behave like κ^t with

$$\kappa = \beta \exp\left[(1-\gamma)\mu + \frac{1}{2}(1-\gamma)^2(\sigma_c + \sigma_z/(1-\rho))^2\right],$$

and U_0 is finite iff $\kappa < 1$. This is the condition of exercise 14.7 with σ_c replaced by the long-run volatility of consumption growth $\sigma_c + \sigma_z/(1-\rho)$. For $\gamma = 1$, $U_0 = \sum_t \beta^t E_0 c_t$ is always finite. `ch14.py` checks that f satisfies $f(z) = \frac{1}{1-\gamma} + \beta E[e^{(1-\gamma)(\mu+z+\sigma_c\varepsilon)} f(\rho z + \sigma_z \varepsilon)]$ (the recursion $U_t = u(C_t) + \beta E_t U_{t+1}$) by quadrature; in the example above $\kappa = .990$.

Exercise 14.10 Long-run risk, II

(a) Under the guess,

$$U_{t+1} = k_0 + k_1(c_t + \mu) + (k_1 + \rho k_2)z_t + (k_1\sigma_c + k_2\sigma_z)\varepsilon_{t+1},$$

so $-\beta\theta \log E_t \exp(-U_{t+1}/\theta) = \beta[k_0 + k_1(c_t + \mu) + (k_1 + \rho k_2)z_t] - \beta(k_1\sigma_c + k_2\sigma_z)^2/(2\theta)$. Matching coefficients: $k_1 = 1 + \beta k_1$, $k_2 = \beta(k_1 + \rho k_2)$, and $(1-\beta)k_0 = \beta k_1\mu - \beta(k_1\sigma_c + k_2\sigma_z)^2/(2\theta)$. Hence

$$k_1 = \frac{1}{1-\beta}, \quad k_2 = \frac{\beta}{(1-\beta)(1-\beta\rho)}, \quad k_0 = \frac{\beta}{(1-\beta)^2} \left[\mu - \frac{\tilde{\sigma}^2}{2\theta(1-\beta)} \right], \quad \tilde{\sigma} \equiv \sigma_c + \frac{\beta\sigma_z}{1-\beta\rho}.$$

This is (14.7.9) with σ_c replaced by $\tilde{\sigma}$, the response of the present value $(1 - \beta) \sum_j \beta^j E_{t+1} c_{t+1+j}$ to the shock ε_{t+1} . Because ρ is close to one, $\tilde{\sigma}$ can be several times σ_c even though σ_z is small.

(b) By (14.7.12), $m_{t+1} = \beta e^{-(c_{t+1}-c_t)} g_{t+1}$ with $g_{t+1} = \exp(-U_{t+1}/\theta)/E_t \exp(-U_{t+1}/\theta) = \exp(w\varepsilon_{t+1} - \frac{1}{2}w^2)$ and

$$w = -\frac{k_1\sigma_c + k_2\sigma_z}{\theta} = -\frac{\tilde{\sigma}}{\theta(1-\beta)} = (1-\gamma)\tilde{\sigma}, \quad \gamma \equiv 1 + \frac{1}{\theta(1-\beta)}.$$

Thus

$$\log m_{t+1} = \log \beta - \mu - z_t - \frac{1}{2}w^2 - (\sigma_c - w)\varepsilon_{t+1}, \quad r_t = -\log \beta + \mu + z_t - \frac{1}{2}\sigma_c^2 + \sigma_c w.$$

The price of risk is $\sigma_c - w = \gamma\sigma_c + (\gamma - 1)\beta\sigma_z/(1 - \beta\rho)$. Unlike the time-separable case of exercise 14.9, where it is $\gamma\sigma_c$, news about long-run growth is priced, and it is amplified by $1/(1 - \beta\rho)$ (the mechanism of Bansal and Yaron). With $\beta = .995$, $\rho = .95$, $\sigma_c = .005$, $\sigma_z = .0005$ and $\gamma = 10$ ($\theta = 22.2$): $\tilde{\sigma} = .01409$, $w = -.127$, and the price of risk is .132, against $\gamma\sigma_c = .050$ without the long-run component. The worst-case model twists ε_{t+1} to $\mathcal{N}(w, 1)$, lowering both current growth and the persistent component z_{t+1} . (ch14.py verifies the value function and $E_t m_{t+1} = e^{-r_t}$ by quadrature.)

Exercise 14.11 *Stochastic volatility*

(a) The stochastic discount factor is the random variable m_{t+1} , measurable with respect to date- $(t+1)$ information, such that the date- t price of any payoff x_{t+1} is $E_t(m_{t+1}x_{t+1})$. For this consumer it is the intertemporal marginal rate of substitution $\beta u'(C_{t+1})/u'(C_t)$.

(b) $m_{t+1} = \beta \exp[-\gamma(\mu + \sigma_c(s_t)\varepsilon_{t+1})]$.

(c) $E_t m_{t+1} = \beta \exp[-\gamma\mu + \frac{1}{2}\gamma^2\sigma_c(s_t)^2]$, so $r(s) = -\log \beta + \gamma\mu - \frac{1}{2}\gamma^2\sigma_c(s)^2$. The risk-free rate is lower in the high-volatility state: a precautionary saving motive. The market price of risk, $[\exp(\gamma^2\sigma_c(s)^2) - 1]^{1/2} \approx \gamma\sigma_c(s)$, is higher there.

(d) Given s_t , ε_{t+1} is independent of s_{t+1} , so

$$E_t(m_{t+1}m_{t+2}) = E_t(m_{t+1}) E_t[E_{t+1}m_{t+2}] = \beta^2 e^{-2\gamma\mu + \frac{1}{2}\gamma^2\sigma_c(s_t)^2} \sum_{s'} P_{s_t s'} e^{\frac{1}{2}\gamma^2\sigma_c(s')^2}.$$

This is the price of a two-period bond: today's short-bond price times the expected price of a short bond tomorrow. There is no risk premium (only a convexity term), because volatility news is uncorrelated with the one-period discount factor.

(e) For any maturity, $q_n(s) = \beta^n e^{-n\gamma\mu} h_n(s)$ with $h_n = (DP)^{n-1} D \mathbf{1}$ and $D = \text{diag}(e^{\gamma^2\sigma_L^2/2}, e^{\gamma^2\sigma_H^2/2})$, so $y_n(s) = -\log \beta + \gamma\mu - \frac{1}{n} \log h_n(s)$. Factoring $e^{\gamma^2\sigma_L^2/2}$ out of D gives the structure of exercise 14.2 with $k = \frac{1}{2}\gamma^2(\sigma_H^2 - \sigma_L^2)$. So yields of *all* maturities are lower in the high-volatility state, the gap shrinks with maturity, and the yield curve slopes up in the high-volatility state (rates are expected to rise as volatility reverts) and down in the low-volatility state. Over time the whole curve shifts when the volatility regime switches, the short end moving most, while very long yields stay put. With $\beta = .995$, $\gamma = 10$, $\mu = .005$, $\sigma_L = .004$, $\sigma_H = .008$, $\pi_0 = .95$, $\pi_1 = .9$ (quarterly), the low-minus-high spread is 24.0, 22.2, 19.1, 7.7 and 1.6 basis points for $n = 1, 2, 4, 20, 100$, and $y_{20} - y_1$ is -5.5 basis points in the low- and +10.8 in the high-volatility state. (The levels, about

5.4% per quarter, reflect the risk-free rate puzzle, $\gamma\mu = .05$; only the differences matter here. A Monte Carlo check matches the two-period yields to 3×10^{-6} .)

(f) s_t becomes a hidden state that the consumer must infer. Let $\hat{\pi}_t = \mathbb{P}(s_t = 1 \mid c^t)$. After observing Δc_{t+1} , Bayes' rule and the chain give

$$\bar{\pi}_t = \frac{\hat{\pi}_t \phi_H(\Delta c_{t+1})}{\hat{\pi}_t \phi_H(\Delta c_{t+1}) + (1 - \hat{\pi}_t) \phi_L(\Delta c_{t+1})}, \quad \hat{\pi}_{t+1} = \bar{\pi}_t \pi_1 + (1 - \bar{\pi}_t)(1 - \pi_0),$$

where ϕ_s is the $\mathcal{N}(\mu, \sigma_s^2)$ density; the filter starts from $\hat{\pi}_0 = \bar{\pi}_0(1)$. The discount factor $\beta e^{-\gamma \Delta c_{t+1}}$ is unchanged as a function of Δc_{t+1} , but its conditional distribution becomes a mixture of normals. By the law of iterated expectations, prices are the full-information prices averaged over the posterior:

$$E_t m_{t+1} = \beta e^{-\gamma \mu} \left[(1 - \hat{\pi}_t) e^{\frac{1}{2} \gamma^2 \sigma_L^2} + \hat{\pi}_t e^{\frac{1}{2} \gamma^2 \sigma_H^2} \right],$$

and likewise for $E_t(m_{t+1} m_{t+2})$. The state becomes the continuous variable $\hat{\pi}_t \in (0, 1)$. The short rate moves smoothly between the two full-information values, and a large surprise $|\Delta c_{t+1} - \mu|$ raises $\hat{\pi}_{t+1}$ and lowers rates.

Exercise 14.12 *Unknown μ*

Remark. The printed law of motion contains a term z_t that is never defined in this exercise (it is carried over from the long-run risk exercises), and the statement mentions $\gamma \geq 0$, which plays no role with log utility. We set $z_t \equiv 0$; exercise 14.13 treats the case with z_t . We read the information assumption as: at date t the consumer has observed $y_s \equiv c_s - c_{s-1}$ for $s = 1, \dots, t$.

(a) With a normal prior and normal observations $y_s = \mu + \sigma_c \varepsilon_s$, the posterior after t observations is $\mu \sim \mathcal{N}(\hat{\mu}_t, \sigma_t^2)$ with

$$\frac{1}{\sigma_t^2} = \frac{1}{\sigma_\mu^2} + \frac{t}{\sigma_c^2}, \quad \hat{\mu}_t = \sigma_t^2 \left(\frac{\hat{\mu}_0}{\sigma_\mu^2} + \frac{\sum_{s \leq t} y_s}{\sigma_c^2} \right),$$

or recursively $\hat{\mu}_{t+1} = \hat{\mu}_t + K_t(y_{t+1} - \hat{\mu}_t)$ with $K_t = \sigma_t^2 / (\sigma_t^2 + \sigma_c^2)$. The predictive distribution is $y_{t+1} \mid y^t \sim \mathcal{N}(\hat{\mu}_t, \lambda_t^2)$ with $\lambda_t^2 \equiv \sigma_c^2 + \sigma_t^2$. With log utility the marginal rate of substitution is $\beta C_t / C_{t+1}$ whatever the beliefs, and the beliefs enter through the expectation used for pricing. Writing $y_{t+1} = \hat{\mu}_t + \lambda_t \tilde{\varepsilon}_{t+1}$, where $\tilde{\varepsilon}_{t+1} \sim \mathcal{N}(0, 1)$ is the standardized forecast error (i.i.d. from the consumer's point of view),

$$m_{t+1} = \beta e^{-y_{t+1}} = \exp\left(-r_t - \frac{1}{2} \lambda_t^2 - \lambda_t \tilde{\varepsilon}_{t+1}\right), \quad r_t = -\log \beta + \hat{\mu}_t - \frac{1}{2} \lambda_t^2.$$

This is the form (14.6.8) with a time-varying drift and a market price of risk λ_t that falls deterministically from $(\sigma_c^2 + \sigma_\mu^2)^{1/2}$ to σ_c . An econometrician who knows μ and imposes rational expectations would instead recover m_{t+1} times the likelihood ratio of the consumer's predictive density to the true one, $\mathcal{N}(y_{t+1}; \hat{\mu}_t, \lambda_t^2) / \mathcal{N}(y_{t+1}; \mu, \sigma_c^2)$, as in section 14.13 and exercise 14.21.

(b) $R_t^f = 1/E_t m_{t+1} = \beta^{-1} \exp(\hat{\mu}_t - \frac{1}{2}(\sigma_c^2 + \sigma_t^2))$. It has two components.

- $\sigma_t^2 \approx \sigma_c^2/t$ falls deterministically, so the precautionary term shrinks and pushes R_t^f up.
- $\hat{\mu}_t$ is revised with every observation. From the consumer's point of view it is a martingale with increments of variance $K_t^2 \lambda_t^2 = \sigma_t^2 - \sigma_{t+1}^2$.

Hence, under the consumer's beliefs,

$$E_t \log R_{t+1}^f - \log R_t^f = \frac{1}{2}(\sigma_t^2 - \sigma_{t+1}^2) > 0, \quad E_t R_{t+1}^f / R_t^f = e^{\sigma_t^2 - \sigma_{t+1}^2} > 1 :$$

the risk-free rate is a submartingale and is expected to rise. But its path is not monotone: bad news about μ lowers it, and the magnitude of the revisions, governed by $K_t \approx 1/t$, dominates the precautionary drift. Under the true distribution, $\hat{\mu}_t \rightarrow \mu$ almost surely and $E^P \hat{\mu}_t$ moves monotonically from $\hat{\mu}_0$ toward μ , so $R_t^f \rightarrow \beta^{-1} \exp(\mu - \sigma_c^2/2)$; whether it drifts up or down on average depends on whether the prior mean was too low or too high. With the parameters of exercise 14.13 the precautionary drift is negligible (the shrinking precautionary term raises $\log R^f$ by only 3.1×10^{-6} between $t = 0$ and $t = 10,000$), and the revisions dominate (Figure S14.2).

Exercise 14.13 Long-run risk, III

Remark. The parameter list omits σ_μ and σ_z . We choose $\sigma_\mu = .0025$ (prior 95% interval $[0, .01]$ for mean quarterly growth, i.e. 0–4% a year) and $\sigma_z = .00005$, the same number as the given σ_{z0} . With $\rho = .99$ this σ_z makes the long-run component as important for long-run risk as the i.i.d. part: $\sigma_z/(1 - \rho) = .005 = \sigma_c$. We simulate data with true values $\mu = .005$ and $z_0 = 0$.

(a) Let $x_t = (\mu, z_t)'$ and $y_{t+1} = c_{t+1} - c_t$. Then

$$x_{t+1} = Ax_t + C\varepsilon_{t+1}, \quad y_{t+1} = Gx_t + D\varepsilon_{t+1}, \quad A = \begin{bmatrix} 1 & 0 \\ 0 & \rho \end{bmatrix}, \quad C = \begin{bmatrix} 0 \\ \sigma_z \end{bmatrix}, \quad G = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad D = \sigma_c,$$

a state-space system in which the same shock drives the state and the observation. Let $x_t | y^t \sim \mathcal{N}(\hat{x}_t, \Sigma_t)$, starting from $\hat{x}_0 = (\hat{\mu}_0, \hat{z}_0)'$ and $\Sigma_0 = \text{diag}(\sigma_\mu^2, \sigma_{z0}^2)$. Conditional on y^t , (x_{t+1}, y_{t+1}) is jointly normal with means $(A\hat{x}_t, G\hat{x}_t)$, variances $A\Sigma_t A' + CC'$ and $\Omega_t \equiv G\Sigma_t G' + D^2$, and covariance $A\Sigma_t G' + CD$. Conditioning on y_{t+1} gives the Kalman filter with correlated noise:

$$K_t = \frac{A\Sigma_t G' + CD}{\Omega_t}, \quad \hat{x}_{t+1} = A\hat{x}_t + K_t(y_{t+1} - G\hat{x}_t), \quad \Sigma_{t+1} = A\Sigma_t A' + CC' - K_t \Omega_t K_t'.$$

The predictive distribution of consumption growth is $y_{t+1} | y^t \sim \mathcal{N}(\hat{\mu}_t + \hat{z}_t, \Omega_t)$. As in exercise 14.12, $m_{t+1} = \beta e^{-y_{t+1}}$, and with $\tilde{\varepsilon}_{t+1} = (y_{t+1} - G\hat{x}_t)/\Omega_t^{1/2}$,

$$m_{t+1} = \exp\left(-r_t - \frac{1}{2}\Omega_t - \Omega_t^{1/2}\tilde{\varepsilon}_{t+1}\right), \quad r_t = -\log \beta + \hat{\mu}_t + \hat{z}_t - \frac{1}{2}\Omega_t, \quad R_t^f = e^{r_t}.$$

With full information the rate would be $R_t^{FI} = \beta^{-1} \exp(\mu + z_t - \frac{1}{2}\sigma_c^2)$. *Checks:* for $t = 30$ the filter reproduces the batch (GLS) conditional mean and covariance of x_{30} , computed directly from the joint normal distribution of $(\mu, z_0, \varepsilon_1, \dots, \varepsilon_{30})$ and $(y_1, \dots, y_{30})$, to 10^{-15} . With $\sigma_z = \sigma_{z0} = 0$ it reproduces the conjugate-normal formulas of exercise 14.12 exactly.

(b) Figure S14.2 shows one simulated path, computed with `ch14.py` (a Python version of the requested program), together with the exercise 14.12 case (same shocks, $z \equiv 0$).

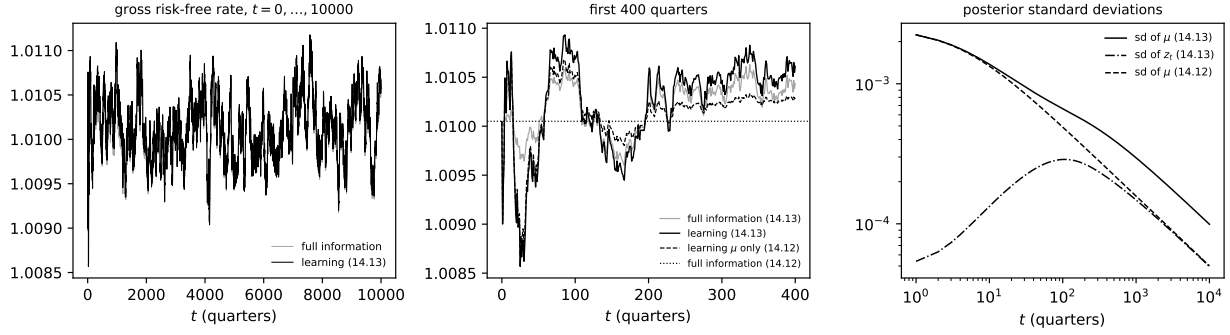


Figure S14.2. Left and middle: gross risk-free rate with learning (black) and with full information (grey), for all 10,001 quarters and for the first 400; dashed: learning about μ only (exercise 14.12). Right: posterior standard deviations.

The rate tracks the consumer's estimate of expected growth, $\hat{\mu}_t + \hat{z}_t$; the precautionary term is negligible ($\Omega_0 - \sigma_c^2 = 6.25 \times 10^{-6}$). Three phases are visible.

- *Early learning.* With few observations, each growth surprise moves $\hat{\mu}_t$ a lot, so the rate is much more volatile than the full-information rate: its standard deviation over the first 100 quarters is 6.9×10^{-4} , against 2.8×10^{-4} . It starts at $R_0^f = 1.01005$ and falls to 1.0090 after one bad draw.
- *Slow learning about μ .* The posterior standard deviation of μ falls only like $t^{-1/2}$: 6.6×10^{-4} at $t = 100$ and 9.9×10^{-5} at $t = 10,000$. That is twice the value when z is absent (5.0×10^{-5}), because a persistent z_t is easily confused with μ : the long-run standard deviation of consumption growth is $\sigma_c + \sigma_z/(1 - \rho) = .01 = 2\sigma_c$. The uncertainty about z_t first grows (from 5×10^{-5} to 2.9×10^{-4} near $t = 100$), because the unknown μ contaminates the inference about z , and then falls.
- *Convergence.* Because a single shock drives both equations, knowing μ would reveal z_t exactly: $z_t = (\rho - \sigma_z/\sigma_c)z_{t-1} + (\sigma_z/\sigma_c)(y_t - \mu)$, and $|\rho - \sigma_z/\sigma_c| = .98 < 1$. So $\Sigma_t \rightarrow 0$ and R_t^f converges to the full-information rate, which fluctuates persistently with z_t (standard deviation 3.3×10^{-4}). For $t \geq 5,000$ the two rates differ by at most 5.2×10^{-5} .

So the risk-free rate is volatile early and becomes a persistent, AR(1)-like process driven by the (eventually learned) long-run growth component. The precautionary component of learning contributes only a tiny upward drift.

Exercise 14.14 *Affine term structure model*

$p_t(1) = E_t m_{t+1} = \exp(-\delta_0 - \delta'_1 z_t)$ by (14.11.5), so $\bar{A}_1 = -\delta_0$ and $\bar{B}_1 = -\delta_1$. (Equivalently, start from $p_t(0) = 1$: $\bar{A}_0 = 0$, $\bar{B}_0 = 0$.) Suppose $p_{t+1}(n) = \exp(\bar{A}_n + \bar{B}'_n z_{t+1})$. Then $p_t(n+1) = E_t[m_{t+1} p_{t+1}(n)]$, and

$$\log(m_{t+1} p_{t+1}(n)) = -\delta_0 - \delta'_1 z_t - \frac{1}{2} \Lambda'_t \Lambda_t + \bar{A}_n + \bar{B}'_n (\mu + \phi z_t) + (C' \bar{B}_n - \Lambda_t)' \varepsilon_{t+1}$$

is conditionally normal. The mean of a lognormal gives

$$\begin{aligned} \log p_t(n+1) &= -\delta_0 - \delta'_1 z_t - \frac{1}{2} \Lambda'_t \Lambda_t + \bar{A}_n + \bar{B}'_n (\mu + \phi z_t) + \frac{1}{2} (C' \bar{B}_n - \Lambda_t)' (C' \bar{B}_n - \Lambda_t) \\ &= \bar{A}_n + \bar{B}'_n \mu + \frac{1}{2} \bar{B}'_n C C' \bar{B}_n - \delta_0 + [\bar{B}'_n \phi - \delta'_1] z_t - \bar{B}'_n C (\Lambda_0 + \Lambda_z z_t), \end{aligned}$$

because the terms $\pm \frac{1}{2} \Lambda'_t \Lambda_t$ cancel. Collecting the constant and the coefficient on z_t gives the two recursions: $\bar{A}_{n+1} = \bar{A}_n + \bar{B}'_n(\mu - C\Lambda_0) + \frac{1}{2} \bar{B}'_n C C' \bar{B}_n - \delta_0$ and $\bar{B}'_{n+1} = \bar{B}'_n(\phi - C\Lambda_z) - \delta'_1$. They are the bond-pricing recursions under the risk-neutral dynamics $z_{t+1} = \mu - C\Lambda_0 + (\phi - C\Lambda_z)z_t + C\varepsilon_{t+1}$ of section 14.12. *Numerical check* (`ch14.py`, a two-factor example): the recursion agrees with (i) the closed form $\log E_t^Q \exp(-\sum_{i \leq n} r_{t+i})$, computed from the Gaussian distribution of the sum under the risk-neutral dynamics, to 10^{-17} for $n \leq 12$; and (ii) a two-dimensional Gauss–Hermite evaluation of $E_t[m_{t+1}p_{t+1}(n)]$ under the physical measure, to 5×10^{-16} .

Exercise 14.15 *Reverse engineering*

Equations (0), (1) and (3) refer to the numbered displays of the exercise in the book.

(a) $\log(m_{t+1}R_{j,t+1}) = -\delta - \mu - \frac{1}{2}w^2 + \eta_{jt} - \frac{1}{2}(\alpha_j^2 + \sigma_j^2) + (w - \sigma_c + \alpha_j)\varepsilon_{t+1} + \sigma_j u_{j,t+1}$, a normal variable with variance $(w - \sigma_c + \alpha_j)^2 + \sigma_j^2$. Setting $E_t(m_{t+1}R_{j,t+1}) = 1$ and simplifying ($\frac{1}{2}(w - \sigma_c + \alpha_j)^2 - \frac{1}{2}w^2 - \frac{1}{2}\alpha_j^2 = -w\sigma_c + \frac{1}{2}\sigma_c^2 + \alpha_j(w - \sigma_c)$):

$$\eta_{jt} = \delta + \mu - \frac{1}{2}\sigma_c^2 + w\sigma_c + \alpha_j(\sigma_c - w) = \delta + \mu - \frac{1}{2}\sigma_c^2(2\gamma - 1) + \gamma\sigma_c\alpha_j.$$

It does not depend on t or on σ_j .

(b) $\eta_{1t} = \delta + \mu - \frac{1}{2}\sigma_c^2(2\gamma - 1)$. Asset 1 has no exposure to the priced shock ε , and its own risk u_1 is independent of m , so its expected return $E_t R_{1,t+1} = e^{\eta_{1t}}$ equals the risk-free rate: $\eta_{1t} = r_t = -\log E_t m_{t+1}$, which is Tallarini's short rate (14.7.22) with $\rho = \delta$.

(c) $\eta_{jt} - \eta_{1t} = \alpha_j(\sigma_c - w) = \alpha_j\gamma\sigma_c$. The log expected excess return equals the exposure α_j to the shock ε times its price $\sigma_c - w = \gamma\sigma_c$. Following section 14.8.6, the price splits into σ_c , the price of consumption risk that a log-utility consumer would charge, and $-w = (\gamma - 1)\sigma_c$, a price of model uncertainty. Exposure to u_j earns nothing: that risk is unpriced even though it raises the variance of the return.

(d) Take a pure exchange (Lucas tree) economy in which the representative consumer eats the endowment, which follows (0), and ranks consumption plans by Tallarini's recursion (14.7.6),

$$U_t = c_t - \beta\theta \log E_t \exp(-U_{t+1}/\theta), \quad \beta = e^{-\delta}, \quad \theta = \frac{-1}{(1-\beta)(1-\gamma)}.$$

By (14.7.12)–(14.7.17) (exercise 14.5), $m_{t+1} = \beta e^{-(c_{t+1}-c_t)} \exp(w\varepsilon_{t+1} - \frac{1}{2}w^2)$ with $w = -\sigma_c/[\theta(1-\beta)] = \sigma_c(1-\gamma)$, which is (1). The assets are claims whose returns satisfy (3), priced by this consumer's Euler equations.

(e) $\gamma = 1$: $w = 0$ and $\theta = \infty$. The recursion becomes $U_t = c_t + \beta E_t U_{t+1}$: time-separable expected log utility, with $m_{t+1} = \beta C_t / C_{t+1}$.

(f) $\gamma > 1$: Epstein–Zin–Weil preferences with a unit elasticity of intertemporal substitution (the aggregator (14.7.3) with $\eta = 1$) and risk aversion γ in the certainty equivalent (14.7.2). The parameter γ governs attitudes toward atemporal risk, and with it the preference for early resolution of uncertainty (exercise 14.5). It does not govern intertemporal substitution: that elasticity is fixed at one, which is why γ barely affects the risk-free rate in (b). Alternatively (section 14.8), the same recursion describes a log-utility consumer who distrusts the model (0) (multiplier preferences). Then

θ , or γ , measures the concern about model misspecification, and $\exp(w\varepsilon - \frac{1}{2}w^2)$ is the likelihood ratio of the worst-case model.

Exercise 14.16 Incomplete markets, I

(a) $m_{t+1}^i = \beta(Y_{t+1}^i/Y_t^i)^{-\gamma} = \exp(-\rho - \gamma\mu - \gamma\sigma\epsilon_{t+1}^i)$. They differ unless $\epsilon^1 = \epsilon^2$. With complete markets consumers would trade until their marginal rates of substitution agreed state by state. Here only the bond is traded, so only the conditional *means* of the m^i , which price the bond, must agree.

(b) Consumer i 's Euler equation for the bond is $1 = R_t E_t[\beta(C_{t+1}^i/C_t^i)^{-\gamma}]$. At $C^i = Y^i$, $E_t m_{t+1}^i = \exp(-\rho - \gamma\mu + \frac{1}{2}\gamma^2\sigma^2)$, the same for both consumers, so the Euler equation holds for both iff $r_t = \rho + \gamma\mu - \frac{1}{2}\gamma^2\sigma^2$. The Euler equation is also sufficient here. Let $\{\tilde{C}_t, \tilde{b}_{t+1}\}$ be any feasible plan with $\tilde{b}_0 = 0$ that satisfies the no-Ponzi condition $\liminf_T E_0 \beta^T u'(C_T) \tilde{b}_T \geq 0$ (no debt left in present value). By concavity and the budget constraint, $u(\tilde{C}_t) - u(C_t) \leq u'(C_t)(\tilde{C}_t - C_t) \leq u'(C_t)(\tilde{b}_t - R_t^{-1}\tilde{b}_{t+1})$. Using the Euler equation $\beta^t u'(C_t) R_t^{-1} = \beta^{t+1} E_t u'(C_{t+1})$, the sum telescopes:

$$E_0 \sum_{t=0}^T \beta^t [u(\tilde{C}_t) - u(C_t)] \leq u'(C_0) \tilde{b}_0 - E_0 \beta^{T+1} u'(C_{T+1}) \tilde{b}_{T+1},$$

whose limsup is ≤ 0 . So no trade is optimal for each consumer at this r_t , and the bond market clears, $b^1 + b^2 = 0$. (Lifetime utility is finite if $\beta \exp[(1-\gamma)\mu + \frac{1}{2}(1-\gamma)^2\sigma^2] < 1$, exercise 14.7.)

(c) The only traded payoff is a unit that is known at t , so every random variable m_{t+1} with $E_t m_{t+1} = e^{-r_t}$ prices all traded assets. That includes m^1 , m^2 , their convex combinations, and $m^i + e_{t+1}$ for any e_{t+1} with $E_t e_{t+1} = 0$. What is unique is the projection of any discount factor on the space of traded payoffs. Here that is the constant $E_t m_{t+1} = e^{-r_t}$, the Hansen–Jagannathan y^* of footnote 10. All discount factors share their conditional mean, i.e. the price of the bond. Different discount factors would assign different prices to a nontraded risky claim, such as a claim to Y_{t+1}^1 , and that is the sense in which markets are incomplete.

Exercise 14.17 Incomplete markets, II

(a) Given x_t , which is known at t , $m_{t+1}^i = \exp(-\rho - \gamma(\mu + x_t) - \gamma\sigma\epsilon_{t+1}^i)$ (the shock u_{t+1} affects growth only from $t+2$ on). So $E_t m_{t+1}^i = \exp(-\rho - \gamma(\mu + x_t) + \frac{1}{2}\gamma^2\sigma^2)$ for both i . The Euler equations hold at the no-trade allocation iff $r_t = \rho + \gamma(\mu + x_t) - \frac{1}{2}\gamma^2\sigma^2$. The sufficiency argument of exercise 14.16 goes through unchanged with a time-varying R_t , and markets clear.

(b) Expected endowment growth, $\mu + x_t$, is common to both consumers and varies over time. When it is high, both would like to borrow against their higher expected future income. Since bonds are in zero net supply, the interest rate must rise, by γ per unit of expected growth (the inverse of the elasticity of intertemporal substitution), until neither wants to borrow. In exercise 14.16 expected growth and its variance were constant, and so was the rate.

Exercise 14.18 Incomplete markets, III

(a) σ_t is known at t , so $E_t m_{t+1}^i = \exp(-\rho - \gamma\mu + \frac{1}{2}\gamma^2\sigma_t^2)$ for both consumers, and the Euler equations hold at no trade iff

$$r_t = \rho + \gamma\mu - \frac{1}{2}\gamma^2\sigma_t^2.$$

Sufficiency and market clearing follow as in exercise 14.16.

(b) Rates are low when endowment uncertainty σ_t is high: the precautionary motive to save is strong, and the rate must fall until nobody wants to save. They are high in calm periods. Because $\log \sigma_t^2$ is a random walk, r_t has no stationary distribution; it wanders, and can become arbitrarily negative.

Remark. With $\gamma \neq 1$ the lifetime utility $E_0 \sum \beta^t u(C_t)$ of the no-trade plan is not finite: already $E_0 C_2^{1-\gamma}$ involves $E \exp[\frac{1}{2}(1-\gamma)^2 \sigma_0^2 e^{\sigma \sigma_{u_1}}] = \infty$, since the exponential of a lognormal variable has no mean. (Two-period bond prices are infinite for the same reason, $E_t \exp(\frac{1}{2} \gamma^2 \sigma_{t+1}^2) = \infty$.) The equilibrium in (a) should therefore be read either with $\gamma = 1$, or as the period-by-period Euler-equation characterization, which involves only one-period-ahead moments and is well defined, or with a volatility process that is bounded above.

Exercise 14.19 *Relative entropy*

Remark. Minor misprints: $\lambda_t = \lambda_0 + \lambda'_1 x_t$ should have an $n \times n$ matrix multiplying x_t ; the twisted mean in (d) is $\mu + \phi x_t - C \lambda_t$ (not μ_x); and in (e) $\log l_{t+1} = -\frac{1}{2} \lambda'_t \lambda_t - \lambda'_t \varepsilon_{t+1}$ (a transpose is missing).

(a) Assume C is nonsingular and define $\varepsilon_{t+1} = C^{-1}(x_{t+1} - \mu - \phi x_t)$. Conditional on x^t (the process is Markov), $\varepsilon_{t+1} \sim \mathcal{N}(0, C^{-1} C C' C'^{-1}) = \mathcal{N}(0, I)$. This conditional distribution does not depend on x^t , so ε_{t+1} is independent of x^t , and hence of $\varepsilon_1, \dots, \varepsilon_t$: the sequence is i.i.d. $\mathcal{N}(0, I)$, and $x_{t+1} = \mu + \phi x_t + C \varepsilon_{t+1}$. (If CC' is singular, apply the construction to its range and complete ε with independent draws.)

(b) Given x_t , λ_t is known and $\log l = -\frac{1}{2} \lambda'_t \lambda_t - \lambda'_t \varepsilon \sim \mathcal{N}(-\frac{1}{2} \lambda'_t \lambda_t, \lambda'_t \lambda_t)$, so $E_t l = \exp(-\frac{1}{2} \lambda'_t \lambda_t + \frac{1}{2} \lambda'_t \lambda_t) = 1$.

(c) $l > 0$ and $E_t l = 1$. Explicitly, with φ the $\mathcal{N}(0, I)$ density,

$$\frac{\varphi(\varepsilon + \lambda_t)}{\varphi(\varepsilon)} = \exp\left(-\frac{1}{2}(\varepsilon + \lambda_t)'(\varepsilon + \lambda_t) + \frac{1}{2}\varepsilon'\varepsilon\right) = \exp\left(-\lambda'_t \varepsilon - \frac{1}{2}\lambda'_t \lambda_t\right) = l.$$

So l_{t+1} is the ratio of the $\mathcal{N}(-\lambda_t, I)$ density of ε_{t+1} (numerator: the twisted, e.g. risk-neutral, conditional distribution) to its $\mathcal{N}(0, I)$ density under the physical measure (denominator), both conditional on x_t .

(d) Write $\varepsilon = C^{-1}(x - \mu - \phi x_t)$. The physical density of x_{t+1} is $(2\pi)^{-n/2} |CC'|^{-1/2} \exp(-\frac{1}{2}\varepsilon'\varepsilon)$. Multiplying by l gives $(2\pi)^{-n/2} |CC'|^{-1/2} \exp(-\frac{1}{2}(\varepsilon + \lambda_t)'(\varepsilon + \lambda_t))$. Now $\varepsilon + \lambda_t = C^{-1}(x - \tilde{m})$ with $\tilde{m} = \mu + \phi x_t - C \lambda_t$, so $(\varepsilon + \lambda_t)'(\varepsilon + \lambda_t) = (x - \tilde{m})'(CC')^{-1}(x - \tilde{m})$: the product is the $\mathcal{N}(\mu + \phi x_t - C \lambda_t, CC')$ density.

(e) By (c), the twisted density of ε_{t+1} is $\mathcal{N}(-\lambda_t, I)$. Hence

$$\text{ent} = E_t(l \log l) = \hat{E}_t \log l = \hat{E}_t \left[-\frac{1}{2} \lambda'_t \lambda_t - \lambda'_t \varepsilon_{t+1} \right] = -\frac{1}{2} \lambda'_t \lambda_t + \lambda'_t \lambda_t = \frac{1}{2} \lambda'_t \lambda_t \geq 0,$$

with equality iff $\lambda_t = 0$. A Monte Carlo check with $\lambda = (.3, -.2, .5)'$ gives $E(l \log l) = .1895$ and $\hat{E} \log l = .1910$, against $\frac{1}{2} \lambda' \lambda = .19$.

Exercise 14.20 Likelihood ratio process

(a) Multiplying the t ratios gives $\xi_t = \xi_0 \prod_{s=1}^t \exp(-\lambda' \varepsilon_s - \frac{1}{2} \lambda' \lambda) = \exp(-\lambda' \sum_{s \leq t} \varepsilon_s - \frac{t}{2} \lambda' \lambda)$. Since $\lambda' \sum_s \varepsilon_s \sim \mathcal{N}(0, t\lambda' \lambda)$, $\log \xi_t \sim \mathcal{N}(-\frac{t}{2} \lambda' \lambda, t\lambda' \lambda)$, and $E\xi_t = \exp(-\frac{t}{2} \lambda' \lambda + \frac{t}{2} \lambda' \lambda) = 1$.

(b) ξ_t is a function of ε^t , $E|\xi_t| = 1 < \infty$, and $E[\xi_{t+1} | \varepsilon^t] = \xi_t E \exp(-\lambda' \varepsilon_{t+1} - \frac{1}{2} \lambda' \lambda) = \xi_t$.

(c) With $a = \lambda' \lambda$, $\log \xi_t \sim \mathcal{N}(m, s^2)$ with $m = -ta/2$, $s^2 = ta$:

$$\begin{aligned} \text{mean } e^{m+s^2/2} &= 1, & \text{median } e^m &= e^{-ta/2}, & \text{mode } e^{m-s^2} &= e^{-3ta/2}, \\ \text{variance } (e^{s^2} - 1)e^{2m+s^2} &= e^{ta} - 1, & \text{std } (e^{ta} - 1)^{1/2}, & & \text{skewness } (e^{ta} + 2)(e^{ta} - 1)^{1/2}. \end{aligned}$$

(d) The program (`lognormal.cdf` in `ch14.py`) evaluates $F(x) = \Phi((\log x - m)/s)$ for $x > 0$ with the error function. It agrees with `scipy.stats.lognorm` to 10^{-12} and with simulated frequencies for $t = 1, 5$. With $\lambda = 1$ ($a = 1$):

t	$\log_{10}$ median	$F(0.01)$	$F(0.5)$	$F(1)$	$\log_{10} \mathbb{P}(\xi_t > 1)$	$\log_{10}$ std	$\log_{10}$ skewness
1	-0.22	0.00002	0.4234	0.6915	-0.51	0.12	0.79
5	-1.09	0.1732	0.7905	0.8682	-0.88	1.08	3.26
100	-21.7	0.999997	1.0000	0.9999997	-6.54	21.7	65.1
1000	-217	1	1	1	-55.9	217	651
10000	-2171	1	1	1	-545	2171	6514

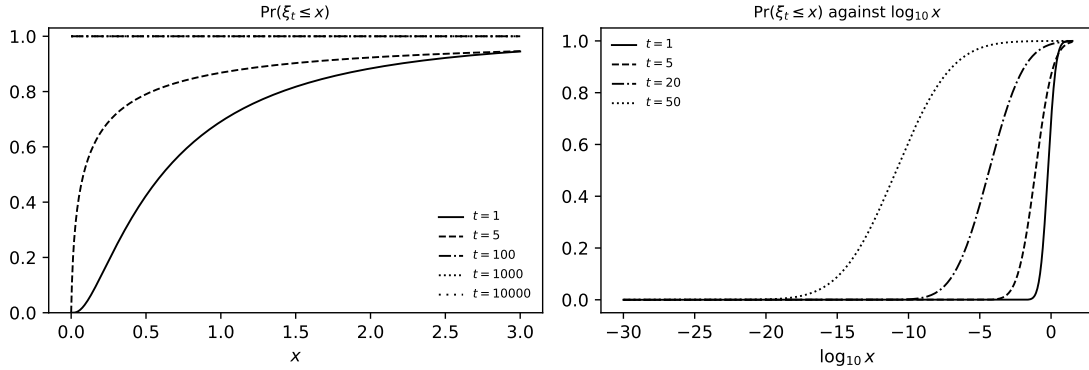


Figure S14.3. C.d.f. of ξ_t ($\lambda = 1$). Left: against x ; for $t \geq 100$ it is indistinguishable from 1 on $(0, 3]$. Right: against $\log_{10} x$, showing the median $e^{-t/2}$ moving left.

For every $x > 0$, $F_t(x) = \Phi((\log x + t/2)/\sqrt{t}) \rightarrow 1$, so $\xi_t \rightarrow 0$ in distribution. In fact $\xi_t \rightarrow 0$ almost surely, since $\frac{1}{t} \log \xi_t \rightarrow -\frac{1}{2}$ by the law of large numbers. Yet $E\xi_t = 1$ for every t . The mean is carried by events of vanishing probability on which ξ_t is astronomically large: $E[\xi_t \mathbf{1}\{\xi_t > 1\}] = \Phi(\sqrt{t}/2)$, which is .69, .87 and $1 - 2.9 \times 10^{-7}$ for $t = 1, 5, 100$, while $\mathbb{P}(\xi_t > 1) = \Phi(-\sqrt{t}/2)$ is $10^{-6.5}$ at $t = 100$. The skewness explodes like $e^{3t/2}$. The family $\{\xi_t\}$ is not uniformly integrable, so almost-sure convergence does not carry over to the means: the martingale converges to $\xi_\infty = 0$ with $E\xi_\infty = 0 \neq 1$. Simulation sees only the typical paths: sample means of 10^6 draws of ξ_t are 1.00, 1.00, 1.01, 0.15 and 8×10^{-6} for $t = 1, 5, 20, 50, 100$. In the language of section 14.8, the two probability models become perfectly distinguishable as the sample grows. Under the baseline

model $\xi_t \rightarrow 0$, while under the alternative model, for which ξ_t is the likelihood ratio, $\log \xi_t$ has mean $+t/2$ and $\xi_t \rightarrow \infty$.

Exercise 14.21 *Distorted beliefs*

Remark. The display defining m_{t+1}^* is unnumbered in the book; we call it (2), as the questions do. Part (c) refers to the discount factor “discovered in part a”; it is found in part (b).

(b) (We answer (b) first.) The subjective density of ε relative to the physical one is the likelihood ratio $g(\varepsilon) = \varphi(\varepsilon - w)/\varphi(\varepsilon) = \exp(w\varepsilon - \frac{1}{2}w^2)$, so $E_t^S(m^*R) = E_t^P(gm^*R)$. An econometrician who imposes rational expectations therefore recovers

$$m_{t+1} = g(\varepsilon_{t+1}) m_{t+1}^* = \exp(-\rho - (c_{t+1} - c_t) + w\varepsilon_{t+1} - \frac{1}{2}w^2) = \exp(-\rho - \mu - \frac{1}{2}w^2 - (\sigma_c - w)\varepsilon_{t+1}).$$

(a) Under the physical measure,

$$E(m) = \exp(-\rho - \mu + \frac{1}{2}\sigma_c^2 - \sigma_c w), \quad \frac{\sigma(m)}{E(m)} = [\exp((\sigma_c - w)^2) - 1]^{1/2} \approx \sigma_c - w.$$

The belief distortion w sets the market price of risk: pessimism ($w < 0$) makes it large even though σ_c is small. It also raises $E(m)$ through $-\sigma_c w$ (a pessimist expects low growth and wants to save, which lowers the risk-free rate), which helps with the risk-free rate puzzle as well. The discount rate ρ moves $E(m)$ without affecting the market price of risk. Together, (w, ρ) can place $(E(m), \sigma(m))$ anywhere above the bound. With the annual data of exercise 14.1 (bills 1.02), the point of the bound with the smallest market price of risk (.2967 at $E(m) = .977$) is attained with $w = -.2555$ and $\rho = .0156$ ($\beta = .9845$), a perfectly ordinary discount factor. The implied distortion of mean log growth, $\sigma_c w = -.0089$, is a quarter of a standard deviation of annual growth. Its entropy is $w^2/2 = .033$ per year, and with 90 annual observations the detection error probability (section 14.8.7) is $\Phi(-\sqrt{90}|w|/2) = .113$ (simulated: .112): a distortion that is hard to detect statistically.

(c) This m_{t+1} is exactly Tallarini’s (14.7.17), with $\beta = e^{-\rho}$ and $w = \sigma_c(1 - \gamma)$, i.e. risk aversion $\gamma = 1 - w/\sigma_c$ (8.30 in the example; $\gamma > 1$ corresponds to $w < 0$), or $\theta = -\sigma_c/[w(1 - \beta)]$. So a log-utility agent with pessimistic beliefs, Tallarini’s risk-sensitive agent (unit elasticity of substitution, risk aversion γ), and the ambiguity-averse agent of section 14.8 whose worst-case model has mean shift w (equation (14.8.11)) are observationally equivalent for asset prices. The difference is the source of w . For Tallarini, and under robustness, $w = -\sigma_c/[\theta(1 - \beta)]$ is endogenous: it moves with the volatility of consumption. With distorted beliefs, w is a primitive.

Exercise 14.22 *Costs of aggregate fluctuations*

Equations (2) and (3) refer to the numbered displays of the exercise in the book.

Remark. In (g) the value should satisfy recursion (3), not (2).

(a) $C_{t+j} = C_t \exp(j\mu + \sigma_c \sum_{i=1}^j \varepsilon_{t+i})$, and $\log(C_{t+j}/C_t) \sim \mathcal{N}(j\mu, j\sigma_c^2)$, so

$$E_t^P C_{t+j} = C_t e^{j(\mu + \sigma_c^2/2)}, \quad \text{std}_t^P C_{t+j} = C_t e^{j(\mu + \sigma_c^2/2)} (e^{j\sigma_c^2} - 1)^{1/2}.$$

(b) Guess $U_t^P = a + bc_t$: $a + bc_t = c_t + \beta(a + b(c_t + \mu))$ gives $b = 1/(1 - \beta)$ and $a = \beta\mu/(1 - \beta)^2$. So $U_t^P = \frac{c_t}{1 - \beta} + \frac{\beta\mu}{(1 - \beta)^2}$, with state $c_t = \log C_t$.

- (c) Under S , $E_t^S(c_{t+1} - c_t) = \mu + \sigma_c w$, so $U_t^S = \frac{c_t}{1 - \beta} + \frac{\beta(\mu + \sigma_c w)}{(1 - \beta)^2}$.
- (d) $e^{g^P} = E^P(C_{t+1}/C_t) = e^{\mu + \sigma_c^2/2}$, so $g^P = \mu + \frac{1}{2}\sigma_c^2$. It rises one for one with μ and increases with σ_c : for a given mean of log growth, more volatility raises the mean of the level (Jensen).
- (e) $g^S = \mu + \sigma_c w + \frac{1}{2}\sigma_c^2$. It rises with μ and with w ($\partial g^S/\partial w = \sigma_c$); $\partial g^S/\partial \sigma_c = w + \sigma_c$, so more volatility *lowers* subjective mean growth when the agent is pessimistic enough, $w < -\sigma_c$.
- (f) A deterministic path has the same value for both types: $U_0^d = \log \tilde{C}_0/(1 - \beta) + \beta g/(1 - \beta)^2$. Equating this, with $g = g^P$, to U_0^P gives

$$\log \tilde{C}_0^P = \log C_0 + \frac{\beta(\mu - g^P)}{1 - \beta} = \log C_0 - \frac{\beta\sigma_c^2}{2(1 - \beta)}, \quad \tilde{C}_0^P = C_0 \exp\left(-\frac{\beta\sigma_c^2}{2(1 - \beta)}\right),$$

formula (14.9.3) with $\gamma = 1$.

- (g) Likewise, $\log \tilde{C}_0^S = \log C_0 + \beta(\mu + \sigma_c w - g^S)/(1 - \beta) = \log C_0 - \beta\sigma_c^2/[2(1 - \beta)]$: the same answer. Each type compares the risky path with a certain path having the growth rate *it* expects, so the distortion w cancels. Both types bear the same cost of *risk*.
- (h) The type- S consumer is indifferent between being type S with C_0 and type P with $C_0 e^{-\Delta}$, where $U_0^P(C_0 e^{-\Delta}) = U_0^S(C_0)$, i.e.

$$\Delta = -\frac{\beta\sigma_c w}{1 - \beta} \quad (> 0 \text{ for } w < 0).$$

Interpretation: type P faces risk only, with a trusted model. Type S evaluates the same risk under a pessimistic distortion: this is how the ambiguity-averse (multiplier) consumer of section 14.8 evaluates plans, with $w = -\sigma_c/[\theta(1 - \beta)]$ the worst-case mean shift. So Δ measures the cost of *model uncertainty*, on top of the common cost of risk from (f)–(g). Two qualifications. First, the multiplier consumer's value adds back the entropy penalty, θ times discounted entropy $w^2/2$, which halves the cost: $\Delta_{\text{mult}} = -\beta\sigma_c w/[2(1 - \beta)]$, and then risk plus model uncertainty add up to $\beta\gamma\sigma_c^2/[2(1 - \beta)]$, Tallarini's (14.9.3) with $\gamma = 1 - w/\sigma_c$. Second, the magnitudes differ enormously. With the quarterly estimates of Table 14.6.1 ($\mu = .004952$, $\sigma_c = .00505$), $\beta = .995$ and $w = (1 - 50)\sigma_c = -.2475$ (Tallarini's $\gamma = 50$), and measuring costs in logs of initial consumption as in (14.9.3): the cost of risk is 0.25% of consumption; the payment Δ is 24.9%; with the entropy penalty it is 12.4%, and risk plus model uncertainty total 12.7%, equal to (14.9.3) with $\gamma = 50$. This is the message of Figure 14.9.1: the cost of risk is small, while the cost of model uncertainty can be large.

Exercise 14.23 *Reverse engineering*

Equation (1) refers to a numbered display of the exercise in the book.

Remark. As printed, (1) cannot be a pricing kernel: the quadratic term enters with a $+$ sign, so the integral diverges. It should be $-(x_{t+1} - x_t - \mu)^2/(2\sigma_x^2)$. Moreover, the normal density written there is a density for x_{t+1} , so the price of $\{X_{t+1} \in \mathcal{A}\}$ is $\int_{\log \mathcal{A}} Q dx_{t+1}$, or $\int_{\mathcal{A}} Q X_{t+1}^{-1} dX_{t+1}$. We use the corrected kernel $Q = e^{-\rho - \gamma(x_{t+1} - x_t)} f(x_{t+1} | x_t)$ with f the $\mathcal{N}(x_t + \mu, \sigma_x^2)$ density of x_{t+1} .

(a) A Lucas tree economy. One perishable good, no production (the only technology is the tree), and an endowment $C_t = X_t$ with $x_{t+1} = x_t + \mu + \sigma_x \varepsilon_{t+1}$, ε i.i.d. $\mathcal{N}(0, 1)$ under the true probabilities. A representative consumer has preferences $E_0 \sum_t \beta^t C_t^{1-\gamma} / (1-\gamma)$ with $\beta = e^{-\rho}$. In a competitive equilibrium with complete Arrow markets, consumption equals the endowment, and the price of the Arrow security paying in state x_{t+1} is $\beta u'(C_{t+1}) / u'(C_t)$ times its probability: $e^{-\rho} e^{-\gamma(x_{t+1}-x_t)} f(x_{t+1} | x_t) = Q$. The same kernel arises with any number of consumers having these identical CRRA preferences: with complete markets they consume constant shares of X_t , and their common marginal rate of substitution is the one above.

(b) With complete markets a representative agent always exists in the sense of chapter 8: Arrow prices equal the marginal rate of substitution of a planner with Negishi weights, whose utility is CRRA with the same γ when individual utilities are identical CRRA. But Q identifies only the product “discount factor $\times$ marginal rate of substitution $\times$ probability,” not its three factors. For example,

$$e^{-\gamma \Delta x} \mathcal{N}(\Delta x; \mu, \sigma_x^2) = e^{-(\gamma-1)\mu + \frac{1}{2}(\gamma-1)^2 \sigma_x^2} e^{-\Delta x} \mathcal{N}(\Delta x; \mu - (\gamma-1)\sigma_x^2, \sigma_x^2),$$

so the same Arrow prices are generated by a log-utility agent with discount rate $\rho + (\gamma-1)\mu - \frac{1}{2}(\gamma-1)^2 \sigma_x^2$ and pessimistic beliefs. They are also generated by Tallarini’s agent with risk aversion γ and unit elasticity of substitution (exercise 14.21 with $w = \sigma_x(1-\gamma)$). The representative agent is not unique.

(c) $\int Q dx_{t+1} = e^{-\rho} E e^{-\gamma \Delta x} = \exp(-\rho - \gamma\mu + \frac{1}{2}\gamma^2 \sigma_x^2)$, so the one-period risk-free yield is $r = \rho + \gamma\mu - \frac{1}{2}\gamma^2 \sigma_x^2$, as in (14.6.9).

(d) With the physical measure $x_{t+1} - x_t \sim \mathcal{N}(\mu, \sigma_x^2)$, the discount factor is $m_{t+1} = Q/f = \exp(-\rho - \gamma(x_{t+1} - x_t))$. *Given* the physical measure it is unique (almost surely): markets are complete, so the Arrow securities span all payoffs, and any m that prices them must equal Q/f state by state. But Q alone does not reveal the physical measure. Any density $\tilde{f}$ with the same support yields a discount factor $\tilde{m} = Q/\tilde{f}$ that prices everything. An extreme case is the risk-neutral measure $\tilde{f} = Qe^r$, i.e. $x_{t+1} - x_t \sim \mathcal{N}(\mu - \gamma\sigma_x^2, \sigma_x^2)$, with the constant discount factor $\tilde{m} = e^{-r}$.

(e) $\log m_{t+1} \sim \mathcal{N}(-\rho - \gamma\mu, \gamma^2 \sigma_x^2)$, so the market price of risk is

$$\frac{\sigma(m)}{E(m)} = [\exp(\gamma^2 \sigma_x^2) - 1]^{1/2} \approx \gamma \sigma_x.$$

If X_t is aggregate consumption, σ_x is about .005 per quarter or .035 per year. The market price of risk is then .051 quarterly and .36 annually for $\gamma = 10$, and .26 quarterly for $\gamma = 50$. With the annual bound of exercise 14.1, which requires at least .297, one needs $\gamma \geq 8.3$, and then a $\beta > 1$ ($\rho < 0$) to put $E(m)$ where the bound is lowest. So the model does not attain the bounds with reasonable parameters: it has the equity premium and risk-free rate puzzles of section 14.6. It could do so only if X were a state far more volatile than consumption, which would contradict the reverse engineering of (a).

Exercise 14.24 Unpriced risk

Remark. The book writes $\eta_t(j) = \eta_0(j) + \alpha_z(j)z_t$ (the matrix should be $\eta_z(j)$, as the next sentence says) and $\eta_t(j)'u_{t+j}$ (the shock is u_{t+1}); the pricing condition in (b) is the conditional one, E_t .

(a) Given z_t , $\log R_{j,t+1}$ is normal with mean $\nu_t - \frac{1}{2}\alpha_t'\alpha_t - \frac{1}{2}\eta_t'\eta_t$ and variance $\alpha_t'\alpha_t + \eta_t'\eta_t$ (ε and u are independent), so $E_t R_{j,t+1} = e^{\nu_t}$.

(b) $\log(m_{t+1}R_{j,t+1}) = \nu_t - r_t - \frac{1}{2}\lambda_t'\lambda_t - \frac{1}{2}\alpha_t'\alpha_t - \frac{1}{2}\eta_t'\eta_t + (\alpha_t - \lambda_t)'\varepsilon_{t+1} + \eta_t'u_{t+1}$, with conditional variance $(\alpha_t - \lambda_t)'(\alpha_t - \lambda_t) + \eta_t'\eta_t$. So $E_t(m_{t+1}R_{j,t+1}) = \exp(\nu_t - r_t - \alpha_t'\lambda_t)$, and

$$\nu_t(j) = r_t + \alpha_t(j)'\lambda_t = \delta_0 + \delta_1'z_t + (\alpha_0(j) + \alpha_z(j)z_t)'(\lambda_0 + \lambda_z z_t),$$

a quadratic function of z_t that does not involve $\eta_t(j)$.

(c) The shocks ε move the discount factor, with loadings $-\lambda_t$. An asset with exposure $\alpha_t(j)$ pays off badly when m is high (marginal utility is high) to the extent that $\alpha_t(j)'\lambda_t > 0$, so it must offer a higher expected return: $\alpha_t(j)'\lambda_t = \sum_i (\text{exposure to } \varepsilon_i) \times (\text{price of } \varepsilon_i)$, as in exercise 14.6.

(d) It does not depend on $\eta_t(j)$ at all. The risk u_{t+1} is independent of the discount factor, so it is not priced: bearing it earns no compensation, however large it is. It is diversifiable, idiosyncratic risk from the point of view of the pricing kernel. The $-\frac{1}{2}\eta_t'\eta_t$ term in the return is only a convexity adjustment that keeps $E_t R$ equal to e^{ν_t} ; it lowers the expected $\log$ return by $\frac{1}{2}\eta_t'\eta_t$ but leaves the expected gross return unchanged. Under the risk-neutral measure of section 14.12, u keeps its physical distribution while the mean of ε shifts to $-\lambda_t$.

Chapter 15

Economic Growth

Solutions to all 6 exercises of Chapter 15 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018), all donated by Rodolfo Manuelli. The numerical illustration in exercise 15.3 comes from `code/ch15.py`.

Notation: $\rho \equiv 1/\beta - 1$ is the rate of time preference, so the modified golden rule is “net marginal product = ρ .”

Exercise 15.1 *Productive government spending*

(a) Substituting the constraints, $c_t = f(k_t, g_t) + (1 - \delta)k_t - k_{t+1} - g_t$. The first-order conditions for g_t and k_{t+1} are

$$f_g(k_t, g_t) = 1, \quad u'(c_t) = \beta u'(c_{t+1}) [f_k(k_{t+1}, g_{t+1}) + 1 - \delta],$$

together with the resource constraint and the transversality condition $\lim_t \beta^t u'(c_t) k_{t+1} = 0$. Government spending is a static input: it is set each period where its marginal product equals its unit cost.

(b) In a steady state, $f_g(k, g) = 1$ and $f_k(k, g) = \rho + \delta$. Let $g(k)$ solve $f_g(k, g) = 1$. It is unique because $f_{gg} < 0$, and it exists if $f_g(k, 0^+) > 1 > \lim_{g \rightarrow \infty} f_g$. Define $\phi(k) = f_k(k, g(k))$. Since $g'(k) = -f_{gk}/f_{gg}$,

$$\phi'(k) = f_{kk} - \frac{f_{kg}^2}{f_{gg}} = \frac{f_{kk}f_{gg} - f_{kg}^2}{f_{gg}} < 0,$$

by strict concavity (a positive Hessian determinant and $f_{gg} < 0$). Under the assumption $\lim_{k \rightarrow 0} \phi(k) > \rho + \delta > \lim_{k \rightarrow \infty} \phi(k)$, there is a unique steady state k^* with $\phi(k^*) = \rho + \delta$, and $g^* = g(k^*)$.

(c) A higher β lowers $\rho + \delta$, and since ϕ is decreasing, k^* rises. Then $dg^*/d\beta = g'(k^*) dk^*/d\beta$ has the sign of f_{gk} : government spending rises with patience if g and k are complements ($f_{gk} > 0$, as with Cobb–Douglas) and falls if they are substitutes.

(d) With $f = zk^\alpha g^\eta$ and $y = f$: $f_g = \eta y/g = 1$ gives $g/y = \eta$, and $f_k = \alpha y/k = \rho + \delta$ gives $k/y = \alpha/(\rho + \delta)$, so investment is $x/y = \delta k/y = \alpha\delta/(\rho + \delta)$. Both ratios are independent of z : technology changes the scale of the economy but not the steady-state shares, because the optimality conditions pin down *shares* of output under Cobb–Douglas technology. The levels scale up: $y = z^{1/(1-\alpha-\eta)} [(\alpha/(\rho + \delta))^\alpha \eta]^{1/(1-\alpha-\eta)}$.

Exercise 15.2 *Productivity and employment*

(a) Steady-state conditions:

$$zf_k(k, n) = \rho + \delta, \quad \frac{u_2(c, 1 - n)}{u_1(c, 1 - n)} = zf_n(k, n), \quad c = zf(k, n) - \delta k - g.$$

Assumptions: f is homogeneous of degree one, so the first equation pins down $\kappa \equiv k/n$ uniquely (using $f_{kk} < 0$ and Inada conditions). The wage $w = zf_n(\kappa, 1)$ and output net of depreciation per worker, $a = zf(\kappa, 1) - \delta\kappa > 0$, are then constants, and $c(n) = an - g$. Uniqueness of n follows if both consumption and leisure are normal goods. Then the marginal rate of substitution u_2/u_1 along $c = an - g$ is increasing in n : more consumption and less leisure both raise the value of leisure. Existence requires $u_2/u_1 < w$ at the lowest feasible n ($c = 0^+$) and $u_2/u_1 \rightarrow \infty$ as $n \rightarrow 1$ (Inada in leisure). Interpretation: normality means extra wealth is spent partly on leisure, so the labor-supply curve crosses the fixed wage only once.

(b) Here $u_2/u_1 = \frac{1-\mu}{\mu} \frac{c}{1-n}$, $\kappa = (z\alpha/(\rho + \delta))^{1/(1-\alpha)}$, $w = z(1-\alpha)\kappa^\alpha$ and $a = z\kappa^\alpha[1 - \alpha\delta/(\rho + \delta)]$. Both w and a are proportional to $z^{1/(1-\alpha)}$. The labor condition $\frac{1-\mu}{\mu}(an - g) = w(1-n)$ gives

$$n = \frac{w + \frac{1-\mu}{\mu}g}{w + \frac{1-\mu}{\mu}a} = \frac{1 + \frac{1-\mu}{\mu}\frac{g}{w}}{1 + \frac{1-\mu}{\mu}\frac{a}{w}}.$$

$a/w = [1 - \alpha\delta/(\rho + \delta)]/(1 - \alpha)$ does not depend on z . So, if $g = 0$, employment is *independent of productivity*: income and substitution effects cancel under these balanced-growth preferences. With $g > 0$ fixed, a higher z lowers g/w and hence *reduces* n : the government's claim becomes relatively smaller, a positive wealth effect. Output per capita $y = z\kappa^\alpha n \propto z^{1/(1-\alpha)}n$ rises with z in either case, one for one in the $z^{1/(1-\alpha)}$ scale when $g = 0$. (σ plays no role in the steady state.)

(c) With constant returns, $k/n = \kappa$ is pinned down by $zf_k(\kappa, 1) = \rho + \delta$ and is *independent of* g . So an increase in g can raise k/n only if f is not constant-returns (e.g. if f_k depends on n separately with $f_{kn} > 0$, more work raises the marginal product of capital and hence k/n). Output per capita $y = zf(\kappa, 1)n$ moves with employment. A higher g is a negative wealth effect ($c = an - g$). If leisure is normal the household works more, so n and output per capita rise (in (b), $\partial n/\partial g > 0$). Output rises, but consumption and welfare fall.

Exercise 15.3 *Vintage capital and cycles*

(a) $\max \sum_t \beta^t u(c_t)$ subject to $c_t + k_{t+1} = zf(K_t)$, $K_t \equiv k_t + k_{t-1}$, given (k_0, k_{-1}) . The state is the pair of surviving vintages. The Euler equation for k_{t+1} (which produces at $t+1$ and $t+2$) is

$$u'(c_t) = \beta u'(c_{t+1})zf'(K_{t+1}) + \beta^2 u'(c_{t+2})zf'(K_{t+2}).$$

(b) In a steady state $u'(c)$ cancels, giving $zf'(2k)(\beta + \beta^2) = 1$. With f' strictly decreasing from ∞ to 0 there is a unique k^* , with $f'(2k^*) = 1/[z\beta(1 + \beta)]$, and $c^* = zf(2k^*) - k^*$.

(c) A *persistent period-two cycle is impossible* (with interior solutions and strictly concave u). In a cycle $(k_o, k_e, k_o, \dots)$, total capital $K_t = k_o + k_e$ is the same every period, so output is constant and $c_o = zf(K) - k_e$, $c_e = zf(K) - k_o$. Write $a = zf'(K) > 0$. The Euler equations at odd and even dates read

$$u'(c_o)(1 - \beta^2 a) = \beta a u'(c_e), \quad u'(c_e)(1 - \beta^2 a) = \beta a u'(c_o).$$

Multiplying them gives $(\frac{\beta a}{1 - \beta^2 a})^2 = 1$, and since both marginal utilities are positive, $\beta a = 1 - \beta^2 a$. The first equation then gives $u'(c_o) = u'(c_e)$, so $c_o = c_e$ and $k_o = k_e$. The “cycle” is the steady state.

But the vintage structure does produce damped oscillations (echo effects) along the transition. With $u = \ln c$, $f(K) = K^3$, $z = 1$, $\beta = .95$ (`ch15.py`), the linearized Euler equation is a fourth-order difference equation with roots $\{-1.936, -0.544, 0.430, 2.446\}$, in reciprocal pairs with products $1/\beta$. The two stable roots are 0.430 and -0.544 , and the negative one makes investment alternate. Starting from unequal vintages $k_{-1} = .5k^*$, $k_0 = 1.5k^*$ (so that $K_0 = K^*$), the optimal path of k_t/k^* is 0.834, 1.132, 0.947, 1.037, 0.983, 1.011, $\dots$, and consumption converges monotonically to c^* (Figure S15.1). The planner smooths consumption by filling in the “hole” left by the small vintage, which creates echoes that die out at rate 0.544 per period.

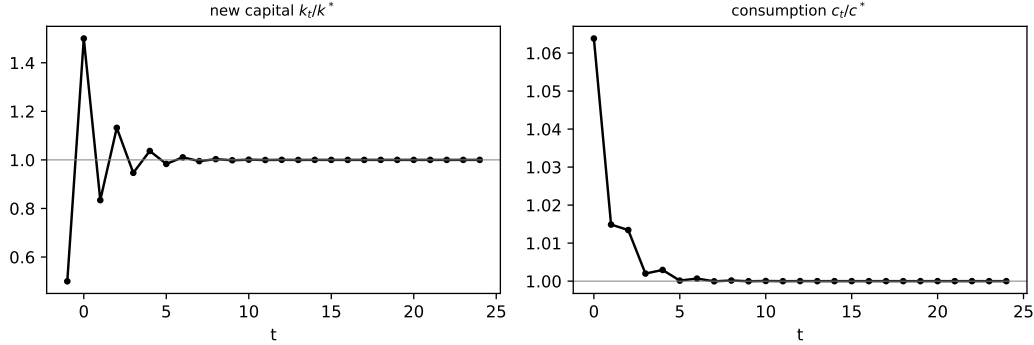


Figure S15.1. Optimal path from unequal initial vintages: damped echoes in investment, smooth consumption.

So the researcher’s claim is wrong for the stated economy: vintage-driven fluctuations are transitional, not permanent.

Exercise 15.4 *Excess capacity*

(a) The planner maximizes $\sum \beta^t u(c_t)$ subject to $c_t + k_{t+1} = z f(\kappa_t) + k_t - \delta \kappa_t$ and $0 \leq \kappa_t \leq k_t$. Using a machine yields $z f'(\kappa)$ but costs δ in depreciation, so utilization is chosen statically:

$$\kappa_t = \min\{k_t, \bar{\kappa}\}, \quad z f'(\bar{\kappa}) = \delta.$$

All machines are used unless there are more than $\bar{\kappa}$, beyond which the marginal machine is worth more idle than in use. The Euler equation for k_{t+1} is

$$u'(c_t) = \beta u'(c_{t+1}) [1 + \lambda_{t+1}], \quad \lambda_{t+1} = \begin{cases} z f'(k_{t+1}) - \delta, & k_{t+1} \leq \bar{\kappa}, \\ 0, & k_{t+1} > \bar{\kappa}. \end{cases}$$

An idle machine earns a zero net return; since $\beta < 1$, holding idle capacity would require consumption to fall over time.

(b) In a steady state $\beta(1+\lambda) = 1$, i.e. $\lambda = \rho > 0$. So capital cannot be partly idle, and $z f'(k^*) = \rho + \delta$. This has a unique solution under the usual Inada conditions, and it satisfies $k^* < \bar{\kappa}$ because $\rho + \delta > \delta$.

(c) Full utilization: $\kappa^* = k^*$. Idle capacity is optimal only in transition, when the economy starts with more capital than $\bar{\kappa}$. Then the planner lets it sit idle (undepreciated) and consumes out of output until the stock falls.

(d) Not in steady state. Every country uses all its capital, whatever its z . Cross-country differences in z show up in k^* and y^* , not in utilization. The model could associate low output with low utilization only for economies far above their steady state, which is not the usual situation of poor countries.

Exercise 15.5 *Heterogeneity and growth*

(a) A competitive equilibrium is a set of prices $\{r_t, w_t\}$ and allocations $\{c_{it}, k_{i,t+1}\}$, $\{k_t\}$ such that: (1) given prices, each household maximizes $\sum \beta^t u_i(c_{it})$ subject to $c_{it} + k_{i,t+1} = w_t + (1 + r_t - \delta)k_{it}$ (plus a no-Ponzi condition); (2) firms maximize profits, $r_t = F_k(k_t, 1)$ and $w_t = F_n(k_t, 1)$, with zero profits under constant returns; (3) markets clear: $k_t = \sum_i k_{it}$, labor = 1, and $\sum_i c_{it} + k_{t+1} = F(k_t, 1) + (1 - \delta)k_t$.

(b)(i) Each household's Euler equation is $u'_i(c_{it}) = \beta(1 + r_{t+1} - \delta)u'_i(c_{i,t+1})$. In a steady state with constant prices, if $\beta(1 + r - \delta)$ exceeded (fell short of) one, *every* household's consumption would grow (shrink) forever, contradicting a constant aggregate. Hence $\beta(1 + r - \delta) = 1$, i.e. $F_k(k^*, 1) = \rho + \delta$. This determines k^* and the interest rate uniquely, independently of the u_i . A is right and B is wrong: the steady-state interest rate depends only on β and δ (and F gives k^*).

(b)(ii) If $k_0 = k^*$, prices are at their steady-state values from the start. With $\beta(1 + r - \delta) = 1$ each household keeps its consumption, and hence its wealth, constant: $c_i = w + (r - \delta)k_{i0}$. Consumption inequality is an affine image of initial capital inequality, $c_i - c_j = (r - \delta)(k_{i0} - k_{j0})$, so A is right. B is wrong. Perfect capital markets let households *smooth* consumption, not equalize it, and nothing pushes wealth to converge. For $k_0 \neq k^*$, households with different intertemporal elasticities respond differently during the transition, so the wealth distribution changes. But long-run consumption still depends on initial wealth (and on the u_i), not on a common level.

(c) In each case aggregate capital and prices stay at their steady-state values, because lump-sum reallocations do not distort anyone's Euler equation and $\beta(1 + r - \delta) = 1$ still holds. Each household immediately moves to a new constant consumption equal to its new permanent income. (i) After the redistribution, $c_i = w + (r - \delta)k_i^{\text{new}}$: consumption is redistributed accordingly, with no aggregate effects. (ii) Permanent lump-sum taxes T on one half and transfers on the other simply shift permanent incomes: the payers' consumption falls by T , the recipients' rises by the corresponding transfer. Aggregates are unchanged, since this is a pure redistribution. (iii) The taxed two-thirds each reduce consumption by their tax. Aggregate consumption falls by g each period, while capital, output and the interest rate are unchanged: the resources absorbed by g come entirely out of the taxpayers' consumption.

Exercise 15.6 *Taxes and growth*

(a) Taking v_t as given, the planner's Euler equation is $u'(c_t) = \beta u'(c_{t+1})[(1 - \tau)f'(k_{t+1}) + 1 - \delta]$. In equilibrium $v_t = \tau f(k_t)$, so the resource constraint is $c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t - g$. A steady state requires $(1 - \tau)f'(k^*) = \rho + \delta$, which has a unique solution (since f' decreases from ∞ to 0), and $c^* = f(k^*) - \delta k^* - g$, which we assume positive. The equilibrium dynamics are those of the standard growth model, with the $\dot{c} = 0$ locus at $k = k^*$ and the $\dot{k} = 0$ locus $c = f(k) - \delta k - g$. Linearizing around the steady state, with $\tilde{R} \equiv f'(k^*) + 1 - \delta > 1$ and $b \equiv \beta(1 - \tau)f''(k^*)u'(c^*)/u''(c^*) > 0$,

gives $dk_{t+1} = \tilde{R} dk_t - dc_t$ and $dc_{t+1} = dc_t - b dk_{t+1}$. The Jacobian $\begin{bmatrix} \tilde{R} & -1 \\ -b\tilde{R} & 1+b \end{bmatrix}$ has characteristic polynomial $P(\lambda) = \lambda^2 - (\tilde{R} + 1 + b)\lambda + \tilde{R}$, with $P(0) = \tilde{R} > 0$ and $P(1) = -b < 0$. So one root lies in $(0, 1)$ and the other exceeds one: a saddle. Paths above the stable arm hit $k = 0$ in finite time, and paths below it violate the transversality condition, so the equilibrium is the stable arm, which converges monotonically to (k^*, c^*) from any $k_0 > 0$.

(b) The new steady state has $(1 - \tau')f'(k^{**}) = \rho + \delta$, so $k^{**} > k^*$. Output $f(k^{**})$ is higher, and consumption $c^{**} = f(k^{**}) - \delta k^{**} - g > c^*$ because both steady states lie below the golden rule ($f' > \delta$), where $f(k) - \delta k$ is increasing. *Transition*: capital is predetermined, so output does not jump at impact. The new stable arm passes below the old steady-state point (at $k = k^* < k^{**}$ it lies below the $\dot{k} = 0$ locus, on which c^* sits). So consumption jumps down and investment jumps up. Thereafter k_t , y_t and c_t rise monotonically to their new steady-state values, and c_t eventually exceeds the old c^* . The transfer $v_t = \tau' f(k_t)$ falls at impact and then rises with output; the second planner treats it as exogenous, which is exactly why the tax distorts saving.

(c) With inelastic labor, the tax on labor income is effectively lump sum, and only the tax on capital income distorts. The household Euler equation is $u'(c_t) = \beta u'(c_{t+1})[1 + (1 - \tau)r_{t+1} - \delta]$ with $r = F_k(k, 1)$, so in a steady state the *after-tax* net return equals ρ : $(1 - \tau)r^* = \rho + \delta$. The before-tax equilibrium interest rate $r^* = (\rho + \delta)/(1 - \tau)$ therefore *falls* with the tax cut, from $(\rho + \delta)/(1 - \tau)$ to $(\rho + \delta)/(1 - \tau')$, as capital accumulates. At impact r_t is unchanged (capital is predetermined), and it then declines monotonically as k_t rises. The after-tax return $(1 - \tau')r_t - \delta$ jumps up above ρ at impact, which is what makes consumption grow along the transition, and falls back to ρ in the new steady state.

Chapter 16

Optimal Taxation with Commitment

Solutions to all 13 exercises of Chapter 16 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Exercises 16.1–16.10 are analytical. Exercises 16.11 and 16.12 leave preferences and the discount factor open, so we pick them. The numbers for 16.5 and 16.11–16.13 come from `code/ch16.py`.

Notation. We use the primal approach of Section 16.5. With the household's first-order conditions substituted into its present-value budget constraint, the constraint becomes the *implementability condition*

$$\sum_{t=0}^{\infty} \beta^t [u_c(t)c_t - u_\ell(t)n_t] = A, \quad A = u_c(0) \{[(1 - \tau_0^k)F_k(0) + 1 - \delta]k_0 + b_0\} \quad (16.5.5) - (16.5.6).$$

Let $V(c, n, \Phi) = u(c, 1 - n) + \Phi[u_c c - u_\ell n]$, and let θ_t be the multiplier on feasibility. The Ramsey first-order conditions are (16.5.9): for $t \geq 1$, $V_c(t) = \beta V_c(t+1)[F_k(t+1) + 1 - \delta]$ and $V_n(t) = -V_c(t)F_n(t)$. At $t = 0$ they contain the extra terms ΦA_c and ΦA_n . A *wedge* is the gap between a marginal rate of substitution and the corresponding marginal rate of transformation.

Exercise 16.1 *A small open economy (Razin and Sadka)*

(a) A unit of capital owned between t and $t + 1$ earns the world gross return $R_t^* = r_{t+1}^* + 1 - \delta$, however it is used. So the household's Arrow–Debreu prices $q_t^0 = \prod_{i < t} (R_i^*)^{-1}$ are exogenous, and its first-order conditions are

$$q_t^0 = \beta^t \frac{u_c(t)}{u_c(0)}, \quad \frac{u_\ell(t)}{u_c(t)} = (1 - \tau_t^n)w_t.$$

Its present-value budget constraint is $\sum_t q_t^0 c_t = \sum_t q_t^0 (1 - \tau_t^n)w_t n_t + a_0$, where a_0 is initial wealth (capital and bonds, including government debt b_0). A domestic firm uses capital k_t^d and sets $F_k(k_t^d, n_t) = (1 + \hat{\tau}_t^k)r_t^*$ and $w_t = F_n(k_t^d, n_t)$. Add the household's budget to the government's, $\sum_t q_t^0 [g_t - \tau_t^n w_t n_t - \hat{\tau}_t^k r_t^* k_t^d] = -b_0$, and use $F = F_k k^d + F_n n$. The result is the *national* present-value resource constraint

$$\sum_t q_t^0 (c_t + g_t) = \sum_t q_t^0 [F(k_t^d, n_t) - r_t^* k_t^d] + a_0 - b_0. \quad (i)$$

Substituting the household's first-order conditions into its budget gives the implementability condition $\sum_t \beta^t [u_c(t)c_t - u_\ell(t)n_t] = u_c(0)a_0$ (ii). The Euler equations $\beta^t u_c(t)/u_c(0) = q_t^0$ (iii) are further restrictions on the allocation, because prices are fixed by the world market. Conversely, any $\{c_t, n_t, k_t^d\}$ that satisfies (i)–(iii) is a competitive equilibrium: set $w_t = F_n(k_t^d, n_t)$, read τ_t^n off $u_\ell/u_c = (1 - \tau_t^n)w_t$ and $\hat{\tau}_t^k$ off $F_k = (1 + \hat{\tau}_t^k)r_t^*$. The government's budget then holds by Walras' law.

So the Ramsey planner maximizes $\sum_t \beta^t u(c_t, 1 - n_t)$ subject to (i)–(iii). The domestic capital stock k_t^d appears *only* in (i), through national net output $F(k_t^d, n_t) - r_t^* k_t^d$. It does not enter (ii), because

any change in the pre-tax wage is absorbed by τ_t^n . Hence for any (c, n) the planner maximizes net output,

$$F_k(k_t^d, n_t) = r_t^* \implies \hat{\tau}_t^k = 0 \quad \text{for all } t \geq 0.$$

This includes $t = 0$: domestic capital is not a fixed factor even at $t = 0$, since firms can rent any amount at r_0^* . This is the Diamond–Mirrlees production-efficiency result. A tax on capital used in domestic production is a tax on an intermediate input. Capital's net return is fixed by the world market, so the whole burden of $\hat{\tau}^k$ falls on the immobile factor, labor. Taxing labor directly raises the same revenue without distorting production.

(b) In the closed economy, capital income may be taxed (or subsidized) during the transition. It is zero only in a steady state (PROPOSITION 1), and a tax on the inelastic k_0 is a lump-sum levy, which is why τ_0^k must be bounded. In the small open economy the supply of capital to domestic production is infinitely elastic at r_t^* . So the optimal capital tax is zero *at every date*, not only asymptotically, and no capital levy is available. Only labor is taxed.

Exercise 16.2 Consumption taxes

(a) With multiplier λ on the budget, the household's first-order conditions are $\beta^t u_c(t) = \lambda q_t^0(1 + \tau_t^c)$ and $\beta^t u_\ell(t) = \lambda q_t^0(1 - \tau_t^n)w_t$. Capital income is untaxed, so no arbitrage gives $q_t^0/q_{t+1}^0 = F_k(t+1) + 1 - \delta$. With $q_0^0 = 1$, $\lambda = u_c(0)/(1 + \tau_0^c)$. Substituting into the budget gives the implementability condition

$$\sum_{t=0}^{\infty} \beta^t [u_c(t)c_t - u_\ell(t)n_t] = A^c, \quad A^c = \frac{u_c(0)}{1 + \tau_0^c} \{ [F_k(0) + 1 - \delta]k_0 + b_0 \}.$$

The Ramsey problem is (16.5.8) with A replaced by A^c and with τ_0^c given. Its first-order conditions are (16.5.9a,b) for $t \geq 1$. At $t = 0$, $V_c(0) - \Phi A_c^c = \beta V_c(1)[F_k(1) + 1 - \delta]$ and $V_n(0) = [\Phi A_c^c - V_c(0)]F_n(0) + \Phi A_n^c$, together with feasibility and implementability. Taxes are recovered from the two wedges. The intertemporal wedge (divide the first-order conditions at t and $t + 1$) gives

$$\frac{1 + \tau_{t+1}^c}{1 + \tau_t^c} = \frac{\beta u_c(t+1)[F_k(t+1) + 1 - \delta]}{u_c(t)}, \quad \frac{1 - \tau_t^n}{1 + \tau_t^c} = \frac{u_\ell(t)}{u_c(t)F_n(t)}.$$

The first equation determines τ_{t+1}^c recursively from τ_0^c , and the second then gives τ_t^n . There are two instruments for two wedges each period, so the Ramsey taxes are unique once τ_0^c is fixed.

(b) In a steady state V_c is constant, so (16.5.9a) gives $\beta(F_k + 1 - \delta) = 1$. With u_c constant, the first displayed formula gives $(1 + \tau_{t+1}^c)/(1 + \tau_t^c) \rightarrow 1$: the *consumption tax converges to a constant*. A constant consumption tax does not distort the intertemporal margin, so this is the counterpart of the zero limiting capital tax. Its level, $1 + \tau_\infty^c = (1 + \tau_0^c) \prod_{t \geq 0} \beta u_c(t+1)[F_k(t+1) + 1 - \delta]/u_c(t)$, reflects the transition. In the limit only the intratemporal wedge $(1 - \tau_\infty^n)/(1 + \tau_\infty^c) = u_\ell/(u_c F_n)$ distorts.

(c) Take $\tau_t^c = \tau^c$ and $\tau_t^n = -\tau^c$ for all t . Then $(1 - \tau_t^n)/(1 + \tau_t^c) = 1$ and $(1 + \tau_{t+1}^c)/(1 + \tau_t^c) = 1$, so the household faces no wedge at all. Its budget becomes $(1 + \tau^c) \sum_t q_t^0(c_t - w_t n_t) = W_0 \equiv [r_0 + 1 - \delta]k_0 + b_0$, and the government collects, in present value,

$$\sum_t q_t^0 [\tau^c c_t + \tau_t^n w_t n_t] = \tau^c \sum_t q_t^0 (c_t - w_t n_t) = \frac{\tau^c}{1 + \tau^c} W_0.$$

This policy is a lump-sum levy of the fraction $\tau^c/(1+\tau^c)$ of initial wealth (capital *and* government debt), and that fraction tends to one as $\tau^c \rightarrow \infty$. A uniform consumption tax with an equal labor subsidy is thus a capital levy in disguise. In the Ramsey problem, $\partial J/\partial \tau_0^c = -\Phi \partial A^c/\partial \tau_0^c = \Phi A^c/(1+\tau_0^c) > 0$ whenever $\Phi > 0$ and $W_0 > 0$. The planner would raise τ_0^c until $\Phi = 0$ and then finance everything without distortion. Only τ_0^c enters A^c , so bounding τ_0^c is what keeps the problem interesting, exactly as bounding τ_0^k does in Section 16.6.

Exercise 16.3 *A specific utility function (Chamley)*

(a) Here $u_c c = c^{1-\sigma}$ and u_ℓ does not depend on c , so $V = u + \Phi[c^{1-\sigma} - v'(\ell)n]$ and

$$V_c = u_c + \Phi(1-\sigma)c^{-\sigma} = [1 + \Phi(1-\sigma)]u_c.$$

The factor $1 + \Phi(1-\sigma)$ is positive, because $V_c(t) = \theta_t$, the shadow value of goods. Dividing (16.5.9a) by it gives, for $t \geq 1$, $u_c(t) = \beta u_c(t+1)[F_k(t+1) + 1 - \delta]$. The household's Euler equation (16.5.4), with $q_t^0/q_{t+1}^0 = u_c(t)/[\beta u_c(t+1)]$, is $u_c(t) = \beta u_c(t+1)[(1 - \tau_{t+1}^k)F_k(t+1) + 1 - \delta]$. Subtracting, $\tau_{t+1}^k F_k(t+1) = 0$, so $\tau_{t+1}^k = 0$ for $t \geq 1$, i.e. $\tau_t^k = 0$ for all $t \geq 2$. Only τ_1^k can differ from zero, because the $t = 0$ condition (16.5.9c) contains ΦA_c .

(b) In the stochastic model, (16.10.4a) gives $V_c(s^t) = \beta E_t V_c(s^{t+1})[F_k(s^{t+1}) + 1 - \delta]$ for $t \geq 1$. With $V_c = [1 + \Phi(1-\sigma)]u_c$ this is $u_c(s^t) = \beta E_t u_c(s^{t+1})[F_k(s^{t+1}) + 1 - \delta]$. The household's condition (16.8.5c) is $u_c(s^t) = \beta E_t u_c(s^{t+1})[(1 - \tau_{t+1}^k(s^{t+1}))F_k(s^{t+1}) + 1 - \delta]$. Subtracting,

$$E_t[u_c(s^{t+1})\tau_{t+1}^k(s^{t+1})F_k(s^{t+1})] = 0 \implies \bar{\tau}_{t+1}^k(s^t) = \frac{E_t u_c(s^{t+1})\tau_{t+1}^k(s^{t+1})r_{t+1}(s^{t+1})}{E_t u_c(s^{t+1})r_{t+1}(s^{t+1})} = 0$$

by (16.9.5) and (16.11.1). The ex ante capital tax is zero for all $t \geq 2$. Realized state-by-state rates are not pinned down (Section 16.9); only their $u_c F_k$ -weighted average is. This is the case $\Lambda = 1 + \Phi(1-\sigma)$ of PROPOSITION 2.

Exercise 16.4 *Two labor inputs (Jones, Manuelli, and Rossi)*

(a) The household sets $u_\ell(t) = u_c(t)(1 - \tau_{it}^n)w_{it}$ for $i = 1, 2$. Both kinds of work cost the same leisure, so after-tax wages are equal: $(1 - \tau_{1t}^n)w_{1t} = (1 - \tau_{2t}^n)w_{2t}$. The implementability condition is $\sum_t \beta^t [u_c c - u_\ell(n_1 + n_2)] = A$, with $A = u_c(0)W_0$ and $W_0 = [(1 - \tau_0^k)F_k(0) + 1 - \delta]k_0 + b_0$. So $V = u(c, 1 - n_1 - n_2) + \Phi[u_c c - u_\ell(n_1 + n_2)]$ depends on (n_1, n_2) only through $n_1 + n_2$, and $V_{n_1} = V_{n_2} \equiv V_n$. In the Lagrangian $J = \sum_t \beta^t \{V_t + \theta_t[F(k_t, n_{1t}, n_{2t}) + (1 - \delta)k_t - c_t - g_t - k_{t+1}]\} - \Phi A$ the first-order conditions for n_{it} , $t \geq 1$, are $V_n(t) + \theta_t F_{n_i}(t) = 0$ for $i = 1, 2$. Hence $F_{n_1}(t) = F_{n_2}(t)$, i.e. $w_{1t} = w_{2t}$, and equal after-tax wages give

$$\tau_{1t}^n = \tau_{2t}^n \quad (t \geq 1).$$

The two labor types are perfect substitutes in utility. Only total hours should be distorted, and the production margin between the two types should not (production efficiency). At $t = 0$ the conditions are $V_n(0) + \theta_0 F_{n_i}(0) - \Phi A_{n_i} = 0$, with $A_{n_i} = -u_{c\ell}(0)W_0 + u_c(0)(1 - \tau_0^k)F_{kn_i}(0)k_0$. The first term is common to $i = 1, 2$, so subtracting,

$$\theta_0 [F_{n_1}(0) - F_{n_2}(0)] = \Phi u_c(0)(1 - \tau_0^k)k_0 [F_{kn_1}(0) - F_{kn_2}(0)].$$

With $F_{kn_1} > 0 > F_{kn_2}$, $\Phi > 0$ and $\tau_0^k < 1$, the right side is positive. So $w_{10} > w_{20}$, and equal after-tax wages then require $\tau_{10}^n > \tau_{20}^n$. The reason is that at $t = 0$ the planner wants to shrink the value A of initial wealth, which includes the untaxed part of capital's rental $F_k(0)k_0$. Discouraging n_1 (a complement of k) and encouraging n_2 (a substitute) lowers $F_k(0)$, which works like a partial levy on the inelastic k_0 . From $t = 1$ on, capital is no longer a fixed factor and the motive disappears.

(b) The household sets $u_{\ell_i}(t) = u_c(t)(1 - \tau_t^n)w_{it}$. The common tax rate is equivalent to $u_{\ell_1}/F_{n_1} = u_{\ell_2}/F_{n_2}$, i.e. to $G_t \equiv u_{\ell_1}(t)F_{n_2}(t) - u_{\ell_2}(t)F_{n_1}(t) = 0$. The implementability condition is $\sum_t \beta^t [u_c c - u_{\ell_1} n_1 - u_{\ell_2} n_2] = A$, and $V = u + \Phi[u_c c - u_{\ell_1} n_1 - u_{\ell_2} n_2]$. With multipliers θ_t on feasibility and η_t on $G_t = 0$, the first-order conditions for $t \geq 1$ are

$$c_t : V_c(t) + \eta_t G_c(t) = \theta_t,$$

$$n_{it} : V_{n_i}(t) + \theta_t F_{n_i}(t) + \eta_t G_{n_i}(t) = 0, \quad i = 1, 2,$$

$$k_{t+1} : \theta_t = \beta \{ \theta_{t+1} [F_k(t+1) + 1 - \delta] + \eta_{t+1} G_k(t+1) \}, \quad G_k = u_{\ell_1} F_{n_2 k} - u_{\ell_2} F_{n_1 k}.$$

The $t = 0$ conditions add the derivatives of $-\Phi A$. Together with feasibility, implementability and $G_t = 0$, these determine the allocation. The taxes then follow from $u_{\ell_1} = u_c(1 - \tau_t^n)F_{n_1}$ and $u_c(t) = \beta u_c(t+1)[(1 - \tau_{t+1}^k)F_k(t+1) + 1 - \delta]$. Note that η_t appears in the capital condition: capital changes the relative wage F_{n_1}/F_{n_2} , and hence how tight the equal-tax restriction is.

(c) In a steady state, the k condition is $1 = \beta(F_k + 1 - \delta) + \beta\eta G_k/\theta$, and the household's Euler equation is $1 = \beta[(1 - \tau^k)F_k + 1 - \delta]$. Subtracting, $\tau^k F_k = -\eta G_k/\theta$. Using $u_{\ell_2} = u_{\ell_1} F_{n_2}/F_{n_1}$ from the constraint,

$$\tau^k = -\frac{\eta u_{\ell_1}}{\theta F_k F_{n_1}} [F_{n_1} F_{n_2 k} - F_{n_2} F_{n_1 k}].$$

If the restriction binds ($\eta \neq 0$), the limiting capital tax is zero if and only if $F_{n_1} F_{n_2 k} = F_{n_2} F_{n_1 k}$, i.e. $\partial(F_{n_1}/F_{n_2})/\partial k = 0$. When capital moves the relative wage, the planner taxes or subsidizes capital to steer relative wages, as a substitute for the differential labor taxation it is not allowed to use. This matches the incomplete-taxation logic of Section 16.7.

Exercise 16.5 *Another specific utility function*

(a) With multiplier $\beta^t \lambda_t$ on the date- t budget, the condition for c_t is $1 = \lambda_t$. The one for b_{t+1} is $\lambda_t q_t = \beta \lambda_{t+1}$, so $q_t = \beta$. The one for leisure is $u_x = 1 - x_t = \lambda_t(1 - \tau_t)$, so $x_t = \tau_t$.

(b) Since $u_c = 1 > 0$, the budget holds with equality. Multiply the date- t budget by β^t , use $q_t = \beta$, and sum from 0 to T . The bond terms telescope, giving $\sum_{t=0}^T \beta^t [c_t - (1 - \tau_t)(1 - x_t)] + \beta^{T+1} b_{T+1} = b_0 = 0$. Let $T \rightarrow \infty$.

(c) Substitute $x_t = \tau_t$ and $c_t = 1 - \tau_t - g_t$. Period utility becomes $1 - \tau_t - g_t - .5(1 - \tau_t)^2 = .5 - .5\tau_t^2 - g_t$, and the budget becomes $\sum_t \beta^t \tau_t(1 - \tau_t) = G \equiv \sum_t \beta^t g_t$. So the planner minimizes $\sum_t \beta^t \tau_t^2$ subject to $\sum_t \beta^t (\tau_t - \tau_t^2) = G$. The Lagrangian condition $-\tau_t + \mu(1 - 2\tau_t) = 0$ gives $\tau_t = \mu/(1 + 2\mu)$, the same at every date whatever g_t is. The constant τ is also the *global* optimum. Let $S_1 = (1 - \beta) \sum \beta^t \tau_t$ and $S_2 = (1 - \beta) \sum \beta^t \tau_t^2$. The constraint is $S_2 = S_1 - \bar{g}$ with $\bar{g} = (1 - \beta)G$, and Jensen's inequality gives $S_2 \geq S_1^2$. Hence $S_1^2 - S_1 + \bar{g} \leq 0$, so $S_1 \geq \tau_-$, the smaller root of $\tau(1 - \tau) = \bar{g}$. Therefore

$S_2 = S_1 - \bar{g} \geq \tau_- - \bar{g} = \tau_-^2$, with equality only if $\tau_t \equiv \tau_-$. Thus

$$\tau(1 - \tau) = (1 - \beta) \sum_{t \geq 0} \beta^t g_t, \quad \tau = \frac{1}{2} [1 - \sqrt{1 - 4(1 - \beta)G}].$$

Here $G = .2\beta/(1 - \beta^2)$, so $(1 - \beta)G = .2\beta/(1 + \beta) = .097436$ and $\tau = .109405$. The other root, .890595, raises the same revenue on the wrong side of the Laffer curve.

(d) Now $G = .2/(1 - \beta^2)$, so $(1 - \beta)G = .2/(1 + \beta) = .102564$ and $\tau = .116026$. This is larger than in (c): the same purchases come one period earlier, so their present value, and the annuity the tax must finance, are higher by the factor $1/\beta$.

(e) The tax rate is constant even though purchases alternate between 0 and .2. Debt absorbs the fluctuations. Since $b_{t+1} = [b_t + g_t - \tau(1 - \tau)]/\beta$, in (c) the government saves in the low-purchase periods ($b_1 = -\tau(1 - \tau)/\beta = -.102564$) and spends the savings in the high ones ($b_2 = 0$), and so on. In (d) it borrows .102564 in each high-purchase period and repays in the next. The deadweight loss is convex in the tax rate ($.5\tau^2$ per period), so it is cheapest to spread it evenly over time: tax smoothing (Barro).

(f) $\tau = 0$ exactly when no revenue is needed, i.e. when $\sum_t \beta^t g_t = 0$: $g_t = 0$ for all t (with $b_0 = 0$). More generally, with initial debt b_0 the condition is $\tau(1 - \tau) = (1 - \beta)(G + b_0)$. Then $\tau = 0$ if the government's initial assets $-b_0$ exactly cover the present value of its purchases.

Exercise 16.6 Yet another specific utility function

Equations (1), (4) and (5) refer to the numbered displays of the exercise in the book.

(a) A competitive equilibrium with taxes is an allocation $\{c_t, n_t\}$, a price system $\{q_t\}$ and a budget-feasible tax sequence $\{\tau_t\}$ (one that satisfies (5)) such that: (i) given $\{q_t, \tau_t\}$, the allocation maximizes (1) subject to (4); and (ii) $n_t = c_t + g_t$ for all t . The household's first-order conditions (interior n_t) are $\beta^t = \lambda q_t$ and $\beta^t(u_1 + u_2 n_t) = \lambda q_t(1 - \tau_t)$. With $q_0 = 1$,

$$q_t = \beta^t, \quad n_t = \frac{1 - u_1 - \tau_t}{u_2} \quad \text{i.e. } \tau_t = 1 - u_1 - u_2 n_t.$$

(b) Because $\tau_t = 1 - u_1 - u_2 n_t$, the planner effectively chooses $\{n_t\}$ to maximize $\sum_t \beta^t [(1 - u_1)n_t - .5u_2 n_t^2 - g_t]$ subject to $\sum_t \beta^t [(1 - u_1 - u_2 n_t)n_t - g_t] = 0$. With multiplier μ , the first-order condition is $(1 - u_1 - u_2 n_t) + \mu(1 - u_1 - 2u_2 n_t) = 0$, so

$$n_t = \bar{n} = \frac{(1 + \mu)(1 - u_1)}{(1 + 2\mu)u_2}, \quad \tau_t = \tau = 1 - u_1 - u_2 \bar{n} = \frac{\mu}{1 + 2\mu}(1 - u_1),$$

constant over time and independent of g_t . The budget $\tau \bar{n}/(1 - \beta) = G \equiv \sum \beta^t g_t$ gives $\tau(1 - u_1 - \tau) = u_2(1 - \beta)G$, so $\tau = \frac{1}{2} [(1 - u_1) - \sqrt{(1 - u_1)^2 - 4u_2(1 - \beta)G}]$. A solution exists iff $(1 - u_1)^2 \geq 4u_2(1 - \beta)G$. As in 16.5, Jensen's inequality shows the constant plan is globally optimal. Write $S_1 = (1 - \beta) \sum \beta^t n_t$, $S_2 = (1 - \beta) \sum \beta^t n_t^2$ and $\bar{g} = (1 - \beta)G$. On the constraint set $(1 - u_1)S_1 - u_2 S_2 = \bar{g}$ the objective equals $(1 - \beta)^{-1} [\bar{g} + .5u_2 S_2] - G$, so the planner maximizes S_2 . Now $S_2 \geq S_1^2$ forces $u_2 S_1^2 - (1 - u_1)S_1 + \bar{g} \leq 0$, i.e. $S_1 \leq n_+$, the larger root. Hence $S_2 = [(1 - u_1)S_1 - \bar{g}]/u_2 \leq n_+^2$, with equality iff $n_t \equiv n_+ = \bar{n}$. Prices are $q_t = \beta^t$, so the gross interest rate is $1/\beta$ at every date. Consumption $c_t = \bar{n} - g_t$ absorbs all fluctuations in g_t . The tax

rate and labor supply are perfectly smoothed, and debt finances the difference between constant revenue $\tau\bar{n}$ and fluctuating purchases.

(c) $G_A = \beta g/(1 - \beta^2)$ and $G_B = \beta g \cdot 1/(1 - \beta^2)$ are equal. Since τ depends on $\{g_t\}$ only through G , both sequences have the *same* Ramsey tax rate, labor supply $\bar{n}$ and interest rates ($q_t = \beta^t$, gross rate $1/\beta$). They differ only in consumption, $c_t = \bar{n} - g_t$, and in the debt path. Under A the government runs a surplus $\tau\bar{n} = \beta g/(1 + \beta)$ at even dates, saving $b_{2j+1} = -g/(1 + \beta)$, and spends the savings at odd dates. Under B it borrows $b_{2j+1} = \beta g/(1 + \beta)$ at even dates and repays at odd dates.

Exercise 16.7 Comparison of tax systems

The book writes the capital-tax revenue as $\tau_t^a(r_{t+1} - \delta)k_t$. Since k_t is the capital used, and rented at r_t , in period t , we read it as $\tau_t^a(r_t - \delta)k_t$.

Remark. The date index r_{t+1} in the book's revenue formula appears to be a misprint for r_t .

(a) Economy 1. The household's budget is $c_t + k_{t+1} + b_{t+1}/R_t = (1 - \tau_t^n)w_t n_t + [1 + (1 - \tau_t^a)(r_t - \delta)]k_t + b_t$. Its first-order conditions are

$$\frac{u_\ell(t)}{u_c(t)} = (1 - \tau_t^n)w_t, \quad u_c(t) = \beta u_c(t+1)[1 + (1 - \tau_{t+1}^a)(r_{t+1} - \delta)],$$

with $q_t^0 = \beta^t u_c(t)/u_c(0)$. Substituting into the present-value budget, with $b_0 = 0$ and $\tau_0^a = 0$, gives the implementability condition

$$\sum_{t \geq 0} \beta^t [u_c(t)c_t - u_\ell(t)n_t] = A \equiv u_c(0)[1 + F_k(0) - \delta]k_0.$$

The restriction $\tau_0^n = 0$ adds the constraint $u_\ell(0) = u_c(0)F_n(k_0, n_0)$: time-0 labor is undistorted. The Ramsey problem maximizes $\sum \beta^t u(c_t, 1 - n_t)$ subject to feasibility, implementability and this date-0 constraint. With multipliers θ_t , Φ and ν , the Lagrangian is $J = \sum_t \beta^t \{V(c_t, n_t, \Phi) + \theta_t[F(k_t, n_t) + (1 - \delta)k_t - c_t - g_t - k_{t+1}]\} - \Phi A + \nu[u_\ell(0) - u_c(0)F_n(0)]$. For $t \geq 1$ its first-order conditions are exactly (16.5.9a,b). The date-0 conditions contain ΦA_c , ΦA_n and the ν terms. Taxes follow from $1 - \tau_t^n = u_\ell(t)/[u_c(t)F_n(t)]$ and $u_c(t) = \beta u_c(t+1)[1 + (1 - \tau_{t+1}^a)(F_k(t+1) - \delta)]$.

(b) In a steady state (16.5.9a) gives $\beta(1 + F_k - \delta) = 1$, and the Euler equation gives $\beta[1 + (1 - \tau^a)(F_k - \delta)] = 1$. Hence $\tau^a(F_k - \delta) = 0$. Since $F_k - \delta = 1/\beta - 1 > 0$, $\tau_\infty^a = 0$.

(c) Economy 2. Let $\tilde{q}_t$ be the date-0 price of pre-tax goods. Consumption is taxed but investment is not. The household's first-order conditions are $\beta^t u_c(t) = \tilde{\lambda} \tilde{q}_t(1 + \tilde{\tau}_t^c)$ and $\beta^t u_\ell(t) = \tilde{\lambda} \tilde{q}_t w_t$, with no-arbitrage $\tilde{q}_{t-1} = \tilde{q}_t[1 + (1 - \tilde{\tau}_t^a)(r_t - \delta)]$. So

$$\frac{u_\ell(t)}{u_c(t)} = \frac{w_t}{1 + \tilde{\tau}_t^c}, \quad \frac{u_c(t)}{1 + \tilde{\tau}_t^c} = \beta \frac{u_c(t+1)}{1 + \tilde{\tau}_{t+1}^c} [1 + (1 - \tilde{\tau}_{t+1}^a)(r_{t+1} - \delta)].$$

With $\tilde{q}_0 = 1$ and $\tilde{\tau}_0^c = 0$, $\tilde{\lambda} = u_c(0)$. The budget $\sum \tilde{q}_t(1 + \tilde{\tau}_t^c)c_t = \sum \tilde{q}_t w_t n_t + [1 + r_0 - \delta]k_0$ becomes $\sum \beta^t [u_c c - u_\ell n] = u_c(0)[1 + F_k(0) - \delta]k_0 = A$, the *same* implementability condition. The restriction $\tilde{\tau}_0^c = 0$ again means $u_\ell(0) = u_c(0)F_n(0)$. So the Ramsey problem of economy 2 has the same objective and the same constraints on the allocation as that of economy 1.

(d) The two Ramsey problems are identical as problems in $\{c_t, n_t, k_{t+1}\}$. So (with a unique solution) $\tilde{c}_t = c_t$, $\tilde{n}_t = n_t$, $\tilde{\ell}_t = \ell_t$ and $\tilde{k}_{t+1} = k_{t+1}$ for all t , and the factor prices r_t, w_t coincide.

(e) Match the two wedges at the common allocation. Intratemporally, $1/(1 + \tilde{\tau}_t^c) = 1 - \tau_t^n$, so

$$\tilde{\tau}_t^c = \frac{\tau_t^n}{1 - \tau_t^n}.$$

Intertemporally, $(1 + \tilde{\tau}_t^c)/(1 + \tilde{\tau}_{t+1}^c) = (1 - \tau_{t+1}^n)/(1 - \tau_t^n)$, so

$$1 + (1 - \tilde{\tau}_{t+1}^a)(r_{t+1} - \delta) = \frac{1 - \tau_t^n}{1 - \tau_{t+1}^n} [1 + (1 - \tau_{t+1}^a)(r_{t+1} - \delta)],$$

i.e. $\tilde{\tau}_{t+1}^a = \{(1 + r_{t+1} - \delta) - \rho_t[1 + (1 - \tau_{t+1}^a)(r_{t+1} - \delta)]\}/(r_{t+1} - \delta)$ with $\rho_t = (1 - \tau_t^n)/(1 - \tau_{t+1}^n)$. Besides the tax rates, this involves only the return $r_{t+1} - \delta$, which is common to the two economies. When $\tau_{t+1}^n = \tau_t^n$ it reduces to $\tilde{\tau}_{t+1}^a = \tau_{t+1}^a$. If labor taxes rise over time ($\rho_t > 1$), economy 2 must tax capital less, because a rising consumption tax already penalizes saving. Asymptotically, $\tilde{\tau}_\infty^c = \tau_\infty^n/(1 - \tau_\infty^n)$ and $\tilde{\tau}_\infty^a = \tau_\infty^a = 0$.

(f) In economy 1, $\text{Gov}_1 \equiv \sum_t q_t [\tau_t^n w_t n_t + \tau_t^a (r_t - \delta) k_t - g_t] = 0$, with $q_t = q_t^0$. By no arbitrage, $q_t \tau_t^a (r_t - \delta) k_t = q_t (1 + r_t - \delta) k_t - q_{t-1} k_t$ for $t \geq 1$ ($\tau_0^a = 0$). By feasibility, $g_t = w_t n_t + (1 + r_t - \delta) k_t - c_t - k_{t+1}$. Substituting, the capital terms telescope (using the transversality condition), and

$$\text{Gov}_1 = \sum_t q_t c_t - \sum_t q_t (1 - \tau_t^n) w_t n_t - (1 + r_0 - \delta) k_0.$$

The same steps in economy 2 give $\text{Gov}_2 \equiv \sum_t \tilde{q}_t [\tilde{\tau}_t^c c_t + \tilde{\tau}_t^a (r_t - \delta) k_t - g_t] = \sum_t \tilde{q}_t (1 + \tilde{\tau}_t^c) c_t - \sum_t \tilde{q}_t w_t n_t - (1 + r_0 - \delta) k_0$. By (d) and (e), $\tilde{q}_t (1 + \tilde{\tau}_t^c) = \beta^t u_c(t)/u_c(0) = q_t$, and hence $\tilde{q}_t = q_t (1 - \tau_t^n)$ and $\tilde{q}_t w_t = q_t (1 - \tau_t^n) w_t$. Therefore $\text{Gov}_2 = \text{Gov}_1$ term by term: economy 1's budget at Ω is economy 2's budget at $\tilde{\Omega}$. (Each also equals minus the slack in the corresponding household budget, which is Walras' law.)

Exercise 16.8 *Taxes on capital, labor, and consumption, I*

(a) The first-order conditions are $\beta^t u_c = \lambda q_t^0 (1 + \tau_t^c)$ and $\beta^t u_\ell = \lambda q_t^0 (1 - \tau_t^n) w_t$, with no arbitrage $q_t^0/q_{t+1}^0 = (1 - \tau_{t+1}^k) r_{t+1} + 1 - \delta$ and $\lambda = u_c(0)/(1 + \tau_0^c)$. The implementability condition is $\sum \beta^t [u_c c - u_\ell n] = A$, with

$$A = \frac{u_c(0)}{1 + \tau_0^c} \{[(1 - \tau_0^k) F_k(0) + 1 - \delta] k_0 + b_0\}.$$

A falls when either τ_0^c or τ_0^k rises, and $\partial J/\partial A = -\Phi < 0$, so both are set at their upper bounds. Given them, the Ramsey allocation solves (16.5.9)–(16.5.10). For $t \geq 1$ the allocation pins down only two wedges per period:

$$\frac{1 - \tau_t^n}{1 + \tau_t^c} = \frac{u_\ell(t)}{u_c(t) F_n(t)}, \quad \frac{1 + \tau_t^c}{1 + \tau_{t+1}^c} [(1 - \tau_{t+1}^k) F_k(t+1) + 1 - \delta] = \frac{u_c(t)}{\beta u_c(t+1)}.$$

There are three instruments for two wedges, so the *Ramsey tax system is not unique*: one instrument per period can be chosen freely. For example, one can hold τ_t^c at its date-0 value and use (τ_t^n, τ_{t+1}^k) as in the text. Or one can set $\tau_t^k = 0$ for $t \geq 1$ and let a time-varying τ_t^c deliver the intertemporal wedge. Debt adjusts, and the government's budget holds by Walras' law.

(b) Let $\hat{r}_t = F_k(\hat{k}_t, \hat{n}_t)$, $\hat{R}_t = (1 - \hat{\tau}^k)\hat{r}_t + 1 - \delta$ and $W(\tau) = [(1 - \tau)\hat{r}_0 + 1 - \delta]k_0 + b_0$. Define

$$1 + \tau_0^c = \frac{W(0)}{W(\hat{\tau}^k)}, \quad 1 + \tau_t^c = (1 + \tau_0^c) \prod_{j=1}^t \frac{\hat{r}_j + 1 - \delta}{(1 - \hat{\tau}^k)\hat{r}_j + 1 - \delta} \quad (t \geq 1),$$

$$1 - \tau_t^n = (1 - \hat{\tau}^n)(1 + \tau_t^c) \quad (t \geq 0), \quad \tau_t^k = 0 \quad (t \geq 0).$$

The household's conditions hold. Intratemporally, $(1 - \tau_t^n)/(1 + \tau_t^c) = 1 - \hat{\tau}^n$, as before. Intertemporally, $[(1 + \tau_t^c)/(1 + \tau_{t+1}^c)](\hat{r}_{t+1} + 1 - \delta) = \hat{R}_{t+1}$, as before. *The budget set is the same.* The new prices of pre-tax goods satisfy $q'_t/q'_{t+1} = \hat{r}_{t+1} + 1 - \delta$, so $q'_t(1 + \tau_t^c) = (1 + \tau_0^c)\hat{q}_t$, where $\hat{q}_t = \prod_{j \leq t} \hat{R}_j^{-1}$ are the old prices, and $q'_t(1 - \tau_t^n) = (1 + \tau_0^c)(1 - \hat{\tau}^n)\hat{q}_t$. The new budget $\sum q'_t(1 + \tau_t^c)c_t = \sum q'_t(1 - \tau_t^n)w_t n_t + W(0)$ is then the old one, $\sum \hat{q}_t c_t = \sum \hat{q}_t(1 - \hat{\tau}^n)w_t n_t + W(\hat{\tau}^k)$, multiplied by $1 + \tau_0^c$. So the household chooses the same allocation. Feasibility is unchanged, and the government's present-value budget holds by Walras' law, with a different debt path. If $\hat{\tau}^k > 0$, the date-0 consumption tax is positive. It replaces the part of the time-0 capital levy that $\hat{\tau}^k$ imposed on k_0 .

(c) With $\hat{\tau}^k > 0$, each factor in the product exceeds one. The equivalent consumption tax rises geometrically (at the steady-state rate $(\hat{r} + 1 - \delta)/\hat{R}$), and $1 - \tau_t^n$ rises with it, so the labor tax falls and becomes an ever larger subsidy. A constant capital tax is thus an ever-growing tax on consumption (and on leisure) at later dates relative to earlier ones. The intertemporal wedge between c_0 and c_t grows without bound as $t \rightarrow \infty$, while the intratemporal wedge between c_t and ℓ_t stays at $1 - \hat{\tau}^n$.

(d) The Ramsey plan satisfies both lessons asymptotically: $\tau_\infty^k = 0$, i.e. $(1 + \tau_{t+1}^c)/(1 + \tau_t^c) \rightarrow 1$ (no intertemporal wedge), and $(1 - \tau_t^n)/(1 + \tau_t^c)$ converges to a constant. The policy of (b) obeys lesson 2 exactly but violates lesson 1 more and more over time, which is why it is not optimal. The lessons concern wedges on the final goods (c_t, ℓ_t) , not particular instruments: a "capital tax" can be relabeled as a rising consumption tax plus a falling labor tax.

Exercise 16.9 *Taxes on capital, labor, and consumption, II*

(a) As in 16.8(a): both date-0 taxes are at their upper bounds $\bar{\tau}_0^k, \bar{\tau}_0^c$, and the allocation solves (16.5.9)–(16.5.10) with A evaluated at the bounds. The tax system is not unique, since there are three instruments for two wedges per period.

(b) This part only introduces the benchmark policy and its allocation; nothing needs to be derived.

(c) Yes, provided the frozen rate is compatible with the date-0 restriction, i.e. $\hat{\tau}^k = \bar{\tau}_0^k$ (the frozen rate uses the capital levy allowed at $t = 0$). Let $\{c_t, n_t, k_{t+1}\}$ be the Ramsey allocation. Set $\tau_0^c = \bar{\tau}_0^c$ and $\tau_t^k = \hat{\tau}^k$, and define recursively

$$1 + \tau_{t+1}^c = (1 + \tau_t^c) \frac{\beta u_c(t+1) [(1 - \hat{\tau}^k)F_k(t+1) + 1 - \delta]}{u_c(t)}, \quad 1 - \tau_t^n = (1 + \tau_t^c) \frac{u_\ell(t)}{u_c(t)F_n(t)}.$$

Then both wedge conditions of 16.8(a) hold at the Ramsey allocation. A has the Ramsey value, so implementability holds, and the allocation is a competitive equilibrium under this policy. It is therefore a Ramsey policy. If instead $\hat{\tau}^k < \bar{\tau}_0^k$, freezing $\tau_0^k = \hat{\tau}^k$ forgoes part of the date-0 levy. Since τ_0^c is already at its bound, A is larger than in the Ramsey plan, and the Ramsey allocation

of (a) cannot be reached. The class then implements only the Ramsey plan of the problem with $\tau_0^k = \hat{\tau}^k$ taken as given. Along the Ramsey path, $\beta(F_k + 1 - \delta) \rightarrow 1$ and u_c converges, so

$$\frac{1 + \tau_{t+1}^c}{1 + \tau_t^c} \rightarrow \frac{(1 - \hat{\tau}^k)F_k + 1 - \delta}{F_k + 1 - \delta} < 1 \quad (\hat{\tau}^k > 0).$$

The consumption tax declines geometrically, with $1 + \tau_t^c \rightarrow 0$, and $1 - \tau_t^n$ falls in proportion, so labor taxes rise toward 100%.

(d) The declining consumption tax subsidizes future consumption at exactly the rate that undoes the capital tax's penalty on saving. So the Ramsey plan's intertemporal wedge on (c_t, c_{t+1}) still vanishes in the limit (lesson 1). The labor tax moves in step with $1 + \tau_t^c$ so that the intratemporal wedge $(1 - \tau_t^n)/(1 + \tau_t^c) = u_\ell/(u_c F_n)$ converges to a constant (lesson 2). Any single instrument can be frozen, as long as the remaining two can still produce both wedges. The lessons restrict wedges, not tax rates.

Exercise 16.10 *The Lucas and Stokey (1983) model*

(a) Here $w_t = 1$, $u_c = 1$ and $u_\ell = H'(\ell)$. The household's conditions are $H'(\ell_t) = 1 - \tau_t^n$ and $p_t(s_{t+1}|s^t) = \beta\pi(s_{t+1}|s^t)$. The implementability condition (16.12.2), with $c_t = 1 - \ell_t - g_t$, reads

$$\sum_t \sum_{s^t} \beta^t \pi_t(s^t) [(1 - \ell_t)(1 - H'(\ell_t)) - g_t] = b_0.$$

The Ramsey problem is to maximize $\sum \beta^t \pi_t [1 - \ell_t - g_t + H(\ell_t)]$ subject to it. Since $u_{cc} = u_{c\ell} = 0$, the date-0 terms $\Phi u_{cc} b_0$ and $\Phi u_{c\ell} b_0$ in (16.12.3c,d) vanish. The first-order condition is the same at every t and s^t , including $t = 0$:

$$(1 + \Phi) [H'(\ell) - 1] = \Phi (1 - \ell) H''(\ell).$$

It involves neither t , s^t nor g , so $\ell_t(s^t) = \bar{\ell}(\Phi)$ is constant (we assume the relevant root is unique, e.g. because $(1 - \ell)(1 - H'(\ell))$ is concave). Then $n_t = \bar{n} = 1 - \bar{\ell}$, and

$$\tau_t^n = \tau = 1 - H'(\bar{\ell}) = \frac{\Phi}{1 + \Phi} \bar{n} |H''(\bar{\ell})|, \quad \text{i.e.} \quad \frac{\tau}{1 - \tau} = \frac{\Phi}{1 + \Phi} \frac{1}{\varepsilon},$$

where $\varepsilon = H'(\bar{n})/|H''(\bar{n})|$ is the elasticity of labor supply with respect to the after-tax wage: an inverse-elasticity rule. Φ is determined by the revenue requirement $\tau \bar{n}/(1 - \beta) = b_0 + E_0 \sum_t \beta^t g_t$ (on the rising side of the Laffer curve). Arrow prices are $\beta\pi(s_{t+1}|s^t)$. Debt is the expected present value of future surpluses, $b_t(s_t|s^{t-1}) = E_t \sum_{j \geq 0} \beta^j (\tau \bar{n} - g_{t+j})$. With u_c constant there is no date-0 incentive to manipulate interest rates, so here the plan is time consistent in allocations: every continuation is again a constant-tax plan.

(b) Leisure, labor and the tax rate do not depend on g_t at all. Consumption $c_t = \bar{n} - g_t$ absorbs purchases one for one. Tax revenue is constant, and all fluctuations in g are financed by state-contingent debt: perfect tax smoothing across dates and states.

(c) Revenue is $R = \tau \bar{n}$ at every date. The budget is $R/(1 - \beta) = b_0 + \alpha \beta^T (.5)/(1 - \beta)$, so $R = (1 - \beta)b_0 + .5\alpha\beta^T$. (With $b_0 = 0$, $R = .5\alpha\beta^T$.) For $t \leq T - 1$ nothing has been revealed, and debt is

$$b_t = \frac{R}{1 - \beta} - \frac{.5\alpha\beta^{T-t}}{1 - \beta}, \quad b_{t+1} - b_t = -.5\alpha\beta^{T-t-1} < 0.$$

The government runs surpluses and steadily reduces its debt, or accumulates claims on the private sector (with $b_0 = 0$, $b_t < 0$ for $1 \leq t \leq T$). At $T - 1$ it rebalances into Arrow securities for date T :

$$b_T(g = .5) = \frac{R - .5}{1 - \beta}, \quad b_T(g = 0) = \frac{R}{1 - \beta}.$$

The government pays $.5/(1 - \beta)$ less in the high-spending state: it insures itself against the spending shock at the actuarially fair price $\beta\alpha$. If $g = .5$ occurs, $b_T < 0$ (for $R < .5$): the government is a creditor, and the interest on its claims covers the permanent deficit $.5 - R$. If $g = 0$ occurs, it owes $R/(1 - \beta)$ and pays interest R forever. In both cases, for $t > T$, debt stays constant and the tax rate never changes.

Exercise 16.11 *Another Lucas–Stokey economy*

Equations (1) and (2) refer to the numbered displays of the exercise in the book.

The exercise leaves H and β open. We use $H(l) = \kappa \ln l$ with $\kappa = .2$, $\beta = .95$ and $b_0 = .2$. Without taxes the household would set $H'(l) = 1$, i.e. $l = .2$.

Remark. The conditions “ $H(0) > 1$, $H(1) > 0$ ” look garbled. What an interior untaxed leisure choice requires is $H'(0) > 1 > H'(1)$; our H has $H'(0^+) = \infty$ and $H'(1) = .2$.

(a) States $1, \dots, 5$ are $(0, g_L), (1, g_L), (2, g_L), (2, g_H), (t \geq 3, g_L)$, with $g(s) = (.1, .1, .1, .2, .1)$, $\pi_0 = (1, 0, 0, 0, 0)$ and

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & .5 & .5 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

(b) $\Pi_1(1, 2) = 1$ and $\Pi_2(1, 2, 3) = \Pi_2(1, 2, 4) = .5$. For $t \geq 3$, $\Pi_t(1, 2, 3, 5, \dots, 5) = \Pi_t(1, 2, 4, 5, \dots, 5) = .5$. Every other history has probability zero.

(c) An Arrow–Debreu equilibrium consists of an allocation $\{c_t(s^t), l_t(s^t)\}$, prices $\{q_t^0(s^t)\}$ and a policy $\{\tau_t^c(s^t)\}$ with the following properties. (i) Given prices and taxes, the allocation maximizes (1) subject to $\sum_{t,s^t} q_t^0[(1 + \tau_t^c)c_t - (1 - l_t)] = b_0$ (the wage is one). (ii) Feasibility holds. (iii) The government’s budget (2) holds. The first-order conditions are $\beta^t \Pi_t = \mu q_t^0(1 + \tau_t^c)$ and $\beta^t \Pi_t H'(l_t) = \mu q_t^0$. Hence $H'(l_t) = 1/(1 + \tau_t^c)$ and $q_t^0(s^t) = \beta^t \Pi_t(s^t) H'(l_t(s^t))/H'(l_0(s_0))$. A unit of goods is worth $u_c/(1 + \tau^c) = H'(l)$ utils.

(d) In the sequential version the household faces, at each (t, s^t) , the budget $(1 + \tau_t^c)c_t + \sum_{s_{t+1}} p_t(s_{t+1}|s^t) a_{t+1}(s_{t+1}|s^t) = 1 - l_t + a_t(s_t|s^{t-1})$, with $a_0 = b_0$ and natural debt limits. The government faces $\tau_t^c c_t + \sum p_t b_{t+1} = g_t + b_t$, and the asset market clears, $a = b$. The household’s first-order conditions give $H'(l_t) = 1/(1 + \tau_t^c)$ and

$$p_t(s_{t+1}|s^t) = \beta P(s_t, s_{t+1}) \frac{H'(l_{t+1})}{H'(l_t)} = \beta P(s_t, s_{t+1}) \frac{1 + \tau_t^c}{1 + \tau_{t+1}^c}.$$

(e) Since $l = \kappa(1 + \tau)$, labor is $n(\tau) = 1 - \kappa(1 + \tau)$. A balanced budget requires $\tau c = g$ with $c = n(\tau) - g$, i.e. $\kappa\tau^2 - (1 - \kappa - g)\tau + g = 0$, and we take the smaller root. For g_L : $\tau = .149219$,

$n = .770156$, $c = .670156$. For g_H : $\tau = .381966$, $n = .723607$, $c = .523607$. The allocation depends only on the current g , and consumption bears most of the spending shock. The nonzero Arrow prices are $p(2|1) = p(5|3) = p(5|5) = .95$, $p(3|2) = .475$, $p(4|2) = .395002$ and $p(5|4) = 1.142400$. The last exceeds one: after $(2, g_H)$ the tax falls, so goods become more valuable in utils. The risk-free gross rates $1/\sum_{s'} p(s'|s)$ are 1.052632 in states 1, 3 and 5, 1.149423 in state 2, and 0.875350 in state 4.

(f) In every state the government owes b_0 and issues $b_{t+1}(s'|s^t) = b_0$ for every successor s' with $P(s, s') > 0$. This is effectively risk-free debt with gross return

$$R(s) = \left[\sum_{s'} p(s'|s) \right]^{-1} = \left[\beta(1 + \tau(s)) \sum_{s'} \frac{P(s, s')}{1 + \tau(s')} \right]^{-1}, \quad \tau(s)c(s) = g(s) + b_0 \left(1 - \frac{1}{R(s)} \right).$$

Taxes pay for purchases plus the net interest on the rolled-over debt. Because $R(s)$ depends on current and future taxes, we solve backward from the absorbing state 5, where $R = 1/\beta$ and $\tau c = g_L + (1 - \beta)b_0$, then states 3 and 4, then 2 and 1. In each state we take the lowest tax that works:

state	1 (0, L)	2 (1, L)	3 (2, L)	4 (2, H)	5 (≥ 3 , L)
$\tau^c, b_0 = 0$	.149219	.149219	.149219	.381966	.149219
$\tau^c, b_0 = .1$	.158161	.166877	.157047	.356764	.157047
$\tau^c, b_0 = .2$	.168138	.180832	.164913	.340898	.164913
$R(s), b_0 = .2$	1.064070	1.111372	1.052632	0.914480	1.052632

Taxes rise with b_0 where $R(s) > 1$. In state 4 the rate falls with b_0 , because $R(4) < 1$: rolling the debt over at a negative net rate earns revenue.

(g) A constant $\bar{\tau}$ fixes $l = \kappa(1 + \bar{\tau})$, so $H'(l)$ is constant and $p(s'|s) = \beta P(s, s')$, $q_t^0 = \beta^t \Pi_t$. The budget (2) is $\bar{\tau}[n(\bar{\tau})/(1 - \beta) - G] - G = b_0$, where $G = E \sum \beta^t g_t = g_L + \beta g_L + \beta^2(.15) + \beta^3 g_L/(1 - \beta)$. The smallest root is $\bar{\tau} = .169079$ ($n = .766184$). The primary surplus $\bar{\tau}n - (1 + \bar{\tau})g(s)$ is .012638 in the low states and $-.104270$ in state 4. Debt owed on entering each state solves $B = (I - \beta P)^{-1} \cdot$ surplus:

$$B = (.200000, .197223, .252755, .135847, .252755).$$

The government sells, in state 1, claims paying .197223 in state 2 (price .95). In state 2 it sells claims paying .252755 in state 3 and .135847 in state 4 (price .475 each). In states 3, 4 and 5 it sells claims paying .252755 in state 5 (price .95). At $t = 1$ the government hedges: it owes $(1 + \bar{\tau})(g_H - g_L) = .116908$ less if g_H occurs.

(h) Along $(1, 2, 3, 5, 5, \dots)$ the government pays .197223 at $t = 1$, .252755 at $t = 2$, and .252755 at every $t \geq 3$, each time refinancing it and paying the interest $(1 - \beta)(.252755) = .012638$ out of the surplus. Along $(1, 2, 4, 5, 5, \dots)$ it pays .197223 at $t = 1$, only .135847 at $t = 2$, and .252755 at every $t \geq 3$.

(i) Substituting (c) into the household's budget gives the implementability condition $\sum_{t,s^t} \beta^t \Pi_t [c_t - H'(l_t)(1 - l_t)] = H'(l_0)b_0$. The Ramsey problem maximizes $\sum \beta^t \Pi_t [1 - l_t - g_t + H(l_t)]$ subject to

it, with multiplier Φ . The first-order conditions in l are

$$\begin{aligned} t \geq 1: & \quad -1 + H'(l) + \Phi[-1 + H'(l) - (1-l)H''(l)] = 0, \\ t = 0: & \quad -1 + H'(l_0) + \Phi[-1 + H'(l_0) - (1-l_0)H''(l_0)] - \Phi H''(l_0)b_0 = 0. \end{aligned}$$

The $t \geq 1$ condition does not involve g or the date. So for *any* H , leisure and the consumption tax are constant from $t = 1$ on, and consumption $c = 1 - \bar{l} - g$ absorbs the spending shock. The date-0 condition has the extra positive term $-\Phi H''(l_0)b_0$. The left side decreases in l , so $l_0 > \bar{l}$ and $\tau_0 > \bar{\tau}$ when $b_0 > 0$. A higher date-0 consumption tax lowers the utility value $H'(l_0)b_0$ of the initial debt: it is a partial levy on the government's creditors, as in 16.2(c). *Algorithm.* (1) Guess Φ . (2) Solve the $t \geq 1$ condition for $\bar{l}(\Phi)$ and the $t = 0$ condition for $l_0(\Phi)$. Set $c(s) = 1 - \bar{l} - g(s)$ and $\tau = 1/H'(l) - 1$. (3) Compute utility-denominated debt $x(s) = H'(l(s))b(s)$ for the states reached at $t \geq 1$ from the linear system $x = (I - \beta P)^{-1}[(1 - H'(l))c - H'(l)g]$ (use $\tau H' = 1 - H'$ in the budget). (4) Check the date-0 budget $H'(l_0)b_0 = (1 - H'(l_0))c_0 - H'(l_0)g_0 + \beta \sum_s P(s_0, s)x(s)$, which is the implementability condition. Raise Φ if the budget falls short, and lower it otherwise. (5) Recover $b(s) = x(s)/H'(l(s))$ and $p(s'|s) = \beta P(s, s')H'(l')/H'(l)$. Our case gives $\Phi = .053533$, $\bar{l} = .233382$ and $\bar{\tau} = .166911$ for $t \geq 1$, and $l_0 = .240537$ with $\tau_0 = .202684$. Debt owed on entering states 2–5 is (.169884, .225312, .108621, .225312), and the date-0 budget closes at $b_0 = .200000$.

(j) The Ramsey plan and the constant-tax plan both smooth taxes across the g states and use state-contingent debt as insurance. The Ramsey rate for $t \geq 1$, .166911, is slightly below the constant rate .169079, because the Ramsey plan front-loads the tax to .202684 at $t = 0$. The roll-over policy of (f) uses non-contingent debt, so its tax must jump to .340898 when g_H occurs. Expected discounted utilities are 7.465804 (Ramsey), 7.465672 (constant tax) and 7.464334 (roll-over).

(k) The time- t , history- s^t continuation of a time-0 Ramsey plan is its tail $\{\tau_j^c(s^j), b_{j+1}(s^{j+1}|s^j)\}_{j \geq t}$. A time- t Ramsey plan is a Ramsey plan that starts from $(s_t, b_t(s_t|s^{t-1}))$ (Section 16.12.3). The continuation is *not* a Ramsey plan here. A planner who re-optimizes at t faces a first-period condition containing $-\Phi_t H''(l_t)b_t$, whereas the continuation uses the $t \geq 1$ condition. So whenever $b_t \neq 0$, the new planner would again raise the current tax to devalue the inherited debt. Numerically, in state 2 with $b_1 = .169884$, the time-1 Ramsey plan sets $\tau_1 = .195336$ (the continuation prescribes .166911) and .165126 afterwards. The Ramsey plan is time inconsistent. (If $b_t(s^t) = 0$ the two first-order conditions coincide, and the continuation is a time- t Ramsey plan with the same Φ .)

Exercise 16.12 Yet another Lucas–Stokey economy

We use $u(c) = \kappa \ln c$ with $\kappa = .8$, $\beta = .95$ and $b_0 = .2$. The household's condition $u'(c)(1 - \tau^n) = 1$ gives $c = \kappa(1 - \tau)$, and $n = c + g$. (With $\kappa = .7$ or .75 the roll-over policy of (f) would be infeasible at $b_0 = .2$, because the revenue required in state $(3, g_H)$ exceeds the peak of that state's Laffer curve.)

(a) The five-state hint is copied from 16.11 and does not work here. With random draws at both $t = 2$ and $t = 3$, the state must record whether a draw is still to come. We need seven states: $1 = (0, L)$, $2 = (1, L)$, $3 = (2, L)$, $4 = (2, H)$, $5 = (3, L)$, $6 = (3, H)$, $7 = (t \geq 4, L)$, with $\pi_0 = e_1$. Transitions: $1 \rightarrow 2$ with probability 1; $2 \rightarrow 3, 4$, $3 \rightarrow 5, 6$ and $4 \rightarrow 5, 6$ with probability .5 each;

and $5 \rightarrow 7$, $6 \rightarrow 7$, $7 \rightarrow 7$ with probability 1. (We read the two draws as independent. If the book means $g_3 = g_2$, the rows of states 3 and 4 become $3 \rightarrow 5$ and $4 \rightarrow 6$ with probability one.)

Remark. The book repeats the five-state hint of 16.11, which cannot generate independent draws at two dates.

(b) $\Pi_1(1, 2) = 1$ and $\Pi_2(1, 2, 3) = \Pi_2(1, 2, 4) = .5$. At $t = 3$ each of $(1, 2, i, j)$, $i \in \{3, 4\}$, $j \in \{5, 6\}$, has probability .25, and for $t \geq 4$ each of these four histories, continued with 7's, keeps probability .25.

(c) As in 16.11(c), with the budget $\sum q_t^0 [c_t - (1 - \tau_t^n)(1 - l_t)] = b_0$ and first-order conditions $q_t^0 = \beta^t \Pi_t u'(c_t)/u'(c_0)$ and $u'(c_t)(1 - \tau_t^n) = 1$.

(d) As in 16.11(d), with the household budget $c_t + \sum p_t a_{t+1} = (1 - \tau_t^n)n_t + a_t$ and $p_t(s_{t+1}|s^t) = \beta P(s_t, s_{t+1})u'(c_{t+1})/u'(c_t)$, which here equals $\beta P c_t/c_{t+1}$.

(e) A balanced budget requires $\tau(c + g) = g$ with $c = \kappa(1 - \tau)$, so $\kappa\tau(1 - \tau) = g(1 - \tau)$ and $\tau = g/\kappa$. (The root $\tau = 1$ would mean $c = 0$.) So $c = \kappa - g$ and $n = \kappa = .8$ in every state: $\tau = .125$, $c = .7$ for g_L and $\tau = .25$, $c = .6$ for g_H . Labor is constant and consumption absorbs g . Arrow prices are $p(s'|s) = \beta P(s, s')(\kappa - g(s))/(\kappa - g(s'))$. The risk-free gross rates are 1.052632 in states 1, 5 and 7, 0.971660 in states 2 and 3, 1.133603 in state 4 and 1.228070 in state 6. Rates are low when consumption may fall next period and high after a bad draw.

(f) As in 16.11(f), the government issues b_0 of claims on every successor state. The gross return is $R(s) = [\beta c(s) \sum_{s'} P(s, s')/c(s')]^{-1}$ and the tax solves

$$\left(1 - \frac{c(s)}{\kappa}\right)(c(s) + g(s)) = g(s) + b_0 \left(1 - \frac{1}{R(s)}\right).$$

Solving backward from state 7 and taking the root with the lowest tax gives:

state	1 (0, L)	2 (1, L)	3 (2, L)	4 (2, H)	5 (3, L)	6 (3, H)	7 (≥ 4 , L)
τ^n , $b_0 = 0$	.125000	.125000	.125000	.250000	.125000	.250000	.125000
τ^n , $b_0 = .1$	.134745	.118547	.114189	.269097	.132202	.288958	.132202
τ^n , $b_0 = .2$	.147923	.121923	.065305	.261245	.139527	.372214	.139527
$R(s)$, $b_0 = .2$	1.084752	0.989308	0.817536	1.034371	1.052632	1.442786	1.052632

The dependence on b_0 is strongly nonlinear. In state 6 the tax climbs steeply (.25 $\rightarrow$.289 $\rightarrow$.372), toward the top of the Laffer curve. In state 3 it *falls* (.125 $\rightarrow$.114 $\rightarrow$.065): claims on state 6, where consumption is low, sell at a high price, so the roll-over at $R(3) < 1$ raises revenue.

(g) With $\bar{\tau}$ constant, $c = \kappa(1 - \bar{\tau})$ is constant and $p = \beta P$. The budget is $\bar{\tau}[\kappa(1 - \bar{\tau})/(1 - \beta) + G] - G = b_0$ with $G = g_L(1 + \beta) + .15(\beta^2 + \beta^3) + \beta^4 g_L/(1 - \beta) = 2.087994$. So $\bar{\tau} = .145122$ and $c = .683903$. The surplus is .013761 in the g_L states and $-.071727$ in the g_H states. Debt owed on entering each state is

$$B = (I - \beta P)^{-1} \text{surplus} = (.200000, .196041, .234617, .149129, .275224, .189736, .275224).$$

At $t = 0$ the government sells claims paying .196041 in state 2 (price .95). At $t = 1$ it sells claims paying .234617 in state 3 and .149129 in state 4, and at $t = 2$ claims paying .275224 in state 5 and .189736 in state 6 (price .475 each). From $t = 3$ on it sells .275224 on state 7 (price .95).

(h) Payouts at $t = 1, 2, 3$ and at every $t \geq 4$ along the four possible histories:

history	$t = 1$	$t = 2$	$t = 3$	$t \geq 4$
(1, 2, 3, 5, 7, 7, ...)	.196041	.234617	.275224	.275224
(1, 2, 3, 6, 7, 7, ...)	.196041	.234617	.189736	.275224
(1, 2, 4, 5, 7, 7, ...)	.196041	.149129	.275224	.275224
(1, 2, 4, 6, 7, 7, ...)	.196041	.149129	.189736	.275224

Each high-spending draw cuts that period's payout by $(1 - \bar{\tau})(g_H - g_L) = .085488$: the extra spending less the extra labor-tax revenue it generates, since $n = c + g$. From $t = 4$ on the government refinances .275224 each period and pays the interest .013761 out of its surplus.

(i) The implementability condition is $\sum \beta^t \Pi_t [u'(c_t)c_t - n_t] = u'(c_0)b_0$ with $n_t = c_t + g_t$. The Ramsey first-order conditions are

$$\begin{aligned} t \geq 1: \quad & u'(c) - 1 + \Phi[u'(c) + u''(c)c - 1] = 0, \\ t = 0: \quad & u'(c_0) - 1 + \Phi[u'(c_0) + u''(c_0)c_0 - 1] - \Phi u''(c_0)b_0 = 0. \end{aligned}$$

Again the $t \geq 1$ condition does not involve g . For any u , consumption and the labor tax are constant from $t = 1$ on, and labor $n = \bar{c} + g$ absorbs the shocks. The algorithm is that of Section 16.12.2 with $S = 7$. (1) Given Φ , get $\bar{c}$ and c_0 from these conditions. (2) Compute $x = (I - \beta P)^{-1}[u'(c)(n - g) - n]$ and $b = x/u'(c)$. (3) Adjust Φ until (16.12.16) holds. Taxes are $1 - \tau = 1/u'(c)$ and prices $p = \beta P u'(c')/u'(c)$. For $u = \kappa \ln c$ the conditions give $\bar{c} = \kappa/(1 + \Phi)$, $\bar{\tau} = \Phi/(1 + \Phi)$ and $(1 + \Phi)c_0^2 - \kappa c_0 - \Phi \kappa b_0 = 0$. Thus $c_0 > \bar{c}$ and $\tau_0 < \bar{\tau}$, and implementability reads $\sum \beta^t E[\kappa - c_t - g_t] = \kappa b_0/c_0$. Numerically, $\Phi = .171677$, $\bar{c} = .682782$, $\bar{\tau} = .146523$, $c_0 = .715545$ and $\tau_0 = .105569$. Debt owed on entering states 2–7 is (.214852, .253366, .168018, .293906, .208558, .293906). The date-0 budget closes at $b_0 = .200000$, and the gross interest rate from 0 to 1 is $\bar{c}/(\beta c_0) = 1.004434 < 1/\beta$.

(j) Figure S16.1 (left) compares the four policies. The Ramsey tax .146523 ($t \geq 1$) is close to the constant tax .145122. The Ramsey plan instead *lowers* the date-0 tax to .105569: higher c_0 raises the price at which the government sells its debt at $t = 0$ (it lowers the interest rate), which reduces the burden of b_0 . The roll-over policy makes taxes volatile, up to .372214 in $(3, H)$. Welfare is -1.844177 (Ramsey), -1.845082 (constant tax) and -1.865712 (roll-over).

(k) Definitions as in 16.11(k). The continuation is not a Ramsey plan. In state 2, a time-1 planner with inherited debt $b_1 = .214852$ would again manipulate the first-period interest rate: $\tau_1 = .103647$ instead of the continuation's .146523, and .147987 afterwards. The planner at t would always use the interest rate between t and $t + 1$ to reduce the value of the debt it inherits.

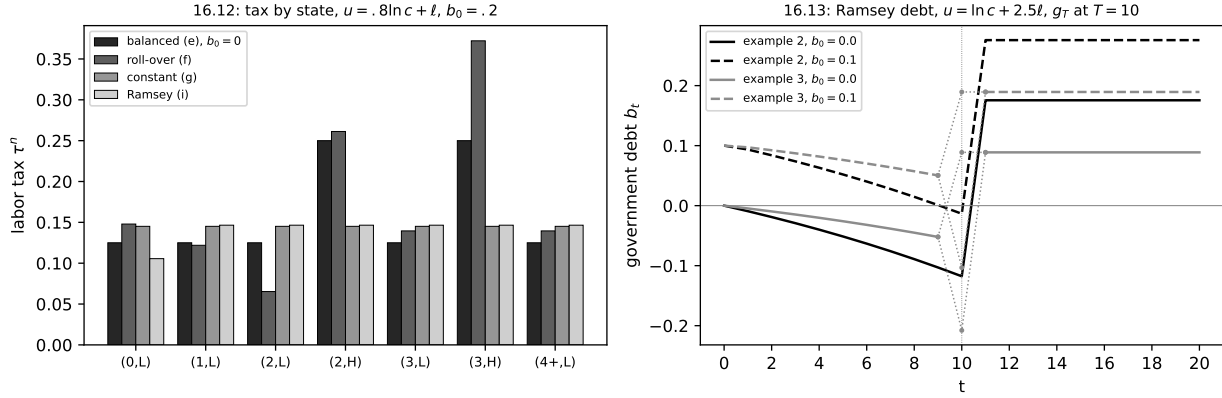


Figure S16.1. Left: labor tax by state in Exercise 16.12 under the balanced-budget ($b_0 = 0$), roll-over, constant-tax and Ramsey policies ($b_0 = .2$). Right: Ramsey debt paths in examples 2 and 3 of Exercise 16.13 for $b_0 = 0$ and $b_0 = .1$. In example 3 the dotted lines fan out to the two date- T states and merge again at $T + 1$.

Exercise 16.13 Positive initial debt

General case. For $t \geq 1$ the Ramsey allocation still solves (16.12.4), so it remains a time-invariant function of g_t , now evaluated at a larger Φ . More debt requires more revenue, so taxes are higher at all $t \geq 1$. Only the date-0 condition (16.12.5) changes. Write it as $D(c_0, g_0) = \Phi b_0(u_{cc} - u_{c\ell})$, where $D(c, g)$ is the left minus the right side of (16.12.4). Suppose $u_{c\ell} \geq 0$ and that D decreases in c (the second-order condition). Then the right side is negative, so $c_0 > \hat{c}(g_0)$, the allocation that (16.12.4) prescribes for g_0 . Since u_ℓ/u_c increases with c along $c + g = 1 - \ell$, the date-0 tax is below the tax the plan levies whenever $g_t = g_0$. As in Section 16.13.4, the planner raises the price $p_0 = \beta u_c(1)/u_c(0)$ at which it sells its debt at $t = 0$, i.e. it lowers the interest rate on the debt it rolls over.

Illustration. Take $u = \ln c + \kappa \ell$ with $\kappa = 2.5$ and $\beta = .95$, and put the spike in examples 2 and 3 at $T = 10$ ($g_T = .3$, or $.3$ with probability $.5$). Then (16.13.1) gives $1/\bar{c} = (1 + \Phi)\kappa$ for every $t \geq 1$ whatever g_t is. So for $t \geq 1$ consumption and the tax rate $\bar{\tau} = 1 - \kappa\bar{c} = \Phi/(1 + \Phi)$ are constant, including at T , and labor $n_t = \bar{c} + g_t$ absorbs the spike. The date-0 values solve $(1 + \Phi)\kappa c_0^2 - c_0 - \Phi b_0 = 0$, and Φ comes from $\sum \beta^t E[1 - \kappa(c_t + g_t)] = b_0/c_0$. For $t \geq 1$ Arrow prices are $\beta \times$ probabilities, and debt is the present value of future surpluses, $b_t = \bar{\tau}\bar{c}/(1 - \beta) - (1 - \bar{\tau})E_t \sum_{j \geq t} \beta^{j-t} g_j$.

example	b_0	Φ	$\bar{\tau}$ ($t \geq 1$)	τ_0	c_0	$\bar{c}$	gross rate $0 \rightarrow 1$
1: $g_t = .1$	0	.33333	.25000	.25000	.30000	.30000	1.05263
	.1	.36895	.26951	.18667	.32533	.29220	0.94542
2: $g_{10} = .3$	0	.02297	.02245	.02245	.39102	.39102	1.05263
	.1	.03708	.03575	.02657	.38937	.38570	1.04270
3: $g_{10} \in \{0, .3\}$	0	.01135	.01123	.01123	.39551	.39551	1.05263
	.1	.02487	.02427	.01809	.39277	.39029	1.04601

Example 1. With $b_0 = 0$ the budget balances every period. With $b_0 = .1$ the tax is lower at $t = 0$ than later. At $t = 0$ the government runs a primary deficit of .0206 and sells debt at the price $p_0 = \beta c_0 / \bar{c} = 1.0577$, a negative net interest rate. From then on it rolls over the constant debt $b_t = .1140$, paying the interest .0057 out of a constant surplus. Positive initial debt thus breaks perfect tax smoothing only at $t = 0$.

Example 2. With $b_0 = 0$ the book's pattern holds: b_t falls from $-.0092$ at $t = 1$ to $-.1177$ at $t = T$ (the government accumulates claims), and after paying g_T it owes .1756 forever. With $b_0 = .1$ the whole path shifts up. The government starts as a debtor ($b_1 = .0935$) and retires its debt with surpluses $\bar{\tau}\bar{c} = .0138$ that exceed the interest. It becomes a small net creditor just before T ($b_T = -.0135$), then issues debt to pay g_T and owes .2758 forever. The tax is still perfectly smooth for $t \geq 1$ (Figure S16.1, right).

Example 3. Before T debt again falls: with $b_0 = 0$ from $-.0047$ to $-.0521$ at $T - 1$, with $b_0 = .1$ from .0972 to .0504. At $T - 1$ the government buys Arrow securities that pay off if $g_T = .3$. The date- T debt is $b_T(.3) = -.2078$ and $b_T(0) = .0888$ when $b_0 = 0$, and $-.1033$ and .1894 when $b_0 = .1$. Afterwards it equals the second number in both states. The insurance position $b_T(0) - b_T(.3) = (1 - \bar{\tau})g_T$ (.2966 and .2927) hardly depends on b_0 . Positive initial debt changes the level of debt, the level of taxes and the date-0 interest-rate manipulation, but not the way the government insures against the spending shock.

Chapter 17

Self-Insurance

Solutions to all 5 exercises of Chapter 17 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Numerical examples and checks come from `code/ch17.py`.

Throughout, $(1+r)\beta = 1$, so with b_t the debt due at t the Euler equation is $u'(c_t) \geq E_t u'(c_{t+1})$, with equality when the borrowing constraint does not bind.

Exercise 17.1 *Deterministic endowments and borrowing limits*

Solving the budget constraint forward with $b_0 = 0$ and $\beta = (1+r)^{-1}$ gives the present-value constraint $\sum_t \beta^t c_t \leq \sum_t \beta^t \lambda^t = 1/(1-\beta\lambda)$. Ignoring borrowing constraints, the Euler equation $u'(c_t) = u'(c_{t+1})$ makes consumption constant at

$$\bar{c} = \frac{1-\beta}{1-\beta\lambda},$$

and the implied debt is the present value of future surpluses, $b_t = \sum_{s \geq t} \beta^{s-t} (y_s - \bar{c}) = \frac{\lambda^t - 1}{1-\beta\lambda}$.

(a) $\lambda < 1$: $b_t = (\lambda^t - 1)/(1-\beta\lambda) \leq 0$ for all t . The unconstrained plan never borrows (the consumer saves early, while income is high), so the no-borrowing constraint is slack and the optimum is

$$c_t = \frac{1-\beta}{1-\beta\lambda}, \quad b_{t+1} = \frac{\lambda^{t+1} - 1}{1-\beta\lambda} \leq 0.$$

(b) $\lambda > 1$: now $\bar{c} > 1 = y_0$, and the unconstrained plan would borrow. With $b_{t+1} \leq 0$, the optimum is $c_t = y_t = \lambda^t$ and $b_{t+1} = 0$ for all t . *Proof*: this plan is feasible, and the Kuhn–Tucker conditions hold with multipliers $\mu_t = u'(c_t) - u'(c_{t+1}) = u'(\lambda^t) - u'(\lambda^{t+1}) > 0$ on the constraints $b_{t+1} \leq 0$. The consumer would like to borrow against rising income every period but cannot. Saving would move resources from poorer to richer dates, which lowers utility, so consuming the endowment is optimal. The problem is concave, so the conditions are sufficient.

(c) The natural limit is the largest debt that can be repaid with zero consumption forever:

$$b_t \leq \sum_{j \geq 0} \beta^j y_{t+j} = \frac{\lambda^t}{1-\beta\lambda}, \quad t \geq 0,$$

finite because $|\beta\lambda| < 1$. It grows at rate λ with the endowment.

(d) $\lambda < 1$: the plan of (a) never borrows, so it satisfies the natural limit and is optimal: $c_t = (1-\beta)/(1-\beta\lambda)$, $b_{t+1} = (\lambda^{t+1} - 1)/(1-\beta\lambda)$.

(e) $\lambda > 1$: the unconstrained plan $c_t = (1-\beta)/(1-\beta\lambda)$, $b_{t+1} = (\lambda^{t+1} - 1)/(1-\beta\lambda) > 0$ satisfies $b_t < \lambda^t/(1-\beta\lambda)$, so the natural limit never binds, and this is the optimum. The consumer borrows against his growing income and consumes a constant amount, the annuity value of lifetime income. Comparing (b) and (e): the ad hoc constraint matters only when the consumer wants to borrow, i.e. when income grows.

Exercise 17.2 *Two draws, then permanence*

(a) From $y_0 \in \{1, 2\}$ the tree branches once, at $t = 1$, into $y_1 = 2$ (probability π) or $y_1 = 1$ (probability $1 - \pi$). After that each branch is a straight line: $y_t = y_1$ for all $t \geq 1$. All uncertainty is resolved at $t = 1$.

(b) Let $s = a_0 - c_0 \geq 0$ be saving at $t = 0$. From $t = 1$ on there is no risk and $(1 + r)\beta = 1$, so the consumer keeps consumption constant and never needs to borrow. Wealth at $t = 1$ is $(1 + r)s + y_1$, and he consumes its annuity value, $c_t = y_1 + (1 - \beta)(1 + r)s = y_1 + rs$ for $t \geq 1$, holding s in assets forever. Using $\beta r / (1 - \beta) = 1$, the date-0 problem is

$$\max_{0 \leq s \leq 2} \ln(2 - s) + \frac{\beta}{1 - \beta} E \ln(y_1 + rs), \quad \text{with FOC} \quad \frac{1}{2 - s} = \frac{\pi}{2 + rs} + \frac{1 - \pi}{1 + rs}.$$

At $s = 0$ the right side, $1 - \pi/2$, exceeds the left side, $\frac{1}{2}$, so saving is strictly positive. Clearing fractions gives a quadratic, whose positive root is

$$s = \frac{-(r + 2 - \pi) + \sqrt{(r + 2 - \pi)^2 + 8r(1 + r)(1 - \pi)}}{2r(1 + r)}.$$

The plan is $c_0 = 2 - s$ and $c_t = y_1 + rs$ for $t \geq 1$, with lending s at every date. For example, $r = .05$, $\pi = .5$ gives $s = 0.632$, $c_0 = 1.368$ and $c_t = y_1 + 0.032$. The consumer saves a third of his high initial endowment as a buffer against the possibility that his income drops permanently.

(c) With $y_0 = 1$, at $s = 0$ the marginal utility of consumption is 1, while the marginal value of saving is $E[1/y_1] = 1 - \pi/2 < 1$: the consumer would like to borrow. The no-borrowing constraint binds, so $s = 0$, $c_0 = 1$ and $c_t = y_1$ for $t \geq 1$: he simply consumes his endowment.

(d) For $t \geq 2$ (indeed $t \geq 1$), $u'(c_t) = 1/c_t$ is constant along each branch:

$$y_0 = 2 : u'(c_t) = \begin{cases} \frac{1}{2+rs} & \text{prob. } \pi, \\ \frac{1}{1+rs} & \text{prob. } 1 - \pi; \end{cases} \quad y_0 = 1 : u'(c_t) = \begin{cases} \frac{1}{2} & \text{prob. } \pi, \\ 1 & \text{prob. } 1 - \pi. \end{cases}$$

(e) With $\beta(1 + r) = 1$, $u'(c_t) \geq E_t u'(c_{t+1})$, so $\{u'(c_t)\}$ is a nonnegative supermartingale and converges almost surely. Here it converges immediately, to the random limits in (d). For $y_0 = 2$ the Euler equation holds with equality at $t = 0$ (a martingale); for $y_0 = 1$ it holds with strict inequality ($1 > 1 - \pi/2$), a strict supermartingale step caused by the binding constraint. Chamberlain and Wilson go further: when income keeps fluctuating, the limit of $u'(c_t)$ must be zero, i.e. consumption diverges to $+\infty$. That step fails here because the endowment becomes *deterministic* after $t = 1$. With no further risk, constant consumption is compatible with the Euler equation, and the limit is a positive random variable that depends on y_1 . So the result conforms to the convergence theorem, but not to the divergence of consumption, whose hypotheses (perpetual, nondegenerate income risk) are violated.

Exercise 17.3 *Natural borrowing limits in the stochastic savings problem*

(a) The worst case for repayment is the lowest endowment forever, and it has positive probability over any finite horizon. Repaying b_{t+1} with $c \equiv 0$ requires $b_{t+1} \leq \sum_{j \geq 0} (1 + r)^{-j} y_1 = \frac{(1+r)y_1}{r}$. So

debt must satisfy $(1+r)^{-1}b_{t+1} \leq y_1/r$, the present value at t of the minimal income stream from $t+1$ on. Any larger debt would, on a path of positive probability, force negative consumption.

(b) Cash on hand is $a_t = y_t - b_t$: the budget $c_t + b_t = y_t + (1+r)^{-1}b_{t+1}$ gives $a_t - c_t = -(1+r)^{-1}b_{t+1}$ and $a_{t+1} = (1+r)(a_t - c_t) + y_{t+1}$. Substituting $b_{t+1} = y_{t+1} - a_{t+1}$ into (a):

$$a_{t+1} - y_{t+1} \geq -\frac{(1+r)y_1}{r} \quad (t \geq 0).$$

(c) *Step 1: the natural limit never binds.* If debt reached the limit, the consumer would have to set $c = 0$ forever on the path where $y = y_1$ forever, which gives $u'(0) = \infty$. So he keeps strictly inside it, and the Euler equation holds with equality: $u'(c_t) = E_t u'(c_{t+1})$. *Step 2: convergence.* $\{u'(c_t)\}$ is a positive martingale, so by the martingale convergence theorem $u'(c_t) \rightarrow M$ almost surely, for a finite random variable $M \geq 0$. *Step 3: $M = 0$.* Suppose $\mathbb{P}(M > 0) > 0$. On $\{M > 0\}$, $c_t \rightarrow \bar{c} = (u')^{-1}(M) < \infty$. With i.i.d. income the problem is stationary in cash on hand, and optimal consumption is $c_t = h(a_t)$ for a continuous, strictly increasing function h (strict concavity). So $c_t \rightarrow \bar{c}$ implies $a_t \rightarrow \bar{a} = h^{-1}(\bar{c})$. But $a_{t+1} = (1+r)(a_t - h(a_t)) + y_{t+1}$, and the first term converges while y_{t+1} is an independent, nondegenerate draw each period. So a_{t+1} cannot converge (by Borel–Cantelli it keeps jumping by at least $y_S - y_1$ infinitely often). This contradiction shows $M = 0$ almost surely, i.e. $u'(c_t) \rightarrow 0$ and $c_t \rightarrow +\infty$. Intuitively, a consumer who never stops facing risk keeps accumulating precautionary assets, because the martingale $u'(c_t)$ can only settle down where marginal utility is zero.

(d) The ad hoc constraint set is contained in the natural one, so the value is lower and, starting from the same position, the ad hoc consumer saves more. He faces a real risk of hitting the constraint, an extra precautionary motive. Along identical endowment histories, the natural-limit consumer consumes more early on, particularly after bad draws (he can borrow), and accumulates assets more slowly. In both cases $u'(c_t)$ is a nonnegative supermartingale, and the argument of (c) shows that consumption diverges to $+\infty$ under either constraint. As wealth grows, both constraints become irrelevant and the two consumption functions approach each other, but the paths do not coincide, because the levels of wealth they have accumulated differ. (Section 17.3 also notes that an ad hoc limit can be represented as a natural limit for a suitably shifted endowment process.)

Exercise 17.4 Trade?

Note that the aggregate endowment $y_{1t} + y_{2t} = 2y_{10}$ is constant, and each individual endowment is a martingale.

(a) A competitive equilibrium consists of prices $\{q_t^0(s^t)\}$ for history-contingent claims (with s^t the history of ϵ 's) and allocations $\{c_{it}(s^t)\}$, $i = 1, 2$, such that: each household maximizes $E_0 \sum \beta^t u(c_{it})$ subject to $\sum_t \sum_{s^t} q_t^0(s^t) c_{it}(s^t) \leq \sum_t \sum_{s^t} q_t^0(s^t) y_{it}(s^t)$; and markets clear, $c_{1t}(s^t) + c_{2t}(s^t) = y_{1t}(s^t) + y_{2t}(s^t)$ for all t and s^t .

(b) Maximize $E_0 \sum \beta^t [u(c_{1t}) + u(c_{2t})]$ subject to $c_{1t} + c_{2t} = 2y_{10}$. With equal weights, equal marginal utilities give $c_{1t} = c_{2t} = y_{10}$ for every history: perfect risk sharing and constant consumption.

(c) Take $c_{it} = y_{10}$ and $q_t^0(s^t) = \beta^t \pi_t(s^t)$, the marginal utility $u'(y_{10})$ normalized by $u'(y_{10})$ at $t = 0$. Each household's wealth is $\sum_t \beta^t E_0 y_{it} = y_{10}/(1-\beta)$, since each endowment is a martingale. At these

prices a constant c exhausting wealth is $c = y_{10}$, and the first-order conditions $\beta^t \pi_t u'(c_{it}) = \mu_i q_t^0$ hold with $\mu_i = u'(y_{10})$. Markets clear, so this is the competitive equilibrium: it implements the equal-weight Pareto allocation because the two households have equal wealth.

(d) *The guess is correct.* Suppose $c_{it} = y_{it}$ and $b_{it} = 0$. Household i 's Euler equation is $u'(c_{it}) = \beta R_t E_t u'(c_{i,t+1})$. With quadratic utility and a martingale endowment,

$$u_1 - u_2 y_{it} = \beta R_t (u_1 - u_2 E_t y_{i,t+1}) = \beta R_t (u_1 - u_2 y_{it}),$$

so $R_t = \beta^{-1}$ for *both* households at every date and history (as long as consumption stays below bliss). Bond demands of zero clear the bond market, the budget constraints hold with $b_{it} = 0$, and the terminal condition holds trivially. So

$$R_t = \beta^{-1}, \quad b_{1t} = b_{2t} = 0 \quad \text{for all } t.$$

Although the two endowments are perfectly negatively correlated, and complete markets would insure each household fully, a risk-free bond is useless. Each household already sees its income as a random walk and wants consumption to be a random walk: the permanent-income consumer consumes his endowment when all shocks are permanent. Welfare is lower than under complete markets, since consumption has variance growing linearly with t instead of zero.

Exercise 17.5 Trade??

(a) *Guess* $b_{t+1} = 0$ and $c_t = y_t$ for all t . The Euler equation $c_t^{-\gamma} = \beta R E_t c_{t+1}^{-\gamma} = E_t c_{t+1}^{-\gamma}$ requires

$$E_t \left(\frac{y_{t+1}}{y_t} \right)^{-\gamma} = E \exp(-\gamma \sigma_y \varepsilon - \gamma \mu) = \exp\left(-\gamma \mu + \frac{\gamma^2 \sigma_y^2}{2}\right) = 1,$$

which holds exactly because $\mu = .5\gamma\sigma_y^2$ (confirmed by quadrature in `ch17.py`). The plan is feasible and satisfies the natural limit. Since $\log y$ has i.i.d. Gaussian increments, future income can fall arbitrarily close to zero, so the natural debt limit is in fact zero debt, and $b_{t+1} = 0$ respects it. The transversality condition holds trivially. The problem is concave, so these conditions are sufficient:

$$c_t = y_t, \quad b_{t+1} = 0 \quad \text{for all } t.$$

(b) Log income is a random walk, so every income shock is permanent. Consuming current income is then consuming permanent income, as in Friedman's hypothesis: a permanent shock changes consumption one for one, and there is no reason to save out of it. (Compare exercise 17.4 and example 1 of section 2.12.3.)

(c) Yes, in the sense that matters. Expected income grows: $E(y_{t+1}/y_t) = e^{\mu + \sigma_y^2/2} > 1$. A certainty-equivalent consumer facing the same expected income path, with $\beta R = 1$, would want constant consumption and would *borrow* against future growth. The prudent CRRA consumer ($u''' > 0$) instead chooses zero borrowing. The drift $\mu = .5\gamma\sigma_y^2$ is exactly the growth rate at which the precautionary motive, which increases with γ and σ_y^2 , offsets the desire to borrow against expected growth. Net saving is zero because precautionary saving and dissaving against expected income growth cancel.

Chapter 18

Incomplete Markets Models

Solutions to all 7 exercises of Chapter 18 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Most of the exercises ask for Bellman equations and definitions of stationary equilibria. `code/ch18.py` (Python instead of the book's Matlab) solves the mobility-cost model of 18.2 with the book's parameters. It also computes the Huggett and Bewley equilibria of 18.6 and the welfare comparison asked for there, and it illustrates the algorithms of 18.1 and 18.3.

Exercise 18.1 *Random discount factor (Bewley–Krusell–Smith)*

To avoid a clash of notation, write P_{ij}^β for the discount-factor chain and $P(s, s')$ for the employment chain, and assume the two chains are independent. Let $v_i(a, s) \equiv v(\bar{\beta}_i, a, s)$. The current β_t discounts next period's value, so for $i = 1, 2, 3$

$$v_i(a, s) = \max_{-\phi \leq a' \leq (1+r)a + ws} \left\{ u((1+r)a + ws - a') + \bar{\beta}_i \sum_{j=1}^3 P_{ij}^\beta \sum_{s'} P(s, s') v_j(a', s') \right\}.$$

These are three coupled Bellman equations. Equation i involves all three value functions, weighted by row i of P^β . If β and s are correlated, replace $P_{ij}^\beta P(s, s')$ by the joint transition probability of (β, s) ; nothing else changes. The stacked operator $T(v_1, v_2, v_3)$ satisfies Blackwell's conditions: it is monotone, and $T(v + c) = Tv + \bar{\beta}_i c \leq Tv + \max_i \bar{\beta}_i c$. So T is a contraction with modulus $\max_i \bar{\beta}_i < 1$ (for bounded u), and the solution is unique.

Algorithm. (1) Discretize assets on a grid $\mathcal{A} = \{-\phi = \bar{a}_1 < \dots < \bar{a}_N\}$ and stack the unknowns into $v_i(\bar{a}_h, \bar{s}_k)$, a vector of length $3Nn$. (2) Start from any guess, e.g. $v_i \equiv 0$. (3) Given $v^{(k)}$, compute the continuation values $W_i(a', s) = \sum_j P_{ij}^\beta \sum_{s'} P(s, s') v_j^{(k)}(a', s')$ for all (i, a', s) . Then maximize $u((1+r)a + ws - a') + \bar{\beta}_i W_i(a', s)$ over $a' \in \mathcal{A}$ to get $v^{(k+1)}$ and the policies $g_i(a, s)$. (4) Iterate to convergence. Howard's policy improvement (evaluate a fixed policy by solving a linear system, then improve) speeds this up. Equivalently, one can iterate on the Euler inequality $u'(c_i(a, s)) \geq \bar{\beta}_i(1+r) \sum_j P_{ij}^\beta \sum_{s'} P(s, s') u'(c_j(a', s'))$, with equality unless $a' = -\phi$, by the endogenous grid method. (5) The policies and the two chains induce a Markov chain on (β, a, s) . Its invariant distribution gives the cross-section; in a Bewley–Aiyagari economy one then adjusts r until average assets equal the supply of the asset.

Illustration. Take $u = -1/c$, $r = .02$, $\bar{\beta} = (.94, .96, .98)$ with $P^\beta = \begin{bmatrix} .90 & .10 & 0 \\ .05 & .90 & .05 \\ 0 & .10 & .90 \end{bmatrix}$, $s \in \{.3, 1\}$ with $P = \begin{bmatrix} .5 & .5 \\ .05 & .95 \end{bmatrix}$, $w = 1$ and $\phi = 0$. The endogenous grid method solves the three coupled equations in a few seconds. In the invariant distribution, mean assets are 1.08, 1.28 and 1.51 for the three discount factors, which have population shares $(.25, .5, .25)$. The most patient quarter holds 29.3% of wealth. Persistent heterogeneity in patience is what Krusell and Smith use to fatten the upper tail of the wealth distribution.

Exercise 18.2 Mobility costs (Bertola)

(a) The state is (A, w) , with w the wage of the current job this period. Staying gives income $RA + w$, and next period's wage is drawn from row w . Moving gives income $RA + w_g - m$, and the new job's next wage is drawn from the good row. Writing $E_w[v(A', \cdot)] = P(w, w_g)v(A', w_g) + P(w, w_b)v(A', w_b)$,

$$\begin{aligned} v(A, w) &= \max\{v^s(A, w), v^m(A)\}, \\ v^s(A, w) &= \max_{0 \leq A' \leq RA+w} \{u(RA + w - A') + \beta E_w[v(A', \cdot)]\}, \\ v^m(A) &= \max_{0 \leq A' \leq RA+w_g-m} \{u(RA + w_g - m - A') + \beta E_{w_g}[v(A', \cdot)]\}. \end{aligned}$$

In the good state, moving gives the same future as staying but m less today, so $v^m(A) < v^s(A, w_g)$ and nobody leaves a good job. The moving decision matters only for $w = w_b$, where the worker trades current income ($w_g - m = .5$ instead of $w_b = 1$) for better wage prospects ($p = .8$ instead of $.2$ chance of w_g next period).

(b) We use value iteration on 601 grid points in $[0, 3]$ (418 iterations to a tolerance of 10^{-10}); see Figure S18.1. Selected decisions:

A	A', c (good wage)	move? (bad wage)	A', c (bad wage)
0	.200, 1.200	no	.000, 1.000
.50	.655, 1.255	no	.350, 1.160
.55	.700, 1.261	yes	.000, 1.061
1.00	1.115, 1.305	yes	.300, 1.220
2.00	2.060, 1.380	yes	1.225, 1.315

The moving rule is a wealth threshold: *a worker in a bad job moves if and only if $A \geq .55$* . The cost must be paid out of current resources and borrowing is not allowed. A poor worker could pay it only by cutting consumption sharply, which is very costly with $\sigma = 4$, so poor workers stay in bad jobs. Wealthier workers use their assets to smooth the cost and move. A worker just above the threshold spends all his wealth on the move. In good jobs workers save, and more so the richer they are. Below about $A = 3.3$ assets rise, a buffer against the bad state and a fund for a future move. In bad jobs they run assets down. The upper end of the grid is almost harmless: on $[0, 8]$ the good-state saving rule has its fixed point at $A = 3.275$, and every statistic below is unchanged to four decimals.

(c) Let $\tilde{w}(A, w) = w_g$ if $w = w_g$ or the worker moves, and $\tilde{w} = w_b$ otherwise. The decision rules induce the Markov chain

$$\mathbb{P}[(A_{t+1}, w_{t+1}) = (A', w') \mid (A_t, w_t) = (A, w)] = \mathbf{1}\{A' = g(A, w)\} P(\tilde{w}(A, w), w'),$$

on the 2×601 states. It has exactly one unit eigenvalue (next largest modulus .910), so the invariant distribution exists and is unique. Its recurrent class has 1188 states and covers practically all of $[0, 3]$, but the mass above $A = 2.5$ is only .002.

(d) Read as a cross-section of many workers, the invariant distribution has mean wealth .711 and median .62. The 25th and 90th percentiles are .20 and 1.55, and 13.3% of workers hold no assets. Only 33.2% of workers are in bad jobs, fewer than the 50% the symmetric wage chain alone would imply, because 11.2% of all workers (a third of those in bad jobs) move each period. Wealth and job quality are correlated: bad-job workers are poorer (Figure S18.1, right). About 30% of them have zero wealth and are stuck, unable to afford a move, while the rest run down their assets or move. This is Bewley's point in miniature: with incomplete markets, the wealth distribution affects aggregate mobility, and poor workers are trapped in bad matches.

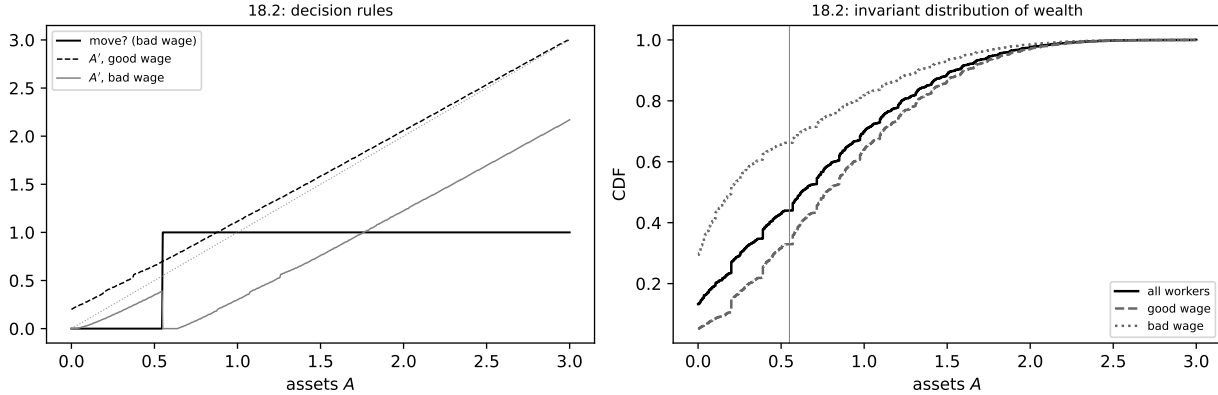


Figure S18.1. Exercise 18.2. Left: saving rules in the good and bad states and the moving decision (1 = move) in the bad state. Right: CDF of wealth in the invariant distribution, overall and by current wage; the vertical line marks the moving threshold $A = .55$.

Exercise 18.3 Unemployment

(a) Use the timing implicit in (b): flows decided at t take effect at $t + 1$. A worker fired at t is unemployed from $t + 1$, and an offer accepted at t starts paying at $t + 1$. Let $V(w)$ be the value of starting a period employed at wage w , and Q the value of starting a period unemployed. Then

$$V(w) = w + \beta[(1 - \lambda)V(w) + \lambda Q], \quad Q = \beta \left[\mu \int \max\{V(w), Q\} dF(w) + (1 - \mu)Q \right].$$

So $V(w) = (w + \beta\lambda Q)/(1 - \beta(1 - \lambda))$ is strictly increasing in w , while rejecting yields the constant Q . The optimal policy is therefore to accept iff $w \geq \bar{w}$, where $V(\bar{w}) = Q$, i.e. $\bar{w} = (1 - \beta)Q$: the reservation-wage property. Since $V(w) - Q = (w - \bar{w})/(1 - \beta(1 - \lambda))$, the second equation becomes

$$\bar{w} = \frac{\beta\mu}{1 - \beta(1 - \lambda)} \int_{\bar{w}}^{\infty} (w - \bar{w}) dF(w).$$

The left side increases and the right side decreases in $\bar{w}$, so $\bar{w}(\mu)$ is unique. It increases with μ : more frequent offers make workers choosier.

(b) A fraction λ of the employed lose their jobs, and a fraction $f(U_t)[1 - F(\bar{w})]$ of the unemployed find acceptable jobs. Hence

$$U_{t+1} = U_t + \lambda(1 - U_t) - f(U_t)[1 - F(\bar{w})]U_t,$$

and a stationary rate satisfies $\lambda(1 - U) = f(U)[1 - F(\bar{w})]U$: inflows equal outflows.

(c) A stationary equilibrium is a pair $(U^*, \bar{w}^*)$ such that (i) $\bar{w}^* = \bar{w}(f(U^*))$ is the optimal reservation wage of a worker who takes $\mu = f(U^*)$ as given, and (ii) $\lambda(1 - U^*) = f(U^*)[1 - F(\bar{w}^*)]U^*$.

(d) Define $H(U) = f(U)[1 - F(\bar{w}(f(U)))]U - \lambda(1 - U)$ on $[0, 1]$. For each trial U , compute $\mu = f(U)$ and solve the reservation-wage equation of (a) for $\bar{w}(\mu)$ by bisection, since its right side is monotone. Then look for zeros of H by scanning a grid and bisecting on sign changes; several equilibria are possible, so all sign changes should be found. Note $H(0) = -\lambda < 0$ and $H(1) = 0$. $U = 1$ is always a degenerate stationary point: nobody is employed and no offers arrive. If $|f'(1)| > \lambda$, then $H < 0$ near 0 and $H > 0$ just below 1 (there $\bar{w}(0) = 0$), so an interior equilibrium exists. For instance, with $f(U) = 1 - U$, F uniform on $[0, 1]$, $\beta = .95$ and $\lambda = .05$, the reservation-wage equation is $\bar{w} = 9.7436 \mu(1 - \bar{w})^2/2$. The unique interior equilibrium is $U^* = .1309$, with $\mu = .8691$ and $\bar{w}^* = .6180$.

Exercise 18.4 *Asset insurance*

First, the Bellman equation should include the non-capital income y :

$$v(k, x) = \max_{0 \leq k' \leq y + \max(x, g)k^\alpha - \tau} \left\{ u(y + \max(x, g)k^\alpha - k' - \tau) + \beta \sum_{x'} P(x, x') v(k', x') \right\}.$$

Remark. The book's Bellman equation omits y , which appears in the budget constraint; it is presumably a typo.

Given $(g, y, \alpha, \beta, P, \bar{x}_1, \bar{x}_2)$, a *stationary equilibrium* is a tax τ , a value function v , a decision rule G and a distribution λ on (k, x) such that:

- (1) given τ , v solves the Bellman equation and $G(k, x)$ attains the maximum;
- (2) λ is stationary under the Markov chain induced by G and P :

$$\lambda(k', x') = \sum_x \sum_{\{k: G(k, x) = k'\}} \lambda(k, x) P(x, x');$$

- (3) the government's budget balances period by period: the per-capita tax equals per-capita insurance payments,

$$\tau = \max(0, g - \bar{x}_2) \sum_k k^\alpha \lambda(k, \bar{x}_2) \quad (= \nu - Y).$$

Output Y and the insured return ν are then implied. The resource constraint $\bar{c} + \bar{k} = y + Y$ holds automatically, since households receive ν , pay τ , and $\nu - \tau = Y$. An equilibrium requires $\tau < y$, so that households with little capital can pay the tax. Computationally one iterates on τ : guess τ , solve the Bellman equation, compute λ , update τ from 3, and repeat. Insurance ($g > \bar{x}_2$) raises the return on capital in the bad state, which encourages investment. Because it is financed by a lump-sum tax, it also redistributes toward households with more capital in the bad state.

Exercise 18.5 *Matching and job quality (Acemoglu and Shimer)*

(a) Write $\mathcal{U}(c, h) = (1 - \theta)^{-1}(c(\bar{h} - h)^\eta)^{1-\theta}$. Let W_g, W_b be the values of holding a good or bad job, and V_1, V_0 the values of being unemployed and eligible or ineligible, all as functions of assets a . Primes denote next period, and $a' = R(a + y - c) \geq 0$. A job found this period starts next period, and

a quit takes effect immediately. With $j \in \{g, b\}$ the type searched for and $0 \leq h \leq \min(\bar{h}, 1/m_j)$:

$$\begin{aligned} W_g(a) &= \max_{c,h} \left\{ \mathcal{U}(c, h) + \beta[(1 - \delta_g)W_g(a') + \delta_g V_1(a')] \right\}, & y &= w_g h(1 - \tau), \\ W_b(a) &= \max \left\{ \max_{c,h} \left\{ \mathcal{U}(c, h) + \beta[(1 - \delta_b)W_b(a') + \delta_b V_1(a')] \right\}, V_0(a) \right\}, & y &= w_b h(1 - \tau), \\ V_1(a) &= \max_{c,h,j} \left\{ \mathcal{U}(c, h) + \beta[m_j h W_j(a') + (1 - m_j h)((1 - \phi)V_1(a') + \phi V_0(a'))] \right\}, & y &= b, \\ V_0(a) &= \max_{c,h,j} \left\{ \mathcal{U}(c, h) + \beta[m_j h W_j(a') + (1 - m_j h)V_0(a')] \right\}, & y &= 0. \end{aligned}$$

The second argument of the max in W_b is the quit option. A good-job worker could also quit, but never gains from it. The four equations are coupled through the transitions. They can be solved jointly by value iteration on a grid for a ; for each unemployed state the discrete choice of j is made by comparing the two maximized values.

(b) A stationary equilibrium is: a tax rate τ ; value functions (W_g, W_b, V_1, V_0) ; decision rules for consumption and hours or search time in every state, the search direction $j_e(a)$ for $e \in \{1, 0\}$ and the quit rule $q(a)$ in bad jobs; and a distribution Γ over (a, e) , $e \in \{g, b, 1, 0\}$, such that: (i) given τ (and the exogenous $R, w_g, w_b, b, m_g, m_b, \delta_g, \delta_b, \phi$), the value functions solve the Bellman equations and the rules attain them; (ii) Γ is invariant under the Markov process on (a, e) generated by the rules and the exogenous job-finding, separation and expiry probabilities; (iii) the insurance budget balances,

$$b \Gamma(\mathbb{R}_+, 1) = \tau \left[w_g \int h_g(a) \Gamma(da, g) + w_b \int h_b(a) \Gamma(da, b) \right].$$

Since R is exogenous (a storage technology or a small open economy), there is no asset-market-clearing condition. Unemployment, the good/bad job mix, output and welfare are statistics of Γ .

(c) Only the bad-job equation and the law of motion change:

$$W_b(a) = \max \left\{ \max_{c,h} \left\{ \mathcal{U}(c, h) + \beta[(1 - \delta_b)(\delta_{up} W_g(a') + (1 - \delta_{up})W_b(a')) + \delta_b V_1(a')] \right\}, V_0(a) \right\}.$$

In the stationary distribution, a fraction $(1 - \delta_b)\delta_{up}$ of bad-job workers move to good jobs each period.

(d) Yes; it would most likely weaken the case for higher benefits without reversing its logic. In Acemoglu and Shimer, benefits raise productivity because they let poorly insured workers take the risk of searching for good jobs (low m_g) instead of bad ones. With upgrades, a bad job is a stepping stone: its value W_b now contains $\delta_{up} W_g$. The private and social gap between searching for a good job and taking a bad one therefore shrinks. Part of the composition gain that more insurance would bring is obtained anyway, through upgrades. So the marginal productivity gain from raising b falls, while its moral-hazard cost (longer spells, more time without work) does not, and the net benefit is smaller. As $\delta_{up} \rightarrow 1$ the direction of search hardly matters, and benefits would be justified only by consumption insurance. Upgrades also make quitting a bad job to search less attractive, which removes one more channel through which benefit design affects job quality. The size of these effects depends on δ_{up} and must be settled by recalibrating the model.

Exercise 18.6 *Gluing stationary equilibria*

(a) Let $A(r) = [-\phi, \infty)$, with $\phi > 0$ an ad hoc limit no looser than the natural one. A Huggett equilibrium is an interest rate r , a value function $v(a, s)$ and policy $g(a, s)$, and a distribution $\Lambda(a, s)$ such that: (i) given r , g solves $v(a, s) = \max_{a' \in A(r)} \{u((1+r)a + ws - a') + \beta \sum_{s'} P(s, s') v(a', s')\}$; (ii) Λ is invariant under the chain induced by g and P ; (iii) the IOU market clears, $\sum_{a,s} \Lambda(a, s) g(a, s) = 0$. IOUs are in zero net supply, so $Ea(r) = 0$.

(b) For the economy to be in this stationary equilibrium from time 0, the initial draws must reproduce Λ . So F must be the marginal CDF of assets under Λ , $F(a) = \sum_s \Lambda(\{\tilde{a} \leq a\}, s)$. Moreover, the joint distribution of $(\tilde{a}(i, 0), s_0(i))$ must be Λ itself, not just have marginals F and the invariant distribution of P : poor households are disproportionately in low-income states.

(c) Figure S18.2 (left) draws the curve for Huggett's (1993) parameters: $\sigma = 1.5$, $\beta = .99322$ per sixth of a year, $s \in \{1, .1\}$, $P = \begin{bmatrix} .925 & .075 \\ .5 & .5 \end{bmatrix}$, $w = 1$ and $\phi = 2$. $Ea(r)$ rises with r and diverges as $r \uparrow \rho = 1/\beta - 1 = .0068$. It crosses zero at $r_H = -.01307$ per period (-7.6% a year), so $r_H < 0$. At $r = 0$ average assets are $Ea(0) = 1.4231 > 0$.

(d) This part sets up the experiment; the transfers are chosen in (f).

(e) With constant M_0 and a constant price level p , currency earns $r = 0$. Private IOUs must then also pay $r = 0$: if they paid more, nobody would hold currency, and if they paid less, nobody would hold IOUs. A stationary Bewley equilibrium is a price level $p > 0$, a policy $g_B(a, s)$ and a distribution $\Lambda_B(a, s)$ such that: (i) g_B solves the household's problem at $r = 0$ with the same limit $A = [-\phi, \infty)$; (ii) Λ_B is invariant; (iii) net asset demand equals the real value of the outside money, $\sum \Lambda_B(a, s) g_B(a, s) = Ea(0) = M_0/p$ (private IOUs net out). It exists iff $Ea(0) > 0$, i.e. iff $r_H < 0$ (the case in (c)), and then $p = M_0/Ea(0)$. G is the marginal CDF of assets under Λ_B . By Proposition 1 of Section 18.6, $Ea(0; \phi) = Ea(0; 0) - \phi$, so the equilibrium requires $\phi < Ea(0; 0)$ ($= 3.42$ here).

(f) Index consumers by their rank in the Huggett distribution. Consumer i holds $a_H(i) = F^{-1}(i)$, where $F^{-1}(u) = \inf\{a : F(a) \geq u\}$ and ties at atoms are broken by i . Let $\tilde{F}$ be the CDF of time-0 wealth $(1 + r_H)a_H$ and set

$$\tilde{T}_{-1}(i) = G^{-1}(i) - \tilde{F}^{-1}(i) = G^{-1}(i) - (1 + r_H)F^{-1}(i).$$

Then $a_B(i) = (1 + r_H)a_H(i) + \tilde{T}_{-1}(i) = G^{-1}(i)$. With i uniform on $(0, 1)$, $\mathbb{P}\{G^{-1}(i) \leq a\} = \mathbb{P}\{i \leq G(a)\} = G(a)$ (the inverse probability integral, or Smirnov, transform), so post-transfer assets have CDF G . Both quantile functions are nondecreasing, so every consumer keeps his rank. Aggregate transfers equal $\int G^{-1} - (1 + r_H) \int F^{-1} = M_0/p - 0$, which is the government's budget. In our example the transfers average 1.4231 and range from $-.026$ to 3.005 . They are slightly negative only for the 0.12% poorest. Those agents are at the debt limit in both equilibria, and $(1 + r_H)(-2) = -1.974 > -2$.

The overall-rank scheme reproduces the marginal G but not the joint distribution Λ_B . In our example $\mathbb{P}(s = \text{low} \mid \text{rank})$ differs between the two equilibria by up to .33, so right after the transfers the economy would not yet be in the stationary Bewley equilibrium. Applying the same

construction *within each employment state*, with $\tilde{T}(i, s) = G_s^{-1}(u) - \tilde{F}_s^{-1}(u)$ where u is i 's rank among consumers with the same s and $G_s, \tilde{F}_s$ are conditional CDFs, maps Λ_H exactly into Λ_B . This works because both have the same s -marginal, the invariant distribution of P . That scheme preserves ranks within each state, and it is the one to use if the Bewley equilibrium is to start immediately.

(g) Not necessarily. Moving to the Bewley equilibrium raises the interest rate from $r_H < 0$ to 0, which helps lenders and hurts borrowers, and it hands out M_0/p in transfers, which helps almost everybody. To sort agents: (1) compute both stationary equilibria and their value functions $v_H(a, s)$ at r_H and $v_B(a, s)$ at $r = 0$; (2) for each (i, s) compute Huggett wealth and post-transfer Bewley wealth; (3) the agent prefers the Bewley equilibrium iff $v_B(a_B(i), s) > v_H(a_H(i), s)$. With CRRA utility the consumption-equivalent gain is $(v_B/v_H)^{1/(1-\sigma)} - 1$. Under the within-state scheme the comparison is exact, since the economy is in the Bewley equilibrium from time 0 on. In our example 99.92% of agents prefer the Bewley equilibrium, with gains of up to 2.7% of consumption. The exceptions are the 0.64% poorest low-income agents, 0.08% of the population, who sit at the debt limit. They lose the subsidy that $r_H < 0$ gives to debtors and receive a slightly negative transfer, a loss of 0.73% of consumption. Figure S18.2 (right) plots the gains. The overall-rank scheme gives practically the same split (99.92%).

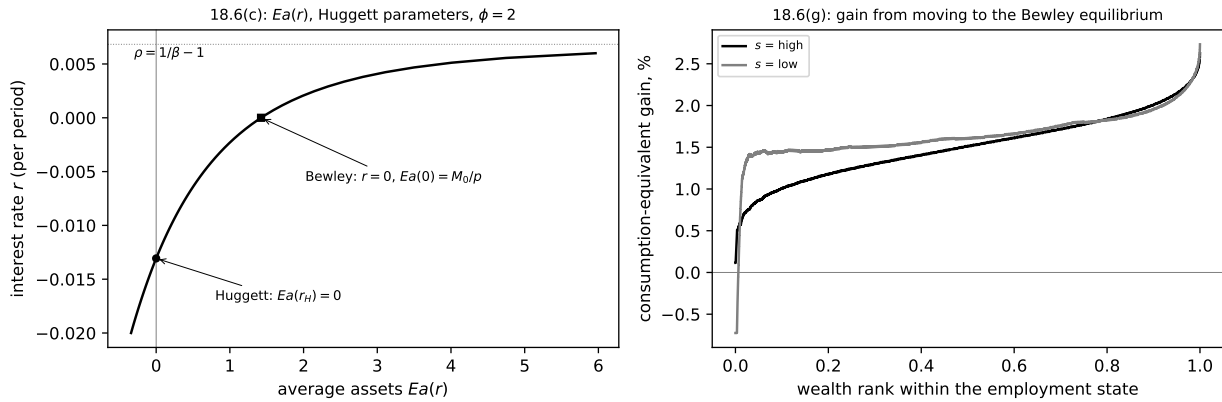


Figure S18.2. Exercise 18.6 with Huggett's parameters. Left: average assets $Ea(r)$, the Huggett equilibrium r_H and the Bewley equilibrium at $r = 0$. Right: consumption-equivalent gain from moving to the Bewley equilibrium under rank-preserving transfers within each employment state.

Exercise 18.7 *Real bills*

Equation (2) refers to a numbered display of the exercise in the book.

(a) Summing $A_{j+1} - A_j = (M_{j+1} - M_j)/p$ over $j = 0, \dots, t$ with $A_0 = 0$ gives $A_{t+1} = (M_{t+1} - M_0)/p$. The constraint $A_{t+1} \geq 0$ means $M_{t+1} \geq M_0$.

(b) With $Ea(0) = M_0/p$ and (a), $A_{t+1} + Ea(0) = (M_{t+1} - M_0)/p + M_0/p = M_{t+1}/p$ for every t and every admissible sequence. At $r = 0$ with a constant price level, currency and IOUs both earn zero, so households are indifferent about the composition of their portfolios. Their decision rules and Λ_B are unchanged. Their net asset position, currency plus IOUs held minus IOUs issued, still

averages $Ea(0)$. The extra currency is matched one for one by the IOUs that households have sold to the monetary authority.

(c) The price level is pinned down by the time-0 condition $Ea(0) = M_0/p$. $Ea(0)$ is a property of household behavior at $r = 0$ and does not depend on money. So a larger M_0 , handed out as in 18.6, raises p in proportion: this is outside money, a net asset of the private sector. Increases of M_{t+1} above M_0 are backed one for one by private IOUs. They raise M_{t+1}/p and A_{t+1} equally, leave the net asset supply $M_{t+1}/p - A_{t+1} = M_0/p$ unchanged, and satisfy (2) at the same p .

(d) M_0 is outside money: unbacked and a net asset of households. $M_{t+1} - M_0$ is inside money: a liability of the monetary authority fully backed by households' own IOUs, so it is not net wealth. The real bills doctrine says that money issued against sound short-term private paper is not inflationary. In this model it holds exactly: the quantity theory applies to outside money only, and any amount of inside money leaves p unchanged.

Chapter 19

Dynamic Stackelberg Problems

Solutions to all 3 exercises of Chapter 19 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Numerical examples and checks come from `code/ch19.py`. Every Stackelberg plan reported below is computed twice, recursively (Riccati equation plus the initial condition for the forward-looking variable) and by brute force as one large quadratic program, and the two agree to 10^{-13} . The script also reproduces the book's numerical example of Section 19.5.3.

Notation follows the chapter: z_t are natural state variables, x_t forward-looking variables, $y_t = (z'_t, x'_t)'$, and the leader minimizes $\sum_t \beta^t (y'_t R y_t + u'_t Q u_t)$ subject to $y_{t+1} = A y_t + B u_t$. Subproblem 1 gives $v(y) = -y' P y$ with

$$P = R + \beta A' P A - \beta^2 A' P B (Q + \beta B' P B)^{-1} B' P A, \quad u_t = -F y_t, \quad F = \beta (Q + \beta B' P B)^{-1} B' P A,$$

and subproblem 2 gives $x_0 = -P_{22}^{-1} P_{21} z_0$, i.e. $\mu_{x0} = 0$ for the multiplier $\mu_{xt} = P_{21} z_t + P_{22} x_t$.

Exercise 19.1 *Optimal monetary policy with commitment*

Equations (1)–(3) refer to the numbered displays of the exercise in the book.

(a) $m_t - p_t$ is the log of real balances and, under perfect foresight, $p_{t+1} - p_t$ is the expected inflation rate, the opportunity cost of holding money (the real return on money is $-(p_{t+1} - p_t)$). Equation (2) is Cagan's demand function: a one-point rise in expected inflation lowers real balances by α percent. Rearranging (2) gives $(1 + \alpha)p_t = m_t + \alpha p_{t+1}$, that is,

$$p_t = \frac{1}{1 + \alpha} m_t + \frac{\alpha}{1 + \alpha} p_{t+1} = (1 - \lambda) m_t + \lambda p_{t+1}, \quad \lambda = \frac{\alpha}{1 + \alpha} \in (0, 1).$$

Iterating T times gives $p_t = (1 - \lambda) \sum_{j=0}^{T-1} \lambda^j m_{t+j} + \lambda^T p_{t+T}$. The homogeneous equation $p_t = \lambda p_{t+1}$ has solutions $c \lambda^{-t}$, so every solution has the form

$$p_t = (1 - \lambda) \sum_{j=0}^{\infty} \lambda^j m_{t+j} + c \lambda^{-t}.$$

The sum converges because $\{m_t\}$ is square summable, hence bounded. Since $\lambda^{-1} > 1$, the term $c \lambda^{-t}$ is an explosive price-level bubble. The only square-summable (indeed the only bounded) solution has $c = 0$:

$$p_t = (1 - \lambda) \sum_{j=0}^{\infty} \lambda^j m_{t+j} = \frac{1 - \lambda}{1 - \lambda L^{-1}} m_t.$$

The price level is a weighted average of current and future money supplies; the weights $(1 - \lambda) \lambda^j$ sum to one.

(b) Normalize the target log price level to zero. The authority wants price stability (p_t^2) but finds rapid changes in money costly (u_t^2). Since $m_0 > 0$, stabilizing prices eventually requires removing money, and doing so quickly is costly. The authority does not set p_t ; money holders do, through (2).

By (a), $p_t = (1 - \lambda) \sum_j \lambda^j m_{t+j}$ depends on the whole future path of money. These equations are *implementability constraints*: a sequence $\{m_t, p_t\}$ can be an equilibrium outcome only if it satisfies them. They make the problem a Stackelberg problem. The authority (the leader) can lower today's price level by promising low future money supplies, and money holders (the followers) react now to those promises. Promises about $u_{t+1}, u_{t+2}, \dots$ thus influence p_t at no adjustment cost today, but they must be kept later.

(c) Take the natural state $z_t = m_t$ and the forward-looking variable $x_t = p_t$, with $y_t = (m_t, p_t)'$. Solving (2) for p_{t+1} gives $p_{t+1} = \frac{1+\alpha}{\alpha} p_t - \frac{1}{\alpha} m_t$, so

$$y_{t+1} = Ay_t + Bu_t, \quad A = \begin{bmatrix} 1 & 0 \\ -1/\alpha & (1+\alpha)/\alpha \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

and the criterion is $-\sum_t \beta^t (y_t' R y_t + u_t^2)$ with $\beta = .95$, $R = \text{diag}(.00001, 1)$, $Q = 1$. The second row is the implementability constraint in the form (19.2.6), $p_t = (1 - \lambda)m_t + \lambda p_{t+1}$, with $\phi_0 = \lambda \in (0, 1)$, so the chapter's Condition A holds. The problem has two Bellman equations:

$$\text{subproblem 1: } v(m, p) = \max_u \left\{ -p^2 - u^2 - .00001m^2 + \beta v(m + u, \frac{1+\alpha}{\alpha} p - \frac{1}{\alpha} m) \right\},$$

$$\text{subproblem 2: } w(m_0) = \max_{p_0} v(m_0, p_0).$$

Subproblem 1 treats p as if it were inherited; its solution is $v(y) = -y' P y$ and $u = -F y$ from the Riccati equation above. Subproblem 2 chooses the initial price level optimally: $p_0 = -P_{22}^{-1} P_{21} m_0$, equivalently $\mu_{p_0} = P_{21} m_0 + P_{22} p_0 = 0$. (The .00001 m^2 term makes R positive definite, which guarantees a unique stabilizing solution of the Riccati equation; it has no visible effect on the numbers.)

Why “dynamic programming squared”? Money holders' behavior is itself the solution of a dynamic optimization problem, summarized by its Euler equation (2), and the authority's dynamic program takes that Euler equation as a constraint. So one dynamic program sits inside another. As in the book's other uses of the phrase (Chapters 1 and 24), the variable carrying the followers' side of the problem plays a double role. The price level p_t summarizes money holders' expectations about the future, so it is forward looking. It is also, for $t \geq 1$, a promise inherited from the past that the authority must keep, so it enters the authority's Bellman equation as a state variable. Accordingly the problem needs two Bellman equations with different state vectors: $v(m, p)$ for a continuation leader and $w(m)$ for the leader at time 0.

(d) Given m_0 , a *plan* is a sequence $(\vec{u}_0, \vec{p}_0, \vec{m}_1) = \{u_t, p_t, m_{t+1}\}_{t \geq 0}$ that satisfies (1), (2) and the no-bubble condition, i.e. $p_t = (1 - \lambda) \sum_j \lambda^j m_{t+j}$ for all t . Equivalently, a plan is a sequence of possibly history-dependent rules $u_t = \sigma_t(m_0, \dots, m_t)$; with no uncertainty, a strategy and the sequence it induces carry the same information. The *continuation* of a plan at date t is its tail $(\vec{u}_t, \vec{p}_t, \vec{m}_{t+1})$. An optimal (Stackelberg) plan maximizes (3) over all plans, given m_0 .

(e) With P and F from (c),

$$p_0 = -P_{22}^{-1} P_{21} m_0, \quad u_t = -F \begin{bmatrix} m_t \\ p_t \end{bmatrix}, \quad \begin{bmatrix} m_{t+1} \\ p_{t+1} \end{bmatrix} = (A - BF) \begin{bmatrix} m_t \\ p_t \end{bmatrix}, \quad t \geq 0.$$

The rule is time invariant in (m_t, p_t) , but p_t is a linear function of the history $m_0, \dots, m_t$, so u_t is history dependent: it is not a function of m_t alone. An equivalent representation uses the multiplier $\mu_{pt} = P_{21}m_t + P_{22}p_t$ as the extra state, with initial condition $\mu_{p0} = 0$: substituting $p_t = P_{22}^{-1}(\mu_{pt} - P_{21}m_t)$ into the rule and the law of motion gives a linear system in (m_t, μ_{pt}) .

For $\alpha = 1$ ($\lambda = .5$), `ch19.py` gives

$$P = \begin{bmatrix} 6.79015 & -9.44112 \\ -9.44112 & 15.06706 \end{bmatrix}, \quad u_t = -2.06958 m_t + 2.40759 p_t, \quad p_0 = .626607 m_0,$$

or, in terms of the multiplier,

$$u_t = -.56097 m_t + .15979 \mu_{pt}, \quad \begin{bmatrix} m_{t+1} \\ \mu_{p,t+1} \end{bmatrix} = \begin{bmatrix} .43903 & .15979 \\ -.32979 & .49138 \end{bmatrix} \begin{bmatrix} m_t \\ \mu_{pt} \end{bmatrix}, \quad \mu_{p0} = 0.$$

The closed-loop eigenvalues are $.465 \pm .228i$, so money and prices converge to zero with damped oscillations. With $m_0 = 1$:

t	m_t	p_t	u_t	μ_{pt}
0	1.00000	.62661	-.56097	.00000
1	.43903	.25321	-.29898	-.32979
2	.14005	.06739	-.12760	-.30685
3	.01246	-.00527	-.03846	-.19697
4	-.02600	-.02299	-.00153	-.10090

Along this path $p_t = (1 - \lambda) \sum_j \lambda^j m_{t+j}$ holds to machine precision, so the recursive plan respects the implementability constraints. The authority promises to overshoot, pushing m_t below zero from $t = 4$; that promise lowers the price level at the start. Figure S19.1 plots the path.

(f) Disagree. The plan is time inconsistent: its continuation from $t \geq 1$ is not a Stackelberg plan. At $t = 0$ the leader chooses p_0 freely (subproblem 2), which is why $\mu_{p0} = 0$. It also takes into account that promises about $u_1, u_2, \dots$ move p_0 through (2). At $t = 1$ the effect of those promises on p_0 is sunk. A leader re-optimizing at $t = 1$ would solve subproblem 2 afresh and set $p_1 = -P_{22}^{-1}P_{21}m_1$, i.e. $\mu_{p1} = 0$. But the continuation of the original plan delivers the promised p_1 , with $\mu_{p1} \neq 0$. With $\alpha = 1$, $m_0 = 1$: $m_1 = .43903$, the promised $p_1 = .25321$ and $\mu_{p1} = -.32979$. A fresh plan would set $p_1 = .27510$, a looser promise, and would be worth $v = -.168519$, against $-.175737$ for continuing. The continuation *is* optimal for the *continuation* leader, who takes the promise p_1 as a state (Bellman's principle applies to subproblem 1), but not for a leader who can choose p_1 anew.

(g) A program needs the following steps.

- (1) Set $\lambda = \alpha/(1 + \alpha)$, $\beta = .95$, and $A, B, R = \text{diag}(.00001, 1)$, $Q = 1$ as in (c).
- (2) Iterate $P_{j+1} = R + \beta A' P_j A - \beta^2 A' P_j B (Q + \beta B' P_j B)^{-1} B' P_j A$ from $P_0 = 0$ to convergence; set $F = \beta (Q + \beta B' P B)^{-1} B' P A$. Check that the eigenvalues of $\sqrt{\beta}(A - BF)$ are inside the unit circle.
- (3) Initial price level: $p_0 = -(P_{21}/P_{22}) m_0$.
- (4) For $t \geq 0$: $u_t = -F_1 m_t - F_2 p_t$, $m_{t+1} = m_t + u_t$, $p_{t+1} = \frac{1+\alpha}{\alpha} p_t - \frac{1}{\alpha} m_t$.

- (5) Multipliers $\mu_t = Py_t$ (with $\mu_{p0} = 0$); continuation values $v(y_t) = -y'_t Py_t$; value of the plan $w(m_0) = -(P_{11} - P_{12}^2/P_{22}) m_0^2$.

As a check, `ch19.py` truncates the problem at $T = 400$ ($u_t = 0$ afterwards) and minimizes the loss directly over $(u_0, \dots, u_{T-1})$, with p_t given by the forward solution of (a). This is a single linear system, and it reproduces the u_t and the loss of the recursive solution to 10^{-13} . Table 19.1 reports results for several α . The larger α is, the more the price level depends on expected future money, and the larger the gain from commitment over the Markov perfect policy of Exercise 19.2.

α	commitment (19.1)				Markov perfect (19.2)		
	F_1	F_2	p_0/m_0	loss/ m_0^2	g	h	loss/ m_0^2
0.5	3.27025	-3.48755	.76303	1.11784	-.61053	.76613	1.12129
1	2.06958	-2.40759	.62661	0.87428	-.56763	.63791	0.88659
2	1.33920	-1.80755	.48033	0.63110	-.48938	.50537	0.65784
5	0.75667	-1.36766	.31553	0.37773	-.36111	.35644	0.42053

Table 19.1. Commitment versus Markov perfect policy ($\beta = .95$). Losses are $-w(m_0)/m_0^2$.

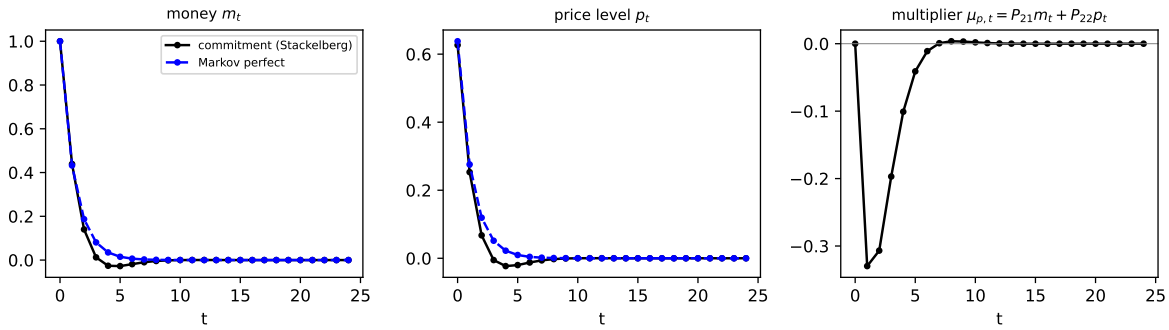


Figure S19.1. $\alpha = 1$, $m_0 = 1$: the Stackelberg plan (solid) and the Markov perfect equilibrium of Exercise 19.2 (dashed). The multiplier μ_{pt} starts at zero and then turns negative: the promises made at $t = 0$ bind later.

Exercise 19.2 Markov perfect policy makers

Equations (2) and (3) refer to the numbered displays of the exercise in the book.

(a) If $m_{t+1} = (1+g)m_t$ and $p = hm$, then $(1-\lambda)m_t + \lambda p_{t+1} = [(1-\lambda) + \lambda(1+g)h]m_t$, which equals hm_t if and only if $h(1-\lambda(1+g)) = 1-\lambda$, i.e. $h = (1-\lambda)/(1-\lambda(1+g))$. When $|\lambda(1+g)| < 1$ this is also the forward (bubble-free) solution of Exercise 19.1(a): $p_t = (1-\lambda) \sum_j \lambda^j (1+g)^j m_t = hm_t$. All equilibria below have $g \in (-1, 0)$, so this condition holds.

(b) Along $u = gm$, $p = hm$, money is $m_{t+j} = (1+g)^j m_t$, so

$$w(m) = -\frac{h^2 + g^2 + .00001}{1 - \beta(1+g)^2} m^2 \equiv -W(g) m^2, \quad \text{provided } \beta(1+g)^2 < 1.$$

$w(m)$ is the value, to an authority in office when money is m , of a world in which (i) it and every successor set money growth by the same Markov rule $u = gm$, and (ii) money holders know this rule and price accordingly, $p = hm$, which is the rational-expectations price level given the rule.

(c) The pricing equation is (2) at $t = 0$, $p_0 = (1 - \lambda)m_0 + \lambda p_1$, with $p_1 = hm_1 = h(m_0 + u_0)$. Money holders see the actual u_0 (whatever it is) and expect every authority from $t = 1$ on to follow the rule g , which makes $p_1 = hm_1$. So the time-0 authority knows that its own action moves p_0 , through its effect on m_1 and hence on expected p_1 , with slope λh . Functional equation (3) describes an authority that controls only u_0 . It takes as given that successors will use $u = gm$ no matter what it does, and it cannot commit them. Its continuation value is therefore $w(m_1)$, the value of the rule g from m_1 on, and not the value of an optimal continuation.

(d) The right side of (3) is strictly concave in u_0 :

$$-[\lambda h(m_0 + u_0) + (1 - \lambda)m_0]^2 - u_0^2 - .00001m_0^2 - \beta W(g)(m_0 + u_0)^2.$$

The first-order condition $\lambda h p_0 + u_0 + \beta W(m_0 + u_0) = 0$ gives $u_0 = G(g)m_0$ with

$$G(g) = -\frac{\lambda^2 h^2 + \lambda(1 - \lambda)h + \beta W}{1 + \lambda^2 h^2 + \beta W}, \quad h = h(g), \quad W = W(g).$$

The book calls the maximizer gm_0 as well; in equilibrium the two coincide. A *Markov perfect equilibrium* is a pair of time-invariant linear rules, $u_t = gm_t$ for policy and $p_t = hm_t$ for prices, together with the value function $w(m) = -W(g)m^2$, such that:

- (1) given that successors use g and future prices obey h , the rule g solves the current authority's problem (3) at every m : $G(g) = g$;
- (2) $h = (1 - \lambda)/(1 - \lambda(1 + g))$: money holders' pricing satisfies their Euler equation (2), without bubbles, given the policy rule;
- (3) w is the value of following the rules, $W = W(g)$.

Timing in period t : m_t is inherited; the authority in office chooses u_t ; money holders, seeing u_t and expecting $p_{t+1} = hm_{t+1}$, set $p_t = (1 - \lambda)m_t + \lambda h(m_t + u_t)$, which equals hm_t in equilibrium; then $m_{t+1} = m_t + u_t$ and the next authority takes office. Each authority assumes that all successors use $u = gm$ and that the public prices by $p = hm$ from then on. Money holders assume that every future authority uses $u = gm$.

(e) The objects to compute are the numbers g , h , W (hence the functions $u = gm$, $p = hm$, $w = -Wm^2$). Algorithm: start from $g_0 = 0$ (any $g_0 \in (-1, 0]$ will do). Given g_j , compute $h_j = (1 - \lambda)/(1 - \lambda(1 + g_j))$, then $W_j = (h_j^2 + g_j^2 + .00001)/(1 - \beta(1 + g_j)^2)$, then $g_{j+1} = G(g_j)$, and stop when $|g_{j+1} - g_j| < 10^{-14}$. Equivalently, solve $G(g) = g$ by bisection. At least one fixed point exists, and all of them lie in $(-1, 0)$. For $g \in [-1, 0]$, $1 - \lambda(1 + g) \geq 1 - \lambda$, so $0 < h \leq 1$. Then $\lambda(1 - \lambda)h < 1$, so the numerator of G is positive and smaller than its denominator, and $-1 < G(g) < 0$. By continuity $G(g) - g$ changes sign on $[-1, 0]$. Admissible $g > 0$ (with $\beta(1 + g)^2 < 1$ and $\lambda(1 + g) < 1$) give $h > 0$ and hence $G(g) < 0 < g$. Admissible $g < -1$ give $0 < h < 1 - \lambda$ and hence $G(g) > -1 > g$. For $\alpha = 1$, the iteration from $g_0 = 0$ converges in 17 steps to

$$g = -.567631, \quad h = .637905, \quad W = .886593,$$

with $|G'(g)| \approx .11$ at the fixed point, and a scan of $(-1, 0]$ finds no other fixed point. Maximizing the right side of (3) numerically at $m_0 = 1$, with W taken as given, returns $u_0 = -.56763063 = g$ and $v(1) = w(1)$, which confirms the equilibrium. Table 19.1 compares the Markov perfect equilibrium

with commitment. The loss is always larger without commitment (.886593 against .874279 for $\alpha = 1$), as it must be: the equilibrium outcome $u_t = gm_t$, $p_t = hm_t$ satisfies all the implementability constraints, so it is one of the plans available to the committed authority of 19.1. Figure S19.1 shows that the Markov perfect authority contracts money monotonically. Unable to promise future tightening, it gets a higher initial price level ($p_0 = .638$ against .627).

(f) Yes. The equilibrium strategies are time-invariant functions of the natural state m_t only. From any date t and any m_t , on or off the equilibrium path, the continuation uses the same rules (g, h), and conditions 1–3 of (d) hold there because they hold at every m . So the continuation is a Markov perfect equilibrium starting from m_t . The equilibrium is time consistent by construction. Contrast Exercise 19.1(f): commitment gives a better outcome but is time inconsistent.

Exercise 19.3 *Duopoly with a Stackelberg leader*

(a) Firm 2 takes the leader's announced sequence $\{Q_{1t}\}$ and $\{v_t\}$ as given, and chooses $\{Q_{2,t+1}\}_{t \geq 0}$, with Q_{20} given, to maximize

$$\sum_{t=0}^{\infty} \beta^t \left\{ (A_0 - A_1(Q_{1t} + Q_{2t}) + v_t)Q_{2t} - eQ_{2t} - .5gQ_{2t}^2 - .5c(Q_{2,t+1} - Q_{2t})^2 \right\}.$$

Unlike the competitive fringe of Section 19.5, firm 2 is not a price taker: it internalizes the effect of its own output on the price. $Q_{2,t+1}$ appears in the adjustment costs of periods t and $t+1$ and in the revenue of period $t+1$. So the first-order condition is, for $t \geq 0$,

$$c(Q_{2,t+1} - Q_{2t}) = \beta[A_0 - e - A_1Q_{1,t+1} - (2A_1 + g)Q_{2,t+1} + v_{t+1}] + \beta c(Q_{2,t+2} - Q_{2,t+1}).$$

In terms of the follower's investment $x_t \equiv Q_{2,t+1} - Q_{2t}$,

$$x_t = \beta x_{t+1} + \frac{\beta}{c} [A_0 - e - A_1Q_{1,t+1} - (2A_1 + g)Q_{2,t+1} + v_{t+1}].$$

This is the counterpart of (19.5.5): marginal revenue $p - A_1Q_2$ replaces the price, so (d, h, A_1) there become $(e, g, 2A_1)$ here. Add the transversality condition that rules out explosive paths. The objective is strictly concave, so the Euler equations and transversality condition characterize the optimum.

To express the link to firm 1 explicitly, divide by βc :

$$Q_{2,t+2} - \phi Q_{2,t+1} + \beta^{-1}Q_{2t} = -\frac{1}{c}f_{t+1}, \quad \phi = 1 + \beta^{-1} + \frac{2A_1 + g}{c}, \quad f_{t+1} = A_0 - e - A_1Q_{1,t+1} + v_{t+1}.$$

The polynomial $z^2 - \phi z + \beta^{-1}$ is positive at 0 and equals $-(2A_1 + g)/c < 0$ at 1. So its roots are $0 < \delta_1 < 1 < \delta_2 = 1/(\beta\delta_1)$. Factoring $(1 - \delta_1 L)(1 - \delta_2 L) = -\delta_2 L(1 - \delta_1 L)(1 - \delta_2^{-1}L^{-1})$ and solving the unstable root forward gives the stable solution

$$Q_{2,t+1} = \delta_1 Q_{2t} + \frac{\beta\delta_1}{c} \sum_{j=0}^{\infty} (\beta\delta_1)^j [A_0 - e - A_1Q_{1,t+1+j} + v_{t+1+j}].$$

Firm 2's output today depends on the whole future path of firm 1's output: a higher $Q_{1,t+1+j}$ lowers $Q_{2,t+1}$ with weight $A_1\beta\delta_1(\beta\delta_1)^j/c$. This is the channel the leader exploits. With the parameters used below, $\delta_1 = .263901$.

(b) Firm 1 chooses $\{u_t\} = \{Q_{1,t+1} - Q_{1t}\}$ to maximize $\sum_t \beta^t \{p_t Q_{1t} - C_{1t}\}$ subject to the demand curve, the law of motion of v_t , and firm 2's Euler equations, which are its implementability constraints. Stack $z_t = (1, v_t, Q_{1t}, Q_{2t})'$ and $x_t = Q_{2,t+1} - Q_{2t}$, so $y_t = (z_t', x_t)'$. As in (19.5.6),

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ A_0 - e & 1 & -A_1 & -(2A_1 + g) & c \end{bmatrix} y_{t+1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & \rho & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & c/\beta \end{bmatrix} y_t + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} u_t.$$

The last row is firm 2's Euler equation. Premultiplying by the inverse of the left matrix gives $y_{t+1} = Ay_t + Bu_t$. Firm 1's profit is $-(y_t' Ry_t + u_t' Qu_t)$ with

$$R = - \begin{bmatrix} 0 & 0 & \frac{A_0 - e}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 \\ \frac{A_0 - e}{2} & \frac{1}{2} & -A_1 - \frac{g}{2} & -\frac{A_1}{2} & 0 \\ 0 & 0 & -\frac{A_1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad Q = \frac{c}{2}.$$

Solution: (i) solve the Riccati equation for P and F (subproblem 1); (ii) set $x_0 = -P_{22}^{-1}P_{21}z_0$ (subproblem 2); (iii) iterate $u_t = -Fy_t$, $y_{t+1} = (A - BF)y_t$. Stability of $\sqrt{\beta}(A - BF)$ ensures that the follower's investment sequence is the forward solution of its Euler equation, so it is firm 2's best response. As in the chapter, the plan is time inconsistent, since $\mu_{x0} = 0$ but $\mu_{xt} \neq 0$ for $t \geq 1$.

Numerical example, with the chapter's parameters $(A_0, A_1, \rho, c, e, g, \beta) = (100, 1, .8, 1, 20, .2, .95)$:

$$u_t = 85.3171 + .8172v_t - .9789Q_{1t} - 2.4873Q_{2t} - 3.3790x_t,$$

$$x_0 = 18.2562 + .1770v_0 - .0871Q_{10} - .7103Q_{20}.$$

The eigenvalues of $A - BF$ are 1, .8 ($= \rho$), .4386, .2639 ($= \delta_1$), .1923. Starting from the static Cournot outputs $Q_{10} = Q_{20} = (A_0 - e)/(3A_1 + g) = 25$ with $v_0 = 0$, the leader expands and the follower contracts at once ($x_0 = -1.679$):

t	Q_{1t}	Q_{2t}	p_t	μ_{xt}
0	25.000	25.000	50.000	0.000
1	29.339	23.321	47.340	-5.022
2	31.729	22.197	46.074	-6.687
3	32.871	21.578	45.551	-7.288
∞	33.803	20.999	45.198	-7.682

Two brute-force checks confirm the plan. (1) Solving firm 2's own sequence problem (a 300-period quadratic program) against the leader's $\{Q_{1t}\}$ reproduces the $\{Q_{2t}\}$ of the plan to 10^{-14} . (2) Writing firm 2's best response as an affine function of the whole path $\{Q_{1t}\}$ and maximizing firm 1's profit over that path directly reproduces the leader's $\{Q_{1t}\}$ to 10^{-12} .

The long run is the *static* Stackelberg equilibrium. Put multiplier $\beta^t \theta_t$ on the follower's Euler equation at t . In a steady state, the leader's first-order condition for Q_{2t} collapses to $-A_1 Q_1 - (2A_1 + g)\theta = 0$, because the adjustment-cost terms $-\beta c - c + c + \beta c$ cancel. So $\theta = -A_1 Q_1 / (2A_1 + g)$.

The condition for Q_{1t} becomes

$$A_0 - e - (2A_1 + g)Q_1 - A_1Q_2 + \frac{A_1^2}{2A_1 + g}Q_1 = 0,$$

which is the first-order condition of a static leader facing the static reaction function $Q_2 = (A_0 - e - A_1Q_1)/(2A_1 + g)$. With our numbers, $Q_1 = 43.636/1.2909 = 33.803$ and $Q_2 = 20.999$, exactly the limits in the table. The limit of μ_{xt} is nonzero, so the leader keeps honoring its promises forever. Figure S19.2 plots the paths.

Remark. In Section 19.5.3 the book prints the large firm's rule as $u_t = [-83.98 \ -0.78 \ 0.95 \ 1.31 \ 2.07] y_t$. Running the same computation (`ch19.py`) reproduces these numbers as F in $u_t = -Fy_t$, so the printed vector has the wrong sign: the rule is $u_t = 83.98 + .78v_t - .95Q_t - 1.31q_t - 2.07i_t$. With the printed sign, $Q_{t+1} = 1.95Q_t + \dots$ would explode. The book's x_0 rule $[31.08 \ 0.29 \ -0.15 \ -0.56]$ is reproduced exactly.

(c) Firm 2 would pay at most the difference between its value as a monopolist that owns both plants and its value as a follower: $WTP(z_0) = V_M(z_0) - V_F(z_0)$.

- *Follower's value.* Firm 2's profit is a quadratic form $-y_t'R_2y_t$ in the leader's state (the adjustment cost $.5cx_t^2$ is in it because x_t is a state). Along the Stackelberg plan $y_{t+1} = (A - BF)y_t$, so $V_F = -y_0'P_Fy_0$, where $P_F = R_2 + \beta(A - BF)'P_F(A - BF)$ (a Lyapunov equation solved by doubling) and $y_0 = (z_0', (-P_{22}^{-1}P_{21}z_0)')'$.
- *Monopoly value.* The merged firm chooses (u_{1t}, u_{2t}) for the two plants with state $z_t = (1, v_t, Q_{1t}, Q_{2t})'$ and no implementability constraint. This is an ordinary LQ problem with $A_M = \text{diag}(1, \rho, 1, 1)$, $B_M = [0 \ 0; 0 \ 0; 1 \ 0; 0 \ 1]$, $Q_M = \frac{\epsilon}{2}I_2$, and $-z'R_Mz = (A_0 - e + v)(Q_1 + Q_2) - A_1(Q_1 + Q_2)^2 - \frac{g}{2}(Q_1^2 + Q_2^2)$. Its Riccati equation gives $V_M = -z_0'P_Mz_0$.
- Firm 1 sells only for at least its leader value $V_L(z_0) = -z_0'(P_{11} - P_{12}P_{22}^{-1}P_{21})z_0$. A deal exists because $V_M \geq V_L + V_F$: the merged firm could replicate the Stackelberg outcome. Any price in $[V_L, V_M - V_F]$ works.

From $Q_{10} = Q_{20} = 25$, $v_0 = 0$: $V_L = 14,648.2$, $V_F = 10,063.5$ (the Lyapunov equation and direct summation of the simulated profits agree), and $V_M = 30,298.1$. So firm 2 would pay up to $V_M - V_F = 20,234.6$, well above firm 1's reservation value 14,648.2, and merging creates a surplus of 5,586.4. The monopolist runs each plant at $(A_0 - e)/(4A_1 + g) = 19.048$ in the long run, and the price rises to 61.90 (Figure S19.2). All values are quadratic in z_0 , so the willingness to pay is $-z_0'(P_M - P_F^z)z_0$, where P_F^z is P_F restricted to $y_0 = [I; -P_{22}^{-1}P_{21}]z_0$. For example, it is 19,328.4 from $Q_{10} = Q_{20} = 0$ and 20,761.3 from $(Q_{10}, Q_{20}, v_0) = (25, 25, 5)$.

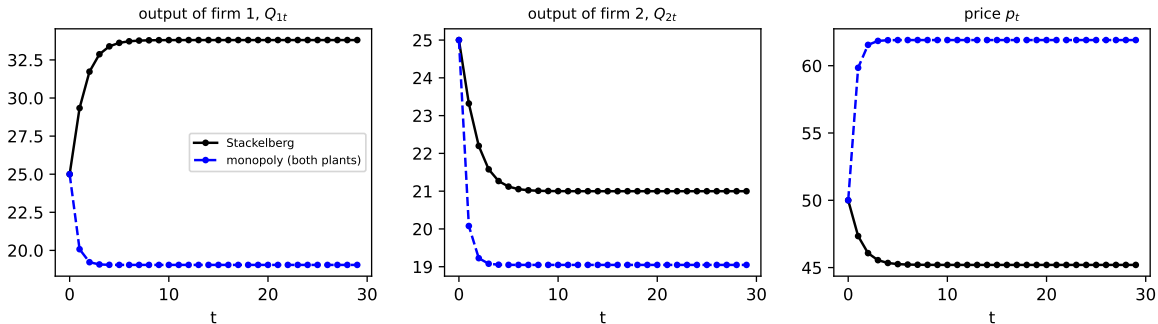


Figure S19.2. Duopoly with firm 1 as Stackelberg leader (solid) versus a monopolist owning both plants (dashed), from $Q_{10} = Q_{20} = 25$, $v_0 = 0$.

Chapter 21

Incentives and Insurance

Solutions to all 5 exercises of Chapter 21 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Numbers and figures come from `code/ch21.py`, which replaces the book's Matlab programs. The closed-form moneylender contract is checked to be a fixed point of the Bellman equation. The Phelan–Townsend version of Exercise 21.4 is solved by value iteration in which each Bellman step is a linear program.

Exercise 21.1 *Thomas and Worrall meet Markov*

(a) In state (v, i) the insurer chooses current consumption c (the transfer is $\tau = c - y_i$) and a promised value w_j for each possible next-period endowment y_j . Then

$$C(v, i) = \min_{c, \{w_j\}} \left\{ c - y_i + \beta \sum_{j=1}^I P_{ij} C(w_j, j) \right\} \quad \text{subject to} \quad u(c) + \beta \sum_j P_{ij} w_j \geq v.$$

The constraint is promise keeping. It binds at the optimum, because C is increasing in v .

(b) Let $\theta \geq 0$ be the multiplier on promise keeping. The first-order conditions are $1 = \theta u'(c)$ and $\beta P_{ij} C_v(w_j, j) = \theta \beta P_{ij}$, and the envelope condition is $C_v(v, i) = \theta$. Hence

$$C_v(w_j, j) = C_v(v, i) = \frac{1}{u'(c)} \quad \text{for all } j :$$

the insurer equalizes the marginal cost of delivering utility across dates and states. Since $C_v(w, j) = 1/u'(c(w, j))$ at every state, next period's consumption equals this period's in every state. So the optimum is full insurance: $c_t = \bar{c}$ for all t and all histories, with $u(\bar{c}) = (1 - \beta)v$. Transfers absorb all endowment risk, $\tau_t = \bar{c} - y_t$, and promised values stay constant, $v_{t+1} = v_t = v$ in every state. One can verify directly that

$$C(v, i) = \frac{u^{-1}((1 - \beta)v)}{1 - \beta} - Y_i, \quad Y = (I - \beta P)^{-1}y,$$

solves the Bellman equation. Here $Y_i = E[\sum_t \beta^t y_t \mid y_0 = y_i]$. With $c = \bar{c}$ and $w_j = v$, promise keeping holds, and the cost is $\bar{c} - y_i + \beta \sum_j P_{ij} (\bar{c}/(1 - \beta) - Y_j) = \bar{c}/(1 - \beta) - Y_i$. Because u^{-1} is convex, the problem is convex, so the first-order conditions are sufficient.

(c) In autarky $c_t = y_t$, so $v_a(i) = u(y_i) + \beta \sum_j P_{ij} v_a(j)$, $i = 1, \dots, I$; in vector form $v_a = (I - \beta P)^{-1}u(y)$.

(d) The household must never want to leave. A promise v made in state i must satisfy $v \geq v_a(i)$, and every continuation promise must satisfy $w_j \geq v_a(j)$. So, for $v \geq v_a(i)$,

$$C(v, i) = \min_{c, \{w_j\}} \left\{ c - y_i + \beta \sum_j P_{ij} C(w_j, j) \right\} \quad \text{s.t.} \quad u(c) + \beta \sum_j P_{ij} w_j \geq v, \quad w_j \geq v_a(j) \quad \forall j.$$

With multipliers θ and $\beta P_{ij} \lambda_j \geq 0$, the first-order conditions give $1 = \theta u'(c)$ and $C_v(w_j, j) = \theta + \lambda_j$, and the envelope condition gives $C_v(v, i) = \theta = 1/u'(c)$. Therefore:

- if the participation constraint in state j is slack ($\lambda_j = 0$), then $C_v(w_j, j) = C_v(v, i)$ and next period's consumption equals this period's;
- if it binds ($\lambda_j > 0$), then $w_j = v_a(j)$ and $1/u'(c') = C_v(v_a(j), j) > 1/u'(c)$, so consumption rises.

Consumption never falls. Let $c^*(j)$ be the consumption the contract gives at state $(v_a(j), j)$. The constraint binds in state j exactly when $c^*(j) > c_t$, so

$$c_{t+1} = \max\{c_t, c^*(y_{t+1})\}, \quad v_{t+1} = \begin{cases} \tilde{w}(c_t, y_{t+1}) & \text{if } c_t \geq c^*(y_{t+1}), \\ v_a(y_{t+1}) & \text{otherwise,} \end{cases}$$

where $\tilde{w}(c, j)$ is the promise that keeps consumption at c in state j . Equivalently, $\tilde{w}(c, j) = V(c, j)$, where $V(c, j) = u(c) + \beta \sum_k P_{jk} V(\max\{c, c^*(k)\}, k)$ is the value of a contract that pays c until a constraint binds.

This is Kocherlakota's "amnesia": whenever a constraint binds, the continuation contract depends only on the current state y_{t+1} , not on history. Consumption is the running maximum of the thresholds $c^*(y_s)$ (and of c_0). Unlike the i.i.d. case of the chapter, promised values need not be monotone. With constant consumption c , the value $V(c, j)$ depends on the state j because the chance of future binding constraints does, so v_t can fall when the endowment moves to a state in which such constraints are less likely.

Exercise 21.2 *Wealth dynamics in the moneylender model*

(a) With $v_0 = v_{aut}$ the contract is the one of Section 21.3.3. When the highest endowment realized so far is y_j , the villager consumes c_j and is promised w_j , where (21.3.25) and the binding participation constraint (21.3.22a) give

$$u(c_j) = u(y_j) - \beta \sum_{k \leq j} \Pi_k [u(y_j) - u(y_k)], \quad w_j = \frac{u(y_j) + \beta v_{aut} - u(c_j)}{\beta}.$$

Here $v_{aut} = \sum_s \Pi_s u(y_s) / (1 - \beta) = -0.173157$ and $E(y) = 13.8238$. The contract is:

s	y_s	Π_s	c_s	w_s	s	y_s	Π_s	c_s	w_s
1	6	.07794	6.0000	-.17316	11	16	.04667	13.0334	-.14470
2	7	.07404	6.9548	-.17130	12	17	.04433	13.4616	-.14223
3	8	.07034	7.8564	-.16859	13	18	.04212	13.8527	-.13989
4	9	.06682	8.7002	-.16550	14	19	.04001	14.2100	-.13767
5	10	.06348	9.4845	-.16229	15	20	.03801	14.5365	-.13557
6	11	.06031	10.2096	-.15908	16	21	.03611	14.8352	-.13358
7	12	.05729	10.8772	-.15595	17	22	.03430	15.1086	-.13169
8	13	.05443	11.4900	-.15294	18	23	.03259	15.3592	-.12990
9	14	.05171	12.0512	-.15006	19	24	.03096	15.5892	-.12820
10	15	.04912	12.5644	-.14731	20	25	.02941	15.8005	-.12658

Table 21.1. The optimal contract with $v_0 = v_{aut}$: consumption c_s and promised value w_s after a record endowment y_s .

So $c_t = c_{s(t)}$ and $v_{t+1} = w_{s(t)}$, where $y_{s(t)} = \max\{y_0, \dots, y_t\}$ (21.3.27). Consumption is constant between record endowments and rises at each new record. The villager keeps y_1 ($c_1 = y_1$, $w_1 = v_{aut}$) and surrenders $y_s - c_s > 0$ in exchange for a higher promise at every other record.

Checks. The two formulas (21.3.22a) and (21.3.24) for w_j agree to 6×10^{-17} , and $\sum_s \Pi_s[u(c_s) + \beta w_s] = v_{aut}$. More importantly, let $P(v)$ be the closed-form value function of Sections 21.3.3–21.3.5 (with constant consumption $u^{-1}((1 - \beta)v)$ for $v \geq w_S$). The script solves the right side of the Bellman equation (21.3.4)–(21.3.8) numerically with this P : forty choice variables (c_s, w_s) , by SLSQP from two starting points. It does so at ten values of v between v_{aut} and beyond w_S , and gets back $P(v)$ to 8×10^{-14} . So the closed form is a fixed point of the Bellman operator, which is a contraction, and hence it is the value function.

(b) Profits follow from (21.3.28): $P(w_S) = (Ey - c_S)/(1 - \beta)$, then backward. Offering v_{aut} earns

$$P(v_{aut}) = P(w_1) = 2.8405,$$

about 10 percent of the present value of the endowment, $Ey/(1 - \beta) = 27.65$. (A simulation of 200,000 villagers gives $2.841 \pm .006$.) The moneylender captures all the gains from insurance, because the villager is left at the autarky value.

(c) By (21.3.38), $F_t(c_j) = (\sum_{k \leq j} \Pi_k)^{t+1}$, a step function with jumps at the c_j (Figure S21.1, left). At $t = 0$ the distribution mirrors the endowment distribution, compressed: mass .078 at $c_1 = 6$ and mean consumption 11.23. It then shifts right as villagers accumulate record endowments: the mean is 14.79 at $t = 5$, 15.28 at $t = 10$, and at $t = 500$ essentially everyone has had y_S and consumes $c_S = 15.80$. The distribution first fans out and then fans in, toward complete insurance at the top level.

(d) Starting from $B_{-1} = 0$, the balance obeys $B_t = \beta^{-1}B_{t-1} + E(y) - \sum_j f_{j,t}c_j$ (21.3.40). With $v_0 = v_{aut}$,

$$B_0 = 2.589, B_1 = 6.017, B_2 = 12.05, B_5 = 92.1, B_{20} = 2.98 \times 10^6, B_{100} = 3.60 \times 10^{30}.$$

The balance explodes at the rate $\beta^{-1} = 2$. Consistently with (21.3.42), $\beta^{100}(B_{100} + \beta P(w_S)) = 2.8405 = P(v_{aut})$: the moneylender banks a positive present value of profits, which then compounds. Eventually each villager is paid $c_S = 15.80 > E(y) = 13.82$, and interest on the balance covers the difference. (Numerical note: with $\beta = .5$, iterating (21.3.40) multiplies rounding errors by 2^t . So `ch21.py` uses the equivalent $B_t = \beta^{-t}P(v_0) - \sum_{s>t} \beta^{s-t}[E(y) - Ec_s]$, which agrees with the recursion to 5×10^{-6} for $t < 30$.)

(e) Since $P(w_S) = -3.953 < 0$, a zero-profit promise exists in $[v_{aut}, w_S]$. Following the recipe of Section 21.3.4: $P(w_{10}) = .289 > 0 \geq P(w_{11}) = -.138$, so $k = 11$. The zero-profit condition $0 = \sum_{j < k} \Pi_j(y_j - \tilde{c}) + \sum_{j \geq k} \Pi_j[y_j - c_j + \beta P(w_j)]$ gives $\tilde{c} = 12.8822$, and (21.3.26) gives

$$v_0 = -0.145523 \in [w_{10}, w_{11}) = [-0.14731, -0.14470).$$

The contract pays $\tilde{c} = 12.88$ until the villager's record endowment exceeds y_{10} , and c_j afterwards. A root finder applied to the closed-form P returns the same v_0 , and a simulation gives $P(v_0) = -.0001 \pm .011$. The bank balance now converges to $(c_S - Ey)/r = 1.9767$ (with $r = \beta^{-1} - 1 = 1$): $B_0 = .355$, $B_5 = 1.235$, $B_{20} = 1.768$, $B_{100} = 1.967$. This is the case of the book's Figure 21.3.3.

Even at zero profits the villager gets less than the pooling value $u(Ey)/(1 - \beta) = -0.144678$, because the participation constraints still bind.

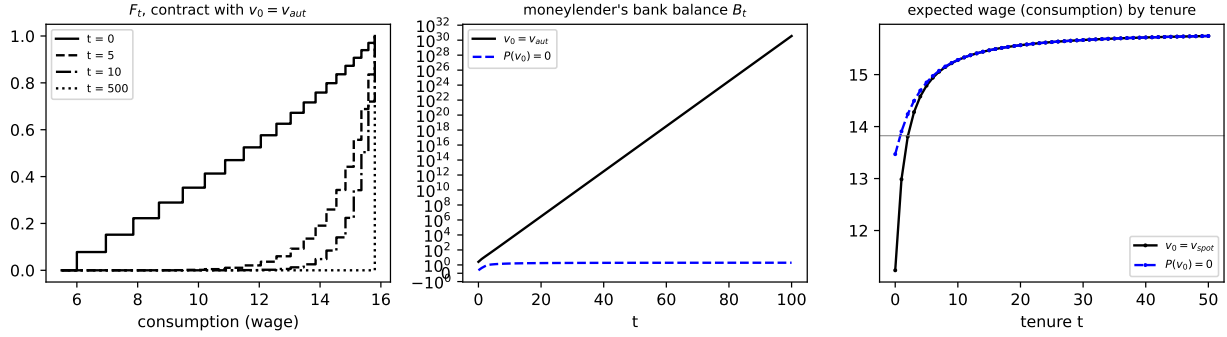


Figure S21.1. Left: F_t for the contract with $v_0 = v_{aut}$. Middle: the moneylender's bank balance (symmetric log scale); it explodes when $v_0 = v_{aut}$ and converges to 1.977 when $P(v_0) = 0$. Right: expected consumption, i.e. the expected wage of Exercise 21.3(d), by tenure.

Exercise 21.3 *Thomas and Worrall (1988)*

This is the moneylender economy of Exercise 21.2 with new labels: productivity w_t is the endowment, the wage ω_t is consumption, the company is the moneylender, and the spot market is autarky ($v_{spot} = v_{aut}$). The parameters are the same, so the numbers are those of Exercise 21.2.

- $\omega_t = c_{s(t)}$ and $v_{t+1} = w_{s(t)}$ from Table 21.1, where $s(t)$ indexes the highest productivity the worker has had so far. The wage never falls: it is constant between productivity records and is raised at each record, when the quit constraint binds. At such a date the worker is paid less than the spot wage ($\omega_t = c_s < w_s$) in exchange for a higher promised value. Wages are rigid downward and backloaded, and the worker is insured against productivity falls.
- $P(v_{spot}) = 2.8405$ per worker.
- $F_t(c_j) = (\sum_{k \leq j} \Pi_k)^{t+1}$, as in Figure S21.1 (left).
- $E\omega_t = \sum_j f_{j,t} c_j$ rises with tenure: 11.23, 12.98, 13.81 at $t = 0, 1, 2$, then 14.79, 15.28, 15.57, 15.74 at $t = 5, 10, 20, 50$, and converges to $c_S = 15.80$ (Figure S21.1, right). The profile is increasing and concave. It is steep early, when new records are frequent, and above the mean productivity 13.82 after about two years: senior workers are paid more than they produce.
- With free entry, competition drives expected profits to zero: $v_0 = -0.145523$ as in Exercise 21.2(e). The entry wage $\tilde{c} = 12.88$ is paid until productivity first exceeds $w_{10} = 15$. The expected wage starts higher (13.47) and reaches the same limit c_S (dashed line in Figure S21.1, right).

Remark. The exercise writes the utility function as $(1 - \gamma)^{-1} w^{1-\gamma}$; it should be $(1 - \gamma)^{-1} \omega^{1-\gamma}$, a function of the wage paid by the company, not of productivity.

Exercise 21.4 *Thomas and Worrall meet Phelan and Townsend*

Equations (4)–(5) refer to the numbered displays of the exercise in the book.

(a) The unknowns are the probabilities $x(b, w, y) \equiv \Pi_v(b, w, y) \geq 0$. Given the numbers $P(w)$, the objective $\sum [-b + \beta P(w)] x(b, w, y)$ is linear in x . Promise keeping is one linear equality whose coefficients $u(y + b) + \beta w$ are numbers once the grids are fixed. Constraints (4)–(5) amount to the S equalities $\sum_{b,w} x(b, w, y) = \Pi(y)$. Using $\Pi_v(b, w | y) = x(b, w, y)/\Pi(y)$, each truth-telling constraint is linear:

$$\frac{1}{\Pi(y)} \sum_{b,w} [u(y + b) + \beta w] x(b, w, y) \geq \frac{1}{\Pi(\tilde{y})} \sum_{b,w} [u(y + b) + \beta w] x(b, w, \tilde{y}).$$

So each Bellman step is a linear program with $N_B N_W S$ variables, $1 + S$ equalities and $S(S - 1)$ inequalities, one program for each promised value $v \in W$. The lotteries convexify the planner's problem. That is Phelan and Townsend's device, and it is what makes linear programming applicable.

(b) *Implementation.* Consumption must exceed a , so for type y the transfers with $y + b \leq a$ are excluded. In the truth-telling constraints, a lie that would push consumption to or below a gets utility $-\infty$ (in the code, -10^4). Such lies are never attractive. W runs from $w_{\min} = -8.3333$ (the value of consuming $y_{\min}$ forever) to $w_{\max} = -0.4167$. Each LP has 4,775 variables, 11 equalities and 90 inequalities, and `ch21.py` solves them with HiGHS.

Remark. As printed, $b_{\min} = 1 - y_{\max} + .33 = -13.67$. Then $y + b_{\min} < a$ for every endowment, so the lowest grid transfers could never be used. The statement just before, $B \subset (a - y_S, \bar{b}]$, indicates that $b_{\min} = a - y_{\max} + .33 = -9.67$ is meant. We use that value.

Which promises can be delivered? Most of W cannot be. To deliver a very low value the contract must give low current utility and low continuation values. But once continuation values are at their floor, truth telling forces every type to receive the same transfer: otherwise the types with smaller transfers would report the type with the largest. The smallest transfer that keeps the lowest endowment above a is the grid point $b^* = -0.335$. Hence current utility is at least $Eu(y + b^*)$, and the smallest deliverable value is

$$\underline{v} = \frac{Eu(y + b^*)}{1 - \beta} = -2.4816.$$

Value iteration finds this set by itself, in the spirit of the APS operator of Chapter 24. Starting from $P = 0$ on all 25 points, each step finds the LP infeasible at the lowest remaining promises, because they would need continuation values that were just ruled out. After 17 steps only the seven grid points $-2.3958, \dots, -0.4167$ remain. From then on we iterate on P over the deliverable points. Value iteration takes 159 steps to reach a step size of 10^{-4} . Then one policy-iteration step (an LP improvement step plus an exact evaluation of the lottery policy, $P = r + \beta MP$) reaches $\max |T(P) - P| = 7 \times 10^{-15}$. In the final contract promise keeping holds to 4×10^{-16} and truth telling to 4×10^{-15} , and the transition matrix $\mathbb{P}(w' | w)$ has unit row sums. On the deliverable points,

$$P(w) = (45.06, 48.65, 47.66, 44.52, 38.57, 27.68, 4.14) \quad \text{for } w = -2.40, -2.07, \dots, -0.42.$$

With only seven usable points the grid is coarse. So we also solve the problem with 25 points spanning the deliverable range $[\underline{v}, w_{\max}]$ (all of them turn out to be deliverable, and the checks

hold to 2×10^{-10}). P is then hump-shaped: it rises from 39.89 at $\underline{w}$ to a maximum of 52.82 near $w = -1.8$ and then falls to 9.93 at $w_{\max}$ (Figure S21.2, right). It increases at the bottom because there all types must get nearly the same transfer, so the planner cannot collect more from high endowments. Slightly higher promises let it spread continuation values across reports, which relaxes truth telling and raises revenue. (The planner would never choose to promise values in that region.) This spreading is the heart of the contract: at w_0 (refined grid) the expected continuation value rises with the reported endowment, from -2.22 for $y = 6$ to -1.67 for $y = 15$. A household that reports a high endowment pays more today and is rewarded with a higher promised value.

(c) From the joint lottery $\Pi_{w_t}(b_t, w_{t+1}, y_t)$,

$$\mathbb{P}(w_{t+1} = w' \mid w_t = w) = \sum_{b \in B, y \in Y} \Pi_w(b, w', y), \quad \mathbb{P}(b_t, y_t \mid w_t) = \sum_{w' \in W} \Pi_{w_t}(b_t, w', y_t).$$

Let $p_t(w) = \mathbb{P}(w_t = w)$. Starting from $p_1 = \delta_{w_0}$, iterate $p_{t+1} = p_t M$, with $M_{ww'} = \mathbb{P}(w' \mid w)$. Consumption is $c_t = y_t + b_t$, so

$$\mathbb{P}(c_t = c) = \sum_w p_t(w) \sum_{(b, w', y): y+b=c} \Pi_w(b, w', y),$$

and F_t is its cumulative sum over the finite support of $y + b$.

(d) With w_0 the grid point closest to -2 (-2.066 on the book's grid, -1.965 on the refined grid):

t	book's grid		refined grid	
	mean c_t	std c_t	mean c_t	std c_t
1	7.138	0.622	7.097	0.328
5	7.385	0.712	7.218	0.467
10	7.548	0.741	7.321	0.535
100	8.170	1.146	7.817	0.738
1000	8.246	1.206	7.945	0.809

Figure S21.2 plots F_t for $t = 1, 5, 10, 100$. The comparison with Exercise 21.2 is striking. In the moneylender model, F_t moves right and collapses onto a single point, c_S : eventually every villager has had the highest endowment and is fully insured. Here the distribution *fans out*. The dispersion of consumption grows with t and settles at a nondegenerate invariant distribution, and there is no convergence to full insurance. To keep households telling the truth, the planner must keep making continuation values respond to reports, so identical households drift apart and inequality is permanent. In the Thomas–Worrall model without bounds this spreading never stops: promised values drift toward $-\infty$ (the immiseration result of Section 21.5.9). On our grids the minimum transfer bounds consumption below, which keeps promised values in $[\underline{w}, w_{\max}]$. Mean consumption rises in this experiment only because $w_0 \approx -2$ lies near the bottom of the deliverable set.

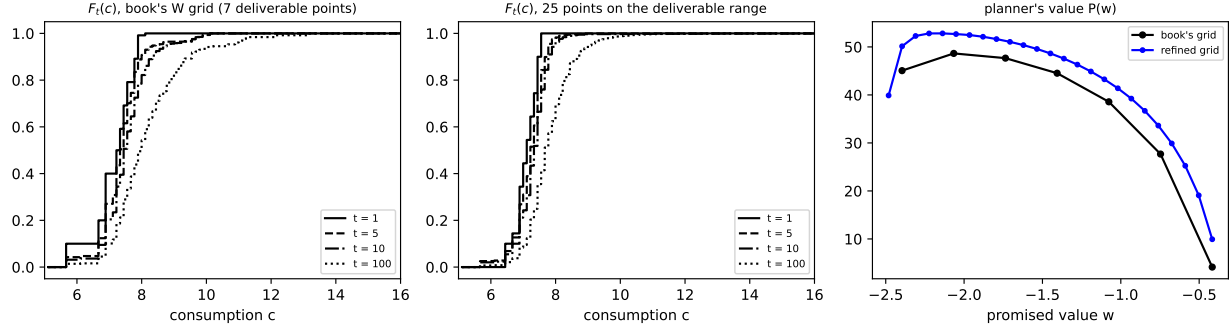


Figure S21.2. Exercise 21.4. Left and middle: F_t for $w_0 \approx -2$ on the book's grid (7 deliverable promised values) and on 25 points spanning the deliverable range. Right: the planner's value $P(w)$.

Exercise 21.5 The IMF

(a) Let v be the expected discounted *loss* promised to the country before g is realized, and $P(v)$ the IMF's maximal expected value. Then

$$P(v) = \max_{\{T_s, w_s\}} \sum_{s=1}^S \pi_s [T_s - g_s + \beta P(w_s)]$$

subject to promise keeping, $\sum_s \pi_s [W(T_s) + \beta w_s] \leq v$, and the participation constraints $W(T_s) + \beta w_s \leq W(g_s) + \beta v_{aut}$ for $s = 1, \dots, S$. P is increasing: a worse deal for the country is a better one for the IMF.

(b) With multipliers μ and $\lambda_s \geq 0$, the first-order conditions are $\pi_s = (\mu \pi_s + \lambda_s) W'(T_s)$ and $P'(w_s) = \mu + \lambda_s / \pi_s$, and the envelope condition is $P'(v) = \mu$. Hence $W'(T_s) P'(w_s) = 1$ in every state.

- If the participation constraint is slack, then $w_s = v$ and $W'(T_s) = 1/P'(v)$: the tax is the same in all such states (tax smoothing), and the promise is unchanged.
- If it binds, then $P'(w_s) > P'(v)$, so $w_s < v$ (a lower future loss) and T_s is lower. (T_s, w_s) solve $W(T_s) + \beta w_s = W(g_s) + \beta v_{aut}$ and $W'(T_s) P'(w_s) = 1$, which do not involve v (amnesia).

Default is tempting when *spending is low*: financing a low g_s alone costs little, while the contract asks the country to pay for insurance. So the constraints bind at low g , the mirror image of the high endowments that tempt the villager of Section 21.3.

Starting from v_{aut} , the contract works exactly like (21.3.27) with the order of states reversed. Let $m(t)$ index the *lowest* spending realized so far. Then $T_t = T_{m(t)}$ and $v_{t+1} = w_{m(t)}$, where, by the argument that gave (21.3.25),

$$W(T_j) = W(g_j) + \beta \sum_{k \geq j} \pi_k [W(g_k) - W(g_j)], \quad w_j = \frac{W(g_j) + \beta v_{aut} - W(T_j)}{\beta}.$$

To derive the first formula, note that at record j the next period either leaves the record unchanged ($g_k \geq g_j$, loss $A_j \equiv W(g_j) + \beta v_{aut}$) or sets a new record $k < j$ (loss A_k). So $w_j = A_j \sum_{k \geq j} \pi_k + \sum_{k < j} \pi_k A_k$, and $\sum_k \pi_k A_k = v_{aut}$; substitute into $W(T_j) = A_j - \beta w_j$. Properties: $T_S = g_S$ (no transfer when spending is highest); $T_j > g_j$ for $j < S$ (a surplus when a new record low occurs);

and $W(T_j) - W(T_{j-1}) = (1 - \beta \sum_{k \geq j} \pi_k)[W(g_j) - W(g_{j-1})] > 0$, so taxes are lower the lower the record. Promise keeping holds: $\sum_s \pi_s [W(T_s) + \beta w_s] = \sum_s \pi_s A_s = v_{aut}$.

(c) For each country, T_t and $W(T_t)$ are nonincreasing step functions of time. They are constant while g_t stays above the lowest level seen so far, and they fall at each new record low. Once g_1 has occurred (which happens eventually with probability one) the tax is constant at T_1 , with $W(T_1) = (1 - \beta)W(g_1) + \beta EW(g)$. Surpluses $T_t - g_t$ fluctuate: the IMF lends when g_t is high and collects when it is low. Across a panel of identical countries the average surplus at tenure t is $\sum_j \mathbb{P}(m(t) = j)T_j - Eg$, with $\mathbb{P}(m(t) \geq j) = (\sum_{k \geq j} \pi_k)^{t+1}$. It is positive at $t = 0$ (since $T_j \geq g_j$) and declines with tenure toward $T_1 - Eg$, whose sign is ambiguous: as $\beta \rightarrow 0$, $T_1 \rightarrow g_1 < Eg$, while as $\beta \rightarrow 1$, $W(T_1) \rightarrow EW(g) \geq W(Eg)$.

Yes, there is a cohort effect. Tax rates, distortions and surpluses depend on the time since a country signed, so at a given calendar date countries that joined earlier have lower taxes and smaller surpluses. Panel averages depend on the mix of signing dates.

Illustration (ch21.py): $W(T) = T^2$, $g \in \{.1, .2, .3, .4\}$ equally likely, $\beta = .9$. The record taxes are $T_j = (.2617, .2797, .3252, .4000)$. The average tax at tenure 0, 1, 2, 5, 10, 30 is .3167, .2879, .2762, .2657, .2625, .2617 against $Eg = .25$, so the average surplus falls from .067 to .012.

(d) Yes: the IMF expects strictly positive profits. The argument mirrors the book's "limited obligations" proof that $P(v_{aut}) > 0$ (Section 21.3.4), with convexity of W in place of concavity of u . By convexity, the formula for $W(T_j)$, whose weights sum to one, implies $T_j \geq (1 - \beta q_j)g_j + \beta \sum_{k \geq j} \pi_k g_k$, where $q_j = \sum_{k \geq j} \pi_k$. The inequality is strict for $j < S$. Consider the obligation that begins with a record g_j and lasts while $g_t \geq g_j$. In its first period it earns $T_j - g_j \geq \beta \sum_{k \geq j} \pi_k (g_k - g_j)$. Afterwards it earns, in expected present value,

$$\frac{\beta q_j}{1 - \beta q_j} \left(T_j - \frac{\sum_{k \geq j} \pi_k g_k}{q_j} \right) \geq -\beta \sum_{k \geq j} \pi_k (g_k - g_j).$$

So each obligation breaks even, and every one with $j < S$ makes a strict profit. The contract is a collection of such obligations, so $P(v_{aut}) > 0$. The country gains nothing, and the IMF appropriates the whole value of the insurance it provides. In the illustration $P(v_{aut}) = .2243$ (exact, from the recursion analogous to (21.3.28) and from the distribution of $m(t)$; a simulation gives $.2251 \pm .0005$).

Chapter 22

Equilibrium without Commitment

Solutions to all 5 exercises of Chapter 22 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Numbers and figures come from `code/ch22.py`. For Exercise 22.3 the script computes the whole constrained Pareto frontier by value iteration, solving each Bellman step through its concave dual. This independently confirms the “amnesia overwhelms memory” logic of Section 22.10.

Exercise 22.1 *A Lagrangian method with two-sided lack of commitment*

(a) In autarky consumer 1 eats y_t and consumer 2 eats $1 - y_t$, so

$$v_1^{aut} = \frac{\sum_s \Pi_s u(y_s)}{1 - \beta}, \quad v_2^{aut} = \frac{\sum_s \Pi_s u(1 - y_s)}{1 - \beta}.$$

They are equal when the distribution of y is symmetric about $\frac{1}{2}$ (as in the chapter), but not in general. After $y_t = y_s$ is observed, the autarky values are $U_1(y_s) = u(y_s) + \beta v_1^{aut}$ and $U_2(y_s) = u(1 - y_s) + \beta v_2^{aut}$.

(b) Following Marcet and Marimon (Section 21.4), fix a Pareto weight $\alpha \in (0, 1)$ and let $\sum_{j \geq 0} \beta^j E_t u(c_i(t + j))$ be consumer i 's continuation value. The planner solves

$$\max_{\{c_1(t), c_2(t)\}} E_0 \sum_{t=0}^{\infty} \beta^t [\alpha u(c_1(t)) + (1 - \alpha) u(c_2(t))]$$

subject to $c_1(t) + c_2(t) \leq 1$ and, for $i = 1, 2$ and every history, $E_t \sum_{j \geq 0} \beta^j u(c_i(t + j)) \geq U_i(y_t)$. With multipliers $\beta^t \psi_t$ (feasibility) and $\beta^t \mu_{it} \geq 0$ (participation), and using the law of iterated expectations to collect the terms that multiply each $u(c_i(t))$, the Lagrangian is

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left\{ \sum_{i=1,2} [M_{it} u(c_i(t)) - \mu_{it} U_i(y_t)] + \psi_t (1 - c_1(t) - c_2(t)) \right\},$$

where

$$M_{it} = M_{i,t-1} + \mu_{it}, \quad M_{1,-1} = \alpha, \quad M_{2,-1} = 1 - \alpha.$$

The cumulative multipliers M_{it} are “co-state” variables. They make the problem recursive in $(M_{1,t-1}, M_{2,t-1}, y_t)$, or, by homogeneity, in the single ratio $x_{t-1} = M_{2,t-1}/M_{1,t-1}$ and y_t .

(c) The first-order conditions for consumption are $M_{1t} u'(c_1(t)) = \psi_t = M_{2t} u'(c_2(t))$, with $c_2(t) = 1 - c_1(t)$. So

$$\frac{u'(c_1(t))}{u'(1 - c_1(t))} = x_t \equiv \frac{M_{2t}}{M_{1t}} = \frac{M_{2,t-1} + \mu_{2t}}{M_{1,t-1} + \mu_{1t}}.$$

The ratio x_t is a time-varying relative Pareto weight, and $c_1(t) = g(x_t)$ with g decreasing. Consequences:

- (1) *No binding constraint*: $\mu_{1t} = \mu_{2t} = 0$, so $x_t = x_{t-1}$ and consumption is unchanged. The contract insures both consumers fully against the endowment realization as long as no participation constraint binds.
- (2) *Consumer 1's constraint binds*: $\mu_{1t} > 0$ raises consumer 1's weight, so $x_t < x_{t-1}$ and $c_1(t) > c_1(t-1)$. The multiplier is just large enough that the constraint holds with equality. The value of a contract that restarts from weight x depends only on x (endowments are i.i.d.), so the new weight is a number $x^{(1)}(y_t)$ that depends on the current state alone (amnesia). It is the largest weight consistent with consumer 1's participation.
- (3) *Consumer 2's constraint binds*: symmetrically, $x_t = x^{(2)}(y_t) > x_{t-1}$, the smallest weight consistent with consumer 2's participation, and $c_1(t)$ falls.
- (4) Only one constraint can bind at a time. If both did, both consumers would be at their autarky values, which is inefficient when there are gains from trade.

Hence the law of motion of the weight is

$$x_t = \min\{x^{(1)}(y_t), \max\{x_{t-1}, x^{(2)}(y_t)\}\},$$

or, in consumption, $c_1(t) = \max\{\underline{c}(y_t), \min\{c_1(t-1), \bar{c}(y_t)\}\}$. Here $[\underline{c}(y_s), \bar{c}(y_s)]$ is the interval of consumption levels of consumer 1 that satisfy both participation constraints in state y_s . The lower end, $c_1(x^{(1)}(y_s))$, makes consumer 1's constraint bind, and the upper end, $c_1(x^{(2)}(y_s))$, makes consumer 2's bind. This is exactly (22.4.3). The thresholds solve the fixed-point conditions $V_1(x^{(1)}(y_s)) = U_1(y_s)$ and $V_2(x^{(2)}(y_s)) = U_2(y_s)$, where the values of a contract that starts the period with weight x satisfy

$$V_i(x) = u(c_i(x)) + \beta \sum_{s'} \Pi_{s'} V_i(\min\{x^{(1)}(y_{s'}), \max\{x, x^{(2)}(y_{s'})\}\}).$$

Properties that follow (Sections 22.4–22.6):

- consumption moves only when a constraint binds, and then to the nearest end of the current interval;
- the intervals are ordered in y and cannot contain one another, and each contains the endowment;
- if all the intervals have a common point $\hat{c}$, full insurance at $\hat{c}$ is eventually reached and kept (temporary imperfect risk sharing);
- otherwise consumption keeps moving between the lower ends and upper ends forever (permanent imperfect risk sharing).

The initial weight α indexes the point on the constrained Pareto frontier, and it matters only until the first binding constraint.

Exercise 22.2 *Dixit, Grossman, and Gul (2000)*

Autarky values. Let $\mathbf{1}_i(x) = 1$ if party i is in power in state x . In autarky the party in power takes everything, so

$$A_i(x) = U(\mathbf{1}_i(x)) + \beta \sum_{x'} P(x, x') A_i(x'), \quad \text{i.e.} \quad A_i = (I - \beta P)^{-1} U(\mathbf{1}_i).$$

Here $U(\mathbf{1}_i(x))$ is $U(1)$ in power and $U(0)$ out of power.

Recursive contract. The state is the current x and the value u_0 promised to party 2 (measured after x is observed). The contract chooses today's share y of party 1 and continuation promises $w(x')$ to party 2 for each state tomorrow:

$$V(u_0, x) = \max_{y \in [0,1], \{w(x')\}} \left\{ U(y) + \beta \sum_{x'} P(x, x') V(w(x'), x') \right\}$$

subject to promise keeping, $U(1-y) + \beta \sum_{x'} P(x, x') w(x') \geq u_0$, and the participation constraints in every next-period state,

$$w(x') \geq A_2(x'), \quad V(w(x'), x') \geq A_1(x') \quad \text{for all } x'.$$

The current state's participation constraints restrict the domain: $u_0 \geq A_2(x)$ and $V(u_0, x) \geq A_1(x)$, i.e. $u_0 \in [A_2(x), \bar{u}(x)]$ with $V(\bar{u}(x), x) = A_1(x)$. In practice only the party in power can be tempted, because the party out of power gets nothing in autarky today. Initial promises u_0 trace the constrained Pareto frontier $V(\cdot, x)$.

Characterization. This is Kocherlakota's two-sided model with a Markov state, and the analysis of Exercise 22.1 carries over. With multiplier θ on promise keeping, $U'(y)/U'(1-y) = \theta = -V_u(u_0, x)$. The share of the party in power stays constant as long as no participation constraint binds. When the party in power threatens to take the whole pie, its share jumps to the least generous level that keeps it in, which depends only on the current state (amnesia). So the contract shares power “gradually”: it rewards a party with a larger share when it comes into power only as much as needed to prevent a grab.

Exercise 22.3 A two-state numerical example of social insurance

Let $u(c) = 4c - c^2/2$, so $u(0) = 0$, $u(1) = 3.5$, $u(2) = 6$. The autarky value is $v_{aut} = \frac{.5[u(2)+u(0)]}{1-\beta} = 15$.

(a) A memoryless contract transfers $\tau \in [0, 1]$ from the agent with 2 goods to the agent with 0, so the high agent eats $2 - \tau$ and the low agent eats τ . Every agent then has the same value before the current draw (22.10.1),

$$v(\tau) = \frac{.5[u(2-\tau) + u(\tau)]}{1-\beta} = \frac{3 + \tau - \tau^2/2}{1-\beta}.$$

The low agent never wants to leave, since $u(\tau) \geq u(0)$ and $v(\tau) \geq v_{aut}$. The high agent stays if $u(2-\tau) + \beta v(\tau) \geq u(2) + \beta v_{aut}$. Using $u(2-\tau) = 6 - 2\tau - \tau^2/2$ and multiplying by $1-\beta$, this reduces to

$$\tau(3\beta - 2) - \frac{1}{2}\tau^2 \geq 0 \quad \Longleftrightarrow \quad \tau \leq 2(3\beta - 2) = 6\beta - 4.$$

Welfare increases in τ up to full insurance ($\tau = 1$), so the planner picks the largest admissible transfer, $\tau^* = \min\{1, \max\{0, 6\beta - 4\}\}$. With $\beta = .8$: $\tau^* = .8$, so the high agent eats 1.2 and the low agent 0.8, with value $v = 17.4$ against $v_{aut} = 15$ and 17.5 under full insurance. Full insurance is not sustainable: at $\tau = 1$ the high agent's constraint fails by .5 utils ($18 + 2 - 2.5 = 17.5 < 18$).

(b) No. In this symmetric two-state economy “amnesia overwhelms memory” (Section 22.10). Every period one of the two agents has the high endowment. If that agent's constraint binds, amnesia fixes the allocation at (1.2, 0.8) whatever the history. So history can matter only until the

first binding constraint. The formal argument uses the constrained Pareto frontier $P(v)$ (the best value for agent 2 given v for agent 1, before the endowment is drawn). P is concave, and at the symmetric point $\bar{v} = 17.4$ it has a kink with one-sided slopes

$$-\frac{u'(0.8)}{u'(1.2)} = -\frac{3.2}{2.8} = -1.143 \quad (\text{right}), \quad -\frac{u'(1.2)}{u'(0.8)} = -0.875 \quad (\text{left})$$

(Section 22.10.1). Since -1 lies between them it is a supergradient, so $P(v) \leq P(\bar{v}) - (v - \bar{v})$, i.e. $v + P(v) \leq 2\bar{v} = 34.8$ for every sustainable allocation. The equally weighted optimum is therefore the memoryless contract of (a). History dependence can move along the frontier, e.g. promise $\hat{v} > \bar{v}$ to an agent who has not yet had a low endowment (the frontier extends to $v_{\max} = [u(2) + u(.8) + \beta\bar{v}]/(2 - \beta) = 19$), but only at a loss in total welfare.

Numerical check. `ch22.py` computes the ex ante frontier by value iteration. It starts from the first-best frontier (an upper bound) and solves each Bellman step through its Lagrangian dual, $P(v) = \min_{\mu \geq 0} [D(\mu) - \mu v]$, which is exact because the step is a concave program. After 97 iterations the frontier lives on $[15, 19]$, as the theory says. $\max_v [v + P(v)] = 34.80000$ is attained at $v = 17.4000$, and the numerical one-sided slopes at $\bar{v}$ are -0.864 and -1.160 (finite differences over .1; theory $-0.875, -1.143$). Figure S22.1 plots the frontier.

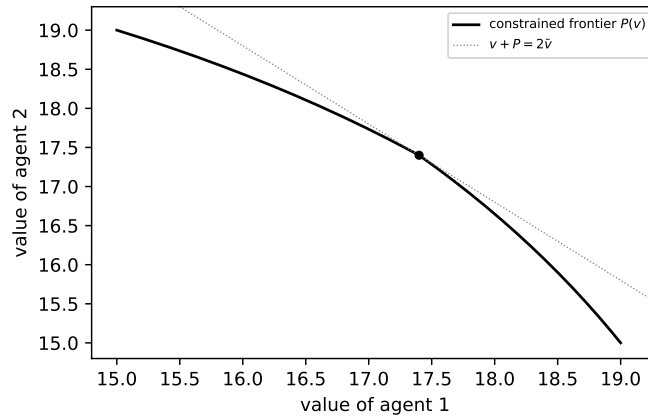


Figure S22.1. Constrained Pareto frontier of Exercise 22.3. It touches the line $v + P = 34.8$ only at the kink $(17.4, 17.4)$, the memoryless contract.

(c) From (a), $\tau^*(\beta) = \min\{1, \max\{0, 6\beta - 4\}\}$. As β rises, the future gains from staying in the arrangement grow relative to the one-period gain from walking away with the high endowment, so more insurance is sustainable. For $\beta \geq 5/6$ the contract gives full insurance ($\tau = 1$), and in particular as $\beta \rightarrow 1$. As β falls, insurance shrinks, and for $\beta \leq 2/3$ no transfer is sustainable: the contract collapses to autarky, and in particular as $\beta \rightarrow 0$. (`ch22.py` confirms the formula at nine values of β by solving the participation constraint numerically.)

Exercise 22.4 *Kehoe–Levine without risk*

Here $u(c) = -1/(c + 5)$ and $u'(c) = (c + 5)^{-2}$, with $u(1.5) = -.153846$, $u(1) = -.166667$, $u(.5) = -.181818$.

(a) Endowments alternate, so

$$v_{aut,h} = \frac{u(1.5) + \beta u(.5)}{1 - \beta^2} = -0.831391, \quad v_{aut,\ell} = \frac{u(.5) + \beta u(1.5)}{1 - \beta^2} = -0.846931.$$

(b) The aggregate endowment is constant at 2 and preferences are identical. With $q_t^0 = \beta^t$, each household's first-order condition $\beta^t u'(c_{it}) = \lambda_i q_t^0$ makes its consumption constant, and constant consumption clears the market. So Arrow–Debreu prices are $q_t^0 = \beta^t$, the one-period gross interest rate is $1/\beta = 1.25$, and each household consumes the annuity value of its endowment. Type 1, rich at $t = 0$, has wealth $(1.5 + .8 \cdot .5)/(1 - .64) = 5.2778$, so

$$c_1 = (1 - \beta) \cdot 5.2778 = 1.055556, \quad c_2 = 0.944444.$$

(c) $v_1^{CE} = u(1.055556)/(1 - \beta) = -0.825688$ and $v_2^{CE} = u(0.944444)/(1 - \beta) = -0.841121$.

(d) At every odd date type 2 has the high endowment. Its continuation value in the competitive allocation is $v_2^{CE} = -0.841121 < v_{aut,h} = -0.831391$, so it would rather renege and eat its endowment. (At even dates both types prefer the competitive allocation: $-0.825688 > -0.831391$ and $-0.841121 > -0.846931$. So the violations occur at odd dates, not at every $t > 0$.)

(e) Let $A_i(t)$ be household i 's autarky value from t on ($v_{aut,h}$ or $v_{aut,\ell}$ according to the parity of t). The planner solves

$$\max_{\{c_{1t}\}} \sum_t \beta^t u(c_{1t}) \quad \text{s.t.} \quad \sum_t \beta^t u(2 - c_{1t}) \geq \tilde{v}_2, \quad \sum_{j \geq 0} \beta^j u(c_{i,t+j}) \geq A_i(t) \quad \forall i, t.$$

Recursively, with the parity $p \in \{\text{even}, \text{odd}\}$ and the promise v_2 to household 2 as states, $V(v_2, p) = \max_{c, v'_2} \{u(c) + \beta V(v'_2, 1 - p)\}$ subject to $u(2 - c) + \beta v'_2 \geq v_2$, $v'_2 \geq A_2(1 - p)$ and $V(v'_2, 1 - p) \geq A_1(1 - p)$.

(f) In the periodic allocation the household with the high endowment eats $\check{c} \geq 1$ and the other eats $2 - \check{c}$, so

$$v_h = \frac{u(\check{c}) + \beta u(2 - \check{c})}{1 - \beta^2}, \quad v_\ell = \frac{u(2 - \check{c}) + \beta u(\check{c})}{1 - \beta^2}.$$

The low household's constraint is slack. The function $g(\check{c}) = u(2 - \check{c}) + \beta u(\check{c})$ is concave, equals $u(.5) + \beta u(1.5)$ at $\check{c} = 1.5$, and is larger at $\check{c} = 1$ ($-.3000$ against $-.3049$). So $g(\check{c}) > u(.5) + \beta u(1.5)$, i.e. $v_\ell > v_{aut,\ell}$, for every $\check{c} \in [1, 1.5)$. The high household's constraint, $v_h \geq v_{aut,h}$, fails at full smoothing ($\check{c} = 1$ gives $-0.833333 < -0.831391$). The function $\check{c} \mapsto u(\check{c}) + \beta u(2 - \check{c})$ is concave and equals $u(1.5) + \beta u(.5)$ at $\check{c} = 1.5$. So the efficient (most smoothing) choice is its other root:

$$\check{c} = 1.168224, \quad v_h = v_{aut,h} = -0.831391, \quad v_\ell = -0.836587.$$

Figure S22.2 compares the three allocations. Autarky has large swings, $1.5, .5, \dots$. Complete markets smooth perfectly, but at levels $(1.056 \text{ and } .944)$ set by initial wealth. With enforcement the swings shrink to $1.168, .832$ and the continuation values alternate. The high-endowment household is held exactly at its autarky value, while the low-endowment household gets strictly more than autarky. Under complete markets the continuation values are constant, which is why type 2's constraint fails in the periods when it is rich.

(g) Without enforcement constraints $R = 1/\beta = 1.25$ every period. With them, the Alvarez–Jermann decentralization prices a one-period bond by the household whose constraint does not bind next period, the one that is rich today:

$$R = \left[\max_i \beta \frac{u'(c_{i,t+1})}{u'(c_{it})} \right]^{-1} = \frac{u'(\check{c})}{\beta u'(2 - \check{c})} = 1.117355$$

at every date (the allocation is periodic). The rate is lower with enforcement. The poor household would like to borrow against its high endowment tomorrow, but it cannot promise to repay, because tomorrow it would default. Its borrowing is restricted, so the rich household, which wants to save, must be induced by a low interest rate to hold the limited supply of claims. Its consumption falls tomorrow, and it values future consumption highly.

(h) As β rises the future matters more, so more smoothing is sustainable and $\check{c}$ falls. Full smoothing becomes sustainable when $(1 + \beta)u(1) \geq u(1.5) + \beta u(.5)$, i.e. for

$$\beta \geq \beta^* = \frac{u(1.5) - u(1)}{u(1) - u(.5)} = 0.846154.$$

For $\beta \geq \beta^*$, $\check{c} = 1$ and both economies have $R = 1/\beta$. For $\beta < \beta^*$ the rate with enforcement is lower: for example 1.1593 against 1.2048 at $\beta = .83$. At β^* the high household's constraint holds with equality at full smoothing but no longer restricts the allocation (a zero multiplier), and for $\beta > \beta^*$ no participation constraint binds. (The competitive allocation of (b) is not the periodic one, but its interest rate is $1/\beta$ at any β .)

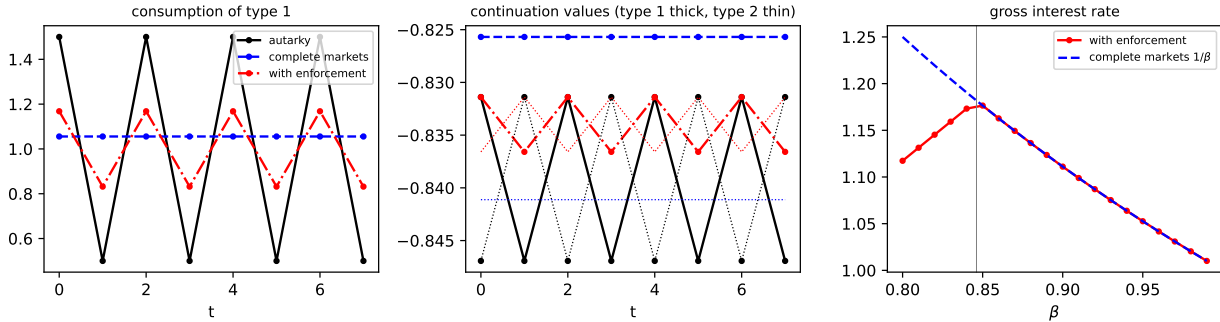


Figure S22.2. Exercise 22.4. Left: consumption of type 1. Middle: continuation values (type 1 thick, type 2 thin). Right: gross interest rates as functions of β ; they coincide for $\beta \geq \beta^* = .846$.

Exercise 22.5 The kink

(a) The aggregate endowment is always 1. With equal weights and strictly concave u , the planner equates consumption: $c_1 = c_2 = .5$ at every date and history.

(b) Let $q_t^0(h^t) = \beta^t \Pi(h^t)$ be the time-0 price of consumption after history h^t , where $\Pi(h^t) = .5^{t+1}$. With constant consumption, each consumer's first-order conditions $\beta^t \Pi(h^t) u'(c_t(h^t)) = \lambda_i q_t^0(h^t)$ hold with $\lambda_i = u'(.5)$. Each consumer's endowment is worth $\sum_t \beta^t E y_t = .5/(1 - \beta)$, which buys .5 per period, so budgets balance and markets clear. A one-period risk-free bond costs $\sum_{y'} \beta = \beta$, so its gross return is $R = 1/\beta$.

(c) In autarky each consumer eats its endowment, so $v_{aut} = .5[u(y) + u(1-y)]/(1-\beta)$ ex ante. After observing the endowment, the autarky value is $u(y) + \beta v_{aut}$ for the high agent and $u(1-y) + \beta v_{aut}$ for the low agent. The ex ante problem maximizes consumer 2's expected utility subject to consumer 1 getting at least a given value, feasibility, and, after every history, each consumer's continuation value being at least its autarky value given its current endowment.

(d) Use the conditional frontiers of Thomas and Worrall (22.3.1). With states $s \in \{1, 2\}$ ($y_1 = 1-y$, $y_2 = y$) and x consumer 1's promised surplus over autarky in the current state,

$$Q_s(x) = \max_{c, \chi_1, \chi_2} \left\{ u(1-c) - u(1-y_s) + \beta \sum_j .5 Q_j(\chi_j) \right\}$$

subject to $u(c) - u(y_s) + \beta \sum_j .5 \chi_j \geq x$, $\chi_j \geq 0$ and $Q_j(\chi_j) \geq 0$ for $j = 1, 2$, with $c \in [0, 1]$. Equivalently, the ex ante frontier $P(v)$ solves $P(v) = \max \sum_s .5[u(1-c_s) + \beta P(w_s)]$ subject to $\sum_s .5[u(c_s) + \beta w_s] \geq v$ and the four participation constraints $u(c_s) + \beta w_s \geq u(y_s) + \beta v_{aut}$, $u(1-c_s) + \beta P(w_s) \geq u(1-y_s) + \beta v_{aut}$.

(e) The Part I allocation gives $u(.5)/(1-\beta)$ forever, and only the high agent's constraint can fail. It is sustainable if and only if

$$\frac{u(.5)}{1-\beta} \geq u(y) + \beta v_{aut} \quad \Longleftrightarrow \quad u(.5) \geq \left(1 - \frac{\beta}{2}\right)u(y) + \frac{\beta}{2}u(1-y).$$

This holds when β is close enough to 1 (the right side tends to $\frac{1}{2}[u(y) + u(1-y)] < u(.5)$), when y is close enough to .5, or when u is sufficiently concave. It fails for small β ($u(.5) < u(y)$).

(f) By the promise-keeping argument of Section 22.10, $v^+ = v^- = v$ with

$$v = \frac{.5[u(c) + u(1-c)]}{1-\beta}.$$

(g) The planner wants as much insurance as possible, i.e. the smallest $c \geq .5$ for the high agent. The low agent's welfare $u(1-c) + \beta v(c)$ is decreasing in c on $[.5, y]$ (Figure S22.3, left). The high agent's constraint $u(c) + \beta v(c) \geq u(y) + \beta v_{aut}$ fails near $c = .5$ (by the assumption of Part II) and holds at $c = y$ with equality. Its left side minus the right side is concave in c (right panel). So the optimum is the smallest root $c \in (.5, y)$, where the curve crosses zero from below, and $v = v(c)$. With the chapter's example $u(c) = c^{1-\gamma}/(1-\gamma)$, $(\beta, \gamma, y) = (.85, 1.1, .6)$: full insurance fails ($u(.5) - [(1-\beta/2)u(y) + \frac{\beta}{2}u(1-y)] = -0.0086 < 0$), and $c = 0.536145$ (the book reports .536), $v = -71.4722$, $v_{aut} = -71.6123$.

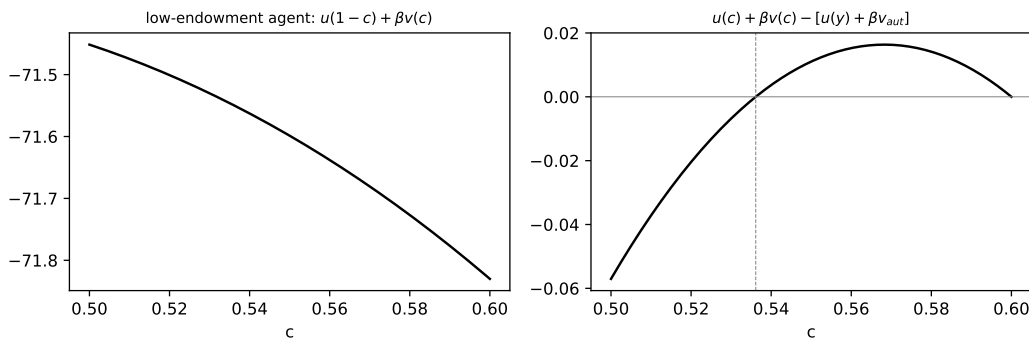


Figure S22.3. The chapter's example ($\beta = .85$, $\gamma = 1.1$, $y = .6$). Left: welfare of the low-endowment agent. Right: the high-endowment agent's participation constraint; the optimal $c = .5361$ is where it first becomes nonnegative.

(h) Raising the constrained agent's continuation value instead of its current consumption would require promising it more in some future state. But in every future state one of the two agents has the high endowment, so one participation constraint binds, and amnesia pins the allocation at $(c, 1 - c)$ whatever was promised. There is no future state with slack in which a promise could be delivered. Formally, the two consumption intervals are disjoint (no first-best sustainable allocation), so the ergodic set is $\{1 - c, c\}$ and every continuation value equals v . Any other promise can only be delivered by moving away from the kink of P , which lowers total welfare, as in Exercise 22.3(b).

(i) If the same agent is rich tomorrow, both marginal rates of substitution equal β , so $q(y | y) = .5\beta$. If the roles switch, today's rich agent (who eats c today and $1 - c$ tomorrow) has the larger ratio, $\beta u'(1 - c)/u'(c) > \beta$, so $q(1 - y | y) = .5\beta u'(1 - c)/u'(c)$. Hence

$$R(y) = R(1 - y) = \frac{1}{.5\beta[1 + u'(1 - c)/u'(c)]} < \frac{1}{\beta},$$

because $c > .5$. The rate is lower than with complete markets and no enforcement problems. The rich agent, who is unconstrained, prices the bond; poor agents cannot borrow as much as they would like against future high endowments, and the scarcity of claims drives their price up. In the example, $q(y | y) = .425$, $q(1 - y | y) = .4984$, and $R = 1.0830$, against $1/\beta = 1.1765$.

Remark. Footnote 14 of the chapter reports a risk-free rate of 1.0146 for this example. With $c = .536$ and the pricing formula above, the rate is 1.0830. A rate of 1.0146 would require $c = .5626$, which is not the optimal c .

Chapter 23

Optimal Unemployment Insurance

Solutions to all 6 exercises of Chapter 23 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). `code/ch23.py` calibrates Hopenhayn and Nicolini's model and solves the Bellman equation (23.2.11) for Exercise 23.1. The other exercises are analytical.

Throughout, V is the promised value of an unemployed worker, $C(V)$ the insurance agency's minimum cost of delivering it, and $V^e = u(w)/(1 - \beta)$ the value of permanent employment. Unless stated otherwise, the agency discounts at β (it borrows and lends at $R = \beta^{-1}$).

Exercise 23.1 *Optimal unemployment compensation*

The parameters are those of Section 23.2.5: one period is a week, $\beta = .999$, $u(c) = c^{1-\sigma}/(1 - \sigma)$ with $\sigma = .5$ (so $u(c) = 2\sqrt{c}$ and $u(0) = 0$), $w = 100$ and $p(a) = 1 - e^{-ra}$. The parameter r is chosen so that the autarky hazard is $p(a^*) = .1$. Then $V^e = u(100)/(1 - \beta) = 20,000$.

(a) In autarky the worker solves (23.2.3) with first-order condition (23.2.4), $\beta r e^{-ra}(V^e - V^u) = 1$. Impose $e^{-ra^*} = .9$. The Bellman equation gives $V_{\text{aut}}(1 - .9\beta) = -a^* + .1\beta V^e$, so $V^e - V_{\text{aut}} = [u(w) + a^*]/(1 - .9\beta)$. Here $u(w) = (1 - \beta)V^e$ and $a^* = \ln(10/9)/r$. Combining with the first-order condition $V^e - V_{\text{aut}} = 1/(\beta r)$ gives the closed form

$$r u(w) = \frac{1 - .9\beta}{.9\beta} - \ln \frac{10}{9} \implies r = 3.4314 \times 10^{-4},$$

with $a^* = 307.05$ and $V_{\text{aut}} = 16,758.70$. The iterative algorithm of Section 23.2.1 (guess V^u , get a from (23.2.4), update V^u from (23.2.3)) converges from $V^u = 0$ in 257 iterations to the same values, with hazard .1000 and expected duration 10 weeks. Continuation values lie in $[V_{\text{aut}}, V^e - 1/(\beta r)] = [16,758.70, 17,082.83]$; the upper bound is the one of Section 23.2.6, which keeps search effort positive.

(b) Following Section 23.2.6, we eliminate c and a . The binding incentive constraint gives $a(V^u) = \max\{0, \ln[r\beta(V^e - V^u)]/r\}$, and promise keeping gives $c = u^{-1}(V + a - \beta[p(a)V^e + (1 - p(a))V^u])$. The Bellman equation (23.2.11) is then a minimization over V^u alone. We solve it by Howard policy iteration on 2001 nodes, clustered at both ends of $[V_{\text{aut}}, 17,082.83]$, with C interpolated linearly. Each improvement step is a golden-section search over V^u . Each evaluation step solves exactly the linear system $C = c + \beta(1 - p)C(V^u(\cdot))$. Convergence takes 24 steps, and the Bellman residual is 7×10^{-6} . The computed C is increasing and convex, which is the property Hopenhayn and Nicolini assume. It satisfies the first-order relations of Section 23.2.4. With $p = 1 - e^{-ra}$, (23.2.8) and the binding (23.2.4) imply $p'/(1 - p) = r$ and $\eta = C(V^u)/[r(V^e - V^u)]$, hence

$$C'(V) = \frac{1}{u'(c)} = \sqrt{c} = C'(V^u) + \frac{C(V^u)}{V^e - V^u}.$$

The computed solution satisfies this to within 0.4%. Since $C(V^u) > 0$, it implies $C'(V^u) < C'(V)$ and $V^u < V$: benefits fall with duration. The optimal policy indeed has $V^u < V$ everywhere,

except in a thin layer within 3 utils of V_{aut} , where C is essentially flat and the minimization is numerically ill-conditioned.

The table reports, for three initial promises: the cost $C(V_0)$; the full-information cost (Section 23.2.2: constant c and a); and the cheapest *constant* benefit b that delivers V_0 when the worker chooses effort freely, with its cost. It then gives the replacement ratio c/w by week of unemployment and search effort in weeks 1 and 50.

V_0	$C(V_0)$	full info	b/w	const. cost	saving	wk 1	5	10	20	30	40	50	a_1	a_{50}
16,850	210.4	197.2	.161	215.6	2.4%	.212	.190	.167	.132	.107	.088	.074	225	259
16,942	849.0	742.8	.466	991.2	14.4%	.861	.691	.542	.358	.254	.190	.148	142	239
17,000	1472.7	1232.1	.628	2196.6	33.0%	1.496	1.124	.826	.502	.337	.242	.182	90	231

The middle row starts at the book's $V_0 = 16,942$. Figure S23.1 reproduces the book's Figure 23.2.1: the replacement ratio falls from .86 to .15 over 50 weeks, while effort rises from 142 toward the autarky level 307. The more generous the initial promise, the steeper the decline. Near the upper bound of V , delivering the promise requires front-loading consumption above the wage. The value of history dependence is large: relative to the best constant benefit, the optimal contract saves 2%, 14% and 33% of the cost. A constant benefit generous enough to deliver V_0 destroys search incentives (hazard .028 instead of .1 in the last case). The full-information contract would be cheaper still. It keeps c constant (e.g. $c/w = .570$ for $V_0 = 16,942$) and asks for constant, higher effort ($a = 230$).

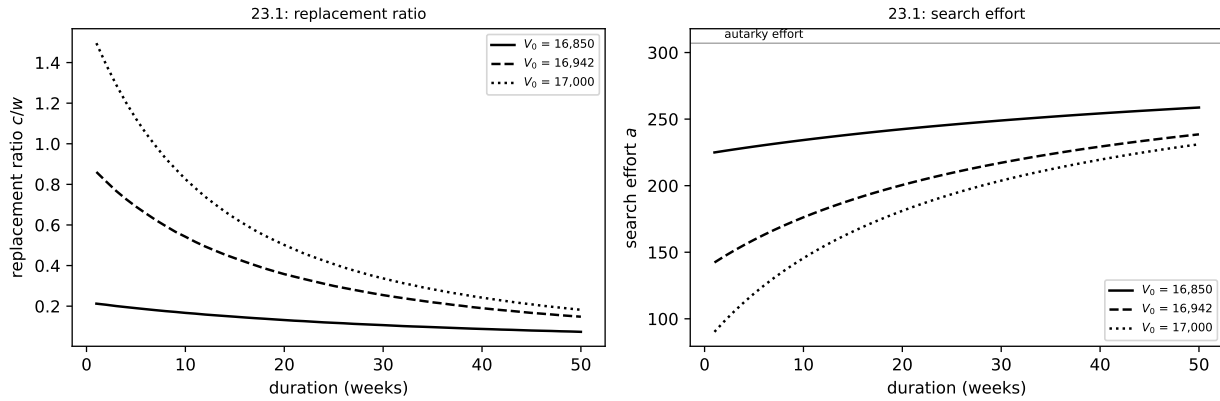


Figure S23.1. Exercise 23.1: optimal replacement ratio and search effort by duration of unemployment for three initial promised values.

Exercise 23.2 *Taxation after employment*

Now the value of employment is a choice variable. A worker who finds a job after the current period gets $V^e = u(w - \tau)/(1 - \beta)$, and the tax τ may depend on the history of the unemployment spell. Since there is no incentive problem on the job, it is optimal to tax the employed worker at a constant rate. Delivering V^e to an employed worker costs the agency

$$C_e(V^e) = -\frac{\tau}{1 - \beta} = \frac{u^{-1}((1 - \beta)V^e) - w}{1 - \beta}, \quad C'_e(V^e) = \frac{1}{u'(w - \tau)}.$$

The Bellman equation becomes

$$C(V) = \min_{c,a,V^u,V^e} \left\{ c + \beta [p(a)C_e(V^e) + (1-p(a))C(V^u)] \right\}$$

subject to promise keeping, $V = u(c) - a + \beta[p(a)V^e + (1-p(a))V^u]$, and the incentive constraint $\beta p'(a)(V^e - V^u) \leq 1$, with equality if $a > 0$. Here V^e now appears in both constraints as a control. With multipliers θ on promise keeping and η on the incentive constraint, the first-order conditions are

$$\frac{1}{u'(c)} = \theta, \quad C'(V^u) = \theta - \eta \frac{p'(a)}{1-p(a)}, \quad \frac{1}{u'(w-\tau)} = C'_e(V^e) = \theta + \eta \frac{p'(a)}{p(a)},$$

plus the condition for a . With $C'(V) = \theta$, the incentive problem ($\eta > 0$) again makes $V^u < V$, so unemployment consumption falls with duration. It also makes $w - \tau > c$: the reward for finding a job is delivered partly through a lower tax on the job. As the promised V falls with duration, the V^e assigned to a worker who finds a job falls as well, so the employment tax is history dependent. In Hopenhayn and Nicolini's computations it rises with the length of the spell, which substantially slows the decline of the replacement ratio (Section 23.2.8). Exercise 23.5(e) proves the rising tax in a two-effort version.

Exercise 23.3 *Optimal unemployment compensation with unobservable wage offers*

Let V be the value of an unemployed worker at the start of a period, before this period's offer is drawn. An accepted offer w is worth $u(w)/(1-\beta)$ and costs the agency nothing.

(a) With a constant benefit, $Q = \int \max\{u(w)/(1-\beta), u(b) + \beta Q\} dF(w)$. The worker accepts iff $w \geq \bar{w}$, where $u(\bar{w})/(1-\beta) = u(b) + \beta Q$. Substituting $\bar{w}$ into the equation for Q and rearranging,

$$(1-\beta)[u(\bar{w}) - u(b)] = \beta \int_{\bar{w}}^{w_H} [u(w) - u(\bar{w})] \frac{dw}{w_H - w_L}.$$

The left side increases in $\bar{w}$ and the right side decreases, so $\bar{w}$ is unique, provided the solution lies in $[w_L, w_H]$; otherwise the worker accepts every offer. The reservation wage rises with b and is constant over the spell.

(b) A contract specifies this period's reservation wage $\bar{w}$, the benefit c if no acceptable offer arrives, and the continuation promise V^u . Then

$$C(V) = \min_{\bar{w}, c, V^u} F(\bar{w})[c + \beta C(V^u)] \quad \text{s.t.} \quad V = \int_{\bar{w}}^{w_H} \frac{u(w)}{1-\beta} dF(w) + F(\bar{w})[u(c) + \beta V^u].$$

With multiplier θ , the first-order conditions are $\theta = 1/u'(c)$ and $C'(V^u) = \theta$, and for $\bar{w}$

$$\frac{u(\bar{w})}{1-\beta} = u(c) + \beta V^u - u'(c)[c + \beta C(V^u)].$$

The envelope condition $C'(V) = \theta$ and convexity give $V^u = V$. So consumption and the reservation wage are constant during the spell: full smoothing, as in Section 23.2.2. The reservation wage is *below* the one the worker would choose given (c, V) , which satisfies $u(\bar{w})/(1-\beta) = u(c) + \beta V$. The agency internalizes the cost $c + \beta C(V)$ of another period of unemployment and makes the worker accept lower offers.

(c) Now the worker chooses his reservation wage. Given (c, V^u) , it satisfies the incentive constraint $u(\bar{w})/(1 - \beta) = u(c) + \beta V^u$, and promise keeping becomes

$$V = \Phi(\bar{w}) \equiv \frac{F(\bar{w})u(\bar{w}) + \int_{\bar{w}}^{w^H} u(w) dF(w)}{1 - \beta}, \quad \Phi'(\bar{w}) = \frac{F(\bar{w})u'(\bar{w})}{1 - \beta} > 0.$$

So the promised value pins down the reservation wage, $\bar{w} = \Phi^{-1}(V)$. The agency can only choose how to deliver the utility $k = u(\bar{w})/(1 - \beta)$ of rejecting. Let $G(k) = \min\{c + \beta C(V^u) : u(c) + \beta V^u = k\}$. Its first-order condition is $1/u'(c) = C'(V^u)$, and $G'(k) = 1/u'(c)$. Then $C(V) = F(\bar{w})G(u(\bar{w})/(1 - \beta))$ with $\bar{w} = \Phi^{-1}(V)$, and differentiating,

$$C'(V) = \frac{f(\bar{w})G + F(\bar{w})G' u'(\bar{w})/(1 - \beta)}{\Phi'(\bar{w})} = G'(k) + \frac{(1 - \beta)f(\bar{w})}{F(\bar{w})u'(\bar{w})} G(k) > G'(k) = C'(V^u).$$

By convexity of C , $V^u < V$: promised values fall during the spell. Along the spell $1/u'(c_t) = C'(V_{t+1})$, and $V_{t+2} < V_{t+1}$ gives $1/u'(c_{t+1}) = C'(V_{t+2}) < C'(V_{t+1}) = 1/u'(c_t)$. So *consumption (the benefit) declines with the duration of unemployment*, and the reservation wage $\bar{w}_t = \Phi^{-1}(V_t)$ declines with it. This is Shavell and Weiss's result. The agency cannot force acceptance, but lowering future benefits lowers the reservation wage and shortens unemployment. This is worth some loss of consumption smoothing. The wedge is larger the higher the density $f(\bar{w})$ of offers at the margin relative to the probability $F(\bar{w})$ of remaining unemployed.

Exercise 23.4 *Full unemployment insurance*

The agency chooses current consumption c , effort a , the continuation promise V^u if no job is found, and the tax τ paid forever by a worker who finds a job now. A tax τ is worth $-\tau/(1 - \beta)$ to the agency. So

$$C(V) = \min_{c, a, V^u, \tau} \left\{ c + \beta \left[-p(a) \frac{\tau}{1 - \beta} + (1 - p(a)) C(V^u) \right] \right\}$$

subject to $V = u(c) - a + \beta[p(a)u(w - \tau)/(1 - \beta) + (1 - p(a))V^u]$. Effort is observed, so there is no incentive constraint.

The name of the exercise comes from the solution. With multiplier θ on promise keeping, the conditions for c and τ are $1 = \theta u'(c)$ and $\beta p/(1 - \beta) = \theta \beta p u'(w - \tau)/(1 - \beta)$. So $u'(w - \tau) = u'(c)$, i.e. $w - \tau = c$: the worker consumes the same whether employed or not. The condition for V^u is $C'(V^u) = \theta = C'(V)$, so $V^u = V$ and c stays constant during the spell. Effort is constant too and set at its social optimum. The first-order condition for a balances its disutility against the fall in the agency's cost when the worker leaves unemployment: $\theta[1 - \beta p'(a)(u(c)/(1 - \beta) - V)] = \beta p'(a)[C(V) + c/(1 - \beta)]$. The worker is fully insured against the unemployment risk.

Exercise 23.5 *Two effort levels*

Write $\pi \equiv \pi(a)$ and $\Delta\pi = \pi(a) - \pi(0) > 0$.

(a)

$$C(V) = \min_{c, V^u} \{c + \beta(1 - \pi)C(V^u)\} \quad \text{s.t.} \quad V = u(c) - a + \beta[\pi V_e + (1 - \pi)V^u]$$

and the incentive constraint. High effort must beat zero effort: $u(c) - a + \beta[\pi V_e + (1 - \pi)V^u] \geq u(c) + \beta[\pi(0)V_e + (1 - \pi(0))V^u]$, i.e.

$$\beta \Delta \pi (V_e - V^u) \geq a \iff V^u \leq \hat{V} \equiv V_e - \frac{a}{\beta \Delta \pi}.$$

Because effort enters utility additively and linearly, the constraint involves only the continuation value.

(b) With multipliers θ and $\eta \geq 0$, the first-order conditions are $1/u'(c) = \theta$ and $\beta(1 - \pi)C'(V^u) = \theta\beta(1 - \pi) - \eta\beta\Delta\pi$. The envelope condition is $C'(V) = \theta$, so

$$C'(V^u) = C'(V) - \eta \frac{\Delta\pi}{1 - \pi}.$$

If the constraint binds ($\eta > 0$), $C'(V^u) < C'(V)$ and $V^u < V$. Next period's benefit then satisfies $1/u'(c_{t+1}) = C'(V_{t+1}) = C'(V_t^u) < C'(V_t) = 1/u'(c_t)$, so $c_{t+1} < c_t$: benefits decline. With two effort levels the decline is a single step. The constraint binds iff $V > \hat{V}$, and then it sets $V^u = \hat{V}$. At $V = \hat{V}$ the unconstrained choice $V^u = V = \hat{V}$ satisfies the constraint with equality, so from then on the promise and the benefit $\hat{c}$ are constant, where $u(\hat{c}) = \hat{V}(1 - \beta(1 - \pi)) + a - \beta\pi V_e$. So a worker promised $V_0 > \hat{V}$ gets $c_0 > \hat{c}$ in the first week of unemployment and $\hat{c}$ afterwards. If $V_0 \leq \hat{V}$ the constraint is slack and benefits are constant.

(c) If $\pi(a) = \pi(0)$, the incentive constraint reads $0 \geq a$, which fails: high effort cannot be induced, and it would be pointless because it does not affect job finding. The agency lets the worker set $a = 0$. Without an incentive constraint, $C'(V^u) = C'(V)$ gives $V^u = V$, and benefits are constant. Benefits decline only because declining benefits provide incentives, as the text notes after (23.2.8c).

(d) With $V_e = u(w - \tau)/(1 - \beta)$ chosen along with (c, V^u) and cost $C_e(V_e) = [u^{-1}((1 - \beta)V_e) - w]/(1 - \beta)$,

$$C(V) = \min_{c, V^u, V_e} \{c + \beta[\pi C_e(V_e) + (1 - \pi)C(V^u)]\}$$

subject to $V = u(c) - a + \beta[\pi V_e + (1 - \pi)V^u]$ and $\beta\Delta\pi(V_e - V^u) \geq a$.

(e) The first-order conditions are

$$\frac{1}{u'(c)} = \theta = C'(V), \quad C'(V^u) = \theta - \eta \frac{\Delta\pi}{1 - \pi}, \quad \frac{1}{u'(w - \tau)} = C'_e(V_e) = \theta + \eta \frac{\Delta\pi}{\pi}.$$

The incentive constraint always binds. If $\eta = 0$, the agency would give full insurance, $V^u = V$ and $w - \tau = c$. Promise keeping would then imply $V_e - V = a/(1 - \beta + \beta\pi)$, and the constraint $V_e - V \geq a/(\beta\Delta\pi)$ would require $\beta\Delta\pi \geq 1 - \beta + \beta\pi(a)$, i.e. $-\beta\pi(0) \geq 1 - \beta$, which is impossible. So $\eta > 0$ at every date. Hence $V_{t+1} = V_t^u < V_t$, and the promise declines strictly and permanently over the spell. The binding constraint fixes $V_{e,t} = V_{t+1} + a/(\beta\Delta\pi)$, so $V_{e,t}$ declines with duration too, i.e. $u(w - \tau_t)$ falls. *The tax on a worker who finds a job rises with the duration of his unemployment spell.* Benefits also keep falling ($c_{t+1} < c_t$), and at every date $w - \tau_t > c_t$. Unlike in subproblem A, where the decline was a single step, the tax instrument makes both benefits and post-employment consumption decline gradually. The agency spreads the punishment for long spells over the worker's whole future.

Exercise 23.6 Partially observed search effort

(a) A contract specifies consumption c , recommended effort a , and continuation values for an unsuccessful searcher: $V_O(\tilde{a})$ if last period's effort $\tilde{a}$ is observed, and V_N if it is not. Punishing an observed deviation as hard as possible is free. So set $V_O(\tilde{a}) = V_{\text{aut}}$ for $\tilde{a} \neq a$: the lowest deliverable value, which costs $C(V_{\text{aut}}) = 0$. Write $V_O \equiv V_O(a)$. The best deviation $\tilde{a}$ solves $\max_{\tilde{a}} \{-\tilde{a} + \beta[p(\tilde{a})V^e + (1 - p(\tilde{a}))(dV_{\text{aut}} + (1 - d)V_N)]\}$, i.e. $\beta p'(\tilde{a})[V^e - dV_{\text{aut}} - (1 - d)V_N] = 1$ if interior. The Bellman equation is

$$C(V) = \min_{c, a, V_O, V_N} \left\{ c + \beta(1 - p(a)) [dC(V_O) + (1 - d)C(V_N)] \right\}$$

subject to promise keeping, $V = u(c) - a + \beta\{p(a)V^e + (1 - p(a))[dV_O + (1 - d)V_N]\}$, and the global incentive constraint

$$\begin{aligned} -a + \beta\{p(a)V^e + (1 - p(a))[dV_O + (1 - d)V_N]\} \\ \geq \max_{\tilde{a}} \left\{ -\tilde{a} + \beta[p(\tilde{a})V^e + (1 - p(\tilde{a}))(dV_{\text{aut}} + (1 - d)V_N)] \right\}, \end{aligned}$$

with $V_O, V_N \geq V_{\text{aut}}$. The first-order approach of the text is inappropriate here. Detection makes the worker's payoff jump at $\tilde{a} = a$, so the constraint must be imposed against the best global deviation.

(b) Let θ and $\eta \geq 0$ be the multipliers and $\tilde{a}$ the best deviation. By the envelope theorem, the derivative of the right side with respect to V_N is $\beta(1 - d)(1 - p(\tilde{a}))$. The first-order conditions and the envelope condition $C'(V) = \theta$ give

$$\frac{1}{u'(c)} = \theta, \quad C'(V_O) = \theta + \eta, \quad C'(V_N) = \theta - \eta \frac{p(a) - p(\tilde{a})}{1 - p(a)}.$$

When the constraint binds, $p(a) > p(\tilde{a})$ and so $C'(V_O) > C'(V) > C'(V_N)$, i.e. $V_O > V > V_N$.

- After a week without a job in which effort is *not* observed, the promise falls ($V_N < V$). As in Hopenhayn and Nicolini, benefits decline with the duration of unmonitored unemployment.
- After a week in which compliant effort *is* observed, the promise rises ($V_O > V$): the contract pays a bonus for verified search. Rewarding verified compliance is a cheap way to provide incentives, because it pays off only in states where the planner knows the worker searched.
- Detected shirking sends the worker to V_{aut} (off the equilibrium path).

Compensation therefore depends on the whole history of verification outcomes, not just on duration. Benefits drift down during unmonitored stretches and jump up at each successful audit, so two workers unemployed for the same time can receive very different benefits. The threat of detection weakens the constraint, since V_{aut} replaces V_N in the deviation payoff with probability d . As $d \rightarrow 0$ we recover Hopenhayn and Nicolini (V_O irrelevant, $V_N < V$). As d grows the constraint relaxes, and for d large enough it no longer binds at the full-information contract ($\eta = 0$). Then $V_O = V_N = V$ and constant consumption, the full-information outcome of Section 23.2.2, is implementable.

Chapter 24

Credible Government Policies, I

Solutions to all 3 exercises of Chapter 24 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Numbers and the figure come from `code/ch24.py`. For the Kydland–Prescott model of Exercise 24.3, the script iterates the APS operator to convergence, which independently checks every best and worst value derived by hand.

Notation follows the chapter. C is the set of competitive equilibria (x, y) with $x = h(y)$, $r(x, y) = u(x, x, y)$ is the government's one-period payoff, and $H(x)$ is its one-period best response. A Nash equilibrium is a pair $(x, y) \in C$ with $y = H(x)$; the Ramsey outcome maximizes r over C . In the repeated economy the government values paths by $(1 - \delta) \sum_{t \geq 1} \delta^{t-1} r(x_t, y_t)$. The lemma of Section 24.6 reduces subgame perfection to one-shot incentive constraints:

$$(1 - \delta)r(x, y) + \delta v' \geq (1 - \delta)r(x, \eta) + \delta \tilde{v}(\eta) \quad \forall \eta,$$

with continuation values v' and $\tilde{v}(\eta)$ that are themselves SPE values.

Exercise 24.1 *Finite repetition with multiple Nash equilibria*

(a) The government's best responses are $H(x_M) = y_M$ ($10 > 4$) and $H(x_H) = y_M$ ($20 > 15$). Of the two competitive equilibria only (x_M, y_M) satisfies $y = H(x)$, so the unique Nash equilibrium is (x_M, y_M) , with $v^N = 10$. The Ramsey outcome is the best competitive equilibrium, (x_H, y_H) , with $v^R = 15$. The government would like to promise y_H , but after the public has set x_H it is tempted by y_M , which pays 20.

(b) No. In the last period any SPE must play a one-period Nash equilibrium after every history. Since (x_M, y_M) is the only one, the second-period outcome, and hence the continuation value, is the same whatever the government does in period 1. Then in period 1 the government simply maximizes its current payoff. Given x_H , it chooses y_M and gains $20 - 15 = 5$ at no future cost. “Reverting to Nash” is no punishment when Nash is also what happens after conformity.

(c) No, in neither period. By backward induction the third period plays (x_M, y_M) after every history. The second period then has a history-independent continuation, so it must also play the stage Nash equilibrium, and the same argument applies to the first period. With a unique one-period Nash equilibrium, the finitely repeated economy has a unique SPE: Nash in every period.

(d) Now $H(x_L) = y_L$ ($3 > 1 > 0$), $H(x_M) = y_M$ ($10 > 7 > 4$) and $H(x_H) = y_M$ ($20 > 15 > 9$). The competitive equilibria (x_L, y_L) and (x_M, y_M) satisfy $y = H(x)$, but (x_H, y_H) does not. So there are two Nash equilibria, a bad one (x_L, y_L) with value 3 and a good one (x_M, y_M) with value 10. Ramsey is still (x_H, y_H) , with value 15.

(e) The multiplicity of Nash equilibria provides the punishment (Benoit and Krishna's idea). Strategy profile: in period 1 play (x_H, y_H) . In period 2 play the good Nash equilibrium (x_M, y_M) if the period-1 outcome was (x_H, y_H) , and the bad one (x_L, y_L) otherwise. Period-2 play is a one-period Nash equilibrium after every history, so it is subgame perfect. In period 1, $x_H = h(y_H)$ is a

competitive equilibrium. The government's best deviation is y_M , and it does not deviate if

$$15 + 10\delta \geq 20 + 3\delta \iff \delta \geq \frac{5}{7} \approx .714.$$

(Multiplying both sides by $1 - \delta$ gives the normalized criterion and changes nothing.) So the profile is an SPE exactly when $\delta \geq 5/7$.

(f) With $\delta = .8$ ($> 5/7$), consider the following profile.

- Period 1: (x_H, y_H) .
- Period 2: (x_H, y_H) if the period-1 outcome was (x_H, y_H) ; otherwise (x_L, y_L) .
- Period 3: (x_M, y_M) if the outcomes of periods 1 and 2 were both (x_H, y_H) ; otherwise (x_L, y_L) .

Period-3 play is a stage Nash equilibrium after every history. In period 2 on the path, conforming gives $15 + .8 \cdot 10 = 23$ and deviating gives $20 + .8 \cdot 3 = 22.4$. Off the path, period 2 plays a stage Nash equilibrium followed by a history-independent continuation. In period 1, conforming gives $15 + .8 \cdot 15 + .64 \cdot 10 = 33.4$ and deviating gives $20 + .8 \cdot 3 + .64 \cdot 3 = 24.32$. So Ramsey is supported in the first two periods. (The period-1 constraint alone would hold for $\delta \geq .347$; the binding constraint is the period-2 one, $\delta \geq 5/7$.)

The SPE is *not unique*, although its outcome path is. On the path, period 3 must play (x_M, y_M) , because otherwise the period-2 constraint $15 + .8w_3^{\text{on}} \geq 20 + .8w_3^{\text{off}}$ fails. After a period-2 deviation, period 3 must play (x_L, y_L) , because $23 \geq 20 + .8w$ requires $w \leq 3.75$. But the punishment after a *period-1* deviation can be chosen in several ways. Its continuation can be (x_L, y_L) then (x_M, y_M) (the deviator gets $20 + 2.4 + 6.4 = 28.8$), or (x_M, y_M) then (x_L, y_L) ($20 + 8 + 1.92 = 29.92$). Both deter the deviation (< 33.4) and give different SPEs with the same outcome path. Punishments (x_M, y_M) twice (34.4) or the good continuation (38.4) would not deter it.

Exercise 24.2 *An economy without a one-period Nash equilibrium (Stokey 1989)*

(a) In a Ramsey plan the government moves first and is bound by its choice. It picks $y \in Y$ knowing that the public responds with the competitive equilibrium $x = h(y)$. So it solves $\max_{(x,y) \in C} r(x, y)$ (Definitions 3–4). Here $h(y_L) = x_L$ and $h(y_H) = x_H$, and $\max\{0, 10\} = 10$, so the Ramsey outcome is (x_H, y_H) , with $v^R = 10$.

(b) A pure-strategy Nash equilibrium is a pair (x^N, y^N) such that $(x^N, y^N) \in C$ (the public has no incentive to deviate given y^N) and $r(x^N, y^N) = \max_{\eta} r(x^N, \eta)$ (the government has no incentive to deviate given x^N), i.e. $y^N = H(x^N)$.

(c) $H(x_L) = y_H$ (since $1 > 0$) and $H(x_H) = y_L$ (since $20 > 10$). The competitive equilibrium (x_L, y_L) has $H(x_L) = y_H \neq y_L$, and (x_H, y_H) has $H(x_H) = y_L \neq y_H$. So no competitive equilibrium is a best response to itself, and there is no pure-strategy Nash equilibrium. Low inflation expectations tempt the government to inflate, and high expectations tempt it to disinflate.

(d) A strategy profile $\sigma = (\sigma^h, \sigma^g)$ maps each history (x^{t-1}, y^{t-1}) into actions (x_t, y_t) . It is an SPE if for every t and every history (i) $(x_t, y_t) = \sigma_t(x^{t-1}, y^{t-1}) \in C$, and (ii) for every $\eta \in Y$, $(1 - \delta)r(x_t, y_t) + \delta V_g(\sigma|_{(x^t, y^t)}) \geq (1 - \delta)r(x_t, \eta) + \delta V_g(\sigma|_{(x^t, y^{t-1}, \eta)})$ (Definition 6).

(e) *Lower bound.* Let $\underline{v}$ be the lowest SPE value, and consider the SPE that attains it, with first-period outcome $(x, y) \in C$. The government can always play its one-period best response, and the worst it can then face is $\underline{v}$, so $\underline{v} \geq (1 - \delta) \max_{\eta} r(x, \eta) + \delta \underline{v}$, i.e. $\underline{v} \geq \max_{\eta} r(x, \eta) \geq \min_y \max_{\eta} r(h(y), \eta)$. Here $\min\{\max_{\eta} r(x_L, \eta), \max_{\eta} r(x_H, \eta)\} = \min\{1, 20\} = 1$. *Attainment.* The bound is reached by starting with the “stick” (x_L, y_L) (payoff 0, temptation 1), with deviations restarting the equilibrium. The binding constraint $\underline{v} = (1 - \delta) \cdot 0 + \delta v_1 = (1 - \delta) \cdot 1 + \delta \underline{v}$ gives $\underline{v} = 1$ and $v_1 = 1/\delta$, which must itself be an SPE value. At the discount factor of part (f) it is (parts (h)–(k)), so the worst SPE value is $\underline{v} = 1$. Part (l) discusses discount factors for which the needed continuation values are not available.

(f) Yes. With the worst SPE as punishment, the incentive constraint for Ramsey is

$$v^R = 10 \geq (1 - \delta) r(x_H, y_L) + \delta \underline{v} = (1 - .8913) \cdot 20 + .8913 \cdot 1 = 3.066.$$

A recursive SPE (Definition 12) has states $v_j = \delta^{-j}$, $j = 0, 1, \dots, 20$. Note $v_0 = 1$ and $v_{20} = \delta^{-20} = 10$ because $\delta^{20} = 1/10$. Its prescriptions are

$$(z^h, z^g)(v_j) = (x_L, y_L) \quad (j < 20), \quad (z^h, z^g)(v_{20}) = (x_H, y_H),$$

$$\mathcal{V}(v_j, x, y) = \begin{cases} v_{\min\{j+1, 20\}} & \text{if } (x, y) = (z^h, z^g)(v_j), \\ v_0 & \text{otherwise,} \end{cases}$$

starting from $v_{20} = 10$. Promise keeping holds: $(1 - \delta) \cdot 0 + \delta v_{j+1} = \delta^{-j} = v_j$ for $j < 20$, and $(1 - \delta)10 + \delta \cdot 10 = 10$. The incentive constraints hold: at v_j , $j < 20$, $v_j \geq (1 - \delta) \cdot 1 + \delta \cdot 1 = 1$; at v_{20} , $10 \geq 3.066$. On the path, Ramsey is played forever. Any deviation starts the worst SPE.

(g) The best SPE value cannot exceed $\max_C r = 10$, and (f) attains it: $\bar{v} = v^R = 10$.

(h) The worst SPE constructed in (f) (state v_0) has the path

$$(x_L, y_L) \text{ for } t = 1, \dots, 20, \quad (x_H, y_H) \text{ for } t \geq 21,$$

with value $\delta^{20} \cdot 10 = 1$. Each period of the stick raises the promised value by the factor $1/\delta$ until it reaches the Ramsey value. The discount factor $(1/10)^{1/20}$ is chosen so that this happens after exactly 20 periods. This is the path generated by the recursion of parts (i)–(k).

(i) Adherence to the worst SPE requires $1 = (1 - \delta) \cdot 0 + \delta v_1$, so $v_1 = \delta^{-1} = 1.12202$. σ_1 starts with the stick again: its path is 19 periods of (x_L, y_L) followed by (x_H, y_H) forever.

(j) Adherence to the first-period outcome (x_L, y_L) of σ_1 requires $v_1 = \delta v_2$, so $v_2 = \delta^{-2} = 1.25893$. The path of σ_2 is 18 periods of (x_L, y_L) , then Ramsey forever.

(k) In general $v_j = \delta^{-j} = 10^{j/20}$ and σ_j plays (x_L, y_L) for $20 - j$ periods and then (x_H, y_H) forever, for $j = 1, \dots, 20$. At $j = 20$, $v_{20} = 10 = v^R$ and σ_{20} is Ramsey forever; for $j \geq 20$, $v_j = 10$. The values are 1.122, 1.259, 1.413, 1.585, 1.778, 1.995, 2.239, 2.512, 2.818, 3.162, $\dots$, 7.943, 8.913, 10. Every σ_j satisfies its incentive constraints, since $v_j \geq 1$ in stick periods and $10 \geq 3.066$ in Ramsey periods.

(1) Ramsey repetition is an SPE if and only if some SPE value w satisfies $10 \geq (1 - \delta)20 + \delta w$, i.e. $w \leq 20 - 10/\delta$. Since every SPE value is at least 1, this requires $20 - 10/\delta \geq 1$, i.e.

$$\delta \geq \frac{10}{19} = .5263.$$

If the value set is an interval, as the chapter's APS construction presumes (e.g. with a public randomization device), the worst value 1 is available. Then $\delta = 10/19$ is the answer: at $\delta = 10/19$ the stick (x_L, y_L) with continuation value $1/\delta = 1.9 \in [1, 10]$ delivers 1.

Remark. With pure strategies and no randomization this bound is not attained, because SPE values are restricted to values of $\{L, H\}$ paths. For $\delta < .6466$ a period of (x_H, y_H) can never be followed by (x_L, y_L) : the constraint $(1 - \delta)10 + \delta v' \geq (1 - \delta)20 + \delta \cdot 1$ fails for every $v' \leq 10\delta$. So every SPE path has the form $L^k H^\infty$, with value $10\delta^k$. Path $L^k H^\infty$ is an SPE (with itself as punishment) if and only if $10\delta^k \geq 1$ and $10 \geq 20(1 - \delta) + \delta \cdot 10\delta^k$. So Ramsey is sustainable if and only if some integer k has $.1 \leq \delta^k \leq 2 - 1/\delta$. The smallest such δ is the root of $\delta + \delta^2 + \delta^3 = 1$, namely $\delta^* = .543689$ (with $k = 3$), and every larger δ also works (ch24.py checks this on a fine grid). At δ^* the worst SPE is three periods of (x_L, y_L) followed by Ramsey, with value $10\delta^{*3} = 1.607 = 20 - 10/\delta^*$. For $\delta < \delta^*$ there is no pure-strategy SPE at all, because L^∞ is worth $0 < 1$.

Exercise 24.3 *Worst and best SPEs in the Kydland–Prescott model*

(a) Given y , a forecaster maximizes $-.5(y - \xi)^2$ by setting $\xi = y$. All forecasters do the same, so $x = \xi = y$, and then $U = U^* - \theta(y - y) = U^*$. Hence $C = \{(x, y) : x = y \in [0, 10]\}$, $h(y) = y$, and $r(h(y), y) = -.5(U^{*2} + y^2)$.

(b) Given expectations x , the best response is $H(x) = \arg \max_{y \in [0, 10]} r(x, y)$. The first-order condition $\theta(U^* - \theta(y - x)) = y$ gives

$$H(x) = \min \left\{ 10, \frac{\theta U^* + \theta^2 x}{1 + \theta^2} \right\},$$

which is always positive; with $U^* = 5$, $\theta = 1$, $H(x) = (5 + x)/2 \in [2.5, 7.5]$. Given expectations, surprise inflation lowers unemployment.

(c) A Nash equilibrium is $(x, y) \in C$ with $y = H(x)$. Setting $x = y$ in $y(1 + \theta^2) = \theta U^* + \theta^2 y$ gives $y^N = x^N = \theta U^*$ (assuming $\theta U^* \leq 10$), with $U = U^*$ and $v^N = -.5(1 + \theta^2)U^{*2}$. For $U^* = 5$, $\theta = 1$: $y^N = 5$ and $v^N = -25$.

(d) The Ramsey problem is $\max_y r(h(y), y) = \max_y -.5(U^{*2} + y^2)$, so $y^R = x^R = 0$ and $v^R = -.5U^{*2}$ (-12.5 for $U^* = 5$). Committing to zero inflation removes the inflation bias and costs no unemployment, because unemployment equals U^* in every competitive equilibrium.

(e) $v^R - v^N = .5\theta^2 U^{*2} > 0$. The Nash inflation bias θU^* buys no unemployment reduction.

(f) As in Definition 6: a profile $\sigma = (\sigma^h, \sigma^g)$ of history-dependent rules such that after every history (i) $x_t = y_t$ (forecasts equal the government's prescribed action, a competitive equilibrium) and (ii) for every $\eta \in [0, 10]$, $(1 - \delta)r(x_t, y_t) + \delta V_g(\sigma|_{(x^t, y^t)}) \geq (1 - \delta)r(x_t, \eta) + \delta V_g(\sigma|_{(x^t, y^{t-1}, \eta)})$.

(g) As in Definition 12: a recursive strategy $(v_1, z^h, z^g, \mathcal{V})$ with $x_t = z^h(v_t)$, $y_t = z^g(v_t)$, $v_{t+1} = \mathcal{V}(v_t, x_t, y_t)$ and $v_1 \in V$ is an SPE if, for every promised value v it can reach, (1) $z^h(v) = z^g(v)$; (2) $\mathcal{V}(v, z^h(v), \eta) \in V$ for all η ; and (3) $v = (1 - \delta)r(z^h(v), z^g(v)) + \delta \mathcal{V}(v, z^h(v), z^g(v)) \geq (1 - \delta)r(z^h(v), \eta) + \delta \mathcal{V}(v, z^h(v), \eta)$ for all η .

(h) Take $v_1 = v^N$, $z^h(v) = z^g(v) = \theta U^*$ and $\mathcal{V} \equiv v^N$. Condition (1) holds, and (2) holds because v^N is the value of this very strategy. Condition (3) reduces to $r(y^N, y^N) \geq r(y^N, \eta)$ for all η , which holds because $y^N = H(x^N)$.

(i) With $U^* = 5$, $\theta = 1$, the deviation payoff from a competitive equilibrium (y, y) is

$$r(y, H(y)) = \max_{\eta} r(y, \eta) = -\left(\frac{5+y}{2}\right)^2.$$

To see this, note that $H(y) = (5+y)/2$ and $U = 5 - (H(y) - y) = (5+y)/2$, so $r = -.5(U^2 + H^2) = -H^2$. The method has two steps.

- (1) *Lower bound.* Let $\underline{v}$ be the worst SPE value and (y, y) the first-period outcome of an SPE that attains it. Deviating to $H(y)$ yields at least $(1-\delta)r(y, H(y)) + \delta\underline{v}$, so $\underline{v} \geq r(y, H(y)) \geq \min_{y \in [0,10]} r(y, H(y)) = -56.25$, attained at $y = 10$.
- (2) *Attainment.* Candidate: the stick $y^\# = 10$ (so $x = 10$, $r(10, 10) = -62.5$), with any deviation restarting the equilibrium. The binding incentive constraint $\underline{v} = (1-\delta)r(10, 10) + \delta v_1 = (1-\delta)r(10, H(10)) + \delta\underline{v}$ gives $\underline{v} = -56.25$ and

$$v_1 = \frac{\underline{v} - (1-\delta)r(10, 10)}{\delta} = \frac{-56.25 + 3.125}{.95} = -55.921.$$

It remains to check that v_1 is an SPE value. The best SPE value, with $\underline{v}$ as punishment, is Ramsey, because $r(0, 0) = -12.5 \geq (1-\delta)r(0, H(0)) + \delta\underline{v} = .05(-6.25) + .95(-56.25) = -53.75$. So $v_1 \in [\underline{v}, \bar{v}] = [-56.25, -12.5]$, and part (j) constructs an SPE that delivers it.

Hence the worst SPE value is -56.25 . As an independent check, `ch24.py` iterates the APS operator on intervals, starting from $[\min_C r, \max_C r] = [-62.5, -12.5]$. It converges to $[-56.25, -12.5]$.

The other values are:

- repeated Ramsey, $v^R = -12.5$ (an SPE here);
- repeated Nash, $v^N = -25$;
- Abreu's stick-and-carrot, with one period of the stick $y = 10$ followed by Ramsey forever: $\tilde{v} = .05(-62.5) + .95(-12.5) = -15$.

Abreu's profile is an SPE, since $\tilde{v} \geq r(10, H(10)) = -56.25$ is its binding self-enforcing constraint. But with $\delta = .95$ one period of stick is too mild: $\tilde{v} = -15$ is *better* for the government than Nash. The worst value -56.25 needs a much longer or harsher stick.

(j) A recursive SPE with two states attains the worst value (the chapter's method 3).

- State $\underline{v} = -56.25$: play $x = y = 10$. If this is observed, move to v_1 ; after any deviation, stay at $\underline{v}$.
- State $v_1 = -55.921$: play $x = y = \tilde{y}$, where $r(\tilde{y}, \tilde{y}) = v_1$, i.e. $\tilde{y} = \sqrt{-2v_1 - 25} = 9.3189$. If this is observed, stay at v_1 ; after any deviation, move to $\underline{v}$.

Start at $\underline{v}$. Promise keeping: $.05(-62.5) + .95v_1 = -56.25$, and $v_1 = r(\tilde{y}, \tilde{y})$ is stationary. Incentives: at $\underline{v}$ the constraint holds with equality ($\underline{v} = r(10, H(10))$). At v_1 , $-55.921 \geq .05r(\tilde{y}, H(\tilde{y})) + .95(-56.25) = .05(-51.26) - 53.44 = -56.000$. The outcome path is $y_1 = 10$, then $y_t = 9.3189$ forever. Method 1 also works: $y = 10$ for 40 periods, with the promised value rising from -56.25

to -13.867 , then $y = 7.393$ once, then Ramsey forever. (At the transition, $-13.867 \geq -55.357$.) So the worst value can be attained with a path that ends in Ramsey.

(k) Ramsey with Nash reversion requires $v^R \geq (1 - \delta)r(0, H(0)) + \delta v^N$, i.e. $-12.5 \geq -6.25(1 - \delta) - 25\delta$, i.e. $18.75\delta \geq 6.25$. So $\delta_c = 1/3$.

(l) With the worst SPE as punishment, the condition is $-12.5 \geq -6.25(1 - \delta) - 56.25\delta$, i.e. $50\delta \geq 6.25$, so $\tilde{\delta}_c = 1/8$. This uses the worst value -56.25 , which is available at $\tilde{\delta}_c$. There the continuation value needed after the stick is $v_1 = (-56.25 + .875 \cdot 62.5)/.125 = -12.5 = v^R$, so Abreu's stick-and-carrot (one period at $y = 10$, then Ramsey) is exactly the worst SPE, and the worst value at $\tilde{\delta}_c$ is -56.25 . For $\delta < 1/8$ the needed v_1 would exceed the best value, the worst value rises above -56.25 , and Ramsey is not sustainable at all (Figure S24.1).

(m) For $\delta = .08 < \tilde{\delta}_c$, neither $y = 10$ as a stick nor Ramsey as a carrot is available. The best and worst SPEs support each other. The worst is a stick y_{lo} followed by the best SPE; the best is a stationary y_{hi} whose deviations are punished by the worst. Both incentive constraints bind, which gives the system (24.13.5)–(24.13.8):

$$\underline{v} = r(y_{lo}, H(y_{lo})), \quad \bar{v} = \frac{1 - \delta}{\delta} [\underline{v} - r(y_{lo}, y_{lo})] + \underline{v}, \quad \bar{v} = r(y_{hi}, y_{hi}), \quad \bar{v} = (1 - \delta)r(y_{hi}, H(y_{hi})) + \delta \underline{v}.$$

With $r(y, y) = -.5(25 + y^2)$ and $r(y, H(y)) = -((5 + y)/2)^2$, the solution is

$$y_{lo} = 8.2, \quad y_{hi} = 1.8, \quad \underline{v} = -6.6^2 = -43.56, \quad \bar{v} = -.5(25 + 1.8^2) = -14.12.$$

Check: $.92(-.5(25 + 8.2^2)) + .08(-14.12) = -43.56$ and $.92(-3.4^2) + .08(-43.56) = -14.12$. The APS iteration gives the same interval $[-43.56, -14.12]$. Figure S24.1 shows how both values depend on δ . Below $1/8$ the set of SPE values shrinks toward the Nash value -25 as $\delta \rightarrow 0$.

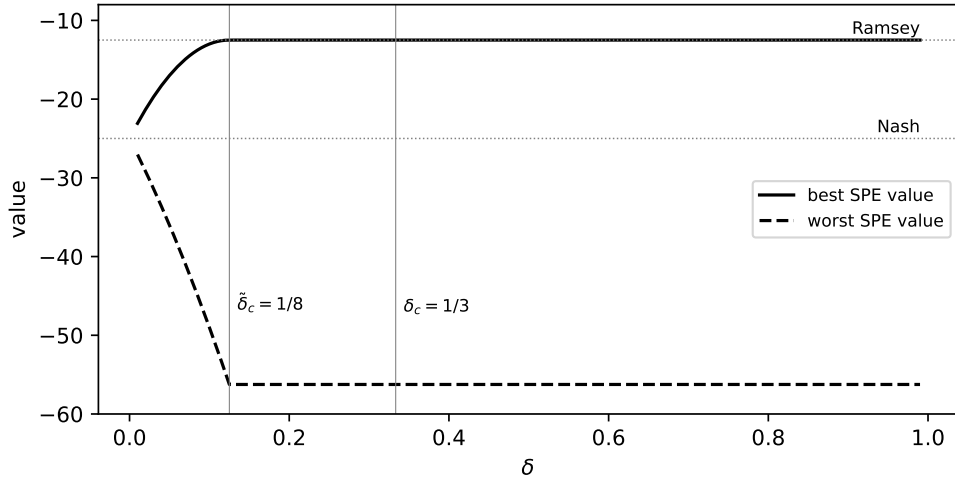


Figure S24.1. Best and worst SPE values in the Kydland–Prescott example ($U^* = 5$, $\theta = 1$), from APS iteration. For $\delta \geq 1/8$ they are the Ramsey value -12.5 and $\min_y r(y, H(y)) = -56.25$.

Chapter 26

Two Topics in International Trade

Solutions to both exercises of Chapter 26 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). Both concern the Bond–Park model of gradualism in trade policy (Section 26.3). Numbers and the figure come from `code/ch26.py`, which also checks the closed-form constrained Pareto frontier by value-function iteration on the Bellman equation (26.3.26).

Recall the setting. Country L sets a tariff $t_L \geq 0$ and country S makes a voluntary transfer $e_S \geq 0$, simultaneously, every period. Payoffs are $r_L = u_L(t_L) + e_S$ and $r_S = u_S(t_L) - e_S$, and deviations are punished by reversion to the static Nash equilibrium $(t_L^N, 0)$, with values $v_i^N = u_i(t_L^N)/(1 - \beta)$. Write $W = u_W(0)/(1 - \beta)$ for the value of free trade to the world. The constrained Pareto frontier $P(v_L)$ solves (26.3.26)–(26.3.27): maximize $u_S(t_L) - e_S + \beta P(y)$ subject to promise keeping $u_L(t_L) + e_S + \beta y \geq v_L$ (multiplier θ), L 's participation constraint $u_L(t_L) + \beta y \geq u_L(t_L^N) + \beta v_L^N = v_L^N$ (multiplier μ_L ; the transfer e_S cancels because a deviating L keeps it), and S 's participation constraint $-e_S + \beta P(y) \geq \beta v_S^N$ (multiplier μ_S).

Exercise 26.1 A quadratic Bond–Park economy

Preliminaries. The small country's payoff is $u_S = u_W - u_L = \frac{1}{8} - \frac{1}{2}t_L$, which is linear and decreasing. The functions have the properties used in Section 26.3: u_L is strictly concave with $u_L'(0) = \frac{1}{2} > 0$, and u_W is strictly concave with $u_W'(0) = 0$ and $u_W'(t) < 0$ for $t > 0$. The parameter γ_S enters only through the upper bound $t_L \leq 1 - \gamma_S = .6$ on tariffs consistent with trade. That bound does not bind, so the Nash tariff is $t_L^N = \min\{\arg \max u_L, 1 - \gamma_S\} = \min\{.5, .6\} = .5$. Then

$$u_L(t_L^N) = 0, \quad u_L(0) = -\frac{1}{8}, \quad u_S(0) = \frac{1}{8}, \quad u_S(t_L^N) = -\frac{1}{8}, \quad u_W(0) = 0,$$

so $v_L^N = 0$, $v_S^N = -\frac{1}{8(1-\beta)}$ and $W = 0$. The unconstrained frontier is $v_S = W - v_L = -v_L$.

(a) By (26.3.23),

$$\beta_c = \sqrt{\frac{u_L(t_L^N) - u_L(0)}{u_S(0) - u_S(t_L^N)}} = \sqrt{\frac{1/8}{1/4}} = \frac{1}{\sqrt{2}} = .70711.$$

For $\beta \in (\beta_c, 1)$, (26.3.21)–(26.3.22) give

$$v_L^* = \frac{v_L^N - u_L(0)}{\beta} = \frac{1}{8\beta}, \quad v_L^{**} = \frac{u_L(0) + \beta(u_S(0) - u_S(t_L^N))}{1 - \beta} = \frac{2\beta - 1}{8(1 - \beta)}.$$

Indeed $v_L^* < v_L^{**}$ if and only if $1 - \beta < \beta(2\beta - 1)$, i.e. $2\beta^2 > 1$, i.e. $\beta > \beta_c$. The corresponding time-invariant transfers (26.3.18) and (26.3.20) are $e_{S\min} = 1/(8\beta)$ and $e_{S\max} = \beta/4$. The largest value L can be promised (reached in part (b)) is $v_L^{\max} = u_L(t_L^N) + \beta(W - v_S^N) = \beta/(8(1 - \beta))$.

β	v_L^*	v_L^{**}	$v_L^{\max}$	v_S^N	$e_{S \min}$	$e_{S \max}$
.75	.16667	.25000	.37500	-.50000	.16667	.18750
.80	.15625	.37500	.50000	-.62500	.15625	.20000
.90	.13889	1.00000	1.12500	-1.25000	.13889	.22500
.95	.13158	2.25000	2.37500	-2.50000	.13158	.23750

(b) The frontier is defined on $[v_L^N, v_L^{\max}]$ and has two pieces.

- *Regions III and I*, $v_L^N \leq v_L \leq v_L^{**}$: $P(v_L) = W - v_L = -v_L$. In both regions tariffs are zero from the start (Section 26.3.9), and transfers alone deliver v_L (constant in region I, rising once in region III), so the frontier coincides with the unconstrained one.
- *Region II*, $v_L^{**} < v_L \leq v_L^{\max}$: here S 's participation constraint binds. Solve (26.3.35) for a given $\theta > 1$. First, (26.3.35a), $-\frac{1}{2} + \theta(\frac{1}{2} - t_L) = 0$, gives

$$t_L = \frac{1}{2}(1 - \theta^{-1}) \in (0, t_L^N).$$

Next, (26.3.35c) and (26.3.35e) give $e_S + \beta y = \beta(W - v_S^N) = \beta/(8(1 - \beta))$; only this sum is determined. Then (26.3.35b) gives $v_L = u_L(t_L) + \beta(W - v_S^N) = v_L^{\max} - \frac{1}{2}(\frac{1}{2} - t_L)^2$, and (26.3.35d), with (26.3.35c), gives $P(v_L) = u_S(t_L) + \beta v_S^N$. Eliminating t_L ,

$$t_L = \frac{1}{2} - \sqrt{2(v_L^{\max} - v_L)}, \quad \theta = \frac{1}{2\sqrt{2(v_L^{\max} - v_L)}}, \quad P(v_L) = v_S^N + \sqrt{\frac{1}{2}(v_L^{\max} - v_L)}.$$

So in region II the frontier is an arc of the parabola $(v_S - v_S^N)^2 = \frac{1}{2}(v_L^{\max} - v_L)$. At v_L^{**} we have $v_L^{\max} - v_L^{**} = \frac{1}{8}$, so $P = v_S^N + \frac{1}{4} = W - v_L^{**}$ and $P' = -1$ (smooth pasting, $\theta = 1$, $t_L = 0$). At $v_L^{\max}$, $t_L = t_L^N$, $P = v_S^N$ and the slope $-\theta$ is $-\infty$. Throughout region II, $P'(v_L) = -\theta$, as the envelope condition requires. The frontier is concave and lies strictly inside the unconstrained frontier on $(v_L^{**}, v_L^{\max}]$ because the period-0 tariff is distorting. For $\beta = .8$:

v_L	.375	.400	.425	.450	.475
t_L (period 0)	0	.0528	.1127	.1838	.2764
θ	1	1.118	1.291	1.581	2.236
$P(v_L)$	-.3750	-.4014	-.4314	-.4669	-.5132

From $t = 1$ on the economy is in region I with $t_L' = 0$ and a constant transfer $e_S' = (1 - \beta)y - u_L(0)$. Any $y \in [v_L^*, v_L^{**}]$ can be used, with $e_S = \beta/(8(1 - \beta)) - \beta y$. For $\beta = .8$ ($e_S + \beta y = .5$): $y = v_L^{**}$ gives constant transfers $e_S = e_S' = .2 = e_{S \max}$, while $y = v_L^*$ gives $e_S = .375$ and $e_S' = .15625 = e_{S \min}$. In every case $e_S' \leq e_S$, as the chapter shows, and `ch26.py` checks promise keeping and both participation constraints.

Numerical check. `ch26.py` iterates on the Bellman operator (26.3.26)–(26.3.27) for $\beta = .8$ on a grid of 401 promised values in $[v_L^N, W - v_S^N]$ and 601 tariffs in $[0, 1 - \gamma_S]$. It starts from the unconstrained frontier, an upper bound on P , and uses the smallest transfer that keeps the promise. The iteration converges after two steps (the optimal continuation values lie in region I, where the starting guess is already correct). It shrinks the domain to $[0, .5] = [v_L^N, v_L^{\max}]$, matches the closed form exactly on regions III and I, and matches it to 4×10^{-4} in region II, where the tariff grid (step .001) limits accuracy. Figure S26.1 plots the frontiers.

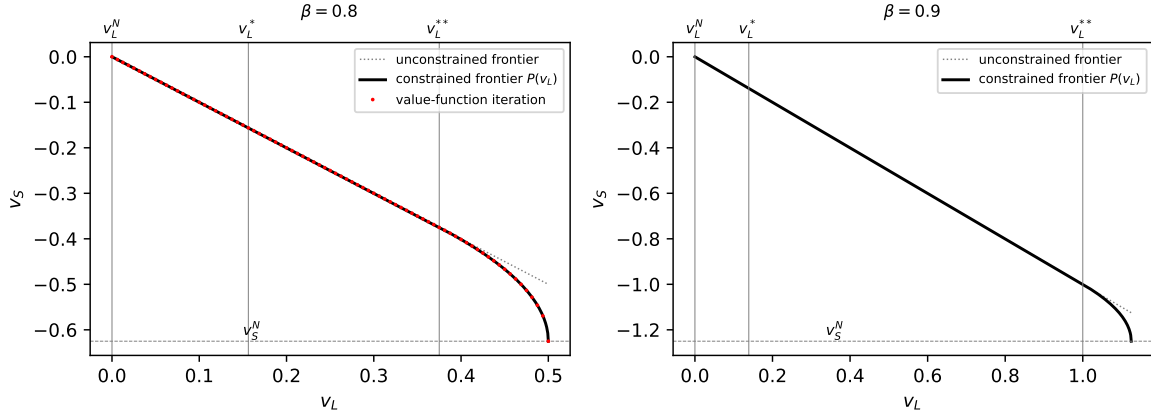


Figure S26.1. Constrained Pareto frontier $v_S = P(v_L)$ for $\beta = .8$ and $\beta = .9$. It is linear, on the unconstrained frontier, up to v_L^* , and then bends down to $(v_L^{\max}, v_S^N)$. The dots for $\beta = .8$ are the value-function iteration.

(c) In region III, $v_L \in (v_L^N, v_L^*) = (0, \frac{1}{8\beta})$. Follow (26.3.36) with $t_L = 0$ from the start. L 's participation constraint holds with equality but with $\mu_L = 0$, so (26.3.36c) gives the continuation value; (26.3.36b) gives the period-0 transfer; and the stationary region-I rule gives the transfer from period 1 on:

$$y = \frac{v_L^N - u_L(0)}{\beta} = v_L^* = \frac{1}{8\beta}, \quad e_S = v_L - u_L(0) - \beta y = v_L - v_L^N = v_L,$$

$$e'_S = (1 - \beta)y - u_L(0) = \frac{1}{8\beta}.$$

Also $P(v_L) = -v_L$ and $P(y) = -\frac{1}{8\beta}$. Transfers rise from $e_S = v_L$ to $e'_S = v_L^* = e_{S\min}$ and then stay constant; $e'_S - e_S = y - v_L > 0$. S 's participation constraints hold: at $t = 0$, $-e_S + \beta P(y) = -v_L - \frac{1}{8} \geq \beta v_S^N = -\frac{\beta}{8(1-\beta)}$ because $v_L < v_L^* < v_L^{**}$, and from $t = 1$ on, $-e'_S + \beta P(y) = -\frac{1+\beta}{8\beta} \geq \beta v_S^N$ because $1 - \beta^2 \leq \beta^2$, i.e. $\beta \geq \beta_c$. For $\beta = .8$ ($v_L^* = .15625$):

v_L	y	e_S	e'_S	admissible y
0	.15625	0	.15625	{.15625}
.05	.15625	.05	.15625	[.15625, .21875]
.10	.15625	.10	.15625	[.15625, .28125]
.15	.15625	.15	.15625	[.15625, .34375]

The formulas above give the baseline policy. As Section 26.3.10 shows, any $y \in [v_L^*, \min\{(v_L - u_L(0))/\beta, v_L^{**}\}]$ (last column) also works, with $e_S = v_L - u_L(0) - \beta y$ and $e'_S = (1 - \beta)y - u_L(0)$.

Remark. In (26.3.23) the denominator should read $u_S(0) - u_S(t_L^N)$; the book prints $u_S(t_i^N)$.

Exercise 26.2 No binding participation constraint for L in region III

Take $v_L \in (v_L^N, v_L^*)$, and suppose an optimum of (26.3.26)–(26.3.27) has multipliers $\theta \geq 0$, $\mu_L > 0$ and $\mu_S = 0$. We use four facts.

- (i) Every feasible pair of values lies weakly below the unconstrained frontier: $v_L + v_S = \sum_t \beta^t u_W(t_{Lt}) \leq W$, since u_W is maximized at 0. In particular $P(y) + y \leq W$ for every y .
- (ii) $P = W - v_L$ on region I, $[v_L^*, v_L^{**}]$: the baseline policy with $t_L = 0$ and a constant transfer attains it (Section 26.3.9). That argument does not use region III.
- (iii) P is concave, and $-\theta$ is a supergradient of P at v_L (the envelope condition $P'(v_L) = -\theta$ when P is differentiable).
- (iv) Complementary slackness, and the first-order conditions (26.3.28): with $\mu_S = 0$, (26.3.28c) reads $\theta \leq 1$, with equality if $e_S > 0$.

Case $e_S = 0$. Since $\mu_L > 0$, L 's participation constraint binds: $u_L(t_L) + \beta y = v_L^N$. Promise keeping with $e_S = 0$ requires $u_L(t_L) + \beta y \geq v_L > v_L^N$, a contradiction.

Case $e_S > 0$. Then $\theta = 1$ by (iv), so promise keeping binds: $v_L = u_L(t_L) + e_S + \beta y$. The tariff condition (26.3.28a) becomes

$$u'_S(t_L) + (1 + \mu_L)u'_L(t_L) = u'_W(t_L) + \mu_L u'_L(t_L) \leq 0, \quad \text{with equality if } t_L > 0.$$

At $t_L = 0$ the left side is $\mu_L u'_L(0) > 0$, so $t_L > 0$: L must be allowed a distorting tariff to satisfy its binding participation constraint.

The pair is not on the frontier. Adding promise keeping and the objective,

$$v_L + P(v_L) = u_W(t_L) + \beta(y + P(y)) \leq u_W(t_L) + \beta W < u_W(0) + \beta W = W,$$

by (i) and because u_W is strictly decreasing on $t \geq 0$ and $t_L > 0$. So $P(v_L) < W - v_L$.

The pair is on the frontier. By (iii) with $\theta = 1$, $P(v) \leq P(v_L) - (v - v_L)$ for all v . Take any $v \in [v_L^*, v_L^{**}]$, where $P(v) = W - v$ by (ii). Then $W - v \leq P(v_L) - v + v_L$, i.e. $P(v_L) \geq W - v_L$.

The two conclusions contradict each other: $(v_L, P(v_L))$ both is and is not on the unconstrained Pareto frontier. Hence $\mu_L > 0$, $\mu_S = 0$ is impossible for $v_L \in (v_L^N, v_L^*)$. The resolution in the text is $\mu_L = 0$ with L 's participation constraint binding only "barely": $t_L = 0$, $\theta = 1$, and L 's continuation value is raised to $y = v_L^*$ through the rising transfers of Exercise 26.1(c).

At the endpoint $v_L = v_L^N$ the argument does not apply, and no contradiction arises. There the baseline policy has $e_S = 0$, and promise keeping and L 's participation constraint are then the same inequality, $u_L(t_L) + \beta y \geq v_L^N$. Only the sum $\theta + \mu_L$ is determined ($= 1$ from (26.3.28b) with $y = v_L^*$), so a split with $\mu_L > 0$ is consistent with the first-order conditions. It describes the same allocation, $t_L = 0$, $y = v_L^*$, on the unconstrained frontier.

Chapter 27

Fiscal-Monetary Theories of Inflation

Solutions to all 8 exercises of Chapter 27 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). The exercises are analytical. `code/ch27.py` checks the numbers and closed forms: the deficit adjustment of 27.1, Brock's example, the Ramsey allocation of 27.3, the anticipated open-market price path of 27.4(e), and the elasticity condition of 27.5.

Notation follows Section 27.2. $R_{mt} = p_t/p_{t+1}$ is the return on currency, R_t the gross real interest rate, $1 + i_t = R_t/R_{mt}$ the gross nominal rate, and $h_t = M_{t+1}/p_t$ real balances. In a stationary equilibrium $R = \beta^{-1}$, money demand is $f(R_m) = F(c, R_m/R)$, and seigniorage is $f(R_m)(1 - R_m)$ (27.2.21)–(27.2.22).

Exercise 27.1 *Why deficits in Italy and Brazil were once extraordinary proportions of GDP*

(a) With $p_{t-1}/p_t = 1$, $\alpha = (1.02 - 1)/(1.02 - 1) = 1$. The nominal and real interest components coincide: $.5(1 - 1/1.02) = .0098$, i.e. 0.98% of GDP.

(b) With 100% inflation, $p_{t-1}/p_t = .5$ and $\alpha = .02/(1.02 - .5) = .02/.52 = .0385$. The reported interest bill is $.5[1 - 1/(1.02 \times 2)] = .2549$, 25.5% of GDP, but the real interest cost is only $.0385 \times 25.5\% = 0.98\%$ of GDP, the same as in (a). The other 24.5 points are the inflation compensation built into nominal rates: they amortize the real value of the debt, which inflation erodes. So they are not a burden that must be covered by taxes or money creation. This is why reported deficits of 30% of GDP in high-inflation Brazil (or very large ones in heavily indebted Italy) overstate the real fiscal problem.

Exercise 27.2 *A strange example of Brock (1974)*

(a) A competitive equilibrium is a positive price sequence $\{p_t\}$, an allocation $\{c_t, m_{t+1}\}$ and a government policy $\{g_t, \tau_t, M_{t+1}\}$ such that: (i) given prices and taxes, $\{c_t, m_{t+1}\}$ maximizes the household's utility subject to its budget constraints (and the transversality condition); (ii) the government's budget holds at every t ; (iii) markets clear: $c_t + g_t = y$ and $m_{t+1} = M_{t+1}$.

With multiplier $\beta^t \lambda_t$ on the date- t budget, $\lambda_t = 1/c_t$, and the condition for m_{t+1} is $\gamma/m_{t+1} = \lambda_t/p_t - \beta \lambda_{t+1}/p_{t+1}$. In equilibrium $c_t = y - g$, so

$$\frac{\gamma}{h_t} = \frac{1 - \beta R_{mt}}{y - g} \iff h_t = \frac{\gamma(y - g)}{1 - \beta R_{mt}}.$$

(b) Let $x = g - \tau$. In a stationary equilibrium, the budget for $t \geq 1$ reads $x = h(1 - R_m)$, and with the money demand above

$$x = S(R_m) \equiv \gamma(y - g) \frac{1 - R_m}{1 - \beta R_m}, \quad S'(R_m) = -\gamma(y - g) \frac{1 - \beta}{(1 - \beta R_m)^2} < 0.$$

Seigniorage decreases strictly in R_m on $(0, \beta^{-1})$. It rises toward $\gamma(y - g)$ as $R_m \downarrow 0$ and toward $-\infty$ as $R_m \uparrow \beta^{-1}$. So a stationary monetary equilibrium exists iff

$$g - \tau < \gamma(y - g),$$

and it is unique. With $k = x/(\gamma(y - g)) < 1$,

$$R_m = \frac{1 - k}{1 - \beta k}, \quad h = \frac{\gamma(y - g)}{1 - \beta R_m} = \frac{\gamma(y - g) - \beta x}{1 - \beta}.$$

The date-0 budget $x = h - M_0/p_0$ gives $p_0 = M_0/(h - x)$, and $h - x = \gamma(y - g)R_m/(1 - \beta R_m) > 0$ exactly when $R_m > 0$. So the same condition guarantees $p_0 > 0$, and $p_t = p_0 R_m^{-t}$. (A surplus, $x < 0$, gives deflation, $R_m \in (1, \beta^{-1})$.) The example is “strange” because there is no Laffer curve. The marginal utility of real balances satisfies $h u_h = \gamma$, so the demand for real balances never collapses, however high inflation gets. Any deficit below $\gamma(y - g)$ is financed at a unique inflation rate, and higher deficits always mean higher inflation. Contrast the two stationary equilibria in Figure 27.2.1.

(c) From the money demand, $\beta R_{mt} = 1 - \gamma(y - g)/h_t$. The budget at $t + 1$ is $x = h_{t+1} - h_t R_{mt}$, so $\beta h_t R_{mt} = \beta(h_{t+1} - x)$. Combining,

$$h_{t+1} = x + \frac{h_t - \gamma(y - g)}{\beta}, \quad t \geq 0,$$

together with the date-0 condition $x = h_0 - M_0/p_0$. Any positive solution with $R_{mt} = (h_{t+1} - x)/h_t > 0$ that satisfies the transversality condition is an equilibrium.

(d) The fixed point is $h^* = [\gamma(y - g) - \beta x]/(1 - \beta)$, which is the h of (b). It is positive, and the implied $R_m = (h^* - x)/h^*$ is positive, iff $x < \gamma(y - g)$, the condition of (b). The difference equation has slope $1/\beta > 1$, so every other solution diverges from h^* . If $h_0 < h^*$, h_t falls until $h_{t+1} < x$, which would require $R_m < 0$. If $h_0 > h^*$, h_t grows like β^{-t} and $R_{mt} \rightarrow \beta^{-1}$, so $\beta^t \lambda_t h_t$ does not vanish and the household’s transversality condition fails. (In the code, a 10^{-3} perturbation reaches $h < x$ after 170 periods, or grows to 34.6 after 200 periods.) The stationary equilibrium is therefore the unique equilibrium, and p_0 is determined.

Exercise 27.3 Optimal inflation tax in a cash-in-advance model

Setup. With $\alpha = 0$, $y_t = n_t$ and the real wage is one. Preferences are $\sum \beta^t (c_t^\gamma / \gamma - n_t)$, $0 < \gamma < 1$. We keep Ireland’s timing: consumption at t must be paid with cash brought into the period (net of bond trades), and wages, taxed at τ_t , are paid in cash at the end of the period:

$$c_t \leq \frac{m_t}{p_t} + b_t - \frac{b_{t+1}}{R_t}, \quad c_t + \frac{b_{t+1}}{R_t} + \frac{m_{t+1}}{p_t} = \frac{m_t}{p_t} + b_t + (1 - \tau_t)n_t.$$

Private bonds are in zero net supply. As the exercise says, the government finances g by taxes and money creation only, $g = \tau_t n_t + (M_{t+1} - M_t)/p_t$. Feasibility is $c_t + g = n_t$.

The implementability condition. With multipliers $\beta^t \mu_t$ (cash constraint) and $\beta^t \lambda_t$ (budget), the first-order conditions are $c_t^{\gamma-1} = \lambda_t + \mu_t$, $1 = \lambda_t(1 - \tau_t)$, $\lambda_t = \beta c_{t+1}^{\gamma-1} p_t / p_{t+1}$, and $1 + i_t = (\lambda_t + \mu_t) / \lambda_t$. Hence

$$1 + i_t = c_t^{\gamma-1} (1 - \tau_t), \quad 1 = \beta c_{t+1}^{\gamma-1} \frac{1 - \tau_t}{\pi_{t+1}}, \quad \pi_{t+1} \equiv \frac{p_{t+1}}{p_t}.$$

A unit of work at t yields $(1 - \tau_t)/\pi_{t+1}$ units of consumption at $t + 1$: *the inflation tax is a tax on labor income*, because wages must be held as cash for a period. When the cash constraint binds, $m_{t+1}/p_t = (1 - \tau_t)n_t$ and $c_{t+1} = m_{t+1}/p_{t+1} = (1 - \tau_t)n_t/\pi_{t+1}$. Multiplying the labor condition by c_{t+1} ,

$$n_t = \beta c_{t+1}^\gamma \implies c_t + g = \beta c_{t+1}^\gamma, \quad t \geq 0. \quad (*)$$

The tax rate has dropped out. The government's budget is implied: $(M_{t+1} - M_t)/p_t = (1 - \tau_t)n_t - c_t$, so it reduces to $c_t + g = n_t$. Any τ_t can be combined with the inflation rate $\pi_{t+1} = \beta(1 - \tau_t)c_{t+1}^{\gamma-1}$, and the effective after-tax real wage $(1 - \tau_t)/\pi_{t+1} = c_{t+1}/n_t$ depends only on the allocation.

The Ramsey problem is to maximize $\sum \beta^t [c_t^\gamma/\gamma - c_t - g]$ subject to $(*)$, choosing c_0 as well, since $p_0 = M_0/c_0$ is free. The stationary points solve $c + g = \beta c^\gamma$. For $g < (\beta\gamma)^{1/(1-\gamma)}(1 - \gamma)/\gamma$ there are two, $c_L < c_H$. The forward map $c_{t+1} = ((c_t + g)/\beta)^{1/\gamma}$ is increasing, with slope $c^{1-\gamma}/(\gamma\beta)$ at a fixed point. That slope exceeds one at c_H and is below one at c_L . Paths starting below c_H stay below it forever and converge to c_L (or turn infeasible). Paths starting above c_H explode, with $n_t \rightarrow \infty$ and utility $\rightarrow -\infty$. Period utility $c^\gamma/\gamma - c - g$ increases on $(0, 1)$, and $c_H^{1-\gamma} < \beta$ gives $c_H < 1$. So the constant path dominates every alternative date by date:

$$c_t = c_H, \quad n_t = c_H + g, \quad \text{effective tax on labor income } 1 - \frac{c_H}{n} = \frac{g}{c_H + g}.$$

The government balances its budget every period through a single effective wedge, and *the split between the labor tax and the inflation tax is indeterminate*. Every $\tau \leq 1 - c_H^{1-\gamma}$ (the bound is $i \geq 0$), paired with $\pi = \beta(1 - \tau)c_H^{\gamma-1}$, supports the Ramsey allocation. At $\tau = 1 - c_H^{1-\gamma}$ we get $\pi = \beta$ and $i = 0$: the Friedman rule is *an* optimal policy, but not the only one. Example ($\gamma = .5$, $\beta = .95$, $g = .1$): $c_H = .68796$, $n = .78796$, and an effective tax of 12.69%. One implementation is the Friedman rule with $\tau = .1706$ and 5% deflation; another is $\tau = 0$ with 14.5% inflation, which gives the same allocation and welfare.

This contrasts with Section 27.5, where money saves shopping time and the Friedman rule is uniquely optimal. Here money is needed only because labor income arrives as cash, so money and labor are taxed on the same base. (Had the government also been allowed to issue bonds, the date-0 price level would tax the initial money stock M_0 lump-sum, and the Ramsey planner would want an initial hyperinflation, as in Section 27.3.8. That is why the problem restricts financing to taxes and seigniorage.) As in any Ramsey problem, the planner selects the equilibrium: for a given tax sequence, paths converging to c_L are also equilibria.

Exercise 27.4 *Deficits, inflation, and anticipated monetary shocks (Manuelli)*

(a) With $h_t = m_{t+1}/p_t$, the budget is $c_t + B_{t+1}/R_t + h_t + \tau_t = y + B_t + h_{t-1}R_{m,t-1}$. Leisure depends on (c_t, h_t) . Write $u = c^\alpha \ell^{1-\alpha}$, so $u_c = \alpha u/c$ and $u_\ell \ell = (1 - \alpha)u$. The first-order conditions are

$$\lambda_t = u_c + u_\ell \frac{\partial \ell}{\partial c} = \frac{u}{c} [\alpha - \eta(1 - \alpha)], \quad \lambda_t = \beta R_t \lambda_{t+1}, \quad \frac{(1 - \alpha)u}{h_t} = \lambda_t - \beta \lambda_{t+1} R_{mt} = \lambda_t \frac{i_t}{1 + i_t}.$$

The assumption $\alpha - \eta(1 - \alpha) > 0$ makes $\lambda_t > 0$. Dividing,

$$\frac{m_{t+1}}{p_t} = \kappa c_t \frac{1 + i_t}{i_t} = \frac{\kappa c_t}{1 - R_{mt}/R_t}, \quad \kappa = \frac{1 - \alpha}{\alpha - \eta(1 - \alpha)} > 0,$$

which decreases in i_t (the derivative of $(1+i)/i$ is $-1/i^2$).

(b) With c constant, $\lambda_t \propto u_t$ depends on ℓ_t , i.e. on real balances. In a stationary equilibrium (constant real balances and R_m , which is the natural outcome of a constant policy), λ_t is constant and $\lambda_t = \beta R \lambda_{t+1}$ gives $R = \beta^{-1}$. (Along a nonstationary path $R_t = \beta^{-1}(\ell_t/\ell_{t+1})^{1-\alpha}$ would vary with real balances; such paths behave like the unstable solutions of 27.2.)

(c) With $R = \beta^{-1}$, $R_m/R = \beta R_m$, and the stationary budget is

$$d = h(1 - R_m) = \kappa(y - g) \frac{1 - R_m}{1 - \beta R_m} \equiv S(R_m), \quad S'(R_m) < 0,$$

the same shape as in Brock's example. Seigniorage rises monotonically with inflation toward $\kappa(y - g)$ as $R_m \rightarrow 0$. So every deficit $d < \kappa(y - g)$ can be financed, the supremum $\kappa(y - g)$ is approached only as inflation becomes infinite, and larger deficits cannot be financed. The economist is right *in this model and in stationary comparisons*. With g and B fixed, a larger d (lower τ) must be financed by seigniorage, and since S is strictly monotone the stationary inflation rate must rise; near the upper bound it rises explosively. The claim relies on the fiscal authority not adjusting taxes later (fiscal dominance, "unpleasant arithmetic"). If a higher d today is financed by issuing debt that is later serviced by higher taxes, inflation need not change. Moreover, a model with a Laffer curve (two stationary equilibria, as in Figure 27.2.1) would have "perverse" comparative statics at the low-return equilibrium.

(d) With $d = 0$ before and after the operation, the stationary equilibrium has zero seigniorage, $R_m = 1$, and real balances $\bar{h} = \kappa(y - g)/(1 - \beta)$ (because $1 + i = \beta^{-1}$). At T the unexpected purchase raises the money stock to M' . The interest saved on the retired bonds is returned through lower taxes from $T + 1$ on, so $d = 0$ again and the new stationary equilibrium has the same $R_m = 1$ and $\bar{h}$. Money demand $M'/p_T = \bar{h} = M/p$ then gives $p_T = (1 + \mu)p$: the price level jumps in proportion to money, and real variables are unchanged.

(e) Now the operation is anticipated. For $t < T$ the money stock is M , and money demand requires $M/p_t = \kappa(y - g)/(1 - \beta p_t/p_{t+1})$, with terminal condition $p_T = (1 + \mu)p$ (from T on the economy is in the stationary equilibrium of (d)). With $q_t = p_t/p$ and $M/p = \bar{h}$, this becomes $1/q_t = (1 - \beta)/(1 - \beta q_t/q_{t+1})$, i.e.

$$q_t = \frac{q_{t+1}}{(1 - \beta)q_{t+1} + \beta}, \quad q_T = 1 + \mu.$$

The map $q \mapsto q/((1 - \beta)q + \beta)$ is increasing, fixes $q = 1$, and satisfies $1 < \cdot < q$ for $q > 1$. Solving backward, $1 < q_0 < q_1 < \dots < q_T = 1 + \mu$. The price level jumps at the announcement ($q_0 > 1$) and then rises gradually. Expected inflation lowers money demand even before money grows. Inflation between $T - 1$ and T is $q_T/q_{T-1} = (1 - \beta)(1 + \mu) + \beta = 1 + (1 - \beta)\mu < 1 + \mu$. Example: with $\mu = .1$, $T = 10$ and $\beta = .95$, prices jump 5.76% at $t = 0$ and rise by only 0.5% between $T - 1$ and T .

Exercise 27.5 *Interest elasticity of the demand for money (Manuelli)*

Remark. The exercise asks about the *income* elasticity, but income (consumption) does not change in this experiment, and the title refers to the *interest* elasticity. The relevant condition is on the interest elasticity.

This is the unpleasant monetarist arithmetic of Section 27.3.4. In a stationary equilibrium for $t \geq 1$, (27.2.22)–(27.2.23) give

$$g - \tau + B \frac{R-1}{R} = S(R_m) \equiv f(R_m)(1 - R_m), \quad \frac{M_0}{p_0} = f(R_m) - (g + B_0 - \tau_0) + \frac{B}{R}.$$

The open-market sale raises B (and lowers M_1) with g, τ, τ_0 fixed. From the first equation, $dR_m/dB = (R-1)/(RS'(R_m))$. On the good side of the Laffer curve ($S' < 0$) the permanent deficit rises and R_m falls: inflation rises permanently. From the second,

$$\frac{d(M_0/p_0)}{dB} = \frac{1}{R} \left[1 + \frac{f'(R_m)(R-1)}{S'(R_m)} \right].$$

The price level p_0 rises iff M_0/p_0 falls, i.e. (with $S' < 0$) iff $f'(R-1) > -S' = f - f'(1 - R_m)$, i.e. $f'(R_m)(R - R_m) > f(R_m)$. Let $e \equiv f'(R_m)R_m/f(R_m) = \partial \ln F / \partial \ln(R_m/R)$ be the elasticity of money demand with respect to the return on money, i.e. minus its elasticity with respect to $1 + i$. Using $R/R_m = 1 + i$, the condition becomes

$$e > \frac{R_m}{R - R_m} = \frac{1}{i} \quad \Longleftrightarrow \quad \left| \frac{\partial \ln F}{\partial \ln i} \right| = \frac{e i}{1 + i} > \frac{1}{1 + i}.$$

We also need $e < R_m/(1 - R_m)$ (when $R_m < 1$) for the good side of the Laffer curve, $S' < 0$. The contraction raises the initial price level iff money demand is sufficiently interest elastic, roughly iff its elasticity with respect to the net nominal rate exceeds one. The higher future inflation needed to pay interest on the extra debt then depresses today's demand for money by more than the fall in M_1 . The code confirms the sign of dp_0/dB on both sides of $e = 1/i$ for constant-elasticity money demand.

Exercise 27.6 Dollarization (Manuelli)

Let $F \equiv F(y - g, \beta R_m)$ be peso real balances and $F^* \equiv F(y - g, \beta R_m^*) > F$ the (larger) real dollar balances A's residents will hold at the lower U.S. inflation rate.

(a) Dollarization hands A's inflation-tax base to the U.S. After the switch, A's residents hold F^* in dollars. They must acquire those dollars once, and then each period buy $F^*(1 - R_m^*)$ more to offset U.S. inflation, paying with exports (goods) to the U.S. Valued at $t = 0$ with $R = \beta^{-1}$, the U.S. gains

$$PV_{US} = F^* + \sum_{t \geq 1} \beta^t F^*(1 - R_m^*) = F^* \frac{1 - \beta R_m^*}{1 - \beta} = \frac{F^*}{1 - \beta} \frac{i^*}{1 + i^*},$$

the present value of the interest that A's residents forgo by holding U.S. currency. This is the most A can extract, and it is what A should ask for: a lump sum PV_{US} , or equivalently a stream of U.S. seigniorage on A's dollar holdings. A's own fiscal loss is the cost of redeeming its currency, F , plus the lost seigniorage $\beta d/(1 - \beta)$: $PV_A = F(1 - \beta R_m)/(1 - \beta)$. That is the minimum A should accept, unless it values the lower inflation its residents enjoy (less shopping time), which is a private gain but not fiscal revenue.

(b) Economist 1 assumes no compensation (at most the redemption of pesos). Then A loses its whole seigniorage flow, and taxes must rise by $d = F(1 - R_m)$ per period. Economist 2 assumes that A negotiates a seigniorage-sharing agreement, under which the U.S. rebates the flow it collects

from A's residents, $F^*(1 - R_m^*)$ per period. Then taxes rise only by $F(1 - R_m) - F^*(1 - R_m^*)$. This is positive because A is on the good side of its Laffer curve and $R_m^* > R_m$. Each statement is correct under its own assumption about compensation. Economist 1 overstates the tax increase whenever A obtains anything. Economist 2's number still omits two items. The one-time stock component (F^* at $t = 0$) is a further gain to the U.S. that A could claim; annuitized, full compensation PV_{US} is worth $F^*(1 - \beta R_m^*)$ per period, more than $F^*(1 - R_m^*)$. And the redemption of the peso stock F is a cost to A. With full compensation for both, the per-period tax increase is $(1 - \beta)(PV_A - PV_{US}) = F(1 - \beta R_m) - F^*(1 - \beta R_m^*)$, which can even be negative.

Exercise 27.7 *Currency boards (Manuelli)*

(a) With p^* constant, $p = ep^*$ is constant and $R_m = 1$. Substitute $M_t = eB_t^*$: the dollar-bond terms become $M_{t+1}/(Rp)$ and M_t/p , and

$$g + B\left(1 - \frac{1}{R}\right) = \tau + \frac{M_{t+1}}{p}\left(1 - \frac{1}{R}\right) \implies \tau = g + (1 - \beta)[B - F(y - g, \beta)].$$

Even without inflation, the board earns interest on the reserves that back the currency. This is the board's seigniorage, $(1 - \beta)F(y - g, \beta)$ per period, and its size is set by the demand for money. When y rises permanently (with g fixed, $c = y - g$ rises), money demand rises by ΔF . Households acquire the extra pesos by bringing dollars (earned by net exports) to the board, which invests them in dollar bonds. The interest on these reserves lets taxes fall permanently by $(1 - \beta)\Delta F$.

(b) With e fixed, $p_t = ep_t^*$ grows at π^* : *domestic inflation equals foreign inflation*, and $R_m = 1/\pi^*$. The dollar-bond term becomes $M_{t+1}/(R\pi^*p_t)$, so

$$\tau = g + (1 - \beta)B - F\left(y - g, \frac{\beta}{\pi^*}\right)\left(1 - \frac{\beta}{\pi^*}\right).$$

The board now earns the U.S. *nominal* rate on its reserves, $i^*/(1 + i^*) = 1 - \beta/\pi^*$ per unit of money, so it collects the inflation tax on domestic money holders. Higher π^* raises this rate but shrinks the base F . Writing $z = \beta/\pi^*$, revenue $F(y - g, z)(1 - z)$ rises with π^* (and taxes fall) iff the elasticity of money demand with respect to z is below $z/(1 - z)$, i.e. on the good side of the corresponding Laffer curve. With very elastic money demand, imported inflation lowers revenue and raises taxes.

(c) A surprise, permanent rise from e to e' raises the price level to $p' = e'p^*$. The existing money stock $M = eB^*$ loses a fraction $1 - e/e'$ of its real value. Real money demand is unchanged if the devaluation is believed to be once and for all. Under the rule $M = e'B^*$ the board's reserves now back $e'B^*$ pesos, so it can issue $(e' - e)B^*$ new pesos. Households want them to rebuild their real balances. This is a one-time levy on money holders worth $(1 - e/e')F$ in real terms. With B fixed, the government can take it as a one-time tax cut of that size at the devaluation date. Alternatively, it can hold the proceeds in extra foreign bonds and lower taxes permanently by $(1 - \beta)(1 - e/e')F$ per period. Recurring taxes are otherwise unaffected, since the board's interest income depends only on real money demand. The catch is credibility. Once devaluations are expected, domestic nominal rates include expected depreciation, money demand falls, and so does the board's future revenue.

Exercise 27.8 Growth and inflation (Manuelli)

(a) Write $h = m_{t+1}/p_t$ and $z = h/c$. Then $H = \psi c/h = \psi/z$, $H_c = \psi/h$, $H_{m/p} = -\psi c/h^2$, $\ell = 1 - \psi/z$, and $u_c/u_\ell = \varphi \ell / ((1 - \varphi)c)$. Multiplying the money-demand condition (27.2.16) by c ,

$$\left(1 - \frac{R_m}{R}\right) \left[\frac{\varphi}{1 - \varphi} \left(1 - \frac{\psi}{z}\right) - \frac{\psi}{z} \right] = \frac{\psi}{z^2},$$

an equation in z and R_m/R alone. So $h_t = z(R_m/R) c_t$: real balances are proportional to consumption. With R_m/R constant, $h_{t+1} = \gamma h_t$.

(b) With ℓ constant (constant z), $\lambda = u_c - u_\ell H_c$ is proportional to $c^{\varphi(1-\sigma)-1}$. By (27.2.14),

$$R = \frac{1}{\beta} \frac{\lambda_t}{\lambda_{t+1}} = \frac{\gamma^{1-\varphi(1-\sigma)}}{\beta},$$

which increases with the growth rate, since $1 - \varphi(1 - \sigma) > 0$. Faster growth means a higher real interest rate.

(c) With $p_t = p$, $R_m = 1$ and $h_t = z(1/R)c_t$ grows at γ , so $M_{t+1} = ph_t$ must grow at γ as well. Seigniorage is $(M_{t+1} - M_t)/p = h_t - h_{t-1} = h_t(1 - 1/\gamma) > 0$ without any inflation. Growth raises the demand for real balances, and the government meets it by printing money. The revenue is a transfer from households who must acquire the new balances by giving up goods. For example, $\gamma = 1.05$ yields seigniorage equal to 4.8% of real balances each period at zero inflation.

(d) After a stabilization, money demand rises for two reasons. Growth raises c_t (part (a)), and lower inflation raises R_m/R , and hence z , a remonetization. The central bank can then expand the nominal money supply substantially while the price level stays constant, because the extra money only meets the larger demand for real balances. The seigniorage from growing money demand also helps the fiscal side of the reform. Rapid money growth after a successful reform is therefore not inflationary, as long as it does not outrun the growth in money demand.

Chapter 28

Credit and Currency

Solutions to all 11 exercises of Chapter 28 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). The exercises are analytical. `code/ch28.py` checks the closed forms with log utility and $\beta = .9$.

Throughout, “odd” agents are endowed at even dates $t = 0, 2, \dots$ and “even” agents at odd dates, unless stated otherwise. $c_0 \in (.5, 1)$ denotes the root of $\beta = u'(c_0)/u'(1 - c_0)$ (28.4.4); with $u = \ln c$, $c_0 = 1/(1 + \beta)$. The stationary monetary allocation (28.4.3) gives the agent endowed at t consumption c_0 and the other $1 - c_0$. For brevity we normalize money per odd–even pair.

Exercise 28.1 *Arrow–Debreu*

The aggregate endowment is constant, so guess constant consumption c^o, c^e with $c^o + c^e = 1$. The first-order conditions $\beta^t u'(c^h) = \mu_h q_t^0$ then give $q_t^0 = \beta^t$ (normalizing $q_0^0 = 1$). Present values of endowments are $\sum_t \beta^t y_t^o = (1 + \beta)/(1 - \beta^3)$ and $\sum_t \beta^t y_t^e = \beta^2/(1 - \beta^3)$. Budget balance, $c^h/(1 - \beta) = \sum \beta^t y_t^h$, gives

$$c^o = \frac{(1 - \beta)(1 + \beta)}{1 - \beta^3} = \frac{1 + \beta}{1 + \beta + \beta^2}, \quad c^e = \frac{\beta^2}{1 + \beta + \beta^2}, \quad q_t^0 = \beta^t.$$

These add to one, so markets clear, and constant consumption satisfies each agent’s first-order conditions. The allocation is Pareto optimal: it is the point of the family (28.3.3) with $u'(c^o)/u'(c^e) = (1 - \theta)/\theta$.

Exercise 28.2 *One-period consumption loans*

(a) A competitive equilibrium is a sequence of gross interest rates $\{R_t\}$ and an allocation $\{c_t^h, b_t^h\}_{h=o,e}$ such that: (i) given $\{R_t\}$, each household maximizes $\sum \beta^t u(c_t^h)$ subject to its budget constraints and a no-Ponzi condition (e.g. the natural debt limit); (ii) goods and loan markets clear, $c_t^o + c_t^e = 1$ and $b_t^o + b_t^e = 0$.

(b) Guess $R_t = \beta^{-1}$ and the Arrow–Debreu allocation (28.3.5), $c^o = 1/(1 + \beta)$ and $c^e = \beta/(1 + \beta)$. Euler equations $u'(c_t) = \beta R_t u'(c_{t+1})$ hold with constant consumption. Odd’s budget: at $t = 0$, $b_0^o = 1 - c^o = \beta/(1 + \beta)$. At $t = 1$, $b_1^o = R b_0^o - c^o = 1/(1 + \beta) - 1/(1 + \beta) = 0$. The cycle then repeats: odd lends $\beta/(1 + \beta)$ at even dates and holds nothing at odd dates. The even agent borrows $\beta/(1 + \beta)$ at even dates and repays $1/(1 + \beta)$ at odd dates. Loan markets clear, debts are bounded, and transversality holds.

(c) Yes. The allocation is the Arrow–Debreu allocation of Section 28.3.2, which is Pareto optimal (constant consumption, a point of (28.3.3)). In this deterministic economy a sequence of one-period loan markets spans everything the complete time-0 markets can do (Section 28.3.4).

Exercise 28.3 Stock market

(a) An equilibrium is a price sequence $\{a_t^o, a_t^e\}$ and an allocation $\{c_t^h, s_t^{ho}, s_t^{he}\}$ such that: (i) each household maximizes utility subject to its budget constraints (and transversality); (ii) $c_t^o + c_t^e = 1$, $s_t^{oo} + s_t^{eo} = 1$ and $s_t^{oe} + s_t^{ee} = 1$.

(b) Price the trees with the Arrow–Debreu prices $q_t^0 = \beta^t$, ex dividend. At even dates the odd tree’s next dividend is two periods away and the even tree’s one period away, so

$$a_t^o = \frac{\beta^2}{1 - \beta^2}, \quad a_t^e = \frac{\beta}{1 - \beta^2} \quad (t \text{ even}); \quad a_t^o = \frac{\beta}{1 - \beta^2}, \quad a_t^e = \frac{\beta^2}{1 - \beta^2} \quad (t \text{ odd}).$$

These satisfy $a_t^j = \beta(a_{t+1}^j + y_{t+1}^j)$, the Euler equation at constant consumption. At $t = 0$ the odd agent diversifies into $1/(1 + \beta)$ shares of each tree and holds them forever; the even agent holds $\beta/(1 + \beta)$ of each. The odd agent’s time-0 wealth is $1 + a_0^o = 1/(1 - \beta^2)$ and his portfolio costs $(\beta^2 + \beta)/[(1 + \beta)(1 - \beta^2)]$, so $c_0^o = (1 - \beta)/(1 - \beta^2) = 1/(1 + \beta)$. Afterwards he consumes the dividends of his portfolio, which are $1/(1 + \beta)$ every period, because at every date exactly one tree pays one unit. Likewise $c_t^e = \beta/(1 + \beta)$. Markets clear and all first-order conditions hold.

(c) The allocation is the Arrow–Debreu allocation (28.3.5). The two trees have linearly independent payoff streams, so trading them each period spans the (deterministic) date-contingent claims: the stock market completes the market.

Exercise 28.4 Inflation

(a) Guess M_t/p_t constant, i.e. $p_t = zp_{t-1}$. The first-order conditions are $u'(c_t)/p_t \geq \beta z u'(c_{t+1})/p_{t+1}$, with equality if $m_t > 0$. Since proportional transfers pay “interest” $z - 1$ on money, the effective gross return on money is $zp_t/p_{t+1} = 1$. For the odd agent at $t = 0$ the condition reads $u'(c_0) = \beta u'(1 - c_0)$, which is (28.4.4): c_0 is the *same* as without inflation. At $t = 0$ the even agent spends zM_{-1} and the odd agent ends the period with all the money. The odd agent’s budget gives $p_0(1 - c_0) = M_0 = zM_{-1}$ (per pair), so

$$p_0 = \frac{zM_{-1}}{1 - c_0}, \quad p_t = z^t p_0 = \frac{M_t}{1 - c_0}.$$

The even agent’s first-order condition holds as an inequality: $u'(1 - c_0) \geq \beta u'(c_0)$, true since $1 - c_0 < c_0$.

(b) The allocation does not depend on z . Transfers proportional to holdings are like a stock split: they change the units in which money is measured but not its real return. Inflation is neither good nor bad, and both odd and even agents are indifferent among economies with different z . This contrasts with Section 28.6, where money is injected by equal lump-sum transfers. There inflation lowers the return on money and moves the allocation along the frontier in Figure 28.6.1.

Exercise 28.5 *A Friedman-like scheme*

With $p_t = (1 + \tau)p_{t-1}$, the odd agent's first-order condition at $t = 0$, with $m_0^o > 0$, is (28.6.2): $1/(1 + \tau) = u'(c_0)/[\beta u'(1 - c_0)]$. Hence

$$\frac{u'(c_0)}{u'(1 - c_0)} = \frac{\beta}{1 + \tau} > 1 \implies c_0 < .5.$$

The even agent spends all his money at $t = 0$ and holds none into $t = 1$. His first-order condition (28.6.3) must hold as an inequality: $1/(1 + \tau) \leq u'(1 - c_0)/[\beta u'(c_0)]$. Substituting (28.6.2) gives $[u'(c_0)/u'(1 - c_0)]^2 \leq 1$, i.e. $c_0 \geq .5$, a contradiction. If money paid more than β^{-1} , the even agent, who consumes $1 - c_0 > .5$ at $t = 0$ and $c_0 < .5$ at $t = 1$, would also want to carry money from $t = 0$ to $t = 1$. That breaks the periodic pattern. So the part of the frontier ABC in Figure 28.6.1 that is reachable by stationary monetary equilibria stops at the Friedman-rule point A, $c_0 = .5$ and $1 + \tau = \beta$.

Exercise 28.6 *Distribution of currency*

(a) An equilibrium with valued currency is a price sequence $0 < p_t < \infty$, an allocation $\{c_t^o, c_t^e\}$ and money holdings $\{m_t^o, m_t^e\} \geq 0$ such that each agent maximizes $\sum \beta^t \ln c_t$ subject to $p_t c_t + m_t \leq p_t y_t + m_{t-1}$, given initial money. Markets clear: $c_t^o + c_t^e = 1$ and $m_t^o + m_t^e = H$ (per pair).

(b) We construct an equilibrium that is stationary from $\bar{t} = 1$ on. For $t \geq 1$ take the stationary equilibrium: the agent without endowment holds all the money and spends it, the endowed agent consumes $c_0 = 1/(1 + \beta)$, and $H/\bar{p} = 1 - c_0 = \beta/(1 + \beta)$, so $\bar{p} = H(1 + \beta)/\beta$. At $t = 0$ the even agent spends all his money, and the odd agent keeps his own αH , sells goods for the even agent's $(1 - \alpha)H$, and enters $t = 1$ with H . The odd agent's Euler equation between 0 and 1, with $c_1^o = \beta/(1 + \beta)$, is $1/(c_0^o p_0) = \beta(1 + \beta)/(\beta \bar{p})$, i.e. $p_0 c_0^o = H/\beta$. His budget gives $p_0(1 - c_0^o) = (1 - \alpha)H$. Hence

$$p_0 = \frac{H[1 + \beta(1 - \alpha)]}{\beta}, \quad c_0^o = \frac{1}{1 + \beta(1 - \alpha)}, \quad c_0^e = \frac{\beta(1 - \alpha)}{1 + \beta(1 - \alpha)},$$

$$p_t = \bar{p} = \frac{H(1 + \beta)}{\beta} \quad (t \geq 1).$$

Check the even agent at $t = 0$. He carries no money into $t = 1$, where he is endowed, and needs $1/(c_0^e p_0) \geq \beta(1 + \beta)/\bar{p} = \beta^2/H$, i.e. $1 \geq \beta^2(1 - \alpha)$, which always holds. From $t = 1$ on all conditions are those of the stationary equilibrium. So an eventually stationary equilibrium exists for every α . For $\alpha = 0$ it is stationary from $t = 0$ ($p_0 = \bar{p}$). A larger α means less money is spent at $t = 0$, a lower initial price level, and a return $p_0/\bar{p} < 1$ between 0 and 1. The initially money-rich odd agents consume more at $t = 0$. At $\alpha = 1$ the even agents own nothing at $t = 0$ and consume nothing. From $t = 1$ on the initial distribution leaves no trace.

Exercise 28.7 *Capital overaccumulation*

(a) In each high period an agent stores k and eats it in the next (low) period. Maximizing $\ln(1 - \varepsilon - k) + \beta \ln(\varepsilon + \delta k)$ gives $\varepsilon + \delta k = \beta \delta (1 - \varepsilon - k)$, i.e.

$$k = \frac{\beta \delta (1 - \varepsilon) - \varepsilon}{\delta(1 + \beta)}, \quad c_H = \frac{\delta(1 - \varepsilon) + \varepsilon}{\delta(1 + \beta)}, \quad c_L = \beta \delta c_H = \frac{\beta[\delta(1 - \varepsilon) + \varepsilon]}{1 + \beta},$$

with $k > 0$ for small ε . Nobody stores in a low period. Each period only the N agents who are currently endowed store, so total storage is Nk per period ($\approx N\beta/(1+\beta)$ for small ε), of which a fraction $1 - \delta$ is lost.

(b) An equilibrium with valued currency is a positive price sequence $\{p_t\}$, allocations $\{c_t^h, k_t^h \geq 0, m_t^h \geq 0\}$ and, for each agent, a solution to $\max \sum \beta^t \ln c_t$ subject to $p_t c_t + p_t k_t + m_t \leq p_t y_t + p_t \delta k_{t-1} + m_{t-1}$. Markets clear: $\sum_h c_t^h + \sum_h k_t^h = 1 + \delta \sum_h k_{t-1}^h$ (per pair) and $\sum_h m_t^h = M$. In a stationary equilibrium with constant p , money returns $1 > \delta$, so nobody stores. The endowed agent sells goods for money, and with $u'(c_H) = \beta u'(c_L)$ we get $c_H = 1/(1+\beta)$ and $c_L = \beta/(1+\beta)$. Real balances are $M/p = 1 - \varepsilon - c_H = \beta/(1+\beta) - \varepsilon > 0$ (per pair), which pins down p . The even agents start with the money and are not endowed at $t = 0$, which is consistent with the pattern from $t = 0$ on. Pareto dominance: in its high period an agent can move goods forward at the gross return 1 with money, instead of $\delta < 1$ with storage. For every $c_H < 1 - \varepsilon$ the monetary budget line gives $c_L = \varepsilon + (1 - \varepsilon - c_H) > \varepsilon + \delta(1 - \varepsilon - c_H)$. The monetary budget set strictly contains the autarkic one, so every odd agent is strictly better off. From $t = 1$ on even agents face the same comparison, and at $t = 0$ they consume $\beta/(1+\beta)$ instead of ε . (With $\beta = .9$, $\delta = .8$ and $\varepsilon = .05$, lifetime utility from a high date rises from -7.85 to -6.92 .) Currency also eliminates the waste of $(1 - \delta)Nk$ goods per period: autarky “overaccumulates” a low-return asset.

(c) With $\delta = 1$, storage and money (at a constant price level) both return 1, so they are perfect substitutes. An eventually stationary equilibrium is an equilibrium in which, from some $\bar{t}$ on, the allocation has period 2 and the price level is constant. Take any constant $p \geq \underline{p} \equiv M/[\beta/(1+\beta) - \varepsilon]$. Every endowed agent consumes $1/(1+\beta)$, and his saving $S = \beta/(1+\beta) - \varepsilon$ is split between real balances M/p and storage $k = S - M/p \geq 0$. At $t = 0$ the even agents spend their money: $c_0^e = \varepsilon + M/p$. Odd agents consume $1/(1+\beta)$ and store k . From $t = 1$ on the allocation is the same period-2 pattern for every p . Every such p , and $p = \infty$ (a nonmonetary equilibrium with storage only), is an equilibrium. There is thus a continuum of equilibrium price levels. They differ in how much storage is used (k) and in the time-0 allocation: $c_0^e = \varepsilon + M/p$ ranges over $[\varepsilon, \beta/(1+\beta)]$ as p falls from ∞ to $\underline{p}$. The real value of the initial currency is indeterminate, because an equally good store of value exists.

Exercise 28.8 *Altered endowments*

Guess the pattern (28.4.3) with a constant price level. The endowed agent's first-order condition is again $u'(c_0) = \beta u'(1 - c_0)$, so consumption is c_0 when endowed and $1 - c_0$ otherwise, the same allocation for every F . The endowed agent sells $1 - F - c_0$ goods for all the money, so

$$p = \frac{M}{1 - F - c_0},$$

which rises with F : the less unequal the endowments, the smaller the trade that money has to finance, and the lower the value of money. The construction requires $1 - F - c_0 > 0$. Equivalently, at autarky the endowed agent must want to save, $u'(1 - F) < \beta u'(F)$: since $u'(c)/u'(1 - c) = \beta$ at $c = c_0$, this holds iff $1 - F > c_0$. So a stationary equilibrium with valued currency exists iff $F < 1 - c_0$. As $F \uparrow 1 - c_0$, $p \rightarrow \infty$. For $F \geq 1 - c_0$ the endowment is already smooth enough that nobody wants to save through money, and the only equilibrium is autarky.

Exercise 28.9 Inside money

(a) A competitive equilibrium is prices $\{p_t\}$ ($0 < p_t < \infty$), interest rates $\{R_t\}$, and an allocation $\{c_t^h, m_t^h \geq 0, b_t^h \geq -F/R_t\}$ such that each household maximizes $\sum \beta^t u(c_t^h)$ subject to its budget constraints and initial positions. Markets clear: $c_t^o + c_t^e = 1$, $m_t^o + m_t^e = M$ (per pair) and $b_t^o + b_t^e = 0$.

(b) If $R_t > p_t/p_{t+1}$, nobody would hold currency: a holder could switch it into loans at a higher return, and a borrower could hold less currency instead of borrowing. Currency would then be valueless. If $R_t < p_t/p_{t+1}$, nobody would lend (currency dominates) while borrowers would like to borrow to hold currency, so the loan market could not clear with positive loans. With both assets held, $R_t = p_t/p_{t+1}$.

(c) With a constant price level, $R = 1$, and only each agent's total real assets $a_t = m_t/p + b_t$ matter. The constraints $m \geq 0$ and $b \geq -F$ amount to $a_t \geq -F$. Guess the allocation (28.4.3). The endowed agent's Euler equation $u'(c_0) = \beta u'(1 - c_0)$ holds with equality. The agent without endowment would like to borrow more, because $u'(1 - c_0) = u'(c_0)/\beta > \beta u'(c_0)$, so he ends his low period at the limit, $a^L = -F$. From the two budget constraints, $a^H - a^L = 1 - c_0$, so the endowed agent ends his period with $a^H = 1 - c_0 - F$. Asset market clearing gives $a^H + a^L = M/p$, hence

$$\frac{M}{p} = 1 - c_0 - 2F, \quad p = \frac{M}{1 - c_0 - 2F}.$$

This is positive iff $F < (1 - c_0)/2$. In terms of portfolios, each period the low agent borrows F , and the high agent repays the F he borrowed a period earlier and lends F . So credit moves $2F$ goods from the endowed to the non-endowed agent every period, and currency moves the remaining $1 - c_0 - 2F$. The allocation is that of the $F = 0$ economy. A larger F raises the price level: private credit (inside money) substitutes for currency, which lowers the real demand for outside money.

Remark. For this equilibrium to be stationary from $t = 0$, the time-0 creditor must be the agent without endowment at $t = 0$. He lent F in his previous, endowed period. So the equilibrium needs even agents to be owed F (and hold M) and odd agents to owe F . The exercise states the reverse. With the debts as printed, the economy reaches the same stationary equilibrium from $t = 1$ after a date-0 adjustment like the one in Exercise 28.10. The odd agent enters $t = 0$ richer by $2F$, and the date-0 price level is lower: with log utility $p_0/p = (1 - F)/(1 + F)$ and $R_0 = (1 - F)/(1 + F)$.

(d) At $F = (1 - c_0)/2$, $M/p = 0$: currency is valueless ($p = \infty$). Private loans alone support the same allocation. Each period the borrower is at the limit, credit transfers $2F = 1 - c_0$ goods, and $R = 1$ satisfies the endowed agent's Euler equation $u'(c_0) = \beta R u'(1 - c_0)$.

(e)–(f) Without valued currency, let R be the equilibrium interest rate. As long as the borrower is at the limit, the low agent receives the repayment F of his earlier loan plus a new loan F/R , so $c_L = F(1 + 1/R)$ and $c_H = 1 - c_L$. R is set by the endowed agent's Euler equation:

$$u'(1 - F(1 + 1/R)) = \beta R u'(F(1 + 1/R)), \quad u'(c_L) \geq \beta R u'(c_H).$$

At $F = (1 - c_0)/2$ the solution is $R = 1$ (part (d)). As F rises, credit smooths more, and R rises until $c_H = c_L = \frac{1}{2}$, which happens at $R = \beta^{-1}$ and $F = 1/(2(1 + \beta))$. So for $F \in ((1 - c_0)/2, 1/(2(1 + \beta)))$ there is a stationary nonmonetary equilibrium with $R \in (1, \beta^{-1})$ and a binding borrowing limit.

With log utility R solves $\beta R^2 = F(1 + R)(1 + \beta R)$; for $\beta = .9$ and $F = .25$, $R = 1.0548$ and $(c_H, c_L) = (.5130, .4870)$. For $F \geq 1/(2(1 + \beta))$ the limit no longer binds in a stationary state. Then $R = \beta^{-1}$, and each agent consumes a constant amount determined by the initial debt, as in the Arrow–Debreu equilibrium with initial claims. The agent who starts owing F consumes $1/(1 + \beta) - (1 - \beta)F$ and the other $\beta/(1 + \beta) + (1 - \beta)F$. In particular, at $F = \beta/(1 + \beta)$ (for $\beta \geq \frac{1}{2}$), $R = \beta^{-1}$ with consumption $(1 - \beta + \beta^2)/(1 + \beta)$ and $\beta(2 - \beta)/(1 + \beta)$; for $\beta = .9$, $(.4789, .5211)$.

Remark. The printed upper end of the interval in (f), $\beta/(1 + \beta)$, does not match this analysis: R reaches β^{-1} already at $F = 1/(2(1 + \beta))$. The two agree only for $\beta = \frac{1}{2}$. So the interest rate lies strictly between 1 and β^{-1} for $F \in ((1 - c_0)/2, 1/(2(1 + \beta)))$ and equals β^{-1} above that. The value $\beta/(1 + \beta)$ is the borrowing needed at even dates to implement the Arrow–Debreu allocation (28.3.5) with no initial debts, under a limit on the amount borrowed ($b \geq -F$) rather than on the repayment.

Exercise 28.10 *Initial conditions and inside money*

(a) From $t = 1$ on use the stationary equilibrium of 28.9(c): $p_t = p = M/(1 - c_0 - 2F)$, $R_t = 1$, and allocation (28.4.3). To enter it, the end-of-period-0 positions must be those of the stationary equilibrium. The even agent (endowed at $t = 1$) has borrowed to the limit, $b_0^e = -F/R_0$, and holds no money. The odd agent holds all the money and the loan, $b_0^o = F/R_0$. Both assets are held, so $R_0 = p_0/p$.

(b) Let $x = p_0/p = R_0$. Then $M/p_0 = (1 - c_0 - 2F)/x$ and $F/R_0 = F/x$. The budgets at $t = 0$ are

$$c_0^e = \frac{M}{p_0} + \frac{F}{R_0} = \frac{1 - c_0 - F}{x}, \quad c_0^o = 1 - \frac{1 - c_0 - F}{x}.$$

The odd agent's Euler equation between 0 and 1, $u'(c_0^o) = \beta R_0 u'(1 - c_0) = x u'(c_0)$, determines x :

$$u'\left(1 - \frac{1 - c_0 - F}{x}\right) = x u'(c_0), \quad x \in (0, 1),$$

since at $x = 1$ the left side is $u'(c_0 + F) < u'(c_0)$. With log utility $x = 1 - F$. Then $p_0 = (1 - F)p$, the return on currency and on loans between 0 and 1 is $R_0 = 1 - F < 1$, $c_0^o = c_0/(1 - F) > c_0$ and $c_0^e = (1 - c_0 - F)/(1 - F)$. For $\beta = .9$ and $F = .1$: $p_0/p = .9$, $c_0^o = .5848$, $c_0^e = .4152$. The borrowing-constrained even agent satisfies $u'(c_0^e) \geq \beta R_0 u'(c_0)$. Compared with 28.9, the even agents start without the claim F , so they are poorer at $t = 0$. They borrow up to the limit at $t = 0$, which needs a low interest rate, and the date-0 price level is lower than later.

Exercise 28.11 *Real bills experiment*

(a) Guess a constant price level, so $R = 1$, no interest accrues to the government, and $\bar{M}_t = \bar{M}$ for $t \geq 0$. The government rolls over its loan of Δ to the non-endowed agent each period. Private agents' problems are unchanged: the low agent still borrows up to F in total, Δ from the government and $F - \Delta \geq 0$ from the endowed agent. The endowed agent's Euler equation gives c_0 from (28.4.4), and the allocation is (28.4.3). The goods transfer from the endowed to the non-endowed agent each period is: repayment $F - \Delta$ of the private loan, plus a new private loan $F - \Delta$, plus the government's net lending of zero, plus currency. So

$$1 - c_0 = 2(F - \Delta) + \Delta + \frac{\bar{M}}{p} \implies \frac{\bar{M}}{p} = 1 - c_0 - 2F + \Delta.$$

(b) Combining with the government's time-0 budget $\bar{M} = M + \Delta p$ gives $M/p = 1 - c_0 - 2F$, i.e.

$$p = \frac{M}{1 - c_0 - 2F} \quad \text{for every } \Delta \in (0, F).$$

The price level does not depend on Δ . The extra currency Δp is matched one for one by an increase in the real demand for currency. Government lending displaces private lending, so the endowed agents hold more of their savings as currency.

(c) No, not for currency issued this way. $\bar{M}$ increases with Δ while p does not change: this is the real bills doctrine, under which money issued against sound private IOUs is not inflationary. The quantity theory holds for the unbacked outside money M : doubling M doubles p . It fails for inside money backed by private IOUs.

Chapter 29

Equilibrium Search, Matching, and Lotteries

Solutions to all 11 exercises of Chapter 29 of Ljungqvist and Sargent, *Recursive Macroeconomic Theory* (4th edition, 2018). `code/ch29.py` solves the island model of 29.1 for an example and the Gomes–Greenwood–Rebelo model of 29.2 (with a simulated unemployment series). It also carries out the numerical studies of 29.4–29.6 and the transition of 29.8. The matching models use the timing of Section 29.3, with production first and separation at the end of the period. Periods are months.

Exercise 29.1 *An island economy (Lucas and Prescott, 1974)*

Equations (1) and (2) refer to the numbered displays of the exercise in the book.

(a) Let $x_1 = x^+(\theta_L)$ be the labor force to which an island that turns bad sheds workers (29.2.4), and $x_2 = x^-(\theta_H)$ the labor force it attracts when it turns good (29.2.5). With labor movements, $x^+(\theta_L) < x^-(\theta_H)$, and the ergodic set of Section 29.2.2 is $X = \{x_1, x_2\}$. An island that turns bad loses workers immediately, down to x_1 . An island that turns good receives workers who arrive next period, up to x_2 . An island whose productivity does not change keeps its labor force.

(b) Order the states as $(L, x_1), (H, x_1), (H, x_2), (L, x_2)$, with x the beginning-of-period labor force. Next period's labor force is set by today's state: x_1 after $(L, \cdot)$ and x_2 after $(H, \cdot)$. So

$$\Gamma = \begin{bmatrix} \pi & 1-\pi & 0 & 0 \\ 0 & 0 & \pi & 1-\pi \\ 0 & 0 & \pi & 1-\pi \\ \pi & 1-\pi & 0 & 0 \end{bmatrix}, \quad n(L, x_1) = x_1, \quad n(H, x_1) = x_1, \quad n(H, x_2) = x_2, \quad n(L, x_2) = x_1.$$

In (L, x_2) , $x_2 - x_1$ workers leave and forgo this period's earnings. In (H, x_1) the newcomers arrive only next period. The stationary distribution is $(\pi, 1-\pi, \pi, 1-\pi)/2$, so $\hat{x} = (x_1 + x_2)/2$, i.e. $x_1 = 2\hat{x} - x_2$.

(c) $v(L, x_2) = \beta v_u$, because some workers leave. In (L, x_1) a worker earns the same wage $\theta_L f'(x_1)$ as a stayer in (L, x_2) and faces the same future ($x' = x_1$), so $v(L, x_1) = v(L, x_2) = \beta v_u$.

(d) Arrivals make the continuation value of (H, x_1) equal to βv_u , so $v(H, x_1) = \theta_H f'(x_1) + \beta v_u$. The departure condition (29.2.4) in (L, x_2) reads $\theta_L f'(x_1) + \beta[\pi v(L, x_1) + (1-\pi)v(H, x_1)] = \beta v_u$, i.e.

$$[\theta_L + \beta(1-\pi)\theta_H]f'(x_1) = \beta(1-\beta)v_u. \quad (\text{A})$$

The arrival condition (29.2.5) is $\pi v(H, x_2) + (1-\pi)\beta v_u = v_u$. With $v(H, x_2) = \theta_H f'(x_2) + \beta[\pi v(H, x_2) + (1-\pi)\beta v_u]$, a little algebra gives $\theta_H f'(x_2) = (1-\beta)v_u/\pi$, i.e.

$$\beta\pi\theta_H f'(x_2) = \beta(1-\beta)v_u. \quad (\text{B})$$

Equating (A) and (B) with $x_1 = 2\hat{x} - x_2$ gives (2). The left side of (2) increases in x_2 , from its value at $x_2 = \hat{x}$ to ∞ as $x_2 \rightarrow 2\hat{x}$ (Inada). The right side decreases. So a solution with $x_2 > \hat{x}$ (labor

movements) exists, and it is unique, iff the left side is smaller at $x_2 = \hat{x}$: $\theta_L + \beta(1 - \pi)\theta_H < \beta\pi\theta_H$, which is (1). (Example: $f = \sqrt{n}$, $\theta = (1, 2)$, $\beta = .95$ and $\hat{x} = 1$. For $\pi = .9$, $x_1 = .6526$ and $x_2 = 1.3474$. For $\pi = .7$ or $.6$, (1) fails and labor does not move.) Moving is worthwhile only if shocks are persistent enough, relative to the output lost in transit.

(e) Let the planner maximize $E \sum \beta^t \sum_i \theta_i f(n_i)$, where moving takes a period without output. Let $\lambda(\theta, x)$ be the social value of an extra worker at an island in state (θ, x) , and μ the value of a worker in transit, who arrives next period wherever the planner sends him. Optimality requires the following. (i) Workers are sent out of an island until the marginal stayer is worth $\beta\mu$, so $\lambda(L, x_1) = \lambda(L, x_2) = \beta\mu$. (ii) Arrivals are sent until the expected marginal value at the destination equals μ : $\pi\lambda(H, x_2) + (1 - \pi)\lambda(L, x_2) = \mu$. (iii) An extra worker in (H, x_1) produces $\theta_H f'(x_1)$ and saves one arrival: $\lambda(H, x_1) = \theta_H f'(x_1) + \beta\mu$. (iv) Marginal values satisfy the same recursions as v , with marginal products in place of wages. These are exactly the equations behind (A) and (B), with v_u replaced by μ . So the planner chooses the same x_2 , defined by (2). Starting from equal labor forces, moving the first worker from an island that has just turned bad to one that has just turned good is worthwhile iff (1) holds. The equilibrium is efficient: wages equal marginal products, and nothing distorts moving decisions.

Exercise 29.2 *Business cycles and search (Gomes, Greenwood, and Rebelo)*

Let $\tilde{\beta} = \beta(1 - \mu)$.

(a) Working at w forever is worth $w/(1 - \tilde{\beta})$. Drawing a new wage is worth $Q = b + \tilde{\beta} \int \max\{w'/(1 - \tilde{\beta}), Q\} dF(w')$, the same for everyone. So a worker holding w (a job or an offer) works iff $w \geq \bar{w} \equiv (1 - \tilde{\beta})Q$, where

$$\bar{w} - b = \frac{\tilde{\beta}}{1 - \tilde{\beta}} \int_{\bar{w}}^M (w - \bar{w}) dF(w).$$

An unemployed worker accepts the offer iff it is at least $\bar{w}$. An employed worker never quits: his wage was acceptable when he took the job, and nothing has changed since.

(b) A higher μ lowers $\tilde{\beta}$, so future wages are worth less relative to current income, and the right side of the reservation-wage equation falls. The reservation wage falls: shorter horizons make the unemployed less choosy.

(c) Survivors who drew a wage below $\bar{w}$ remain unemployed, and entrants start unemployed. So $U_{t+1} = \mu + (1 - \mu)F(\bar{w})U_t$, and

$$U = \frac{\mu}{1 - (1 - \mu)F(\bar{w})}.$$

Unemployment rises with turnover μ (entrants must search), holding $\bar{w}$ fixed, and with the reservation wage. So it rises with b and with the option value of waiting ($\tilde{\beta}$, the spread of F), and falls when offers improve.

(d) States are i.i.d. and draws do not depend on the state, so the value of drawing, $Q = b + \tilde{\beta}E_{s'}E_{w'}V_{s'}(w')$, is the same in booms and recessions. Working pays $W_B(w) = W_R(w) + z$, where $W_R(w) = w + \tilde{\beta}E_{s'}V_{s'}(w)$ and $V_s = \max\{W_s, Q\}$. So the reservation wages satisfy $W_B(w_B) = W_R(w_R) = Q$ and $w_B < w_R$. Searching is more costly in a boom, because the forgone income includes the bonus. (The gap $w_R - w_B$ is less than z , since the continuation value also rises with

w .) Workers at wages in $[w_B, w_R)$ keep working in booms and *quit when a recession starts*, to search while search is cheap. In a recession unemployed workers accept only offers above w_R , and in a boom anything above w_B .

(e) At the start of t each survivor holds a wage (his job's or last period's offer), and entrants hold none. If t is a boom,

$$U_t = \mu + (1 - \mu)F(w_B)U_{t-1}, \quad G_t = (1 - \mu)[G_{t-1} + (F(w_R) - F(w_B))U_{t-1}].$$

If t is a recession,

$$U_t = \mu + (1 - \mu)[F(w_R)U_{t-1} + G_{t-1}], \quad G_t = 0.$$

(f) Unemployment jumps at the onset of a recession, when the stock G_{t-1} of marginal jobs accumulated during booms is dissolved by quits. The jump is larger the longer the preceding boom. In a boom unemployment falls, as the unemployed accept offers down to w_B . During a long recession it declines slowly toward $\mu/[1 - (1 - \mu)F(w_R)]$. The code reproduces this pattern (our Figure S29.1) with F uniform on $[0, 1]$, $b = .5$, $z = .05$, $\beta = .99$ and $\mu = .01$. Then $w_B = .8475$ and $w_R = .8718$ (without cycles $\bar{w} = .8764$), and unemployment moves between 6.6% and 7.8%. In this model recessions are times of intensive search: unemployment rises because workers choose to reallocate when the opportunity cost is low.

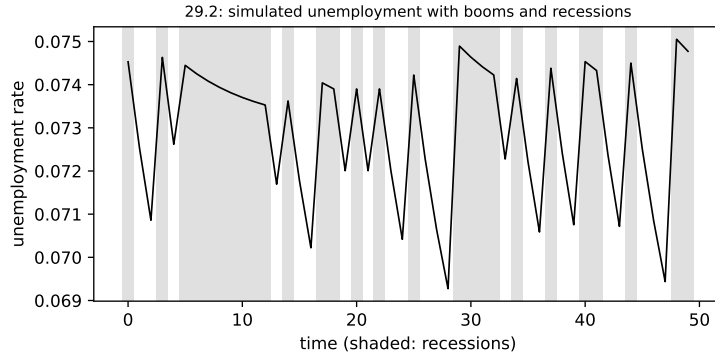


Figure S29.1. Exercise 29.2: simulated unemployment with i.i.d. booms and recessions (shaded).

Exercise 29.3 *Business cycles and search again*

(a) Let $F_R = F$ and $F_B = F^2$ (the c.d.f. of the better of two draws), and $\tilde{\beta} = \beta(1 - \mu)$. A worker holding wage w at the start of state s has

$$V_s(w) = \max \left\{ \underbrace{w + \tilde{\beta} E_{s'} V_{s'}(w)}_{W(w)}, \underbrace{b + \tilde{\beta} \int E_{s'} V_{s'}(w') dF_s(w')}_{Q_s} \right\}.$$

The value of working, $W(w)$, is the same in both states. The value of searching depends on the state only through the offer distribution.

(b) $F^2 \leq F$, so F_B first-order dominates F_R . Since $E_{s'} V_{s'}$ is nondecreasing (and increasing above the reservation wages), $Q_B > Q_R$. With W increasing, the reservation wages $W(\bar{w}_s) = Q_s$ satisfy $\bar{w}_B > \bar{w}_R$. Employed workers with wages in $[\bar{w}_R, \bar{w}_B)$ keep their jobs in recessions but *quit when a boom arrives*, because jobs are more plentiful then. Workers above $\bar{w}_B$ never quit. Quits are

procyclical here, the opposite of Exercise 29.2, where the bonus z made searching expensive in booms. Unemployed workers are also choosier in booms. Whether unemployment rises or falls in a boom depends on how the extra draw raises job finding relative to the quits it triggers.

Exercise 29.4 *Skill-biased technological change (Mortensen and Pissarides)*

(a) Apply (29.3.14) to market i with $y = h_i$, $z = b$ and vacancy cost ch_i , and divide by h_i :

$$1 - \frac{b}{h_i} = \frac{r + s + \varphi\theta_i q(\theta_i)}{(1 - \varphi)q(\theta_i)} c, \quad u_i = \frac{s}{s + \theta_i q(\theta_i)}.$$

The right side increases in θ_i . With $b > 0$, θ_i therefore increases with h_i , and u_i falls with skill: vacancy costs scale with productivity but the outside option b does not, so the surplus share left after b grows with skill. The market for skill $h_i \leq b$ shuts down ($\theta_i = 0$, $u_i = 1$). With $b = 0$, θ and u do not depend on h . (We use the book's timing. The exercise's separations at the start of a period change only the flow-balance details.)

(b) Parameters (monthly): $r = .004$, $s = .02$, $\alpha = \varphi = .5$, $A = .35$ and $c = .65$, which give $u = 6\%$ at $h = 1$, $b = .4$. Skills are uniform on $[1 - D, 1 + D]$, so D indexes mean-preserving spreads.

aggregate u (%)	$D = 0$	$D = .2$	$D = .4$	$D = .6$
$b = 0$	4.64	4.64	4.64	4.64
$b = .2$	5.19	5.21	5.24	5.34
$b = .4$	6.00	6.05	6.22	7.61
$b = .6$	7.36	7.55	9.62	24.05

Since u_i is a decreasing, convex function of h_i when $b > 0$, spreads raise aggregate unemployment. The effect grows sharply with b , as the least skilled markets approach $h_i = b$ (Figure S29.2, left). Skill-biased change interacts with generous benefits to produce European-style unemployment.

(c) With $b_i = \rho h_i$ the left side becomes $1 - \rho$ for every skill. Tightness and unemployment are then the same in all markets, and mean-preserving spreads have no effect ($u = 6.00\%$ for $\rho = .4$). The interaction in (b) arises because a fixed benefit is relatively more generous for low-skill workers.

Exercise 29.5 *Dispersion of match values (Marimon and Zilibotti)*

(a) The match surplus is $S(p) = [p\bar{h} - (1 - \beta)U]/(1 - \beta(1 - s))$, where $(1 - \beta)U$ is the flow value of unemployment. Matches form iff $p \geq p^*$, with $p^*\bar{h} = (1 - \beta)U$. Nash bargaining gives $E(p) - U = \varphi S(p)$ and $J(p) = (1 - \varphi)S(p)$, and the wage $w(p) = \varphi p\bar{h} + (1 - \varphi)p^*\bar{h}$. Free entry, with vacancy cost $c\bar{h}$, and the value of unemployment give

$$q(\theta) = \frac{c(r + s)}{(1 - \varphi) \int_{p^*}^{\infty} (p - p^*) dG(p)}, \quad p^* = \frac{b}{\bar{h}} + \frac{\varphi\theta c}{1 - \varphi}.$$

These two equations determine (p^*, θ) . (The second is the analogue of (29.3.12), and it follows from $(1 - \beta)U = b + \beta\theta q\varphi \int_{p^*}^{\infty} S dG$ and free entry.) Steady-state unemployment is $u = s/[s + \theta q(\theta)(1 - G(p^*))]$: job finding requires both a meeting and an acceptable draw.

(b) Take $\bar{h} = 1$, $\log p \sim N(-\sigma^2/2, \sigma^2)$ (mean one), and the matching parameters of 29.4.

u (%)	$\sigma = .05$	$\sigma = .3$	$\sigma = .55$	$\sigma = .8$
$b = 0$	4.71	9.87	15.17	20.29
$b = .4$	6.31	13.80	19.28	23.88
$b = .6$	8.13	17.20	22.12	26.05
$b = .8$	12.87	22.59	25.72	28.53

Greater dispersion raises the option value of search: $\int_{p^*} (p - p^*) dG = E(p - p^*)^+$ rises under mean-preserving spreads, because $(p - p^*)^+$ is convex in p . So the reservation productivity rises, more meetings are rejected, and unemployment rises even though tightness may increase. Higher benefits raise p^* further. Over the empirically relevant range of moderate dispersion, the rise in unemployment from $\sigma = .05$ to $.3$ is 5.2 points with $b = 0$ but 9.1 with $b = .6$ (Figure S29.2, middle). The employed earn more, since only good matches survive, but at the cost of longer unemployment.

Exercise 29.6 *Idiosyncratic shocks to human capital (Ljungqvist and Sargent)*

(a) The states are (w, h) for the employed and (b, h) for the unemployed, where b is the benefit. Workers survive each period with probability $1 - \delta$ and are replaced by unemployed newcomers with $(0, h_1)$, and $\tilde{\beta} = \beta(1 - \delta)$. We assume quitters forfeit benefits. Then

$$\begin{aligned}
 W(w, h) &= \max\{wh + \tilde{\beta}K(w, h), U(0, h)\}, \\
 K(w, h) &= (1 - s)[\pi_e W(w, h) + (1 - \pi_e)W(w, h_2)] + s[\pi_u U(\gamma wh, h) + (1 - \pi_u)U(\gamma wh, h_1)], \\
 U(b, h) &= b + \tilde{\beta} \int \max\{W(w', h), U(b, h)\} dG(w').
 \end{aligned}$$

An unemployed worker accepts w' iff $w' \geq \bar{w}(b, h)$, where $W(\bar{w}, h) = U(b, h)$. The reservation wage rises with b . A stationary equilibrium is a solution of these equations together with the invariant distribution over employed (w, h) and unemployed (b, h) that the decision rules and shocks induce. The key state is a worker who was laid off from a good job, lost skills, and draws benefits γwh_2 while he can now earn only $w'h_1$.

(b) Parameters (monthly): $\beta = .995$, $\delta = .002$ (a 40-year working life), $s = .02$, $\pi_e = .99$ (slow reskilling), $1 - \pi_u = .8$, G uniform on $[0, 1]$ (100 points) and $h_2 = 1$. Steady-state unemployment in percent:

h_1	1	.9	.8	.7	.6	.5	.4
$\gamma = 0$	9.5	9.6	9.6	9.6	9.6	9.5	9.3
$\gamma = .3$	11.3	11.3	11.3	11.3	11.3	11.4	11.4
$\gamma = .5$	13.1	13.2	13.3	13.4	13.7	14.4	16.8
$\gamma = .7$	16.6	16.7	17.4	19.4	61.3	72.3	74.7

Without benefits, turbulence (a lower h_1) barely affects unemployment: skill losers simply accept lower earnings. With generous earnings-related benefits it is explosive. Once γwh_2 exceeds what a skill loser can earn, laid-off high earners rarely accept any job and stay unemployed until they retire (Figure S29.2, right). In calm times (h_1 near h_2) the welfare state works well. In turbulent times the same benefits produce mass long-term unemployment.

(c) All three models attribute high European unemployment to generous benefits in a more dispersed environment, through different channels. 29.4: a fixed benefit shuts down low-skill markets. The low tail of earnings is missing among the employed, which compresses measured earnings dispersion, while low-skill unemployment soars. Every unemployed worker has a constant hazard, so declining hazards appear only through composition across skill groups. 29.5: benefits raise the reservation match quality, truncating the wage distribution from below (less dispersion) and lengthening spells. Again there is no true duration dependence. 29.6 explains both facts best. Skill losers with benefits tied to past earnings are extremely choosy, so they accept few low-paying jobs, and wage dispersion among the employed rises less than in the U.S., where turbulence is absorbed by lower wages. They also pile up in the unemployment pool. The pool's composition shifts toward them with duration, so the aggregate hazard falls sharply with duration: long-term unemployment.

(d) In 29.4 and 29.5 a worker's situation is stationary: his prospects do not change with age or with his employment history. A finite life only adds a constant hazard of exit, which acts like a slightly lower discount factor, and steady states are essentially unchanged. In 29.6, benefits indexed to past earnings create (nearly) absorbing states: a skill loser whose benefit exceeds any wage he can earn never works again. With infinite lives these workers accumulate forever, and steady-state unemployment would be determined entirely by the mass that eventually falls into such states (in the limit, everyone). With finite lives they leave through retirement. Unemployment then depends on working-life length and on age: older workers with fewer years left to recover skills are even choosier. Life-cycle structure is therefore essential for the quantitative results.

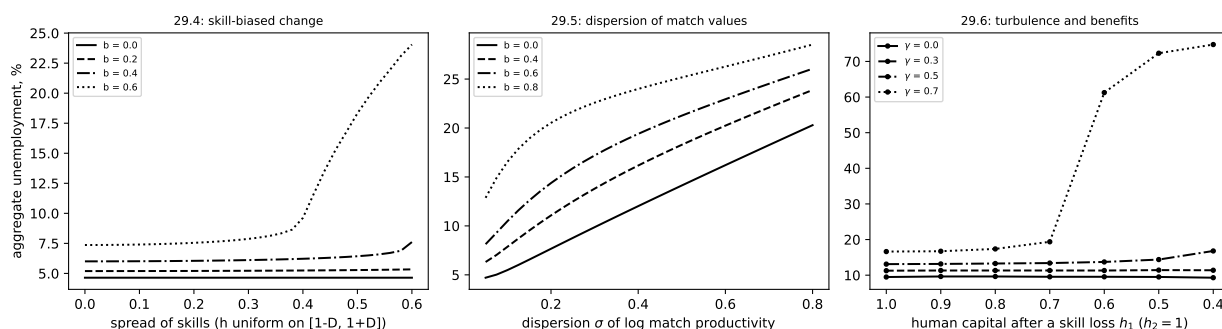


Figure S29.2. Steady-state unemployment in Exercises 29.4 (mean-preserving spreads of skills), 29.5 (dispersion of match productivity) and 29.6 (turbulence: lower human capital after a skill loss), for several benefit levels.

Exercise 29.7 *Temporary jobs and layoff costs*

(a) Let Q be the value of drawing a new wage (unemployed this period). A job at w that is never quit is worth $E(w) = w + \beta[(1 - \mu)E(w) + \mu Q]$, i.e. $E(w) = (w + \beta\mu Q)/(1 - \beta(1 - \mu))$. A worker holding w works iff $E(w) \geq Q$, i.e. iff $w \geq \bar{w} = (1 - \beta)Q$. From $Q = b + \beta \int \max\{E(w'), Q\} dF(w')$,

$$\bar{w} = b + \frac{\beta}{1 - \beta(1 - \mu)} \int_{\bar{w}}^M (w - \bar{w}) dF(w).$$

Unemployed workers accept offers above $\bar{w}$ and employed workers never quit. Shorter jobs (higher μ) lower the reservation wage.

(b) No. With both types, $E(w, \mu) = (w + \beta\mu Q)/(1 - \beta(1 - \mu)) \geq Q$ iff $w \geq (1 - \beta)Q$, for either μ . (Here Q averages over both types of offers.) A job is worth taking iff its flow w beats the flow value of search $(1 - \beta)Q$. Its duration only scales how long that gain lasts, and when the job ends the worker is back to searching anyway. Duration affects the value of a good job but not whether it is good.

(c) With the layoff cost, a job that is never quit is worth $E(w) = [w + \beta\mu(Q - \tau)]/(1 - \beta(1 - \mu))$.

(i) An unemployed worker holding an offer accepts iff $E(w) \geq Q$, i.e. $w \geq \bar{w}_u = (1 - \beta)Q + \beta\mu\tau$. (ii) An employed worker continues iff $E(w) \geq Q - \tau$, since quitting costs τ , i.e. $w \geq \bar{w}_e = (1 - \beta)(Q - \tau)$. So $\bar{w}_u - \bar{w}_e = \tau(1 - \beta + \beta\mu) > 0$. Unemployed workers are choosier because accepting a job commits them to a future layoff cost. Employed workers are more reluctant to quit because quitting costs τ now. Since accepted wages exceed $\bar{w}_u > \bar{w}_e$, no one quits.

(d) $\bar{w}_u(\mu) = (1 - \beta)Q + \beta\mu\tau$ increases with μ , so $\bar{w}_s = \bar{w}_u(\mu_s) > \bar{w}_l = \bar{w}_u(\mu_l)$. A short-lasting job brings the layoff cost sooner in expected present value, so it must pay more to be accepted. Layoff costs thus tilt acceptance toward long-lasting jobs.

(e) Employed workers at the start of $t + 1$ lose their jobs with probability μ_i . Unemployed workers at t draw an offer that starts at $t + 1$ if accepted, and a new hire does not face the loss draw in her first period. Hence

$$N_{s,t+1} = (1 - \mu_s)N_{st} + \pi_s[1 - F(\bar{w}_s)]U_t, \quad N_{l,t+1} = (1 - \mu_l)N_{lt} + \pi_l[1 - F(\bar{w}_l)]U_t,$$

$$U_{t+1} = \mu_s N_{st} + \mu_l N_{lt} + \left[1 - \pi_s(1 - F(\bar{w}_s)) - \pi_l(1 - F(\bar{w}_l))\right]U_t,$$

which preserve $N_{st} + N_{lt} + U_t = 1$.

Exercise 29.8 *Productivity shocks, job creation, and job destruction (Manuelli)*

(a) A filled job is worth $J = y - w + \beta[sV + (1 - s)J]$, and a vacancy $V = -c + \beta[qJ + (1 - q)V]$. With $V = 0$, $J = c/(\beta q)$. Then $(y - w)/(1 - \beta(1 - s)) = c/(\beta q)$, and since $(1 - \beta(1 - s))/\beta = r + s$, $w = y - (r + s)c/q(\theta)$.

(b) This is (29.3.13). Nash bargaining gives $E - U = \varphi S$ and $J = (1 - \varphi)S$. The worker's value of unemployment satisfies $rU/(1 + r) = z + \varphi\theta c/(1 - \varphi)$ (29.3.12), and the wage is the annuity value of unemployment plus the share φ of the one-period surplus. Together these give $w = z + \varphi(y - z + \theta c)$.

(c) Equating the two wage expressions gives (29.3.14), $y - z = [(r + s + \varphi\theta q(\theta))/((1 - \varphi)q(\theta))]c$. The right side increases continuously in θ (because $q' < 0$ and θq increases). So there is a unique θ , increasing in $y - z$ and decreasing in c , r , s and φ . Unemployment then follows from $u = s/(s + \theta q(\theta))$.

(d) A higher y raises the left side, so $\theta' > \theta$. In the dynamic economy, free entry must hold every period, $c = \beta q(\theta_t)J_{t+1}$, with $J_t = y' - w_t + \beta(1 - s)J_{t+1}$ and a Nash wage that depends on θ_t . None of these equations involves the unemployment rate. With constant returns to matching, the value of a vacancy depends only on the ratio θ , not on the levels of u and v . Vacancies can be opened or closed instantly. Unemployment is the only slow-moving state variable, and it does not feed back into θ . The only non-explosive solution of the forward-looking free-entry recursion is therefore $\theta_t = \theta'$ from $t = 0$ on: tightness jumps, and vacancies jump to $v_0 = \theta' u_0$.

(e) Before the shock, $JC = JD = s(1 - u_0)$ with $u_0 = s/(s + \theta q(\theta))$. At $t = 0$ job destruction is unchanged, $JD_0 = s(1 - u_0)$, while job creation jumps to $JC_0 = \theta' q(\theta') u_0 > s(1 - u_0)$. Afterwards $u_{t+1} = u_t + s(1 - u_t) - f' u_t$ with $f' = \theta' q(\theta')$, so

$$u_t - u' = (1 - s - f')^t (u_0 - u'), \quad u' = \frac{s}{s + f'}.$$

Unemployment falls monotonically to u' . Job creation $JC_t = f' u_t$ falls after its initial spike, and job destruction $JD_t = s(1 - u_t)$ rises as employment grows. Both converge to $s(1 - u')$, with $JC_t > JD_t$ throughout the transition. Vacancies $v_t = \theta' u_t$ overshoot on impact and then decline with u_t . (Example: $y = 1 \rightarrow 1.05$, $z = .4$ and the calibration of 29.4. Then θ jumps from .800 to .872, the job-finding rate from .313 to .327, and u falls from 6.00% to 5.77%. JC jumps from .0188 to .0196 and then declines, while JD rises slowly from .0188.)

Exercise 29.9 *Workweek restrictions, unemployment, and welfare (Manuelli)*

(a) A match produces $yx + z(1 - x)$, compared with z for an unmatched worker, so its flow surplus is $(y - z)x$. Since x is observed, the pair chooses it to maximize the surplus, and Nash bargaining then splits it: $x = 1$ because $y > z$. The equilibrium is that of 29.8: θ solves (29.3.14), $w = z + \varphi(y - z + \theta c)$, and $u = s/(s + \theta q(\theta))$.

(b) With $q = A\theta^{-\alpha}$, the elasticity of q is $-\alpha$, and the planner's condition is (29.3.19). The market equilibrium is efficient iff $\varphi = \alpha$ (the Hosios condition). If $\varphi > \alpha$ there are too few vacancies; if $\varphi < \alpha$, too many.

(c) The cap binds ($x = x^*$), since the surplus increases in x . Free entry now reads $(y - w)x^* = (r + s)c/q(\theta)$, where w is the hourly wage. The same steps as in 29.8, applied to the flow surplus $(y - z)x^*$, give $w = z + \varphi(y - z) + \varphi\theta c/x^*$ and

$$y - z = \frac{r + s + \varphi\theta q(\theta)}{(1 - \varphi)q(\theta)} \cdot \frac{c}{x^*}.$$

The cap acts like raising the vacancy cost from c to c/x^* : each vacancy now brings a smaller flow of output.

(d) The right side increases in θ , so tightness falls. The job-finding rate $\theta q(\theta)$ falls, unemployment $u = s/(s + \theta q(\theta))$ rises, and the number of jobs $1 - u$ falls. The hourly wage $w = z + \varphi(y - z) + \varphi\theta c/x^*$ also falls. With $q = A\theta^{-\alpha}$, the equilibrium condition gives $(y - z)(1 - \varphi) = (\theta c/x^*)[(r + s)\theta^{\alpha-1}/A + \varphi]$, and as c/x^* rises θ falls, so $\theta^{\alpha-1}$ rises and $\theta c/x^*$ must fall. Weekly earnings $w x^*$ fall by more. The cap does not share work. It destroys match surplus, so fewer jobs are created, and welfare falls. The equilibrium is not optimal: the planner would choose $x = 1$. Given the cap, the allocation is constrained efficient only if $\varphi = \alpha$.

Exercise 29.10 *Costs of creating a vacancy and optimality (Manuelli)*

(a) Firms take p_A as given, so free entry gives $y - w = (r + s)C'(v)/q(\theta)$. The Nash wage is $w = z + \varphi(y - z + \theta C'(v))$, because the cost of a vacancy enters the worker's outside option as in (29.3.12). In a steady state $v = \theta u = s\theta/(s + \theta q(\theta))$, which increases in θ . Hence θ solves

$$y - z = \frac{r + s + \varphi\theta q(\theta)}{(1 - \varphi)q(\theta)} C'\left(\frac{s\theta}{s + \theta q(\theta)}\right),$$

whose right side increases in θ , so the solution is unique. A higher y or lower z raises θ and vacancies and lowers unemployment. A steeper (or higher) C' lowers θ . Because the supply of vacancies is less than perfectly elastic, shocks to $y - z$ move θ and u less than with a constant c , and part of each shock shows up in the price p_A .

(b) The planner maximizes $\sum \beta^t [y n_t + z(1 - n_t) - C(v_t)]$ subject to $n_{t+1} = (1 - s)n_t + q(v_t/(1 - n_t))v_t$. With $q = A\theta^{-\alpha}$ (so $q + q'\theta = (1 - \alpha)q$ and $q'\theta^2 = -\alpha\theta q$), the stationary first-order conditions give

$$y - z = \frac{r + s + \alpha\theta q(\theta)}{(1 - \alpha)q(\theta)} C'(v).$$

This is the market condition with α in place of φ : the marginal social cost of a vacancy, $C'(v)$, equals its market price. The equilibrium is efficient iff $\varphi = \alpha$, the Hosios condition, and inefficient otherwise.

Exercise 29.11 *Financial wealth, heterogeneity, and unemployment (Manuelli)*

Each period an unemployed worker draws an offer, which he can take up at once. Otherwise he consumes a out of wealth. With linear utility and $1 + r = \beta^{-1}$, the timing of consumption is irrelevant, and a worker's value is his wealth plus the present value of his earnings. So rejecting an offer costs exactly one period of wage income.

(a) A type-0 worker has no wealth and cannot finance the minimum consumption for a period of search. He must take the first offer, so $w^*(0) = a$, the bottom of the support.

(b) A type-1 worker can afford one period of search and then becomes type 0. Accepting w yields $w/(1 - \beta)$ in earnings; waiting yields $\beta\bar{w}/(1 - \beta)$. Hence $w^*(1) = \beta\bar{w} > a$ by assumption. A type-2 worker who waits becomes type 1 (his wealth $a\beta$ grows to a). So $w^*(2) = \beta E \max\{w, w^*(1)\} > \beta E w = w^*(1)$, strictly because $F(w^*(1)) > 0$. Wealthier workers are choosier: across workers, financial wealth and unemployment duration are positively related, and employment probabilities fall with wealth. Wealth finances search, which liquidity constraints otherwise rule out. The poor take the first job, and the wealthy hold out for better wages.

(c) Count as unemployed in period t everyone without a job at the start of t . These are the newborns nN_{t-1} , the type-1 and type-2 workers born at $t - 1$ who rejected ($F_1 = F(w^*(1))$, $F_2 = F(w^*(2))$), and the type-2 workers born at $t - 2$ who rejected twice and must now accept. With $\varphi_2 = 1 - \varphi_0 - \varphi_1$ and $N_t = (1 + n)N_{t-1}$,

$$u_t = \frac{U_t}{N_t} = \frac{n}{1 + n} \left[1 + \frac{\varphi_1 F_1 + \varphi_2 F_2}{1 + n} + \frac{\varphi_2 F_2 F_1}{(1 + n)^2} \right],$$

which is constant: the population grows at a constant rate, and each cohort's unemployment profile is the same.

(d) Moving mass from type 0 to type 1 raises u by $\frac{n}{(1+n)^2} F_1$ per unit. Moving mass from type 1 to type 2 raises it by $\frac{n}{1+n} \left[\frac{F_2 - F_1}{1+n} + \frac{F_2 F_1}{(1+n)^2} \right] > 0$ per unit. So redistributing wealth toward the richer types raises unemployment, while it also raises the average accepted wage. Transfers to the poorest (φ_0 down) raise measured unemployment and wages, since they give workers the means to search. Redistribution from type 2 to type 0 lowers unemployment, but it also lowers wages. *Extra credit.* A type-2 worker is unemployed for $D = 1$ period with probability $1 - F_2$ (he accepts at birth),

$D = 2$ with probability $F_2(1 - F_1)$, and $D = 3$ with probability F_2F_1 (after two rejections he must accept). So $ED = 1 + F_2 + F_2F_1$, and the hazard of leaving unemployment rises with duration, to one in the third period, as his wealth runs down.